Solomonoff Axiom Induction:
Definitions, Basic Theorems, and What a Theory Is and Does

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Abstract

We study Solomonoff axiom induction (SAI), a Solomonoff-style prior over axiom systems. A universal machine reads a random tape and emits first-order sentences; the prior probability of sentences labelled true or false is the probability that the emitted theory is consistent and proves every datum. Its predictions form a belief function with three outcomes: proved, refuted, independent. We develop its basic theory and relate it exactly to Solomonoff operator induction (SOI): version by version (coin-free, one-tape, Kolmogorov), SAI agrees with SOI over consistent labellers up to constants, and on labelled atoms of a fresh predicate it is SOI on strings. Logic adds a lottery: data on which the one-tape inducer has bounded loss although no consistent r.e. theory proves them; in arithmetic, adversarially chosen true data can concentrate the limit posterior on false theories. For Boolean laws and bounded reasoners, the least length of a usable written theory depends exponentially on the pair (reasoner, vocabulary), which program length cannot see. On literal data with charged deduction, the reply to Hempel's theoretician's dilemma that Craigian axiomatisations are unmanageable becomes a theorem about Craig's construction, not about theoretical vocabulary. SAI prices approximately true laws by their calibrated error models, not by their adequacy for a purpose; Kepler's laws are worked through. The upshot: on data labelled true or false the idealised formalism cannot see the difference between a theory and a program; in the settings where we find that shape matters, it is relative to a reasoner and a vocabulary.

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*Claude is an AI system made by Anthropic. Claude produced the proofs, the computations and the text, under Kaarel Hänni's direction and starting from his questions and working notes (Hänni, 2026). The results were checked by separate Claude instances acting as referees, which re-derived the proofs of this paper's results and re-implemented its main computations with independent code. No human has yet independently checked all of the mathematics, and nothing was verified in a proof assistant.

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1 Introduction

What does a scientific theory do, and what is it? On one view, behind Solomonoff's theory of inductive inference (Solomonoff, 1964), a theory predicts observations and is a program, weighted in a Bayes mixture by its length. On another, a theory is a logical theory, a set of sentences closed under consequence (the received view; Suppe, 1977), from which answers to questions posed in some language are derived; on van Fraassen's account, explanations are such answers to why-questions (van Fraassen, 1980, ch. 5). Others hold that theories explain (Hempel and Oppenheim, 1948) or solve problems (Laudan, 1977), that they are classes of models (Suppes, 1960), or that scientific knowledge rests on exemplars and tacit skills (Kuhn, 1970; Polanyi, 1958).[1] We put the second view into the form of the first: we study Solomonoff axiom induction (SAI), a universal prior over axiom systems, sets of first-order sentences produced by programs, and use it to make precise, and to test, claims about what theories do and what they are.

The object. Fix a first-order language $\mathcal{L}$ and a decidable consistent background theory B; the data are sentences labelled true or false. A universal machine reads a random tape and emits sentences, and the prior probability $M_A(D)$ of a data set D is the probability that the emitted theory is consistent with B and proves every datum, a sentence labelled false through its negation (Definition 2.4): a hypothesis must explain each datum, not merely tolerate it. At a new sentence the posterior splits three ways: the random theory proves it, refutes it or leaves it open, and the predictions form a belief function, updated by geometric conditioning (Remark 2.6 and Proposition 3.2).

Raw data and questions. Science predicts much more particular things than a stream of observations: where Mars is at a given time, to within two minutes of arc; the period of a

[1] Several of the questions studied here first appeared in Hänni's working notes (Hänni, 2026), among them whether theories are better seen as answering questions than as predicting raw data, and how a written theory relates to the reasoner who uses it.

planet, given the size of its orbit; whether an orbit is planar. We order views of what a theory does on a ladder: (1) predicting raw data, such as pixels; (2) answering questions posed with available abstractions, such as positions, periods and orbits, which may be highly theoretical; (3) extending the language with theoretical terms that pay for themselves, the data staying in the base language; (4) revising the base abstractions, as when a concept changes its meaning or a question is shown to be ill-posed. SAI over a fixed language sits on step 2. Its base language is not an observation language in the logical empiricists' sense, so the classical objections to a theory-neutral observation language do not apply to it in the same way, although theory-ladenness biases comparisons between theories that disagree about the base language (Section 6.1). We take step 2 to be a step in the right direction, but not the end point: an account of what theories do needs steps 3 and 4 as well. We argue that it differs from step 1 in the target, the data type, logical coherence and partiality, not in the inference principle (Section 6.3 and Table 3): version by version SAI agrees, up to constants, with Solomonoff operator induction (SOI; Hänni and Claude (Anthropic), 2026c), developed in a companion paper, restricted to consistent labellers (Section 4). We have partial results on step 3 (Sections 7 and 8) and none on step 4.

What a theory is. On the second question the idealised formalism is neutral, on data labelled true or false; this rests on theorems, and the reading is ours. Every consistent labeller of sentences is an axiom system at a cost of $O(1)$ bits: coin-free SAI agrees with coin-free consistent SOI through Craig's construction inside the universal prior, and one-tape with one-tape, tape by tape (Theorems 4.2 and 4.3); new vocabulary costs $O(1)$ bits, and program length sees no reasoner (Corollary 3.6 and Theorem 7.2). So, on data labelled true or false, SAI cannot see the difference between a theory and a program; when "independent" is itself observed, partiality separates them (Section 6.4). That textbook theories are declarative (laws, definitions and equations, not procedures that print predictions) is an observation, with two qualifications: declarative is not sentential, and theories in use have procedural parts (Section 6.2). Why that shape is good is open; Craig's construction rules out an answer based only on what can be represented, and our best formal answer locates the large effects in the pair (reasoner, vocabulary), not in logical form (Section 7 and Question 6.1).

A preview: Kepler's third law. Kepler's third law K3 says that the squares of the planets' periods are proportional to the cubes of the semi-major axes of their orbits. In a language of heliocentric directions it adds nothing to the first two laws (Proposition 10.2); in a geocentric one it fixes, given the other elements of the orbits of the Earth and Mars, the size of the orbit of Mars and with it every geocentric direction of Mars (Propositions 10.3 and H.2). A computation puts what K3 thus saves on the elements of Mars at 11.3 to 14.7 bits for precisions from $8'$ to $1'$, against a law text of 375 bits in our arithmetic syntax and 165 or 115 bits in two Polish-notation codes; so in none of these codes does compression alone select K3 at Kepler's precision (an extrapolation from Mars to the other planets, conditional on the code; Example H.4). Kepler's laws are also false: against the plan94 ephemeris a fixed ellipse misses Mars by $1.6'$ to $1.9'$ over 12 to 30 years (Section 10.3); Section 9 treats such laws.

1.1 Contributions

The main results, roughly in order of importance (Section 12 says what is new):

(1) *SAI agrees with consistent SOI, version by version:* coin-free and one-tape within constant factors, Kolmogorov within an additive constant; on labelled atoms of a fresh predicate it is

SOI on strings, so the binary-label results of the companion paper that hold up to additive constants transfer (Theorems 4.2 to 4.4 and Corollary B.12).

(2) *Random theories no r.e. theory can replace:* a stream with bounded one-tape loss that no consistent r.e. theory proves, the limit posterior living on random theories (Theorem 5.1); over a decidable true background containing Robinson arithmetic Q, adversarial true data that can concentrate it on false theories (Corollary 5.5), and, on Π_1 data, a mixture equal up to a constant factor to one random true theory (Theorem 5.6).

(3) *Shape is relative to a reasoner and a vocabulary:* for Boolean laws program length sees neither, while the least written length of a usable theory depends exponentially on the pair (Theorem 7.2 and Corollary 7.6); procedural form costs a parity law a factor m, its number of query directions, over the base variables or with one round, and any law at most a factor 9 in written length with intermediate concepts and unbounded rounds (Theorem E.26).

(4) *The theoretician's dilemma:* with free deduction, theoretical vocabulary changes observational predictions only by a constant factor; on literal data with charged deduction, padded (Craig-type) axiomatisations with polynomial-time membership fit exactly the polynomial-time regularities, and vocabulary is eliminable at polynomial cost; and, with an empty background, a finite theory with only unary theoretical predicates and constants, compatible with the theory of the named objects, fits exactly the regular sets, while one binary theoretical relation fits more, and so does quantification over unnamed objects, at the price, for a theory without theoretical vocabulary, of proving a sentence false in the named structure (Theorem 8.3).

(5) *Approximate laws:* over deterministic theories, proof-requiring compatibility prefers a false exact law to its true tolerance form; on data from a law with independent errors, the best loss rate is that of the law with its error distribution (attained by M_A if computable), which is no account of approximate correctness (Propositions 9.2 and G.10 and Section 9.3).

(6) *Basic theory:* dominance; a loss at most the complexity up to deductive equivalence of any consistent r.e. theory proving the data, plus a constant (semimeasure convention); an inconsistency probability that is Solovay complete and random; no coding theorem (Theorems 3.3, 3.4, 3.7 and 3.9).

(7) *Where SAI and SOI separate:* in the semimeasure convention logical closure separates them, and so does an observed "independent" label (Propositions 4.5 and 4.6); time-bounded separations come from free deduction, not from axioms (Theorem 4.7 and Proposition 4.8); with deduction charged, on hypotheses closed under renaming of parameters, SAI is certificate SOI up to polynomial changes of the budgets (Theorem B.31).

(8) *Gödel phenomena:* over a background containing Q, belief or independent mass on a Π_1 sentence certifies its truth; over one containing Peano arithmetic PA, the coin-free posterior never learns Σ_1-soundness; the ω-gap is the ω-incompleteness of the background plus the instances (Propositions 5.8 and 5.11 and Theorem 5.9).

(9) *Best systems and explanatory power:* the best system is the same for every balance of simplicity against strength in an interval exactly when the hull of length against fit has a kink there, one aspect of what Lewis calls a kind nature; explanatory power is a per-bit log-likelihood ratio whose growth with the number of systems singles out laws (Proposition 9.3 and Definition G.29).

(10) *The orbit of Mars:* among five historical forms, on noise-free data the syntax of the law texts

decides among the forms compatible with the data; with graded fit the data decide every step but one within tens of observations (Table 5 and Example H.3).

What we do not claim. We do not claim that SAI shows a theory to be a logical theory rather than a program (on data labelled true or false it cannot see the difference), nor that the declarative shape of theories is good (Section 6.4 and Question 6.1); the results of Section 7 hold in a stated model and are not theorems about textbooks (Question 7.7). We say nothing realist about the theories an inducer favours: its limit posterior can live on random theories and, with true but adversarially chosen data, on false ones (Theorem 5.1 and Corollary 5.5). SAI is an uncomputable ideal of the evaluation of theories, not of their discovery, and its proof-requiring core is the noise-free special case of the confrontation of theory with data (Section 11). Our philosophical conclusions are arguments; Section 6.1 says which rest on theorems.

Organisation. Sections 2 to 5 develop the formalism, Sections 6 to 10 what theories do and are, and Sections 11 to 13 limitations, related work and open problems; proofs are in the appendices.

2 Setting

Hypotheses are axiom systems produced by programs; data are sentences labelled true or false, and a hypothesis is compatible with the data when it is consistent and proves every datum. Randomness enters as in the one-tape version of Solomonoff operator induction in the companion paper (Hänni and Claude (Anthropic), 2026c). Two further works are cited by name: the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d) and our earlier report (Hänni and Claude (Anthropic), 2026a). Proofs for this section and the next are in Appendix A.

2.1 Language, background and data

Definition 2.1 (Setting and data)**.** $\mathcal{L}$ is a countable first-order language with equality and computable syntax, $\mathrm{Sent} := \mathrm{Sent}_{\mathcal{L}}$, and $\vdash$ is derivability in a fixed sound and complete calculus. The *background* $B \subseteq \mathrm{Sent}$ is decidable and consistent; by default $B = \emptyset$. Put $\mathrm{Cn}(\Gamma) := \{\varphi : B \cup \Gamma \vdash \varphi\}$. "$\Gamma$ proves φ" means $B \cup \Gamma \vdash \varphi$; Γ is *consistent* if $B \cup \Gamma$ is; Γ *decides* φ if it proves φ or $\neg\varphi$; $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

A *data set* is a finite partial function $D : \mathrm{Sent} \rightharpoonup \{0,1\}$ (label 1 means true); a *three-valued* data set maps into $Y_3 := \{1, 0, \iota\}$, where ι means "independent". Put $D^{\pm} := \{\varphi^{D(\varphi)} : D(\varphi) \in \{0,1\}\}$, and let ψ_D be the conjunction of $D^{\pm}$ in a fixed canonical order ($\top$ if $D^{\pm} = \emptyset$); D is *consistent* if $B \cup D^{\pm}$ is. $D + (\varphi, y)$ is defined for $\varphi \notin \mathrm{dom}\, D$. A *reveal sequence* $(\varphi_t, y_t)_{t \le n}$ is consistent if $D_n := \{(\varphi_t, y_t) : t \le n\}$ is a partial function; D_t are its prefixes. An infinite stream D_∞ has $D_\infty^{\pm} := \bigcup_n D_n^{\pm}$. For a map ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, \varphi) := \nu(D + (\varphi, y))/\nu(D)$ for $\varphi \notin \mathrm{dom}\, D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, \varphi) := \mathbf{1}[y = D(\varphi)]$ for $\varphi \in \mathrm{dom}\, D$ (the repeated-input convention of the companion paper).

An r.e. background would serve as well, since by Craig's trick it has a decidable axiomatisation with the same consequences.

Algorithmic notation. V is a fixed universal prefix machine, K prefix complexity, $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \ge 2^{-K(i)}$ and $\Omega_V := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}$; W_e is the e-th r.e. set of sentences. Put $\Omega_V^{\ge \ell} := \sum\{2^{-|p|} : p \in \mathrm{dom}\, V,\ |p| \ge \ell\}$, the tail of Ω_V, which tends to 0 as $\ell \to \infty$; and

$K^{\mathrm{pad}}(n) := \min_{n' \geq n} K(n')$, which is nondecreasing and unbounded. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, $[w]$ the cylinder of a finite string w, and $\omega_{1:n}$ the first n bits of ω. Logarithms are binary and losses are in bits. The constants $c, c', \ldots$ depend only on the fixed universal machines, the coding of syntax and B; "lsc" means lower semicomputable. We use two standard facts (see Li and Vitányi, 2008): $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor; and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a, z) and $\sum_a P(a \mid z) \leq 1$, then $K(a \mid z) \leq -\log P(a \mid z) + K(P) + c$.

Definition 2.2 (Literal setting; fresh atoms)**.** In the *literal setting*, $\mathcal{L}$ contains a unary relation symbol R that does not occur in B, and closed terms t_w ($w \in \{0,1\}^*$) that some model of B interprets as pairwise distinct elements. *Fresh atoms* are sentences $R(t_w)$ for pairwise distinct words w; so $R(t_w)$ and $\neg R(t_w)$ are never two fresh atoms. The standard instance is the *literal language*: a constant $\mathtt{e}$, unary function symbols $\mathtt{s}_0, \mathtt{s}_1$, and the unary relation symbol R, with $t_w := \mathtt{s}_{w_1}(\cdots \mathtt{s}_{w_k}(\mathtt{e}) \cdots)$ and $\varphi_w := R(t_w)$, of $|w| + 2$ symbols; its *named structure* $\mathfrak{T}$ is the free term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$. For a set of words $X \subseteq \{0,1\}^*$, D^X lists $(\varphi_w, \mathbf{1}[w \in X])$ in length-lexicographic order and f_X is the literal labeller of X. The *fit class* of an inducer is the set of X on whose sequence D^X it has bounded loss.

Every result stated for fresh atoms or on literal data assumes this setting. Two facts carry all such uses (Lemma A.14): a set of literals of R with at most one literal per word is consistent with B, and a consistent such set proves exactly its own members among the literals.

2.2 Theory machines and their laws

A hypothesis of Solomonoff axiom induction is produced by a machine that reads random bits and writes sentences. A function machine of Solomonoff operator induction must give each input at most one label; a theory machine may write anything, and consistency is a property of its output, a random event.

Definition 2.3 (Theory machine)**.** A *theory machine* is a c.e. set $G \subseteq \{0,1\}^* \times \mathrm{Sent}$ ("after reading w from the tape, emit χ"). For finite w put $A^G_w := \{\chi : (v, \chi) \in G,\ v \sqsubseteq w\}$, and for a tape ω let $A^G_\omega := \bigcup_n A^G_{\omega_{1:n}}$ be the axioms emitted on ω. Let

$$O^G_\chi := \{\omega : B \cup A^G_\omega \vdash \chi\}, \qquad \mathrm{Cons}^G := \{0,1\}^\infty \setminus O^G_\bot, \qquad E^G_D := O^G_{\psi_D} \cap \mathrm{Cons}^G.$$

The *law* of G is $\nu_G(D) := \mathbb{U}(E^G_D)$, the probability that the random theory is consistent and proves every datum. The *three-valued law* is $\nu^3_G(D) := \mathbb{U}(E^{3,G}_D)$ with $E^{3,G}_D := E^G_D \cap \bigcap_{D(\varphi)=\iota} (\{0,1\}^\infty \setminus (O^G_\varphi \cup O^G_{\neg\varphi}))$: in addition, the theory decides no ι-labelled sentence. On Cons^G the *trichotomy function* $\mathrm{tri}^G_\omega : \mathrm{Sent} \to Y_3$ is 1 at the sentences $B \cup A^G_\omega$ proves, 0 at those it refutes and ι elsewhere. (G_i) is a fixed effective enumeration of the c.e. subsets of $\{0,1\}^* \times \mathrm{Sent}$; $K(G) := \min\{K(i) : G_i = G\}$ and, for a law ν, $K(\nu) := \min\{K(i) : \nu_{G_i} = \nu\}$.

These events are measurable: O^G_χ is effectively open uniformly in χ and an index of G, Cons^G is a Π^0_1 class, and $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\bot)$ (Lemma A.1); examples are in Appendix A.1. A theory machine can complete a law at random inside what the law allows; such randomised theories are how approximate laws enter Solomonoff axiom induction (Proposition G.4).

Definition 2.4 (Solomonoff axiom induction)**.** The *universal theory machine* is $G_U := \{(pw, \chi) : p \in \mathrm{dom}\, V,\ (w, \chi) \in G_{V(p)}\}$: it reads a self-delimiting program and runs that machine on the rest of the tape. Write $A_\omega := A^{G_U}_\omega$, $O_\chi := O^{G_U}_\chi$, $\mathrm{Cons} := \mathrm{Cons}^{G_U}$, $E_D := E^{G_U}_D$, $E^3_D := E^{3,G_U}_D$ and $\mathrm{tri}_\omega := \mathrm{tri}^{G_U}_\omega$.

- *SAI* is the one-tape proof-requiring law $M_A := \nu_{G_U}$. Put $M_A^3 := \nu_{G_U}^3$, $L_A := -\log M_A$ and $\mathrm{Inc}_0 := \mathbb{U}(O_\perp)$, the probability that the random theory is inconsistent.
- The *posterior* after D with $M_A(D) > 0$ is the law of A_ω under $\mathbb{U}(\cdot \mid E_D)$; three-valued, under $\mathbb{U}(\cdot \mid E_D^3)$.
- *Trichotomy.* $\mathrm{Bel}_D(\chi) := \mathbb{U}(E_D \cap O_\chi)/\mathbb{U}(E_D)$ is the posterior probability that the random theory proves χ; $\mathrm{Dis}_D(\chi) := \mathrm{Bel}_D(\neg\chi)$; $\mathrm{Indep}_D(\chi) := 1 - \mathrm{Bel}_D(\chi) - \mathrm{Dis}_D(\chi)$.
- The *prior probability of provability* $\mathrm{Bel}_0(\chi) := M_A(\{(\chi, 1)\}) = \mathbb{U}(O_\chi) - \mathrm{Inc}_0$ is not normalised: $\mathrm{Bel}_0(\top) = 1 - \mathrm{Inc}_0$. Its normalisation is $\widehat{\mathrm{Bel}}_0 := \mathrm{Bel}_0/(1 - \mathrm{Inc}_0) = \mathrm{Bel}_\emptyset$. The *plausibility* is $\mathrm{Pl}_0(\chi) := \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\chi)$, and $\mathrm{Pl}_D(\chi) := 1 - \mathrm{Bel}_D(\neg\chi)$.
- *Class probabilities.* For a class $\mathcal{C}$ of deductively closed theories, $\mathbb{U}_A(\mathcal{C}; D) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \vdash \psi_D, \mathrm{Cn}(A_\omega) \in \mathcal{C}\}$ and $\mathbb{U}_A(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; \emptyset)$. For a theory T, $\mathbb{U}_A[T] := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)\}$ and $\mathbb{U}_A(\mathrm{Ext}T) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)\}$.

If $\omega = p\omega'$ with $p \in \mathrm{dom}\, V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$, so G_U reads one tape in two parts, a program and that program's coins; a tape with no prefix in $\mathrm{dom}\, V$ emits nothing. By Proposition 3.1, $M_A(D) = \mathrm{Bel}_0(\psi_D)$, $M_A(b \mid D, \varphi) = \mathrm{Bel}_D(\varphi^b)$ and $M_A^3(\iota \mid D, \varphi) = \mathrm{Indep}_D(\varphi)$ for $\varphi \notin \mathrm{dom}\, D$: the predictions of SAI are posterior probabilities that the random theory proves, refutes, or leaves open the queried sentence.

2.3 Conventions and the two defects

Definition 2.5 (Loss conventions)**.** • *Semimeasure convention* (primary): the cumulative loss of a consistent reveal sequence is $\sum_{t \le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t)$, which equals $L_A(D_n) - L_A(\emptyset)$ (Proposition 3.1(d)); for the coin-free version it is $-\log W_A(D_n) + \log W_A(\emptyset)$.

- *Normalised convention*: each conditional is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$.
- *Three-valued convention*: labels in Y_3, scored by M_A^3, whose conditionals sum to 1 (Proposition 3.1(f)).
- The consistency-requiring predictor (Definition 2.9) has its own loss, $\sum_t -\log \mathrm{Bel}^{\mathrm{nc}}_{D_{t-1}}(\varphi_t^{b_t})$.

Upper bounds proved in the semimeasure convention hold in the normalised convention, since each normalised conditional is at least the raw one. Lower bounds do not transfer: a lower bound in the semimeasure convention can fail in the normalised convention by an unbounded amount (Proposition C.16), and the normalised loss depends on the reveal order (Theorem A.33). Every lower bound below names its convention; statements about K_A, K_{fin} and the posterior need none.

The missing mass of M_A has two sources, and we keep both rather than renormalise.[2]

Remark 2.6 (Two defects). M_A has a *global* defect $1 - M_A(\emptyset) = \mathrm{Inc}_0 > 0$ and a *local* defect $\mathrm{Indep}_D(\varphi)$ at each query (Proposition 3.1(b), (e)). The one-tape law M_F of SOI has only the local one ($M_F(\emptyset) = 1$), because function machines are consistent by definition. (The correspondence of the outcomes with those of function machines is in Appendix A.1.) Removing the independent mass and renormalising, or splitting it equally between the two labels, gives incoherent predictions (Examples 2.15, 2.16 and Proposition 2.17 of the earlier report); keeping it, there is a probability on the complete consistent extensions of B that lies between Bel_D and Pl_D (Proposition 3.2(c)).

[2]The proof-requiring and consistency-requiring variants, the three outcomes true, false and independent, and the choice to keep the defect of the semimeasure rather than renormalise first appeared in Hänni's working notes (Hänni, 2026).

2.4 Versions and complexities of theories

Definition 2.7 (Versions)**.** As for SOI, the one-tape law has a coin-free version, a mixture over deterministic programs, and a Kolmogorov version, the shortest compatible program. For a data set D:

- the coin-free version, over enumerators: $W_A(D) := \sum_{p\in\operatorname{dom} V} 2^{-|p|}\mathbf{1}[B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash \psi_D]$, which equals $\sum_e \mathbf{m}(e)\mathbf{1}[B\cup\mathsf{W}_e$ consistent and $\vdash \psi_D]$;
- $K_A(D) := \min\{|p| : p \in \operatorname{dom} V,\ B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash \psi_D\}$ (∞ if there is none; the Kolmogorov version);
- W_A^{dec}, K_A^{dec}: the same over *deciders*, the $p \in \operatorname{dom} V$ such that $V(p)$ is an index of a total $\{0,1\}$-valued function d, read as the axiom set $\{\chi : d(\chi) = 1\}$ (Definition 6.2 of the earlier report);
- $W_A^{\mathrm{gen},t}$: the same over enumerators of the delay form (Definition E.10 of the earlier report), which write each axiom a within $O(t(|a|))$ steps of the previous one, and $K_A^{\mathrm{gen},t}(D)$, the least length of such an enumerator compatible with D;
- $K_{\mathrm{fin}}(D) := \min\{K(\psi) : \psi \in \mathrm{Sent},\ B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D\}$ and $W_{\mathrm{fin}}(D) := \sum_\psi \mathbf{m}(\psi)\mathbf{1}[B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D]$ (finite axiomatisations).

Deciders and enumerators give the same weights up to a constant factor, polynomial-delay enumerators too under the self-simulation assumption of the earlier report (Assumption B.5), and $K_A \le K_{\mathrm{fin}} + c \le K(\psi_D) + c$ on consistent data, where $K_{\mathrm{fin}} - K_A$ is unbounded (Remarks A.2 and F.29).

Definition 2.8 (Complexities of theories)**.** For an r.e. theory T: $[T] := \{e : \mathrm{Cn}(\mathsf{W}_e) = \mathrm{Cn}(T)\}$; $K_{\equiv}(T) := \min_{e\in[T]} K(e)$, the complexity of T up to deductive equivalence; $\mathbf{m}_{\equiv}(T) := \sum_{e\in[T]}\mathbf{m}(e)$; $\mathrm{Ext}(T) := \{e : B\cup\mathsf{W}_e$ consistent, $\mathrm{Cn}(\mathsf{W}_e)\supseteq\mathrm{Cn}(T)\}$ and $\mathbf{m}(\mathrm{Ext}T) := \sum_{e\in\mathrm{Ext}T}\mathbf{m}(e)$ (their relations are in Remark A.3).

2.5 Variants and questions

A weaker requirement than proof is consistency with the data; Section 3.6 explains why we take the proof-requiring law as primary.

Definition 2.9 (Variants)**.** Let $\varepsilon_\emptyset := \mathbb{U}\{\omega : A_\omega = \emptyset\}$, the mass of the empty theory; $\varepsilon_\emptyset \ge 2^{-|p_\emptyset|} > 0$ for a V-program $p_\emptyset$ of an index of the empty machine. The *consistency-requiring* 0/1 weight is $M_A^{\mathrm{nc}}(D) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent$\}$. Its posterior is the law of A_ω given that event, and its predictor uses the hypothesis together with the data: $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent and $\vdash\varphi\}/M_A^{\mathrm{nc}}(D)$ and $\mathrm{Dis}_D^{\mathrm{nc}}(\varphi) := \mathrm{Bel}_D^{\mathrm{nc}}(\neg\varphi)$.

A graded consistency-requiring weight $M_A^{1/2}$ and a generative law Ξ of unlabelled data are defined in Definitions A.20 and A.36.

Definition 2.10 (Questions)**.** A *question* is a finite set $\mathcal{Q} = \{\varphi_1,\dots,\varphi_k\}$ of pairwise B-incompatible sentences whose disjunction B proves, its *direct answers*; a yes/no question is $\{\varphi,\neg\varphi\}$. A theory *answers* $\mathcal{Q}$ if it proves some φ_i. A *conditional question* is a pair (α,ℓ), read "given α, is ℓ the case?". Under the *encoding* (E3), a theory T is compatible with the conditional datum (α,ℓ) iff $B\cup T\cup\{\alpha\}$ is consistent and proves ℓ; equivalently, $T\vdash\alpha\to\ell$ and $T\nvdash\neg\alpha$.

This follows the tradition of erotetic logic (cf. Belnap and Steel, 1976); results that need only pairwise incompatibility say so. (E3) makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents (more in Appendix A.1).

3 Basic theorems

This section develops the theory of M_A; proofs not given in place are in Appendix A.

3.1 Structure and updating

Recall from the companion paper (Hänni and Claude (Anthropic), 2026c) that a map ν from data sets to $[0,1]$ is a *weak semimeasure* if $\nu(\emptyset) \le 1$ and the masses of the one-datum extensions of D at an input sum to at most $\nu(D)$ (its Definition 3.2), and that the *containment functional* of a random partial function F is $D \mapsto \Pr[F \supseteq D]$ (its Definition 3.4).

Proposition 3.1 (Structure)**.** *Let G be a theory machine. (a) If $\nu_G(D) > 0$, then D is consistent; $M_A(D) > 0$ iff D is consistent, and $M_A^3(D) > 0$ iff moreover $B \cup D^{\pm}$ decides no ι-labelled sentence. (b) For two-valued D and $\varphi \notin \operatorname{dom} D$, $\nu_G(D+(\varphi,1))+\nu_G(D+(\varphi,0))+\nu_G^3(D+(\varphi,\iota)) = \nu_G(D)$; so ν_G is a weak semimeasure, and for M_A the defect at φ divided by $M_A(D)$ is $\mathrm{Indep}_D(\varphi)$. (c) $\nu_G(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$; if $B \cup D^{\pm} \vdash \varphi^b$ and $\varphi \notin \operatorname{dom} D$, then $\nu_G(D+(\varphi,b)) = \nu_G(D)$ and $\nu_G(D+(\varphi,1-b)) = 0$: entailed data cost nothing and contradicting data are fatal. (d) For every consistent reveal sequence, in any order and with repeats, $\sum_{t \le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t) = L_A(D_n) - L_A(\emptyset)$, which depends only on $\mathrm{Cn}(D_n^{\pm})$. (e) $0 < M_A(\emptyset) = 1 - \mathrm{Inc}_0 < 1$. (f) For three-valued D and $\varphi \notin \operatorname{dom} D$, $\sum_{y \in Y_3} \nu_G^3(D + (\varphi, y)) = \nu_G^3(D)$, and $\nu_G^3(\emptyset) = \nu_G(\emptyset)$; so the conditionals of M_A^3 are probability distributions on Y_3. (g) $\nu_G^3(D) = \mathbb{U}\{\omega \in \mathrm{Cons}^G : \mathrm{tri}_\omega^G \supseteq D\}$, also for two-valued D and ν_G. So, when $\nu_G(\emptyset) > 0$, $\nu_G^3/\nu_G(\emptyset)$ is the containment functional of the random total function tri_ω^G under $\mathbb{U}(\cdot \mid \mathrm{Cons}^G)$, and $\nu_G/\nu_G(\emptyset)$ is that of the random partial function obtained by reading ι as "undefined".*

By (g), M_A is a (sub-probability) containment functional, which is strictly stronger than being a weak semimeasure (Example 3.5 of the companion paper; Proposition A.22(e)).

Proposition 3.2 (Proof-requiring updating is geometric conditioning)**.** *(a) If $M_A(D) > 0$, then $\mathrm{Bel}_D(\chi) = \mathrm{Bel}_0(\psi_D \wedge \chi)/\mathrm{Bel}_0(\psi_D) = \widehat{\mathrm{Bel}}_0(\psi_D \wedge \chi)/\widehat{\mathrm{Bel}}_0(\psi_D)$ for every χ. (b) Bel_0 and every Bel_D are totally monotone on the Lindenbaum algebra of B: $\mathrm{Bel}_D(\bigvee_{i \le k} \chi_i) \ge \sum_{\emptyset \ne I \subseteq [k]} (-1)^{|I|+1} \mathrm{Bel}_D(\bigwedge_{i \in I} \chi_i)$ for all $\chi_1, \dots, \chi_k$. (c) There is a probability P_D on the complete consistent extensions of B (a Borel probability on the Stone space) with $\mathrm{Bel}_D(\chi) \le P_D(\chi) \le \mathrm{Pl}_D(\chi)$ for every χ.*

So Bel_D is a belief function (Shafer, 1976), Bel_0 is the probability of provability, as Pearl (1988) puts it, and (a) is the *geometric rule* of conditioning of $\widehat{\mathrm{Bel}}_0$, usually attributed to Suppes and Zanotti (1977); (b) and (c) are the one-tape forms of Propositions 2.18 and 2.19 of our earlier report (Hänni and Claude (Anthropic), 2026a), stated there for countable classes of theories.

Consistency-requiring updating is Dempster's rule of conditioning (Dempster, 1967; Shafer, 1976) (Proposition A.4). With focal sets $F_\omega := \mathrm{Mod}(B \cup A_\omega)$, proof-requiring updating keeps the F_ω contained in $\mathrm{Mod}(\psi_D)$, while Dempster's rule replaces each F_ω by $F_\omega \cap \mathrm{Mod}(\psi_D)$ and drops the empty ones. Under proof-requiring updating a datum must be explained by the hypothesis; under consistency-requiring updating it is added to it.

3.2 Computability

The function $D \mapsto \mathbb{U}(O_{\psi_D})$ is left-c.e. uniformly in D, and $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathrm{Inc}_0$ with Inc_0 left-c.e. (Proposition A.5); but $\mathbb{U}(O_{\psi_D})$ is not a semimeasure: it gives an inconsistent theory credit for every datum. The correction that turns it into M_A is one real, and that real is as complex as a left-c.e. real can be.

Theorem 3.3 (The inconsistency probability is an Ω-like real)**.** *(a)* Inc_0 *is not computable. Hence* $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ *is right-c.e. and not left-c.e., and* $D \mapsto M_A(D)$ *is not lower semicomputable. (b) For every left-c.e.* $\alpha \in [0,1]$ *there are a* V*-program* p_α *and a left-c.e. real* γ_α *with* $\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha$*; consequently* $\alpha \leq_S \mathrm{Inc}_0$ *for every left-c.e.* α*, and* Inc_0 *is Solovay complete. (c)* Inc_0 *is Martin-Löf random.*

A machine that emits $\perp$ on the tapes whose binary real lies below α has inconsistency probability α; splitting the tapes by their first program gives (b), and (a), (c) follow with the halting set and $\alpha = \Omega_V$ (Lemma A.6). The theorem is a new application of the standard argument that Ω_V is Solovay complete. Machine laws of SOI are lsc and M_F has $\emptyset'$-computable conditionals (Propositions 3.7 and A.10 of the companion paper); M_A keeps the second property and loses the first by exactly one real: $M_A + \mathrm{Inc}_0$ is lsc (Proposition A.7).

3.3 Dominance and invariance

Theorem 3.4 (Mixture identity and dominance)**.** *For every two-valued* D*,* $M_A(D) = \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\nu_{G_{V(p)}}(D) + (1 - \Omega_V)\mathbf{1}[B \vdash \psi_D] = \sum_i \mathbf{m}(i)\nu_{G_i}(D) + (1 - \Omega_V)\mathbf{1}[B \vdash \psi_D]$*, and the same holds for* M^3_A*, the* $\nu^3_{G_i}$*, and the indicator that* B *proves* ψ_D *and decides no* ι*-labelled sentence. Hence* $M_A \geq 2^{-K(\nu)}\nu$ *for every theory-machine law* ν*, and* $M^3_A \geq 2^{-K(G)}\nu^3_G$ *for every theory machine* G*. In particular: (i)* $M_A(D) \geq 2^{-K(e)-c}\mathbf{1}[B \cup \mathsf{W}_e$ *consistent,* $\vdash \psi_D]$ *for every index* e*; consequently, for every r.e.* T *with* $B \cup T$ *consistent,* $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T) \geq 2^{-K_{\equiv}(T)-c}$ *and* $\mathbb{U}_A(\mathrm{Ext}T) \geq 2^{-c}\mathbf{m}(\mathrm{Ext}T)$*; (ii)* $M_A(D) \geq 2^{-K(f)-c}\mathbf{1}[f \supseteq D]$ *for every partial computable* $f : \mathrm{Sent} \rightharpoonup \{0,1\}$ *with* $B \cup \Gamma_f$ *consistent, where* $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$*; (iii)* $M_A(D) \geq 2^{-K(v)-c}\sum_e v(e)\mathbf{1}[B \cup \mathsf{W}_e$ *consistent,* $\vdash \psi_D]$ *for every lsc semimeasure* v *on indices; in particular* $M_A \geq 2^{-c}W_A$*; (iv) for an r.e.* T *and a computable family* (α_j)*, the law of* T *plus the fair-coin facts* $\alpha_j^{\omega_j}$ *(the example of a theory plus coin facts in Appendix A.1) is dominated with constant* $2^{-K(T,(\alpha_j))-c}$*.*

The mixture and dominance arguments are the standard ones for universal mixtures (Zvonkin and Levin, 1970; Theorem 3.8 of the companion paper); what is new is the object. In particular $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$: SAI is the best two-part code over theory machines, randomised theories included (Proposition G.5). The universal machine matters only up to a constant factor (Proposition A.8), and building a background B' into B gains at most $K(B') + c$ bits on any data, while the converse fails (Proposition A.9). A change of language costs a constant too, if the translation preserves what is proved:

Theorem 3.5 (Faithful translations)**.** *Let* $(\mathcal{L}, B)$*,* $(\mathcal{L}', B')$ *be two settings and* $\tau : \mathrm{Sent}_{\mathcal{L}} \to \mathrm{Sent}_{\mathcal{L}'}$ *computable with (T1)* $B \cup \Gamma \vdash \varphi$ *iff* $B' \cup \tau[\Gamma] \vdash \tau(\varphi)$*, for all* $\Gamma \subseteq \mathrm{Sent}_{\mathcal{L}}$ *and* φ*; and (T2)* $B' \vdash \tau(\neg\varphi) \leftrightarrow \neg\tau(\varphi)$ *for all* φ*. For* D *on* $\mathcal{L}$ *let* $M'_A(\tau D) := \mathbb{U}\{\omega : B' \cup A'_\omega$ *consistent and* $\vdash \tau(\varphi)^{D(\varphi)}$ *for all* $\varphi \in \mathrm{dom}\, D\}$*, with* A' *the universal theory machine of* $\mathcal{L}'$*, and define* $W'_A(\tau D)$*,* $K'_A(\tau D)$ *likewise. Then* $2^{-K(\tau)-c}M_A(D) \leq M'_A(\tau D) \leq 2^{K(\tau)+c}M_A(D)$ *for every* D*, the same holds for* W_A*, and* $|K'_A(\tau D) - K_A(D)| \leq K(\tau) + c$*, with* c *depending on both settings.*

Corollary 3.6 (Adding vocabulary)**.** *Let $\mathcal{L} \subseteq \mathcal{L}'$, keep B, and let τ be the inclusion. Then (T1) and (T2) hold, and for data in the old vocabulary $M_A^{\mathcal{L}'}(D)$ and $M_A^{\mathcal{L}}(D)$ agree up to a factor $2^{\pm c}$, W_A likewise, $K_A^{\mathcal{L}'}(D) = K_A^{\mathcal{L}}(D) \pm c$, and predictive probabilities of $\mathcal{L}$-sentences agree up to $2^{\pm 2c}$.*

So in the unbounded universal setting new vocabulary buys at most a constant, since the pull-back of a theory to the old vocabulary is r.e.: this is the effective content of Craig's theorem (Craig, 1953) inside the prior. Theorem F.3 extends it class by class, and Section 8 studies where vocabulary does pay.

3.4 Loss bounds

Theorem 3.7 (Loss against a theory that proves the data)**.** *Let $(\varphi_t, b_t)_{t \le n}$ be any consistent reveal sequence, chosen in any way (in advance, adaptively, adversarially or non-computably; in any order; with repeats), and let T be an r.e. theory with $B \cup T$ consistent and $B \cup T \vdash \varphi_t^{b_t}$ for all $t \le n$. Then in the semimeasure convention*

$$\begin{aligned} L_A(D_n) - L_A(\emptyset) \;\le\; L_A(D_n) \;&\le\; -\log \mathbb{U}_A(\mathrm{Ext}T) \\ &\le\; -\log \mathbf{m}(\mathrm{Ext}T) + c \;\le\; -\log \mathbf{m}_{\equiv}(T) + c \;\le\; K_{\equiv}(T) + c, \end{aligned}$$

and also $L_A(D_n) \le -\log W_A(D_n) + c \le K_A(D_n) + c$.

Proof. Every consistent extension of T proves every datum, so E_{D_n} contains every tape ω with $B \cup A_\omega$ consistent and $\mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)$. Then use Theorem 3.4(i), $[T] \subseteq \mathrm{Ext}T$, $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$ and $L_A(\emptyset) \ge 0$; the left side is the cumulative loss by Proposition 3.1(d). The last chain is Theorem 3.4(iii) and $W_A \ge 2^{-K_A}$. □

The bound favours the weakest theory that decides the data: a theory such as PA has small $K_{\equiv}$ although every axiomatisation of it is infinite. Compared with the bound $K(f)+c$ of Theorem 5.1(b) of the companion paper for data from a partial computable f, a program is replaced by a theory up to equivalence, which need decide only the sentences asked.

A law pays for its listed exceptions, positions included, and only one bit per exception when the exceptions are atoms it is silent about (Proposition A.10). The *Bayes predictor*, which outputs the most probable of $1, 0, \iota$, makes at most $K_{\equiv}(T) + c$ mistakes under the hypotheses of Theorem 3.7; the *Kolmogorov predictor*, which predicts with the first compatible program in a well-order of programs, errs or abstains at most 2^k times, $k := \sup_n K_A(D_n)$, when the order refines length, and this hypothesis is needed (Proposition A.11). With three-valued data labelled by the random theory of a theory machine G with $\nu_G(\emptyset) = 1$, and questions chosen from the history and from randomness independent of the tape, the expected divergences between the predictions of G and of M_A^3 sum to less than $K(G)$; and in any order the three-valued loss on data labelled by the trichotomy of an r.e. theory T with $B \cup T$ consistent is at most $K_{\equiv}(T) + c$ (Theorem A.12 and Corollary A.13).

3.5 No coding theorem

Proposition 3.8 (Coding lower bound and the chain)**.** *(a) If S is a finite set of pairwise B-incompatible sentences and $\sigma \in S$, then $-\log \mathrm{Bel}_0(\sigma) \ge K(\sigma \mid S) - c$. (b) $L_A(D) \ge K(b_D \mid \mathrm{dom}\, D) - c$ for every D (also Proposition A.15). (c) For every consistent D and every finite set S of pairwise B-incompatible sentences with $\psi_D \in S$, $2^{-K_A(D)} \le W_A(D) \le 2^c M_A(D) \le 2^{K(S)+c} 2^{-K_A(D)}$. With S the $2^{|\mathrm{dom}\, D|}$ label conjunctions on the two-valued part of* $\mathrm{dom}\, D$*, $K(S) \le K(\mathrm{dom}\, D) + c$.*

The slack $K(S)$, in place of the $K(\operatorname{dom} D)$ of Theorem 4.4 of the companion paper, can be much smaller: for "which of m incompatible laws holds" it is $K(m)+O(1)$ when S is computable from m. It cannot be removed:

Theorem 3.9 (No coding theorem for axiom systems)**.** *In the literal setting, for every $d \geq 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(R(t_{x_1}), 0), (R(t_{x_2}), 1)\}$ satisfies (i) $K_A(D) \geq \ell_d > \log d - 2$, with $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, and $W_A(D) \leq \Omega_V^{\geq \ell_d}$; (ii) $-\log W_A(D) \leq 2 + K(\lceil \log(d+1) \rceil) + c$; (iii) $-\log M_A(D) \leq 2 + c$. Parts (ii) and (iii) hold for every data set of two literals on distinct words of length at most d. Hence $K_A(D) + \log W_A(D) \geq \log d - K(\lceil \log(d+1) \rceil) - c - 4 \to \infty$ and $M_A(D)/W_A(D) \geq 2^{-c-2}/\Omega_V^{\geq \ell_d} \to \infty$: no two of 2^{-K_A}, W_A, M_A agree up to a constant factor.*

Short programs sort the words of length d into fewer than 2^d classes by what they prove about each word's literal, while a coin per atom proves a given labelling of two atoms with probability $\frac{1}{4}$. This is Theorem 4.9 of the companion paper for axiom systems; since D is the single sentence ψ_D, no coding theorem holds for single sentences. For whole theories $K_{\equiv}$ and $-\log \mathbf{m}_{\equiv}$ agree up to a factor 3 and logarithmic terms (Proposition A.16).

3.6 Why proof-requiring compatibility is primary

The weaker requirements of Definitions 2.9 and A.20 fail as follows; let $H_\omega := B \cup A_\omega \cup D^{\pm}$.

Proposition 3.10 (The 0/1 consistency-requiring weight does not discriminate)**.** *(a) For $\varphi \notin \operatorname{dom} D$, $M_A^{\mathrm{nc}}(D + (\varphi, 1)) + M_A^{\mathrm{nc}}(D + (\varphi, 0)) = M_A^{\mathrm{nc}}(D) + \mathbb{U}(N_\varphi) \geq M_A^{\mathrm{nc}}(D)$, where N_φ is the set of tapes ω such that H_ω is consistent and decides neither φ nor $\neg\varphi$; so M_A^{nc} is not a semimeasure; it is the plausibility $\mathrm{Pl}_0(\psi_D)$, the dual of a belief function. (b) $M_A^{\mathrm{nc}}(D) \geq \varepsilon_\emptyset$ for every consistent D. The coin-free consistency-requiring weight is at least $2^{-|p_\emptyset|}$, and the Kolmogorov version $\min\{|p| : B \cup \mathsf{W}_{V(p)} \cup D^{\pm}$ consistent$\}$ is at most $|p_\emptyset|$, on all consistent D. (c) The consistency-requiring posterior after D is the prior restricted to the tapes whose theory is consistent with D: data eliminate hypotheses but never change the odds between survivors.*

Part (c) is the one-tape form of Proposition 2.22(b) of the earlier report. The inductive work is left to adding the data to the hypothesis, and the proper predictor of this weight, Dempster's rule, pays for it: at least $\log \frac{1}{1-\varepsilon_\emptyset}$ bits for every datum that the earlier data do not entail, and on suitable data of fresh atoms its cumulative loss depends on the reveal order, with a gap linear in the number of data (Proposition A.21).

The graded weight $M_A^{1/2}$ of Definition A.20 is a weak semimeasure with $M_A^{1/2} \geq M_A$, but it charges for entailed data, so it is the law of no theory machine, and its advantage over M_A measures how the data are packaged, not induction (Propositions A.22 and A.24). Its Möbius sum on the inputs $\varphi, \neg\varphi$ is $-\frac{1}{2}$ times the mass of the consistent random theories that decide neither (negative when B does not decide φ), against $+1$ times that mass for M_A, so $M_A^{1/2}/M_A^{1/2}(\emptyset)$ is not even a containment functional; and the consistency-requiring predictor is a mixture of the proof-requiring one and a component of weak theories plus the data (Proposition A.23). The distinction exists only because the random theory is incomplete:

Proposition 3.11. *If $B \cup T$ is complete and consistent, then $B \cup T \vdash \psi_D$ iff $B \cup T \cup D^{\pm}$ is consistent, for every D. Hence for any prior concentrated on complete theories, proof-requiring and consistency-requiring compatibility coincide, the trichotomy has no independent outcome, and conditioning on D restricts to the complete theories that contain $D^{\pm}$.*

Proof. A complete consistent $B \cup T$ proves ψ_D or $\neg\psi_D$, and is consistent with ψ_D iff it does not prove $\neg\psi_D$. □

Priors of Demski's type complete their random theory, and so cannot see the difference between explaining a datum and tolerating it (Proposition A.39).

3.7 Limits and identification

On a consistent stream the posterior converges in total variation to the prior restricted to the consistent extensions of the data if these have positive prior mass, and the loss is then bounded; otherwise it diverges (Proposition A.26). Data presenting a consistent r.e. T^* with $B \cup T^*$ complete concentrate the posterior on the class of T^*; if $B \cup T^*$ is incomplete, two-valued data leave each sentence independent of $B \cup T^*$ with limit belief and disbelief in $(0, 1)$, and three-valued data identify the class of T^* (Proposition A.27). As with Gold's text and informant (cf. Gold, 1967), no learner, computable or not, identifies from two-valued data every theory of a simple family of sets of fresh atoms (Proposition A.28). If $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory is almost surely incomplete (Proposition A.31).

3.8 Order dependence and the Kolmogorov predictor

The normalised loss depends on the reveal order, by a number of bits linear in n on some $2n$ data built from fresh atoms (Theorem A.33); on logically independent data this is open (Question A.34). For every m and well-order of programs, the Kolmogorov predictor errs or abstains $2^m - 1$ times on some reveal sequence with $K_A(D) \leq m + K(m) + c$, where the Bayes predictor makes at most $K_A(D) + c$ mistakes (Theorem A.35). A *random consistent theory* Θ is a Borel random variable with consistent, deductively closed values, and $\beta(\chi) := \Pr[\chi \in \Theta]$ its *provability function*.

Question 3.12. Does M_A dominate every lower semicomputable provability function, that is, is $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc?

Domination holds with the slack $K(S)$ of Proposition 3.8, and for the closure of a theory machine's output; on random sets of literals of a fresh predicate the question is equivalent to Question 3.14 of the companion paper for binary labels (Proposition A.40).

4 Relation to Solomonoff operator induction

Solomonoff operator induction, the subject of the companion paper (Hänni and Claude (Anthropic), 2026c), puts the universal prior on programs that label inputs. On sentences, a labeller consistent with B is the program-shaped counterpart of a theory, and we compare SAI with SOI over such labellers.[3] Table 1 collects the relations, and Appendix B has the proofs. Losses are in the semimeasure convention unless stated; $\asymp$ is equality within $2^{\pm c}$ on every finite data set.

4.1 The objects and four maps

Definition 4.1 (The operator-induction side)**.** Let $\mathcal{L}$ and B be as in Definition 2.1.

[3]The comparison of SAI with consistent SOI, the example in which plain SOI beats SAI (Remark B.14), and the axiom schema of Remark B.3 first appeared in Hänni's working notes (Hänni, 2026).

- A *function machine* is a c.e. set $F \subseteq \mathrm{Sent} \times \{0,1\}^* \times \{0,1\}$ such that $(\varphi, w, y), (\varphi, w', y') \in F$ with w, w' comparable implies $y = y'$ (Definition 3.6 of the companion paper). On a tape ω it computes the partial labeller f^F_ω: $f^F_\omega(\varphi) := y$ if $(\varphi, w, y) \in F$ for a prefix w of ω, and $f^F_\omega(\varphi)$ is undefined ($\perp$) otherwise; its law is $\nu_F(D) := \mathbb{U}\{\omega : f^F_\omega \supseteq D\}$. F_U is the universal function machine (read $p \in \mathrm{dom}\, V$, then run $F_{V(p)}$ on the rest of the tape), $f_\omega := f^{F_U}_\omega$, and $M_F := \nu_{F_U}$.
- For a partial $f : \mathrm{Sent} \rightharpoonup \{0,1\}$ let $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$. The labeller f is *consistent* if $B \cup \Gamma_f$ is consistent, and *closed* if moreover $f(\varphi) = b$ iff $B \cup \Gamma_f \vdash \varphi^b$, for all φ and b.
- *One-tape consistent and closed SOI*: $M^{\mathrm{cons}}_F(D) := \mathbb{U}\{\omega : f_\omega \supseteq D,\ f_\omega \text{ consistent}\}$, and $M^{\mathrm{cl}}_F(D)$ likewise with "closed and consistent"; ν^{cons}_F, ν^{cl}_F are the same for one machine F.
- *Random deciders* are function machines F with f^F_ω total for every ω, read as theory machines with axiom set $A^{\mathrm{dec}}_F(\omega) := \{\chi : f^F_\omega(\chi) = 1\}$; $M^{\mathrm{dec}}_A(D)$ is the sum of $2^{-|p|}\,\mathbb{U}\{\omega : B \cup A^{\mathrm{dec}}_{F_{V(p)}}(\omega) \text{ consistent and } \vdash \psi_D\}$ over the $p \in \mathrm{dom}\, V$ for which $F_{V(p)}$ is a random decider.
- *Coin-free and Kolmogorov*: $W^{\mathrm{cons}}(D)$ is the sum of $2^{-|p|}$ over the $p \in \mathrm{dom}\, V$ for which the partial computable labeller $\{V(p)\}$ is consistent and extends D, and $K^{\mathrm{cons}}_F(D)$ is the least such $|p|$; W^{cl}, K^{cl}_F likewise with closed labellers; W^{all} drops the consistency condition (plain coin-free SOI on sentences). W and K_F are the companion paper's coin-free and Kolmogorov versions on strings.

Four maps. The equivalences rest on four computable maps, each a constant-size prefix in front of a program (Lemma B.2): *proof search* turns a theory machine into a labeller that, where the theory is consistent, is closed and decides every sentence as the theory does; *emission* turns a labeller f into the theory Γ_f; *Craig's map* turns f into a random decider of padded copies of Γ_f; *closure* makes a consistent labeller closed.

4.2 Equivalence, version by version

Theorem 4.2 (Coin-free equivalence)**.** *There is c, depending only on V, the coding of syntax and B, such that for every finite data set D:*

(a) $2^{-c}W_A(D) \le W^{\mathrm{cons}}(D) \le 2^c W_A(D)$, $2^{-c}W_A(D) \le W^{\mathrm{dec}}_A(D) \le 2^c W_A(D)$, *and* $2^{-c}W^{\mathrm{cons}}(D) \le W^{\mathrm{cl}}(D) \le W^{\mathrm{cons}}(D)$*;*

(b) $K_A(D)$, $K^{\mathrm{dec}}_A(D)$, $K^{\mathrm{cons}}_F(D)$ *and* $K^{\mathrm{cl}}_F(D)$ *agree up to* $\pm c$*, and all are finite iff D is consistent;*

(c) for every $\varphi \notin \mathrm{dom}\, D$ *and* $o \in \{1, 0, \perp\}$, $2^{-c}W_A(D; \varphi \mapsto o) \le W^{\mathrm{cl}}(D; \varphi \mapsto o) \le 2^c W_A(D; \varphi \mapsto o)$*, where $W_A(D; \varphi \mapsto o)$ is the weight of the enumerators compatible with D whose theory proves φ ($o = 1$), proves $\neg\varphi$ ($o = 0$) or decides neither ($o = \perp$), and $W^{\mathrm{cl}}(D; \varphi \mapsto o)$ is the weight of the closed labellers extending D with $f(\varphi) = o$.*

Hence the semimeasure losses of W_A, W^{dec}_A, W^{cons} and W^{cl} agree within $2c$ on every data set, in every order.

Proof search and emission give $W_A \asymp W^{\mathrm{cons}}$; Craig's map is needed only for deciders, and for them (a) is Theorem 6.3 of our earlier report (Hänni and Claude (Anthropic), 2026a). The construction is Craig's; apart from that report we have not found the weight equivalence stated in the literature. A route through an axiom schema works with two sorts or with bounded witnesses but fails in one sort over PA (Remark B.3).

Theorem 4.3 (One-tape equivalence)**.** *There is c, depending only on V, the coding and B, such that for every finite D, the empty one included:*

(a) $2^{-c}M_A(D) \le M_F^{\mathrm{cons}}(D) \le 2^c M_A(D)$ *and* $2^{-c}M_A(D) \le M_A^{\mathrm{dec}}(D) \le 2^c M_A(D)$*;*

(b) tape by tape: for every theory machine G and tape ω, if $B \cup A_\omega^G$ is consistent and proves ψ_D, then $f_\omega^{F_G}$ is consistent and extends D; for every function machine F and tape ω, if f_ω^F is consistent and extends D, then G_F and Craig's random decider of F are compatible with D on ω.

Over closed labellers the three outcome weights agree as well, so the trichotomy of SAI is the three-way split of one-tape SOI over closed labellers (Corollary B.4); over all consistent labellers it is not (Proposition 4.5).

Penalties on the axiom side. Charging a hypothesis for the time it takes to check or to generate its axioms, with deduction free ($M_A^{\mathrm{chk},t}$, $M_A^{\mathrm{gen},t}$ and the coin-free $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ of Definition B.7), at any time-constructible budget $t(m) \ge m^{a^*}$ (a^* from Assumption B.5), changes nothing up to constant factors, also class by class: one-tape under an assumption ($\mathrm{S}_{\mathrm{tape}}$) of efficient self-simulation of the universal machine, coin-free under its tape-free form (S), neither of which we prove for a particular machine (Assumptions B.5 and B.6, Theorem B.9, and Propositions B.10 and B.28).

4.3 Positive literals: SAI is SOI

Theorem 4.4 (Transfer principle)**.** *Let $\mathcal{I} \subseteq \{0,1\}^*$ be the companion paper's (infinite, decidable) input set and $x \mapsto t_x$ a computable injective map from $\mathcal{I}$ to closed $\mathcal{L}$-terms, with decidable range and computable inverse. Let R be a unary relation symbol that does not occur in B, and suppose some model of B interprets the terms t_x as pairwise distinct elements. For D with* $\operatorname{dom} D \subseteq \{R(t_x) : x \in \mathcal{I}\}$ *put* $D^{\mathrm{str}} := \{(x,b) : (R(t_x), b) \in D\}$*. Then there is c with*

$$|K_A(D) - K_F(D^{\mathrm{str}})| \le c, \qquad W_A(D) \asymp W(D^{\mathrm{str}}), \qquad M_A(D) \asymp M_F(D^{\mathrm{str}}) \quad (D \neq \emptyset),$$

where K_F, W, M_F are the companion paper's objects over $\mathcal{I}$ with labels $\{0,1\}$; the same holds for K_A^{dec}, W_A^{dec}, M_A^{dec}, and for the three outcome weights of a literal query $R(t_y)$ (proves, refutes, decides neither; label 1, 0, *undefined at y).*

Distinct terms are needed: if $B \vdash t_x = t_y$, the data $(R(t_x), 1), (R(t_y), 0)$ are inconsistent for SAI but not for SOI.

So every result of the companion paper on binary labels that holds up to additive constants in K, $-\log W$ and $-\log M_F$ holds for SAI on literal data, among them the absence of a coding theorem with its two-point rate; normalised cumulative statements, computability statements, and statements about private randomness or weak semimeasures do not transfer in general (Corollary B.12). The content of SAI beyond SOI lies in logically structured data. Plain coin-free SOI can beat SAI by n bits in expectation on literals with random signs, all labelled true, through the inconsistent labeller "always 1": an encoding effect (Remark B.14).

4.4 What breaks

Proposition 4.5 (Logical closure)**.** *Let D be consistent and $\chi \notin \operatorname{dom} D$.*

pair	relation (hypotheses)	where
Equivalences		
W_A, W_A^{dec} – $W^{\mathrm{cons}}, W^{\mathrm{cl}}$	$\asymp$, also the outcome weights against W^{cl}; K_A, K_A^{dec}, K_F^{cons}, K_F^{cl} agree $\pm c$, finite iff D consistent	Thm. 4.2
M_A, M_A^{dec} – $M_F^{\mathrm{cons}}, M_F^{\mathrm{cl}}$	$\asymp$, tape by tape; outcome weights against M_F^{cl}	Thm. 4.3, Cor. B.4
$M_A^{\mathrm{chk},t}, M_A^{\mathrm{gen},t}$, $W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}$	$\asymp$ unpenalised, also class by class ($t \geq m^{a^*}$; ($\mathrm{S_{tape}}$), (S))	Thm. B.9, Prop. B.10
$W_A^{\mathrm{Kt},t}$, under (S)	between $2^{-c}W^1$ and $2^c t(m_0)W^1$; a labeller pays $\log(\lvert f\rvert + 1) + O(1)$	Prop. B.23
positive literals – strings	$\asymp$, $\pm c$, version by version (fresh predicate, distinct terms)	Thm. 4.4, Cor. B.12
all – observational axioms	$\asymp$ in every version, class by class (free deduction)	Thm. F.3
r.e. in $\mathcal{L}$ – finite in $\mathcal{L}^+$	$K_{\mathrm{fin}} = K_A \pm c$, $W_{\mathrm{fin}} \asymp W_A$, $\mathcal{L}^+$ three-sorted; not one-tape ($\mathcal{L}$ finite)	Thm. F.33
charged SAI– certificate SOI	size, material: $\asymp$ up to $O(1)$ bits and polynomial budget changes	Thm. B.31
finite axiom sets – circuits	verifier circuits: polynomially related; Bayes over them (literal data)	Thm. B.43, Cor. B.44
Gaps between versions		
2^{-K_A} – M_A, two atoms	no coding theorem: $K_A \geq \log d - 2 - c$, $-\log M_A \leq 2 + c$	Thm. 3.9, Cor. B.12
W_A – M_A, literals, $\lvert w\rvert \leq d$	gap $\leq \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c$, δ_D the labels' randomness deficiency	Prop. B.13
W^{cons} – M_F^{cons}, W_A – M_A	on the lottery M_F^{cons} is bounded, W^{cons} unbounded, W^{all} bounded	Cor. C.8
M_A – W_A on Π_1 data	a coding theorem holds (B true, $\supseteq \mathsf{Q}$)	Cor. 5.7
M_F^{cons} – machine-wise M_F^{mw}	$\asymp$ if finite consistency is decidable; open in general	Prop. B.21, Q. B.22
computability	none lsc if finite consistency is not r.e.; $\emptyset'$ resp. $\emptyset''$	Prop. B.19
Separations		
SAI– W^{all}	W^{all} wins by n bits on coin literals labelled 1 (encoding effect)	Rem. B.14
entailed query	SAI certain; consistent SOI keeps undefined mass $\geq 2^{-K(\chi)-c}$	Prop. 4.5
observed ι, $\mathsf{Q} \subseteq \mathrm{Cn}(B)$	one-sided; gap unbounded on one datum, bounded on r.e. families	Prop. 4.6, Prop. B.17
$W^{\mathrm{cons},\tau}$, $M_F^{\mathrm{cons},\tau}$; B true	regret $\leq c_\tau$; reverse $\geq n - 1.37 - c_\perp$ on a computable sequence	Cor. B.25, Thm. 4.7
SAI– $W^{\mathrm{all},\tau}$	incomparable	Prop. B.26
W^{cons}, M_F^{cons}, diagonal data	unpenalised loss $\leq \lvert u\rvert + c$: the gap is free deduction	Prop. 4.8, Thm. B.29
clocked SOI, Presburger	SAI bounded; clocked unbounded for $\tau \leq 2^{2^{c'm}}$ (coin-free), $\tau \leq 2^{c'm}$ (one-tape); doubly exponential one-tape conditional	Prop. B.37, Prop. B.40
charged SAI, Presburger	size, material: doubly exp.; (a1): exp., doubly exp. open; (a2), (b2) open	Prop. B.41, Q. B.42
Fit classes under charged deduction (literal data, polynomial budgets)		
readings (b); (a1); (a2), (b2)	$\mathsf{NP} \cap \mathsf{coNP}$; $\mathsf{EXPTIME}$; every decidable set	Cor. B.32
observational, run-padded, finite	see Cor. F.8; finite theories need a binary relation or entities ($B = \emptyset$)	Cor. F.8, Thm. 8.3

Table 1: SAI and SOI. Semimeasure convention unless stated; $\asymp$ is equality within $2^{\pm c}$ on every finite D; B is decidable and consistent; rows with Q or "B true" assume $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$; charged rows assume (F1); only "open" and "conditional" entries are not theorems.

(a) *If $B \cup D^{\pm} \vdash \chi$, every hypothesis of SAI compatible with D proves χ: $M_A(1 \mid D, \chi) = 1$, $M_A(0 \mid D, \chi) = 0$ and $\mathrm{Indep}_D(\chi) = 0$; the same for W_A.*

(b) *If $B \cup D^{\pm} \vdash \chi$, then $W^{\mathrm{cons}}(0 \mid D, \chi) = 0$ and $W^{\mathrm{cons}}(1 \mid D, \chi) \geq 2^{-2c}$; the same for M_F^{cons}.*

(c) *For every $\chi \notin \mathrm{dom}\, D$, entailed or not, the consistent labellers extending D that are undefined at χ have weight at least $2^{-K(\chi)-c}\, W^{\mathrm{cons}}(D)$, and the tapes on which f_ω is consistent, extends D and is undefined at χ have measure at least $2^{-K(\chi)-c}\, M_F^{\mathrm{cons}}(D)$.*

(d) *Over closed labellers, the three predictive probabilities agree with those of SAI within $2^{\pm 2c}$.*

So in the semimeasure convention consistent SOI never becomes certain of a logical consequence: its undefined mass on a fixed simple χ stays above $2^{-K(\chi)-c}$ whatever the data (in the normalised convention its prediction on an entailed χ is 1, by (b)). The equivalences control the losses, not the undefined mass, which mixes independence with laziness.

An observed "independent" label. A labeller f with values in Y_3 is *consistent* if $B \cup \Gamma_f$ is consistent (Γ_f formed from the labels 1, 0) and decides none of the sentences f labels ι. $W^3_A(D)$ is the weight of the enumerators whose theory is consistent, proves the 1/0 data of D and decides no ι-labelled sentence of D; $W^{3,\mathrm{cons}}(D)$ is the weight of the consistent three-valued labellers extending D; M^3_A is as in Definition 2.4, and $M^{3,\mathrm{cons}}_F$ is M^{cons}_F with labels in Y_3. Ind_B is the set of sentences that B decides neither way.

Proposition 4.6 (An observed independent label breaks the equivalence)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.*

(a) $2^{-c}W^{3,\mathrm{cons}}(D) \le W^3_A(D)$ *for every three-valued D, and* $M^{3,\mathrm{cons}}_F(D) \le 2^c M^3_A(D)$ *for every three-valued $D \neq \emptyset$.*

(b) For every n there is $\sigma_n \in \mathrm{Ind}_B$ with $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \le \Omega_V^{\ge n+1}$ *while* $W^3_A(\{(\sigma_n,\iota)\}) \ge 2^{-c}$*; and there is $\sigma'_n \in \mathrm{Ind}_B$ with* $M^{3,\mathrm{cons}}_F(\{(\sigma'_n,\iota)\}) \le 2^{1-n} + \Omega_V^{\ge n+1}$ *while* $M^3_A(\{(\sigma'_n,\iota)\}) \ge 2^{-c}$*.*

(c) For every $\sigma \in \mathrm{Ind}_B$, $W^3_A(\{(\sigma,\iota)\}) \le 2^{K(\sigma)+c}\, W^{3,\mathrm{cons}}(\{(\sigma,\iota)\})$ *and* $M^{3,\mathrm{cons}}_F(\{(\sigma,\iota)\}) \ge 2^{-K(\sigma)-c}$*.*

The mechanism: a theory leaves σ undecided, a Π_1 property, without committing to anything, while a labeller must output ι after finitely many steps. The sentences that short consistent labellers label ι form an r.e. subset of the productive set Ind_B (Lemma B.16), while an r.e. family of independent sentences is labelled ι by one consistent labeller, so no such family shows a large gap (Proposition B.17).

Consistency constraints of unbounded arity separate the versions of consistent SOI (Corollary C.8). None of the versions is lower semicomputable when consistency of finite data sets is not r.e. (Proposition B.19).

4.5 Time bounds with free deduction

On the labeller side a labeller runs for a bounded time: for a time-constructible τ and a string f, let $f^\tau(\varphi) := U_0(f,\varphi)$ if a fixed plain universal machine U_0 halts within $\tau(|\varphi|)$ steps with a label, and let f^τ abstain otherwise; f has weight $w(f) := 2^{-|E(f)|}$ for a prefix code E whose weights sum to 1 with computable tails. *Clocked consistent* SOI is $W^{\mathrm{cons},\tau}(D) := \sum_f w(f)\mathbf{1}[f^\tau$ consistent and $\supseteq D]$; $W^{\mathrm{all},\tau}$ drops consistency, and $M^{\mathrm{cons},\tau}_F$, $M^{\mathrm{all},\tau}_F$ are the one-tape forms (Theorem B.29). Let $c_\perp := -\log w(f_\perp)$ for the always-abstaining $f_\perp$. A *computable predictor* P maps D and $\varphi \notin \mathrm{dom}\, D$ to a pair (q_0,q_1), $q_0+q_1 \le 1$, computable to any precision; its loss is $\ell_P(D_n) := \sum_{j\le n} -\log q_{b_j}(D_{j-1},\varphi_j)$.

On every data set the losses of W_A and W^{dec}_A exceed that of $W^{\mathrm{cons},\tau}$, and the loss of M_A that of $M^{\mathrm{cons},\tau}_F$, by at most a constant c_τ; under (S) the same holds for the penalised versions (Corollary B.25). The reverse fails on sentences that encode a diagonal argument. Let $\mathrm{Out}(e,x,y)$ be a Σ_1 formula without $<$ that holds in $\mathbb{N}$ iff program e halts on input x with output y (by the MRDP theorem; Lemma E.14 of the earlier report), and let DET be the true Π_1 sentence saying that no program has two outputs on one input.

Theorem 4.7 (Diagonal separation)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true in an expansion of $\mathbb{N}$. For every computable predictor P there is a program e such that, with $s_n := \mathrm{Out}(\underline{e},\underline{n},S0)$ labelled by truth: e is total with values in $\{0,1\}$; $\mathsf{Q} \cup \{\mathrm{DET}\}$ decides every s_n; $\ell_P(D_n) \ge n - \sum_j \log(1+2^{-j-2}) \ge n - 1/(4\ln 2) \ge n - 0.361$; and W_A, $W^{\mathrm{chk},t}_A$, $W^{\mathrm{gen},t}_A$ lose at most $\log(1/\omega_0)$, where $\omega_0 > 0$ is the least of the weights of a fixed decider and a fixed enumerator of $\mathsf{Q} \cup \{\mathrm{DET}\}$*

in these mixtures, independent of P. Moreover, on the sequence built in this way against the computable predictor $P^*(D) := (W^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$, $W^{\mathrm{all},\tau}$ loses at least $n - 1.37$ and $W^{\mathrm{cons},\tau}$ at least $n - 1.37 - c_\perp$.

One-tape, $M_F^{\mathrm{all},\tau}$ and $M_F^{\mathrm{cons},\tau}$ lose at least $n-1.37$ and $n-1.37-c_\perp$ on a computable sequence on which M_A loses at most $\log(1/\omega_0)$ (Theorem B.29).

Clocked plain $W^{\mathrm{all},\tau}$, which may be inconsistent, and SAI are incomparable (Proposition B.26): SAI is not stronger than every time-bounded predictor, only strictly stronger than clocked *consistent* SOI. Nor is this gain one of axioms over programs:

Proposition 4.8 (The separation comes from free deduction)**.** *(a) Let u be the labeller that, on a sentence* $\mathrm{Out}(\underline{e}, \underline{n}, S0)$*, runs e on n and outputs* 1 *if the result is* 1 *and* 0 *if it is anything else (diverging if e diverges), and is undefined on all other sentences. Then $\Gamma_u \subseteq \mathrm{Th}(\mathbb{N})$, so u is consistent with every B true in an expansion of $\mathbb{N}$, and $|u|$ is a constant. (b) On every sequence of Theorem 4.7,* unpenalised W^{cons} *(and W^{all}) has loss at most $|u| + c$.*

What the time bounds show. Penalising the axiom side changes nothing, even class by class; penalising the labeller side makes the relation one-sided, and the gap is the free deduction of SAI, which unpenalised consistent SOI has too. So we read the time-bounded separations as showing not that axioms are better hypotheses than programs, but that free deduction is free computation. (The defeating sequence is computable within a slightly larger time bound; we only sketch this, Remark B.27.)

4.6 Charged deduction

The natural repair is to charge deduction, in the setting of Definition E.18 of the earlier report (Appendix B.6): a hypothesis decides its axiom set in bounded time, and each datum needs a derivation of bounded size, or one in which every axiom instance (reading (a1)) or every nonlogical instance ((a2)) is short, or one of bounded material, the total size of the distinct instances used ((b)), or of bounded nonlogical material ((b2)). Under the renaming condition (F1) of that report (Assumption B.30) and with time-constructible nondecreasing budgets, derivation-charged SAI is certificate SOI (labellers defined by time-bounded verifiers of short certificates), up to $O(1)$ bits and polynomial changes of the budgets (Theorem B.31): a derivation is a certificate. On literal sequences at polynomial budgets with charged degrees this fixes the fit classes: $\mathsf{NP} \cap \mathsf{coNP}$ under (b), $\mathsf{EXPTIME}$ under (a1), every decidable set under (a2) and (b2) (Corollary B.32).

What is charged. Every reading bounds the size of a derivation that *exists* and none charges *finding* it, so charged deduction, of the type $\mathsf{NP} \cap \mathsf{coNP}$, cannot express the point that a written theory is far from sufficient for a simple reasoner; a deterministic bounded reasoner can (Theorem E.31).

4.7 A natural separation: Presburger arithmetic

A natural theory separates as well. Let $\mathcal{L}_{\mathrm{Pr}} := \{0, 1, +\}$ (with equality), $B = \emptyset$, $\mathrm{PrA} := \mathrm{Th}(\mathbb{N}, 0, 1, +)$, and let D^{Pres} list all $\mathcal{L}_{\mathrm{Pr}}$-sentences in length-lexicographic order, labelled by truth. By the bound of Fischer and Rabin (1974), under a proviso on their measure of length, there is $c > 0$ such that for all large m every decision procedure for PrA, deterministic or nondeterministic, takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols (Remark B.36).

On D^{Pres}, W_A, W_A^{dec}, M_A, and $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$, $M_A^{\mathrm{chk},t}$ for a large enough polynomial t, have bounded loss through the decider of Presburger's axioms (Presburger, 1929); given the Fischer–Rabin bound, the losses of $W^{\mathrm{cons},\tau}$ and $W^{\mathrm{all},\tau}$ tend to infinity for clocks $\tau(m) \leq 2^{2^{c'm}}$ (m large, $c' < c$), and those of their one-tape forms for $\tau(m) \leq 2^{c'm}$ (Proposition B.37). Under a Σ_2-time lower bound for PrA and a time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$, the one-tape losses are unbounded for doubly exponential clocks as well (Proposition B.40).

With deduction charged, SAI needs doubly exponential budgets on these data under the size and material readings, and at least exponential instance sizes under (a1); whether (a1) needs doubly exponential ones, and what (a2) and (b2) need, is open (Proposition B.41 and Question B.42). So on a natural theory, as on the diagonal sequences, the separation from clocked SOI is the free deduction, for the readings in which the question is settled (Appendix B.7).

4.8 Summary

How to read the table. The equivalences hold version by version, also with the axiom side penalised; charged deduction is certificate SOI. On literals SAI and SOI coincide; the separations we know come from logical closure, observed independence, consistency constraints of unbounded arity, or deduction that is free. In the idealised setting, on data labelled true or false, every consistent labeller is an axiom system at a cost of $O(1)$ bits and every theory a closed labeller, so the formalism cannot see the difference in shape between a theory and a program; Sections 7 and 8 locate where shape becomes visible.

5 Logic: random theories, Π_1 truth and Gödel phenomena

Up to constant factors, every version of SAI is the corresponding version of consistent SOI (Section 4). This section studies what is specific to data that are sentences of a logic. Random theories can succeed where every r.e. theory fails (Sections 5.1 and 5.2); on Π_1 data the mixture is, up to a constant factor, one random true theory (Section 5.3); Gödel's theorems leave fixed marks on the posterior (Section 5.4); given all its instances, a universal sentence gets belief 1 only when the background and the instances prove it (Section 5.5); and the loss on true arithmetic grows at least polynomially (Section 5.5). Proofs are in Appendix C.

Conventions. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$ with numerals $\underline{n}$; Δ_0, Σ_1, Π_1 are syntactic classes, and Π_1^* consists of the finite conjunctions of Π_1 sentences and negations of Σ_1 sentences. "Q-equivalent" means provably equivalent in Q; "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$; "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$. $\mathcal{U}$ is the class of Σ_1-unsound deductively closed theories (those proving a false Σ_1 sentence), $\overline{\mathcal{U}}$ its complement, and $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$. (G2) is Gödel's second theorem for r.e. extensions of PA; other standard facts are in Appendix C.1. Losses are in the semimeasure convention unless stated otherwise.

5.1 The lottery

The one-tape loss on a stream is bounded iff $\mathbb{U}(E_\infty) > 0$ (Proposition A.26). For SOI on strings, bounded one-tape loss implies that some partial computable labeller extends the stream (Proposition C.3(b)). Does bounded one-tape loss of SAI imply that some consistent r.e. theory proves every datum? It does not; the obstruction is a consistency constraint of unbounded arity.

Let $\mathcal{L}$ contain 0, S and a binary relation symbol Win, and let B be decidable and consistent, not mention Win, and have a model in which the numerals denote pairwise distinct elements (for

example $B = \emptyset$, Q or PA). With $N_k := 2^{k+1} - 1$, the *lottery sentences*

$$\mathrm{Lot}_k := \exists x\, \mathrm{Win}(\underline{k}, x) \ \wedge\ \forall x \left(\mathrm{Win}(\underline{k}, x) \to x = \underline{0} \vee \cdots \vee x = \underline{N_k}\right)$$

say that lottery k has a winner among its tickets $0, \ldots, N_k$. The *random lottery theory* G_{lot} reads the tape in consecutive blocks of $k+1$ bits, reads block k as a number $J_k \in \{0, \ldots, N_k\}$, and emits Lot_k, $\mathrm{Win}(\underline{k}, \underline{J_k})$, and $\neg\mathrm{Win}(\underline{k}, \underline{j})$ for $j \le N_k$, $j \neq J_k$. Its theory is consistent with B on every tape.

Theorem 5.1 (The lottery)**.** *There is an infinite stream D_∞ of labelled sentences, computable from $\emptyset'$, such that:*

(a) for every n, $M_A(D_n) \ge 2^{-K(G_{\mathrm{lot}})-c} \prod_{k\ge 0}(1 - 2^{-(k+1)})$; so $\mathbb{U}(E_\infty) > 0$, the cumulative loss is at most $K(G_{\mathrm{lot}}) - \log \prod_{k\ge 0}(1 - 2^{-(k+1)}) + c$, and the posterior converges in total variation to P_∞;

(b) no consistent r.e. theory proves (together with B) every member of $D^{\pm}_\infty$;

(c) $W_A(D_n) \to 0$ and $K_A(D_n) \to \infty$, so the coin-free loss is unbounded; and $K^{\mathcal{L}'}_{\mathrm{fin}}(D_n) \to \infty$ for every language $\mathcal{L}' \supseteq \mathcal{L}$ with computable syntax;

(d) the posterior probability that the tape's program is a deterministic machine (one that emits only under the empty prefix) tends to 0, and no tape in E_∞ has a theory with r.e. deductive closure: the limit posterior lives on random theories whose closure is not r.e.

The product equals $0.288788\ldots$ (a computation; code in the accompanying repository). The stream is a diagonal: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, it cannot refute every ticket of lottery e, and the stream reports as a loser the least ticket j^*_e that it does not refute. So every consistent r.e. theory fails at its own lottery, while a random lottery theory dies at lottery k only if its winner is the reported loser, with probability $2^{-(k+1)}$. Such a stream is neither r.e. nor, for the lottery classes, co-r.e., and the phenomenon needs proof-requiring compatibility (Remark C.2). The random lottery theories share the law $\{\mathrm{Lot}_k\}_k$ and differ only in random contingent facts, the winners: a forced form of the split between laws and random initial conditions (Section 9).

Where it cannot happen. The lottery's minimal inconsistent sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \le N_k\}$ have unbounded size, and at every threshold the majority-vote theory refutes every ticket of the late lotteries, as in Kyburg's lottery paradox (Kyburg, 1961). Bounded arity removes the obstruction: if $\mathbb{U}(E_\infty) > 0$ and the data lie in a decidable sentence class in which every B-inconsistent finite set has a B-inconsistent subset with at most r elements, the majority vote above a Lebesgue density point of E_∞, at a threshold $1 - 1/(r+1)$ that is exact for the argument, is a consistent r.e. theory proving the stream, and the coin-free loss is bounded too (Proposition C.3(a); cf. de Leeuw et al., 1956).

Corollary 5.2 (Bounded loss and r.e. theories)**.** $\mathbb{U}(E_\infty) > 0$ *does not in general imply that a consistent r.e. theory proves every datum of D_∞ (Theorem 5.1; inside true arithmetic, Theorem 5.4). It does when the inconsistency arity of the data's sentence class is bounded (Proposition C.3), for streams of Π_1 type over a true background $B \supseteq \mathsf{Q}$ (Corollary C.14), and for co-r.e. streams over block-decidable sentence classes, which include the lottery classes (Proposition C.5); and if a positive-measure set of true random theories proves the stream, a true r.e. theory does (Proposition C.4).*

We do not claim that these are the only cases. Since consistent SOI agrees with SAI version by version up to constant factors (Theorems 4.2 and 4.3), the lottery separates one-tape from coin-free consistent SOI, while plain coin-free SOI on sentences has bounded loss on it, through an inconsistent labeller (Corollary C.8).

On the lottery stream the coin-free loss diverges more slowly than any computable unbounded nondecreasing function, and $K_A(D_n) \le \lceil n/2 \rceil + O(\log n)$ (Corollary C.7, through the proof horizon of Proposition C.6).

Question 5.3. Is there a stream with bounded one-tape loss on which $-\log W_A(D_n) \ge f(n)$ for some computable unbounded nondecreasing f? And does bounded one-tape loss on a co-r.e. stream imply, in general, that a consistent r.e. theory proves it?

Relatives. Random axiom systems drawn by a universal machine are not new (Section 12 and Appendix C.2); to our knowledge, the comparison of the one-tape law with its own coin-free and Kolmogorov versions, the identification of the succeeding hypotheses as random theories r.e. in the tape, the arity boundary and the arithmetic versions are new.

5.2 Lotteries in arithmetic

Inside arithmetic, self-referential formulas behave like independent atoms with respect to consistency with a true background, with prescribed truth values in $\mathbb{N}$ (Lemma C.9, a variant of the flexible formulas of Kripke, 1962). With them the lottery needs no fresh symbol, and survives even with true data. Let $\langle k, j \rangle$ be a Δ_0 pairing with $k, j \le \langle k, j \rangle$.

Theorem 5.4 (Lotteries in arithmetic)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B \supseteq \mathsf{Q}$. For a formula $\Upsilon(x)$ put $\mathrm{Tick}_{k,j} := \Upsilon(\underline{\langle k, j \rangle})$ and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. (a) There are a formula Υ_0 and a stream $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k \ge 0}$ satisfying (a)–(d) of Theorem 5.1 in which every $\mathrm{Tick}_{k,j}$ is Q-equivalent to a false Σ_1 sentence; so the data are Q-equivalent to false Σ_1 sentences labelled true and to true Π^*_1 sentences. (b) There are a formula Υ_1 and a stream of the same form satisfying (a)–(d) of Theorem 5.1 in which every datum is true in $\mathbb{N}$ and Q-equivalent to a Boolean combination of Σ_1 sentences.*

In (b), Υ_1 is chosen by self-reference so that the true winner of lottery k is the least ticket that $B \cup \mathsf{W}_k$ refutes (0 if none); the diagonal reports as the loser a different ticket, one that $B \cup \mathsf{W}_k$ does not refute whenever $B \cup \mathsf{W}_k$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_k$, and that report is true.

Corollary 5.5 (True data, false theories)**.** *On the stream of Theorem 5.4(b): (i) every datum is true, $L_A(D_n)$ is bounded, and $K_A(D_n) - L_A(D_n) \to \infty$; so no coding theorem of the form $K_A \le L_A + f(L_A)$ holds for true data beyond $\Sigma_1 \cup \Pi_1$; (ii) $P_\infty\{\omega : B \cup A_\omega \subseteq \mathrm{Th}(\mathbb{N})\} = 0$: the limit posterior is concentrated on false theories, although every datum is true; (iii) the random lottery theory is Σ_1-unsound on every tape on which, for some k, its winner J_k differs from the true winner of lottery k; conditionally on its survival this has probability* 1.

On this stream of true sentences the one-tape inducer predicts with bounded loss and comes to believe, with probability one, false theories: a random theory that guesses winners is right where the data check it and wrong where they do not. The data are adversarial (a Δ_2 diagonal), so this is a possibility result in an idealised setting, not evidence about actual science. Whether also $P_\infty(\overline{\mathcal{U}}) = 0$ is open (Question C.10).

5.3 Π_1 data: derandomisation, a coding theorem, Chaitin's theorem

On Π_1 data positive probability of provability certifies truth. Throughout, $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. A data set D is *of* Π_1 *type* if each literal in $D^{\pm}$ is B-equivalent to a Π_1^* sentence or to a true Σ_1 sentence; then $B \vdash \psi_D \leftrightarrow \pi_D$ for some $\pi_D \in \Pi_1^*$, and $M_A(D) = \mathrm{Bel}_0(\pi_D)$ (Appendix C.4).

Theorem 5.6 (The threshold machine)**.** *Let G_{thr} be the theory machine that, after reading s tape bits u, emits every $\pi \in \Pi_1^*$ for which some lower approximation of $\mathbb{U}(O_\pi)$ exceeds $0.u + 2^{-s}$ ($0.u$ the dyadic rational with digits u). For a tape r that is not a dyadic rational, read as a real, $A_r^{G_{\mathrm{thr}}} = \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > r\}$; this theory is inconsistent if $r < \mathrm{Inc}_0$ and true if $r > \mathrm{Inc}_0$. For every data set D of Π_1 type, $M_A(D) \le \nu_{G_{\mathrm{thr}}}(D) \le 2^{K(G_{\mathrm{thr}})+c} M_A(D)$.*

So on Π_1 data the universal mixture is, up to a constant factor, one random true theory: every Π_1^* sentence π with $\mathbb{U}(O_\pi)$ above a uniformly random level. The tape helps only by choosing the level, and that costs only what the probability of the datum already charges. Fixing $k+2$ bits of the level just above Inc_0 gives a true r.e. theory T_k of complexity at most $k + K(k) + c$ that proves every Π_1^* sentence π with $\mathrm{Bel}_0(\pi) \ge 2^{-k}$ (Theorem C.11).

Corollary 5.7 (A coding theorem for Π_1 data)**.** *For every D of Π_1 type, $L_A(D) \le -\log W_A(D) + c \le K_A(D) + c \le L_A(D) + K(\lceil L_A(D) \rceil) + c'$.*

Likewise, if every finite part of a stream is of Π_1 type and $\sup_n L_A(D_n) \le k$, the single true theory $B \cup T_k$ proves every datum (Corollary C.14). For literals no coding theorem holds (Theorem 3.9). For W_A the bound improves to $-\log W_A \le L_A + K^{\mathrm{pad}}(\lceil L_A \rceil + 1) + c$ (Remark C.12); whether $K_A \le L_A + O(1)$ is open (Question C.13).

Code strings by length-lexicographic rank, so that those of length at most m are the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$, and let $\mathrm{Comp}(p, t, y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on p, with output y". The Π_1 sentence $\mathrm{Kgt}_m(x) := \forall t \forall p\, (p \le \underline{\mathrm{rk}_m} \to \neg \mathrm{Comp}(p, t, \underline{x}))$ is true iff $K(x) > m$. If $K(x) > m$, the single true datum $(\mathrm{Kgt}_m(x), 1)$ costs between $m - 2\log m - O(\log\log m)$ and $m + K(m) + c$ bits in the semimeasure and three-valued conventions, and one true theory of complexity at most $m + K(m) + c$ proves all true $\mathrm{Kgt}_m(x)$ at once (Corollary C.15): Chaitin's theorem for random theories. In the normalised convention, with B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$ and for every $j \ge 0$, the datum costs at most $\log(1 + 2^{-j})$ for all but at most $2^{2m+K(m)+j+c} + 2^{m+1}$ strings x of each length (Proposition C.16): the m bits are the price of deciding the sentence, not of predicting its truth value.

The boundary of the coding theorem is exact: with $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, for every $d \ge 2$ there are data D_d with $K_A(D_d) > \log d - 2$ and $L_A(D_d) \le 2 + c$ made of one datum Q-equivalent to a false Σ_1 sentence and one to a true Π_1^* sentence, and others made of two true data Q-equivalent to Boolean combinations of Σ_1 sentences (Propositions C.18 and C.19); on a stream, the gap is bounded against unbounded (Corollary 5.5). To our knowledge, the threshold just above the unknown Inc_0, the single random threshold theory and the coding theorem for an inductive inference rule are new.

5.4 Gödel phenomena

Proposition 5.8 (The arithmetical shape of the trichotomy)**.** *Let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, and let D be consistent. (a) If σ is a true Σ_1 sentence, then $\mathrm{Bel}_D(\sigma) = 1$. (b) If $\pi \in \Pi_1^*$ is false, then $(\mathrm{Bel}_D, \mathrm{Dis}_D, \mathrm{Indep}_D)(\pi) = (0, 1, 0)$; equivalently, $\mathrm{Bel}_D(\pi) > 0$ or $\mathrm{Indep}_D(\pi) > 0$ implies that π is true. (c) If $B \cup D^{\pm}$ is true, $\pi \in \Pi_1^*$ is true and $B \cup D^{\pm} \nvdash \pi$, then*

$\mathrm{Bel}_D(\pi)$, $\mathrm{Dis}_D(\pi)$, $\mathrm{Indep}_D(\pi)$ *are all positive. (d) If* $B \cup D^{\pm}$ *is true,* σ *is a false* Σ_1 *sentence and* $B \cup D^{\pm} \nvdash \neg\sigma$*, then all three values of* σ *are positive.*

The inducer errs one-sidedly: on Σ_1 sentences it may believe a falsehood but never disbelieves a truth; on Π_1 sentences (consistency statements, Gödel sentences, $\mathrm{Kgt}_m(x)$) the reverse. So "independent" on a Π_1 question certifies truth, the inducer's form of "the Gödel sentence is true because it is unprovable"; the normalised convention discards exactly this share.

Theorem 5.9 (Soundness is never learned, coin-free)**.** *Let* B *be decidable and consistent with* $\mathsf{PA} \subseteq \mathrm{Cn}(B)$*. There is* c *such that* $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$ *for every consistent* D*. Hence the coin-free posterior probability of* Σ_1*-soundness, and a fortiori of truth, is at most* $1/(1 + 2^{-c})$ *for every* D*, in particular for every finite part of true arithmetic; and in the Kolmogorov version the shortest compatible* Σ_1*-unsound theory is at most* c *bits longer than the shortest compatible* Σ_1*-sound one.*

Every consistent W_e has a companion $\mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, consistent by (G2), unsound, and only $O(1)$ bits longer. One-tape it gives $P_D(\mathcal{U}) \geq 2^{L_A(D) - K_A(D) - c}$ (Proposition C.20): confidence in soundness needs a large coding gap. On data of Π_1 type the bound is uniform: one-tape, over a decidable true background containing PA, the posterior probability of Σ_1-soundness is at most $1 - 2^{-c}$, with c independent of the data (Theorem C.21, given (G2)); whether this holds for all data is open (for partial answers see Appendix C.5):

Question 5.10. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, is $P_D(\overline{\mathcal{U}})$ bounded away from 1 uniformly over all data D?

On data presenting an r.e. T with $B \cup T$ true and $\mathsf{PA} \subseteq \mathrm{Cn}(B \cup T)$, $\mathrm{Con}(B \cup T)$ keeps positive limit shares on all three answers, and which of "true" and "false" is favoured is a choice of universal machine (Proposition C.22); without truth this fails (for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $T = \emptyset$, $\lim \mathrm{Dis}(\mathrm{Con}(B)) = 1$). Checking instances of a decidable property, the classic case of induction in mathematics,[4] confirms nothing when deduction is free: data entailed by B leave the posterior equal to the prior, for M_A, W_A and K_A alike, so with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and a Δ_0 formula $P(x)$, consistent data $\{(P(\underline{n}), 1) : n \leq N\}$ leave $\mathrm{Bel}(\forall x\, P(x))$ at its prior value for every N (Proposition C.23): the problem of old evidence and logical omniscience (cf. Hacking, 1967; Garber, 1983). Instances of a fresh predicate do confirm, up to the ω-gap; induction in mathematics needs a reasoner for whom computing $P(\underline{n})$ is costly, such as a logical inductor (Garrabrant et al., 2016a) or a derivation-charged inducer (Appendix C.5).

5.5 The ω-gap

Proposition 5.11 (Anatomy of the ω-gap)**.** *Let* $\mathcal{L}$ *have a nonempty set* CT *of closed terms,* B *be decidable and consistent,* $\varphi(x)$ *have one free variable, and* $I := B \cup \{\varphi(t) : t \in \mathrm{CT}\}$ *be consistent. Let the stream list* $(\varphi(t), 1)$ *for all* $t \in \mathrm{CT}$*, in any order. Then* $\mathbb{U}(E_\infty) > 0$*, the posterior converges in total variation to the prior restricted to the consistent extensions of* I*, and the limit trichotomy of* $\forall x\, \varphi$ *is (i)* $(1, 0, 0)$ *if* $I \vdash \forall x\, \varphi$*; (ii)* $(0, 1, 0)$ *if* $I \vdash \neg\forall x\, \varphi$ *(*I *is* ω*-inconsistent at* φ*); (iii) otherwise in* $(0, 1)^3$*, with* $\lim \mathrm{Dis} = P_\infty$*(the theory is* ω*-inconsistent at* φ*) and* $\lim \mathrm{Indep} = P_\infty$*(the theory proves every instance and decides* $\forall x\, \varphi$ *neither way).*

[4] The question of how checking instances can confirm a universal mathematical statement first appeared in Hänni's working notes (Hänni, 2026).

claim	our assessment (an argument)
D answering	true in comparative form; no new inference principle; survives the objection from statistical practice
S logical form	an observation: textbooks are declarative; why that is good: open, relative to a reasoner
E explanation	true in the unification sense, with at least five gaps
L the ladder	SAI is on step 2, with partial results on step 3; step 4 has no formal handle
P progress	replaced by a list of formal differences; in the idealised setting only partiality concerns shape

Table 2: The argument map. The theorems that the assessments rest on are named in Section 6.1.

So the Bayesian ω-gap is the ω-incompleteness of the background plus the instances, weighted by prior shares. In particular Bel satisfies the Gaifman condition (Gaifman, 1964) exactly when $I \vdash \forall x\, \varphi$, and otherwise misses it by at least $2^{-K(I)-c}$ (Proposition C.24), the provability analogue of Proposition B.10 of our earlier report (Hänni and Claude (Anthropic), 2026a). That condition is a desideratum for credences about truth (Hutter et al., 2013); Bel is a probability of provability, and its failure is not a defect. The universal prior does not close the gap even for true Π_1 sentences with trivially checkable instances: over a decidable true background containing PA, for every k there is such a sentence whose belief stays below $2^{-k+c'}$ after any instance data (Theorem C.25, given (G2)). In arithmetic the gap closes under the ω-rule (with $B \supseteq \mathsf{Q}$) and under full second-order consequence, and in the generative variant only with the universal sentence among the data (Remark C.26). A bound on the number of lines of derivations does not close it either: under a full-support prior the limit belief is then 1 iff a Kreisel-type property holds, and that fails already for a fresh predicate and nine lines (Proposition C.27 and Example C.28).

True arithmetic. On true arithmetic, listed by length in a syntax with short binary numerals, the loss in the semimeasure convention grows at least like n^{ξ} up to logarithmic terms, for an exponent $\xi > 0$ of the syntax ($\xi \geq 0.0823$ for a stated syntax, by an exact computation; code in the accompanying repository), and every true r.e. theory leaves at least $C'^{-\xi} n^{\xi}/(2 \log n)$ of the first n sentences undecided for large n, for a constant C' (Theorem C.30 and Corollary C.31); whether the loss is $o(n)$ is open (Question C.33).

6 What a theory does and what it is

What does a scientific theory do? It may predict observations (Solomonoff, 1964); answer questions posed in some language (van Fraassen, 1980, ch. 5); explain (Hempel and Oppenheim, 1948); or solve problems (Laudan, 1977). What is it? A program that predicts observations; a set of sentences closed under consequence (Suppe, 1977); a class of models (Suppes, 1960; van Fraassen, 1980); a core of models with intended applications (Sneed, 1971; Balzer et al., 1987); or, on another line of thought, scientific knowledge rests on exemplars and tacit skills (Kuhn, 1970, 1977; Polanyi, 1958). The *axiom view* answers: questions posed in a given language, and a logical theory; the *raw-data view*: predictions of observations, and a program.

6.1 An argument map

We separate five claims (Table 2). **D**: a theory answers questions posed in a given language rather than predicting raw data (read comparatively). **S**: a theory is a logical theory, not a program. **E**: theories explain propositions. **L**: the ladder of Section 1. **P**: SAI is progress over raw-data prediction, in what a theory does and in what it is.

What rests on theorems. Proved, in the stated settings: on data labelled true or false, the idealised formalism cannot see the difference between a theory and a program (Section 6.4); answering questions differs from raw-data prediction in target, data type, coherence and partiality, not in the inference principle (Propositions 4.5, 4.6, D.2, D.3, D.5 and D.6; the theorems proved, this reading of them an argument); in the model of Section 7, with a cost on the written theory, usefulness depends exponentially on the pair (reasoner, vocabulary), and on declarative against procedural form only by a constant factor once intermediate concepts are available and the reasoner can chain through them (Corollary 7.6 and Theorem E.26; open beyond that model); theoretical vocabulary is dispensable in the ideal mixture, while relations or entities pay for finite theories and under charged deduction (Theorems 8.3 and F.3); and in SAI's mixture an approximately true law is priced as a law with a calibrated error model, carried by the randomness of the theory machine rather than stated as an axiom (Proposition 9.2), a pricing that sees a purpose only through the questions asked (Proposition G.42). Computed, conditional on the pool of forms and the codes: on noise-free interval data simplicity among compatible forms is language-relative, and with graded fit the data decide (Section 10). A theorem plus an argued identification: Hempel's inductive-systematisation reply fails in the idealised setting (theoretical vocabulary changes the cumulative semimeasure loss on observational data by at most a constant number of bits, which depends on the universal machine and the setting; Section 8.6). Arguments only: the descriptive observation; why declarative shape is good; that D survives the objection from statistical practice; claims about explanation beyond the toys (Propositions D.7, G.36 and G.37); that the axiom view misses procedural, statistical and dynamic sides of theories; that approximate correctness, which depends on a purpose, remains a limitation of SAI (Section 11); and our refusal, given a possibility result (Corollary 5.5), to be realist about induced theories.

The ladder. We accept L. Step 2 is a step in the right direction, not the end point: an account of what theories do needs steps 3 and 4 as well. Step 4 is needed because theory-ladenness (Hanson, 1958) biases comparisons between theories that disagree about the base language: a theory that revises it is scored in a language it rejects.

6.2 The descriptive observation and the explanatory question

Claim S combines a description and an explanation. *The observation:* textbook presentations of fundamental theories are declarative, stating laws, definitions and models given by equations, not procedures that print predictions. Two qualifications. *Declarative is not sentential:* physics texts write something closer to a class of models specified by equations (Suppes, 1960, 1962) than to first-order axioms. The semantic view takes a theory to be such a class of models (on the relation of the two views, cf. Halvorson, 2012; Lutz, 2017); Section 10.4 shows where the choice of sentences enters our computation, and how much of its result survives without it. We use sentences because data are sentences and the prior of SAI is over texts; the question-answering view itself is neutral, since in partition semantics a question is a partition of logical space (cf. Groenendijk and Stokhof, 1984). *Theories in use have procedural parts,* such as numerical schemes and simulation codes, so the observation concerns textbook presentations, not every use of a theory.

The question, why the declarative shape is good and whether an ideal inducer should produce it, is open. Craig's construction rules out an answer from representability alone (Section 6.4); any answer must be relative to a reasoner with costs, or to a community of reasoners.

Question 6.1 (Why declarative shape?)**.** Is there a formal setting for a community of bounded

reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs?

A natural argument for S is that a theory answers in many directions while a program computes one cheaply; the second half holds of particular programs only: a program with internal variables answers every direction of a parity law on m variables with $O(m)$ operations. In the model of Section 7 directions matter only relative to a fixed weak reasoner and a cost on what is written or on the time to write it: for one parity law, forward chaining needs exactly m times as many rules as unit propagation needs clauses, over the base variables or with one round, and at most 9 times the written length of the least clausal theory with intermediate concepts and unbounded rounds (Theorem E.26).

6.3 What a theory does: questions against raw data

Questions are defined in Definition 2.10, as in the logic of questions (cf. Belnap and Steel, 1976; Hintikka, 1999); a theory *answers* a question $\{\varphi_1, \ldots, \varphi_k\}$ if it proves, with B, some φ_i, and *partially* if it proves a disjunction of some but not all of them (Definition D.1). A raw-data predictor is scored on one question, "what is the next observation?", and assigns probabilities to every event about future observations; it has no native notion of a question left open, and it answers a question about a law only as a question about all future data (item 5 of Table 3). SAI answers every question the base language can pose, and reports how much of its posterior answers none: the answer masses of a k-way question and its unanswered mass sum to 1 (Proposition D.2).

The conservation-law world. In a toy world a micro-state $x \in \{0,1\}^m$ moves by a uniformly random permutation of each weight class, so its weight is conserved; the raw data are the trajectory, the questions "what is $|x_t|$?". Until the trajectory returns to its start, the law saves a Bayes predictor of the raw data between $\frac{1}{2}\log\frac{\pi m}{2} + o(1)$ bits and one bit more per step (for $t \le Z/2$, Z the size of the weight class), a share $O(\log m/m)$ of its step loss; on $t_{\max}+1$ questions it saves, by construction, the share $t_{\max}/(t_{\max}+1)$ of the no-law loss. Yet by the chain rule the raw saving per step is the loss of the raw no-law predictor's own answer to "what is $|x_{t+1}|$?", and in each task a Bayes mixture pays at most one bit to learn the law (Proposition D.3). The tasks differ in their *target*, which losses are counted, not in what is learned or what the law is worth per step.

Answers never cost more description than raw modelling plus an *extractor* that reads them off the raw data: for a computable law μ of the raw data and an extractor η whose answers are μ-almost surely consistent with B, with μ_η the induced law of the answers, $L_A(D) \le -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$, with c depending also on the family of question sentences (Proposition D.5). The extractor is the bridge from raw data to the question language, and its cost $K(\eta)$ is not small in real cases: it contains whatever recognises a planet in an image.

The objection from statistical practice. Much of physics confronts theory with near-raw data through generative models and information-theoretic losses, as in gravitational-wave searches, microwave-background cosmology and detector simulation. D survives in comparative form, we think, because these pipelines exist to answer a few questions (a mass, a spin, whether a particle exists); but the generative model is theory plus instrument plus noise, and the proof-requiring core of SAI, with no native noise model, sits far from this practice (Section 11).

Purpose and logarithmic loss. The purpose fixes what is graded and with which tolerance, in SAI which propositions are data; scoring credences on them is a different job. The logarithmic

score is, under regularity conditions and up to positive affine transformation, the only strictly proper local scoring rule on more than two outcomes (a result attributed to Bernardo, 1979), but which proper score is privileged is contested (Leitgeb and Pettigrew, 2010; Pettigrew, 2016), and we do not claim that it is; inside SAI even the loss convention can change the price of a datum without bound (Proposition C.16).

6.4 What a theory is, after Craig

The idealised formalism cannot see shape. Coin-free and one-tape, SAI $\asymp$ consistent SOI, by Craig's construction inside the universal prior (Theorems 4.2 and 4.3). Adding vocabulary, restricting coin-free r.e. axiom induction to finite theories in a three-sorted extension, naming a reasoner, and reformulating the flagpole theories of Section 6.5, with any number of laws, each cost $O(1)$ bits (Corollary 3.6, Theorems 7.2, F.3 and F.33, and Proposition D.9); the diagonal separations from SOI under time bounds come from free deduction, not from axioms (Proposition 4.8). So, on data labelled true or false, the idealised formalism cannot see the difference between a theory and a program. Nor does D support S as a claim about what an ideal inducer should produce: labellers carry no theory and are equivalent to theories.

The one respect in which the idealised formalism separates theories from labellers is partiality. When "independent" is itself observed, a theory can leave a sentence open without committing to anything, while a labeller must output "independent" after finitely many steps, and the gap is unbounded on one datum (Proposition 4.6); similarly, in the semimeasure convention a theory is certain of the consequences of the data and a consistent labeller never is (Proposition 4.5). The witnesses of the first gap are arithmetical sentences independent of a background containing Q, and no r.e. family of them shows a large gap (Proposition B.17). We read this as a formal trace of partiality, one of the candidate advantages of logical form, not as support for claim S about empirical theories.

In the idealised setting, on data labelled true or false, the shape of a theory is invisible: every consistent labeller is an axiom system at a cost of $O(1)$ bits. In the settings we study, shape becomes visible only when something other than program length is charged (the written theory, the time to produce it, or the derivations) or the class of hypotheses is restricted (to finite theories); there it is relative to a reasoner and a vocabulary, and for theoretical terms what pays is relations and entities, not vocabulary as such (Theorem 8.3). Whether this holds beyond these settings is open (Question 7.7). We find it plausible that the declarative shape of written theories is good, and not merely common, but our results do not show it. Where shape is visible in our models, the large effects lie in the pair (reasoner, vocabulary), and declarative against procedural form costs at most a constant factor once intermediate concepts are available and the reasoner can chain through them (Theorem E.26); nothing in our results favours sentences over classes of models. What is missing is a formal setting in which that shape provably pays; communication and piecemeal checking by communities of reasoners are the candidates we see (Question 6.1).

6.5 Explanation

In the axiom view T explains φ when $B \cup T \vdash \varphi$, the deductive-nomological pattern (Hempel and Oppenheim, 1948), and an explanation is better the more a short theory derives, as in unification accounts (Friedman, 1974; Kitcher, 1981); the explanatory power of Definition G.29 gives one precise measure of this. But derivability is symmetric where explanation is not: a pole's height and the sun's elevation entail the shadow's length, and conversely (Bromberger, 1966, as attributed by Glymour, 1978). Formally (Proposition D.7): over real closed fields, four formulations of the

	difference	support
Advantages for claims D and E		
1	*Data type:* labelled sentences of a chosen language; questions are sets of sentences.	Defs. 2.10 and D.1
2	*Logical coherence:* consequences of the data get posterior probability 1, unlike for consistent SOI in the semimeasure convention; Bel is monotone under entailment and closed under conjunction up to additive error.	Props. 4.5 and D.6
3	*Partiality:* "independent" is a native third value (its mass is the semimeasure defect); a k-way question's answer masses and unanswered mass sum to 1.	Props. 3.1 and D.2
4	*An observed "independent" label* separates theories from labellers by an unbounded amount on one datum (coin-free and one-tape; $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, $\mathsf{Q} \subseteq \mathrm{Cn}(B)$).	Prop. 4.6
5	*Laws have belief values:* $\mathrm{Bel}_D(\forall x\, \varphi)$ is not the probability that all future data conform.	Prop. 5.11
6	*Answers cost at most raw modelling plus an extractor,* and can cost far less.	Props. D.3 and D.5
Neutral differences		
7	*Piecemeal confirmation:* credence in single laws; SAI inherits the Bayesian treatment of Duhem's problem (Dorling, 1979), which Strevens (2001) calls question-begging, and allocates blame by the prior, so by the syntax.	definitions; an argument
8	*Truth from unprovability:* if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, belief or independent mass on a Π_1 sentence certifies its truth; a curiosity of arithmetic data.	Prop. 5.8
9	*Explanatory direction needs questions the language can pose:* interventional data separate causal theories that the old vocabulary cannot.	Prop. D.7(c), a toy
Liabilities		
10	*Random theories, false theories:* on the lottery stream bounded one-tape loss is achieved by random theories and by no consistent r.e. theory; in true arithmetic with true (adversarial) data the limit posterior can live on false theories, a possibility result.	Thm. 5.1 and Cor. 5.5
11	*Language-relative simplicity, within bounds:* faithful translations change weights by $2^{\pm(K(\tau)+c)}$; within that bound the primitives in which laws are written decide real choices, such as the preferred orbit of Mars on noise-free interval data.	Thm. 3.5; § 10

Table 3: What formally separates SAI from raw-data prediction. Items 7 and 9 and the sorting are arguments, the others results. Items 2 and 4 compare SAI with operator induction over consistent labellers, the program-shaped counterpart of a theory; they bear on claim S as well as on D.

shadow law, with the height or the reference shadow as the hub, are deductively equivalent, so the coin-free posterior keeps their prior odds on every compatible data set; under derivation charges which quantity is the hub is a property of the formulation, not of the world; and one interventional datum separates two causal versions that agree in the old vocabulary.

What the toys show. The explanation relation of SAI is direction-free. Kitcher's answer to the asymmetry (Kitcher, 1989), in its usual summary, appeals to other laws in which the height figures: with k of them, the formulation with the shadow as hub is longer by an amount growing with k in a literal code without definitions (if the reference shadow is not primitive or some objects cast no shadow), by a constant with one definition, and by $O(1)$ under the universal prior (Proposition D.9): Kitcher's answer is a whole-theory simplicity effect that SAI reproduces only under a listing code. Direction enters cleanly with interventional questions, but their semantics was written in by hand, so the toy is consistent with the interventionist account (Woodward, 2003) and gives it no support over rivals such as the causal-process account (Salmon, 1984).

We conclude that derivation from a short theory is explanation in the unification sense, with at least five gaps; we do not claim that derivation is explanation outright. The gaps are *asymmetry*, as above; *explanation by approximate correction*: Newton's theory does not entail Kepler's laws but, in an exact toy, proves the honest old law on every window and refutes the exact one

(Proposition G.37), so explaining must include deriving an honest approximation with an error bound on a domain; *irrelevance* (hexed salt; a man taking birth-control pills): the length prior penalises an irrelevant axiom in a whole theory, but a derivation does not mark which axioms it used; the *conjunction problem* (L is derived from $L \wedge L'$), which the length prior and the explanatory power address in part; and *statistical explanation*: the approximate laws that SAI prefers come with error models (Proposition 9.2) and make their data probable rather than derive them (cf. the statistical-relevance model of Salmon, 1971). Unification can register an explanatory gain without a gain in accuracy, as with Copernicus against Ptolemy (Proposition G.36).

6.6 What separates SAI from raw-data prediction

We replace claim P by the formal differences of Table 3. Neither the inference principle nor, in the idealised setting, the shape of hypotheses separates SAI from raw-data prediction, except through partiality. This is the progress claim we believe: for what a theory does, SAI is a step in the right direction; for what a theory is, the idealised formalism is neutral on data labelled true or false.

SAI is Bayesian confirmation with a universal prior in which a theory either proves the data or is eliminated, and it inherits that framework's problems in specific forms: the dependence of the prior on the machine and the language, bounded by constants that are not small at the scale of our examples (Proposition A.8, Theorem 3.5, and Section 10.4); logical omniscience and the problem of old evidence (Proposition C.23); and the allocation of credit among the axioms of a theory (item 7 of Table 3).

7 Theory shape relative to a reasoner

The axioms of a mechanics textbook do not yield the solutions of its exercises by a simple mechanical procedure: the reader brings methods, lemmas and understanding that are not written down.[5] Program length cannot express this: a short program can contain a generic solver, and by Craig's construction every consistent assigner of truth values is an axiom set at constant cost (Theorem 4.2 and Section 6.4). In a simple model with a reasoner, program length sees neither the reasoner nor the vocabulary, the written size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Table 4), and declarative against procedural form matters only by a constant factor once intermediate concepts are available and the reasoner can chain through them.

7.1 The model

Laws constrain the joint values of m observed quantities, and questions give some values and ask for another (the conditional questions of Definition 2.10, with literals as sentences).

Definition 7.1 (Theories, laws, reasoners, inducers)**.** Fix base variables $x_1, \dots, x_m$ and auxiliary variables $y_1, y_2, \dots$, the *intermediate concepts* (0-ary relation symbols of a first-order language $\mathcal{L}_m$ with equality). (a) An *explicit theory* Φ is a finite set of non-tautological clauses, a *linear theory* a finite set of equations over GF(2), in these variables, up to renaming of the auxiliary ones; $\operatorname{proj}\Phi := \{x \in \{0,1\}^m : \exists y\ (x,y) \models \Phi\}$. The hypothesis classes are $\mathcal{E}_m^{\text{base}}$ (no auxiliary variables) and $\mathcal{E}_m$ (any); likewise for linear theories. $|\Phi|$ counts clauses (equations), the *written length* $\|\Phi\|$ literal occurrences. (b) A *law* is a nonempty $\mathcal{S} \subseteq \{0,1\}^m$. $\mathcal{Q}(\mathcal{S})$ is the set of conditional questions

[5] That written theories are far from sufficient for a simple reasoner, and are aids for a reasoner with much tacit understanding, is argued in Hänni's working notes (Hänni, 2026).

(α, ℓ), α a partial assignment to the base variables consistent with $\mathcal{S}$ and ℓ a literal on a base variable outside $\operatorname{dom}\alpha$, with $\mathcal{S} \wedge \alpha \models \ell$; a *conditional data set* is a finite set of such pairs. (c) A *reasoner* $\mathcal{R}$ maps a theory and an antecedent α to a set of base literals or to a conflict $\perp$; it is *sound* if $\ell \in \mathcal{R}(\Phi, \alpha)$ implies $\Phi \cup \alpha \models \ell$, and $\perp$ occurs only if $\Phi \cup \alpha$ is unsatisfiable. The named reasoners are sound: UP_k, k *parallel* rounds of unit propagation (in each round every clause whose literals are all falsified but one, that one unassigned, assigns it; opposite derivations or a falsified clause give $\perp$); $\mathrm{UP} := \mathrm{UP}_\infty$; Gaussian elimination GE on linear theories and unbounded proof search PS_∞, which return exactly the entailed base literals. (d) Φ is $\mathcal{R}$-*compatible* with D if for every $(\alpha, \ell) \in D$, $\Phi \cup \alpha$ is satisfiable and $\ell \in \mathcal{R}(\Phi, \alpha)$; $\mathcal{R}$-*usable* for $\mathcal{S}$ if $\operatorname{proj}\Phi = \mathcal{S}$ and Φ is $\mathcal{R}$-compatible with $\mathcal{Q}(\mathcal{S})$. (e) The inducers range over the $\mathcal{R}$-compatible $\Phi \in \mathcal{E}_m$ (linear theories for GE); a superscript base restricts them to $\mathcal{E}_m^{\mathrm{base}}$. *Program length*: $K_A^{\mathcal{R}}(D) := \min|p|$ over the p with $V(p) = \langle\Phi\rangle$, and $W_A^{\mathcal{R}}(D) := \sum 2^{-|p|}$ over the same p. *Listing*: $K_{\mathrm{lst}}^{\mathcal{R}}(D) := \min \ell_{\mathrm{lst}}(\Phi)$ and $W_{\mathrm{lst}}^{\mathcal{R}}(D) := \sum 2^{-\ell_{\mathrm{lst}}(\Phi)}$, for the prefix code $\ell_{\mathrm{lst}} \geq \|\cdot\|$ of Equation (4), which lists the clauses literal by literal. *Time-charged*: $Kt_A^{\mathcal{R}}(D) := \min(|p| + \log t_p)$ over the p of $K_A^{\mathcal{R}}$, t_p the running time of V on p.

With PS_∞ and (E3) (the two-valued encoding is degenerate, Proposition E.3), the program-length inducers are the SAI inducers K_A and W_A on question data, up to $O(1)$ for r.e. hypotheses and $m + O(\log m)$ bits for finite ones (Proposition E.4). An *affine law* is $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$, $A \in \mathrm{GF}(2)^{r \times m}$ of rank $r \geq 1$. For affine laws and the implication chain on $m \geq 3$ variables, $\mathcal{R}$-compatibility with $\mathcal{Q}(\mathcal{S})$, for a sound $\mathcal{R}$, already forces $\operatorname{proj}\Phi = \mathcal{S}$ (Lemma E.2).

7.2 Program length sees neither the reasoner nor the vocabulary

The prime implicates $\mathrm{PI}(\mathcal{S})$ of a law are its inclusion-minimal implicates among the clauses over the base variables; one round of unit propagation on them answers every question of $\mathcal{Q}(\mathcal{S})$ (Lemma E.5).

Theorem 7.2 (Reasoner invariance under program length)**.** *Let $\mathcal{R}'$ be a sound reasoner whose hypotheses are explicit or linear theories, of either class. For every conditional data set D, $K_A^{\mathrm{UP}_1,\mathrm{base}}(D) \leq K_A^{\mathcal{R}'}(D) + c$ and $W_A^{\mathrm{UP}_1,\mathrm{base}}(D) \geq 2^{-c} W_A^{\mathcal{R}'}(D)$; and $K_A^{\mathrm{UP}_1} = K_A^{\mathrm{UP}} = K_A^{\mathrm{PS}_\infty} \pm c$ for finite explicit theories of either class.*

Proof idea (Appendix E.2). If Φ' is $\mathcal{R}'$-compatible with D, soundness makes $\mathrm{PI}(\operatorname{proj}\Phi')$ UP_1-compatible with D, and a fixed prefix computes it from a program for Φ'.

This is Craig's collapse again, now for reasoners and vocabularies: the know-how a weak reasoner lacks can live in a generator of the written theory, at constant cost. In this model it becomes visible when what is written, or the time to write it, is charged.

Theorem 7.3 (Where the reasoner and the vocabulary become visible)**.** *Let $\mathcal{S}$ be an affine law, $\mathcal{R}$ a sound reasoner, and take either class. (a) $K_{\mathrm{lst}}^{\mathcal{R}}(\mathcal{Q}(\mathcal{S}))$ is the least $\ell_{\mathrm{lst}}(\Phi)$ over the $\mathcal{R}$-usable Φ of the class. (b) $K_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) = K(\mathcal{S}) \pm c$ for $\mathcal{R} \in \{\mathrm{UP}_k \ (k \geq 1), \mathrm{UP}, \mathrm{PS}_\infty, \mathrm{GE}\}$.*

Proof idea (Appendix E.2). (a) is Lemma E.2; for (b), a compatible Φ gives $\operatorname{proj}\Phi = \mathcal{S}$, which gives $\mathrm{PI}(\mathcal{S})$.

When the time to write the theory is charged instead, $Kt_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S}))$ is at least the logarithm of the least written length of an $\mathcal{R}$-usable theory, and for UP_1, UP and GE it exceeds Levin's $Kt(A, b)$ by at most $\log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, $O(r + \log m)$ and $O(\log m)$ bits (Proposition E.6).

7.3 Over the base variables: the exact minimum

How large must a theory be for unit propagation if it may use only the observed quantities? A *derived equation* of an affine law is $u = yA$, with $u \cdot x = b_u := y \cdot b$ on $\mathcal{S}$; d_* is the least weight of a nonzero one; a *circuit* is a nonzero derived equation whose support contains the support of no other. Let $C_{Z,\beta}$ be the clause over the variable set Z falsified exactly by the assignment β; the prime implicates are the $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ (Proposition E.8). A circuit u is *decomposable* if $u = u' + u''$ for nonzero derived equations whose supports meet in a single base variable outside $\mathrm{supp}(u)$: two derived equations chained through one intermediate base quantity. Let $\Phi_{\mathrm{ind}}(\mathcal{S}) := \{C_{\mathrm{supp}(u),\beta} : u \text{ an indecomposable circuit, } u \cdot \beta \neq b_u\}$. For every affine law the clauses contained in every UP-usable explicit theory over the base variables are exactly those of $\Phi_{\mathrm{ind}}(\mathcal{S})$ (Proposition E.12), and:

Theorem 7.4 (The unique minimum when $d_* \geq 3$)**.** *If $d_* \geq 3$, $\Phi_{\mathrm{ind}}(\mathcal{S})$ is* UP*-usable. Hence an explicit theory over the base variables equivalent to $\mathcal{S}$ is* UP*-usable if and only if it contains $\Phi_{\mathrm{ind}}(\mathcal{S})$, which is the unique minimum, with $\sum_{u \text{ indecomposable}} 2^{|u|-1}$ clauses.*

Proof idea (Appendix E.3). From an antecedent that fixes all of an indecomposable circuit but one variable, the first round must fire a clause supported on that circuit, since a clause through another variable would decompose it. For sufficiency, induct on the weight: a decomposable circuit is answered in two steps through its intermediate variable, and $d_* \geq 3$ makes both parts lighter.

So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one base quantity: base quantities already act as intermediate concepts (Corollary E.13 says when the given equations suffice; Remark E.15 treats $d_* \leq 2$). For the *simplex family*, whose matrix A has as its columns the $m = 2^r - 1$ nonzero vectors of $\mathrm{GF}(2)^r$ ($r \geq 3$), the unique minimum over the base variables has $m2^{(m-1)/2}$ clauses against $r = \log(m+1)$ equations for GE (Corollary E.14); with intermediate concepts and unbounded propagation $m(2m-4)$ clauses suffice for $r \geq 4$ (Remark E.19), so the family separates the reasoners exponentially only over the base variables or for one round.

7.4 Intermediate concepts help only a reasoner that can chain through them

One parity law $\mathrm{Par}_m^b := \{x : x_1 \oplus \cdots \oplus x_m = b\}$, $m \geq 3$, needs all its 2^{m-1} clauses for unit propagation over the base variables (Corollary E.13). With the partial sums $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ as intermediate concepts, $4m-6$ clauses suffice for unit propagation when $m \geq 4$ (Proposition E.16), but only for a reasoner that can chain. One round cannot use intermediate concepts: every UP_1-usable theory for a law, with any auxiliary variables, contains all its prime implicates, 2^{m-1} clauses for one parity law (Proposition E.17). With k rounds and any intermediate concepts, one parity law needs written length $2^{\Omega(m^{1/(2k-1)})}$ for each fixed k, given Håstad's lower bound for parity (Håstad, 1986), while $2\lceil \log m \rceil - 1$ rounds suffice for the $4m-6$ clauses of a balanced tree of partial sums (Proposition E.24).

7.5 Systems of laws, and the regimes

For systems of laws, neither intermediate concepts nor depth need help. In the following law the observed quantities are fixed linear functionals of an unobserved state. Let Λ_q be the set of $v = (u, a) \in \mathrm{GF}(2)^{q+1}$ with $u \in \mathrm{GF}(2)^q$ of weight 1 or 3 and $a \in \{0, 1\}$, and $\mathbf{e}_\star := (0^q, 1)$. The *hidden-state law* on the $m = |\Lambda_q| + 1$ base variables $(x_v)_{v \in \Lambda_q}$, $x_\star$ is $\mathcal{H}_q := \{((z \cdot v)_{v \in \Lambda_q}, z \cdot \mathbf{e}_\star) : z \in$

reasoner $\mathcal{R}$	parity, base	parity, interm.	simplex, interm.	hidden state, interm.
UP_1	2^{m-1}	2^{m-1}	$m2^{(m-1)/2}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, k fixed	2^{m-1}	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{1/(2k-1)})}$	$2^{\Omega(m^{\varepsilon/3})}$
UP_k, $k \geq 2\lceil \log m \rceil - 1$; UP	2^{m-1}	$\leq 4m-6$	$\leq m(2m-4)$	$2^{\Omega(m^{\varepsilon/3})}$
GE	1 equation	1 equation	$\log(m+1)$ equations	$\leq m$ equations

Table 4: The regimes (Cor. 7.6): the least size of a theory usable by the reasoner of the row, with the round budget the row states, in the class of the column (base: no auxiliary variables; interm.: any intermediate concepts). The inducer is $K^{\mathcal{R}}_{\mathrm{lst}}$ (superscript base in the first column), whose value on $\mathcal{Q}(\mathcal{S})$ is the listing length of a least usable theory (Thm. 7.3(a)): at least its written length, which is at least the number of clauses given here, and at most a factor $O(w \log \bar{m})$ larger for clauses of width w over $\bar{m}$ variables. Entries without a sign are exact and count clauses (equations for GE); $\leq$ marks the size of an explicit usable theory; $2^{\Omega(\cdot)}$ bounds the written length from below, given the bound of Håstad (1986) for fixed k and that of Göös et al. (2019) in the last column. Under program length every entry is $K(\mathcal{S}) \pm c$ (Thm. 7.3(b)).

$\mathrm{GF}(2)^{q+1}\}$. With the state $z = (y, z_0)$ as intermediate concept it is written as the m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, each of weight at most 5.

Theorem 7.5 (Systems of laws need exponential written length)**.** *The m equations above are* GE*-usable for $\mathcal{H}_q$. There are $\varepsilon > 0$ and q_0 such that for every $q \geq q_0$, every* UP*-usable explicit theory Φ for $\mathcal{H}_q$, with any auxiliary variables, clause widths and number of rounds, has $\|\Phi\| \geq 2^{q^\varepsilon/2 - O(1)} - m = 2^{\Omega(m^{\varepsilon/3})}$.*

Proof idea (Appendix E.4). Setting a set S of the x_v to 0 determines $x_\star$ exactly when the XOR system $\{u \cdot y = a : (u,a) \in S\}$ is unsatisfiable, so unit propagation on a usable Φ, read as a monotone circuit (Lemma E.22), decides this with $O((\|\Phi\| + m)^2)$ gates. The monotone 3-XOR-SAT function of Göös et al. (2019) is a projection of it and needs monotone circuits of size 2^{q^ε}.

Through the monotone-circuit view of unit propagation (Bessiere et al., 2009), this improves the superpolynomial bound of Gwynne and Kullmann (2014) for arc-consistent representations of XOR systems (Lemma E.21 and Remark E.23); we did not find it stated, and it is probably known to specialists. Private intermediate concepts add nothing (Proposition E.20); a solver whose unit propagation does linear algebra (Beame and Sun, 2026) is a reasoner for which such laws are cheap.

Corollary 7.6 (The regimes)**.** *For one parity law ($m \geq 3$), the simplex family and the hidden-state law, the least sizes of theories usable by* UP_1*, by* UP_k *(k fixed, or $k \geq 2\lceil \log m \rceil - 1$), by* UP *and by* GE*, over the base variables or with any intermediate concepts, are as stated in Table 4.*

So the exponential separations are between pairs (reasoner, vocabulary). For the hidden-state law no vocabulary helps unit propagation at any depth, while elimination needs only the natural one.

7.6 Directions, know-how and learning

Directionality. Logical form might matter because a law can be used in every direction, while a program computes one. Take as the procedural counterpart a *rule base*: sound directed rules

$P \Rightarrow \ell$ over the base variables (P consistent with $\mathcal{S}$, $\mathcal{S} \wedge P \models \ell$), used by forward chaining. For one parity law the only usable rule base has all $m2^{m-1}$ sound rules, m times the clauses of the least UP-usable theory over the base variables, a factor m (the number of query directions) that also holds for one round with any vocabulary. But for every law and round budget the factor is at most the largest clause width of a smallest usable theory, with intermediate concepts and unbounded rounds at most 9 in written length, and under program length it vanishes (Theorem E.26): the dependence on directions is real but cheap.

Where the know-how lives. Know-how can be paid for in the reasoner (the r equations with GE), in written lemmas, or in a computation that the cost model does not charge: a generator under program length, or a membership test with free search for derivations, such as a polynomial-time decider of $\mathrm{PI}(\mathcal{S})$ of length $K^q(A, b) + c$, K^q being description length under a suitable polynomial time bound q (Proposition E.29). So the charged-deduction readings of Theorem B.31, extended to conditional data, give the full lemma library a cost within $K^q(A, b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible, given a derivation bound that we sketch (Theorem E.31 and Remark E.32). They do not charge finding a derivation (Section 4.6), so they cannot express the textbook point, which concerns a reasoner that must find the answer.

Learning. For one parity law ($m \geq 3$), an inclusion-minimal UP-compatible theory over the base variables fails on some question of every unseen *pattern* (the non-solution obtained by flipping the asked variable), and on data made of whole patterns the listing inducer pays at least the listing length $\mathrm{len}_m \approx m \log m$ of a full clause per pattern (semimeasure convention); with GE, or with intermediate concepts and unbounded propagation, it pays $O(m \log m)$ bits in all (Theorem E.34).

Depth against size. For the implication chain $x_1 \to \cdots \to x_m$ the least UP_k-usable theory over the base variables has as many clauses as a minimum k-hop transitive-closure spanner of the path (Theorem E.36), $\Theta(m\, \mathrm{ia}_k(m))$ with ia_k the k-th inverse-Ackermann row (Alon and Schieber, 1987; Bhattacharyya et al., 2012): $\binom{m}{2}$ at $k = 1$, $\Theta(m \log m)$ at $k = 2$; at depth 2 intermediate concepts and wide clauses save at most a factor of about 2 (Theorem E.39).

7.7 What this shows and does not show

Question 7.7. Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws: to real-valued laws with numerical solvers, to probabilistic programs with conditioning, to simulation codes?

This is the precise form of the open question why theories are shaped like logical theories (Question 6.1).

Reading. Whether a written theory suffices depends on the reader. Paired with a reasoner that can do algebra, a linear law's few equations suffice. Paired with one that only applies rules, the theory must spell out exponentially many derived laws, unless it is written with new intermediate quantities and the reader can chain many steps through them. Even then, some systems of laws need exponentially long written theories for such a reader, whatever quantities are introduced. The idealised inducer cannot see any of this, because it charges a theory by the length of a program that may compute the missing lemmas. The model does not show that logic, rather than probabilistic programs with conditioning or classes of models, is the right formal home for

theories: the large separations are between pairs (reasoner, vocabulary). It does not touch the gap between a problem as posed and its formalisation, nor tacit skill (Section 11). Its laws are linear or implicational, its reasoners unit propagation, forward chaining and elimination, and its cost the written length. It is not a theorem about textbooks.

8 Theoretical terms: Craig, Ramsey and the theoretician's dilemma

Theories introduce vocabulary that their data do not contain: field, entropy, gene. Hempel's theoretician's dilemma (Hempel, 1958) runs: if theoretical terms serve to systematise the observable facts, they are dispensable, by Craig's theorem (Craig, 1953) or by Ramsey's method (Ramsey, 1931); if not, they are pointless. In Solomonoff axiom induction hypotheses may use vocabulary that the data lack, so one can ask in which hypothesis classes theoretical terms change the posterior, and what pays: vocabulary, relations or entities (the third step of the ladder of Section 6.1). Theorem 8.3 answers; Appendix F has the proofs.

8.1 Setting and the dilemma theorem

Definition 8.1 (Observational restriction)**.** Let $\mathcal{L} = \mathcal{L}_O \cup \mathcal{L}_T$ be a countable language with equality and computable syntax (finite where derivations are charged), with *observational* part $\mathcal{L}_O$; B may contain theoretical symbols. The *observational content* of $A \subseteq \mathrm{Sent}_{\mathcal{L}}$ is $T_O(A) := \mathrm{Cn}(A) \cap \mathrm{Sent}_{\mathcal{L}_O}$, and D is *observational* if $\mathrm{dom}\, D \subseteq \mathrm{Sent}_{\mathcal{L}_O}$. Let $G_U^O := \{(w,\chi) \in G_U : \chi \in \mathrm{Sent}_{\mathcal{L}_O}\}$, $M_A^{\mathcal{L}_O} := \nu_{G_U^O}$, $W_A^{\mathcal{L}_O}(D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup (\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O})$ consistent and $\vdash \psi_D]$, and let $K_A^{\mathcal{L}_O}(D)$ be the least $K(e)$ over the same e.

Definition 8.2 (Literal data and charged deduction)**.** *Literal data* are those of Definition 2.2, in the fresh-atom setting, with $\mathcal{L}_O = \{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R\}$. *Charged deduction* is that of Section 4.6: a hypothesis decides an axiom set A_p within time t, satisfies Assumption B.30, is consistent, and derives each datum in the calculus K with every axiom instance, logical ones included, of size at most h (reading (a1)), every nonlogical instance of size at most h ((a2)), material at most g ((b)), or nonlogical material at most g ((b2)); the mixtures are $W_A^{a1}[t,h]$, $W_A^{a2}[t,h]$, $W_A^{b}[t,g]$, $W_A^{b2}[t,g]$. At *polynomial budgets* the degrees of membership time and budget are charged in the prior, as in Definition E.18 of our earlier report (Hänni and Claude (Anthropic), 2026a); then X is in the fit class iff one hypothesis is compatible with every D_n^X (Lemma E.11 there).

A theory in a language containing $\mathcal{L}_O$ is *about the named objects* if it has a model whose $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$-reduct is the named structure $\mathfrak{T}$, and *compatible with the named structure* if it is consistent with the first-order theory $\mathrm{Th}_{\mathfrak{T}}$ of $(\mathfrak{T}; \mathsf{e}, \mathsf{s}_0, \mathsf{s}_1)$ (Definition F.1); "named object" is our formal stand-in for "observable thing".

Theorem 8.3 (The theoretician's dilemma in Solomonoff axiom induction)**.** *Data are observational sentences; in (ii)–(iv), literal data under the readings of Definition 8.2 at polynomial budgets. In the fit-class statements of (iii), and in (iv) and (v), $B = \emptyset$.*

(i) In principle. *With deduction free, restricting hypotheses to the observational vocabulary changes the prior weight of every Borel class of observational contents, the predictive probability of every observational sentence and the cumulative loss by at most a constant, in the*

one-tape, coin-free and Kolmogorov versions, and also when the time to check or to generate axioms is penalised (under Assumptions B.5 and B.6). Tape by tape, the observational projection of a random theory has exactly the same law on observational data.

(ii) Craig's construction in practice. *When derivations are charged, Craig's axiomatisation of a theory whose observational consequences have derivations of size* ds *needs derivations of size at least* 2^{ds}. *Every axiomatisation by padded observational literals with polynomial-time membership fits exactly the polynomial-time decidable regularities, under every reading. Axiom sets with theoretical symbols fit* $\mathsf{NP} \cap \mathsf{coNP}$ *(total material),* $\mathsf{EXPTIME}$ *(every instance short) and every decidable regularity (only the theory's own axioms short); so Craig's construction loses unconditionally under the last two readings and, under the first, iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$.

(iii) Eliminating the vocabulary in practice. *Theoretical vocabulary can be eliminated at polynomial cost: per hypothesis, with* $O(1)$ *bits and a polynomial change of budget, under total material, by storing proofs instead of run times; and at the level of fit classes under (a1), (a2), (b2), by storing the eliminated relations in objects that no observational term names. Observational hypotheses, and already finite observational theories, fit what all hypotheses fit.*

(iv) What is indispensable. *A finite theory in the observational vocabulary with any unary theoretical predicates and constants, compatible with the first-order theory of the named objects, fits exactly the regular regularities, under every reading and also with free deduction. One binary theoretical relation over the named objects restores every decidable regularity ((a2), (b2)),* $\mathsf{NP} \cap \mathsf{coNP}$ *((b)) and* $\mathsf{EXPTIME}$ *((a1)). So does quantifying over unnamed objects; but a finite theory that does so without theoretical vocabulary, and fits a non-regular regularity, must prove some* $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$*-sentence false in the named structure. For the theory built in Theorem F.14 that sentence is* $\exists z\, \mathsf{s}_1\mathsf{s}_1 z = z$*, true of no named object.*

(v) Free deduction, finite theories. *With deduction free, finite theories gain without bound from a theoretical function (infinity), from theoretical sorts (the theorems of* PA*) and from a binary relation over the named objects; and r.e. axiom induction is finite-theory induction with theoretical sorts up to* $O(1)$*, coin-free.*

Proof. (i) is Theorem F.3. (ii) is Lemma F.5, Theorem F.6, and Corollary F.8, given Corollary B.32, the time hierarchy theorem and $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981). (iii) is Propositions F.9 and F.11 and Theorem F.14, under Assumption B.30, given the one-tape simulation and $\mathsf{APSPACE} = \mathsf{EXPTIME}$. (iv) is Theorems F.14, F.16 and F.17 and Proposition F.20, given the Ehrenfeucht–Fraïssé theorem, the one-tape simulation and, for (a1), $\mathsf{APSPACE} = \mathsf{EXPTIME}$. (v) is Theorems F.28, F.31 and F.33 and Corollary F.19; the clause on PA is given Ryll-Nardzewski's theorem and Con(PA), the clause on sorts the Σ_1-completeness of Q. □

8.2 In principle

Part (i) is Theorem F.3, which states it for each version named in (i) and each Borel class of observational contents, and proves the identity of laws tape by tape.

Remark 8.4 (Inductive systematisation in the mixture). By Theorem F.3, the posterior predictive probability of every observational sentence, given observational data, is the same up to a factor $2^{\pm 2c}$, c the constant of that theorem, whether or not hypotheses may use theoretical vocabulary. So in the universal mixture, with deduction free, theoretical terms are not needed for inductive

inference among observational sentences. This is a statement about the mixture, not about a single theory (Section 8.6).

8.3 In practice: Craig's construction and better eliminations

Craig's construction pads each consequence with the time at which it was found: the literal φ found after k steps becomes the axiom $\varphi^{\wedge(k+1)}$, a conjunction of $k+1$ copies. When the time is that of canonical proof search, on literal data every derivation from the Craig set has a line of size at least $2^{\mathrm{ds}}|\varphi^b|$, ds the size of a shortest derivation of φ^b from the original theory (Lemma F.5). For a seven-sentence Horn theory, in a small proof system for ground literals and at a word of length 24, a computation from closed forms gives a Craig axiom of more than $10^{204.6}$ symbols, against 795 for an axiom that carries a certificate (Example F.7). At polynomial budgets, every *run-padded* axiomatisation (each member a power $\psi^{\wedge m}$ of a literal $\psi \in \{\varphi_w, \neg\varphi_w\}$) with polynomial-time membership fits exactly the polynomial-time regularities, under every reading (Theorem F.6); so the classical reply holds of Craig's construction, and we did not find this complexity-theoretic form of it in the literature.

Vocabulary itself can be eliminated at polynomial cost. Under reading (b), the certificate axioms $\theta_\gamma \to \varphi^b$ of Theorem B.31, observational because θ_γ uses only logical symbols, replace derivations hypothesis by hypothesis at $O(1)$ bits and a polynomial change of budget (Proposition F.9). Under the other readings the content goes into objects that no observational term names: in a structure of words and formal pairs, the pairs being the elements z with $\mathtt{s}_1\mathtt{s}_1 z = z$, an observational formula $\theta_2(a, b)$, with a fresh bound variable y_θ, says that R holds of the pair of a and b, and it can replace the atoms of a binary theoretical relation (Lemma F.10 and Proposition F.11). With empty background, for every *verifier* $\mathcal{V}$, a halting machine that accepts or rejects pairs (word, certificate) consistently, there are a finite theory in $\mathcal{L}_O \cup \{\mathsf{C}\}$, C binary, about the named objects and a finite set of $\mathcal{L}_O$-sentences that prove exactly the literals that $\mathcal{V}$ certifies, by derivations whose length and line size are linear in the time and space of a one-tape simulation of $\mathcal{V}$ plus the lengths of the input and the certificate; with a variant for alternating machines, finite theories of either kind fit what all hypotheses fit, under every reading (Theorem F.14). With deduction free the existence part is classical: as stated in Pakhomov and Visser (2019), every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952), even one into which every model expands (Craig and Vaught, 1958). To our knowledge the derivation bounds, hence the fit classes under charged deduction, and the observational translation are new.

8.4 Vocabulary or ontology

Without objects outside the named structure unary theoretical vocabulary does not help: with empty background, a finite theory in $\mathcal{L}_O$ plus unary predicates and constants, compatible with the named structure, that proves the literals of X forces X to be regular (Theorem F.16, given the Ehrenfeucht–Fraïssé theorem), and every regular X has such a theory (Theorem F.17); one binary relation over the named objects, or quantification over unnamed objects in $\mathcal{L}_O$ alone, lets finite theories fit what all finite theories fit, and with deduction free the contrast holds for complexities too (Corollaries F.18 and F.19). The second way has a price: a finite theory without theoretical vocabulary that fits a non-regular regularity proves some $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-sentence false in the named structure (for the observational theory built above, $\exists z\, \mathtt{s}_1\mathtt{s}_1 z = z$; Proposition F.20), while the theory with the binary relation is conservative over the theory of the named objects plus the data (Proposition F.21). One binary observable relation breaks Theorem F.16 (Proposition F.22).

8.5 Finite theories with free deduction

With deduction free and $B = \emptyset$, finite theories, the single sentences of K_{fin} and W_{fin} (Definition 2.7), can gain from theoretical vocabulary without bound. On the data "there are at least n things", the least complexity of a consistent sentence in pure equality plus unary predicates that proves the data is $K^{\text{pad}}(n) \pm c$, which tends to infinity, though more slowly than every computable unbounded nondecreasing function, while with a unary function symbol f the sentence saying that f is injective and not surjective gives $O(1)$ (Theorem F.28). Its Ramsey sentence is finite but second-order, the observational axiomatisation of its $\{=\}$-consequences is first-order but infinite, and no finite first-order theory in pure equality, or in pure equality with unary predicates added, has the same consequences (Proposition F.35). On the theorems of PA as data, finite $\mathcal{L}_{\text{ar}}$-theories diverge, given the theorem of Ryll-Nardzewski (1952), while theoretical sorts give bounded complexity (Theorem F.31); and for finite $\mathcal{L}$, r.e. axiom induction is, up to a constant and coin-free, finite-theory induction in a three-sorted extension with numbers, assignments and satisfaction (Theorem F.33), an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958) of which only the Bayesian consequence is new. The equivalence is not one-tape: on the lottery stream of Theorem 5.1, random r.e. theories have bounded loss and random finite theories with any theoretical apparatus do not (Remark F.34).

A concept can also pay by factoring inferences.[6] If each of n_A unary predicates implies each of n_B others, a theory of atom-to-atom implications needs $n_A n_B$ members, with a new intermediate predicate exactly $\min(n_A n_B, n_A + n_B)$; so the predicate pays in this format iff $(n_A - 1)(n_B - 1) > 1$, and in full first-order logic only by a constant factor (Proposition F.36). This is the elementary instance, with the exact gain, of the compression argument for predicate invention (cf. Muggleton and Buntine, 1988).

8.6 What this does and does not show

What bounded deduction and finiteness make indispensable is theoretical *relations* or theoretical *entities* (for the theory we build, objects outside the named structure), not vocabulary ((iii), (iv)); and in the idealised setting nothing theoretical is indispensable, because the r.e. prior already is a prior over finite theories with theoretical vocabulary and entities ((i), (v)). The results concern one poor observational language and literal data.

Hempel's replies. *Impracticality*: Craig's replacement axiomatisation is unmanageable. This is a theorem, but only about Craig's construction and run-padded axiomatisations under charged derivations; vocabulary as such is eliminable at polynomial cost (Theorem 8.3(ii), (iii)). *Inductive systematisation*: Hempel argued that theoretical terms may be needed for inductive, not deductive, systematisation (Hempel, 1958); the debate is surveyed by Niiniluoto (1972). Solomonoff axiom induction is an inductive systematisation, and in its universal mixture with free deduction the reply fails: theoretical terms add no inductive power among observational sentences beyond a constant factor (Remark 8.4). This is a statement about the mixture; Hempel's own notion is relative to a single theory, and our statement is not a reconstruction of it. The reply survives, in corrected form, where something theoretical pays: for finite theories and under charged derivations, and there by relations or entities, not by vocabulary.

[6] The idea of introducing a concept that factors a complete bipartite pattern of inferences first appeared in Hänni's working notes (Hänni, 2026).

9 Approximate laws and the best system

Kepler's laws are false: against a modern planetary theory a fixed ellipse misses Mars by one or two arcminutes over a few decades and the Moon by about a degree within a month (Table 8), yet the laws were, and are, good science. An exact law that is slightly wrong is refuted by any datum precise enough to show its error. We ask what SAI does with an approximately true law, and what becomes of Lewis's best system and of explanatory power.[7] M_A also mixes randomised theories, and under SAI's pricing an approximately true law is a law with a calibrated error model (Proposition 9.2); that pricing is not an account of approximate correctness (Section 9.3).

9.1 Setting

Definition 9.1 (Answered data, exception codes, approximate laws)**.** (a) *Answered data.* A datum $(\mathcal{Q}, a)$ is a question (Definition 2.10) and its true answer, read as the two-valued datum $(a, 1)$; a data set D has pairwise distinct questions and $n := |D|$ data. T *proves* (*refutes*) the datum if $B \cup T \vdash a$ ($B \cup T \vdash \neg a$), and is *silent* otherwise; $\gamma(\mathcal{Q}) := \log|\mathcal{Q}|$ is the *literal cost*. (b) *Codes.* $\|\Gamma\| := \sum_{\chi\in\Gamma} \ell(\chi)$, with $\ell(\chi) := 1 + |c_S(\chi)|$ for a prefix-free code c_S, for a finite theory or *module* Γ. If $B \cup T \cup D^{\pm}$ is consistent, the *exceptions* of T, the data it does not prove, cost $\mathrm{Exc}^{Q}_D(T) := \sum \gamma(\mathcal{Q})$ (code Q), $\mathrm{Exc}^{F}_D(T) := \sum \ell(a)$ (code F) or $\mathrm{Exc}^{R}_D(T) := -\log P^R_T(D)$, $P^R_T(D)$ the probability of the data when the law's interval is completed uniformly at random (code R, Definition G.1); otherwise $\mathrm{Exc}_D(T) := \infty$. Unmarked, Exc is code Q. The minimisers of $J_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T)$, $\lambda > 0$, are the λ-*MAPs* (Definition 3.6 of the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d)); T^{-Z} withdraws T at the times in a finite set Z, at about c_{time} bits each. (c) *Interval laws.* Over $B :=$ RCF with constants q_t for times t in a decidable set I of numerals, a *pointwise interval law* is $\{\underline{f}(t) \le q_t \le \overline{f}(t) : t \in I\}$ for computable $\underline{f} \le \overline{f}$; the *exact law* T_0 has $\underline{f} = \overline{f} = f$, the *honest law* T^ε has $f \mp \varepsilon$. *Interval data* answer questions that partition a range $[0, \mathsf{R})$ into cells. (d) *Errors.* $e(t) := f(t) - q(t)$ for a real quantity q, and $\varepsilon^* := \operatorname{ess\,sup}|e|$ at a data time. *Regime G*: intervals of half-width δ around measurements with errors at most $\delta_0 \le \delta$. *Regime R*: the errors $e_j := e(t_j)$ at fresh times t_j are i.i.d., and datum j is the cell of width Δ containing $q(t_j)$ in a uniformly dithered grid on a circle of circumference R; $\gamma = \log(\mathsf{R}/\Delta)$, $p_\Delta := \mathbb{E}\min(|e|/\Delta, 1)$, $s_\Delta := \min(2\varepsilon^*/\Delta, 1)$; e is Euler's number. (e) *Peakedness.* $\mathrm{pk}(e) := \varepsilon^*/\mathbb{E}|e|$ and $\mathrm{pk}_\varepsilon(e) := \varepsilon/\mathbb{E}|e|$; reported values name their fit (minimax or mean).

9.2 Deterministic theories and precision

Over deterministic theories proof-requiring compatibility is monotone in logical strength: $T_0 \vdash T^\varepsilon$, data compatible with T^ε are compatible with T_0, and with prior weights $2^{-\|T\|}$ the posterior odds of T_0 against T^ε stay at least 2, although T_0 is false when $e \not\equiv 0$ (Proposition G.10). In regime G, T_0 is almost surely compatible with the data at every sample size when $\delta \ge \varepsilon^* + \delta_0$ and, under mild conditions on the measurement error, eventually refuted when $\delta < \varepsilon^* + \delta_0$ (Proposition G.12). In regime R the exact law survives n data with probability $(1 - p_\Delta)^n$ and is almost surely refuted eventually iff $\mathbb{P}[e \ne 0] > 0$, however coarse the cells; with a sinusoidal error of amplitude $1'$ it is refuted by positions rounded to a dithered grid of 1° after about 94 data (Proposition G.13).

[7] The questions of approximately true laws and of the domain of applicability of a model are discussed in Hänni's working notes (Hänni, 2026).

9.3 Randomised theories and laws with error models

A theory machine reading a random tape answers each question at random within the law, so $L_A(D) \le \|T\| + \mathrm{Exc}_D^R(T) + c$ for a pointwise interval law T on interval data consistent with it (Proposition G.4). SAI is the best two-part code over theory machines, $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$ (Proposition G.5); *SAI's pricing* of a form is the weight $2^{-(K(G)-\log \nu_G(D))}$ of a theory machine G in M_A's mixture. Between the honest law T^{ε^*} and the exact law with exceptions T_0^{-Z}, for $\Delta \ge 2\varepsilon^*$, spread errors favour the first and peaked ones the second, with a threshold on $\mathrm{pk}(e)$ set by the prices of an exception and of silence (Propositions G.17 and G.18); band-confined completions move this comparison inside the band. For $\varepsilon \ge \varepsilon^*$ and a computable distribution $\mathcal{G}$ on $[0,1]$, the *band-confined completion* $\mathrm{BC}_\mathcal{G}$ states T^ε and draws each value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$, $\hat{w}_i \sim \mathcal{G}$, from the tape; at fresh times its law is $\prod_j \mathcal{G}(C'_j)$, C'_j the cell of datum j with the band mapped affinely onto $[0,1]$. Uniform $\mathcal{G}$ gives the *uniform band*, and $\mathcal{G} = (1-\rho)\delta_{1/2} + \rho\,\mathrm{Unif}[0,1]$ gives $\mathrm{HY}(\rho)$: exact except in a share ρ of cases and then within ε.

Proposition 9.2 (Laws with error models)**.** *In regime R let $\varepsilon \ge \varepsilon^*$, and let $\mathcal{F}$ be the distribution function of $w_j := (\varepsilon - e_j)/(2\varepsilon)$, the position of the true value in the band.*

(a) If $\Delta \ge 2\varepsilon$, the rate of $\mathrm{BC}_\mathcal{G}$ is $(2\varepsilon/\Delta)\, c_\mathcal{G}$, with

$$c_\mathcal{G} := \mathbb{E}_w \int_0^1 \left(\mathbf{1}[v < w](-\log(1 - \mathcal{G}(v))) + \mathbf{1}[v > w](-\log \mathcal{G}(v))\right) dv.$$

So band-confined completions are ranked in the same way at every such Δ.

(b) $c_\mathcal{G} \ge c_\mathcal{F} := \int_0^1 H_2(\mathcal{F}(v))\, dv$, H_2 the binary entropy, with equality iff $\mathcal{G} = \mathcal{F}$ almost everywhere on $[0,1]$. In particular the uniform band is the best band-confined completion iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) For every theory machine G', G_U included, almost surely $\liminf_n \frac{1}{n}(-\log \nu_{G'}(D_n)) \ge r_\mathcal{F}(\Delta) := \mathbb{E}[-\log \mathcal{F}(C'_1)]$, the conditional entropy rate of the answers given the questions, while $-\frac{1}{n}\log \nu_{\mathrm{BC}_\mathcal{F}}(D_n) \to r_\mathcal{F}(\Delta)$. If $\mathcal{F}$ is computable, then $L_A(D_n)/n \to r_\mathcal{F}(\Delta)$ almost surely. For $\Delta \ge 2\varepsilon$, $r_\mathcal{F}(\Delta) = (2\varepsilon/\Delta)\, c_\mathcal{F}$.

(d) $c_{\mathrm{HY}(\rho)}$ is convex in $\rho \in (0,1]$, and some $\mathrm{HY}(\rho)$ with $\rho < 1$ beats the uniform band iff $\mathbb{E}\ln(1 + |e|/\varepsilon) < 2\ln 2 - 1$. This requires $\mathrm{pk}_\varepsilon(e) > \ln 2/(2\ln 2 - 1) \approx 1.79$, and holds whenever $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4 - \mathrm{e}) \approx 2.12$.

Proof idea. A boundary falls in the band with probability $2\varepsilon/\Delta$, and the datum costs $-\log$ of the $\mathcal{G}$-mass on the true side; (b) is Gibbs' inequality, (c) the Borel–Cantelli lemma (Appendix G.4).

So under SAI's pricing the preferred form is a law with a calibrated error model, neither a silent partial theory nor an exact law with listed exceptions. That a model includes its noise distribution is standard in MDL (cf. Rissanen, 1978; Grünwald, 2007); what is specific is that for a proof-requiring inducer the honest partial theory is one completion among many. Selection among finitely many forms is almost sure, and for the randomised theories that M_A mixes humility is calibration (Theorem G.22, Corollary G.23, and Proposition G.24).

What the pricing does not see. Proposition 9.2 describes the credence- and compression-best forecast, not approximate correctness. For questions about one quantity the loss is the log score of the cell that the answers determine: a question family fixes a resolution, while the cost of a miss is set by each hypothesis's own error model, not by a purpose (Proposition G.42). SAI's pricing ranks rival laws by a dispersion of the error for coarse cells and, for fine ones, by the entropy of

the error rather than its size (Theorem G.45); it can rank a law above one that is closer to the truth at every time; and other proper scores rank rival laws differently, under misspecification and even without it (Examples G.47 and G.48). A tolerance on a domain is a sentence, in which SAI holds a credence; a graded loss is not (Proposition G.49).

Kepler's residuals. For seven bodies and six windows each (three for the Moon) we fitted one Kepler ellipse four ways and kept the fit with the least largest longitude error (the minimax fit) and the one with the least mean error (the mean fit), with errors against the plan94 and moon98 ephemerides (Table 7). A computation at eleven cell widths (code in the accompanying repository) finds that among the Kepler-based theory machines compared a law with a 40-bin error histogram, fitted and scored on disjoint halves of the times, has the least rate in all 429 cells, and that in the coarse regime the two-point form beats the uniform band for 34 of 39 body–window pairs, exactly where Proposition 9.2(d) says. Over the four fits, maximum, RMS and mean absolute error disagree on the best fit, the calibrated pricing agrees with none, and its ranking changes with cell width for 11–16 of 234 pairs, by up to 6.8% of the rate (Appendix G.9).

9.4 Sound reasoning with approximate models

True honest models of one quantity are jointly consistent over a true background and prove only truths about it; when the background decides the comparisons (as RCF does for semialgebraic models), exact approximations that differ anywhere on their common domain are jointly inconsistent and so prove everything (Proposition G.28). Honest dynamical laws have derivation horizons, up to which exact approximations are guaranteed correct (Propositions G.25 and G.27); after 100 steps with Lipschitz constant 1 the propagated tolerance is 100ε, against a mean deviation of about 7.96ε for independent random-sign errors (a computation, Remark G.26). The device does not say how to choose a model, a domain or a tolerance (Section 11).

9.5 The best system as a steeper MAP

Lewis's laws are the theorems of the system that best balances simplicity against strength (cf. Lewis, 1983, 1994). By analogy (best systems are a metaphysics of laws, SAI is inference from finite data) we take strength to be fit to data in a fixed base language, the balance to be λ, and laws with private literal facts as candidates. The points $(\|T\|, \mathrm{Exc}_D(T))$ have a convex hull with vertices v_i in order of length and edge slopes $-\sigma_i$, $\infty =: \sigma_0 > \sigma_1 > \cdots > 0$; for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are the theories at v_i (Lemma 3.8 of the companion paper on steeper priors). A point $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ of some theory is a *zero-window* (σ, τ)*-kink* (Definition 5.2 there) if theories shorter by $x > 0$ have fit at least $L_{\mathrm{r}} + \sigma x$ and theories longer by $x > 0$ fit at least $L_{\mathrm{r}} - \tau x$.

Proposition 9.3 (Robustness to the balance is a kink)**.** *For a family of candidate theories satisfying the hypotheses of the hull lemma (Proposition G.8) and $0 \le \tau < \sigma \le \infty$, the following are equivalent: (a) the set of λ-MAPs is the same for every $\lambda \in (\tau, \sigma)$; (b) some hull vertex v_i has $\sigma_{i+1} \le \tau$ and $\sigma \le \sigma_i$; (c) the point family has a zero-window (σ, τ)-kink at a point that is a hull vertex.*

Proof idea. The λ-MAP set is fixed between consecutive edge slopes and changes at each; (b)⇔(c) is Corollary 5.4(c) of the companion paper on steeper priors (Appendix G.6).

So the best system is the same for every balance in an interval exactly when the data's hull has a kink there. This makes precise robustness to the balance of simplicity against strength,

with the measures of simplicity and of fit held fixed, one aspect of what Lewis calls a kind nature (cf. Lewis, 1994); robustness to the measure of simplicity itself fails among the compatible Mars forms in the noise-free regime (Section 10.4). On a solar-system toy a computation finds one best system for all $\lambda \in (7.03, 135.37)$ (Appendix G.6); Lewis's worry about a predicate true of exactly the actual things is blocked up to the complexity of a language chosen after the data (Proposition G.31).

The *explanatory power* $\mathrm{EP}_D(\mathcal{A}; T)$ of a module $\mathcal{A} \subseteq T$ is the bits of data it explains per bit of its statement, in the context T (Definition G.29). Every module of a λ-MAP has EP $\geq \lambda$, no addition to it has EP $> \lambda$, and for $\lambda > 1$ a fact used once never enters (Propositions G.30 and G.32). So the model–randomness split of the companion paper on steeper priors separates facts reused across many data from facts used once, not laws from initial conditions; what singles out a law is that its EP grows with the number of systems it serves. Kepler's third law, which shortens the stated elements, is a law about initial conditions (Proposition G.33; for its worth to Kepler, Section 10), and in the question format elements never enter a λ-MAP (Proposition G.34).

9.6 Explanation and correction

$\|\mathcal{A}\| \, \mathrm{EP}_D(\mathcal{A}; T)$ is the log-likelihood ratio of T against $T \setminus \mathcal{A}$ for $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$. The support a hypothesis gets from joint data $D_1 \cup D_2$ splits into the supports from the parts plus a term U, the change in the mutual information of D_1 and D_2 that the hypothesis brings about, of the form of Myrvold's measure of unification (Myrvold, 2003): $U = 0$ for laws that are products over questions with stated parameters; with a k-bit parameter integrated out, U is k within one system when each part of the data identifies the parameter, 0 across systems with independent parameters, and the bits of a shared constant across systems that share one and each identify it (Proposition G.35). Copernicus against Ptolemy is unification without better prediction (Proposition G.36).

Newton's theory does not entail Kepler's laws; with the planets' masses and initial conditions it contradicts their exact form (cf. Duhem, 1954) and derives corrected statements. In an exact rational toy of perturbed circular motion the successor proves the honest old law with uniform tolerance ε on a window iff ε is at least the largest distance between the two theories' trajectories on it, and refutes the exact old law whenever the perturbation changes the first step (Proposition G.37): the old law's domain of applicability is derived, not stipulated.

9.7 Domains of applicability and retention

Domains are learned as tolerances. A domain restriction leaves a system's data literal (Proposition G.39). Under SAI's pricing an honest band of half-width ε^* costs at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$ bits per datum when $\Delta \leq 2\varepsilon^*$, below the literal $\log(\mathsf{R}/\Delta)$ iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, and at most $(2\varepsilon^*/\Delta) \log \mathrm{e} < \log(\mathsf{R}/\Delta)$ when $2\varepsilon^* \leq \Delta < \mathsf{R}/\mathrm{e}$; so the Moon ($\varepsilon^* = 2.6^\circ$ over a year, minimax fit) is kept in Kepler's law with its own tolerance at every cell width computed, while literal pricing at $10'$ and $60'$ excludes it by a cheap domain clause. We conclude that under SAI's pricing a domain of applicability shows up as a tolerance and an error distribution that depend on the system and the window, not as a sharp boundary; a superseded theory is retained where its derivation savings exceed its loss of fit (Proposition G.40).

10 A worked example: Kepler and Newton

We work through Kepler's laws, the theories of Mars that preceded the ellipse, and what Newton adds. What a law says depends on the data language (Section 10.2); in a proof-requiring posterior

over the historical theories of Mars the data decide compatibility and, on noise-free data, the language of the law texts decides among compatible theories (Section 10.4); a successor's gain need not lie in a better fit to the old data (Section 10.5). The data are computed with the ERFA library, so *every positional error in this section is against the plan94 ephemeris (moon98 for the Moon), not against the sky.* Appendix H has proofs and tables; the code is in the accompanying repository.

10.1 The language and the data

Definition 10.1 (The Kepler languages and theories)**.**

(i) *Data languages.* Sorts bodies and reals; body constants Sun, Me, Ve, Ea, Ma, Ju, Sa, Mo; a predicate Planet. The *heliocentric language* $\mathcal{L}_{\mathrm{hel}}$ has the heliocentric ecliptic longitude, latitude and distance $\theta(b,t), \beta(b,t), r(b,t)$ of body b at time t (for the Moon, geocentric); its direction-only part $\mathcal{L}_{\mathrm{dir}}$ omits r. The *geocentric language* $\mathcal{L}_{\mathrm{geo}}$ has $\theta_g(b,t), \beta_g(b,t)$, giving the direction $\mathrm{dir}_g(b,t) \in S^2$: a modern idealisation of Tycho's data.

(ii) *Background.* B: the axioms of real closed fields with sin, cos, π and enough true analysis to prove the numerical bounds used, a sound decidable fragment of the undecidable theory of the reals with sin (Appendix H.1); each result names the true sentences it needs.

(iii) *Kepler's first two laws* K12, in the theoretical vocabulary of heliocentric positions $\mathbf{r}(b,t) \in \mathbb{R}^3$ and the elements $a, e, i, \Omega, \varpi, M_0, P$ (functions from bodies to reals: semi-major axis, eccentricity, inclination, node, longitude of perihelion, mean anomaly at the epoch, period): for every planet b and every t, with $M := M_0(b) + 2\pi t/P(b)$, there is ψ with $\psi - e(b)\sin\psi = M$ and $\mathbf{r}(b,t) = \mathrm{Rot}(b)(a(b)(\cos\psi - e(b)),\ a(b)\sqrt{1-e(b)^2}\sin\psi,\ 0)$, where $\mathrm{Rot}(b)$ is the rotation given by $i(b), \Omega(b), \varpi(b)$; moreover $0 \le e(b) < 1$, $a(b), P(b) > 0$ and $\mathbf{r}(\mathrm{Sun}, t) = 0$. Over $\mathcal{L}_{\mathrm{geo}}$, K12 also contains the frame convention $i(\mathrm{Ea}) = 0$ and the *bridge*: $\mathrm{dir}_g(b,t)$ is the unit vector of $\mathbf{r}(b,t) - \mathbf{r}(\mathrm{Ea},t)$ if this is nonzero, and the north ecliptic pole $\mathbf{z}$ otherwise.

(iv) *Kepler's third law* K3: $P(b)^2 a(b')^3 = P(b')^2 a(b)^3$ for all planets b, b'.

(v) *Elements* (initial conditions): numerals, or intervals, for the elements of a body. *Data*: true interval sentences $\mathrm{Near}(b,t,\mathbf{v},\delta)$, "the direction of b at t lies within angle δ of $\mathbf{v}$" (Near_g for geocentric directions), with numerals for $t, \mathbf{v}, \delta$, and, where available, $r(b,t) \in [r_1, r_2]$. *Questions* (Definition 2.10): covers by such sentences ("where is Mars at t, to within 2′?"), and structural questions, which the laws answer without elements (is the orbit planar?).

Laws alone answer no position question;[8] laws with elements do. With the elements in the antecedents of conditional questions the laws alone answer them, and in the model of Proposition G.34 no steeper MAP then contains elements: this is where laws and initial conditions part formally.

10.2 What the laws say, and in which language

Proposition 10.2 (No content in heliocentric directions)**.** *For every B that contains the axioms of real closed fields,* K12 *and* K12 ∪ K3 *have the same $\mathcal{L}_{\mathrm{dir}}$-consequences.*

[8] Even the Ramsey sentence of K12 ∪ K3, with any background true in $\mathbb{R}$, entails (in second-order semantic consequence) no datum $\mathrm{Near}(b,t,\mathbf{v},\delta)$ with $\delta < \pi$ (Proposition H.1).

In a geocentric language K3 has content: K12 ∪ K3, with two true sentences of the real field, proves that Mars is not permanently in opposition, and K12 with any background true in $\mathbb{R}$ does not (Proposition H.2). The natural form of that content is this:

Proposition 10.3 (What K3 does in geocentric directions)**.** *Let* Num^- *be numerals for all elements of the Earth and Mars except* $a(\mathrm{Ma})$*, with* $a(\mathrm{Ea}) = 1$*. Let* $\mathbf{F}(t)$ *be Mars's heliocentric position computed from* Num^- *with semi-major axis* 1*, so that* $\mathbf{r}(\mathrm{Ma}, t) = \rho\,\mathbf{F}(t)$ *with* $\rho := a(\mathrm{Ma})$*, and let* $\mathbf{u}(t)$ *be the unit vector of* $\mathbf{r}(\mathrm{Ea}, t)$*.*

(a) $\mathrm{K12} \cup \mathrm{K3} \cup \mathrm{Num}^-$ *fixes* $\rho = (P(\mathrm{Ma})/P(\mathrm{Ea}))^{2/3}$ *and hence every geocentric direction of Mars: it proves* $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ *whenever the computed direction lies within* δ *of* $\mathbf{v}$ *and* B *proves the numerical bound involved.*

(b) $\mathrm{K12} \cup \mathrm{Num}^-$ *leaves* ρ *free. At every* t *at which* $\mathbf{r}(\mathrm{Ea}, t)$ *and* $\mathbf{F}(t)$ *are not parallel, the geocentric directions of Mars at* t *in the models of* $\mathrm{K12} \cup \mathrm{Num}^-$ *over* $\mathbb{R}$ *(with* ρ *ranging over* $(0, \infty)$*) form the open great-circle arc from* $-\mathbf{u}(t)$*, the direction of the Sun, to* $\mathbf{F}(t)/|\mathbf{F}(t)|$*; its length* $\alpha(t) \in (0, \pi)$ *is the angle between these two directions. So for every* B *true in* $\mathbb{R}$*,* $B \cup \mathrm{K12} \cup \mathrm{Num}^-$ *proves no datum* $\mathrm{Near}_g(\mathrm{Ma}, t, \mathbf{v}, \delta)$ *with* $\delta < \alpha(t)/2$*.*

A computation on the daily geocentric longitudes of Mars over 1580–1600 puts what K3 saves on the elements of Mars at 11.3, 12.3, 13.4 and 14.7 bits at $\delta = 8'$, $4'$, $2'$, $1'$, against a text of 375 bits in our arithmetic syntax and 165 or 115 bits in two Polish-notation codes; so in none of these codes does compression alone select K3 at Kepler's precision, an extrapolation from Mars to the other planets that is conditional on the code (Example H.4).

K3 is thus a law about initial conditions, worth what it saves on the elements of the bodies it relates. This depends on Kepler's kinematic parametrisation, with the period and the semi-major axis both free; in a state-space parametrisation K3 fixes the mean motion from the energy, a different question, and in Newton's theory the third law, with its mass correction, follows from the one law that does all the other work and costs nothing extra.

10.3 How wrong Kepler's laws are

Kepler's laws are approximately true, with an error that grows with the domain: one fixed least-squares ellipse misses Mars by $1.6'$–$1.9'$ over 12–30 years and $51.8'$ over 2000 years, and the Moon by 1.1° within a month (Table 8). The third-law "constant" $q := (2\pi/P)^2 a^3/k^2$ (k Gauss's constant) differs between planets at the 10^{-3} level, yet an exact K3 calibrated on the Earth misplaces Jupiter and Saturn geocentrically by at most $0.23'$ and $0.19'$ over 1580–1600, well below Tycho's precision. By the accuracy quoted in the documentation of the ERFA routines `eraPlan94` and `eraMoon98` (which pyerfa wraps), plan94's own error in the heliocentric longitude of Mars over 1580–1600 is at most about $0.43'$ (Appendix H.2).

10.4 The proof-requiring posterior over the historical Mars pool

The pool and the data. We take a hand-picked pool of five forms of theory for Mars, as heliocentric reconstructions: the circle about the Sun (2 elements), the eccentric circle (4), Ptolemy's bisected equant (4), Kepler's vicarious hypothesis (5) and the ellipse K12 (4) (Appendix H.3). The data are the 7306 daily plan94 heliocentric longitudes v_i of Mars over 1580.0–1600.0, as sentences "$\theta(t_i)$ lies within δ of v_i": the noise-free case of the guaranteed-tolerance regime of Proposition G.12. A form proves every datum, for suitable elements, iff δ is at least its *minimax error*, the least over elements of its largest error: $640.79'$, $23.05'$, $8.17'$, $1.41'$ and $1.07'$ in this order

Table 5: The proof-requiring posterior over the Mars pool: law-text lengths and Occam factors (bits; "–": δ below the minimax error; all values in App. H.3). A form's weight is $\propto 2^{-(\text{text+Occam factor})}$. K3's text: 375 bits in S1 and S3, 450 in S2.

	circle	eccentric	equant	vicarious	ellipse
text, S1	155	445	990	990	545
text, S2	186	342	450	468	654
text, S3	155	445	990	990	260
$\text{Occ}_{\mathsf{F}}(720')$	20.93	20.24	21.20	22.77	21.18
$\text{Occ}_{\mathsf{F}}(30')$	–	45.38	40.35	42.83	39.65
$\text{Occ}_{\mathsf{F}}(9')$	–	–	57.34	51.78	46.81
$\text{Occ}_{\mathsf{F}}(1.6')$	–	–	–	74.54	59.93

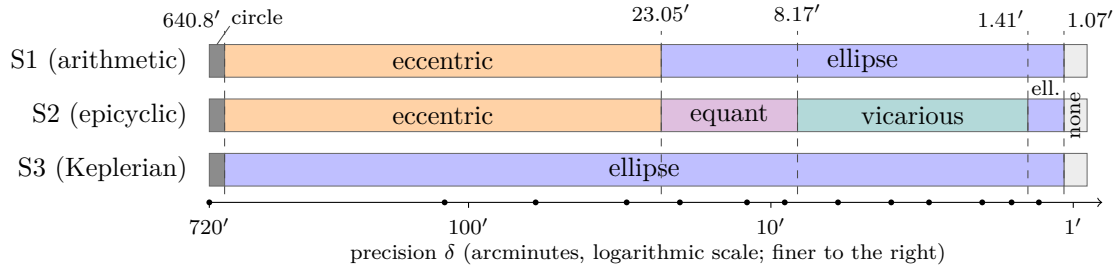


Figure 1: The MAP form of the proof-requiring posterior over the Mars pool against the precision δ of the data, in the three syntaxes, in the noise-free common-tolerance regime. Dashed lines: the minimax errors of Table 9, below which a form is incompatible with the data; below $1.07'$ no form in the pool is compatible. Dots: the precisions at which the posterior was computed; at each of them the runner-up has at most $2^{-12.4}$ of the MAP's weight. Between dots each switch is drawn at the minimax error of the form that drops out; that form stops being the MAP somewhere between this line and the nearest dot to its left.

(upper bounds witnessed by elements, only the circle's proved optimal; Table 9). The equant's $8'$ remains against an exact ellipse, close to the famous eight-minute discrepancy of Kepler's *Astronomia Nova* (Kepler, 1992); the $0.34'$ between the vicarious hypothesis and the ellipse is of the size of plan94's error, so against the sky their ordering rests on the errors against an exact ellipse.

The posterior. A theory is a law text in one of three syntaxes plus fixed-length 64-bit numerals for its elements; the proof-requiring rule of M_A (Definition 2.4), with the universal machine replaced by this code, gives it weight $2^{-\text{length}}$ if it proves every datum and 0 otherwise. Summing over numerals, form F has weight proportional to $2^{-(|\text{LAW}_{\mathsf{F}}|_S + \text{Occ}_{\mathsf{F}}(\delta))}$, with $|\text{LAW}_{\mathsf{F}}|_S$ the length of its law text in syntax S and $\text{Occ}_{\mathsf{F}}(\delta) := -\log(\text{Vol}_{\mathsf{F}}(\delta)/\prod \text{ranges})$ its *Occam factor*, $\text{Vol}_{\mathsf{F}}(\delta)$ being the volume of the elements with which F proves all the data (the term is MacKay's (MacKay, 1992); Jefferys and Berger, 1992 read Bayes factors this way for Mercury's perihelion). The syntaxes are *S1*, arithmetic $(+, -, \cdot, /, \sin, \cos, \text{atan2}, \sqrt{\ })$, the numerals $0, 1, 2$, quantifiers); *S2*, epicyclic (S1 with primitives for unit vectors, directions, and the point where a ray meets a circle); and *S3*, Keplerian (S1 with the true anomaly as a function of the eccentricity and the mean anomaly).

What the data decide and what the syntax decides (Table 5 and Figure 1). (i) *The data decide compatibility:* as δ falls the forms drop out in the historical order, in every syntax. (ii) *Among compatible forms the data contribute only the Occam factors, which on their own point weakly to the true form:* with a uniform prior over the five forms and no text prior, the MAP is the ellipse at every $\delta \leq 60'$, by 0.20 to 14.61 bits. (iii) *The text prior decides the rest:* law-text differences are 100–835 bits for every pair of forms except the equant and the vicarious hypothesis, whose texts are 18 bits apart in the epicyclic syntax, where the Occam factors move the margin by up to 5.6 bits, so the text still decides, and of equal length in S1 and S3, where both lose to the ellipse by more than 400 bits. So in this regime the syntax sets the sequence of MAP forms: with epicyclic primitives every form in the historical order, switching only when a form becomes incompatible; with plain arithmetic the eccentric, then the ellipse from $23'$; with a Keplerian primitive, the circle, then the ellipse from $641'$. A variable-length numeral code, Polish notation or a natural form outside the pool changes no MAP form (Appendix H.3).

"The simplest theory compatible with the data" is thus, in this regime, relative to the language of the law texts. A semanticist will call this relativity an artefact of a prior on texts. Our computation is mostly semantic: compatibility is a condition on a region of element space and the Occam factor is its measure, so only the law-text prior is syntactic; with a presentation-free prior, uniform over the five families and by volume on their elements, the data alone select the ellipse at every $\delta \leq 60'$, by small margins ((ii) above). But a prior over infinitely many families of models needs an enumeration of them, which is a presentation, and a volume prior depends on the parametrisation (halving the eccentricity ranges moves the vicarious hypothesis by one bit against the four-element forms), so the relativity is relocated rather than removed. The universal prior bounds this relativity by the cost $K(\tau)$ of a faithful translation (Theorem 3.5), but that cost, here the definitions of the extra primitives in S1, is a few hundred bits (an estimate), as large as the text differences. Best-system accounts of laws face the same relativity (Lewis requires natural properties (Lewis, 1983); Cohen and Callender accept relativity to a chosen set of kinds (Cohen and Callender, 2009)), by analogy only: best systems are a metaphysics of laws, SAI is inference from finite data. And the noise-free regime is special: a form proves all the data or none, so the power of the data to discriminate among compatible forms does not grow with their number.

Graded fit and distances. With graded fit (a Gaussian error model with $\sigma = 2'$, roughly Tycho's precision) the data decide every step within tens of observations, except the epicyclic syntax's choice between the vicarious hypothesis and the ellipse on longitudes, which needs about 5,300 (Example H.3). With distances within 1% as data, compatibility alone eliminates the vicarious hypothesis below a longitude tolerance that is at most $5.12'$, in every syntax, which reproduces the abandonment of circles, not Kepler's step from an oval to the ellipse (Table 10).

10.5 What Newton adds

Newton's theory adds masses, the constant G and accelerations, with the law $m_b \ddot{\mathbf{r}}_b = \sum_{b' \neq b} G m_b m_{b'} (\mathbf{r}_{b'} - \mathbf{r}_b)/|\mathbf{r}_{b'} - \mathbf{r}_b|^3$, in the same data language; masses are theoretical in Sneed's sense (Sneed, 1971; Balzer et al., 1987). We see five points, gains of Newtonian mechanics over about 150 years, not of Newton alone (Newton accounted for about half of the motion of the lunar apsides, Clairaut for the rest in 1749 (Nauenberg, 2015; Bodenmann, 2010)). (1) *No gain on the old data:* on 20 years of longitudes at Kepler's precision the masses buy nothing measurable, so we think an SAI posterior on such data alone would not prefer Newton's theory. (2) *Unification,* where the gain lies: planets, Moon, falling bodies, tides and comets under one law

(some with more base vocabulary), and the value $q = 1 + m/M$ of the third-law constant, which Jupiter's fitted q matches. (3) *Corrections in the old domain:* in the classical decomposition (cf. Clemence, 1947), Newtonian planetary perturbations give about $531''$ of the fixed-frame advance of Mercury's perihelion, about $574''$ per century ($575.4''$ in our fits), and general relativity the remaining $43''$ or so; Laplace explained the great inequality of Jupiter and Saturn (Wilson, 1985). (4) *New questions:* in the old language, where Neptune is; in a new one, the masses of the Sun and of the planets with satellites. (5) *No deductive entailment:* Newton's axioms do not entail K12 or K3, and with the planets' masses they contradict their exact form, as Duhem (1954) observed; with masses and initial conditions they entail honest tolerance forms, "K12 holds within $\varepsilon(b, W)$ on the window W" (Proposition G.37 is a provable toy version). Formally deriving even one such bound needs an existence-and-uniqueness theorem and a validated integration or perturbation estimate, a very long derivation that no textbook writes out: a concrete case of the computational gap between a written theory and the reasoner who uses it (Section 11).

11 Limitations

The axiom view, and SAI as its formal model, leave out much of what scientific theories are and do. For each limitation we say whether it is partially addressable, and by what, or a genuine limit of a declarative view. The judgements are arguments, except where a theorem is cited.

Statistical confrontation with data. In physics a theory meets data statistically: with instrument and noise models it assigns likelihoods to data, often near-raw (Section 6.3), and statistical theories answer with probabilities. The proof-requiring core of SAI has no native noise model: a datum is proved or not. Graded fit enters only through randomised theories in M_A (Propositions 9.2 and G.4), through the generative variant, whose updating we prove given a support condition and whose identification we only sketch (Proposition A.37 and Remark A.38), or through sentences about probabilities, whose relation to frequencies is not deductive. It matters: on noise-free interval data the syntax of the law texts decides among the forms compatible with the Mars data, while with graded fit the data decide (Table 5 and Example H.3). So the proof-requiring core is the noise-free special case; its natural extension inside M_A is a law with a calibrated error model; a native treatment is open.

Written theories and their readers. A written theory joined to a simple reasoner does not solve the problems the theory is used for; we see three gaps. *The bridge gap:* a problem posed in ordinary terms becomes formal only through idealisations and bridge principles that the theory does not contain (cf. Cartwright, 1983; Morgan and Morrison, 1999; an extractor prices part of the bridge, Proposition D.5). *The computational gap:* theory and mathematics suffice logically, but finding the answer needs methods and lemmas (Section 10.5). In the model of Section 7 this gap is a theorem, visible only when what is written, or the time to write it, is charged: program length sees neither reasoner nor vocabulary (Theorem 7.2), written size depends exponentially on both (Corollary 7.6), and charged deduction cannot express the gap (Theorem E.31; derivation sizes sketched in Remark E.32). So insufficiency for a simple reasoner is a property of the pair (theory, reasoner) and of the vocabulary, in the model of Section 7, when what is written, or the time to write it, is charged; this is not a theorem about textbooks, nor about what people carry in their heads (Section 7.7). *The tacit gap:* skills (Polanyi, 1958, 1966) and exemplars (Kuhn, 1970, 1977) are not sentences, and in SAI exemplars are only data: a genuine limit.

Approximate truth. "Kepler's laws are only approximately true" is a limitation of SAI, not only of proof-requiring compatibility with deterministic theories. Its mixture prices an approximately true law as a law with a calibrated error model (Proposition 9.2); but whether such a law is good enough depends on a purpose, which this pricing sees only through the questions asked (Section 9.3). Adequacy for a purpose needs a loss on answers, either as a decision layer on the posterior, which ranks laws by purpose when the posterior predicts well, or inside learning when the pool of hypotheses is restricted (Remark G.52 and Question G.53). So the limitation is shared by any account that ranks approximate laws by posterior weight or code length, but not by Bayesian inference as such. A law at least as close to the truth on every question of a family is at least as good for every purpose that rewards closeness (Proposition G.49); beyond that a canonical score for approximate predictions, which we think does not exist, and a general comparative notion of truthlikeness (cf. Popper, 1963; Miller, 1974; Tichý, 1974; Niiniluoto, 1987) are open, and the choice of quantities is not neutral (cf. Miller, 1975). Our notion is local: a question family, a tolerance, a domain.

Domains, several models, and inconsistent theories in use. Practice uses mutually inconsistent models of one system, each in its domain (the Earth as flat, as a sphere, as an oblate spheroid), and proof-requiring SAI gives an inconsistent theory weight 0. A sound device exists: honest forms "q lies within ε of f on the domain" of true models are jointly consistent over a true background; exact ones can prove everything (Section 9.4). But physicists rarely state or propagate worst-case bounds on the error of an approximate model (measurement uncertainties are propagated routinely, and statistically); worst-case propagation becomes vacuous over long chains; and Lanford's theorem, the rigorous honest form of the Boltzmann equation as described in expositions (Lanford, 1975), holds only for a fraction of the mean free time in the low-density limit, while practice rightly uses the equation far beyond. How to choose a model, a domain or a tolerance is left to tacit understanding; under SAI's pricing a domain is usually a tolerance with an error distribution varying with the system (Section 9.7), and knowing when a model applies is a recognizer, a genuine limit.

Concepts. Table 6 lists features of human concepts that an account of concepts as pieces of code, combined to generate predictions, misses, and whether the axiom view captures them.[9] We accept them as features of human concepts, the first only comparatively (Section 6.1), with two further qualifications: inferential role is one major account of meaning (cf. Sellars, 1953), disputed by referential accounts (cf. Kripke, 1980); and natural types we find plausible. One feature supports the axiom view: a notion with different recognizers has its identity fixed otherwise.

For analogy there is a bound: with $K(T)$ the least $K(e)$ such that $\mathsf{W}_e = T$, for every r.e. theory T_1 and computable map τ from sentences to sentences

$$K\big(T_1 \cup \tau[T_1]\big) \;\leq\; K(T_1) + K(\tau) + c, \tag{1}$$

since an index of $T_1 \cup \tau[T_1]$ is computable from indices of T_1 and τ, and $K(x,y) \leq K(x)+K(y)+c$. Structure mapping (Gentner, 1983) and models in science (cf. Hesse, 1963) account for analogy; patchy or stretching concepts (cf. Wilson, 2006; Lakatos, 1976) and interpretation as abduction (Hobbs et al., 1993) are outside the view.

[9] The list follows, with changes, an unpublished draft essay of Hänni's on concepts.

feature of concepts	axiom view	how captured, or what is missing
for questions and problems, not predictions	yes/no	yes for questions posed in a language (§ 6.3); no for problems needing search or action
inert alone; parts of a mind	a limit	worth depends on reasoner and vocabulary (Cor. 7.6); free deduction hides it
inferential role constitutive	yes	axioms are inferential relations (meaning postulates, Ramsey–Lewis)
compositional	yes/no	yes for contents, as formulas; no for recognizers, which are procedures
learning by new links	partly	a persisting symbol, but a global mixture over whole theories
soft; meaning change	partly	"within ε of a sphere" is a true partial theory; not meaning change
analogy	partly	partial interpretations, rewarded as they compress (1); not finding them
new concepts from old	partly	definitions cost $O(1)$ (Cor. 3.6), help a chaining reasoner (Prop. E.24)
methods attached	partly	lemmas for a bounded reasoner, or axioms via Craig at exponential cost (Lem. F.5); not motor methods
recognizers	no	a procedure on situations; the extractor of Prop. D.5, taken as given
one notion, many recognizers	yes	identity is the symbol with its axioms; recognizers are bridges
built from pointers	yes	a constant fixed by one axiom, which gathers further axioms
purposes	no	axioms can state functions; their use in search is not modelled
clarification; facing action	no	not in the inducer
natural types	partly	sorts and arities; their emergence is not modelled

Table 6: Features of human concepts, and whether the axiom view captures them (an assessment).

Other limitations. *Science predicts very particular things, in set-ups it builds.* SAI captures half, with selected data and partial theories; which questions are worth asking (value-weighted questions would be a handle), and the making of phenomena (cf. Hacking, 1983), are outside it. *Concept formation and discovery.* Step 3 of the ladder (Section 6.1) is partially addressable (Theorem 8.3 and Corollary 7.6); step 4, the revision of the base abstractions, is a genuine limit for now; and SAI evaluates theories in a fixed space of hypotheses without discovering them. *Theory-laden and fallible data.* A base language may presuppose a theory (heliocentric longitudes presuppose Copernican geometry; Definition 10.1; Section 6.1), which biases comparisons between theories that disagree about it; and a deterministic theory must prove even a mislabelled datum, by an exception axiom; a noise component in the likelihood, as in our earlier report (Hänni and Claude (Anthropic), 2026a), is a partial remedy. *Purposes and technology* are outside the view, and *science as a social structure* is outside a model with one inducer: genuine limits, the last perhaps where Question 6.1 finds its answer. *The mathematics is free, and large.* Building a background B' into B gains at most $K(B') + c$ bits on any data (Proposition A.9), a constant that does not grow with the data but is not small for the mathematics physics uses, and that matters when rival theories need different mathematics. More generally, all our comparisons of theories are relative to the universal machine, up to constants that are not small at the scale of real theories (Proposition A.8). With free deduction, data the background entails carry no information (Proposition C.23), so induction in mathematics needs a reasoner for whom deduction is costly.

What the axiom view captures. It captures the static, declarative and inferential side of theories and concepts (contents, inferential relations and their composition, learning by weighing axioms globally, pointers, partiality, approximate truth at a tolerance, priced with error models) and misses the procedural, the statistical and the dynamic side, and graded adequacy for a purpose. We offer a model of the declarative core of scientific theories, in which the procedural side reappears formally as the dependence of a written theory's worth on its reasoner and its

vocabulary (Section 7).

12 Related work

Universal and operator induction. Solomonoff induction predicts sequences with a universal prior over programs (Solomonoff, 1964, 1978; Li and Vitányi, 2008). Operator induction predicts the answer to a new question from an unordered set of question–answer pairs (Solomonoff, 1999, 2008); the companion paper (Hänni and Claude (Anthropic), 2026c) and the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d) develop its one-tape universal law and its versions, and by Section 4 SAI agrees, version by version and up to constants, with operator induction over consistent labellers of sentences, in which deduction ties the answers together. Whether a universal prior justifies a preference for simplicity is disputed (Sterkenburg, 2016, 2019). Our theorems do not depend on it; our use of SAI as an ideal of the evaluation of theories does, and we take the universal prior as a modelling choice whose dependence on the machine and the language is bounded by constants that, at the scale of our examples, are not small (Proposition A.8, Theorem 3.5, and Section 10.4). The computable, finite-class precursors are our earlier report (Hänni and Claude (Anthropic), 2026a), on a Bayesian axiom inducer with template priors and derivation likelihoods and on coin-free axiom induction under time penalties, which Section 4 extends, and a second report (Hänni and Claude (Anthropic), 2026b), on learning axioms and schemas from their instances.

Probabilities on sentences. Logical probabilities on sentences go back to Carnap (1950); coherent probabilities on first-order sentences were studied by Gaifman (1964), Scott and Krauss (1966) and Gaifman and Snir (1982); Hutter et al. (2013) extend given probabilities of some sentences to all sentences of a higher-order logic under desiderata that include the Gaifman condition; Christiano (2014) gives a tractable algorithm; and logical inductors (Garrabrant et al., 2016a) refine probabilities of a given theory's sentences over time. These are probabilities of truth, and logical inductors are given the axioms; SAI is given labelled sentences and is uncertain about the axioms. Bel is a belief function (Dempster, 1967; Shafer, 1976) and satisfies the Gaifman condition exactly when the instances prove the universal sentence (Proposition C.24). Sawin and Demski (2013) show that computable distributions that satisfy natural coherence conditions and come to believe the true Π_1 sentences must give some short true Π_2 sentences arbitrarily low limiting probability; Theorem 5.6 and Corollary 5.5 show a parallel divide for the ideal SAI posterior, not a consequence of theirs. Demski's prior (Demski, 2012), as usually described, draws random sentences and keeps each one consistent with those kept; the modification due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), draws theories as outputs of a universal machine on random bits, which is the sampling stage of M_A. Those processes iterate draws until the theory is complete; M_A uses a single draw, whose theory is incomplete with positive probability; for complete theories the two compatibility notions coincide (Propositions 3.11 and A.39).

Learning theories from data. Formal learning theory studies identification in the limit (Gold, 1967; Putnam, 1965b), also of first-order theories (Osherson et al., 1991; Kelly, 1996; Martin and Osherson, 1998); Proposition A.28 is a Bayesian form of Gold's distinction between text and informant. Shapiro's model inference infers theories from facts by conjectures and refutations (Shapiro, 1981, 1991); it outputs one conjecture at a time, without a prior. Inductive logic programming (cf. Muggleton and De Raedt, 1994; Cropper and Dumančić, 2022) asks a hypothesis

to entail the positive examples and to be consistent with the negative ones, a mixture of the two compatibility notions of Definition 2.9, invents predicates (Muggleton and Buntine, 1988), and justifies theories by compression (Muggleton, 1995; Conklin and Witten, 1994). In cognitive science, intuitive theories are modelled as logical laws from a probabilistic grammar, learned by Bayesian inference over a finite space (Kemp et al., 2008, 2010; Ullman et al., 2012).

Random axioms and Gödel phenomena. What a probabilistic machine produces with positive probability can be produced deterministically (cf. de Leeuw et al., 1956); the lottery (Theorem 5.1) is where this averaging fails for random theories, since majority votes over consistency constraints of unbounded arity can be inconsistent (cf. the lottery paradox, Kyburg, 1961). Its closer relative is that randomness helps for mass problems: a class of positive measure, such as a class of bounded diagonally non-computable functions, can have no computable member (cf. Kučera, 1985; Bienvenu and Porter, 2016). That randomisation can succeed where determinism cannot, below a success-probability threshold, is familiar for learners (Pitt, 1989; Ambainis, 1999); here the random objects are the hypotheses themselves. Bienvenu et al. (2014) add randomly chosen true complexity axioms to PA and show that they do not help to prove new interesting theorems but can shorten proofs unless $\mathsf{NP} = \mathsf{PSPACE}$; the threshold theories of Theorem 5.6 use a soundness threshold of a related kind, placed just above the unknown inconsistency probability of the universal random theory. Section 5 also meets Chaitin's theorem (Chaitin, 1974; Raatikainen, 1998), flexible formulas (cf. Kripke, 1962; Woodin, 2011), Kreisel's conjecture (cf. Hrubeš, 2009) and the problem of old evidence (Garber, 1983).

Craig, Ramsey, Hempel. Craig's theorem (Craig, 1953, 1956) is the background of Hempel's theoretician's dilemma (Hempel, 1958) and of its replies (Craigian axiomatisations are unmanageable; theoretical terms serve inductive systematisation, cf. Niiniluoto, 1972; science does not aim only at prediction, cf. Putnam, 1965a), the first two of which Section 8 sets against our results. In monadic inductive logic theoretical concepts can change inductive probabilities (cf. Niiniluoto and Tuomela, 1973); in the universal mixture they cannot, beyond a constant factor, that is, by more than a bounded number of bits over all data (Theorem F.3). Every r.e. theory has a polynomial-time axiomatisation by a variant of Craig's trick (Krajíček, 2025), and every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952; Craig and Vaught, 1958, as reported by Pakhomov and Visser, 2019), of which Theorem F.33 is an effective many-sorted form. Ramsey sentences and Lewis's definitions (Ramsey, 1931; Lewis, 1970) appear in Proposition F.35.

Usable theories. Knowledge compilation studies the size of theories a weak reasoner can use (Selman and Kautz, 1996; Darwiche and Marquis, 2002; del Val, 1994; Babka et al., 2013); the declarative/procedural question is old in AI (McCarthy, 1959; Winograd, 1975; Kowalski, 1979).

Laws, approximation, explanation, and what a theory is. Lewis's best-system account (Lewis, 1983, 1994) and its language-relative version (Cohen and Callender, 2009) are the counterpart of Section 9.5, by analogy only. That a model includes its noise distribution and that measurement precision enters the code length are standard in MDL and MML (Rissanen, 1978; Wallace, 2005; Grünwald, 2007); algorithmic statistics studies the split into model and noise (Gács et al., 2001; Vereshchagin and Vitányi, 2004); Akaike-based accounts give a statistical form of the dependence of selection on precision (Forster and Sober, 1994). Under misspecification, estimation error or non-nested information, rankings of point forecasts can depend on the consistent

loss used (Patton, 2020); for proper scoring rules see Gneiting and Raftery (2007), and for the assessment of models by their adequacy for particular purposes Parker (2009, 2020). A purpose loss can replace log loss in learning through generalised Bayesian updating (Bissiri et al., 2016) or the aggregating algorithm (Vovk, 1990, 2001), which with universal weights gives predictive complexity (cf. Vovk and Watkins, 1998; a form over theory machines is in Appendix G.9). On truthlikeness, explanation and what a theory is, see Sections 6 and 11. Discovery systems (BACON on Kepler's third law, Langley et al., 1987; symbolic regression, Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020; the Apperception Engine, Evans et al., 2021) search; SAI evaluates.

What is new. To our knowledge the following are new: the single-draw proof-requiring law M_A and its basic theory (Section 3); the one-tape and Kolmogorov equivalences with consistent SOI and the transfer principle (Theorems 4.2 to 4.4); the lottery, its comparison of versions and its arithmetic versions with true data (Theorem 5.1 and Corollary 5.5); the threshold theories on Π_1 data and the coding theorem for an inductive inference rule (Theorem 5.6 and Corollary 5.7); the reasoner-relative separations inside a universal inducer (Section 7), where the bound for systems of laws (Theorem 7.5) is a short combination of known results that we did not find stated and that is probably known to specialists; a complexity-theoretic form of the impracticality reply to Hempel, which we did not find in the literature, and the derivation-bounded constructions and fit classes of Section 8; and, for a proof-requiring inducer, the choice between honest laws, laws with exceptions and laws with error models (Propositions 9.2 and G.18). Not new are random axiom systems drawn by a universal machine, the mixture and dominance arguments, Craig's construction (the coin-free weight equivalence is that construction inside the universal prior, proved for deciders in the earlier report), the rules of conditioning for belief functions, and the reading of a law with its noise distribution as a two-part code.

13 Discussion

What the formalism shows. Solomonoff axiom induction makes precise the view that theories answer questions posed in a given language, and the view survives in comparative form: it changes the target, the data type, logical coherence and partiality (Section 6.3 and Table 3), while the inference principle is unchanged: version by version and up to constant factors, SAI agrees with operator induction over consistent labellers (Section 4). On data labelled true or false the idealised formalism cannot see the difference between a theory and a program. In the settings we study, shape becomes visible only when something other than program length is charged (the written theory, the time to produce it, or the derivations) or the class of hypotheses is restricted (to finite theories); there it is relative to a reasoner and a vocabulary, and for theoretical terms what pays is relations and entities, not vocabulary as such. Whether this holds beyond these settings is open (Question 7.7). That textbook theories are declarative is an observation we accept, and we find it plausible that this shape is good, but in our models declarative against procedural form buys at most a constant factor once intermediate concepts are available and the reasoner can chain through them; what is missing is a formal setting in which that shape provably pays. Finally, "Kepler's laws are only approximately true" remains a limitation: SAI prices an approximately true law as a law with a calibrated error model (Proposition 9.2), but whether a law is good enough depends on a purpose, which its log loss sees only through the questions asked (Section 11).

What it cannot show. The axiom view models the declarative core of scientific theories (contents, inferential relations, partiality, approximate truth at a tolerance) and misses their proce-

dural, statistical and dynamic sides and graded adequacy for a purpose (Section 11). It supports no general realist conclusion about the theories it favours: it identifies a complete r.e. theory that the data present (Proposition A.27), but its limit posterior can live on random theories and, with adversarially chosen true data, on false ones (Theorem 5.1 and Corollary 5.5). It evaluates theories and does not find them, and it is not computable.

13.1 Open problems

(1) Does M_A dominate every lower semicomputable provability function of a random consistent theory (Question 3.12)?

(2) Is there a stream with bounded one-tape loss on which the coin-free loss diverges computably fast, and does bounded one-tape loss on a co-r.e. stream imply that a consistent r.e. theory proves it (Question 5.3)?

(3) Over a true background containing PA, is the one-tape posterior probability of Σ_1-soundness bounded away from 1 uniformly over all data (Question 5.10)?

(4) Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws, to real-valued laws with numerical solvers, probabilistic programs with conditioning and simulation codes (Question 7.7)?

(5) Is there a formal setting for communities of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs (Question 6.1)?

(6) Can a purpose enter SAI, as a decision layer on its belief-function predictions or as a loss on answers inside learning, and is there then an upper semicomputable universal element (Question G.53)?

The further questions are Questions A.19, A.34, B.20, B.22, B.42, C.10, C.13, C.29, C.33, E.28, E.35, E.41, F.23, F.25, F.26, G.7 and G.41.

Directions. The directions we find most worth pursuing are step 4 of the ladder, for which we have no formal handle; the setting of problem (5), where the declarative shape of written theories may find its explanation; a costed-deduction mixture over unknown axiom systems, combining SAI, uncertain about the axioms, with logical induction (Garrabrant et al., 2016a), uncertain about their consequences, as induction in mathematics needs (Proposition C.23); and a native treatment of statistical confrontation inside the proof-requiring core (Section 11). Smaller ones: whether explanatory asymmetry is interventional in general, and a derivation of Kepler's laws within stated tolerances from Newton's theory in a prover for validated numerics (Section 11).

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Code and data

All computations reported here were done with seeded or deterministic Python scripts (Python 3.11 with NumPy, SciPy and mpmath; the planetary computations use pyerfa, the Python interface to the ERFA library). The scripts, their saved outputs and independent re-implementations of the main computations are in the folder `solomonoff-axiom-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each reported number and how to re-run it. Re-running the scripts reproduces the saved outputs, apart from lines that report run times; the longest computations, on the Mars pool, take more than half an hour each, and their figures were also reproduced by independent re-implementations.

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A Proofs for the setting and the basic theorems

This appendix proves the results of Sections 2 and 3 in their order, and states and proves the results that the main text only mentions. Throughout, G is a theory machine and the notation is that of Definitions 2.3 and 2.4.

A.1 The setting

Lemma A.1 (Measurability and the basic form)**.** *(a)* $O^G_\chi = \bigcup\{[w] : B \cup A^G_w \vdash \chi\}$ *is an effectively open class, uniformly in an index of* G *and in* χ*; hence* $\mathbb{U}(O^G_\chi)$ *is left-c.e., uniformly. (b)* Cons^G *is a* Π^0_1 *class, and* $O^G_\perp \subseteq O^G_\chi$ *for every* χ*. (c)* $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\perp)$*, and* $E^{3,G}_D$ *is Borel.*

Proof. (a) If $B \cup A^G_\omega \vdash \chi$, a derivation uses finitely many axioms $\chi_1, \dots, \chi_r \in A^G_\omega$, each emitted under some prefix of ω. The prefixes of ω are linearly ordered, so the longest of them, w, has $\chi_1, \dots, \chi_r \in A^G_w$, and every $\omega' \sqsupseteq w$ has $B \cup A^G_{\omega'} \vdash \chi$. Conversely $[w] \subseteq O^G_\chi$ whenever $B \cup A^G_w \vdash \chi$. The set of pairs (w, χ) with $B \cup A^G_w \vdash \chi$ is c.e.: enumerate G and all derivations, using that B is decidable. (b) Cons^G is the complement of an effectively open class, and an inconsistent set proves everything. (c) $E^G_D = O^G_{\psi_D} \setminus O^G_\perp$ with $O^G_\perp \subseteq O^G_{\psi_D}$; and $E^{3,G}_D$ is E^G_D intersected with finitely many closed classes. □

Examples of theory machines. (i) A deterministic r.e. theory W_e: $G = \{(\epsilon, \chi) : \chi \in \mathsf{W}_e\}$, with $\nu_G(D) = \mathbf{1}[B \cup \mathsf{W}_e$ consistent and $\vdash \psi_D]$. (ii) Fair-coin facts on a family $(\alpha_j)_{j \geq 1}$ of sentences: $G = \{(w, \alpha_j^{w_j}) : |w| = j\}$. (iii) A theory plus coin facts: the union of (i) and (ii). (iv) A finite mixture of finite theories $T_1, \dots, T_r$: $G = \{(w_j, \chi) : \chi \in T_j\}$ for a prefix-free set $\{w_1, \dots, w_r\}$, which gives T_j weight $2^{-|w_j|}$; a tape with no prefix among the w_j emits nothing, and its theory is the empty one, consistent since B is.

Remark A.2 (Deciders, enumerators and polynomial-delay enumerators agree)*.* There is c such that for every D: (a) $2^{-c} W^{\mathrm{dec}}_A(D) \leq W_A(D) \leq 2^c W^{\mathrm{dec}}_A(D)$ and $|K_A(D) - K^{\mathrm{dec}}_A(D)| \leq c$; (b) under Assumption B.5, for t time-constructible, nondecreasing and superadditive with $t(m) \geq m$ and $t(m) \geq m^{a^*}$ (a^* fixed by the universal machine): $2^{-c} W_A(D) \leq W^{\mathrm{gen},t}_A(D) \leq 2^c W_A(D)$, and $|K_A(D) - K^{\mathrm{gen},t}_A(D)| \leq c$; (c) $K_A(D) \leq K_{\mathrm{fin}}(D) + c \leq K(\psi_D) + c$ if D is consistent; if D is inconsistent, $K_A(D) = K_{\mathrm{fin}}(D) = \infty$, while $K(\psi_D)$ is finite.

Proof. (a) *Deciders to enumerators.* By universality of V there is a string r_1 such that, whenever p is a decider, $V(r_1 p)$ is an index of the r.e. set $\{\chi : \{V(p)\}(\chi) = 1\}$. The map $p \mapsto r_1 p$ is injective, adds $|r_1|$ to the length and preserves the axiom set. *Enumerators to deciders (Craig's trick).* Write $\chi^{\wedge 1} := \chi$ and $\chi^{\wedge(m+1)} := \chi \wedge \chi^{\wedge m}$. For an index e fix the step-by-step enumeration of W_e and let $A^{\mathrm{C}}_e := \{\chi^{\wedge(s+1)} : \chi$ is enumerated at step $s\}$. A sentence θ has at most two readings $\theta = \chi^{\wedge m}$: $(\chi, m) = (\theta, 1)$, and, if θ is a conjunction, χ its left conjunct with m the number of times χ can be stripped off. (If $\theta = \chi^{\wedge m}$ with $m \geq 2$, then χ is the left conjunct of θ, and stripping stops exactly at χ, because χ is not of the form $\chi \wedge X$.) So A^{C}_e is decided by the procedure: find the readings, and for each reading (χ, m) run the enumeration for exactly $m - 1$ steps and check

whether χ appears at step $m-1$. It halts on every input, also when W_e is finite. Since $\chi^{\wedge m}$ is logically equivalent to χ, $\mathrm{Cn}(A_e^{\mathrm{C}}) = \mathrm{Cn}(\mathsf{W}_e)$. A fixed string r_2 makes $V(r_2 p)$ an index of this decider for $e = V(p)$; the map $p \mapsto r_2 p$ is injective and adds $|r_2|$. Both maps preserve consistency and what is proved, so summing over programs, and taking shortest programs, gives (a) with $c := \max(|r_1|, |r_2|)$. This is the argument of Theorem 6.3 of our earlier report (Hänni and Claude (Anthropic), 2026a), here between two kinds of axiom hypotheses rather than between axiom and labeller hypotheses.

(b) Under Assumption B.5, Theorem 6.14(a) of the earlier report shows that $W_A^{\mathrm{gen},t}$ is within a constant factor of the coin-free consistent mixture W^{cons} of labellers, and its Theorem 6.3 shows $W^{\mathrm{cons}} \asymp W_A^{\mathrm{dec}}$; combined with (a) this gives the claim for $W_A^{\mathrm{gen},t}$. For the Kolmogorov versions, $|K_A(D) - K_A^{\mathrm{gen},t}(D)| \le c$, we use the map behind it. An enumerator p of A is sent to the padded enumerator of the semi-decider of A ("yes on χ iff χ is enumerated") of Lemma E.12 of the earlier report; that lemma, proved there under Assumption B.5, gives an enumerator of the delay form with the same consequences as A, injectively and with length at most $|p| + c$. Conversely, an enumerator of the delay form is an enumerator. The translation between the universal machine of the earlier report and programs of V that print indices costs a constant. We do not re-derive Assumption B.5.

(c) If ψ witnesses $K_{\mathrm{fin}}(D)$, a fixed prefix r_3 turns a shortest program for ψ into a program for an index of $\{\psi\}$, which is consistent with B and proves ψ_D; so $K_A \le K_{\mathrm{fin}} + |r_3|$. For consistent D, ψ_D itself is a candidate for K_{fin}. If D is inconsistent, no consistent set proves ψ_D. □

Remark A.3 (Complexities of consistent theories). If $B \cup T$ is consistent, then $[T] \subseteq \mathrm{Ext}T$, so $2^{-K_{\equiv}(T)} \le \mathbf{m}_{\equiv}(T) \le \mathbf{m}(\mathrm{Ext}T)$, and $\mathbb{U}_A[T] \ge 2^{-c}\mathbf{m}_{\equiv}(T)$, $\mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (Theorem 3.4(i)). The hypothesis is needed: for $T = \{\bot\}$, $\mathbb{U}_A[T] = 0$ and $\mathrm{Ext}T = \emptyset$, while $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)} > 0$.

The two defects and function machines. The outcomes of a theory machine at a sentence φ correspond to those of a function machine at an input: "proves φ" is a Σ_1 property of the tape, like "halts with output 1"; "independent" is Π_1, like "diverges"; "inconsistent" corresponds to "halts with both outputs", which a function machine cannot do.

The encoding (E3). (E3) is not a single datum of M_A^3: it combines "the label of $\alpha \to \ell$ is 1" with "the label of $\neg\alpha$ is not 1", a union of two M_A^3-events. The two-valued encoding by the datum $(\alpha \to \ell, 1)$ is degenerate, since a single world satisfying the law fits all of its question data (Proposition E.3); it is never used for conditional data.

A.2 Structure and updating

Proof of Proposition 3.1. (a) If D is inconsistent, every set proving ψ_D is inconsistent, so $E_D^G = \emptyset$. If D is consistent, the deterministic machine $G_D := \{(\epsilon, \psi_D)\}$ has $\nu_{G_D}(D) = 1$, so $M_A(D) \ge 2^{-K(G_D)} > 0$ by Theorem 3.4. Three-valued: $B \cup \{\psi_D\}$ decides exactly what $B \cup D^{\pm}$ decides, so if $B \cup D^{\pm}$ decides no ι-labelled sentence then $\nu_{G_D}^3(D) = 1$; conversely, if $B \cup D^{\pm}$ decides an ι-labelled sentence, so does every set proving ψ_D.

(b) O_χ^G depends only on the B-equivalence class of χ, and $\psi_{D+(\varphi,b)}$ is equivalent to $\psi_D \wedge \varphi^b$. On E_D^G the random theory is consistent, so exactly one of three things happens: it proves φ, it proves $\neg\varphi$, or it decides neither. The three events are $E_{D+(\varphi,1)}^G$, $E_{D+(\varphi,0)}^G$ and $E_{D+(\varphi,\iota)}^{3,G}$. For G_U, dividing by $\mathbb{U}(E_D)$ shows that the third term is $M_A(D)(1 - \mathrm{Bel}_D(\varphi) - \mathrm{Dis}_D(\varphi))$.

(c) $B \cup A \vdash \psi_D$ iff $\mathrm{Cn}(D^{\pm}) \subseteq \mathrm{Cn}(A)$. If $B \cup D^{\pm} \vdash \varphi^b$, then $\mathrm{Cn}(D^{\pm} \cup \{\varphi^b\}) = \mathrm{Cn}(D^{\pm})$, and $D + (\varphi, 1-b)$ is inconsistent, so (a) applies to it.

(d) At a step with $\varphi_t \notin \operatorname{dom} D_{t-1}$ the factor is $M_A(D_t)/M_A(D_{t-1})$; at a repeat (necessarily with the stored label, since the sequence is consistent) it is 1 and $D_t = D_{t-1}$. The product telescopes to $M_A(D_n)/M_A(\emptyset)$. If some $M_A(D_t) = 0$, the first such step has factor 0, and both sides are $+\infty$ by the monotonicity in (b). Dependence only on $\mathrm{Cn}(D_n^{\pm})$ is (c).

(e) $M_A(\emptyset) = \mathbb{U}(\mathrm{Cons})$. The tapes beginning with a V-program for an index of the empty machine have $A_\omega = \emptyset$, consistent since B is; the tapes beginning with a program for the machine $\{(\epsilon, \bot)\}$ are inconsistent. Both sets have positive measure.

(f) As (b), with $E_D^{3,G}$ in place of E_D^G; and $\nu_G^3(\emptyset) = \mathbb{U}(\mathrm{Cons}^G) = \nu_G(\emptyset)$.

(g) By definition $E_D^{3,G} = \{\omega \in \mathrm{Cons}^G : \mathrm{tri}_\omega^G \supseteq D\}$, and for two-valued D this is E_D^G. □

Proof of Proposition 3.2. (a) A theory proves ψ_D and χ iff it proves $\psi_D \wedge \chi$, so $E_D \cap O_\chi = O_{\psi_D \wedge \chi} \setminus O_\bot$, and both sides equal $\mathbb{U}(O_{\psi_D \wedge \chi} \setminus O_\bot)/\mathbb{U}(O_{\psi_D} \setminus O_\bot)$. The normalising constant $1 - \mathrm{Inc}_0$ cancels.

(b) Let $P := \mathbb{U}(\cdot \mid E_D)$ and $Q_i := O_{\chi_i}$. Then $\bigcap_{i \in I} Q_i = O_{\bigwedge_{i \in I} \chi_i}$, since on one tape a theory proves each χ_i iff it proves their conjunction, and $\bigcup_i Q_i \subseteq O_{\bigvee_i \chi_i}$. Inclusion–exclusion for $P(\bigcup_i Q_i)$ gives the inequality. Bel_0 is the unnormalised case, with P the restriction of $\mathbb{U}$ to Cons.

(c) Fix an enumeration (σ_k) of Sent. For $\omega \in \mathrm{Cons}$ let $T_0(\omega) := B \cup A_\omega$, and $T_{k+1} := T_k \cup \{\sigma_k\}$ if that set is consistent, $T_{k+1} := T_k \cup \{\neg\sigma_k\}$ otherwise. $T_\infty(\omega) := \bigcup_k T_k$ is a complete consistent extension of B: it is consistent by compactness, and complete and deductively closed because it is maximal consistent. Each choice depends on whether $B \cup A_\omega \cup F$ is consistent for a finite set F, that is, on membership of ω in the open class $O_{\neg \bigwedge F}$; so $\omega \mapsto T_\infty(\omega)$ is Borel. Let P_D be its law under $\mathbb{U}(\cdot \mid E_D)$. If $B \cup A_\omega \vdash \chi$ then $\chi \in T_\infty$, and if $\chi \in T_\infty$ then $B \cup A_\omega \nvdash \neg\chi$. So $\mathrm{Bel}_D \le P_D \le \mathrm{Pl}_D$. □

Proposition A.4 (Consistency-requiring updating is Dempster conditioning)**.** *Let* $\psi := \psi_D$ *and* $M_A^{\mathrm{nc}}(D) > 0$. *Then* $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi)$ *and* $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \dfrac{\mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)}{\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi)} = \dfrac{\widehat{\mathrm{Bel}}_0(\varphi \vee \neg\psi) - \widehat{\mathrm{Bel}}_0(\neg\psi)}{1 - \widehat{\mathrm{Bel}}_0(\neg\psi)}$.

Proof. For consistent $B \cup A$: $B \cup A \cup \{\psi\}$ is consistent iff $B \cup A \nvdash \neg\psi$, and $B \cup A \cup \{\psi\} \vdash \varphi$ iff $B \cup A \vdash \varphi \vee \neg\psi$ (the deduction theorem; ψ is a sentence). So $M_A^{\mathrm{nc}}(D) = \mathbb{U}(\mathrm{Cons}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\psi) = \mathrm{Pl}_0(\psi)$. Since $B \cup A \vdash \neg\psi$ implies $B \cup A \vdash \varphi \vee \neg\psi$, the numerator of $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi)$ is $\mathbb{U}(\mathrm{Cons} \cap O_{\varphi \vee \neg\psi}) - \mathbb{U}(\mathrm{Cons} \cap O_{\neg\psi}) = \mathrm{Bel}_0(\varphi \vee \neg\psi) - \mathrm{Bel}_0(\neg\psi)$. Dividing numerator and denominator by $1 - \mathrm{Inc}_0$ gives the form with $\widehat{\mathrm{Bel}}_0$. □

A.3 Computability

Proposition A.5. $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathrm{Inc}_0$, *where* $\mathbb{U}(O_{\psi_D})$ *is left-c.e. uniformly in* D *and* $\mathrm{Inc}_0 = \mathbb{U}(O_\bot)$ *is left-c.e.*

Proof. Lemma A.1(a), (c) for $G = G_U$. □

The following lemma is a form of a theorem of Demuth (Demuth, 1975), as it is attributed in the literature: a left-c.e. real plus a random left-c.e. real is random (the converse is attributed to Downey et al., 2002). We prove the form we need, for completeness. For a real $x \in (0,1)$, $x_{1:n}$

denotes its first n binary digits; we use the Levin–Schnorr characterisation (standard; see Li and Vitányi, 2008): x is Martin-Löf random iff $K(x_{1:n}) \geq n - c_x$ for all n.

Lemma A.6 (Randomness passes upwards)**.** *Let α, γ be left-c.e. reals with α Martin-Löf random, let $k \geq 0$, and let $\beta := 2^{-k}\alpha + \gamma < 1$. Then $K(\alpha_{1:n}) \leq K(\beta_{1:n}) + k + c$ for all n, where c depends on k and on fixed approximating programs for α and γ. Hence β is Martin-Löf random.*

Proof. Let $\alpha_s \uparrow \alpha$ and $\gamma_s \uparrow \gamma$ be computable rational approximations and $\beta_s := 2^{-k}\alpha_s + \gamma_s$. Given the string $u := \beta_{1:n}$ (which also gives n), put $q := 0.u - 2^{-n} < \beta$ and find the first s with $\beta_s > q$. Then $\beta - \beta_s < \beta - q \leq 2^{1-n}$, hence $\alpha - \alpha_s \leq 2^k(\beta - \beta_s) < 2^{k+1-n}$, because $\gamma - \gamma_s \geq 0$. Since α is random, it is irrational, so $\alpha_{1:n}$ is the index of the dyadic interval of length 2^{-n} that contains α, one of the at most $2^{k+1} + 1$ such intervals that meet $[\alpha_s, \alpha_s + 2^{k+1-n})$. So $\alpha_{1:n}$ is determined by u and a $(k+2)$-bit index, and $K(\alpha_{1:n}) \leq K(\beta_{1:n}) + k + c$. By the Levin–Schnorr characterisation $K(\alpha_{1:n}) \geq n - c'$ for all n; hence $K(\beta_{1:n}) \geq n - k - c - c'$ for all n, and β is random by the same characterisation. □

Proof of Theorem 3.3. The machine G_α. Let $\alpha_s \uparrow \alpha$ be a computable nondecreasing sequence of rationals, and let $G_\alpha := \{(w, \bot) : \exists s\ 0.w + 2^{-|w|} \leq \alpha_s\}$, where $0.w$ is the dyadic rational with binary digits w. Then $A_\omega^{G_\alpha} \subseteq \{\bot\}$, and $\bot \in A_\omega^{G_\alpha}$ iff some prefix w of ω has $[w] \subseteq [0, \alpha_s]$ for some s, read as reals. This holds for every ω with $0.\omega < \alpha$ (take s with $\alpha_s > 0.\omega$ and a long prefix) and fails for every ω with $0.\omega > \alpha$. So the inconsistency probability of G_α is α.

The decomposition. Let p_α be a V-program for an index of G_α. Splitting the tapes by their first block in $\operatorname{dom} V$ (tapes with no such block have $A_\omega = \emptyset$, which is consistent),

$$\mathrm{Inc}_0 = 2^{-|p_\alpha|}\alpha + \gamma_\alpha, \qquad \gamma_\alpha := \sum_{p \in \operatorname{dom} V \setminus \{p_\alpha\}} 2^{-|p|}\, \mathbb{U}(O_\bot^{G_{V(p)}}),$$

and γ_α is left-c.e., because $\operatorname{dom} V \setminus \{p_\alpha\}$ is c.e. and each term is left-c.e. uniformly (Lemma A.1(a)).

(a) Let $\mathcal{H}$ be the halting set and $\alpha := \sum_{n \in \mathcal{H}} 2^{-n-1}$, which is left-c.e. Its binary expansion is the characteristic sequence of $\mathcal{H}$ (unique, since $\mathcal{H}$ is infinite and co-infinite), so α is not computable. If Inc_0 were computable, $\alpha = 2^{|p_\alpha|}(\mathrm{Inc}_0 - \gamma_\alpha)$ would be right-c.e.; being also left-c.e., it would be computable. Since a real that is both left-c.e. and right-c.e. is computable, the right-c.e. real $1 - \mathrm{Inc}_0$ is not left-c.e.; as $M_A(\emptyset) = 1 - \mathrm{Inc}_0$ is a value of M_A, M_A is not lower semicomputable.

(b) *Solovay reducibility* (Solovay, 1975; we use the definition of Downey and Hirschfeldt, 2010): $\alpha \leq_S \beta$ iff there are a constant k and a partial computable function f on the rationals such that for every rational $q < \beta$, $f(q)$ is defined, $f(q) < \alpha$, and $\alpha - f(q) \leq 2^k(\beta - q)$. Let $\beta := \mathrm{Inc}_0$, with approximations $\beta_s := 2^{-|p_\alpha|}\alpha_s + \gamma_{\alpha,s} \uparrow \beta$. If α is computable the reduction is trivial: given $q < \beta$, find s with $\beta_s > q$ and output a rational in $(\alpha - (\beta_s - q), \alpha)$. Otherwise $\alpha_s < \alpha$ for all s; let $f(q) := \alpha_s$ for the first s with $\beta_s > q$. Then, since $\gamma_\alpha - \gamma_{\alpha,s} \geq 0$,

$$\alpha - \alpha_s = 2^{|p_\alpha|}\big[(\beta - \beta_s) - (\gamma_\alpha - \gamma_{\alpha,s})\big] \leq 2^{|p_\alpha|}(\beta - \beta_s) < 2^{|p_\alpha|}(\beta - q).$$

(c) Ω_V is Martin-Löf random (Chaitin, 1975). Apply the decomposition with $\alpha := \Omega_V$, and Lemma A.6 with $k = |p_\alpha|$ and $\gamma = \gamma_\alpha$; here $\mathrm{Inc}_0 < 1$ by Proposition 3.1(e), and k and γ_α are fixed by V, so the constant of the lemma is a machine constant. □

Probabilities of other machine behaviours are studied by Becher et al. (2006) and Barmpalias et al. (2016). By Kučera and Slaman (2001) and Calude et al. (2001), Inc_0 is then an Ω-number; we do not use this.

Proposition A.7 (Computability relative to the halting problem)**.** *(a) $M_A(D)$ and the conditionals $M_A(b \mid D, \varphi)$, $M^3_A(y \mid D, \varphi)$ and $\mathrm{Bel}_D(\chi)$ are computable relative to $\emptyset'$, uniformly. (b) $D \mapsto M_A(D) + \mathrm{Inc}_0$ is lower semicomputable; so M_A is lower semicomputable relative to the single real Inc_0.*

Proof. (a) Every left-c.e. real is $\emptyset'$-computable (decide "$\exists s\ a_s > q$" for rationals q), and differences and ratios of $\emptyset'$-computable reals are $\emptyset'$-computable once the denominator is known to be positive. By Proposition 3.1(a), $M_A(D) > 0$ iff D is consistent, a Π^0_1 question, which $\emptyset'$ decides. The three-valued events are Boolean combinations of effectively open classes, with $O_a \cap O_b = O_{a \wedge b}$, so their measures are sums and differences of left-c.e. reals by inclusion–exclusion. (b) is Proposition A.5. □

A.4 Dominance and invariance

Proof of Theorem 3.4. $\mathrm{dom}\, V$ is prefix-free, so the cylinders $[p]$, $p \in \mathrm{dom}\, V$, are disjoint, and for $\omega = p\omega'$ we have $A_\omega = A^{G_{V(p)}}_{\omega'}$ with ω' uniform given $[p]$. Hence $\mathbb{U}(E_D \cap [p]) = 2^{-|p|}\nu_{G_{V(p)}}(D)$. The tapes with no prefix in $\mathrm{dom}\, V$ have measure $1 - \Omega_V$ and $A_\omega = \emptyset$, which proves ψ_D iff B does. Grouping the p with $V(p) = i$ gives the weight $\mathbf{m}(i) \geq 2^{-K(i)}$, and taking i with $\nu_{G_i} = \nu$ and $K(i) = K(\nu)$ gives $M_A \geq 2^{-K(\nu)}\nu$. The three-valued case is identical.

(i) The deterministic machine $G'_e := \{(\epsilon, \chi) : \chi \in \mathsf{W}_e\}$ has an index $g(e)$ with g computable and injective, so $\mathbf{m}(g(e)) \geq 2^{-c}\mathbf{m}(e)$: the image of $\mathbf{m}$ under g is an lsc semimeasure, dominated by $\mathbf{m}$. For $e \in [T]$, the tapes beginning with a program for $g(e)$ have $A_\omega = \mathsf{W}_e$, with $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)$, and $B \cup A_\omega$ consistent because $B \cup T$ is. These sets of tapes are disjoint for distinct e; summing gives $\mathbb{U}_A[T] \geq 2^{-c}\mathbf{m}_{\equiv}(T)$, and $\mathbf{m}_{\equiv}(T) \geq 2^{-K_{\equiv}(T)}$. The same argument over $e \in \mathrm{Ext}T$ gives $\mathbb{U}_A(\mathrm{Ext}T) \geq 2^{-c}\mathbf{m}(\mathrm{Ext}T)$.

(ii) $G_f := \{(\epsilon, \varphi^b) : f(\varphi) = b\}$ enumerates Γ_f, which is consistent with B and proves ψ_D when $f \supseteq D$; and $K(G_f) \leq K(f) + c$.

(iii) The map $i \mapsto \sum_{g(e)=i} v(e)$ is an lsc semimeasure of complexity at most $K(v) + c$, so $\mathbf{m}(g(e)) \geq 2^{-K(v)-c}v(e)$. With $v = \mathbf{m}$ this gives $M_A \geq 2^{-c}W_A$.

(iv) The machine of Example (iii) in Appendix A.1 has an index computable from indices of T and of the family. □

Proposition A.8 (Choice of the universal machine)**.** *Call a theory machine G'* universal *if for every theory machine G there is a string p_G with $A^{G'}_{p_G\omega} = A^G_\omega$ for all ω. If G' is universal, then $2^{-c}M_A \leq \nu_{G'} \leq 2^c M_A$, and the same for the three-valued laws, with c depending only on G_U and G'.*

Proof. G_U is universal, with p_G a V-program for an index of G. $M_A \geq 2^{-K(G')}\nu_{G'}$ by Theorem 3.4. Conversely, the tapes $p_{G_U}\omega'$ give G' the axiom set $A_{\omega'}$, so $\nu_{G'}(D) \geq 2^{-|p_{G_U}|}M_A(D)$. The three-valued laws are handled in the same way. So posteriors change by at most a constant factor, and losses in the semimeasure convention by an additive constant. □

Proposition A.9 (The background is a prior shift)**.** *Let B, B' be decidable and consistent with $B \subseteq \mathrm{Cn}_\emptyset(B')$ (plain logical consequence), and write M^B_A, $M^{B'}_A$ for SAI with these backgrounds. Then $M^B_A(D) \geq 2^{-K(B')-c}M^{B'}_A(D)$ for every D, with $K(B')$ the complexity of an index of B'; likewise $W^B_A \geq 2^{-K(B')-c}W^{B'}_A$ and $K^B_A \leq K^{B'}_A + K(B') + c$. The converse fails: a D consistent with B and inconsistent with B' has $M^{B'}_A(D) = 0 < M^B_A(D)$.*

Proof. The machine $G_{B'} := \{(\epsilon, \chi) : \chi \in B'\} \cup G_U$ emits $B' \cup A_\omega$ on ω. Under the background B, the set $B \cup B' \cup A_\omega$ is equivalent to $B' \cup A_\omega$, so the law of $G_{B'}$ under the background B is $M_A^{B'}$; apply Theorem 3.4. (Both laws use the same G_U, which does not depend on the background.) The same map on single programs gives the statements for W_A and K_A. The converse fails as stated, since D is consistent with B (Proposition 3.1(a)). □

Proof of Theorem 3.5. A preliminary fact: $B' \vdash \neg\tau(\bot)$. By (T1) with $\Gamma = \emptyset$, $B \vdash \neg\bot$ gives $B' \vdash \tau(\neg\bot)$, and (T2) converts this.

Push forward. The machine "run G_U and emit $\tau(\chi)$ for every emitted χ" emits $\tau[A_\omega]$. By (T1), $B' \cup \tau[A] \vdash \tau(\bot)$ iff $B \cup A \vdash \bot$; with the preliminary fact, $B' \cup \tau[A]$ is consistent iff $B \cup A$ is. By (T1) and (T2), $B' \cup \tau[A] \vdash \tau(\varphi)^b$ iff $B \cup A \vdash \varphi^b$ (for $b = 0$: $B' \cup \tau[A] \vdash \neg\tau(\varphi)$ iff it proves $\tau(\neg\varphi)$ iff $B \cup A \vdash \neg\varphi$). So the event of this machine for τD is exactly E_D, and Theorem 3.4 in $\mathcal{L}'$ gives $M'_A(\tau D) \geq 2^{-K(\tau)-c} M_A(D)$.

Pull back. For $A' \subseteq \mathrm{Sent}_{\mathcal{L}'}$ let $\tau^*(A') := \{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B' \cup A' \vdash \tau(\varphi)\}$. The set $G^* := \{(w, \varphi) : B' \cup A'_w \vdash \tau(\varphi)\}$ is c.e., and by compactness, as in Lemma A.1, $A_\omega^{G^*} = \tau^*(A'_\omega)$. If $B' \cup A'$ is consistent, so is $B \cup \tau^*(A')$: if $B \cup \Gamma \vdash \bot$ for a finite $\Gamma \subseteq \tau^*(A')$, then $B' \cup \tau[\Gamma] \vdash \tau(\bot)$ by (T1), and $B' \cup A'$ proves every member of $\tau[\Gamma]$, so $B' \cup A' \vdash \tau(\bot) \wedge \neg\tau(\bot)$. If $B' \cup A' \vdash \tau(\varphi)^b$, then $\varphi^b \in \tau^*(A')$ (for $b = 0$ via (T2)). So the event of G^* for D contains the event for τD, and $M_A(D) \geq 2^{-K(\tau)-c} M'_A(\tau D)$.

The same two maps act on single enumerator programs, with a prefix of length $K(\tau) + c$; this gives the statements for W_A and K_A. □

Proof of Corollary 3.6. (T1) holds for the inclusion because every $\mathcal{L}$-structure expands to an $\mathcal{L}'$-structure, so by completeness $B \cup \Gamma \vdash_{\mathcal{L}'} \varphi$ iff $B \cup \Gamma \vdash_{\mathcal{L}} \varphi$ for $\mathcal{L}$-sentences; (T2) is trivial. Theorem 3.5 with $K(\tau) = O(1)$ gives the weights, and a predictive probability is a ratio of two weights, each changed by a factor $2^{\pm c}$. □

A.5 Loss bounds

Proposition A.10 (Laws plus exceptions)**.** *(a)* $L_A(D_n) \leq -\log \nu(D_n) + K(\nu)$ *for every theory-machine law* ν*. (b) For every r.e.* T *and every* $E \subseteq D_n$ *with* $B \cup T \cup E^{\pm}$ *consistent and* $B \cup T \cup E^{\pm} \vdash \psi_{D_n}$*:* $L_A(D_n) \leq K_{\equiv}(T) + K(E) + c$*. (c) Let* $(\alpha_j)_{j \geq 1}$ *be a computable family of sentences and* T *r.e. with* $B \cup T \cup \{\alpha_j^{x_j} : j \geq 1\}$ *consistent for every* $x \in \{0,1\}^{\mathbb{N}}$*. If every datum of* D_n *is proved by* T *or is one of the* α_j *with some label, and* m *distinct* j *occur, then* $L_A(D_n) \leq K(T, (\alpha_j)) + m + c$*.*

Section 9 extends (c) from atoms a law is silent about to answers it only constrains.

Proof. (a) is Theorem 3.4. (b) For $e \in [T]$, the set $\mathsf{W}_e \cup \{\psi_E\}$ has an index computable from (e, E); it is consistent with B, since $B \cup T \cup E^{\pm}$ is, and it proves ψ_{D_n}. As $K(e, E) \leq K(e) + K(E) + c$, Theorem 3.4(i) gives $L_A(D_n) \leq K(e) + K(E) + c'$; minimise over $e \in [T]$. (c) Consider the machine that emits T and $\alpha_j^{\omega_j}$ for every j. By the hypothesis its theory is consistent on every tape, and T decides no α_j. Its theory proves every datum proved by T, and proves the α_j-data whenever ω agrees with their m labels, which has probability 2^{-m}. The machine has an index computable from those of T and of the family; apply (a). □

Proposition A.11 (Mistake bounds)**.** *(a) Under the hypotheses of Theorem 3.7, the predictor that outputs the most probable of* $1, 0, \iota$ *under* $(\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep})_{D_{t-1}}(\varphi_t)$ *(ties broken arbitrarily; the stored label at a repeat) makes at most* $-\log(\mathbb{U}_A(\mathrm{Ext}T)/M_A(\emptyset)) \leq K_{\equiv}(T) + c$ *mistakes, a mistake being any output other than* b_t*,* ι *included. (b) Fix a well-order of* $\mathrm{dom}\, V$ *that refines*

length. The Kolmogorov predictor *takes at step t the first program p_t with $B \cup \mathsf{W}_{V(p_t)}$ consistent and proving $\psi_{D_{t-1}}$, predicts 1 (resp. 0) if $\mathsf{W}_{V(p_t)}$ proves φ_t (resp. $\neg\varphi_t$), and abstains otherwise. If $k := \sup_n K_A(D_n) < \infty$ (for instance $k \le K_{\equiv}(T) + c$ as in Theorem 3.7), it errs or abstains at most 2^k times. (c) The order hypothesis in (b) is needed: for every m there are a well-order of* $\operatorname{dom} V$ *of type ω and a data set D with $K_A(D) \le K(m) + c \le 2\log m + c'$ on which the Kolmogorov predictor with that order errs or abstains $2^m - 1$ times.*

The bound of (b) is essentially attained even for length-refining orders, on data where the Bayes predictor makes at most $K_A(D) + c$ mistakes (Theorem A.35): it is the selection rule that fails, not the prior.

Proof of (a) and (b). (a) At a repeat the predictor outputs the stored label and makes no mistake. Otherwise the posterior trichotomy sums to 1, so if $M_A(b_t \mid D_{t-1}, \varphi_t) = \mathrm{Bel}_{D_{t-1}}(\varphi_t^{b_t}) > \frac{1}{2}$, then b_t is the unique maximiser. Each mistake therefore costs at least one bit of cumulative loss in the semimeasure convention, and by Theorem 3.7 the cumulative loss is $L_A(D_n) - L_A(\emptyset) \le -\log \mathbb{U}_A(\mathrm{Ext}T) + \log M_A(\emptyset)$. This is the halving argument of Corollary 5.4 of the companion paper (Hänni and Claude (Anthropic), 2026c).

(b) At an error or abstention, p_t does not prove $\varphi_t^{b_t}$, so it is incompatible with D_t and with every later $D_{t'}$, and it is never chosen again. (At a repeat p_t proves the stored label, since it is compatible with D_{t-1}.) Since the order refines length, each p_t is a shortest compatible program, of length $K_A(D_{t-1}) \le k$. Kraft's inequality for the prefix-free set $\operatorname{dom} V$ allows at most 2^k programs of length at most k. □

Part (c) is proved in Appendix A.9, after Theorem A.35.

Theorem A.12 (Merging for three-valued data)**.** *Let G be a theory machine with $\nu_G(\emptyset) = 1$, $\omega \sim \mathbb{U}$ and $A := A_\omega^G$. Let (u_t) be random variables independent of ω, with any joint law, and let the questions be $\varphi_t := s_t(h_{t-1}, u_t)$ for arbitrary measurable rules s_t, where $h_{t-1} = (\varphi_1, y_1, \ldots, \varphi_{t-1}, y_{t-1})$ and $y_t \in Y_3$ is the trichotomy label of φ_t under A. Let $D_t := \{(\varphi_s, y_s) : s \le t\}$ and $d_t := \mathrm{KL}(\nu_G^3(\cdot \mid D_{t-1}, \varphi_t) \,\|\, M_A^3(\cdot \mid D_{t-1}, \varphi_t))$ in bits (0 at a repeat). Then for every n, $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} \log \frac{\nu_G^3(D_n)}{M_A^3(D_n)} + \log M_A(\emptyset) \le K(G) + \log M_A(\emptyset) < K(G)$. Hence $d_t \to 0$ almost surely, and $\sum_t \mathbb{E}\|\nu_G^3(\cdot \mid D_{t-1}, \varphi_t) - M_A^3(\cdot \mid D_{t-1}, \varphi_t)\|_1^2 \le 2 \ln 2 \cdot K(G)$.*

With two-valued data the question rule would have to see A for the two-valued conditionals to be the data law, and the expected bound can fail (as in the remark after Theorem 5.3 of the companion paper); the pathwise bound of Proposition A.10(a) still holds.

Proof. Step 1: the conditional law of the next label. Histories take countably many values. For a value h of the history and a value of $u_{<t}$, the event $\{h_{t-1} = h\}$ is the intersection of an event in $u_{<t}$ (the rules produce the questions of h from its labels) with the event $\{\mathrm{tri}_\omega^G \supseteq D_h\}$ (Proposition 3.1(g)), where D_h is the data set of h. Since $\nu_G(\emptyset) = 1$, the theory is consistent and tri_ω^G is defined on almost every tape. Since (u_t) is independent of ω, given $(h_{t-1}, u_{\le t})$ the tape has the law $\mathbb{U}(\cdot \mid \mathrm{tri}_\omega^G \supseteq D_{t-1})$, and $y_t = \mathrm{tri}_\omega^G(\varphi_t)$ has the law $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$. (Almost surely $\nu_G^3(D_{t-1}) > 0$.)

Step 2: telescoping. Let $Z_t := \log \nu_G^3(y_t \mid D_{t-1}, \varphi_t) - \log M_A^3(y_t \mid D_{t-1}, \varphi_t)$ (0 at a repeat) and $S_n := \sum_{t \le n} Z_t$. Since $\nu_G^3(\emptyset) = 1$ and $M_A^3(\emptyset) = M_A(\emptyset)$, $S_n = \log \nu_G^3(D_n) - \log M_A^3(D_n) + \log M_A(\emptyset)$.

Step 3: integrability. By Theorem 3.4, $M_A^3 \ge 2^{-K(G)} \nu_G^3$, so $S_n \le K(G) + \log M_A(\emptyset)$ pointwise. Since $-\log M_A^3(D_n) \ge 0$, $S_n \ge \log \nu_G^3(D_n) + \log M_A(\emptyset)$. And $\mathbb{E}[-\log \nu_G^3(D_n)] = \sum_{t \le n} \mathbb{E}[H_t] \le n \log 3$, where H_t is the entropy of $\nu_G^3(\cdot \mid D_{t-1}, \varphi_t)$, a distribution on three points: the summands

$-\log \nu^3_G(y_t \mid D_{t-1}, \varphi_t)$ are nonnegative, so the tower property applies by Step 1 (Tonelli). Hence S_n is integrable, and so is each $Z_t = S_t - S_{t-1}$.

Step 4. By Step 1, $\mathbb{E}[Z_t \mid h_{t-1}, u_{\le t}] = d_t$, which is nonnegative because the conditionals of M^3_A are probability distributions (Proposition 3.1(f)). Taking expectations and summing gives $\sum_{t\le n} \mathbb{E} d_t = \mathbb{E} S_n$, which is the identity and the bound. Pinsker's inequality in bits, $\mathrm{KL} \ge \|P - Q\|_1^2/(2\ln 2)$, gives the last claim, and $\mathbb{E}\sum_t d_t < \infty$ gives $d_t \to 0$ almost surely. □

Corollary A.13 (Three-valued loss)**.** *If D_n is three-valued and labelled by the trichotomy of an r.e. T with $B \cup T$ consistent, then in any reveal order the loss in the three-valued convention is $-\log M^3_A(D_n) + \log M_A(\emptyset) \le -\log \mathbb{U}_A[T] \le K_{\equiv}(T) + c$.*

Proof. The tapes with $B \cup A_\omega$ consistent and $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)$ lie in $E^3_{D_n}$. Order independence holds for M^3_A with the proof of Proposition 3.1(d), using (f) in place of (b); then apply Theorem 3.4(i). Here $[T]$ replaces $\mathrm{Ext}T$: three-valued data penalise theories that decide more than T. □

A.6 Coding bounds and the absence of a coding theorem

Lemma A.14 (Literals of a fresh predicate)**.** *In the literal setting: (L1) every set Λ of literals $R(t_w)^b$ with at most one literal per word is consistent with B; (L2) if $B \cup \Lambda$ is consistent, then $B \cup \Lambda \vdash R(t_y)^b$ only if $R(t_y)^b \in \Lambda$. The same holds for finitely many fresh predicates. Moreover a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and B together with such combinations proves another such combination iff they imply it propositionally.*

Proof. (L1) Expand the model of B in which the t_w are pairwise distinct by interpreting R as the set of the elements t_w with $R(t_w) \in \Lambda$; B does not mention R, so it stays true. (L2) If $R(t_y)^{1-b} \in \Lambda$, a proof of $R(t_y)^b$ would make $B \cup \Lambda$ inconsistent. If neither literal of t_y is in Λ, then $\Lambda \cup \{R(t_y)^{1-b}\}$ has at most one literal per word and is consistent by (L1). For the last sentence: every valuation of finitely many fresh atoms is realised in an expansion of the model of B, so a propositional combination of fresh atoms is consistent with B iff it is propositionally satisfiable, and the implication statement follows by applying this to the premises together with the negated conclusion. □

Proof of Proposition 3.8. (a) Let $Q_S(\sigma)$ be the probability, over ω, that the following procedure outputs σ: at stage s, read $\omega_{1:s}$, run s steps of the enumeration of G_U on it, and test all derivations of size at most s from B and the axioms emitted so far, for every member of S in a fixed order; output the first member found. The output events are disjoint open sets, so $\sum_{\sigma\in S} Q_S(\sigma) \le 1$, and $Q_S(\sigma)$ is lsc uniformly in (S, σ). If $B \cup A_\omega$ is consistent and proves σ, it proves no other member of S, so the procedure outputs σ; hence $\mathrm{Bel}_0(\sigma) \le Q_S(\sigma)$. The conditional coding theorem (Section 2.1) gives $K(\sigma \mid S) \le -\log Q_S(\sigma) + c$. (So the coding theorem is applied to Q_S, which is lsc, and not to M_A, which is not; Theorem 3.3.)

(b) Apply (a) with S the label conjunctions on $\mathrm{dom}\, D$, which are pairwise logically incompatible and computable from $\mathrm{dom}\, D$, and $\sigma := \psi_D$, so that $M_A(D) = \mathrm{Bel}_0(\psi_D)$ (Proposition 3.1(c)). From $\mathrm{dom}\, D$ and ψ_D one computes b_D, so $K(b_D \mid \mathrm{dom}\, D) \le K(\psi_D \mid S) + c$.

(c) The first inequality is one term of W_A; the second is Theorem 3.4(iii). For the third, $K_A(D) \le K(\psi_D) + c$ (Remark A.2(c)) $\le K(S) + K(\psi_D \mid S) + c'$, and (a) bounds $K(\psi_D \mid S)$ by $L_A(D) + c$. □

Proposition A.15 (Lower bound)**.** *$L_A(D) \geq K(b_D \mid \operatorname{dom} D) - c$ for every D, where b_D is the vector of two-valued labels in canonical order. So in the semimeasure convention, on fair-coin labels of any fixed n sentences, $\mathbb{E}L_A \geq n - c$.*

Proof. The inequality is Proposition 3.8(b). For the expectation, the weights $2^{-K(b|\operatorname{dom} D)}$ sum to at most 1 (Kraft's inequality), so Gibbs' inequality gives $\mathbb{E}_b K(b \mid \operatorname{dom} D) \geq n$ for uniform $b \in \{0,1\}^n$. With logically dependent sentences, inconsistent labellings have $L_A = \infty$, and the bound holds trivially for them. □

Proof of Theorem 3.9. (i) Let $\ell := \ell_d$, so that $2^\ell < d/\log 3$ and $3^{2^\ell} < 2^d$. There are fewer than 2^ℓ programs of length less than ℓ. Colour each $x \in \{0,1\}^d$ by the vector, over these programs p, of the following symbols: "+" if $B \cup \mathsf{W}_{V(p)}$ is consistent and proves $R(t_x)$; "−" if it is consistent and proves $\neg R(t_x)$; "∗" otherwise. There are at most $3^{2^\ell - 1} < 2^d$ colours, so two distinct words x_1, x_2 share a colour. A program p with $|p| < \ell$ that is compatible with D would colour x_1 "−" and x_2 "+". So every compatible program has length at least ℓ, which gives both claims of (i); and $\ell \geq \log(d/\log 3) - 1 > \log d - 2$, since $\log\log 3 < 0.67$.

(ii) Let $j(x)$ be the position of x in the length-lexicographic order ($j(\epsilon) = 0$), so that $j(x) \leq 2^{d+1} - 2$ for $|x| \leq d$. Let $s := \lceil \log(d+1) \rceil$ and $L := 2^{2^s} \geq 2^{d+1}$. For $u \in \{0,1\}^L$ let $\Lambda_u := \{R(t_x)^{u_{j(x)+1}} : j(x) < L\}$; it is consistent by (L1) of Lemma A.14 and proves $R(t_x)^b$ iff $u_{j(x)+1} = b$, by (L2). A fixed prefix machine reads a V-program for s, computes L, reads L further bits u and outputs an index of Λ_u; by universality of V a fixed string r makes $r\,q\,u \in \operatorname{dom} V$ a program for that index, for q a shortest program for s. These 2^L programs are distinct, form a prefix-free family (since q fixes L), and have length $|r| + K(s) + L$. For a two-point D on distinct words of length at most d, exactly a quarter of the strings u give a compatible Λ_u. So $W_A(D) \geq 2^L \cdot \frac{1}{4} \cdot 2^{-|r|-K(s)-L}$.

(iii) The theory machine $G_{\text{coin}} := \{(w, R(t_x)^{w_{j(x)+1}}) : |w| = j(x)+1\}$ emits $\{R(t_x)^{\omega_{j(x)+1}} : x \in \{0,1\}^*\}$, which is consistent by (L1); its theory proves D iff ω agrees with the labels of D at two positions (L2), which has probability $\frac{1}{4}$. Apply Theorem 3.4.

The consequences: $K_A + \log W_A \geq \ell_d - 2 - K(s) - c$, and $\Omega_V^{\geq \ell}$ is the tail of a convergent series, with $\ell_d \to \infty$. □

The bounds of Theorem 3.7 are stated with $K_{\equiv}(T)$, the least complexity of an index of a theory equivalent to T (Definition 2.8); for whole theories the question is how $K_{\equiv}$ compares with the mixture weight $\mathbf{m}_{\equiv}$. For an r.e. set $Y \subseteq \text{Sent}$ let $K_{\text{enum}}(Y) := \min\{|p| : p \in \operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$ and $\mathbf{m}_{\text{enum}}(Y) := \sum\{2^{-|p|} : p \in \operatorname{dom} V,\ \mathsf{W}_{V(p)} = Y\}$, its enumeration complexity and the corresponding discrete semimeasure. A theorem of Solovay (Solovay, 1977), in the form stated by Vereshchagin (2007), says that $K_{\text{enum}}(Y) \leq 3I(Y) + O(\log I(Y))$ for every r.e. set Y, where $I(Y)$ is the negative logarithm of the probability that a machine with a random input tape enumerates exactly Y.

Proposition A.16 (Coding for whole theories)**.** *(a) For every r.e. T: $K_{\equiv}(T) = K_{\text{enum}}(\mathrm{Cn}T) \pm c$ and $2^{-c}\mathbf{m}_{\text{enum}}(\mathrm{Cn}T) \leq \mathbf{m}_{\equiv}(T) \leq 2^c \mathbf{m}_{\text{enum}}(\mathrm{Cn}T)$. (b) For every r.e. T: $-\log \mathbf{m}_{\equiv}(T) \leq K_{\equiv}(T) \leq 3\,(-\log \mathbf{m}_{\equiv}(T)) + O\big(\log(-\log \mathbf{m}_{\equiv}(T))\big) + c$.*

Proof. (a) Let κ be computable and injective with $\mathsf{W}_{\kappa(e)} = \mathrm{Cn}(\mathsf{W}_e)$ (the s-m-n theorem with padding). Every e' with $\mathsf{W}_{e'} = \mathrm{Cn}T$ lies in $[T]$, so $K_{\equiv}(T) \leq K_{\text{enum}}(\mathrm{Cn}T) + c$ and $\mathbf{m}_{\equiv}(T) \geq 2^{-c}\mathbf{m}_{\text{enum}}(\mathrm{Cn}T)$ (grouping programs by the index they print). Conversely κ maps $[T]$ into $\{e' : \mathsf{W}_{e'} = \mathrm{Cn}T\}$, with $K(\kappa(e)) \leq K(e) + c$ and $\mathbf{m}(\kappa(e)) \geq 2^{-c}\mathbf{m}(e)$, since the image of $\mathbf{m}$ under κ is an lsc semimeasure. (b) The left inequality holds because $2^{-K_{\equiv}(T)}$ is one term of $\mathbf{m}_{\equiv}(T)$. For the right one, the probability that a machine with a random input tape enumerates exactly Y is at

least $2^{-c}\mathbf{m}_{\mathrm{enum}}(Y)$, since a deterministic program is a randomised algorithm that ignores its tape; so Solovay's inequality also holds with $-\log \mathbf{m}_{\mathrm{enum}}(Y)$ in place of $I(Y)$. Apply it to $Y = \mathrm{Cn}T$ and use (a). □

Remark A.17 (Factor two). Vereshchagin (2007) replaces Solovay's constant 3 by 2 for finite sets, and Shekhovtsov and Zakharov (2023) show that a bound of this kind for finite sets transfers to all sets up to logarithmic terms. So the factor 3 in Proposition A.16(b) and in Proposition A.18(c) can be replaced by 2, up to logarithmic terms whose form we do not state.

Whether the random tape helps for whole theories is the question whether a probabilistic enumerator of a set can be much more likely than any short deterministic one. Let $\mathbb{U}_{\mathrm{enum}}(Y) := \mathbb{U}\{\omega : A_\omega = Y\}$, the probability that the universal theory machine emits exactly Y; for sets of strings, let $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}$, $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}$ and $K^{\mathrm{str}}_{\mathrm{enum}}$ be the same quantities for a universal probabilistic enumerator of sets of strings and for programs of V printing indices of r.e. sets of strings.

Proposition A.18 (The random tape and whole theories)**.** *(a) For every r.e. T with $B \cup T$ consistent: $2^{-c}\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \le \mathbb{U}_A[T] \le 2^c\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$, and $\mathbb{U}_A[T] \ge 2^{-c}\mathbf{m}_{\equiv}(T)$. (b) In the literal setting: $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for all consistent r.e. T iff $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y)$ for all r.e. $Y \subseteq \{0,1\}^*$; and $K_{\equiv} \le -\log \mathbf{m}_{\equiv} + O(1)$ iff $K^{\mathrm{str}}_{\mathrm{enum}} \le -\log \mathbf{m}^{\mathrm{str}}_{\mathrm{enum}} + O(1)$. (c) $-\log \mathbb{U}_A[T] \le -\log \mathbf{m}_{\equiv}(T) + c \le K_{\equiv}(T) + c \le 3\,(-\log \mathbb{U}_A[T]) + O\big(\log(-\log \mathbb{U}_A[T])\big) + c$ for every r.e. T with $B \cup T$ consistent.*

Proof. (a) $\{\omega : A_\omega = \mathrm{Cn}T\} \subseteq \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and both sets consist of consistent theories, so $\mathbb{U}_A[T] \ge \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$. Conversely, the machine G_{cl} that runs G_U and emits $\mathrm{Cn}(A_\omega)$ emits exactly $\mathrm{Cn}T$ on $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}T\}$, and the tapes $p_{G_{\mathrm{cl}}}\omega'$, with $p_{G_{\mathrm{cl}}}$ a V-program for an index of G_{cl}, show $\mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T) \ge 2^{-|p_{G_{\mathrm{cl}}}|}\mathbb{U}_A[T]$. The last claim is Theorem 3.4(i).

(b) ($\Leftarrow$) By (a) and Proposition A.16(a), $\mathbb{U}_A[T] \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T)$ and $\mathbf{m}_{\equiv}(T) \asymp \mathbf{m}_{\mathrm{enum}}(\mathrm{Cn}T)$, and coding sentences as strings changes the enumeration quantities by constant factors. ($\Rightarrow$) For $Y \subseteq \{0,1\}^*$ let $T_Y := \{R(t_x) : x \in Y\}$. By (L2) of Lemma A.14, $Y = \{x : R(t_x) \in \mathrm{Cn}(T_Y)\}$. A probabilistic enumerator of exactly Y yields one of exactly $\mathrm{Cn}(T_Y)$ (close under Cn) and conversely (extract the words x), with constant overhead, for probabilistic and for deterministic programs alike; and distinct Y give distinct $\mathrm{Cn}(T_Y)$. So $\mathbb{U}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbb{U}_{\mathrm{enum}}(\mathrm{Cn}T_Y) \asymp \mathbb{U}_A[T_Y]$ and $\mathbf{m}^{\mathrm{str}}_{\mathrm{enum}}(Y) \asymp \mathbf{m}_{\equiv}(T_Y)$; likewise for the complexities.

(c) The first two inequalities are (a) and $\mathbf{m}_{\equiv}(T) \ge 2^{-K_{\equiv}(T)}$. For the last, Solovay's theorem bounds $K_{\mathrm{enum}}(Y)$ by $3I(Y) + O(\log I(Y))$, and $I(Y) = -\log \mathbb{U}_{\mathrm{enum}}(Y) \pm c$ for the universal probabilistic enumerator; apply it to $Y = \mathrm{Cn}T$, and use (a) and Proposition A.16(a). □

Question A.19. Does the random tape help for whole theories? That is, is $\mathbb{U}_A[T] \asymp \mathbf{m}_{\equiv}(T)$ for every consistent r.e. T, and is $K_{\equiv}(T) \le -\log \mathbf{m}_{\equiv}(T) + O(1)$? By Proposition A.18(b) these are, in the literal setting, the questions whether randomness helps to enumerate a set of strings.

The question matters for identification: the limit loss on a complete presentation of T^* is $-\log \mathbb{U}_A[T^*] + \log M_A(\emptyset)$ (Proposition A.27(a)), and Question A.19 asks whether this is $-\log \mathbf{m}_{\equiv}(T^*)$ up to a constant.

A.7 The consistency-requiring variants

Definition A.20 (Graded consistency-requiring weight)**.** For consistent A let $\mathrm{gr}_A(\varphi) := 1$ if $B \cup A \vdash \varphi^{D(\varphi)}$ and $\mathrm{gr}_A(\varphi) := \frac{1}{2}$ otherwise, and

$$\nu^{1/2}_A(D) := \mathbf{1}[B \cup A \cup D^{\pm} \text{ consistent}] \prod_{\varphi \in \mathrm{dom}\, D} \mathrm{gr}_A(\varphi), \qquad M^{1/2}_A(D) := \int_{\mathrm{Cons}} \nu^{1/2}_{A_\omega}(D)\, d\mathbb{U}(\omega).$$

The graded weight is the symmetric form of the graded score S_g of Definition 2.12 of the earlier report with $g \equiv 1$ on proved data and $g_\infty = \frac{1}{2}$: S_g grades only the data labelled true, $\nu^{1/2}$ grades the data labelled false too. The results below use only data labelled true, where the two agree.

Throughout, $H_\omega := B \cup A_\omega \cup D^{\pm}$; when $M_A^{\mathrm{nc}}(D) > 0$, write $\Pr_D := \mathbb{U}(\cdot \mid H_\omega \text{ consistent})$, so that $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \Pr_D[H_\omega \vdash \varphi]$.

Proof of Proposition 3.10. (a) For consistent H, the set $H \cup \{\varphi^b\}$ is consistent iff $H \nvdash \varphi^{1-b}$; so both labels are possible iff H decides neither φ nor $\neg\varphi$, and exactly one is possible otherwise; for inconsistent H neither is. That $M_A^{\mathrm{nc}}(D) = \mathrm{Pl}_0(\psi_D)$ is Proposition A.4. (b) If $A_\omega = \emptyset$, then $H_\omega = B \cup D^{\pm}$ is consistent; and these tapes include those beginning with $p_\emptyset$. For the coin-free and Kolmogorov versions, the program $p_\emptyset$ is a witness on every consistent D. (c) holds by definition. □

Proposition A.21 (The consistency-requiring predictor)**.** *(a) For every consistent D and $\varphi \notin \mathrm{dom}\, D$ with $B \cup D^{\pm} \nvdash \varphi^b$, $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi^b) \le 1 - \varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \le 1 - \varepsilon_\emptyset$. So on every consistent reveal sequence the cumulative loss $\sum_t -\log \mathrm{Bel}_{D_{t-1}}^{\mathrm{nc}}(\varphi_t^{b_t})$ is at least $\log \frac{1}{1-\varepsilon_\emptyset}$ times the number of steps whose datum is not entailed by the earlier data. (b) In the literal setting, let $a_i, s_{i,j}$ $(i, j \ge 1)$ be fresh atoms, computable in (i, j), $\zeta_{i,j} := \neg s_{i,1} \wedge \cdots \wedge \neg s_{i,j-1} \wedge s_{i,j}$, $b_{i,j} := a_i \vee \zeta_{i,j}$, and $c_R := |p_R|$ for a V-program p_R of an index of the deterministic machine emitting $\forall x\, R(x)$. (b1) For $D_j := \{(a_1, 1), (b_{1,j}, 1)\}$ and all but at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j, the consistency-requiring probability of D_j is strictly larger in the order $(a_1, b_{1,j})$ than in the order $(b_{1,j}, a_1)$. (b2) For every n there are $j_1, \ldots, j_n \le \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$ such that, for $D_n := \{(a_i, 1), (b_{i,j_i}, 1) : i \le n\}$, the cumulative loss in the order $(b_{1,j_1}, a_1, b_{2,j_2}, a_2, \ldots)$ exceeds that in the order $(a_1, b_{1,j_1}, a_2, b_{2,j_2}, \ldots)$ by at least $n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$ bits.*

On a stream $(\alpha_j, 1)_{j \ge 1}$ of fresh atoms no datum is entailed, so (a) gives a loss linear in n, while $\{\forall x\, R(x)\}$ proves the stream and the proof-requiring loss is $O(1)$ (Theorem 3.7). In (b) revealing the stronger datum first is more probable; under normalised proof-requiring prediction the weaker datum first is (Theorem A.33).

Proof. (a) The tapes with $A_\omega = \emptyset$, of mass $\varepsilon_\emptyset$, lie in the event $\{H_\omega \text{ consistent}\}$ that normalises $\mathrm{Bel}_D^{\mathrm{nc}}$, and not in its numerator, since $B \cup D^{\pm} \nvdash \varphi^b$.

(b) *Step 1: one pair, after any revealed set.* Let D be consistent with $M_A^{\mathrm{nc}}(D) > 0$, let a, ζ be sentences and $b := a \vee \zeta$; write $\Pr := \Pr_D$ and $H := H_\omega$. Put

$$\kappa := \Pr[H \cup \{b\} \text{ consistent}], \qquad \theta := \Pr[H \cup \{b\} \text{ consistent},\ H \vdash b \to a],$$
$$r := \Pr[H \cup \{\zeta \wedge \neg a\} \text{ consistent}], \quad q := \Pr[H \vdash b,\ H \nvdash a].$$

The consistency-requiring probability of revealing $(a, 1)$ and then $(b, 1)$ after D is $P_{(a,b)} := \mathrm{Bel}_D^{\mathrm{nc}}(a)\mathrm{Bel}_{D+(a,1)}^{\mathrm{nc}}(b) = \Pr[H \vdash a]$, because $a \vdash b$, so that $H \cup \{a\}$ proves b whenever it is consistent. The probability of the other order is $P_{(b,a)} := \mathrm{Bel}_D^{\mathrm{nc}}(b)\mathrm{Bel}_{D+(b,1)}^{\mathrm{nc}}(a) = \Pr[H \vdash b] \cdot \theta/\kappa$: by the deduction theorem $H \cup \{b\} \vdash a$ iff $H \vdash b \to a$, and the ratio of the $\mathbb{U}$-measures of two events inside $\{H \text{ consistent}\}$ equals the ratio of their Pr-measures. Two identities hold. First, $\kappa - \theta = r$: $H \nvdash b \to a$ iff $H \cup \{b, \neg a\}$ is consistent, which implies that $H \cup \{b\}$ is consistent, and $\{b, \neg a\}$ is equivalent to $\{\zeta \wedge \neg a\}$. Second, $\Pr[H \vdash b] = \Pr[H \vdash a] + q$, since $H \vdash a$ implies $H \vdash b$. If $\Pr[H \vdash a] > 0$, then $\theta \ge \Pr[H \vdash a] > 0$ and $\Pr[H \vdash b] > 0$, and substituting,

$$\frac{P_{(a,b)}}{P_{(b,a)}} = \frac{\Pr[H \vdash a]\,(\theta + r)}{(\Pr[H \vdash a] + q)\,\theta} = 1 + \frac{z}{\Pr[H \vdash b]\,\theta}, \qquad z := \Pr[H \vdash a]\, r - q\,\theta \ \ge\ \Pr[H \vdash a]\, r - q.$$

Step 2: disjointness. For fixed i and D, the events $\{H \vdash b_{i,j},\ H \nvdash a_i\}$ are pairwise disjoint in j. Indeed $\zeta_{i,j}$ and $\zeta_{i,j'}$ $(j < j')$ are incompatible, since $\zeta_{i,j}$ asserts $s_{i,j}$ and $\zeta_{i,j'}$ denies it; so $H \vdash a_i \vee \zeta_{i,j}$ and $H \vdash a_i \vee \zeta_{i,j'}$ give $H \vdash a_i \vee (\zeta_{i,j} \wedge \zeta_{i,j'})$, hence $H \vdash a_i$. So $\sum_j q_{i,j} \le 1$, where $q_{i,j}$ is q for the pair $(a_i, b_{i,j})$.

Step 3: lower bounds. Let D be $\emptyset$ (for (b1)) or $D_{i-1} := \{(a_{i'}, 1), (b_{i',j_{i'}}, 1) : i' < i\}$ (for (b2)); then $D^{\pm}$ is implied by $\{a_{i'} : i' < i\}$.

- $r_{i,j} \ge \varepsilon_\emptyset$. On the tapes with $A_\omega = \emptyset$ we have $H = B \cup D^{\pm}$, and $H \cup \{\zeta_{i,j} \wedge \neg a_i\}$ is consistent by (L1) of Lemma A.14: it is implied by the literal set $\{a_{i'} : i' < i\} \cup \{\neg a_i\} \cup \{\text{the literals of } \zeta_{i,j}\}$, which has one literal per word. These tapes have Pr-mass $\varepsilon_\emptyset / M_A^{\mathrm{nc}}(D) \ge \varepsilon_\emptyset$.
- $\Pr[H \vdash a_i] \ge 2^{-c_R}$ for (b2). On the tapes beginning with p_R, $H = B \cup \{\forall x\, R(x)\} \cup D^{\pm}$ is consistent (B does not mention R, so interpret R as the whole domain; $D^{\pm}$ consists of the $a_{i'}$ and disjunctions containing them) and proves a_i; their Pr-mass is $2^{-c_R} / M_A^{\mathrm{nc}}(D) \ge 2^{-c_R}$.
- For (b1), $D = \emptyset$, $\Pr = \mathbb{U}(\cdot \mid \mathrm{Cons})$ and $\Pr[H \vdash a_1] = \mathrm{Bel}_0(a_1)/(1 - \mathrm{Inc}_0) \ge \mathrm{Bel}_0(a_1) > 0$, by the one-axiom theory $\{a_1\}$.

Conclusion. (b1) By Steps 1 and 3, $z_j \ge \mathrm{Bel}_0(a_1)\varepsilon_\emptyset - q_{1,j}$, which is positive unless $q_{1,j} \ge \mathrm{Bel}_0(a_1)\varepsilon_\emptyset$; by Step 2 that happens for at most $1/(\mathrm{Bel}_0(a_1)\varepsilon_\emptyset)$ indices j. For the others $P_{(a,b)}/P_{(b,a)} > 1$. (b2) Choose $j_1, j_2, \ldots$ in turn. At step i, with $D = D_{i-1}$ and $J := \lceil 2^{c_R+1}/\varepsilon_\emptyset \rceil$, Step 2 gives some $j_i \le J$ with $q_{i,j_i} \le 1/J \le 2^{-c_R}\varepsilon_\emptyset/2$, and then $z_i \ge 2^{-c_R}\varepsilon_\emptyset - 2^{-c_R}\varepsilon_\emptyset/2 = 2^{-c_R}\varepsilon_\emptyset/2$ by Step 3. Since $0 < \Pr[H \vdash b]\,\theta \le 1$, $\log(P_{(a,b)}/P_{(b,a)}) \ge \log(1 + z_i) \ge \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. The conditionals of the consistency-requiring predictor depend on the revealed data only as a set, and after $i-1$ completed pairs both orders have revealed D_{i-1}. So the loss of the order with b first minus that of the order with a first is $\sum_{i \le n} \log(P^{(i)}_{(a,b)}/P^{(i)}_{(b,a)}) \ge n \log(1 + 2^{-c_R}\varepsilon_\emptyset/2)$. □

For a single pair the sign of z is not determined by universality; the counting over j in Step 2 is what makes (b) hold for M_A itself. The streams of (b2) are not computable (the j_i depend on M_A); they use at most J candidate atoms per block. The consistency-requiring posterior is order independent, since it depends on D as a set; its predictive probabilities do not chain, because $\mathrm{Bel}^{\mathrm{nc}}_D(\varphi) \ne M_A^{\mathrm{nc}}(D + (\varphi, 1))/M_A^{\mathrm{nc}}(D)$ in general.

An explicit law. For a single law the order effect is easy to see. Let a, χ be fresh atoms and $G := \{(1, a)\}$, whose theory is $\{a\}$ on the tapes beginning with 1 and $\emptyset$ otherwise, and let $D := \{(a, 1), (a \vee \chi, 1)\}$. In the order $(a, a \vee \chi)$ the consistency-requiring probability of D under G is $\frac{1}{2}$: only $\{a\}$ proves a, and then $a \vee \chi$ is entailed. In the order $(a \vee \chi, a)$ it is $\frac{1}{4}$: the first factor is $\frac{1}{2}$, and given $a \vee \chi$ both theories remain consistent and only $\{a\} \cup \{a \vee \chi\}$ proves a, so the second factor is $\frac{1}{2}$.

Proposition A.22 (Graded consistency-requiring weight)**.** *(a) For consistent A, $\nu_A^{1/2}$ is an order-independent weak semimeasure with $\nu_A^{1/2}(\emptyset) = 1$; hence $M_A^{1/2}$ is a weak semimeasure with $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0$, and the conditionals of $M_A^{1/2}/M_A^{1/2}(\emptyset)$ form an order-independent predictor. (b) $M_A^{1/2} \ge M_A$. (c) For fresh atoms p, q and $D := \{(p, 1)\}$, $M_A^{1/2}(D + (p \vee q, 1)) \le M_A^{1/2}(D) - \varepsilon_\emptyset/4$, although $B \cup D^{\pm} \vdash p \vee q$; so $M_A^{1/2}$ is not the law of any theory machine. (d) $M_A(\{(p \wedge q, 1)\}) = M_A(\{(p, 1), (q, 1)\})$, but $M_A^{1/2}(\{(p \wedge q, 1)\}) - M_A^{1/2}(\{(p, 1), (q, 1)\}) \ge \varepsilon_\emptyset/4$. (e) $M_A^{1/2}/M_A^{1/2}(\emptyset)$ is not a containment functional, not even of a random partial function on* Sent *that ignores logic: for every φ, its Möbius sum on the inputs $\varphi, \neg\varphi$ (Proposition A.2 of the companion paper) equals*

$-\frac{1}{2}\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ *decides neither* φ *nor* $\neg\varphi\}$, *which is at most* $-\frac{1}{2}\varepsilon_\emptyset < 0$ *when* B *does not decide* φ*; for* M_A *the same sum is* $\mathbb{U}\{\omega \in \mathrm{Cons} : A_\omega$ *decides neither*$\} \geq 0$.

Proof of Proposition A.22. (a) Let $B \cup A \cup D^{\pm}$ be consistent; otherwise every term is 0. If $B \cup A \vdash \varphi^{b^*}$, then $D + (\varphi, b^*)$ keeps the value (it is consistent with A and has factor 1), and $D + (\varphi, 1 - b^*)$ is inconsistent with A. If A decides neither φ nor $\neg\varphi$, each extension has factor at most $\frac{1}{2}$. Integrating over Cons preserves the inequality, and dividing by $M_A^{1/2}(\emptyset) = 1 - \mathrm{Inc}_0 > 0$ gives a weak semimeasure with value 1 at $\emptyset$, to which Proposition 3.3 of the companion paper applies. (b) For a consistent A that proves ψ_D, $\nu_A^{1/2}(D) = 1$. (c) For each consistent A the new datum keeps consistency (it is entailed) and multiplies the value by 1 or $\frac{1}{2}$; for $A = \emptyset$ the value drops from $\frac{1}{2}$ to $\frac{1}{4}$. (d) Per consistent theory: if it proves $p \wedge q$, both values are 1; if it proves exactly one of p, q and is consistent with the other, both are $\frac{1}{2}$; if it proves neither and is consistent with $p \wedge q$, they are $\frac{1}{2}$ and $\frac{1}{4}$; otherwise both are 0. Integrate; the empty theory contributes at least $\varepsilon_\emptyset/4$ to the difference. (e) The Möbius sum on the inputs $\varphi, \neg\varphi$ at $\emptyset$ is $\nu(\emptyset) - \sum_b \nu(\{(\varphi, b)\}) - \sum_b \nu(\{(\neg\varphi, b)\}) + \sum_{b,b'} \nu(\{(\varphi, b), (\neg\varphi, b')\})$, which we compute per consistent theory A for $\nu = \nu_A^{1/2}$. If A decides φ, say $B \cup A \vdash \varphi^{b^*}$, the singletons consistent with A are (φ, b^*) and $(\neg\varphi, 1 - b^*)$, both proved (value 1), and the only pair consistent with A is $\{(\varphi, b^*), (\neg\varphi, 1 - b^*)\}$ (value 1); the sum is $1 - 1 - 1 + 1 = 0$. If A decides neither, all four singletons are consistent and unproved (value $\frac{1}{2}$ each); of the four pairs, $\{(\varphi, 1), (\neg\varphi, 0)\}$ and $\{(\varphi, 0), (\neg\varphi, 1)\}$ are consistent and unproved (value $\frac{1}{4}$ each), and the other two are inconsistent; the sum is $1 - 2 + \frac{1}{2} = -\frac{1}{2}$. Integrate over Cons; the empty-theory tapes decide nothing that B does not decide. For M_A the same computation gives 0 for a deciding theory and $1 - 0 - 0 + 0 = 1$ for an undeciding one, since an undeciding theory proves no singleton datum on these inputs; so its sum is the undecided mass. By Proposition A.2 of the companion paper, a weak ν with $\nu(\emptyset) = 1$ is a containment functional iff all its Möbius sums are nonnegative. □

Proposition A.23 (The consistency-requiring predictor is a mixture)**.** *For every* D *with* $M_A^{\mathrm{nc}}(D) > 0$ *and every* φ*,*

$$M_A^{\mathrm{nc}}(D)\,\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) = \mathbb{U}(E_D \cap O_\varphi) + \mathbb{U}\{\omega : H_\omega \text{ consistent},\ B \cup A_\omega \nvdash \psi_D,\ H_\omega \vdash \varphi\},$$

and $\mathbb{U}(E_D \cap O_\varphi) = M_A(D)\mathrm{Bel}_D(\varphi)$ *when* $M_A(D) > 0$*. So the consistency-requiring predictor is the mixture, with weight* $M_A(D)/M_A^{\mathrm{nc}}(D)$*, of the proof-requiring predictor and a predictor that uses weak theories together with the data.*

Proof. Split the event $\{H_\omega$ consistent, $H_\omega \vdash \varphi\}$ by whether $B \cup A_\omega \vdash \psi_D$. If it does, H_ω is equivalent to $B \cup A_\omega$: the consistency of H_ω is that of $B \cup A_\omega$, and $H_\omega \vdash \varphi$ iff $B \cup A_\omega \vdash \varphi$. So that part of the event is $E_D \cap O_\varphi$. □

The weak component contains the empty theory, of mass at least $\varepsilon_\emptyset$, which predicts from the data alone; it is the source of Proposition A.21(a).

Proposition A.24 (How much the graded weight can exceed M_A)**.** *(a) For every consistent* D*:* $0 \leq \log\left(M_A^{1/2}(D)/M_A(D)\right) \leq K(\mathrm{dom}\, D) + |D| + c$*. (b) Let* $m \geq 3$ *and* $k \geq 1$*, let* $a_{j,i}$ *(*$j \leq k$*,* $i < m$*) be fresh atoms, and put* $\chi_i^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,i-1} \wedge a_{j,i}$ *for* $1 \leq i < m$ *and* $\chi_m^{(j)} := \neg a_{j,1} \wedge \cdots \wedge \neg a_{j,m-1}$*. For* $\vec{\imath} \in [m]^k$ *let* $D_{\vec{\imath}} := \{(\chi_{i_j}^{(j)}, 1) : j \leq k\}$*. Then some* $\vec{\imath}$ *has*

$$\log \frac{M_A^{1/2}(D_{\vec{\imath}})}{M_A(D_{\vec{\imath}})} \;\geq\; k \log \frac{m}{2} + \log \varepsilon_\emptyset, \qquad |D_{\vec{\imath}}| = k, \qquad K(\mathrm{dom}\, D_{\vec{\imath}}) \leq k \log m + K(k, m) + c.$$

For $m = 3$ the gap is $k \log \frac{3}{2} - c$, linear in $|D|$; for $k = 1$ it is $\log m - 1 - c$ on a single datum. The lower bound of (b) is within $2|D| + K(k, m) + c$ of the upper bound of (a). The source is the per-datum grading of Proposition A.22(d): a datum whose content is a long conjunction costs one bit under the graded weight and its full content under M_A; m pairwise incompatible single data, each worth $\frac{1}{2}$ to the empty theory, have total graded mass $m/2 > 1$.

Proof. (a) The lower bound is Proposition A.22(b). For the upper bound, $M_A^{1/2} \leq 1$, and $M_A(D) \geq 2^{-K(\psi_D)-c}$ by the one-axiom theory $\{\psi_D\}$, with $K(\psi_D) \leq K(\operatorname{dom} D) + |D| + c$, since the labels are $|D|$ bits given $\operatorname{dom} D$. (b) Each $D_{\vec{\imath}}$ is consistent, since each $\chi_i^{(j)}$ is a consistent conjunction of literals and the blocks use disjoint atoms (L1), and B proves none of its data (L2 with $\Lambda = \emptyset$). The empty theory gives $\nu_\emptyset^{1/2}(D_{\vec{\imath}}) = 2^{-k}$, so $M_A^{1/2}(D_{\vec{\imath}}) \geq \varepsilon_\emptyset 2^{-k}$. Two different $\vec{\imath}$ differ in some block, where their sentences are incompatible; so the m^k conjunctions $\psi_{D_{\vec{\imath}}}$ are pairwise incompatible, $\sum_{\vec{\imath}} M_A(D_{\vec{\imath}}) \leq M_A(\emptyset) \leq 1$, and some $\vec{\imath}$ has $M_A(D_{\vec{\imath}}) \leq m^{-k}$. Its gap is at least $\log(\varepsilon_\emptyset 2^{-k} m^k)$. Finally $\operatorname{dom} D_{\vec{\imath}}$ is computable from (k, m) and the position of $\vec{\imath}$ among the m^k values. □

A.8 Limits and identification

The following lemma is the averaging argument associated with de Leeuw, Moore, Shannon and Shapiro (cf. de Leeuw et al., 1956), in the form we need. It is used again in Section 5.

Lemma A.25 (Majority vote)**.** *Let Z be a decidable set of sentences, M an oracle machine that on tape ω and input $\sigma \in Z$ outputs $\mathsf{M}^\omega(\sigma) \in \{0,1\}$ or diverges, $S \subseteq Z$, $\kappa : S \to \{0,1\}$, and $\mathcal{G}$ a Borel set of tapes with $\mathbb{U}(\mathcal{G}) > 0$ such that for every $\omega \in \mathcal{G}$, $\mathsf{M}^\omega(\sigma)$ is defined for all $\sigma \in Z$ and equals $\kappa(\sigma)$ for $\sigma \in S$. Then some total computable $h : Z \to \{0,1\}$ extends κ.*

Proof. By outer regularity there is an open $U \supseteq \mathcal{G}$ with $\mathbb{U}(U) < \frac{4}{3}\mathbb{U}(\mathcal{G})$; writing U as a disjoint union of cylinders, some cylinder $[u]$ has $\mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $b \in \{0,1\}$ let $Q_b(\sigma) := \{\omega \in [u] : \mathsf{M}^\omega(\sigma) = b\}$. These classes are effectively open uniformly in σ (a halting computation reads a finite prefix) and disjoint for $b = 0, 1$, so their measures are left-c.e., with computable approximations from below. Let $h(\sigma)$ be the b with the larger approximated measure at the first stage at which the two approximations sum to more than $\frac{1}{2}\, 2^{-|u|}$. That stage exists for every $\sigma \in Z$, because $\mathbb{U}(Q_0(\sigma) \cup Q_1(\sigma)) \geq \mathbb{U}(\mathcal{G} \cap [u]) > \frac{3}{4}\, 2^{-|u|}$. For $\sigma \in S$ the wrong class $Q_{1-\kappa(\sigma)}(\sigma)$ lies in $[u] \setminus \mathcal{G}$, of measure less than $\frac{1}{4}\, 2^{-|u|}$; so at that stage the approximation of the right class exceeds $\frac{1}{4}\, 2^{-|u|}$, more than that of the wrong one. □

Throughout the rest of this subsection D_∞ is a consistent infinite stream, $E_n := E_{D_n}$ (a decreasing sequence), $E_\infty := \bigcap_n E_n = \{\omega : B \cup A_\omega \text{ consistent, } \operatorname{Cn}(A_\omega) \supseteq D_\infty^{\pm}\}$, $a_n := \mathbb{U}(E_n)$, $a := \mathbb{U}(E_\infty)$ and $P_n := \mathbb{U}(\cdot \mid E_n)$.

Proposition A.26 (The limit posterior)**.** *(a) If $a > 0$, then $\sup_{\mathcal{X}} |P_n(\mathcal{X}) - P_\infty(\mathcal{X})| \leq (a_n - a)/a_n \to 0$, the supremum over Borel sets of tapes, with $P_\infty := \mathbb{U}(\cdot \mid E_\infty)$. Hence $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathrm{Bel}_\infty(\chi)| \to 0$ with $\mathrm{Bel}_\infty(\chi) := P_\infty(O_\chi)$, likewise for* Dis *and* Indep*, and the cumulative loss in the semimeasure convention converges to $-\log(a/M_A(\emptyset))$. (b) If $a = 0$, then $M_A(D_n) \to 0$ and the cumulative loss tends to ∞. (c) $a \geq \mathbb{U}_A(\mathrm{Ext}T) \geq 2^{-K_{\equiv}(T)-c}$ whenever some r.e. T with $B \cup T$ consistent proves every datum of D_∞. (d) For every set Γ of sentences with $\operatorname{Cn}(\Gamma) = \operatorname{Cn}(D_\infty^{\pm})$, complete or not, $E_\infty = \{\omega : B \cup A_\omega$ consistent, $\operatorname{Cn}(A_\omega) \supseteq \operatorname{Cn}(\Gamma)\}$; so, if $a > 0$, P_∞ is the prior restricted to the tapes whose theory is a consistent extension of Γ.*

Proof. (a) For a Borel $\mathcal{X}$ let $x := \mathbb{U}(\mathcal{X} \cap E_\infty) \le a$ and $y_n := \mathbb{U}(\mathcal{X} \cap (E_n \setminus E_\infty)) \le a_n - a$. Then

$$\Big|\frac{x+y_n}{a_n} - \frac{x}{a}\Big| = \frac{|y_n a - x(a_n - a)|}{a\, a_n} \le \frac{\max(y_n a,\, x(a_n-a))}{a\, a_n} \le \frac{a_n - a}{a_n},$$

and $a_n \downarrow a$ by continuity from above. The loss is $L_A(D_n) - L_A(\emptyset) = -\log(a_n/M_A(\emptyset))$ by Proposition 3.1(d). (b) Continuity from above. (c) As in the proof of Theorem 3.7. (d) A theory has $D^{\pm}_\infty$ among its consequences iff it contains $\mathrm{Cn}(D^{\pm}_\infty) = \mathrm{Cn}(\Gamma)$. □

Proposition A.27 (Limits)**.** *Let T^* be consistent and r.e. (a) If $B \cup T^*$ is complete (hence decidable) and $\mathrm{Cn}(D^{\pm}_\infty) = \mathrm{Cn}(T^*)$, then $P_n\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \to 1$, $\sup_\chi |\mathrm{Bel}_{D_n}(\chi) - \mathbf{1}[T^* \vdash \chi]| \to 0$, $\sup_\chi \mathrm{Indep}_{D_n}(\chi) \to 0$, and the cumulative loss converges to $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \le K_{\equiv}(T^*) + c$. If T^* is axiomatised by one sentence τ and $(\tau, 1) \in D_n$, the first probability is 1 from then on. (b) If $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_\infty) = \mathrm{Cn}(T^*)$, then P_∞ is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$:*

$$\lim_n \mathrm{Bel}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1), \qquad \lim_n \mathrm{Dis}_{D_n}(\psi) = \frac{\mathbb{U}_A(\mathrm{Ext}(T^* + \neg\psi))}{\mathbb{U}_A(\mathrm{Ext}T^*)} \in (0,1),$$
$$\lim_n \mathrm{Indep}_{D_n}(\psi) \ge \frac{\mathbb{U}_A[T^*]}{\mathbb{U}_A(\mathrm{Ext}T^*)} > 0.$$

The most probable limit class can be a strict extension: in the literal setting, with fresh atoms α_i on a computable sequence of distinct words, $U := \{\alpha_i : i \in \mathbb{N}\}$ and $T_j := U \setminus \{\alpha_j\}$, for all but finitely many j the limit posterior on data presenting T_j gives the class of U more probability than the class of T_j; a most probable class exists, and it strictly extends T_j. (c) If every sentence appears with its T^*-trichotomy label, then $\mathbb{U}(\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\} \mid E^3_{D_n}) \to 1$.

Proof of Proposition A.27. (a) A complete consistent theory has no proper consistent extension, so $E_\infty = \{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ by Proposition A.26(d), and this event has positive measure by Theorem 3.4(i). On it the trichotomy is $\mathbf{1}[T^* \vdash \chi]$ for all χ at once. Apply Proposition A.26(a); the limit loss is $-\log(\mathbb{U}_A[T^*]/M_A(\emptyset)) \le K_{\equiv}(T^*) + c$ by Theorem 3.4(i). If T^* is axiomatised by τ and $(\tau, 1) \in D_n$, then every consistent theory proving ψ_{D_n} contains $\mathrm{Cn}(T^*)$, so $E_n = E_\infty$.

(b) E_∞ is as stated by Proposition A.26(d), and $\lim_n \mathrm{Bel}_{D_n}(\psi) = P_\infty(O_\psi)$ by Proposition A.26(a), likewise for Dis and Indep. Since $T^* + \psi$, $T^* + \neg\psi$ and T^* are consistent r.e. theories, all three ratios are positive by Theorem 3.4(i); each of the first two is less than 1 because the other is positive and the two events are disjoint; and the tapes with $\mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)$ lie in the independent part. For the example: $T_j \nvdash \alpha_j$ by (L2) of Lemma A.14, while $T_{j'} \vdash \alpha_j$ for $j' \ne j$; so the classes $[T_j]$ are pairwise distinct and their events disjoint, and $\sum_j \mathbb{U}_A[T_j] \le 1$, so $\mathbb{U}_A[T_j] \to 0$. U is r.e. with an index of complexity $O(1)$ and is consistent by (L1), so $\mathbb{U}_A[U] \ge 2^{-c}$. The limit posterior odds of $[U]$ against $[T_j]$ are $\mathbb{U}_A[U]/\mathbb{U}_A[T_j]$, which exceed 1 for all but finitely many j. *A most probable class exists:* under P_∞ the classes partition E_∞, so only finitely many classes have limit probability above the positive number $\mathbb{U}_A[U]/\mathbb{U}_A(\mathrm{Ext}T_j)$, which is the limit probability of $[U]$; a maximiser exists among them, lies in $\mathrm{Ext}(T_j)$, and is not $[T_j]$, which $[U]$ beats; so it is a strict extension of T_j.

(c) A consistent theory that proves every theorem of T^* and decides no sentence independent of $B \cup T^*$ has the same consequences as T^*: a theorem $\sigma \notin \mathrm{Cn}(T^*)$ of it would be refutable from T^* (then the theory is inconsistent) or independent of T^* (then σ is labelled ι and decided). So the three-valued analogue of E_∞ is $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$, of positive measure by Theorem 3.4(i), and the proof of Proposition A.26(a) applies to M^3_A unchanged. □

Since $\sum_j \mathbb{U}_A[T_j] \le 1$, the same disjointness argument bounds $\mathbb{U}_A[T]$ for all but finitely many members of any infinite family of pairwise inequivalent theories with a common simple consistent extension. Two-valued data cannot do better than Proposition A.27(b), whatever the learner. Call a sequence (σ_t) of sentences a *text* for T if $\mathrm{Cn}(\{\sigma_t : t \ge 1\}) = \mathrm{Cn}(T)$; it presents T by the two-valued data $(\sigma_t, 1)$, a datum $(\varphi, 0)$ being the datum $(\neg\varphi, 1)$. A function F from finite sequences of sentences to deductive classes *identifies* $[T]$ on a text if its values on the prefixes of the text are eventually $[T]$.

Proposition A.28 (Text against informant)**.** *In the literal setting, let α_i be fresh atoms on a computable sequence of distinct words, $\Lambda_k := \{\alpha_i : i < k\}$ and $\Lambda_\infty := \{\alpha_i : i \in \mathbb{N}\}$. No function F from finite sequences of sentences to deductive classes, computable or not, identifies on every text every member of $\{[\Lambda_k] : k \ge 1\} \cup \{[\Lambda_\infty]\}$. In particular no learner identifies all consistent r.e. theories from two-valued data, while three-valued data identify every consistent r.e. T^* in the limit posterior (Proposition A.27(c)), non-computably.*

Proof. Build a text for Λ_∞ in stages. Stage $k \ge 1$ appends α_{k-1} repeatedly until F outputs $[\Lambda_k]$. This happens, because the sequence built so far, continued by α_{k-1} forever, is a text for Λ_k, on which F eventually outputs $[\Lambda_k]$. The resulting infinite sequence contains every α_i, so it is a text for Λ_∞; and F outputs $[\Lambda_k] \ne [\Lambda_\infty]$ for infinitely many k, since $\Lambda_k \nvdash \alpha_k$ by (L2) of Lemma A.14. □

This is Gold's distinction between learning from text and from an informant (cf. Gold, 1967): a class that contains all finite languages and one infinite language is not identifiable from text, by the same locking argument. So the failure of two-valued SAI to identify incomplete truths is not an artefact of its prior.

Proposition A.29 (Computable structures; the basic ω-gap)**.** *Let $\mathfrak{M}$ be a computable structure in which every element is the value of a closed term, let $\mathrm{Diag}(\mathfrak{M})$ be its atomic diagram (the closed atomic sentences and negations of closed atomic sentences true in $\mathfrak{M}$; a decidable set), and let the data list $\mathrm{Diag}(\mathfrak{M})$ in a computable order, labelled true. (a) If B is true in $\mathfrak{M}$, then $L_A(D_n) \le K_{\equiv}(\mathrm{Diag}\,\mathfrak{M}) + c \le K(\mathfrak{M}) + c$, with $K(\mathfrak{M})$ the complexity of an index of $\mathfrak{M}$. (b) P_∞ is the prior restricted to the consistent extensions of $\mathrm{Diag}(\mathfrak{M})$. For $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $\mathfrak{M} = (\mathbb{N}, 0, S, +, \cdot, <)$, $B = \emptyset$ (or any B true both in $\mathbb{N}$ and in the model $\mathbb{N} \sqcup \{\star\}$ of the proof) and $\vartheta := \forall x\,(x + 0 = x)$, the limits $\lim_n \mathrm{Bel}_{D_n}(\vartheta)$ and $\lim_n \mathrm{Dis}_{D_n}(\vartheta)$ both lie in $(0, 1)$: instances never settle the universal sentence. (If instead $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, then $B \vdash \vartheta$ and $\lim_n \mathrm{Bel}_{D_n}(\vartheta) = 1$.) (c) If $\mathrm{Th}(\mathfrak{M})$ is decidable and the data present it, Proposition A.27(a) applies.*

Proof. (a) Theorem 3.7 with $T = \mathrm{Diag}(\mathfrak{M})$, which is consistent with B, proves every datum, and has an index computable from one of $\mathfrak{M}$. (b) Proposition A.27(b) with $\psi = \vartheta$; it needs $\mathrm{Diag}(\mathbb{N}) \cup \{\vartheta\}$ and $\mathrm{Diag}(\mathbb{N}) \cup \{\neg\vartheta\}$ to be consistent with B. The first holds in $\mathbb{N}$. For the second take $\mathbb{N} \sqcup \{\star\}$ with $S(\star) := \star$, $\star + y := 0$ for every y (including $y = \star$), $y + \star := \star$ for $y \in \mathbb{N}$, $\star \cdot y = y \cdot \star := \star$ for every y, and $\star$ incomparable under $<$ with everything. Closed terms take values in $\mathbb{N}$, which is closed under the operations, so the diagram holds; and $\star + 0 = 0 \ne \star$. □

This is the one-tape form of the coin-free ω-gap of Theorem 3.9 and Proposition B.8 of the earlier report; Section 5 studies its anatomy.

Proposition A.30 (Complete truths that are not decidable)**.** *(a) If $\mathrm{Cn}(D^{\pm}_\infty)$ is complete and not decidable (for example $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, and every sentence labelled by its truth in $\mathbb{N}$), then $a = 0$: $M_A(D_n) \to 0$, and the cumulative loss in the semimeasure convention is*

unbounded. (b) If (σ_t) is a computable enumeration of all sentences and D_n labels $\sigma_1, \dots, \sigma_n$ by a complete consistent theory, with label vector $b_{1:n}$, then $K(b_{1:n} \mid n) - c \le L_A(D_n) \le n + K(n) + c$.

Proof. (a) On E_∞ the theory is consistent and contains the complete set $\mathrm{Cn}(D^{\pm}_\infty)$, so it has exactly these consequences. Let $\mathsf{M}^\omega(\sigma)$ be 1 or 0 according as σ or $\neg\sigma$ is found first to be provable from $B \cup A_\omega$. On E_∞ it is total and decides $\mathrm{Cn}(D^{\pm}_\infty)$. If $a > 0$, Lemma A.25, with $Z = S :=$ Sent, $\mathcal{G} = E_\infty$ and κ the characteristic function of $\mathrm{Cn}(D^{\pm}_\infty)$, gives a decision procedure for it, a contradiction. ($\mathrm{Th}(\mathbb{N})$ is not decidable, for instance by the inseparability step in the proof of Proposition A.31.) (b) The lower bound is Proposition 3.8(b), since $\mathrm{dom}\, D_n$ is computable from n. For the upper bound, $K_A(D_n) \le K(\psi_{D_n}) + c \le K(n) + n + c'$, since given n the labels are n bits; then use $L_A \le K_A + c$ (Theorem 3.7). □

The coin-free analogues are Proposition 7.11 and Lemma E.11 of the earlier report; Section 5 gives the growth rate on $\mathrm{Th}(\mathbb{N})$.

For the next proposition, F is a function machine in the sense of Definition 3.6 of the companion paper on the inputs $\mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ with labels $\{0,1\}$, and f^F_ω its random partial labeller. We use two standard facts about Q (see Boolos et al., 2007): it strongly represents every computable set, and it satisfies the diagonal lemma.

Proposition A.31 (Random theories above Q are almost surely incomplete)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$. (a) For every theory machine G, the set of tapes ω such that $B \cup A^G_\omega$ is consistent, proves every axiom of Q, and decides every $\mathcal{L}_{\mathrm{ar}}$-sentence has measure 0. In particular, if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, the consistent random theory of M_A is almost surely incomplete, and every posterior gives probability 0 to complete theories. (b) For every function machine F as above: $\mathbb{U}\{\omega : f^F_\omega$ is total, labels every Q-provable sentence 1 and every Q-refutable sentence $0\} = 0$; likewise with PA, or any consistent r.e. extension of Q, in place of Q.*

Proof. Inseparability. No total computable $h : \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}} \to \{0,1\}$ is 1 on the theorems of Q and 0 on its refutable sentences. Otherwise the set $\{\ulcorner\sigma\urcorner : h(\sigma) = 1\}$ is strongly represented in Q by some formula $\rho_h(x)$, and the diagonal lemma gives σ with $\mathsf{Q} \vdash \sigma \leftrightarrow \neg\rho_h(\ulcorner\sigma\urcorner)$. If $h(\sigma) = 1$, then $\mathsf{Q} \vdash \rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \neg\sigma$, and h is 1 on a refutable sentence. If $h(\sigma) = 0$, then $\mathsf{Q} \vdash \neg\rho_h(\ulcorner\sigma\urcorner)$, so $\mathsf{Q} \vdash \sigma$, and h is 0 on a theorem. A separator for a consistent r.e. $T \supseteq \mathsf{Q}$ separates the pair of Q too.

(a) On the event, $\mathsf{M}^\omega(\sigma)$, the first of $1, 0$ found according as σ or $\neg\sigma$ is provable from $B \cup A^G_\omega$, is total on $\mathcal{L}_{\mathrm{ar}}$-sentences and separates the theorems of Q from its refutable sentences: a consistent theory that proves Q proves every theorem of Q and refutes every refutable sentence. If the event had positive measure, Lemma A.25 (with $Z := \mathrm{Sent}_{\mathcal{L}_{\mathrm{ar}}}$ and S the sentences Q decides) would give a computable separator. (b) The same, with $\mathsf{M}^\omega(\sigma) := f^F_\omega(\sigma)$, computed by running F on σ with tape ω. □

The proposition is a new application of the fact that separating sets of effectively inseparable pairs have measure zero (for effective inseparability, cf. Smullyan, 1961); Lemma A.25 makes the proof self-contained. Part (b) says that a random tape does not help a labeller to get every Q-decided sentence right.

Corollary A.32 (Local defects above Q)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let G be a theory machine with $\nu_G(\emptyset) > 0$ whose random theory proves Q almost surely on Cons^G (for instance, any G when $\mathsf{Q} \subseteq \mathrm{Cn}(B)$). Then ν_G has positive local defect at $D = \emptyset$ at some $\mathcal{L}_{\mathrm{ar}}$-sentence φ: $\nu_G(\{(\varphi, 1)\}) + \nu_G(\{(\varphi, 0)\}) < \nu_G(\emptyset)$. In particular, above Q there is no theory machine whose law is "a theory plus a fair coin on every sentence it leaves undecided" with zero local defect everywhere.*

Proof. By Proposition 3.1(b) at $D = \emptyset$, the defect at φ is $\nu_G^3(\{(\varphi, \iota)\})$, the measure of the set of $\omega \in \mathrm{Cons}^G$ such that $B \cup A_\omega^G$ decides neither φ nor $\neg\varphi$. If it were 0 for every $\mathcal{L}_{\mathrm{ar}}$-sentence, then, by a countable union, almost every $\omega \in \mathrm{Cons}^G$ would have a theory that decides every $\mathcal{L}_{\mathrm{ar}}$-sentence; with the hypothesis, the event of Proposition A.31(a) would have measure $\mathbb{U}(\mathrm{Cons}^G) = \nu_G(\emptyset) > 0$, a contradiction. □

A.9 Order dependence and the Kolmogorov predictor

In the normalised convention (Definition 2.5) each conditional of M_A is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$, so the loss of a reveal sequence is no longer a function of the final data set.

Theorem A.33 (Order dependence in the normalised convention)**.** *(a) Let a, χ be sentences such that $B \cup \{a\}$ is consistent, and $B \cup \{\neg a\}$ is consistent and decides neither χ nor $\neg\chi$. Let $D := \{(a, 1), (a \vee \chi, 1)\}$, and for $x, y \in Y_3$ let $p_{xy} := \mathbb{U}\{\omega \in \mathrm{Cons} : \mathrm{tri}_\omega(a) = x,\ \mathrm{tri}_\omega(a \vee \chi) = y\}$. Then $p_{10} = p_{1\iota} = p_{\iota 0} = 0$, and the normalised probabilities q_1 and q_2 of D in the orders $(a, a \vee \chi)$ and $(a \vee \chi, a)$ satisfy*

$$\frac{q_2}{q_1} = 1 + \frac{(p_{11} + p_{01})\, p_{0\iota} + p_{\iota 1}\, (p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})\, (p_{11} + p_{01})} \ > \ 1.$$

So the normalised cumulative loss of M_A on D is strictly smaller in the order $(a \vee \chi, a)$, while the loss in the semimeasure convention is the same in both orders. (b) Let a_i, a_i' $(i \geq 1)$ be fresh atoms on pairwise distinct words. For $y \in \{0,1\}^n$ put $\alpha_i := a_i^{y_i}$, $\beta_i := \alpha_i \vee a_i'$ and $D_n(y) := \{(\alpha_i, 1), (\beta_i, 1) : i \leq n\}$, and compare the orders $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$ and $(\beta_1, \alpha_1, \beta_2, \alpha_2, \dots)$. For every y the normalised loss of the first order exceeds that of the second by at least $\sum_{i \leq n} \log(1 + v_i(y)) \geq 0$, where $v_i(y)$ is the posterior probability, after the first $i - 1$ pairs, that the random theory refutes α_i and does not decide a_i'. The average of this difference over $y \in \{0,1\}^n$ is at least $2^{-c_1 - 1} n$, where c_1 is the length of a V-program for the machine $G_{\mathrm{ca}} := \{(w, a_j^{w_j}) : |w| = j\}$ (fair coins on the atoms a_j). So for every n some data set of $2n$ data has two reveal orders whose normalised losses differ by at least $2^{-c_1 - 1} n$ bits.

Proof. (a) If the random theory proves a, it proves $a \vee \chi$; if it refutes $a \vee \chi$, it refutes a. This gives the three zero cells. In the order $(a, a \vee \chi)$ the first normalised factor is $\mathrm{Bel}_\emptyset(a)/(\mathrm{Bel}_\emptyset(a) + \mathrm{Dis}_\emptyset(a)) = p_{11}/(p_{11} + p_{01} + p_{00} + p_{0\iota})$, and the second is 1, since $a \vee \chi$ is then entailed (Proposition 3.1(c)). In the order $(a \vee \chi, a)$ the first factor is $(p_{11} + p_{01} + p_{\iota 1})/(p_{11} + p_{01} + p_{\iota 1} + p_{00})$; given $(a \vee \chi, 1)$ the posterior is the prior restricted to $\mathrm{tri}_\omega(a \vee \chi) = 1$, on which $\mathrm{Bel}(a) \propto p_{11}$ and $\mathrm{Dis}(a) \propto p_{01}$, so the second factor is $p_{11}/(p_{11} + p_{01})$. (The normalising constant $1 - \mathrm{Inc}_0$ cancels in each factor.) Dividing,

$$\frac{q_2}{q_1} = \frac{(p_{11} + p_{01} + p_{\iota 1})(p_{11} + p_{01} + p_{00} + p_{0\iota})}{(p_{11} + p_{01} + p_{\iota 1} + p_{00})(p_{11} + p_{01})},$$

and the numerator minus the denominator is $(p_{11} + p_{01})p_{0\iota} + p_{\iota 1}(p_{00} + p_{0\iota})$. For strictness: $p_{11} > 0$ (the tapes beginning with a program for $\{a\}$), and $p_{0\iota} > 0$ (the tapes beginning with a program for $\{\neg a\}$, which are consistent, refute a, and do not decide $a \vee \chi$, because $B \cup \{\neg a\} \vdash a \vee \chi$ iff $B \cup \{\neg a\} \vdash \chi$, and $B \cup \{\neg a\} \vdash \neg(a \vee \chi)$ iff $B \cup \{\neg a\} \vdash \neg\chi$).

(b) After the first $i - 1$ pairs both orders have revealed the same set $D_{i-1}(y)$, and the normalised conditionals depend only on the current set and the queried sentence. So the difference is the sum over i of $\log(q_2^{(i)}/q_1^{(i)})$, computed as in (a) under the posterior $P_{i-1} := \mathbb{U}(\cdot \mid E_{D_{i-1}(y)})$ in place of the prior restricted to Cons; the zero cells of (a) hold for every consistent theory. Write

$m_1 := p_{11}+p_{01}$, $m_2 := p_{\iota 1}$, $m_3 := p_{00}$ and $m_4 := p_{0\iota}$ for the cell masses under P_{i-1}; they sum to at most 1, and $m_1 > 0$ (the tapes of G_{ca} that agree with y up to i). The ratio is $1+[m_1m_4+m_2(m_3+m_4)]/[(m_1+m_2+m_3)m_1] \geq 1+m_4/(m_1+m_2+m_3) \geq 1+m_4 = 1+v_i(y)$. Next, $\log(1+v) \geq v$ for $v \in [0,1]$. The tapes $p\,\omega'$, with p the program for G_{ca}, $\omega'_{<i} = y_{<i}$ and $\omega'_i \neq y_i$, lie in $E_{D_{i-1}(y)}$ (their theory proves α_j, hence β_j, for $j < i$), refute α_i, and do not decide a'_i (by (L2) of Lemma A.14); they have measure 2^{-c_1-i}. So $v_i(y) \geq 2^{-c_1-i}/M_A(D_{i-1}(y))$. Over uniform y, Jensen's inequality gives $\mathbb{E}[1/M_A(D_{i-1})] \geq 1/\mathbb{E}[M_A(D_{i-1})]$. Moreover $M_A(D_{i-1}(y)) = \mathrm{Bel}_0(\bigwedge_{j<i}\alpha_j)$, and these conjunctions are pairwise incompatible for distinct $y_{<i}$, so $\sum_{y_{<i}} M_A(D_{i-1}(y)) \leq 1$ and $\mathbb{E}[M_A(D_{i-1})] \leq 2^{-(i-1)}$. Hence $\mathbb{E}\,v_i \geq 2^{-c_1-1}$; sum over $i \leq n$. □

The mechanism is the failure of the two-point criterion of Proposition A.7 of the companion paper: whether the random theory decides $a \vee \chi$ depends on its answer to a, since a theory proving a always decides $a \vee \chi$ and one refuting a need not. The witnesses use logical structure: in one order the second datum is entailed. On logically independent data the question stays open.

Question A.34. Is the normalised loss of M_A order dependent on logically independent data, for instance on positive literals of a fresh predicate? This is Question A.8 of the companion paper in another guise: on such data $M_A \asymp M_F$ (Theorem 4.4), but constant factors do not control normalised conditionals.

The single shortest compatible theory can be a poor predictor where the mixture is good. Let $a_1, \ldots, a_m$ be fresh atoms, and for $w \in \{0,1\}^m$ let $\mathrm{desc}(w) := \bigwedge_{i \leq m} a_i^{w_i}$.

Theorem A.35 (The Kolmogorov predictor can be exponentially worse than Bayes)**.** *Fix any well-order $\prec$ of $\mathrm{dom}\,V$, and let the Kolmogorov predictor of Proposition A.11(b) use the $\prec$-first compatible program. There are a data set D of 2^m-1 sentences, all of the form $\neg\mathrm{desc}(v)$ labelled true, and a reveal order of D, such that: (i) the Kolmogorov predictor errs or abstains at every one of the 2^m-1 steps; (ii) $K_A(D) \leq m + K(m) + c$, and the Bayes predictor of Proposition A.11(a) makes at most $K_A(D)+c \leq m + K(m) + c'$ mistakes on the same sequence. If $\prec$ refines length, then $m \leq K_A(D) \leq m + K(m) + c$; so with $k := K_A(D)$ the number of errors or abstentions is at least $2^{k-K(m)-c} - 1 \geq 2^{k-2\log k - c'} - 1$, while the Bayes predictor makes at most $k + c$ mistakes. The upper bound of Proposition A.11(b) is 2^k.*

Proof. For each w let p_w be the $\prec$-first $p \in \mathrm{dom}\,V$ with $B \cup \mathsf{W}_{V(p)}$ consistent and proving $\mathrm{desc}(w)$. It exists, since $\{\mathrm{desc}(w)\}$ is consistent (Lemma A.14) and $\prec$ is a well-order. A consistent theory proves at most one $\mathrm{desc}(w)$, so $w \mapsto p_w$ is injective. Let w^* be the w whose p_w is $\prec$-last among these 2^m programs. Starting from $D = \emptyset$, the adversary repeats the following as long as the $\prec$-first program p compatible with the current data does not prove $\mathrm{desc}(w^*)$.

- p is consistent and $B \vdash \bigvee_v \mathrm{desc}(v)$, so the set $\mathrm{Open}(p)$ of v with $B \cup \mathsf{W}_{V(p)} \cup \{\mathrm{desc}(v)\}$ consistent is nonempty. It contains some $v \neq w^*$: otherwise p proves $\neg\mathrm{desc}(v)$ for all $v \neq w^*$, hence $\mathrm{desc}(w^*)$.
- Present $(\neg\mathrm{desc}(v), 1)$ for such a v.

Then: (1) every presented sentence is true in the valuation w^*, so the data stay consistent and $\{\mathrm{desc}(w^*)\}$ proves them all. (2) p does not prove $\neg\mathrm{desc}(v)$, since v is open for p; so the predictor errs (if p proves $\mathrm{desc}(v)$) or abstains. (3) The sentence $\neg\mathrm{desc}(v)$ was not presented before, since p is compatible with the earlier data. (4) For $w \neq w^*$, p_w proves $\neg\mathrm{desc}(v)$ for every $v \neq w$, so it stays compatible until $\neg\mathrm{desc}(w)$ is presented. While some p_w with $w \neq w^*$ is compatible, the $\prec$-first compatible program is $\preceq p_w \prec p_{w^*}$, and, being consistent and $\prec$-before p_{w^*}, it does not

prove desc(w^*); so the loop continues. By (3) the loop runs at most $2^m - 1$ times, and by (4) it does not stop before every $\neg\mathrm{desc}(w)$, $w \neq w^*$, has been presented. So it runs exactly $2^m - 1$ times, and by (2) every step is an error or an abstention.

(ii) $\bigwedge_{v \neq w^*} \neg\mathrm{desc}(v)$ is equivalent to desc(w^*), so $K_A(D) \leq K(\mathrm{desc}(w^*)) + c \leq m + K(m) + c'$. For the Bayes predictor, apply Proposition A.11(a) with $T := \mathsf{W}_{V(p)}$ for a shortest program p compatible with D: $K_{\equiv}(T) \leq |p| + c = K_A(D) + c$, and T proves every datum.

If $\prec$ refines length, every program compatible with D proves desc(w^*), and the $\prec$-first of them is p_{w^*}, so $K_A(D) = |p_{w^*}|$. The 2^m distinct programs p_w all have length at most $|p_{w^*}|$, and Kraft's inequality gives $2^m \leq 2^{|p_{w^*}|}$. Finally $K(m) \leq 2\log m + c \leq 2\log k + c$. □

The stream is not computable, since it depends on the programs p_w. The construction is logical: in SOI on strings the analogous decoys would be beaten by the constant labeller "true", which is not available here because $\{\neg\mathrm{desc}(v) : v \in \{0,1\}^m\}$ is inconsistent. As in the conjunction ladder of Theorem 9.10 of the companion paper, it is the selection rule that fails, not the prior.

Proof of Proposition A.11(c). For each $w \neq 0^m$ fix a program $q_w \in \mathrm{dom}\, V$ for an index of $\{\mathrm{desc}(w)\}$. Let $\prec$ list the $2^m - 1$ programs q_w first, in any order, and then the rest of $\mathrm{dom}\, V$ by length; this is a well-order of type ω. A consistent theory proves at most one desc(w), so $p_w = q_w$ for $w \neq 0^m$ (notation of the proof of Theorem A.35), and p_{0^m} lies in the tail; hence $w^* = 0^m$. Theorem A.35, which holds for every well-order, gives $2^m - 1$ errors or abstentions on a data set D whose conjunction is equivalent to desc(0^m), so $K_A(D) \leq K(\mathrm{desc}(0^m)) + c \leq K(m) + c'$; and $K(m) \leq 2\log m + c''$. □

A.10 The generative variant

Definition A.36 (Generative law)**.** Take $B = \emptyset$, or include B among the axioms. Fix the likelihood L_1 of Definition 2.9 of the earlier report: a derivation grammar with positive rule probabilities, a full-support instantiation grammar, and admissible parameters. For a set A of axioms with an enumeration $(\chi_1, \chi_2, \dots)$, give χ_j the citation weight proportional to 2^{-j} (normalised; no axiom is cited if $A = \emptyset$), and let P_A be the resulting law on sentences. The data are unlabelled sentences $\sigma_1, \sigma_2, \dots$. In the one-tape form, split the tape into $\omega^{(0)}, \omega^{(1)}, \dots$ by a computable interleaving; the theory is $A_\omega := A^{G_U}_{\omega^{(0)}}$, enumerated in the order of emission, and the i-th datum σ_i is sampled from P_{A_ω} with the coins $\omega^{(i)}$. The universal law of unlabelled data streams is $\Xi(\sigma_{1:n}) := \int \prod_{i \leq n} P_{A_\omega}(\sigma_i)\, d\mathbb{U}(\omega)$, and the generative posterior ϱ_n after n data has density proportional to $\prod_{i \leq n} P_{A_\omega}(\sigma_i)$ with respect to $\mathbb{U}$.

Consider the *support condition*: $\mathrm{supp}\, P_A = \mathrm{Cn}(A)$ for every set of axioms A. Lemma 2.10(c) of the earlier report proves it for finite template theories; we use it here as a hypothesis.

Proposition A.37 (Generative updating, given the support condition)**.** *Assume the support condition. (a)* Reweighting. *On data $\sigma_1, \dots, \sigma_n$, a consistent A has positive likelihood iff $A \vdash \sigma_i$ for all i, which is proof-requiring compatibility with $D := \{(\sigma_i, 1) : i \leq n\}$. So on consistent theories the generative posterior is the proof-requiring posterior reweighted by $\prod_i P_A(\sigma_i) \leq 1$. Inconsistent theories have positive likelihood, since their support is every sentence. (b)* Question conditioning. *For a yes/no question $\mathcal{Q} = \{\varphi, \neg\varphi\}$ and a consistent A: $P_A(\mathcal{Q}) > 0$ iff A decides φ, and then $P_A(\varphi^b \mid \sigma \in \mathcal{Q}) = \mathbf{1}[A \vdash \varphi^b]$. Hence, on the consistent theories, the generative predictive probability that the next datum answers $\mathcal{Q}$ by φ^b, given that it answers $\mathcal{Q}$, is*

$$\frac{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})\, \mathbf{1}[A_\omega \vdash \varphi^b]}{\int \varrho_n(d\omega)\, P_{A_\omega}(\mathcal{Q})},$$

the proof-requiring belief under a posterior reweighted by how likely each theory was to produce an answer to $\mathcal{Q}$; inconsistent theories contribute their own $P_A(\varphi^b \mid \sigma \in \mathcal{Q})$. The outcome "independent" gets probability 0.

Proof. (a) By the support condition, $P_A(\sigma_i) > 0$ iff $\sigma_i \in \mathrm{Cn}(A)$. (b) $\operatorname{supp} P_A = \mathrm{Cn}(A)$ contains at most one of $\varphi, \neg\varphi$, by consistency, and contains one of them iff A decides φ; then apply Bayes' rule. □

Inconsistent theories pay the size principle: by Lemma 2.21 of the earlier report, a theory with mass ε outside the support of the data law loses at least $\log \frac{1}{1-\varepsilon}$ bits per datum in expectation. The generative variant thus models which sentences get stated; the outcome "independent" disappears because the model presupposes that asked questions are answered.

Remark A.38 (Supports and identification). We expect the following two statements to hold, and give only proof sketches. (i) *Supports*: the support condition holds for every axiom set in which every axiom has positive citation weight. (ii) *Identification*: if $\sigma_1, \sigma_2, \ldots$ are independent draws from P_{T^*}, the law of the deterministic machine that emits a consistent r.e. T^*, then the generative posterior probability of $\{\omega : \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T^*)\}$ tends to 1 almost surely; by contrast, the proof-requiring posterior on two-valued data presenting T^* reaches only the prior restricted to the consistent extensions of T^* (Proposition A.27(b)).

Proof sketch. We do not write out the support condition for r.e. axiom sets, nor the measurability of $\omega \mapsto P_{A_\omega}$ that (ii) needs. For (i), the argument of Lemma 2.10(c) of the earlier report applies to such axiom sets: each derivation tree that cites finitely many axioms has positive probability. For (ii), $\operatorname{supp} P_{A_\omega} = \mathrm{Cn}(A_\omega)$ is almost surely a function of the infinite sample (the set of sentences that occur), so Doob's consistency theorem (Theorem 4.2 of the earlier report; Doob, 1949) gives posterior consistency for it at $\mathbb{U}$-almost every ω; the tapes whose theory has the law P_{T^*} form a set of positive measure, so the statement holds almost surely under P_{T^*}.

A.11 Priors that complete their random theory

We compare M_A with two processes that draw sentences or theories until the theory is complete. We define them explicitly and prove their properties for the processes as defined. Process 1 is Demski's construction (Demski, 2012) as it is usually described. Process 2 is, at the level of the published descriptions, the modification of Demski's prior due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), which samples machines on random bits and reads their outputs as sets of sentences; its sampling stage is that of G_U. We use nothing about either construction beyond the definitions below.

- *Process 1 (sentence draws).* Let μ be a probability on Sent with $\mu(\varphi) > 0$ for every φ. Draw sentences independently from μ, and keep a drawn sentence iff it is consistent with B and the sentences kept so far. Let S_t be the set kept after t draws, $T_\infty := \mathrm{Cn}(\bigcup_t S_t)$ and $P_\mu(\varphi) := \Pr[\varphi \in T_\infty]$.
- *Process 2 (theory draws).* At each step t draw an independent tape $\omega^{(t)}$ and the theory $A_{\omega^{(t)}}$; keep it iff it is consistent together with B and what has been kept. Let T'_∞ be the deductive closure of everything kept. (In a variant, the sentences of a drawn theory are kept one by one, as long as they are consistent with what has been kept.)

Proposition A.39 (Completing processes against M_A)**.** *(a) T_∞ is consistent and complete with probability* 1*. Hence P_μ is a probability on the Lindenbaum algebra of B ($P_\mu(\varphi) + P_\mu(\neg\varphi) = 1$),*

and $\Pr[\mathrm{Cn}(T_\infty) = \mathrm{Cn}(T)] = 0$ *for every incomplete* T*. (b)* T'_∞ *is consistent and complete with probability* 1*, also in the variant; so Process 2 also gives a coherent probability over completions, and gives every incomplete theory class probability* 0*. (c) Under* M_A*, by contrast: for every* φ *undecided by* B*, the prior independent mass* $\mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\varphi) - \mathrm{Bel}_0(\neg\varphi)$ *is at least* $\varepsilon_\emptyset > 0$*; every r.e.* T *with* $B \cup T$ *consistent, complete or not, has* $\mathbb{U}_A[T] \geq 2^{-K_{\equiv}(T)-c} > 0$*; and if* $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ *and* $\mathsf{Q} \subseteq \mathrm{Cn}(B)$*, the random theory is almost surely incomplete (Proposition A.31). (d) For any prior concentrated on complete theories, such as those of (a) and (b), proof-requiring and consistency-requiring compatibility coincide, and the trichotomy has no independent outcome (Proposition 3.11).*

Proof. (a) Fix φ. With probability 1 it is drawn at some step, since $\mu(\varphi) > 0$. At the first such step t, either $B \cup S_{t-1} \cup \{\varphi\}$ is consistent and φ is kept, or $B \cup S_{t-1} \vdash \neg\varphi$; either way T_∞ decides φ. There are countably many sentences. Every S_t is consistent with B, so their union is, by compactness. (b) Fix φ. At each step the drawn theory is $\{\varphi\}$ with probability at least $2^{-K(\varphi)-c} > 0$ (the tapes that begin with a program for the machine emitting φ), independently across steps; so with probability 1 some step draws $\{\varphi\}$. At the first such step, either $\{\varphi\}$ is consistent with what has been kept, and is kept, or what has been kept proves $\neg\varphi$. Either way T'_∞ decides φ. Take the countable union; consistency follows by compactness as in (a). In the variant, the same step decides φ. (c) The tapes with $A_\omega = \emptyset$ have mass $\varepsilon_\emptyset$ and decide only what B decides; the rest is Theorem 3.4(i) and Proposition A.31. (d) is Proposition 3.11. □

Running either process needs a consistency test at every step: already for $B = \emptyset$, a language with a binary relation symbol and one draw, it asks whether a sentence is satisfiable, which is undecidable by Church's theorem in Kalmár's form (Church, 1936; Kalmár, 1937). Process 1 is computable relative to $\emptyset'$, since consistency of a finite set is Π^0_1. Process 2 needs a consistency test relative to the tapes, which is Π^0_1 in them. Neither is computable, since a single draw of Process 2 can output one sentence. A theory machine emits sentences unconditionally, and consistency enters M_A only once, as the event measured (Proposition A.5). So Process 2 and M_A share the sampling stage, and with it the program-length credit for infinite schemas and dominance over the laws of machines; that stage is not new here. They differ in completion (iterated draws to a complete theory against a single draw, whose theory is incomplete with positive probability), in what a prediction is (a coherent probability over completions against a belief function of provability with an independent outcome), and in what is conditioned on (for complete theories the two compatibility notions coincide; for M_A they differ, and the proof-requiring one carries the learning guarantees of Section 3).

A.12 Domination of provability functions

Recall from Section 3.8 that a random consistent theory Θ has the provability function $\beta(\chi) = \Pr[\chi \in \Theta]$, and that Question 3.12 asks whether $\mathrm{Bel}_0 \geq 2^{-c_\beta}\beta$ whenever β is lsc. Here "consistent" means consistent with B.

Proposition A.40 (Partial domination)**.** *(a) For every lsc provability function* β*, every finite set* S *of pairwise* B*-incompatible sentences and every* $\sigma \in S$*:* $\mathrm{Bel}_0(\sigma) \geq 2^{-K(\beta)-K(S)-c}\beta(\sigma)$*. In particular* $M_A(D) \geq 2^{-K(\beta)-K(\mathrm{dom}\, D)-c}\beta(\psi_D)$*. (b) If* Θ *is the deductive closure of the output of a theory machine* G *whose output is almost surely consistent, then* $\mathrm{Bel}_0 \geq 2^{-K(G)}\beta$*. (c) In the literal setting, Question 3.12 restricted to random sets of literals of* R *with at most one literal per word (with a lower semicomputable containment functional on literal data, and domination required on literal data) is equivalent to Question 3.14 of the companion paper for labels* $\{0, 1\}$*: whether its*

one-tape law M_F dominates every lower semicomputable law of a random partial function, up to a constant factor.

Proof. (a) Θ contains at most one member of S, by consistency, so $\sum_{\sigma\in S}\beta(\sigma)\le 1$, and β restricted to S is a semimeasure, lsc uniformly in S. The conditional coding theorem (Section 2.1) gives $K(\sigma \mid S)\le -\log\beta(\sigma)+K(\beta)+c$. If $\beta(\sigma)>0$, then σ is consistent with B, and the one-axiom theory $\{\sigma\}$ gives $\mathrm{Bel}_0(\sigma)\ge 2^{-K(\sigma)-c}\ge 2^{-K(S)-K(\sigma|S)-c'}$. For D, take S to be the label conjunctions on $\operatorname{dom} D$. (b) Then $\beta(\chi)=\nu_G(\{(\chi,1)\})$, and Theorem 3.4 gives $\mathrm{Bel}_0(\chi)=M_A(\{(\chi,1)\})\ge 2^{-K(G)}\nu_G(\{(\chi,1)\})$. (c) A random set Λ of literals with at most one literal per word corresponds to the random partial function h_Λ with $h_\Lambda(x)=b$ iff $R(t_x)^b\in\Lambda$, and $\Pr[\Lambda\supseteq D^{\pm}]=\Pr[h_\Lambda\supseteq D^{\mathrm{str}}]$, with D^{str} the string data set of D; and $M_A(D)\asymp M_F(D^{\mathrm{str}})$ on literal data (Theorem 4.4). □

So Question 3.12 is at least as hard as the binary case of Question 3.14 of the companion paper, and for general sentences it is open. The obstacle is the same: a lower semicomputable law of a random object need not be realisable by a machine that reads one tape. Note that Bel_0 itself is not lower semicomputable (Theorem 3.3), while $\mathrm{Bel}_0+\mathrm{Inc}_0$ is; and by Corollary A.32, a theory-machine law above Q always has a local defect somewhere, so the question is not trivial.

B Proofs for the relation to operator induction

This appendix proves the results of Section 4 and states those that the main text mentions in one sentence. Results of our earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion paper (Hänni and Claude (Anthropic), 2026c) are cited by their printed numbers and re-proved only where the setting changes. Throughout, $c, c', \dots$ are machine constants. For a computable map h on indices, universality of V gives a string r_h with $V(r_h s)=h(V(s))$ for all $s\in\operatorname{dom} V$; so $p\mapsto r_h p$ is injective, maps $\operatorname{dom} V$ into $\operatorname{dom} V$ and adds $|r_h|$ to the length. "Tape by tape" means: for every tape ω.

B.1 Decompositions and the four maps

Measurability. $\{\omega : f_\omega\supseteq D\}$ is open. "f_ω is inconsistent" is open: an inconsistency uses finitely many labels, each certified under a prefix of ω, and these prefixes are comparable. "f_ω is closed" is a countable intersection of Borel conditions. For theory machines $\{\omega : B\cup A_\omega\vdash\psi\}$ is open and consistency is closed (Lemma A.1). All events below are Borel.

Lemma B.1 (Mixture decompositions)**.** *For every finite $D\neq\emptyset$,*

$$M_F^{\mathrm{cons}}(D)=\sum_{p\in\operatorname{dom} V}2^{-|p|}\nu^{\mathrm{cons}}_{F_{V(p)}}(D), \tag{2}$$

and the same holds for ν^{cl} (with M_F^{cl}) and for ν (with M_F). For $D=\emptyset$ add $1-\Omega_V$ on the right for ν^{cons} and ν, and nothing for ν^{cl}. For every finite D, the empty one included,

$$M_A(D)=\sum_{p\in\operatorname{dom} V}2^{-|p|}\nu_{G_{V(p)}}(D)+(1-\Omega_V)\,\mathbf{1}[B\vdash\psi_D]. \tag{3}$$

Proof. $\operatorname{dom} V$ is prefix-free, so every tape has at most one prefix $p\in\operatorname{dom} V$. On a tape $p\omega'$, $f^{F_U}_{p\omega'}=f^{F_{V(p)}}_{\omega'}$ and $A^{G_U}_{p\omega'}=A^{G_{V(p)}}_{\omega'}$, with ω' uniform given p. On the tapes with no prefix in $\operatorname{dom} V$

(measure $1-\Omega_V$), f_ω is nowhere defined, so it extends D only if $D=\emptyset$, and then it is consistent, B being consistent; it is never closed, because B proves every tautology and the empty labeller labels none. On these tapes $A_\omega=\emptyset$, so the theory is B, which is consistent and proves ψ_D iff $B\vdash\psi_D$. □

The weight of an index i is $\mathbf{m}(i)=\sum_{V(p)=i}2^{-|p|}$, so (2) reads $M_F^{\mathrm{cons}}=\sum_i \mathbf{m}(i)\nu_{F_i}^{\mathrm{cons}}$ on nonempty D.

Lemma B.2 (The four maps)**.** *(i)* Proof search. *For a theory machine G let F_G answer a query φ as follows: in stages $s=1,2,\dots$ read the tape up to bit s, run s steps of the enumeration of G restricted to pairs (w,χ) with w a prefix of the bits read, and test all derivations of size at most s from B and the sentences emitted so far; halt with* 1 *at the first derivation of φ and with* 0 *at the first derivation of $\neg\varphi$ (within a stage φ is tested first). Then F_G is a function machine, and on every tape ω with $B\cup A_\omega^G$ consistent, $f_\omega^{F_G}$ is consistent and closed and decides every sentence exactly as $B\cup A_\omega^G$ does.*

(ii) Emission. *For a function machine F let $G_F:=\{(w,\varphi):(\varphi,w,1)\in F\}\cup\{(w,\neg\varphi):(\varphi,w,0)\in F\}$. Then $A_\omega^{G_F}=\Gamma_{f_\omega^F}$ for every ω. So if f_ω^F is consistent and extends D, then G_F is compatible with D on ω; if f_ω^F is closed and consistent, $B\cup A_\omega^{G_F}$ decides every sentence exactly as f_ω^F does.*

(iii) Craig's map with the tape. *For a function machine F, read as a machine that reads the tape one-way, let C_F answer a query χ as follows. Find the at most two decompositions $\chi=\psi^{\wedge m}$ (the right-nested m-fold conjunction: $m=1$, or ψ the left conjunct of χ and m fixed by $|\chi|$). For each, test both readings: (1) run F on ψ for exactly $m-1$ steps, reading the tape as F does, and accept if F halts at step $m-1$ with label* 1*; (2) if $\psi=\neg\varphi'$, run F on φ' for exactly $m-1$ steps and accept if it halts at step $m-1$ with label* 0*. Answer* 1 *if some test accepts and* 0 *otherwise. Then C_F is a random decider, and $\mathrm{Cn}(A_{C_F}^{\mathrm{dec}}(\omega))=\mathrm{Cn}(\Gamma_{f_\omega^F})$ for every ω.*

(iv) Closure. *For a partial computable labeller f let $\bar f$ search for proofs of φ and of $\neg\varphi$ from $B\cup\Gamma_f$, dovetailing the enumeration of Γ_f. If f is consistent, $\bar f$ extends f, is consistent and closed, and $\mathrm{Cn}(\Gamma_{\bar f})=\mathrm{Cn}(\Gamma_f)$.*

Proof. (i) The run on a query is a deterministic function of the bits read, and a halting run that has read w runs identically on every tape extending w; so the triples (φ,w,y) of halting runs form a consistent c.e. set. Let $B\cup A_\omega^G$ be consistent. If it proves φ^b, a derivation uses finitely many axioms, each emitted under a prefix of ω; at a late enough stage all of them have been enumerated and the derivation has size at most s, so the run halts, and it cannot first find a derivation of φ^{1-b}, since none exists. If the theory decides neither, no derivation is found and the label is undefined. So $f_\omega^{F_G}(\varphi)=b$ iff $B\cup A_\omega^G\vdash\varphi^b$. Then $\Gamma_f\subseteq\mathrm{Cn}(A_\omega^G)$ is consistent with B, and f is closed: $B\cup\Gamma_f\vdash\varphi^b$ implies $B\cup A_\omega^G\vdash\varphi^b$, which implies $f(\varphi)=b$; the converse holds as $\varphi^b\in\Gamma_f$. (ii) G_F is c.e. and emits φ^y exactly under the prefixes that certify $f(\varphi)=y$, so $A_\omega^{G_F}=\Gamma_{f_\omega^F}$. If f is closed, $B\cup\Gamma_f\vdash\varphi^b$ iff $f(\varphi)=b$. (iii) A decomposition $\chi=\psi^{\wedge m}$ with $m\geq 2$ has ψ the left conjunct and m fixed by $|\chi|$, since $|\psi^{\wedge m}|=m|\psi|+3(m-1)$ in primitive syntax (Lemma E.12(b) of the earlier report); the only other candidate is $m=1$. Every run is bounded by $m-1$ simulated steps, so C_F halts on every query and every tape: it is a random decider. Its answer depends only on the tape bits read by the simulated runs, so its triples are consistent. Its accepted set on ω is $\{\varphi^{\wedge(k+1)}:F$ halts on φ with 1 in exactly k steps$\}\cup\{(\neg\varphi)^{\wedge(k+1)}:F$ halts on φ with 0 in exactly k steps$\}$: reading (1) gives the first set (also when φ is itself a negation), reading (2) the second. Each

member is equivalent to a member of $\Gamma_{f^F_\omega}$, and each member of $\Gamma_{f^F_\omega}$ has one, so Cn is preserved. (Reading (1) alone would lose the members $(\neg\varphi')^{\wedge(k+1)}$ that come from $f(\neg\varphi') = 1$, and Cn would change.) (iv) As (i), with the r.e. set Γ_f in place of A^G_ω; $f(\varphi) = b$ puts $\varphi^b \in \Gamma_f$, so $\bar{f}(\varphi) = b$. □

Only inclusions of events are claimed. The inclusion given by proof search can be strict: on a tape on which the theory is inconsistent, the labeller may already have committed to a consistent set of labels before the inconsistency appeared. A computation on a finite propositional analogue (code in the accompanying repository) finds many such tapes.

B.2 The equivalences

Proof of Theorem 4.2. We use coin-free forms of the maps of Lemma B.2: deterministic machines ignore the tape. *Enumerators to labellers* (proof search): $p \mapsto r_{\mathrm{ps}}p$; if the enumerator p is compatible with D, the proof-search labeller is consistent and closed, extends D, and decides each φ as the theory does. *Labellers to enumerators* (emission): $p \mapsto r_{\mathrm{em}}p$, the enumerator of $\Gamma_{\{V(p)\}}$; if the labeller is consistent and extends D, the theory is consistent and proves $D^{\pm}$; if it is closed, it decides as the labeller does. *Labellers to deciders* (Craig's map without tape): the decider of Craig's set of the labeller, which halts on every input even when the labeller is partial, since every run is bounded by the exponent of the candidate conjunction. *Deciders to enumerators*: enumerate the decided set; the theory is the same. *Consistent to closed*: the closure map. All maps are injective on programs and add a constant to the length. Summing $2^{-|p|}$ over compatible programs gives (a) and (c), where (c) uses proof search and emission restricted to closed labellers, which preserve the decision of every sentence; taking shortest programs gives (b). Finiteness in (b): if D is consistent, the lookup table of D is a consistent labeller, its closure a closed one, and the decider of the finite set $D^{\pm}$ is compatible; if not, no consistent theory proves $D^{\pm}$ and no consistent labeller extends D. For the losses, each of the two terms of $-\log W(D) + \log W(\emptyset)$ changes by at most c. For deciders this is Theorem 6.3 of the earlier report, stated there for programs that read their input; the change of program format costs a constant. □

With weights $2^{-\lambda|p|}$, $\lambda \geq 1$, as in the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d), the same maps give the equivalence within $2^{\pm\lambda c}$.

Remark B.3 (The schema route). A different route to Theorem 4.2 turns a labeller f into the axiom schema $A_f := \{\mathrm{Acc}_f(\ulcorner\varphi\urcorner) \to \varphi\} \cup \{\mathrm{Rej}_f(\ulcorner\varphi\urcorner) \to \neg\varphi\}$, with Σ_1 formulas Acc_f, Rej_f, written without $<$, that express that f accepts or rejects.

(a) *Two sorts.* Let $B = B_A \cup B_L$, with arithmetic axioms $\mathsf{Q} \subseteq B_A \subseteq \mathrm{Th}(\mathbb{N})$ on one sort, axioms B_L on the other and no mixed axioms. If $B_L \cup \Gamma_f$ is consistent, so is $B \cup A_f$, and the sentences of the non-arithmetic sort's language that $B \cup A_f$ proves are exactly those that $B_L \cup \Gamma_f$ proves (Proposition 6.4 of the earlier report; it uses the Σ_1-completeness of Q).

(b) *One sort over* PA. Assuming $\mathrm{Con}(\mathsf{PA})$, the labeller f that accepts $\neg\mathrm{Con}(\mathsf{PA})$ and, on $\mathrm{Con}(\mathsf{PA})$, searches for a PA-proof of $\bot$ has $\Gamma_f = \{\neg\mathrm{Con}(\mathsf{PA})\}$, which is consistent with PA, while $\mathsf{PA} \cup A_f$ is inconsistent (Proposition 6.5 there). So the argument through the schema fails in one sort; the conclusion of Theorem 4.2 does not.

(c) *Bounded witnesses.* In one sort, for every decidable $B \supseteq \mathsf{Q}$, true or not, let A^{bd}_f use the forms $\mathrm{Acc}^{\leq k}_f$, $\mathrm{Rej}^{\leq k}_f$ whose witnesses are bounded by a binary numeral k. Then $\mathrm{Cn}(A^{\mathrm{bd}}_f) = \mathrm{Cn}(\Gamma_f)$ for every labeller f, and membership in A^{bd}_f is decidable in polynomial time without running f (Proposition E.17 there, given the MRDP theorem): each axiom is Q-equivalent to a

member of Γ_f or is a Q-theorem. In substance this is Craig's idea, the bound playing the role of the padding.

(d) *Templates.* No non-ground template of that report's template class has all its instances in the acceptance part of A_f; but for every Σ_n-sound labeller f one ground reflection sentence of length at most $a|f| + b_n$ reproduces f on Σ_n sentences over PA (Propositions 6.6 and 6.7 there, given partial truth predicates).

The one-tape equivalence avoids the schema altogether: a theory machine is an enumerator that reads a tape, so it may emit Γ_{f_ω} itself (emission); neither decidability, nor arithmetic, nor a schema is needed. The schema, like Craig's padding, matters only for the decider format.

Proof of Theorem 4.3. (b) is Lemma B.2(i)–(iii). (a), M_A against M_F^{cons}, for $D \neq \emptyset$: by (3), (b) and (2), with $p \mapsto r_{\mathrm{ps}}p$ (an index of $F_{G_{V(p)}}$) and $p \mapsto r_{\mathrm{em}}p$ (an index of $G_{F_{V(p)}}$),

$$\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq \sum_p 2^{-|p|}\nu^{\mathrm{cons}}_{F_{V(r_{\mathrm{ps}}p)}}(D) \leq 2^{|r_{\mathrm{ps}}|}M_F^{\mathrm{cons}}(D),$$
$$M_F^{\mathrm{cons}}(D) \leq 2^{|r_{\mathrm{em}}|}\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq 2^{|r_{\mathrm{em}}|}M_A(D).$$

The remaining term $(1-\Omega_V)\mathbf{1}[B \vdash \psi_D]$ of (3) is at most $\mathbf{1}[B \vdash \psi_D]$; if $B \vdash \psi_D$, the deterministic proof-search labeller of B is closed and consistent and extends D on every tape, so $M_F^{\mathrm{cons}}(D) \geq 2^{-c_B}$. Hence $M_A \leq (2^{|r_{\mathrm{ps}}|} + 2^{c_B})M_F^{\mathrm{cons}}$. For $D = \emptyset$ both $M_A(\emptyset)$ and $M_F^{\mathrm{cons}}(\emptyset)$ lie in $[1-\Omega_V, 1]$ (the tapes without a program give the consistent theory B and the empty labeller), so their ratio is at most $1/(1-\Omega_V)$. M_A against M_A^{dec}: a random decider is a theory machine (emit χ under the prefix that certifies the answer 1), so $M_A^{\mathrm{dec}} \leq 2^c M_A$; and Craig's map $C_{F_{V(p)}}$ gives $M_A^{\mathrm{dec}} \geq 2^{-c}M_F^{\mathrm{cons}} \geq 2^{-2c}M_A$ on $D \neq \emptyset$; for $D = \emptyset$ the decider of $\emptyset$ gives $M_A^{\mathrm{dec}}(\emptyset) \geq 2^{-c}$. □

Corollary B.4 (Closed labellers and the trichotomy)**.** *For every finite D and $\varphi \notin \operatorname{dom} D$, the three outcome weights $M_A(D + (\varphi, 1))$, $M_A(D + (\varphi, 0))$, $M_A^3(D + (\varphi, \iota))$ (the theory proves φ, proves $\neg\varphi$, decides neither) agree within $2^{\pm c}$ with the measures of the tapes on which f_ω is closed, consistent, extends D and has $f_\omega(\varphi) = 1$, 0, $\perp$ respectively; and $2^{-c}M_A \leq M_F^{\mathrm{cl}} \leq M_F^{\mathrm{cons}}$.*

Proof. Proof search maps tapes with a consistent theory to tapes on which F_G is closed and decides as the theory does; emission maps tapes with a closed consistent labeller to tapes on which the theory decides as the labeller does. The program-less tapes (theory B) are covered by the closed deterministic labeller "proof search from B", of weight 2^{-c_B}, which decides as B does. Sum as in the proof of Theorem 4.3. The outcome weights of M_A are those of the corollary: $M_A(D + (\varphi, 1))$ is the measure of the tapes whose theory is consistent and proves ψ_D and φ, and likewise for the other two. □

Penalised versions. One may charge a hypothesis for the time it takes to check its axioms (a decider) or to generate them (an enumerator), with deduction still free (Definition B.7). The results on these versions rest on an assumption about the universal machine U whose programs are the hypotheses, taken from Appendix E.6 of the earlier report; it is standard for multitape machines, and we do not prove it, or its form with a tape, for a particular U.

Assumption B.5 ((S): efficient self-simulation)**.** Penalised hypotheses are programs of a fixed universal prefix machine U, read to their end before the input. There are constants $\alpha, a \geq 1$ such that the fixed wrappers used in the proofs read, copy and measure a program q within

$\alpha(|q|+1)$ steps; parse, compare or copy strings of total length L within $\alpha(L+1)^2$ steps; and run the *dovetailing schedule* of q (at stage s, start q on the s-th sentence and advance every unfinished run by one simulated step) while keeping a *work counter* T (the simulated steps, plus the lengths of the sentences started, plus $|q|$), within $\alpha(T+1)^a$ steps up to work T, writing a string of length L at cost at most αL. Put $a^* := \max(a, 3)$.

Assumption B.6 (($\mathrm{S}_{\mathrm{tape}}$): self-simulation with a tape)**.** There are constants $\alpha, a \geq 1$ with the properties of Assumption B.5 when q also reads a one-way random tape: all runs of the dovetailing schedule read the same tape ω one-way from its start, the work counter also counts tape reads, and up to work T the schedule costs at most $\alpha(T+1)^a$ steps and reads at most T tape bits.

We use the format of Appendix E.6 of the earlier report: hypotheses are programs of U, read to their end before the input, now with access to a one-way read-only random tape.

Definition B.7 (Penalised one-tape SAI)**.** A *random decider program* q halts on every pair (χ, ω) with yes or no; A^q_ω is its yes-set on ω, and $\mu^{\mathrm{dec}}_q(D) := \mathbb{U}\{\omega : B \cup A^q_\omega \text{ consistent and } \vdash \psi_D\}$. A *random enumerator program* q has no input and writes sentences $a_1(\omega), a_2(\omega), \ldots$, completing a_i at step $\mathrm{st}_i(\omega)$ ($\mathrm{st}_0 := 0$). For time-constructible t:

- $M^{\mathrm{chk},t}_A(D) := \sum_q 2^{-|q|}\mu^{\mathrm{dec}}_q(D)$, the sum over the random decider programs q whose running time on every query χ and every tape is at most a constant multiple of $t(|\chi|)$, that is, $\sup_{\chi,\omega} \mathrm{time}_q(\chi,\omega)/t(|\chi|) < \infty$;
- $M^{\mathrm{gen},t}_A(D) := \sum_q 2^{-|q|}\nu^{\mathrm{gen}}_q(D)$, with $\nu^{\mathrm{gen}}_q(D) := \mathbb{U}\{\omega : B \cup \{a_i(\omega)\}_i \text{ consistent and } \vdash \psi_D\}$, the sum over the random enumerator programs that meet the *delay form* $\sup_{\omega,i}(\mathrm{st}_i - \mathrm{st}_{i-1})/t(|a_i|) < \infty$; in the *strict form* the condition is on $\mathrm{st}_i/t(|a_i|)$, in the *cumulative form* on $\mathrm{st}_i/t(|a_1|+\cdots+|a_i|)$.

The coin-free $W^{\mathrm{chk},t}_A$, $W^{\mathrm{gen},t}_A$ and the Levin-type $W^{\mathrm{Kt},t}_A$ are those of Definition E.10 of the earlier report, which also assumes t nondecreasing and superadditive with $t(m) \geq m$.

Remark B.8 (On Assumption B.6). *Proof sketch.* We do not write this out for a concrete U; like Assumption B.5, the assumption is used, not proved. A tape-reading program can be simulated by keeping the bits read so far on a work tape and one pointer per run of the schedule; a simulated read copies a stored bit or reads the next tape bit, at cost $O(T)$ up to work T. So the exponent a of Assumption B.5 should grow by at most 1.

In this format the unpenalised one-tape laws $M^U_A := \sum_q 2^{-|q|}\nu_{G_q}$, over all tape-reading programs read as theory machines, and the corresponding M^{cons}_F agree with M_A and M^{cons}_F within constant factors, since each universal mixture contains the other's universal machine behind a constant prefix.

Theorem B.9 (Axiom-side penalties change nothing)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible. There is c such that, for every finite D and in the strict, delay and cumulative forms: (a) under Assumption B.6, $2^{-c}M^{\mathrm{cons}}_F(D) \leq M^{\mathrm{chk},t}_A(D) \leq 2^c M_A(D)$ and $2^{-c}M^{\mathrm{cons}}_F(D) \leq M^{\mathrm{gen},t}_A(D) \leq 2^c M_A(D)$, the empty D included; so $M^{\mathrm{chk},t}_A \asymp M^{\mathrm{gen},t}_A \asymp M_A \asymp M^{\mathrm{cons}}_F$; (b) under Assumption B.5, $2^{-c}W^{\mathrm{cons}}(D) \leq W^{\mathrm{chk},t}_A(D) \leq W^{\mathrm{dec}}_A(D) \leq 2^c W^{\mathrm{cons}}(D)$ and $2^{-c}W^{\mathrm{cons}}(D) \leq W^{\mathrm{gen},t}_A(D) \leq 2^c W^{\mathrm{cons}}(D)$.*

Proof. (a) *Lower bounds.* Let p be a tape-reading program for a function machine F. Let q_F say yes on χ if F labels χ with 1, or if $\chi = \neg\varphi$ and F labels φ with 0 (dovetailing the two runs on the same tape). On every tape its yes-set is $\Gamma_{f^F_\omega}$, and $|q_F| \leq |p| + c$. The padded random decider d, a

fixed wrapper followed by q_F, answers χ as in Lemma E.12(b) of the earlier report: it finds the at most two candidates $\chi = \psi^{\wedge m}$, rejects if $m-1 < |q_F|$, and otherwise runs the dovetailing schedule of q_F on the tape, without writing, until the work counter exceeds $m-1$, accepting iff q_F is first detected saying yes on ψ exactly at counter $m-1$. Under Assumption B.6 this takes at most $\alpha(|q_F|+1)+3\alpha(|\chi|+1)^{a'}$ steps, $a' := \max(a, 2)$, on every tape, uniformly in ω (the argument of that lemma, verbatim with the tape); as $(|\chi|+1)^{a'} \le 2^{a'}|\chi|^{a^*} \le 2^{a'}t(|\chi|)$, the membership bound holds. The answer depends only on the tape bits read by the schedule, so d is a random decider. On each tape its yes-set is $\mathrm{Pad}_\omega := \{\psi^{\wedge(T+1)} : q_F \text{ is first detected saying yes on } \psi \text{ at counter } T \text{ on } \omega\}$; distinct detections have distinct counters, so each $\psi \in \Gamma_{f^F_\omega}$ has exactly one padded copy and $\mathrm{Cn}(\mathrm{Pad}_\omega) = \mathrm{Cn}(\Gamma_{f^F_\omega})$ (Lemma E.12(a) there, per tape). So on every tape on which f^F_ω is consistent and extends D, d is compatible with D. The padded enumerator, a fixed wrapper followed by q_F, writes $\psi^{\wedge(T+1)}$ at each detection; Lemma E.12(c) there gives, on every tape, delay at most $2\alpha|a_i|^a$, strict completion at most $5\alpha|a_i|^{a^*}$ and cumulative completion at most $2\alpha(|a_1|+\cdots+|a_i|)^a$, uniformly in ω. Both maps are injective and add a constant to the length. Summing over p gives the lower bounds for $D \neq \emptyset$. For $D = \emptyset$, the decider that rejects every sentence runs in linear time with theory B, which is consistent, and the enumerator that writes nothing meets every delay condition vacuously; so $M^{\mathrm{chk},t}_A(\emptyset), M^{\mathrm{gen},t}_A(\emptyset) \ge 2^{-c} \ge 2^{-c}M^{\mathrm{cons}}_F(\emptyset)$. *Upper bounds.* A random decider program q is read as the theory machine that emits χ under the tape prefix read by a run of q on χ that answers yes; a random enumerator program as the theory machine that emits the i-th written sentence under the prefix read when it is completed. Each reading is a fixed computable map on programs, realised by a constant-size wrapper, which is injective and keeps the theory on every tape. So $M^{\mathrm{chk},t}_A, M^{\mathrm{gen},t}_A \le 2^c M^U_A \le 2^{c'} M_A$ for every D. With Theorem 4.3 this gives the chain of equivalences. (b) This is Theorem 6.14(a) of the earlier report: the coin-free version of the construction above, with the padded sets of Lemma E.12 there. The inequality $W^{\mathrm{chk},t}_A \le W^{\mathrm{dec}}_A$ holds because the penalised deciders are deciders with the same weights, and $W^{\mathrm{dec}}_A \le 2^c W^{\mathrm{cons}}$ by Theorem 4.2. □

The equivalence also holds class by class (coin-free, Proposition B.28).

Proposition B.10 (Class posteriors, one tape)**.** *Let $t(m) \ge m^{a^*}$ be time-constructible and X be $M^{\mathrm{chk},t}_A$ or $M^{\mathrm{gen},t}_A$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C}; D)$ be the measure, under X's mixture, of the tapes on which the hypothesis is compatible with D and the deductive closure of its axioms is in $\mathcal{C}$. Under Assumption B.6 there is c with $2^{-c}\mathbb{U}_A(\mathcal{C}; D) \le X(\mathcal{C}; D) \le 2^c\mathbb{U}_A(\mathcal{C}; D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of theories (genuine axiom systems and disguised labellers alike) under X and under M_A differ by a factor at most 2^{2c}.*

Proof. Lower bound: for a theory machine G with program p, let q_G say yes on χ iff G emits χ on the tape, and pad as in the proof of Theorem B.9. On every tape the padded hypothesis has the same deductive closure as $B \cup A^G_\omega$, so it lies in $\mathcal{C}$ and is compatible with D exactly when G's theory is. The program-less tapes of M_A (theory B) are matched by the decider of $\emptyset$. Upper bound: a penalised hypothesis, read as a theory machine, has the same theory. For the odds, divide by $X(\mathrm{all}; D)$ and $\mathbb{U}_A(\mathrm{all}; D) = M_A(D)$. □

Proposition B.11 (The chain for the consistent versions)**.** *For every consistent $D \neq \emptyset$,*

$$2^{-K^{\mathrm{cons}}_F(D)} \le W^{\mathrm{cons}}(D) \le 2^c M^{\mathrm{cons}}_F(D) \le 2^{K(\mathrm{dom}\, D)+c}\, 2^{-K^{\mathrm{cons}}_F(D)},$$

and $\log(M_F^{\mathrm{cons}}(D)/W^{\mathrm{cons}}(D)) \le K^{\mathrm{pad}}(L_{1/2}^{\mathrm{cons}}(D)) + c$*, where* $L_{1/2}^{\mathrm{cons}}(D)$ *is the least* L *with* $\mathbb{U}\{\omega :$ *every label of* D *is certified within* $\omega_{1:L}$ *and* f_ω *is consistent*$\} \ge M_F^{\mathrm{cons}}(D)/2$*. By Theorems 4.2 and 4.3 the same holds for* K_A*,* W_A*,* M_A *with a larger* c*.*

Proof. A shortest consistent program extending D is a term of W^{cons}. A consistent deterministic labeller, read as a function machine that ignores the tape, is consistent on every tape; with (2) and the wrapper "treat $\{V(p)\}$ as a machine", $W^{\mathrm{cons}} \le 2^c M_F^{\mathrm{cons}}$. Next, $M_F^{\mathrm{cons}} \le M_F \le 2^{-K(D|\mathrm{dom}\, D)+c}$ by Theorem 3.21(a) of the companion paper, which holds for the input set Sent; the lookup table of D is consistent, so $K_F^{\mathrm{cons}}(D) \le K(D) + c \le K(\mathrm{dom}\, D) + K(D \mid \mathrm{dom}\, D) + c$. For the K^{pad} bound, take $L' \ge L$ with $K(L') = K^{\mathrm{pad}}(L)$, and for $w \in \{0,1\}^{L'}$ the stored-prefix labeller $\varphi \mapsto y$ iff some $(\varphi, w', y) \in F$ has $w' \sqsubseteq w$. It is a sub-function of f_ω^F for every $\omega \sqsupseteq w$, so it is consistent whenever some extension of w is a consistent tape. The count in the proof of Proposition 4.6(a) of the companion paper, restricted to such w, gives, with r the constant-size wrapper, $W^{\mathrm{cons}}(D) \ge 2^{-|r|-K(F)-K^{\mathrm{pad}}(L)}\mathbb{U}\{\omega : D$ certified within $\omega_{1:L}$, f_ω^F consistent$\}$; take $F = F_U$ and $L = L_{1/2}^{\mathrm{cons}}(D)$. □

B.3 Positive literals

Proof of Theorem 4.4. By Theorems 4.2 and 4.3 it suffices to compare consistent SOI on sentences with plain SOI on $\mathcal{I}$. *Strings to sentences:* for a partial computable g on $\mathcal{I}$ let $g^P(R(t_x)) := g(x)$, undefined off the literals, and likewise for a function machine on $\mathcal{I}$ (decode x from t_x). Γ_{g^P} is a set of literals $\pm R(t_x)$, at most one per x, and it is consistent with B: in a model of B in which the t_x are pairwise distinct, the symbol R, which B does not mention, can be interpreted as $\{t_x : g(x) = 1\}$. So g^P, and the machine version on every tape, is consistent, extends D iff g extends D^{str}, and costs a constant. *Sentences to strings:* $f^{\mathrm{str}}(x) := f(R(t_x))$, at a constant cost; $f \supseteq D$ implies $f^{\mathrm{str}} \supseteq D^{\mathrm{str}}$, and consistency is not needed. Sum over the injective maps, and take shortest programs. For the three outcome weights of $R(t_y)$: a consistent set of literals of this kind proves exactly its members (for $R(t_y)^b$ not in the set, reinterpret R at t_y), so the theory $\mathrm{Cn}(\Gamma_{g^P})$ decides $R(t_y)$ iff $g(y)$ is defined, with the same value; and the proof-search labeller of a theory, restricted to literals, labels y exactly as the theory decides $R(t_y)$. □

Corollary B.12 (What transfers)**.** *In the setting of Theorem 4.4, every result of the companion paper about binary labels that holds up to additive constants in* K*,* $-\log W$ *and* $-\log M_F$ *holds for SAI on literal data, with the same quantities. In particular:*

(a) no coding theorem for axiom systems*: for every* $d \ge 2$ *there are distinct* $x_1, x_2 \in \{0,1\}^d$ *such that* $D := \{(R(t_{x_1}), 0), (R(t_{x_2}), 1)\}$ *has* $K_A(D) \ge \log d - 2 - c$*,* $-\log W_A(D) \le 2 + K(\lceil \log(d+1) \rceil) + c$ *and* $-\log M_A(D) \le 2 + c$*; and over two-point literal data* $\log d - c \le \max_D(K_A(D) - L_A(D)) \le \log d + O(\log\log d)$ *(Theorem 4.9 and Corollary 4.10 of the companion paper); Theorem 3.9 proves the first three bounds directly;*

(b) random theories beat deterministic ones*:* $\sup_D M_A(D)/W_A(D) = \infty$ *over two-point literal data, and* $W_A(D) \ge 2^{-|D|-K^{\mathrm{pad}}(2^{d+1})-c}$ *for every literal data set over words of length at most* d *(Propositions 4.5(a) and 4.6 there);*

(c) shared randomness is not a mixture of private coins*, read as a comparison of* M_A *on literal data* D *with the companion paper's product mixture* M_{prod} *on* D^{str}*:* M_A/M_{prod} *is unbounded (Proposition 4.7 there); no product law internal to SAI is defined here;*

(d) Solomonoff induction*: with unary numerals* $t_i := S^i 0$ *and* $B \in \{\emptyset, \mathsf{Q}\}$*,* $M_A(\{(R(S^i 0), x_{i+1}) : i < n\})$ *is within* $2^{\pm c}$ *of the universal semimeasure of* $x \in \{0,1\}^n$ *(Proposition 3.24 there).*

Proof. Theorem 4.4 changes K, $-\log W$ and $-\log M_F$ by at most a constant, so every statement that holds up to additive constants in these quantities transfers. (a) is Theorem 4.9 and Corollary 4.10 of the companion paper applied to D^{str}; (b) is its Propositions 4.5(a) and 4.6; (c) is its Proposition 4.7, read through $M_A(D) \asymp M_F(D^{\mathrm{str}})$, so that it compares M_A with M_{prod} on D^{str}; (d) is its Proposition 3.24 with $t_i := S^i 0$, which are pairwise distinct in $\mathbb{N}$, a model of $B \in \{\emptyset, \mathsf{Q}\}$ (with R interpreted arbitrarily). The statements that do not transfer are of the following kinds: labels from a set larger than $\{0,1\}$ (a fresh unary predicate carries one bit); cumulative statements in the normalised convention (each normalised conditional transfers only within $2^{\pm 2c}$, so n steps cost $2cn$; single-step quantities do transfer up to a constant factor); private randomness per input or per query, and repeated inputs with fresh noise (no mixture over theory machines with a private coin per atom is defined here, and conflicting labels on one sentence are inconsistent data); history-aware and weak-semimeasure variants, which have no analogue here; computability statements, which are not "up to constants"; and other objects (time-bounded mixtures, circuits), which we compare directly (Theorems B.29 and B.43). □

Proposition B.13 (The price of the coin on literal data)**.** *In the setting of Theorem 4.4, for every literal data set D over words of length at most d,*

$$\log \frac{M_A(D)}{W_A(D)} \le \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c \le \delta_D + K(\lceil \log(d+1) \rceil) + c', \qquad \delta_D := |D| - K(D^{\mathrm{str}} \mid \operatorname{dom} D^{\mathrm{str}}).$$

Proof. $W_A(D) \ge 2^{-c} W(D^{\mathrm{str}}) \ge 2^{-|D| - K^{\mathrm{pad}}(2^{d+1}) - c'}$ by Theorem 4.4 and Proposition 4.6(b) of the companion paper, and $M_A(D) \le 2^c M_F(D^{\mathrm{str}}) \le 2^{-K(D^{\mathrm{str}} | \operatorname{dom} D^{\mathrm{str}}) + c''}$ by its Theorem 3.21(a). □

So on literal data whose labels are algorithmically random given the inputs ($\delta_D = O(1)$), the coin-free version loses at most $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ bits against the one-tape version, however many labels are predicted.

Remark B.14 (An encoding effect). Let χ_i be $R(t_{w_i})$ or $\neg R(t_{w_i})$ by independent fair coins, for distinct words w_i, with data $(\chi_i, 1)$, $i \le n$. Plain coin-free SOI on sentences, W^{all}, has bounded loss, through the labeller "always 1", while SAI and consistent SOI, coin-free or one-tape, lose at least n bits in expectation (Proposition E.1 of the earlier report). This is an encoding effect: the map $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ turns these data into fair-coin labels on positive literals and leaves M_A and W_A unchanged, since they depend on D only through $\mathrm{Cn}(D^{\pm})$ (Proposition 3.1); there SAI is SOI on strings, which loses at least $n - c$ bits in expectation on fair coins. The labeller "always 1" is inconsistent; consistency is what ties $\neg R(t_w)$ to $R(t_w)$.

Proof. The lower bound for SAI and consistent SOI is Proposition E.1 of the earlier report: on every tape the theory (or the consistent labeller) cannot accept both $R(t_w)$ and $\neg R(t_w)$, so the two predictive probabilities sum to at most 1, and convexity of $-\log$ gives at least one bit per step in expectation; the argument applies tape by tape to M_A and M_F^{cons}. The re-encoding $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ is a bijection of data sets that keeps $D^{\pm}$, so it keeps M_A and W_A (Proposition 3.1). On positive literals SAI is SOI on strings (Theorem 4.4), and on fair-coin labels the latter loses at least $n - c$ bits in expectation, by Theorem 3.21(a) of the companion paper and Gibbs' inequality. □

B.4 What breaks

Proof of Proposition 4.5. (a) A consistent theory proving ψ_D proves χ; and M_A depends on D only through $\mathrm{Cn}(D^{\pm})$ (Proposition 3.1). (b) A labeller $f \supseteq D$ with $f(\chi) = 0$ has $D^{\pm} \cup \{\neg\chi\} \subseteq \Gamma_f$,

which is inconsistent with B. By Theorem 4.2, $W^{\mathrm{cons}}(D+(\chi,1)) \geq 2^{-c}W_A(D+(\chi,1)) = 2^{-c}W_A(D) \geq 2^{-2c}W^{\mathrm{cons}}(D)$; for one tape use Theorem 4.3. (c) Let q_χ be a shortest V-program for χ. The prefix machine that reads q_χ, then $s \in \operatorname{dom} V$, and outputs an index of "$\{V(s)\}$, made undefined at $V(q_\chi)$" is simulated by V with a prefix r. The map $s \mapsto rq_\chi s$ is injective; removing one label preserves consistency, and the restricted labeller still extends D because $\chi \notin \operatorname{dom} D$. So the consistent labellers extending D and undefined at χ have weight at least $2^{-|r|-K(\chi)}W^{\mathrm{cons}}(D)$. For one tape, apply the same map to function machines and use (2); on the program-less tapes the empty labeller is already undefined at χ. (d) is Theorem 4.2(c) and Corollary B.4. □

Remark B.15 (The three outcomes are a measure for SAI). For $\varphi \notin \operatorname{dom} D$, $M_A(D+(\varphi,1)) + M_A(D+(\varphi,0)) + M^3_A(D+(\varphi,\iota)) = M_A(D)$ (Proposition 3.1), and likewise for W_A: each consistent theory proving ψ_D falls in exactly one of the three classes. For SOI the corresponding third mass is the defect, which collects abstention and divergence alike.

Productivity of the independent sentences. Ref_B and Thm_B are the sets of B-refutable and B-provable sentences. We use three textbook facts about Q (standard; see Boolos et al., 2007): (R) every decidable relation on numbers is strongly represented in Q, by a formula $\theta(\bar{x})$ with $\mathsf{Q} \vdash \theta(\underline{\bar{n}})$ when the relation holds of $\bar{n}$ and $\mathsf{Q} \vdash \neg\theta(\underline{\bar{n}})$ when it does not; (O) for every n, $\mathsf{Q} \vdash \forall z\,(z \leq \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ and $\mathsf{Q} \vdash \forall y\,(y \leq \underline{n} \vee \underline{n} \leq y)$, with $\leq$ the usual abbreviation; (Diag) for every formula $\vartheta(x)$ there is a sentence ρ, computable from ϑ, with $\mathsf{Q} \vdash \rho \leftrightarrow \vartheta(\ulcorner\rho\urcorner)$.

Lemma B.16 (Productivity of the independent sentences)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$. From an index of an r.e. set $S \subseteq \mathrm{Ind}_B$ one computes an $\mathcal{L}_{\mathrm{ar}}$-sentence $\rho \in \mathrm{Ind}_B \setminus S$. In particular Ind_B is productive, hence not r.e., and Ref_B is r.e. and not decidable.*

The lemma is classical: it is a form of the effective inseparability of the provable and the refutable sentences of consistent r.e. extensions of Q (cf. Smullyan, 1961). We include the proof, which is Rosser's argument relative to S.

Proof. Let $\mathsf{Neg}(y,x)$ be the decidable relation "y codes a B-derivation of the negation of the sentence coded by x, or x is enumerated into S within y steps", and $\mathsf{Prf}(y,x)$ the decidable relation "y codes a B-derivation of the sentence coded by x" (B is decidable). Let $\theta_{\mathrm{neg}}, \theta_{\mathrm{prf}}$ strongly represent them (R), and let ρ satisfy $\mathsf{Q} \vdash \rho \leftrightarrow \exists y\,(\theta_{\mathrm{neg}}(y,\ulcorner\rho\urcorner) \wedge \forall z \leq y\ \neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner))$ (Diag); ρ is computed uniformly from the index of S. $B \nvdash \rho$. Suppose k codes a B-derivation of ρ. Then $\rho \notin \mathrm{Ind}_B \supseteq S$, and $B \nvdash \neg\rho$ by consistency, so $\mathsf{Neg}(n,\ulcorner\rho\urcorner)$ fails for every n and $\mathsf{Q} \vdash \neg\theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner)$ for every n (R); also $\mathsf{Q} \vdash \theta_{\mathrm{prf}}(\underline{k},\ulcorner\rho\urcorner)$. Argue in Q: given y with $\theta_{\mathrm{neg}}(y,\ulcorner\rho\urcorner) \wedge \forall z \leq y\,\neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner)$, by (O) either $\underline{k} \leq y$, which contradicts $\theta_{\mathrm{prf}}(\underline{k},\ulcorner\rho\urcorner)$, or $y \leq \underline{k}$, so that y is one of $\underline{0},\ldots,\underline{k}$, which contradicts $\neg\theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner)$. So $\mathsf{Q} \vdash \neg\rho$, hence $B \vdash \neg\rho$, and B is inconsistent, a contradiction. $B \nvdash \neg\rho$ *and* $\rho \notin S$. Otherwise $\mathsf{Neg}(n,\ulcorner\rho\urcorner)$ holds for some n, while $\mathsf{Prf}(m,\ulcorner\rho\urcorner)$ fails for every m by the previous step. By (R) and (O), $\mathsf{Q} \vdash \theta_{\mathrm{neg}}(\underline{n},\ulcorner\rho\urcorner) \wedge \forall z \leq \underline{n}\,\neg\theta_{\mathrm{prf}}(z,\ulcorner\rho\urcorner)$, so $\mathsf{Q} \vdash \rho$ and $B \vdash \rho$, against the previous step. So $\rho \in \mathrm{Ind}_B \setminus S$. A productive set is not r.e. (take S to be the set itself). Ref_B is r.e. since B is decidable; if it were decidable, so would be $\mathrm{Thm}_B = \{\varphi : \neg\varphi \in \mathrm{Ref}_B\}$, hence Ind_B, which is not even r.e. □

Proof of Proposition 4.6. (a) Coin-free: map a consistent three-valued labeller f to the enumerator of Γ_f (its 1/0 part); the theory $\mathrm{Cn}(\Gamma_f)$ proves the 1/0 data and, by the consistency of f, decides none of the sentences f labels ι, which include those of D. One tape: emission, tape by tape, on the tapes on which f^F_ω is consistent and extends D, and (2). (b) Coin-free: let $\mathcal{F}_n$ be

the finite (not computable) set of the $p \in \operatorname{dom} V$ with $|p| \le n$ for which $\{V(p)\}$ is a consistent three-valued labeller, and I_n the union of their ι-sets. It is a finite union of r.e. sets, hence r.e., and $I_n \subseteq \mathrm{Ind}_B$, since $f(\sigma) = \iota$ for a consistent f implies $B \cup \Gamma_f \nvdash \sigma, \neg\sigma$. Let $\sigma_n \in \mathrm{Ind}_B \setminus I_n$ (Lemma B.16). Every consistent labeller extending $\{(\sigma_n, \iota)\}$ labels σ_n with ι, so its program is not in $\mathcal{F}_n$ and is longer than n; hence $W^{3,\mathrm{cons}}(\{(\sigma_n, \iota)\}) \le \Omega_V^{\ge n+1}$. The enumerator of $\emptyset$ (theory B) is compatible with $\{(\sigma_n, \iota)\}$, so $W_A^3(\{(\sigma_n, \iota)\}) \ge 2^{-c}$. One tape: fix n and $\varepsilon := 2^{-n}$. For $p \in \operatorname{dom} V$ with $|p| \le n$ let $F_p := F_{V(p)}$ and Cons_p the (Borel) set of tapes on which $f_\omega^{F_p}$ is consistent. If $\mathbb{U}(\mathrm{Cons}_p) = 0$, p contributes nothing. Otherwise, by the Lebesgue density theorem (Lemma C.1) almost every $\omega \in \mathrm{Cons}_p$ has a prefix w with $\mathbb{U}(\mathrm{Cons}_p \cap [w]) > (1-\varepsilon)2^{-|w|}$. Let $\mathcal{W}_p$ be the set of minimal such prefixes, a prefix-free set with $\mathbb{U}(\mathrm{Cons}_p \setminus \bigcup_{w \in \mathcal{W}_p} [w]) = 0$, and choose a finite $\mathcal{W}_p' \subseteq \mathcal{W}_p$ with $\sum_{w \in \mathcal{W}_p \setminus \mathcal{W}_p'} 2^{-|w|} \le \varepsilon$. For $w \in \mathcal{W}_p'$ put

$$J_w := \{\sigma : \mathbb{U}([w] \cap \{\omega : f_\omega^{F_p}(\sigma) = \iota\}) > \varepsilon 2^{-|w|}\}.$$

J_w is r.e., uniformly in (p, w) and the rational ε, because the measure of the open set $[w] \cap \{f(\sigma) = \iota\}$ is lower semicomputable. And $J_w \subseteq \mathrm{Ind}_B$: if $\sigma \in J_w$, the set $[w] \cap \{f(\sigma) = \iota\}$ has measure larger than $\varepsilon 2^{-|w|} > \mathbb{U}([w] \setminus \mathrm{Cons}_p)$, so it meets Cons_p; on such a tape f_ω is consistent and labels σ with ι, so $B \cup \Gamma_{f_\omega}$, and hence B, decides neither σ nor $\neg\sigma$. Let $I_n' := \bigcup_{|p| \le n} \bigcup_{w \in \mathcal{W}_p'} J_w$, a finite union of r.e. subsets of Ind_B, and $\sigma_n' \in \mathrm{Ind}_B \setminus I_n'$ (Lemma B.16). For $|p| \le n$, since $\sigma_n' \notin J_w$,

$$\begin{aligned} \nu_{F_p}^{3,\mathrm{cons}}(\{(\sigma_n', \iota)\}) &\le \sum_{w \in \mathcal{W}_p'} \mathbb{U}([w] \cap \{f(\sigma_n') = \iota\}) + \sum_{w \in \mathcal{W}_p \setminus \mathcal{W}_p'} 2^{-|w|} + \mathbb{U}\Big(\mathrm{Cons}_p \setminus \bigcup_{w \in \mathcal{W}_p} [w]\Big) \\ &\le \sum_{w \in \mathcal{W}_p'} \varepsilon 2^{-|w|} + \varepsilon \le 2\varepsilon. \end{aligned}$$

By (2), $M_F^{3,\mathrm{cons}}(\{(\sigma_n', \iota)\}) \le \sum_{|p| \le n} 2^{-|p|} 2\varepsilon + \Omega_V^{\ge n+1} \le 2^{1-n} + \Omega_V^{\ge n+1}$. The deterministic theory machine that emits nothing (theory B) is compatible with $\{(\sigma_n', \iota)\}$ on every tape, so $M_A^3(\{(\sigma_n', \iota)\}) \ge 2^{-c}$. (c) For $\sigma \in \mathrm{Ind}_B$ the labeller "ι at σ, undefined elsewhere" is consistent and has length $K(\sigma) + c$; with $W_A^3 \le 1$ this gives the coin-free bound, and as a function machine that ignores the tape it gives the one-tape bound. □

The coin-free part of (b) is the special case of machines that ignore the tape: then J_w is the labeller's ι-set and no density argument is needed. One tape, the obstacle is that the set of sentences a random machine labels ι on consistent tapes need not be r.e., since it depends on the positivity of the measure of a Π_1^0 class; the density point restores an r.e. subset of Ind_B that carries almost all of the consistent ι-mass. A computation on finite analogues (code in the accompanying repository), with machines consistent on most tapes and noise that labels B-decided sentences ι, confirms the density step and the summed bound in every case tested.

Proposition B.17 (Large gaps have no r.e. witnesses)**.** *In the setting of Proposition 4.6: (a) for every r.e. $S \subseteq \mathrm{Ind}_B$ and every $\sigma \in S$, $W^{3,\mathrm{cons}}(\{(\sigma, \iota)\}) \ge 2^{-K(S)-c}$ and $M_F^{3,\mathrm{cons}}(\{(\sigma, \iota)\}) \ge 2^{-K(S)-c}$, with $K(S)$ the complexity of an index of S; so no computable sequence $\sigma_n \in \mathrm{Ind}_B$ has $W^{3,\mathrm{cons}}(\{(\sigma_n, \iota)\}) \to 0$. (b) If $n(\sigma)$ is the length of a shortest program of a consistent three-valued labeller that labels σ with ι, then $2^{-n(\sigma)} \le W^{3,\mathrm{cons}}(\{(\sigma, \iota)\}) \le \Omega_V^{\ge n(\sigma)}$.*

Proof. (a) The labeller "ι on every member of S (enumerate S), undefined elsewhere" has $\Gamma = \emptyset$ and labels only B-independent sentences, so it is consistent; it has length $K(S) + c$. As a function machine that ignores the tape it gives the one-tape bound. (b) Every program counted has length at least $n(\sigma)$, and one has length $n(\sigma)$. □

Remark B.18 (Shared versus private randomness). If a product law (private coins per input, Proposition 3.9(a) of the companion paper) gives positive probability to labelling φ true and to labelling ψ true, with $B \cup \{\varphi, \psi\}$ inconsistent, the sampled labeller is inconsistent with probability at least the product of the two. The consistent part of a product law is not a product law: for a B-independent φ, on the two inputs φ, $\neg\varphi$ with fair independent labels, the consistent outcomes are $(1,0)$ and $(0,1)$, each of mass $\frac{1}{4}$, while a product would need $P(1,1)P(0,0) = P(1,0)P(0,1) \neq 0$. (If B decides φ, one of these two outcomes is inconsistent too.) The shared tape is what makes a random labeller a random *theory*; private randomness is harmless on logically independent inputs such as positive literals (Theorem 4.4). Product laws are machine laws, so M_F^{cons} dominates their consistent parts.

Proposition B.19 (Computability of the versions)**.** *(a) Suppose the set of finite data sets D with $B \cup D^{\pm}$ consistent is not r.e. (for instance if $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, since then $\{\varphi : \{(\varphi, 1)\}$ is consistent$\}$ is the complement of Ref_B, which is r.e. and not decidable by Lemma B.16; or if $B = \emptyset$ and $\mathcal{L}$ has a binary relation symbol, by Church's theorem, Church, 1936). Then none of W_A, W_A^{dec}, W^{cons}, W^{cl}, M_A, M_A^{dec}, M_F^{cons}, M_F^{cl} is lower semicomputable.*

(b) In every setting, $M_A(\emptyset)$ and $M_F^{\mathrm{cons}}(\emptyset)$ are not left-c.e.; so M_A and M_F^{cons} are not lower semicomputable.

(c) W_A, W^{cons}, M_A and M_F^{cons} are computable relative to $\emptyset'$, uniformly in D.

(c′) W_A^{dec}, M_A^{dec}, W^{cl} and M_F^{cl} are computable relative to $\emptyset''$, uniformly in D.

Whether the four versions of (c′) are computable relative to $\emptyset'$ is not settled here. Being closed and being a (random) decider are Π_2 conditions, and for closedness the condition is Π_2-complete: with $\chi_0, \chi_1, \ldots$ an injective enumeration of the B-decided sentences, the labeller that labels χ_n with B's decision once program e halts on n is consistent, and it is closed iff e is total.

Proof. (a) For a function ν that is lower semicomputable uniformly in D, the support $\{D : \nu(D) > 0\}$ is r.e. Each object listed has support exactly the consistent data sets: positive weight needs a consistent hypothesis that proves or extends D; conversely, for a consistent D the lookup table of D (as a labeller, a finite theory, a decider, or a machine that ignores the tape) has positive weight in W_A, W_A^{dec}, W^{cons}, M_A, M_A^{dec} and M_F^{cons}, and for the closed versions W^{cl}, M_F^{cl} its closure "proof search from $B \cup D^{\pm}$", which is closed, consistent and extends D, has positive weight (the lookup table itself is never closed). (b) For M_A this is Theorem 3.3. For M_F^{cons}: for a left-c.e. real ζ with approximations $\zeta_s \uparrow \zeta$, let F_ζ label the sentence $\exists x\, x \neq x$ with 1 under a prefix w as soon as some stage s has $0.w + 2^{-|w|} \leq \zeta_s$, and be undefined elsewhere; its labelling is inconsistent with probability $\mathbb{U}\{\omega : 0.\omega < \zeta\} = \zeta$. With p_ζ a V-program for F_ζ, the probability that f_ω is inconsistent is $2^{-|p_\zeta|}\zeta + \kappa$ with κ left-c.e. If $M_F^{\mathrm{cons}}(\emptyset)$, which is one minus that probability, were left-c.e., the probability would be computable and ζ right-c.e.; take $\zeta = \Omega_V$. (c) W_A and W^{cons} are sums $\sum_p 2^{-|p|}\mathbf{1}[\cdot]$ of Δ_2 indicators ("proves ψ_D" and "extends D" are Σ_1, consistency is Π_1), and the tails $\Omega_V^{\geq \ell}$ are $\emptyset'$-computable; so $\emptyset'$ computes each value to any precision. For one tape, $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathbb{U}(O_\perp)$ (Proposition A.5), and $M_F^{\mathrm{cons}}(D) = \mathbb{U}(O) - \mathbb{U}(O \setminus C)$ with $O := \{\omega : f_\omega \supseteq D\}$ effectively open and $C := \{\omega : f_\omega$ consistent$\}$ effectively closed: differences of left-c.e. reals, uniformly in D. (c′) Coin-free: "$V(p)$ is a total $\{0,1\}$-valued program" and "$\{V(p)\}$ is closed" $(\forall \varphi, b : B \cup \Gamma \vdash \varphi^b \to \{V(p)\}(\varphi) = b)$ are Π_2, so $\emptyset''$ decides each indicator and computes the sums. One tape, deciders: "$F_{V(p)}$ halts on every query and every tape" is Π_2, because for each query, halting on all tapes is, by compactness of Cantor space, equivalent

to the Σ_1 condition that some n and s make every tape prefix of length n halt within s steps. For a random decider $\mu_F^{\mathrm{dec}}(D)$ is a difference of left-c.e. reals uniformly in an index, as in (c); so $\emptyset''$ computes $M_A^{\mathrm{dec}}(D)$. One tape, closed: "f_ω is closed" is the countable intersection over (φ, b) of the sets $\{\omega : B \cup \Gamma_{f_\omega} \nvdash \varphi^b\} \cup \{\omega : f_\omega(\varphi) = b\}$, each the union of an effectively closed and an effectively open set, hence an effective G_δ. So the event in $M_F^{\mathrm{cl}}(D)$ is an effective Π_2^0 class $\bigcap_k U_k$ with U_k uniformly effectively open and decreasing; its measure $\inf_k \mathbb{U}(U_k)$ is an infimum of uniformly left-c.e., hence uniformly $\emptyset'$-computable, reals, so it is right-c.e. relative to $\emptyset'$ and computable from $\emptyset''$. □

Question B.20. Can W_A be lower semicomputable when consistency of finite sets of sentences with B is decidable (for instance for a propositional language, or a monadic language without function symbols and $B = \emptyset$)?

There the support argument of Proposition B.19(a) says nothing, and the argument of (b) does not carry over directly to the coin-free version, because the weight of the inconsistent programs is not separated from Ω_V by a computable amount.

Pathwise and machine-wise consistency. M_F^{cons} conditions on the event that the *sampled* labeller is consistent. The machine-wise alternative is

$$M_F^{\mathrm{mw}}(D) := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\, \mathbf{1}[F_{V(p)} \text{ is consistent on every tape}]\, \nu_{F_{V(p)}}(D),$$

and M_A^{mw} is defined likewise over the theory machines whose theory is consistent on every tape.

Proposition B.21 (Pathwise against machine-wise consistency)**.** *(a) $W^{\mathrm{cons}} \le 2^c M_F^{\mathrm{mw}} \le 2^c M_F^{\mathrm{cons}}$ on nonempty D, and $M_F^{\mathrm{mw}} \asymp M_A^{\mathrm{mw}}$.*

(b) For every consistent $D \neq \emptyset$, $M_F^{\mathrm{cons}}(D) \le 2^{\min\{K^{\mathrm{pad}}(L_{1/2}^{\mathrm{cons}}(D)),\, K(\mathrm{dom}\, D)\}+c}\, M_F^{\mathrm{mw}}(D)$, with $L_{1/2}^{\mathrm{cons}}$ as in Proposition B.11.

(c) If consistency of finite sets with B is decidable (an algorithm decides, for finite $\Lambda \subseteq \mathrm{Sent}$, whether $B \cup \Lambda$ is consistent; for example if $\mathrm{Cn}(B)$ is decidable, for propositional languages, and for monadic languages without function symbols with $B = \emptyset$, by the decidability of monadic first-order logic, Löwenheim, 1915; Behmann, 1922), then $M_F^{\mathrm{cons}} \asymp M_F^{\mathrm{mw}}$ and $M_A \asymp M_A^{\mathrm{mw}}$.

Proof. (a) A deterministic consistent labeller is consistent on every tape, and a machine consistent on every tape has $\nu = \nu^{\mathrm{cons}}$. Proof search and emission preserve "consistent on every tape", since on each tape proof search of a consistent theory is consistent and emission of a consistent labelling is consistent. (b) Proposition B.11 with (a). (c) *The global-schedule filter.* For a function machine F, read as a machine that reads the tape one-way, let F' answer a query φ by running the *global schedule* of F: stage s reads tape bit s, starts F on the s-th sentence and advances every started run by one step; certification events (a run halting with a label) are ordered by stage, ties by sentence index. When φ's event occurs, F' checks whether the set of labels certified up to and including it is consistent with B (decidable by hypothesis); if so it outputs φ's label, otherwise it diverges. The answer is a deterministic function of the bits read, so F' is a function machine, and $K(F') \le K(F) + c$. On a tape ω let $l_1, l_2, \ldots$ be the labelled sentences in event order. F' outputs the label of l_i iff $\{l_1, \ldots, l_i\}$ is consistent; consistency of these prefixes is monotone, so the outputs form an initial segment all of whose finite subsets are consistent, hence a consistent

set, by compactness. So F' is consistent on every tape, and $F' = F$ on every tape on which F is consistent. Hence $\nu_{F'}(D) \geq \nu_F^{\mathrm{cons}}(D)$ for every D, and summing with (2) gives $M_F^{\mathrm{mw}} \geq 2^{-c} M_F^{\mathrm{cons}}$. The theory-machine version filters emissions in the same way. □

With two unary function symbols, consistency of finite sets is not decidable, so in (c) "monadic" must exclude function symbols.

Question B.22. Is $M_F^{\mathrm{cons}}/M_F^{\mathrm{mw}}$ bounded in general, for instance for $B = \mathsf{Q}$, or for $B = \emptyset$ and a language with a binary relation symbol?

A machine consistent on every tape must avoid inconsistency, which is only semi-decidable, so its support is an r.e. family of consistent data sets; a pathwise-conditioned machine can exploit the Π_1^0 event that nothing inconsistent ever happens. The lottery does not separate the two: the lottery machine is consistent on every tape (Theorem 5.1).

B.5 Time bounds with free deduction

As in Appendix E.6 of the earlier report, the penalised coin-free hypotheses are programs of the machine U of Assumption B.5, so $\mathrm{time}_p(x) \geq |p|$, and $a' := \max(a, 2)$. Clocked strings, their weights $w(f)$ and $c_\perp$ are as in Section 4.5, with E the Elias-δ code of $|f| + 1$ followed by f; c_1 is the length of the fixed wrapper that runs a clocked string and diverges when it abstains. Simulations between U_0, U and multitape machines have polynomial overhead (standard; used in the same form as there).

Proposition B.23 (A Levin penalty costs a logarithm)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible, m_0 the least length of a sentence, $W_A^{\mathrm{Kt},t}$ the mixture over all deciders p with weight $2^{-|p|}/c_p$, $c_p := \sup_\chi \mathrm{time}_p(\chi)/t(|\chi|)$ (Definition E.10 of the earlier report), and $W^1(D) := \sum_f 2^{-|f|}/(|f|+1)$ over the programs f of consistent labellers extending D. Under Assumption B.5, $W_A^{\mathrm{Kt},t}(\emptyset) \leq t(m_0)$, and there is c with $2^{-c} W^1(D) \leq W_A^{\mathrm{Kt},t}(D) \leq 2^c t(m_0) W^1(D)$ for every D; so a labeller f pays $\log(|f| + 1) + O(1)$ bits beyond its unpenalised weight.*

Proof. This is Proposition E.13 of the earlier report. Since $\mathrm{time}_p \geq |p|$, $c_p \geq |p|/t(m_0)$, so p has weight at most $t(m_0) 2^{-|p|}/|p|$, and the total is at most $t(m_0)$ by Kraft's inequality. For the lower bound, the padded decider of a consistent labeller f (as in the proof of Theorem B.9) has $c_d \leq \beta(|f| + 1)$ for a constant β, by Lemma E.12(b) there and $(|\chi| + 1)^{a'} \leq 2^{a'} t(|\chi|)$, so it has weight at least $2^{-|f|-2c}/(\beta(|f|+1))$; the map is injective. For the upper bound, map a compatible decider p to its proof-search labeller f_p, with $|p| \leq |f_p| \leq |p| + c$; the weight of p is at most $t(m_0)(c + 2) 2^c \, 2^{-|f_p|}/(|f_p| + 1)$, and $p \mapsto f_p$ is injective. □

Theorem B.24 (Domination lemma)**.** *Let X be a coin-free mixture and v_X a weight on consistent labellers such that $W_X(D) \geq 2^{-c_X} \sum_g v_X(g)$, the sum over the consistent labellers g extending D, and $W_X(\emptyset) \leq \Omega_X$. Let Y be a mixture with $W_Y(D) = \sum_h v_Y(h, D)$, the sum over the hypotheses h compatible with D, with an injective map $h \mapsto g_h$ to consistent labellers that compute the same partial map as h, with $v_Y(h, D) \leq k_Y \, v_X(g_h)$, and with $W_Y(\emptyset) \geq \kappa_Y > 0$. Then $\ell_X(D) \leq \ell_Y(D) + c_X + \log(k_Y \Omega_X/\kappa_Y)$ for every D.*

Proof. A labeller g_h computes the same partial map as h, so it extends D iff h is compatible with D. Hence $W_X(D) \geq 2^{-c_X} \sum_h v_X(g_h) \geq 2^{-c_X} k_Y^{-1} W_Y(D)$, and $\ell_X(D) = -\log W_X(D) + \log W_X(\emptyset) \leq -\log W_Y(D) + c_X + \log(k_Y \Omega_X) = \ell_Y(D) - \log W_Y(\emptyset) + c_X + \log(k_Y \Omega_X)$. □

The weaker hypothesis $\Gamma_{g_h} = \Gamma_h$ does not suffice: a labeller that rejects ψ and one that accepts $\neg\psi$ and abstains on ψ have the same Γ but differ on the datum $(\psi, 0)$.

Corollary B.25 (A one-sided equivalence)**.** *For every time-constructible τ there is c_τ such that, on every data set D, the losses of W_A and W_A^{dec} exceed that of $W^{\mathrm{cons},\tau}$ by at most c_τ, and the loss of M_A exceeds that of $M_F^{\mathrm{cons},\tau}$ by at most c_τ. Under Assumption B.5 the same holds for $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$, $t(m) \geq m^{a^*}$ time-constructible.*

Proof. Coin-free (Theorem 6.14(b) of the earlier report): apply Theorem B.24 with $X \in \{W_A, W_A^{\mathrm{dec}}, W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}\}$ and $Y = W^{\mathrm{cons},\tau}$. By Theorem 4.2 and, for the penalised versions under Assumption B.5, Theorem B.9(b), $W_X \geq 2^{-c} W^{\mathrm{cons}}$, and $W_X(\emptyset) \leq 1$ by Kraft's inequality. For a clocked string h let g_h be the wrapper that runs h^τ and diverges when it abstains, followed by $E(h)$; it computes the same partial map as h^τ, $|g_h| = |E(h)| + c_1$, and $h \mapsto g_h$ is injective. $W^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$, the weight of the always-abstaining string (B is consistent). So $c_\tau := c + c_1 + c_\perp$ works. One tape: the clocked function machine of a string f is a function machine whose V-program is a fixed wrapper followed by $E(f)$, injectively, so (2) gives $M_F^{\mathrm{cons}} \geq 2^{-c_1} M_F^{\mathrm{cons},\tau}$ on $D \neq \emptyset$; Theorem 4.3 gives $M_A \geq 2^{-c} M_F^{\mathrm{cons}}$; and $M_A(\emptyset) \leq 1$, $M_F^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$. □

Proof of Theorem 4.7. This is Theorem 6.15 of the earlier report; we recall the argument. By the MRDP theorem (Matiyasevich, 1970; Davis, 1973) there are polynomials E, E' with natural-number coefficients such that program e halts on x with output y iff $\mathbb{N} \models \exists \bar{u}\, (E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}))$; $\mathrm{Out}(e,x,y)$ is this formula (Lemma E.14(a) there), and $\mathrm{DET} := \forall e\, \forall x\, \forall y\, \forall y'\, \forall \bar{u}\, \forall \bar{u}'\, (E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}) \wedge E(e,x,y',\bar{u}') = E'(e,x,y',\bar{u}') \to y = y')$, a true Π_1 sentence. Let $H_0 := \mathsf{Q} \cup \{\mathrm{DET}\}$. *Construction.* Let $\mathcal{G}(e', n)$ be the program that, for $j = 1, \ldots, n$, forms $\sigma_j := \mathrm{Out}(\underline{e'}, \underline{j}, S0)$, asks P for approximations (a_0, a_1) within 2^{-j-3} of its pair on the history $((\sigma_1, \beta_1), \ldots, (\sigma_{j-1}, \beta_{j-1}))$ and the sentence σ_j, and sets $\beta_j := 0$ if $a_0 \leq a_1$ and $\beta_j := 1$ otherwise; it outputs β_n. $\mathcal{G}$ is total, as P is. By Kleene's recursion theorem (standard) there is e with $e(n) = \mathcal{G}(e, n)$ for all n. The values β_j computed inside $\mathcal{G}(e, n)$ do not depend on n, so $e(j) = \beta_j \in \{0, 1\}$ and $\sigma_j = s_j$; and $\mathbb{N} \models s_n$ iff $e(n) = 1$, so the label b_n is β_n, and P is queried exactly on the true histories D_{j-1}. *H_0 decides each s_n.* If $b_n = 1$, s_n is a true existential equation, which Q proves (Lemma E.14(c) there). If $b_n = 0$, $\mathrm{Out}(\underline{e}, \underline{n}, 0)$ is one, so Q proves it, and with s_n and DET it gives $S0 = 0$, which Q refutes; so $H_0 \vdash \neg s_n$. *The loss of P.* With $q_b := q_b(D_{j-1}, s_j)$ and $\varepsilon := 2^{-j-3}$, the tie rule gives $q_{b_j} \leq a_{b_j} + \varepsilon \leq a_{1-b_j} + \varepsilon \leq q_{1-b_j} + 2\varepsilon$, and with $q_0 + q_1 \leq 1$ this gives $q_{b_j} \leq \frac{1}{2} + 2^{-j-3}$. So $-\log q_{b_j} \geq 1 - \log(1 + 2^{-j-2})$, and $\sum_j \log(1 + 2^{-j-2}) \leq \sum_j 2^{-j-2} / \ln 2 = 1/(4 \ln 2) < 0.361$. *Axiom induction.* H_0 is true in $\mathbb{N}$ and B in an expansion of $\mathbb{N}$, so $B \cup H_0$ is consistent, and H_0 is compatible with every D_n. It is finite, so it has a decider running in linear time and an enumerator writing finitely many axioms; let $\omega_0 > 0$ be the least of their weights in W_A, $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$. Then $W(D_n) \geq \omega_0$, $W(\emptyset) \leq 1$, and the loss is at most $\log(1/\omega_0)$, whatever P is. *Clocked SOI.* The last claim concerns the sequence built against one predictor. Put $P^*(D) := (W^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$ and $q_b(D, \varphi) := P^*(D + (\varphi, b))/P^*(D)$. Then $q_0 + q_1 \leq 1$, since $W^{\mathrm{all},\tau}$ is a semimeasure and the uniform part halves at each step; P^* is computable, since clocked compatibility is decidable and the weights have computable tails; $P^*(D) \geq 2^{-|D|-1} > 0$; and $W^{\mathrm{all},\tau}(\emptyset) = 1$, so $P^*(\emptyset) = 1$ and $\ell_{P^*}(D_n) = -\log P^*(D_n)$. On the sequence built against P^*, $W^{\mathrm{cons},\tau}(D_n) \leq W^{\mathrm{all},\tau}(D_n) \leq 2P^*(D_n)$, so $\ell_{W^{\mathrm{all},\tau}}(D_n) \geq n - 1.37$ and $\ell_{W^{\mathrm{cons},\tau}}(D_n) \geq \ell_{P^*}(D_n) - 1 + \log W^{\mathrm{cons},\tau}(\emptyset) \geq n - 1.37 - c_\perp$. □

Proposition B.26 (Incomparability with time-bounded predictors that may be inconsistent)**.** *On the data of Remark B.14, $W^{\mathrm{all},\tau}$ has bounded loss for every τ that covers the constant running*

time of "always 1", while every consistent inducer (SAI and consistent SOI, coin-free or one-tape) loses at least n bits in expectation; on a computable sequence (Theorem 4.7, applied to P^*), $W^{\mathrm{all},\tau}$ loses at least $n - 1.37$ while SAI loses $O(1)$. So $W^{\mathrm{all},\tau}$ and SAI are incomparable.

Proof. Once τ covers its running time, the clocked string "always 1" is compatible with every prefix of the data of Remark B.14, so the loss of $W^{\mathrm{all},\tau}$ is at most $-\log w(\text{always 1})$. The other two claims are Remark B.14 and Theorem 4.7, the latter for the sequence built against P^*. □

Proof of Proposition 4.8. This is Proposition E.15(a), (b) of the earlier report. (a) $\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true iff e outputs 1 on n; if e halts on n with another output, $\neg\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true, as a program has at most one output. So every label of u is true in $\mathbb{N}$. (b) In Theorem 4.7 e is total with values in $\{0, 1\}$, so u labels every s_n correctly and extends every D_n; hence $W^{\mathrm{cons}}(D_n) \geq 2^{-|u|-c}$, where c pays for turning u into a V-program for a labeller index, and with $W^{\mathrm{cons}}(\emptyset) \leq 1$ the loss is at most $|u| + c$. The same holds for $W^{\mathrm{all}} \geq W^{\mathrm{cons}}$. □

Remark B.27 (A time-hierarchy effect)*. Proof sketch.* We do not write out the version of the diagonal theorem that gives the compared predictors the same reuse of state, nor the dependence on the machine model. The claim is that under a time bound the sequence that defeats a time-bounded predictor is itself computable within a slightly larger bound. For the predictors S_p of Proposition E.15(c) of the earlier report, which spend time $p(n) \log n$ per step and reuse their state, the labels that defeat S_p are produced by a predictor with per-step time $O(p(n) \log n)$ and state reuse that predicts its own labels with certainty; so $S_{p'}$, for a slightly larger p' (for example $np(n)$), has bounded loss on the sequence that defeats S_p.

Proposition B.28 (Disguised labellers keep their weight)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible and X be $W_A^{\mathrm{chk},t}$ or $W_A^{\mathrm{gen},t}$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C}; D)$ and $W_A^{\mathrm{dec}}(\mathcal{C}; D)$ be the weights of the hypotheses compatible with D whose theory is in $\mathcal{C}$. Under Assumption B.5 there is c with $2^{-c} W_A^{\mathrm{dec}}(\mathcal{C}; D) \leq X(\mathcal{C}; D) \leq 2^c W_A^{\mathrm{dec}}(\mathcal{C}; D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of positive weight, genuine and disguised alike, move by at most 2^{2c}, and a slow consistent labeller f keeps prior weight at least $2^{-|f|-c}$, as a padded Craig set.*

Proof. This is Proposition E.16 of the earlier report: the padded sets of Lemma E.12 there keep the theory and meet the penalty, which gives the lower bound; for the upper bound the penalised deciders are deciders with the same weights, and an enumerator q maps to the padded decider of "yes iff q writes the input", which keeps the theory. Divide by the total weights. □

Theorem B.29 (One tape and the companion paper's time-bounded mixtures)**.** *Let $M_F^{\mathrm{cons},\tau} := \sum_f w(f) \nu_{F_f^\tau}^{\mathrm{cons}}$, where F_f^τ runs f on U_0 with a tape-reading instruction for at most $\tau(|\varphi|)$ steps and abstains otherwise, and let $M_F^{\mathrm{all},\tau}$ drop the consistency restriction. Parts (b)–(d) assume that $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and that B is decidable and true in an expansion of $\mathbb{N}$.*

- *(a) $\ell_{M_A}(D) \leq \ell_{M_F^{\mathrm{cons},\tau}}(D) + c_\tau$ for every D.*
- *(b) On a computable sequence of arithmetic sentences, $\ell_{M_F^{\mathrm{all},\tau}} \geq n - 1.37$ and $\ell_{M_F^{\mathrm{cons},\tau}} \geq n - 1.37 - c_\perp$, while $\ell_{M_A} \leq \log(1/\omega_0)$.*
- *(c) The length-activated and round-activated mixtures of the companion paper (Definition 9.1 and Proposition 9.4 there), with computable weights whose tails are computable, are computable predictors, so each loses at least $n - 0.361$ on a computable sequence on which SAI loses*

$O(1)$. *Conversely, on the coin literals of Remark B.14, all labelled true, the round-activated mixture has bounded loss, while SAI and consistent SOI lose at least n bits in expectation. So the companion paper's time-bounded SOI and SAI are incomparable.*

(d) Unpenalised W^{cons} and M_F^{cons} have loss at most $|u|+c$ on the diagonal sequences of (b) and (c).

Proof. (a) is the one-tape part of the proof of Corollary B.25. (b) Each query of F_f^τ reads at most $\tau(|\varphi|)$ tape bits, so $\nu_{F_f^\tau}(D)$ is a finite union of cylinders, computable exactly; with computable tails, $M_F^{\mathrm{all},\tau}$ is computable, a semimeasure with value 1 at $\emptyset$, and $P(D) := (M_F^{\mathrm{all},\tau}(D)+2^{-|D|})/2$ is a computable predictor; run Theorem 4.7 against P, as in its last step. The theory machine that emits H_0 is compatible with every D_n on every tape. (c) Theorem 4.7 needs only computability of the predictor. The converse is Proposition 9.4 of the companion paper with the constant comparator "label true with probability 1", which is in its class $\mathcal{C}_t$, and Proposition E.1 of the earlier report, applied tape by tape (Remark B.14). (d) Proposition 4.8, with (2) for M_F^{cons}. On the coin literals of (c), by contrast, every consistent inducer, W^{cons} and M_F^{cons} included, loses at least n bits in expectation. □

B.6 Charged deduction

We use the setting of Definition E.18 of the earlier report: $\mathcal{L}$ is finite, B is decidable in polynomial time, derivations are in the Hilbert calculus K of that report, with sizes counting symbols, and a hypothesis is a program p that decides an axiom set A_p in time $O(t)$. Each datum φ must have a derivation of φ^b from $B \cup A_p$ whose size is at most $s(|\varphi|)$ ($W_A[t,s]$); in which every axiom instance, logical ones included, has at most $h(|\varphi|)$ symbols (reading (a1), $W_A^{a1}[t,h]$), or every nonlogical instance does (reading (a2), $W_A^{a2}[t,h]$); or whose *material*, the total size of the distinct instances used, is at most $g(|\varphi|)$ (reading (b), $W_A^b[t,g]$), or whose nonlogical material is (reading (b2), $W_A^{b2}[t,g]$). *Certificate* SOI, $W^{\mathrm{cert}}[t',s']$, weighs the consistent labellers $f_{\mathcal{V}}$ defined by verifiers $\mathcal{V}$ running in time $O(t')$: $f_{\mathcal{V}}(\varphi)$ is 1 (resp. 0) if some certificate of length at most $s'(|\varphi|)$ makes $\mathcal{V}$ accept (resp. reject). All charged results assume the renaming condition of that report; without it the index of a single parameter, which costs one symbol, could carry a certificate of any length (Remark B.34).

Assumption B.30 ((F1): renaming closure)**.** For every hypothesis p of a charged reading, $B \cup A_p$ is closed under permutations of the parameters.

Theorem B.31 (Charged deduction is certificate SOI)**.** *Under Assumption B.30, with time-constructible nondecreasing budgets: (a) in a normal derivation of φ every line sits at its own node of an axiom instance used or of φ, up to abstracting parameters; so its size is at most $(M+|\varphi|)(M+|\varphi|+1)/2$, with M its material, and the factor $\frac{1}{2}$ is attained; (b) with $G(m) := (g+m+1)(g+m+2)/2$, $t^+(m) := m^{k_0}(t(m)+1)$ and $t'^{\#}(m) := m^{k_0}+t'(m)$, for constants k_0, β and c: $W_A[t,g] \le W_A^b[t,g] \le W_A[t,G]$ hypothesis by hypothesis; $W_A^b[t,g] \le 2^c\, W^{\mathrm{cert}}[t^+, \beta G \log(G+2)]$; and $W^{\mathrm{cert}}[t',s'] \le 2^c$ times each of $W_A^b[t'^{\#}, 2s'+m+31]$, $W_A^{a1}[t'^{\#}, s'+m+16]$ and $W_A[t'^{\#}, 2s'+2m+31]$.*

Its classical core is the correspondence of propositional proof systems with NP (Cook and Reckhow, 1979): a derivation is a certificate; conversely, a certificate ζ becomes the axiom $\theta_\zeta \to \varphi^b$, where θ_ζ is a tautology of size $|\zeta|+14$ that encodes ζ.

Proof. This is Theorem 6.16 of the earlier report, with Lemma E.19 and Proposition E.21 there, whose budgets are time-constructible and nondecreasing. Part (a) is the subformula lemma (Lemma E.19(b), (d) there): in a normal derivation every line sits at its own formula node of an instance used or of the conclusion, up to abstracting the parameters of a chain of Gen steps, and a family with k vacuous Gen steps attains the factor $\frac{1}{2}$. Part (b): a derivation of size at most g has material at most g, and by (a) material at most g gives size at most G; this is the first chain, hypothesis by hypothesis. Under Assumption B.30 the parameters of a derivation of size at most G can be renamed to $p_0, \dots, p_{G-1}$, so the derivation is a certificate of at most $\beta G \log(G+2)$ bits that a verifier checks in time t^+ (membership in A_p in time t per line); this is the second inequality. For the third, let Z be a closed tautology of size 4 and C_ζ a sentence of size $|\zeta|+4$ that encodes the certificate ζ, and put $\theta_\zeta := Z \to (C_\zeta \to Z)$, a tautology of size $|\zeta|+14$. The axiom set $\{\theta_\zeta \to \varphi^b : \zeta \text{ a certificate on which } \mathcal{V} \text{ accepts } (b=1) \text{ or rejects } (b=0)\ \varphi\}$ is decidable in time $t'^{\#}$, satisfies Assumption B.30, and has the same consequences as $\Gamma_{f_\mathcal{V}}$. The derivation θ_ζ, $\theta_\zeta \to \varphi^b$, φ^b has material $2|\zeta|+29+|\varphi^b|$, largest instance $|\zeta|+15+|\varphi^b|$ and size $2|\zeta|+29+2|\varphi^b|$, with $|\varphi^b| \le |\varphi|+1$; these give the three stated budgets. All maps are injective and add a constant to the length. □

Corollary B.32 (Fit classes)**.** *Under Assumption B.30, on literal sequences D^X, at polynomial budgets whose degrees are charged in the prior, with standard simulation overheads:*

(i) reading (b) fits exactly the $X \in \mathsf{NP} \cap \mathsf{coNP}$; it beats polynomial-time consistent SOI, $W^{\mathrm{cons,poly}}$, on some sequence if $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$, and fits the same sequences if $\mathsf{P} = \mathsf{NP}$; with degrees uncharged, $W^{\mathrm{cons,poly}}$ fails to dominate it, unconditionally; and, under Assumption B.5, it beats $W^{\mathrm{cons},\tau}$ at a budget a hierarchy gap above τ;

(ii) reading (a1) fits exactly the $X \in \mathsf{EXPTIME}$; at any computable budgets some decidable X defeats it; and every predictor computable in time polynomial in the number of data loses at least $n - 0.361$ on a sequence on which (a1) at a linear budget loses $O(1)$;

(iii) readings (a2) and (b2) fit every decidable X, by a finite set of universally closed Horn sentences.

Proof. This is Corollary 6.17(a)–(c) of the earlier report, with its proofs there. (ii) uses $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981) through Proposition E.23 there, and the comparisons use the time hierarchy theorems (standard; cf. Hartmanis and Stearns, 1965, and Seiferas et al., 1978 for nondeterministic time). □

Proposition B.33 (Soft charges)**.** *Let a hypothesis pay a factor $2^{-\kappa\ell}$ per datum, ℓ the bit length of a shortest derivation of the datum (soft SAI), and let a verifier pay likewise for its shortest certificate (soft certificate SOI). The two correspond only up to a constant factor in the rate κ; no pair of rates gives two-sided constant regret; and every soft loss diverges on every infinite sequence, so there are no soft fit classes.*

Proof. Proposition E.26 of the earlier report, which gives the explicit rates. □

Remark B.34 (The role of the renaming condition). Without Assumption B.30 the index of one parameter, which costs one symbol, can carry a certificate of any length (Proposition E.27 of the earlier report). The coarse statements (Theorem 6.9 there, and the inclusion of the fit class of reading (b) in $\mathsf{NP} \cap \mathsf{coNP}$) survive in the reading that counts symbols, by truncating parameter indices (Lemma E.28 there); the fine bounds ($2^{O(h \log h)}$ for (a1), certificate lengths) are not claimed without it.

Remark B.35 (One tape). *Proof sketch.* We do not write out the bookkeeping of a verifier that reads a tape, which needs Assumption B.6-style accounting. The maps of Theorem B.31(b) act tape by tape: a random decider with time $O(t)$ on every tape gives a verifier that reads the tape to answer membership queries, and certificate axioms $\theta_\zeta \to \varphi^b$ can be emitted from a random verifier. So a one-tape form of the sandwich should hold with the same budgets.

B.7 Presburger arithmetic

Remark B.36 (Lengths). (i) The length $|\varphi|$ of a sentence is its symbol count: every occurrence of a connective, quantifier, nonlogical symbol, parenthesis or variable counts 1, whatever the variable's index. This is the size measure of the calculus K of the earlier report, in which a variable or parameter counts one symbol. Machines read φ as a bit string of length $b(\varphi) \le \kappa_0|\varphi|\log(|\varphi|+2)$, since a sentence with m symbols has at most m distinct variables, which can be renamed $x_1, \dots, x_m$ and written in binary. Clocks τ, derivation budgets s, g, h and the length m in the lower bounds are functions of $|\varphi|$. (ii) If a clock were a function of the bit length b, which can exceed the symbol count by a factor $\log m$, a clock $2^{2^{c'b}}$ could exceed every $2^{2^{cm}}$, and the contradictions below would fail. For clocks in bit length the results hold with $\tau(b) \le 2^{2^{c'b/\log(b+2)}}$ in place of $\tau(m) \le 2^{2^{c'm}}$, with $c' < c/\kappa_0$ (and likewise for the singly exponential clocks), since $b/\log(b+2) \le \kappa_0|\varphi|$. (iii) *Proviso:* Fischer and Rabin's length counts every symbol occurrence, variables included, at least once, and their nonlogical symbols are among $0, 1, +, =, <$. Under the proviso every sentence of their length n has an equivalent $\mathcal{L}_{\mathrm{Pr}}$-sentence with at most κn symbols, computable in polynomial time (rewrite $x < y$ as $\exists z\,(x + z + 1 = y)$ with z fresh, and copy every other symbol). *Padding:* every $\mathcal{L}_{\mathrm{Pr}}$-sentence ψ has an equivalent sentence with exactly m symbols for every $m \ge |\psi| + c_0$, computable in polynomial time, because there are two equivalence-preserving paddings with coprime increments (in the size measure of K, $\psi \wedge 0 = 0$ adds 4 and $\psi \wedge \neg 0 = 1$ adds 5; every standard syntax has such a pair). *Consequence:* let Q be a decision procedure for PrA in our syntax, deterministic or nondeterministic. For $j = 0, \dots, \kappa - 1$ let P_j decide their sentences of length n by translating, padding to exactly $\kappa n + c_0 + j$ symbols and running Q. Each P_j is a decision procedure in their syntax, with the same reading of nondeterministic time as Q; so by their theorem, for all large n, Q takes more than $2^{2^{c_{\mathrm{FR}} n}} - \mathrm{poly}(n)$ steps on some sentence with exactly $\kappa n + c_0 + j$ symbols. As j runs through all residues, for all large m, Q takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols, for every $c < c_{\mathrm{FR}}/\kappa$; c denotes such a constant. The exact-length form is what the "for all large m" arguments below use.

We use two known results. ($\mathrm{F_P}$) Presburger's axioms for successor and addition, with the induction schema for $\mathcal{L}_{\mathrm{Pr}}$-formulas, form a decidable, consistent and complete axiomatisation of PrA (Presburger, 1929), with membership decidable in polynomial time. ($\mathrm{F_{FR}}$) The bound of Fischer and Rabin (1974), in the form of Remark B.36(iii). Its statement does not say how the time of a nondeterministic procedure is measured. We distinguish two readings: (max) the time is the maximum over the computation paths, all of which halt, every output being correct and some path giving an output; (short) the time on a sentence is the length of the shortest computation that outputs the correct answer. (short) implies (max). Where only (max) is needed nothing is assumed; the one place that uses (short) says so.

Proposition B.37 (Presburger arithmetic separates SAI from clocked SOI)**.** *(a) W_A, W_A^{dec} and M_A, and the penalised $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ and $M_A^{\mathrm{chk},t}$ for a large enough polynomial t, have bounded loss on D^{Pres}. (b) If τ is time-constructible and $\tau(m) \le 2^{2^{c'm}}$ for large m, with $c' < c$, the losses*

of $W^{\mathrm{cons},\tau}$ and $W^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity. (c) If $\tau(m) \le 2^{c'm}$ for large m, with $c' < c$, the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.

Proof. (a) The decider of Presburger's axioms is compatible with every prefix of D^{Pres} (the axiomatisation is complete and consistent), runs in polynomial time and is a deterministic theory machine; with an enumerator of the same axioms it meets the axiom-side penalties for a large enough polynomial t. The losses are at most $-\log$ of its weight. (b) In a clocked mixture compatibility is inherited by prefixes, so the compatible strings form a decreasing family and bounded loss gives one τ-clocked f that labels every sentence correctly (Lemma E.11 of the earlier report). By the standard simulation overhead it yields a multitape decision procedure for PrA that runs in time $q(C_f\tau(m)+b)$, with q a fixed polynomial and $b \le \kappa_0 m \log(m+2)$ the input's bit length. When $c' < c$ this is at most $2^{2^{cm}}$ for large m, which contradicts $\mathrm{F_{FR}}$ in the deterministic case. (c) Write $M := \sum_f w(f)\nu_f$ for either one-tape mixture, each $\nu_f(D_n^{\mathrm{Pres}})$ being non-increasing in n. Bounded loss gives $\inf_n M(D_n^{\mathrm{Pres}}) > 0$; by dominated convergence some f has $\lim_n \nu_f(D_n^{\mathrm{Pres}}) > 0$, which by continuity from above is $\mathbb{U}(\mathcal{G}_f)$ for the set $\mathcal{G}_f$ of tapes on which F_f^τ labels every sentence correctly (and consistently, for the consistent version). By the Lebesgue density theorem (Lemma C.1) there is w_0 with $\mathbb{U}(\mathcal{G}_f \cap [w_0]) > \frac{3}{4}\,2^{-|w_0|}$, so for every sentence the tapes in $[w_0]$ on which F_f^τ answers correctly have conditional measure greater than $\frac{3}{4}$. A query of length m reads at most $\tau(m)$ tape bits, so the procedure "for every v of length $\max(0, \tau(m) - |w_0|)$ run F_f^τ with tape $w_0 v$ and output the majority answer" (w_0 hard-coded) decides PrA deterministically within $2^{\tau(m)} q(C_f\tau(m)+b) \le 2^{2^{cm}}$ steps for large m: a contradiction. The majority vote over the tape completions at a density point is the technique of de Leeuw et al. (1956); the argument applies to every mixture of machines whose tape use per query is bounded by τ. □

A computation with illustrative values of the constants (code in the accompanying repository; c is not known explicitly) gives the lengths from which these contradictions apply: for $c = 1$ and $c' = 0.9$, from about 11–26 symbols in (b) and 6–9 in (c). The separations are asymptotic: below these lengths a clocked hypothesis may label every short sentence correctly.

Assumption B.38 ((B*): a Σ_2-time lower bound for Presburger arithmetic)**.** There is $c_B > 0$ such that, for all large m, no alternating machine with at most two alternations (a Σ_2 machine) decides PrA within $2^{2^{c_B m}}$ steps on all sentences with m symbols.

Assumption B.39 ((SGL): time-bounded $\mathsf{BPP} \subseteq \Sigma_2$)**.** For time-constructible $T(m) \ge m$ and large T, $\mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$.

We do not prove either assumption. PrA is complete for alternating doubly exponential time with linearly many alternations (cf. Berman, 1980); (B*) would follow from the hardness part if the reductions were of linear size, and we do not assume that they are. If they are only of polynomial size, a bound $2^{2^{c_B m^\epsilon}}$ follows for some $\epsilon \in (0,1]$, and the clock condition below must be weakened to $\tau(m) \le 2^{2^{c' m^\epsilon}}$. (SGL) is the time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$ (Sipser, 1983; Lautemann, 1983): amplifying by $O(\log T)$ repetitions to $R = O(T \log T)$ random bits and error below $1/(2R)$, a Σ_2 machine guesses R shifts of R bits, universally chooses R bits, and runs the amplified machine R times, in time $O(T^2 \log^2 T)$.

Proposition B.40 (One tape, doubly exponential clocks)**.** *Under Assumptions B.38 and B.39, if $\tau(m) \le 2^{2^{c'm}}$ for large m with $c' < c_B$, then the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.*

Proof. As in the proof of Proposition B.37(c), obtain w_0 such that on $[w_0]$ the conditional probability of a correct answer exceeds $\frac{3}{4}$ for every sentence. The algorithm "draw v of length $\tau(m)$ at random, run F_f^τ with tape $w_0 v$, and output its answer" errs with probability below $\frac{1}{4}$ on every sentence and runs in time $T(m) := q(C_f\tau(m)+b)$, so $\mathrm{PrA} \in \mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$ by Assumption B.39. Since $T(m)^3 \le 2^{2^{c_B m}}$ for large m when $c' < c_B$, this contradicts Assumption B.38. □

A nondeterministic lower bound alone does not finish the doubly exponential case by this route: separating $\mathsf{NTIME}(2^{2^n})$ from Σ_2-time below it is not known in general.

Proposition B.41 (Charged deduction on Presburger data)**.** *Let p be a hypothesis of a charged reading over $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$, deciding A_p in polynomial time and satisfying Assumption B.30, and compatible with every prefix of D^{Pres}. Let $c'' < c$. (a) Size reading, budget s: $s(m) > 2^{2^{c''m}}$ for infinitely many m, and for all large m if s is time-constructible. (b) Material reading (b), budget g: the same with g in place of s. (c) Reading (a1), with h nondecreasing and time-constructible: $h(m) > 2^{c''m}$ for all large m.*

A mixture of such hypotheses has bounded loss on D^{Pres} only if one of them is compatible with every prefix.

Proof. Since compatibility is inherited by prefixes, it suffices to treat one hypothesis p, deciding A_p in time $O(m^{k_p})$ and compatible with every prefix of D^{Pres}. Then $B \cup A_p = A_p$ is consistent and proves every true sentence, so it proves exactly PrA. (a) *Infinitely often.* Fix $c'' < c$ and suppose $s(m) \le \tau''(m) := 2^{2^{c''m}}$ for all $m \ge m_1$; τ'' is time-constructible. The nondeterministic procedure N: on a sentence φ with $m < m_1$ symbols, look φ up in a table; otherwise compute $\tau''(m)$, guess a derivation of φ or of $\neg\varphi$ of size at most $\tau''(m)$ whose parameter numbers have at most $K+1+\lceil\log(\tau''(m)+1)\rceil$ bits, $K = O(\tau''(m)^{k_p})$ (if a derivation of size at most $\tau''(m)$ exists, such a one exists, by Lemma E.28 of the earlier report), check it as in the proof of Theorem 6.9 there (each line a logical axiom, by matching, a member of A_p, in time $O(\tau''^{k_p})$, or a step of modus ponens or generalisation), and output the label it proves; a path whose guess is not a valid derivation outputs nothing. Every output is correct, every sentence has a path with an output, and every path halts within $\tau''(m)^{c_k}$ steps, c_k depending only on k_p and the checking overhead of K. Since $\tau''(m)^{c_k} = 2^{c_k 2^{c''m}} \le 2^{2^{cm}}$ once $m \ge \log(c_k)/(c-c'')$, N contradicts $\mathrm{F_{FR}}$ (nondeterministic case, reading (max)). So $s(m) > 2^{2^{c''m}}$ for infinitely many m. *For all large m, s time-constructible.* Suppose $s(m) \le \tau''(m)$ for infinitely many m, the *good* lengths. The procedure N'': on a sentence with m symbols, compute $s(m)$ with a clock of $\tau''(m)$ steps; if the computation finishes with $s(m) \le \tau''(m)$, run N's guess-and-check with the budget $s(m)$; otherwise run a deterministic decision procedure for PrA, which exists by $\mathrm{F_P}$. Every output is correct and every path halts, and at each good length every path halts within $O(\tau''(m)) + \tau''(m)^{c_k} \le 2^{2^{cm}}$ steps beyond the thresholds. By $\mathrm{F_{FR}}$ (reading (max)), for all large m some sentence with m symbols takes N'' more than $2^{2^{cm}}$ steps; a good length beyond the thresholds gives a contradiction. (For an arbitrary s the same conclusion follows if $\mathrm{F_{FR}}$ holds in reading (short): let a path whose guess is not a valid derivation run the deterministic procedure; at each good length every sentence then has a correct output within $\tau''(m)^{c_k} + O(1)$ steps.) (b) By the subformula lemma (Theorem B.31(a)), material at most g gives a derivation of size at most $(g+m)(g+m+1)/2$. If $g(m) \le 2^{2^{c''m}}$ for all large m, this is at most $2^{2^{c'''m}}$ for large m, for any $c''' \in (c'', c)$, against (a) with c'''. The form for all large m transfers in the same way. (c) Proposition E.23(a) of the earlier report gives the labeller g_p: with

$H(\varphi) := \max(h(|\varphi|), |\varphi| + 1)$ and Der_H the formulas with a derivation from $B \cup A_p$ all of whose lines have size at most H, it accepts φ if $\varphi \in \mathrm{Der}_{H(\varphi)}$, rejects it if $\neg\varphi \in \mathrm{Der}_{H(\varphi)}$ and $\varphi \notin \mathrm{Der}_{H(\varphi)}$, and abstains otherwise. It runs in time $2^{O(H \log H)}(1+t(H))^{O(1)}$, by closing the at most $2^{O(H \log H)}$ canonical formulas of size at most H under the rules, using Assumption B.30; and it labels correctly every datum of every D with which p is (a1)-compatible, since a normal (a1)-derivation of φ^b has every line of size at most $\max(h(|\varphi|), |\varphi^b|) \le H(\varphi)$ (the subformula lemma) and $B \cup A_p$ is consistent. Nothing in this argument uses the form of φ. On D^{Pres}, g_p therefore decides PrA deterministically (it never abstains, since every sentence is a datum) in time $T(m) \le 2^{C_a H \log H}$ with $H = \max(h(m), m+1)$ and C_a depending on k_p (computing $h(m)$ costs $O(h(m))$). By $\mathrm{F_{FR}}$, $T(m) > 2^{2^{cm}}$ for all large m, so $C_a H \log H > 2^{cm}$, hence $H > 2^{cm}/(C_a cm) > 2^{c''m}$ for all large m; as $m + 1 < 2^{c''m}$ for large m, $H = h(m)$. □

Question B.42. (i) Under reading (a1), does every hypothesis compatible with every prefix of D^{Pres} need doubly exponential instance budgets h? (ii) Does some hypothesis of reading (a1) in $\mathcal{L}_{\mathrm{Pr}}$ fit D^{Pres} at exponential budgets? (iii) What do readings (a2) and (b2) need on D^{Pres}, in either direction?

(i) would follow from a deterministic lower bound $\mathrm{PrA} \notin \mathsf{DTIME}(2^{2^{2^{c_3 m}}})$ for some $c_3 > 0$, by the argument of Proposition B.41(c), which would then give $h(m) > 2^{2^{c_3' m}}$ infinitely often for every $c_3' < c_3$; no such bound is known to us. For (ii), Proposition E.23(b) of the earlier report simulates alternating space-bounded machines, but it needs the literal language with two extra binary relation symbols, which $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$ lacks. For (iii), logical instances are free under (a2) and (b2), so neither the guess-and-check route nor the enumeration route gives a bound, and Corollary B.32(iii) concerns literal data in the literal language, not D^{Pres}.

Each charged reading matches one version of SOI. Once deduction is charged, the budget SAI needs on D^{Pres} is that of the matching version of SOI: under the size and material readings SAI is certificate SOI (Theorem B.31), and the nondeterministic Fischer–Rabin bound forces doubly exponential budgets on both; under (a1) SAI behaves like deterministic SOI with an exponential clock (Proposition E.23 and Corollary 6.17(b) of the earlier report; the converse direction is proved there for literal data only, so for sentence data we offer this as a reading), and the deterministic bound forces exponential instance sizes, which the clock turns into doubly exponential time. So on a natural theory, as on the diagonal sequences, the separation from clocked SOI is the free deduction, for every reading in which the derivation is charged and the conclusion is proved; for (a2) and (b2) the question is open.

B.8 The circuit form of the sandwich

The sandwich of Theorem B.31 has a routine circuit form, which connects finite axiom sets with the circuit Bayes theorem of the companion paper. Fix $d \ge 1$; the data are the literals $\varphi_x := R(t_x)$, $x \in \{0,1\}^d$, labelled by a partial $f : \{0,1\}^d \rightharpoonup \{0,1\}$.

A *verifier circuit* $\mathcal{V}$ of size $|\mathcal{V}|$ is a list of $|\mathcal{V}|$ two-input gates (any of the 16 Boolean functions) over the inputs $x_1, \ldots, x_d$, certificate bits $z_1, \ldots, z_k$ (without loss of generality $k \le 2|\mathcal{V}|$), the constants and earlier gates; its last two gates are the outputs a (accept) and r (reject). It is *consistent* if no x has certificates ζ, ζ' with $\mathsf{a}(x,\zeta) = 1$ and $\mathsf{r}(x,\zeta') = 1$, and it defines $f_{\mathcal{V}}(x) := 1$ if some ζ has $\mathsf{a}(x,\zeta) = 1$, 0 if some ζ has $\mathsf{r}(x,\zeta) = 1$, and undefined otherwise. $\mathrm{vs}(f)$ is the least size of a consistent verifier circuit with $f_{\mathcal{V}} \supseteq f$; it is at most the circuit size of f in the sense of Definition 9.7 of the companion paper plus a constant. The *fixed language* is $\mathcal{L}_{\mathrm{circ}} :=$

$\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, \mathrm{Dig}, \mathrm{Gate}, R\}$, with Dig binary and Gate ternary. Index terms $\mathbf{n}_i$ and gate terms $\mathbf{g}_j$ are the word terms $t_{\beta(i)}$, $t_{\beta(j)}$, with $\beta(i)$ the binary expansion of $i+1$ without its leading 1; so $|\mathbf{n}_i| = O(\log i)$, the terms of each family are pairwise distinct, and the language does not grow with the circuit. $\mathrm{ax}_s(f)$ is the least total size of a finite set T of sentences of $\mathcal{L}_{\mathrm{circ}}$ (or of any fixed finite language containing $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R$) with $T \cup \Gamma_f$ consistent such that every datum $\varphi_x^{f(x)}$, $x \in \mathrm{dom}\, f$, has a K-derivation from T of size at most s.

The translation $T_{\mathcal{V}}$ (all sentences universally closed; $\leftrightarrow$ abbreviates $\neg((A \to B) \to \neg(B \to A))$). Digits: $\forall y\, \mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_1 y)$, $\forall y\, \neg\mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_0 y)$, and $\forall y\, (\mathrm{Dig}(\mathbf{n}_{i+1}, \mathsf{s}_a y) \leftrightarrow \mathrm{Dig}(\mathbf{n}_i, y))$ for $i < \max(d,k) - 1$ and $a \in \{0,1\}$; so $\mathrm{Dig}(\mathbf{n}_i, t_w)$ holds iff the $(i+1)$-st symbol of w is 1. Gates: a gate j that reads x_i (resp. z_i) has the axiom $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{Dig}(\mathbf{n}_{i-1}, x))$ (resp. with z in place of x), and an internal gate j with operation op on gates a, b has $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{op}(\mathrm{Gate}(\mathbf{g}_a, x, z), \mathrm{Gate}(\mathbf{g}_b, x, z)))$. Outputs: $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{a}}, x, z) \to R(x))$ and $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{r}}, x, z) \to \neg R(x))$.

Theorem B.43 (The circuit sandwich)**.** *(i) For every consistent verifier circuit $\mathcal{V}$ with k certificate bits, $T_{\mathcal{V}}$ has total size at most $C_1(|\mathcal{V}| + d + k)\log(|\mathcal{V}| + d + k + 2)$, $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$ is consistent, and every $x \in \mathrm{dom}\, f_{\mathcal{V}}$ has a derivation of $\varphi_x^{f_{\mathcal{V}}(x)}$ from $T_{\mathcal{V}}$ of size at most $s_{\mathcal{V}} := C_2(|\mathcal{V}|(\log(|\mathcal{V}| + 2) + d + k) + (d+k)^3)$, with absolute constants C_1, C_2. Hence, using $k \le 2|\mathcal{V}|$, $\mathrm{ax}_s(f) \le C_1'(\mathrm{vs}(f) + d)\log(\mathrm{vs}(f) + d + 2)$ for $s = C_2'(\mathrm{vs}(f)(\log \mathrm{vs}(f) + d) + (d + \mathrm{vs}(f))^3)$. (ii) Conversely, $\mathrm{vs}(f) \le P(\mathrm{ax}_s(f), s, d)$ for a fixed polynomial P that depends only on the language and K.*

This is the routine circuit form of the sandwich, not a separate result of independent interest; we include it because it carries the circuit Bayes theorem of the companion paper over to verifier circuits (Corollary B.44).

Proof. (i) *Size.* There are $O(d+k)$ digit axioms of size $O(\log(d+k))$, one axiom of size $O(\log(|\mathcal{V}| + d + k))$ per gate (two or three gate terms and at most two index terms), and two output axioms. *Consistency.* Take the free term algebra on $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1$, with equality the identity; every element is a word term. Read off the bits of an element y (the symbols of the word it spells, 0 beyond its length). Let $\mathrm{Dig}(t, y)$ hold iff $t = \mathbf{n}_i$ for some $i < \max(d,k)$ and bit i of y is 1; let $\mathrm{Gate}(t, x, z)$ hold iff $t = \mathbf{g}_j$ and gate j of $\mathcal{V}$ outputs 1 on the first d bits of x and the first k bits of z; and let $R(x)$ hold iff some ζ has $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$. The digit and gate axioms hold by construction. The first output axiom holds by the definition of R; the second holds because $\mathcal{V}$ is consistent: $\mathsf{r}(\mathrm{bits}_d(x), z) = 1$ excludes $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$ for every ζ. On word terms of length d, $R(t_x)$ holds iff $f_{\mathcal{V}}(x) = 1$, and $f_{\mathcal{V}}(x) = 0$ forces $\neg R(t_x)$; so the structure satisfies $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$. *Derivations.* For $x \in \mathrm{dom}\, f_{\mathcal{V}}$ take a certificate ζ for its label. Instantiate every gate axiom with $x := t_x$, $z := t_\zeta$ (two instances of the quantifier axiom and two modus ponens steps per axiom, each line of size $O(\log|\mathcal{V}| + d + k)$). Derive each needed digit, $\mathrm{Dig}(\mathbf{n}_i, t_w)$ or its negation, from an instance of the base axiom and i instances of the successor axioms (each of size $O(d+k)$), substituting at each step a fixed propositional derivation for biconditional elimination: $O((d+k)^2)$ per digit, $O((d+k)^3)$ in all. Then, gate by gate in order, derive $\mathrm{Gate}(\mathbf{g}_j, t_x, t_\zeta)$ or its negation from the ground biconditional and the literals of its inputs, by a substitution instance of a fixed propositional derivation (the propositional fragment of K is complete, and substitution instances of derivations are derivations of size linear in the substituted formulas, as in the proof of Theorem 6.16 of the earlier report). Finish with the output instance and modus ponens. The total is $O(|\mathcal{V}|(\log|\mathcal{V}| + d + k) + (d+k)^3)$. (ii) A certificate is the code of a derivation of size at most s. Since T is a set of sentences, Assumption B.30 holds, and the parameters can be renamed to $p_0, \ldots, p_{s-1}$, so the code has $O(s \log s)$ bits (Proposition E.21(b) of the earlier report). The checker, with T hard-wired, verifies

each line (logical axioms by matching, quantifier axioms by first-order matching, members of T by comparison, modus ponens, generalisation) and the last line against φ_x or $\neg\varphi_x$, in time polynomial in $s+|T|+d$; it accepts if the code is a derivation of φ_x and rejects if it is one of $\neg\varphi_x$. Simulating it by circuits (standard; size $O(T\log T)$ for time T, Pippenger and Fischer, 1979) gives a verifier circuit of size $P(|T|,s,d)$. It is consistent because T is consistent, and $f_{\mathcal{V}} \supseteq f$ because every datum has a derivation within s and T proves no wrong label, $T\cup\Gamma_f$ being consistent. □

A computation (code in the accompanying repository) checks the translation on random consistent verifier circuits with $d\in\{2,3\}$: every axiom of $T_{\mathcal{V}}$ is true in the intended structure on a finite fragment, forward propagation through the ground instances derives the right label for every $x\in\operatorname{dom} f_{\mathcal{V}}$, every corruption of one gate that changes $f_{\mathcal{V}}$ is detected; and, for sizes from 33 to 1099 gates with d from 3 to 32, $|T_{\mathcal{V}}|$ stays between 7.86 and 9.03 times $(|\mathcal{V}|+d)\log(|\mathcal{V}|+d)$.

Corollary B.44 (Bayes over verifier circuits)**.** *Let W_{ver} be the mixture over consistent verifier circuits with prior $w_{\mathrm{ver}}(\mathcal{V}) := 2^{-|\mathcal{V}|}\,(2|\mathcal{V}|+1)^{-1}\,\Lambda^{-1}_{|\mathcal{V}|,k}$, where $k\le 2|\mathcal{V}|$ is uniform and $\Lambda_{j,k} := \prod_{i=1}^{j} 16(d+k+1+i)^2$ is the number of gate lists of size j. For a nonempty $J\subseteq\{0,1\}^d$, an ordering of J and the labels of f, let $L := -\log W_{\mathrm{ver}}(f|_J)$, $\ell_\emptyset := -\log W_{\mathrm{ver}}(\emptyset)$, and* Err $:= \sum_i \epsilon_i$, *with ϵ_i the posterior mass at step i that is wrong* or undefined *at x_i. Then: (a) some consistent verifier circuit of size at most $\lceil L\rceil$ agrees with f on J, and $L\le \mathrm{vs}_J+\log(2\mathrm{vs}_J+1)+\log\Lambda_{\mathrm{vs}_J,2\mathrm{vs}_J} = O(\mathrm{vs}_J\log(\mathrm{vs}_J+d))$, with $\mathrm{vs}_J := \mathrm{vs}(f|_J)$; (b)* Err $\le(\ln 2)(L-\ell_\emptyset)$ *and $L-\ell_\emptyset\le 2\,$*Err $(\mathrm{gad}(\lceil L\rceil)+3)$, *with* $\mathrm{gad}(j) := \mathrm{gg}\,(5+2\log(d+3j+\mathrm{gg}+1))+\log(\mathrm{gg}+1)$ *and* $\mathrm{gg} := 2d+2$.

So Bayes over nondeterministic verifier circuits makes few posterior mistakes on f exactly when f has a small consistent verifier circuit, and, by Theorem B.43, exactly when f has a small finite axiomatisation in a fixed language whose data have polynomially short derivations, up to polynomial changes.

Proof. (a) The prior has an exponential tail in the size (the total weight of sizes above S is at most 2^{-S}), and the verifier with no active output disagrees with f at every size, so the argument of Theorem 9.8(a) of the companion paper applies; the bound on L is the weight of one smallest consistent verifier circuit for $f|_J$. (b) The first inequality is $\epsilon_i\le -\ln(1-\epsilon_i)$ with telescoping; now $W_{\mathrm{ver}}(\emptyset)<1$, hence $\ell_\emptyset$. For the second we use an exception gadget for verifiers: given $\mathcal{V}$, a point a and a label b, append a circuit for $[x=a]$ (at most $2d-1$ gates), its negation, and new outputs $\mathsf{a}' := \mathsf{a}\vee[x=a]$, $\mathsf{r}' := \mathsf{r}\wedge\neg[x=a]$ if $b=1$ (symmetrically if $b=0$): at most $\mathrm{gg}=2d+2$ gates. The new circuit $\mathcal{V}'$ is consistent (at a only the b-output can fire, and elsewhere it is $\mathcal{V}$), is defined at a with label b, agrees with $\mathcal{V}$ off a, and $\mathcal{V}\mapsto\mathcal{V}'$ is injective, with $w_{\mathrm{ver}}(\mathcal{V}')/w_{\mathrm{ver}}(\mathcal{V})\ge 2^{-\mathrm{gad}(|\mathcal{V}|)}$. It maps the hypotheses of size at most S that are wrong or undefined at x_i injectively to hypotheses that are right at x_i and still compatible with D_{i-1}, whose inputs differ from x_i. The rest of the proof of Theorem 9.8(b) of the companion paper is unchanged. □

Remark B.45 (The axiom-set prior itself)*. Proof sketch.* We do not write out the converse direction for a prior over finite axiom sets; the obstacles are listed here. For a mixture over finite axiom sets with a derivation budget, the upper bounds $L=O(a\log a)$, with a the size of a smallest compatible axiom set, and Err $\le(\ln 2)L$ hold directly (Theorem B.43(i) and telescoping). The converse needs an exception gadget for *every* hypothesis T: replace R by a fresh unary predicate R' in T and add $\forall x\,(x\ne t_a\to(R(x)\leftrightarrow R'(x)))$ and $R(t_a)^b$. This works if the language supplies fresh predicate symbols (so it is not a fixed finite language, and the symbol's index costs $O(\log)$ bits), if B contains the free-algebra axioms needed to derive $t_x\ne t_a$ (of size $O(d^2)$), and if the

budget is charged in the prior, so that the gadget's extra derivation size $O(d^2)$ moves a hypothesis to a nearby budget at a cost of $O(\log)$ bits.

C Proofs for random theories, Π_1 data and Gödel phenomena

This appendix proves the results of Section 5 in their order and states and proves the results that the main text only mentions. The notation is that of Definitions 2.3, 2.4 and 2.7.

C.1 Standing facts and conventions

Facts from Section 3. We use four facts throughout.

- *(Dom)* If $\omega = p\omega'$ with $p \in \operatorname{dom} V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$ (Definition 2.4). So for every theory machine G, every event about the random theory of G is matched, at a cost of a factor $2^{-K(G)}$, by the same event for G_U: the tapes $p\omega'$ with p a shortest program for an index of G carry it. In particular $M_A \geq 2^{-K(\nu)}\nu$ for every theory-machine law ν, and $M_A(D) \geq 2^{-K(e)-c}$ whenever $B \cup \mathsf{W}_e$ is consistent and proves ψ_D (Theorem 3.4).
- *(Chain)* $2^{-K_A(D)} \leq W_A(D) \leq 2^c M_A(D)$ for consistent D (Proposition 3.8(c)).
- *(Lim)* If $\mathbb{U}(E_\infty) > 0$, the posterior converges in total variation to $P_\infty = \mathbb{U}(\cdot \mid E_\infty)$ and the cumulative loss is bounded; if $\mathbb{U}(E_\infty) = 0$, the loss tends to ∞. For every Γ with $\mathrm{Cn}(\Gamma) = \mathrm{Cn}(D^\pm_\infty)$, complete or not, $E_\infty = \{\omega : B \cup A_\omega \text{ consistent, } \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(\Gamma)\}$ (Proposition A.26).
- *(Old)* $M_A(D)$ depends on D only through $\mathrm{Cn}(D^\pm)$, and data entailed by B change nothing (Proposition 3.1(c)). The same holds for W_A and K_A, whose definitions mention D only through "proves ψ_D".

$\mathbb{U}(O_\varphi \cap [w])$ is left-c.e. uniformly in φ and w (Lemma A.1). For a set $\mathcal{X}$ of tapes and a finite string w we write $\mathbb{U}(\mathcal{X} \mid [w]) = 2^{|w|}\mathbb{U}(\mathcal{X} \cap [w])$. For a class $\mathcal{C}$ of deductively closed theories, $P_D(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; D)/M_A(D)$ is the posterior probability that the random theory lies in $\mathcal{C}$, and $W_A(\mathcal{C}; D) := \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_D, \ \mathrm{Cn}(\mathsf{W}_e) \in \mathcal{C}]$.

Arithmetic. $\mathcal{L}_{\mathrm{ar}} = \{0, S, +, \cdot, <\}$ with numerals $\underline{n} = S^n 0$. The classes Δ_0, Σ_1, Π_1 are syntactic; Π_1^* is the class of finite conjunctions of Π_1 sentences and negations of Σ_1 sentences; a sentence that is equivalent to a member of a class only provably in Q is called Q-*equivalent* to one. "B is true" means that B has a model whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$ (for $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$: $B \subseteq \mathrm{Th}(\mathbb{N})$), likewise "$B \cup T$ is true"; "$B \supseteq \mathsf{Q}$" means $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, likewise for PA. A theory is Σ_1-*sound* if it proves no false Σ_1 sentence. Binary strings are coded by numbers through their length-lexicographic rank (the empty string is 0, then $0, 1, 00, 01, \ldots$); the coding is injective across lengths, the length of a string is computable from its code, and the strings of length at most m are exactly the codes at most $\mathrm{rk}_m := 2^{m+1} - 2$. We use the following standard facts (see Smoryński, 1977; Boolos et al., 2007).

- *(Σ_1 C)* Q proves every true Σ_1 sentence (in particular every true Δ_0 sentence) and refutes every false Δ_0 sentence.
- *(Π_1 T)* Hence, if $B \supseteq \mathsf{Q}$ and $B \cup A$ is consistent, every Π_1^* sentence that $B \cup A$ proves is true: if a conjunct were false, its negation would be a true Σ_1 sentence, provable in Q.
- *(O)* $\mathsf{Q} \vdash \forall z\,(z \leq \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ for every n; so Q unfolds bounded quantifiers with a numeral bound into finite conjunctions and disjunctions.

- *(G2)* For every consistent r.e. presentation Z with $\mathsf{PA} \subseteq \mathrm{Cn}(Z)$, $Z \nvdash \mathrm{Con}(Z)$, where $\mathrm{Con}(Z)$ is the canonical consistency statement of the presentation, a Π_1 sentence.
- *(Diag)* For every formula $\vartheta(z, x)$ there is a formula $\Upsilon(x)$ with $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \vartheta(\ulcorner \Upsilon \urcorner, x))$.
- *(Uniq)* Let $\mathrm{Comp}(p, t, y)$ be Kleene's Δ_0 predicate "t codes the halting computation of V on the program p, with output y". Each halting computation has a unique code, and $\mathsf{PA} \vdash \forall p, t, t', y, y'\,(\mathrm{Comp}(p, t, y) \wedge \mathrm{Comp}(p, t', y') \to t = t' \wedge y = y')$.

Loss conventions. As Definition 2.5 requires, every loss statement in Section 5 names its convention. Unless stated otherwise losses are in the semimeasure convention: the cumulative loss of D_n is $L_A(D_n) - L_A(\emptyset)$, and $-\log W_A(D_n) + \log W_A(\emptyset)$ for the coin-free version. The lower bounds Theorem 5.1(c), Corollary C.7, Corollary C.15 and Theorem C.30 are statements in this convention. Upper bounds transfer to the normalised convention, since each normalised conditional is at least the raw one; lower bounds do not, and Proposition C.16 shows that the lower bound of Corollary C.15 fails in the normalised convention for most strings. In the three-valued convention the conditional probability of the label 1 on a query φ after two-valued data D is $\mathrm{Bel}_D(\varphi)$, so a lower bound on $-\log \mathrm{Bel}_D(\varphi)$ is a lower bound on the three-valued cost of a true label. Statements about K_A, K_{fin} and the posterior (P_∞, Bel, Dis, Indep) depend on no convention.

C.2 The lottery

Lemma C.1 (Lebesgue density on Cantor space)**.** *Let $\mathcal{X}$ be a Borel set of tapes with $\mathbb{U}(\mathcal{X}) > 0$ and let $\varepsilon \in (0, 1)$. Then there is a finite string w with $\mathbb{U}(\mathcal{X} \mid [w]) > 1 - \varepsilon$.*

Proof. By outer regularity there is an open $O \supseteq \mathcal{X}$ with $\mathbb{U}(O) < \mathbb{U}(\mathcal{X})/(1-\varepsilon)$. Write $O = \bigsqcup_i [w_i]$ with $\{w_i\}$ prefix-free. If $\mathbb{U}(\mathcal{X} \cap [w_i]) \le (1-\varepsilon)2^{-|w_i|}$ for every i, summing gives $\mathbb{U}(\mathcal{X}) \le (1-\varepsilon)\mathbb{U}(O) < \mathbb{U}(\mathcal{X})$, a contradiction. □

Proof of Theorem 5.1. The random lottery theory is consistent. Take a model $\mathfrak{M}$ of B in which the numerals denote pairwise distinct elements, and for a tape ω put $\mathrm{Win}^{\mathfrak{M}} := \{(\underline{k}^{\mathfrak{M}}, \underline{J_k}^{\mathfrak{M}}) : k \in \mathbb{N}\}$. Because the numerals are distinct, the only Win-partner of $\underline{k}^{\mathfrak{M}}$ is $\underline{J_k}^{\mathfrak{M}}$, so every sentence G_{lot} emits holds; B does not mention Win. So $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent for every ω.

The stream. Let $\mathsf{W}_0, \mathsf{W}_1, \ldots$ enumerate the r.e. sets of $\mathcal{L}$-sentences. For each e define $j_e^* \in \{0, \ldots, N_e\}$: if $B \cup \mathsf{W}_e$ is consistent and proves Lot_e, let j_e^* be the least $j \le N_e$ with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j})$; otherwise $j_e^* := 0$. In the first case such a j exists: if $B \cup \mathsf{W}_e$ proved $\neg\mathrm{Win}(\underline{e}, \underline{j})$ for all $j \le N_e$, then with the second conjunct of Lot_e and the equality axioms it would prove $\forall x\,\neg\mathrm{Win}(\underline{e}, x)$, contradicting the first conjunct and its consistency. Put

$$D_\infty := (\mathrm{Lot}_0, 1),\ (\mathrm{Win}(\underline{0}, \underline{j_0^*}), 0),\ (\mathrm{Lot}_1, 1),\ (\mathrm{Win}(\underline{1}, \underline{j_1^*}), 0),\ \ldots$$

"$B \cup \mathsf{W}_e$ is consistent" is Π_1, and "$B \cup \mathsf{W}_e \vdash \mathrm{Lot}_e$" and "$B \cup \mathsf{W}_e \vdash \neg\mathrm{Win}(\underline{e}, \underline{j})$" are Σ_1, uniformly; so $e \mapsto j_e^*$ and the stream are computable from $\emptyset'$.

(a) Let $\mathrm{Surv} := \{\omega : J_k(\omega) \ne j_k^* \text{ for all } k\}$. The J_k are independent and uniform on $\{0, \ldots, N_k\}$, so by continuity from above $\mathbb{U}(\mathrm{Surv}) = \prod_k (1 - 2^{-(k+1)})$, which is positive because $\sum_k 2^{-(k+1)} < \infty$. For $\omega \in \mathrm{Surv}$ the theory $B \cup A_\omega^{G_{\mathrm{lot}}}$ is consistent, emits every Lot_k, and emits $\neg\mathrm{Win}(\underline{k}, \underline{j_k^*})$ because $j_k^* \le N_k$ and $j_k^* \ne J_k$. So $\nu_{G_{\mathrm{lot}}}(D_n) \ge \mathbb{U}(\mathrm{Surv})$ for all n, and (Dom) gives the bound. The rest is (Lim), with $M_A(\emptyset) \le 1$ for the bound on the cumulative loss.

(b) If $B \cup \mathsf{W}_e$ is consistent and proves every member of $D^{\pm}_{\infty}$, it proves Lot_e, so j^*_e was chosen with $B \cup \mathsf{W}_e \nvdash \neg\mathrm{Win}(\underline{e}, \underline{j^*_e})$; but that sentence is in $D^{\pm}_{\infty}$.

(c) $W_A(D_n) = \sum_e \mathbf{m}(e)\mathbf{1}[B \cup \mathsf{W}_e \text{ consistent}, \vdash \psi_{D_n}]$. Each indicator is non-increasing in n and, by (b), eventually 0; dominated convergence ($\sum_e \mathbf{m}(e) \leq 1$) gives $W_A(D_n) \to 0$, and $K_A \geq -\log W_A$ by (Chain). For a single sentence ψ of $\mathcal{L}'$ with $B \cup \{\psi\}$ consistent, its $\mathcal{L}$-content $\mathrm{Cont}(\psi) := \{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\psi\} \vdash \varphi\}$ is r.e., because $\mathcal{L}'$ has computable syntax, and $B \cup \mathrm{Cont}(\psi)$ is consistent; so by (b) $\mathrm{Cont}(\psi) \not\supseteq D^{\pm}_{\infty}$, and every consistent ψ fails some datum. Only finitely many ψ have $K(\psi) \leq j$, so $K^{\mathcal{L}'}_{\mathrm{fin}}(D_n) \to \infty$.

(d) Let P_{det} be the set of $p \in \mathrm{dom}\, V$ with $G_{V(p)} \subseteq \{\epsilon\} \times \mathrm{Sent}$, and $A^{(p)}$ the set that $G_{V(p)}$ emits. Then

$$\mathbb{U}\{\omega \in E_{D_n} : \omega \text{ begins with some } p \in P_{\mathrm{det}}\} = \sum_{p \in P_{\mathrm{det}}} 2^{-|p|}\mathbf{1}[B \cup A^{(p)} \text{ consistent}, \vdash \psi_{D_n}] \to 0$$

by (b) and dominated convergence; divide by $M_A(D_n)$, which is bounded below by (a). If $\omega \in E_{\infty}$ and $\mathrm{Cn}(B \cup A_{\omega})$ were r.e., it would be a consistent r.e. theory proving every member of $D^{\pm}_{\infty}$, contrary to (b). □

A computation (code in the accompanying repository) confirms the survival probabilities $\prod_{k<K}(1 - 2^{-(k+1)})$ for $K \leq 5$ by enumerating winner vectors, and runs the diagonal against random lists of theories and against an adversarial list in which each listed theory survives until its own lottery.

Remark C.2 (Which streams, which variant). (i) A stream with $\mathbb{U}(E_{\infty}) > 0$ that no consistent r.e. theory proves is not r.e.: otherwise $D^{\pm}_{\infty}$ itself would be such a theory. For the lottery classes it is not co-r.e. either (Proposition C.5). The streams constructed here are Δ_2; an arbitrary stream with these properties need not be arithmetical (for instance, a diagonal against a non-arithmetical list). (ii) The phenomenon needs proof-requiring compatibility: the r.e. theory $\{\mathrm{Lot}_k : k \in \mathbb{N}\}$ is consistent with every finite part of D_{∞}, so under the consistency-requiring variant it stays compatible forever. Proof-requiring compatibility demands that each reported loser be *derived*. (iii) On the stream of Theorem 5.1 every random *finite* theory, in any extended language with computable syntax, fails, because its content is r.e.; so random r.e. theories strictly beat random finite theories there.

Relatives of the lottery. The closest relative in computability theory is that randomness helps for mass problems: a class of positive measure can have no computable member. For a computable bound q, the q-bounded diagonally non-computable functions ($f(n) < q(n)$, and $f(n)$ differs from the value at n of a fixed partial computable diagonal function) form a Π^0_1 class with no computable member; under the uniform product measure on $\prod_n\{0, \ldots, q(n)-1\}$ it has measure at least $\prod_n(1 - 1/q(n))$, which is positive exactly when $\sum_n 1/q(n) < \infty$. Every Martin-Löf random sequence computes a diagonally non-computable function (Kučera, 1985; see Bienvenu and Patey, 2017), and Bienvenu and Porter (2016) show that the bounded classes are not tt-negligible when $\sum_n 1/q(n) < \infty$ and are deep when the sum diverges. The lottery's survival condition $\sum_k 1/(N_k+1) = \sum_k 2^{-(k+1)} < \infty$ is this convergence condition, and the diagonal ticket j^*_k plays the role of the diagonal value. The differences: the lottery's diagonal is computed from $\emptyset'$, against r.e. theories rather than partial computable functions; what fails are consistent r.e. theories proving the stream, not computable members of a class; and the class of surviving tapes is the inducer's compatibility event, so its measure is the inducer's likelihood. The modified Demski prior (Garrabrant, 2015; Garrabrant et al., 2016b) also draws random axiom systems

from a universal machine, and, for the process as described in those works, it too gives the lottery stream positive probability: with probability at least $2^{-K(G_{\mathrm{lot}})-c}\prod_k(1-2^{-(k+1)})$ its first draw is a surviving lottery theory, every later draw is kept only if consistent with it, and the final theory extends it. Its completed theories containing the stream are not r.e., but completions of extensions of Q never are, so the contrast between random and r.e. hypotheses is not visible in that prior.

Proposition C.3 (Majority vote)**.** *(a) Let $Z \subseteq \mathrm{Sent}$ be decidable and $r \geq 1$, and suppose that every B-inconsistent finite subset of Z contains a B-inconsistent subset with at most r elements. If $D^{\pm}_{\infty} \subseteq Z$ and $\mathbb{U}(E_\infty) > 0$, then some consistent r.e. $T \subseteq Z$ contains $D^{\pm}_{\infty}$; hence $\inf_n W_A(D_n) > 0$, and the coin-free loss is bounded too. (b) For the one-tape law M_F of SOI with inputs X and labels Y: if $D_\infty : X \rightharpoonup Y$ is infinite and $\lim_n M_F(D_n) > 0$, some partial computable f extends D_∞, so the coin-free version W also has bounded loss on D_∞.*

Proof. (a) The argument is the majority vote of Lemma A.25, with a threshold adapted to the arity. Put $\varepsilon := 1/(r+1)$ and choose w by Lemma C.1 with $\mathbb{U}(E_\infty \mid [w]) > 1-\varepsilon$. Let $T := \{\varphi \in Z : \mathbb{U}(O_\varphi \mid [w]) > 1-\varepsilon\}$; it is r.e., because $\mathbb{U}(O_\varphi \mid [w])$ is left-c.e. uniformly in φ. For $\varphi \in D^{\pm}_{\infty}$ we have $O_\varphi \supseteq E_\infty$, so $\varphi \in T$. Suppose $B \cup T$ is inconsistent. By compactness and the hypothesis, some $\varphi_1, \ldots, \varphi_j \in T$ with $j \leq r$ are jointly B-inconsistent. Then $\mathbb{U}(\bigcap_i O_{\varphi_i} \mid [w]) > 1 - j\varepsilon \geq 1 - r\varepsilon = \varepsilon$, and every tape in that intersection has an inconsistent theory, so $\mathbb{U}(O_\bot \mid [w]) > \varepsilon$. But $O_\bot \cap E_\infty = \emptyset$ gives $\mathbb{U}(O_\bot \mid [w]) < \varepsilon$, a contradiction. So T is consistent, and $\inf_n W_A(D_n) \geq \mathbf{m}(e) > 0$ for an index e of T.

(b) $\mathbb{U}\{\omega : f_\omega \supseteq D_\infty\} = \lim_n M_F(D_n) > 0$ by continuity from above. Choose w with $\mathbb{U}(\{\omega : f_\omega \supseteq D_\infty\} \mid [w]) > \frac{1}{2}$ and let $f(x) := y$ iff $\mathbb{U}(\{\omega : f_\omega(x) = y\} \mid [w]) > \frac{1}{2}$. The events $\{f_\omega(x) = y\}$ are effectively open and disjoint in y, so f is a well-defined partial computable function; for $x \in \operatorname{dom} D_\infty$ the event $\{f_\omega(x) = D_\infty(x)\}$ contains $\{f_\omega \supseteq D_\infty\}$, so $f(x) = D_\infty(x)$.

The threshold is exact for the argument. Let $\{\varphi_1, \ldots, \varphi_s\}$ be a minimal B-inconsistent set, and consider a law with inconsistent mass a and mass $(1-a)/s$ on each of s consistent types, type i proving exactly the φ_j with $j \neq i$ (and their consequences). Then every φ_j is proved with probability $a + (s-1)(1-a)/s$, so the majority set at threshold $1-\varepsilon$ contains all of $\varphi_1, \ldots, \varphi_s$ iff $a > 1 - s\varepsilon$, while the consistent mass exceeds $1-\varepsilon$ iff $a < \varepsilon$. Both hold for some a iff $\varepsilon > 1/(s+1)$. So $1/(r+1)$ is the largest threshold at which the argument is guaranteed to return a consistent set, and a family of minimal inconsistent sets of unbounded size defeats every threshold. A computation (code in the accompanying repository) confirms the critical value for $s \leq 12$, and a linear program over all laws, for each of the 166 nonempty families of minimal inconsistent sets on four sentences, finds the same critical value. □

Instances. Literal data of a fresh predicate ($r = 2$, by Lemma A.14). Plain SOI, by (b). For the lottery class the sets $\{\mathrm{Lot}_k\} \cup \{\neg\mathrm{Win}(\underline{k}, \underline{j}) : j \leq N_k\}$ are minimal inconsistent, of size $N_k + 2$, so no r works; and for every threshold $1-\varepsilon$ the majority set refutes every ticket of each lottery k with $2^{k+1} > 1/\varepsilon$, so it is inconsistent: this is the formal content of Kyburg's lottery paradox.

Proposition C.4 (Majority over true theories)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable. If $\mathcal{T}_{\mathrm{true}} := \{\omega : B \cup A_\omega \subseteq \mathrm{Th}(\mathbb{N}),\ B \cup A_\omega \vdash D^{\pm}_{\infty}\}$ has positive measure, then some r.e. $T \subseteq \mathrm{Th}(\mathbb{N})$ has $B \cup T \vdash D^{\pm}_{\infty}$.*

Proof. $\mathcal{T}_{\mathrm{true}}$ is Borel: $\{\omega : A_\omega \subseteq \mathrm{Th}(\mathbb{N})\} = \bigcap_{\chi\ \mathrm{false}}\{\omega : \chi \notin A_\omega\}$ is closed, and the proving condition is a countable intersection of open sets. Choose w with $\mathbb{U}(\mathcal{T}_{\mathrm{true}} \mid [w]) > \frac{1}{2}$ by Lemma C.1 and let $T := \{\varphi : \mathbb{U}(O_\varphi \mid [w]) > \frac{1}{2}\}$, an r.e. set. It contains $D^{\pm}_{\infty}$, since $O_\varphi \supseteq \mathcal{T}_{\mathrm{true}}$ for $\varphi \in D^{\pm}_{\infty}$.

For $\varphi \in T$, $\mathbb{U}(O_\varphi \cap \mathcal{T}_{\text{true}} \mid [w]) > 0$, so a true theory proves φ and φ is true. Hence $T \subseteq \mathrm{Th}(\mathbb{N})$ and $B \cup T$ proves $D^{\pm}_{\infty}$. □

Truth is a constraint of arity one; this is why the majority argument transfers to it, while Σ_1-soundness, which is not a constraint of arity one, escapes it (Question C.10).

Proposition C.5 (Co-r.e. streams over block-decidable classes)**.** *Let $Z \subseteq \mathrm{Sent}$ be decidable and computably partitioned into finite blocks $Z_0, Z_1, \ldots$ (each given by a canonical index) such that (i) for every finite $X \subseteq Z$, $B \cup X$ is consistent iff $B \cup (X \cap Z_k)$ is consistent for every k, and (ii) consistency of $B \cup X$ for finite $X \subseteq Z_k$ is decidable, uniformly in k. If $D^{\pm}_{\infty} \subseteq Z$ is co-r.e. and $B \cup D^{\pm}_{\infty}$ is consistent, then some consistent decidable $T \subseteq Z$ contains $D^{\pm}_{\infty}$.*

Proof. $Z_k \setminus D^{\pm}_{\infty}$ is r.e. uniformly in k; let C^s_k be Z_k minus what has been enumerated into it by stage s. As Z_k is finite, $C^s_k = D^{\pm}_{\infty} \cap Z_k$ for large s, and that set is consistent with B; so by (ii) the least s_k with $B \cup C^{s_k}_k$ consistent exists and is computable from k. Put $T := \bigcup_k C^{s_k}_k$, a decidable set. $C^s_k \supseteq D^{\pm}_{\infty} \cap Z_k$ for all s, so $T \supseteq D^{\pm}_{\infty}$, and $B \cup T$ is consistent by compactness and (i). □

Instances. (1) The lottery class $Z = \{\mathrm{Lot}_k\} \cup \{\mathrm{Win}(\underline{k}, \underline{j})^b : j \le N_k,\ b \in \{0,1\}\}$ with blocks by k: under the hypotheses of Theorem 5.1 a finite $X \subseteq Z$ is consistent with B iff no block contains a complementary pair, or Lot_k together with all $N_k + 1$ negative literals of lottery k (choose the Win-row of each $\underline{k}$ separately in a model with distinct numerals); so (i) and (ii) hold. A computation (code in the accompanying repository) checks this criterion against satisfiability in all structures with at most five elements, for $B = \emptyset$ and two blocks. (2) The arithmetic lottery classes of Theorem 5.4: by Lemma C.9(i) every propositionally satisfiable finite set of Boolean combinations of the tickets is consistent with B, so consistency is a propositional, decidable and blockwise property. Hence every stream of these kinds that answers the question of Corollary 5.2 negatively is neither r.e. (Remark C.2) nor co-r.e. Whether some co-r.e. stream over a sentence class on which consistency is not decidable has $\mathbb{U}(E_\infty) > 0$ and no consistent r.e. theory proving it is open (it is the second part of Question 5.3). The adversary in the proof waits, in each block, for a stage at which the remaining data are certifiably consistent; a counterexample must deny it any such certificate at every finite stage.

Proof of Corollary 5.2. Theorem 5.1(a),(b) give the negative answer, and Theorem 5.4 gives one inside true arithmetic. The positive cases are Proposition C.3(a), Corollary C.14 (for a stream of Π_1 type, bounded loss gives $\sup_n L_A(D_n) \le k$ for some k), Proposition C.5 (with $B \cup D^{\pm}_{\infty}$ consistent because $\mathbb{U}(E_\infty) > 0$), and Proposition C.4. □

For a theory machine G and a finite string u, A^G_u is the set of axioms emitted under prefixes of u (Definition 2.3).

Proposition C.6 (The proof horizon bounds the coin-free loss)**.** *Let G be a theory machine and D_∞ a stream with*

$$a_G := \mathbb{U}\{\omega : B \cup A^G_\omega \text{ consistent},\ B \cup A^G_\omega \vdash D^{\pm}_{\infty}\} > 0.$$

Let the proof horizon *$h_G(n)$ be the least m with $\mathbb{U}\{\omega : B \cup A^G_\omega$ consistent, $B \cup A^G_{\omega_{1:m}} \vdash \psi_{D_n}\} \ge a_G/2$. Then, in the semimeasure convention, for every n,*

$$-\log W_A(D_n) \ \le\ K^{\mathrm{pad}}(h_G(n)) + K(G) + \log(2/a_G) + c.$$

Proof. $h_G(n)$ exists: a derivation of ψ_{D_n} uses finitely many axioms, all emitted under some finite prefix, so the events in the definition increase in m to a set containing the one that defines a_G. Fix $m \geq h_G(n)$ and let Good_m be the set of $u \in \{0,1\}^m$ such that $B \cup A_u^G \vdash \psi_{D_n}$ and $[u]$ contains a tape ω with $B \cup A_\omega^G$ consistent. For $u \in \mathrm{Good}_m$, $A_u^G \subseteq A_\omega^G$ for such an ω, so $B \cup A_u^G$ is consistent and proves ψ_{D_n}; and $|\mathrm{Good}_m| 2^{-m} \geq a_G/2$. The set A_u^G is r.e. with an index $e(G,u)$ computable from (G,u) and injective in u (write u into the index). Hence, using $K(e(G,u)) \leq K(G)+K(u)+c$ and $K(u) \leq m + K(m) + c$ for $|u| = m$,

$$W_A(D_n) \geq \sum_{u \in \mathrm{Good}_m} \mathbf{m}(e(G,u)) \geq \sum_{u \in \mathrm{Good}_m} 2^{-K(G)-m-K(m)-c} \geq \tfrac{a_G}{2}\, 2^{-K(G)-K(m)-c}.$$

Minimise $K(m)$ over $m \geq h_G(n)$. □

Corollary C.7 (Rate on the lottery stream)**.** *On the stream of Theorem 5.1, in the semimeasure convention:*

(a) $-\log W_A(D_n) \leq K^{\mathrm{pad}}(n) + c$ *for all* n*. Hence the coin-free loss diverges, but more slowly than any computable unbounded nondecreasing function: for every such* f*,* $-\log W_A(D_n) \leq \frac{2}{3} f(n) + c_f$*.*

(b) $K_A(D_n) \leq \lceil n/2 \rceil + O(\log n)$*, and* $K_A(D_n) \to \infty$*.*

(c) Batch lottery. *Give lottery* k *the* 2^{2k+1} *tickets* $0, \ldots, 2^{2k+1}-1$ *and diagonalise in it against every* $e < 2^k$ *at once. Then Theorem 5.1(a)–(d) hold with the same product,* $W_A(D_n) \leq \sum_{e \geq 2^k} \mathbf{m}(e)$ *once lottery* k *is complete, and again* $-\log W_A(D_n) \leq K^{\mathrm{pad}}(n) + c$*.*

The restriction to nondecreasing functions in (a) is needed: a computable unbounded function that vanishes on the odd numbers is not eventually above K^{pad}.

Proof. (a) For $G = G_{\mathrm{lot}}$, every tape gives a consistent theory, and $a_G = \mathbb{U}(\mathrm{Surv}) > 0$. For $\omega \in \mathrm{Surv}$ the axioms needed for D_n are emitted once the first $\lceil n/2 \rceil$ blocks are read, that is, by the prefix of length $m(n) := \sum_{k < \lceil n/2 \rceil} (k+1)$; so $h_{G_{\mathrm{lot}}}(n) \leq m(n)$, which is computable from n, and $K^{\mathrm{pad}}(h_{G_{\mathrm{lot}}}(n)) \leq K(m(n)) \leq K(n) + c$. Proposition C.6 gives $-\log W_A(D_n) \leq K(n) + c'$. As $D_n \subseteq D_{n'}$ for $n \leq n'$, $W_A(D_n) \geq W_A(D_{n'})$, so $-\log W_A(D_n) \leq \min_{n' \geq n} K(n') + c' = K^{\mathrm{pad}}(n) + c'$. *Slow growth.* For a computable nondecreasing unbounded g let $\eta_g(j)$ be the least n' with $g(n') \geq j$, a computable function. Then $\eta_g(g(n)+1) > n$, so $K^{\mathrm{pad}}(n) \leq K(\eta_g(g(n)+1)) \leq 2\log(g(n)+1) + c$. For a computable unbounded nondecreasing f, apply this to $g := \lfloor 2^{f/3} \rfloor$, which gives $K^{\mathrm{pad}}(n) \leq \frac{2}{3} f(n) + c_f$. Divergence is Theorem 5.1(c).

(b) Let $K := \lceil n/2 \rceil$. The tickets j_k^*, $k < K$, are determined by the truth values of the Σ_1 facts "$B \cup \mathsf{W}_k \vdash \mathrm{Lot}_k$" and "$B \cup \mathsf{W}_k \vdash \neg\mathrm{Win}(\underline{k}, \underline{j})$" ($j \leq N_k$): j_k^* is the least unrefuted ticket if W_k proves Lot_k and some ticket is unrefuted, and 0 otherwise (an inconsistent W_k refutes every ticket). There are $2^{K+1} + K - 2$ such facts. Given n and the number of true ones ($K+2$ bits, given K), enumerate proofs until that many facts are found true. So $K(\psi_{D_n}) \leq K + O(\log n)$, and $K_A(D_n) \leq K(\psi_{D_n}) + c$ by Remark A.2(c). $K_A(D_n) \to \infty$ is Theorem 5.1(c).

(c) The data of lottery k are $(\mathrm{Lot}_k, 1)$ and $(\mathrm{Win}(\underline{k}, \underline{j_{k,e}^*}), 0)$ for $e < 2^k$, where Lot_k now names the tickets $0, \ldots, 2^{2k+1}-1$ and $j_{k,e}^*$ is the least ticket that $B \cup \mathsf{W}_e$ does not refute if $B \cup \mathsf{W}_e$ is consistent and proves Lot_k, and 0 otherwise. The random theory reads a block of $2k+1$ bits for lottery k. At most 2^k tickets are reported, so it survives lottery k with probability at least $1 - 2^k/2^{2k+1} = 1 - 2^{-(k+1)}$, and the proof of Theorem 5.1 goes through with the same product. After lottery k every $e < 2^k$ has failed, which gives the bound on W_A. At most n lotteries are

complete after n data, so the horizon is at most $\sum_{k<n}(2k+1) = n^2$, which is computable from n, and (a)'s argument applies. □

A computation (code in the accompanying repository) checks the bookkeeping of (b) and the survival probabilities of the batch lottery for three lotteries.

Corollary C.8 (The lottery separates the versions of consistent SOI)**.** *On the stream of Theorem 5.1: (a) $M_F^{\mathrm{cons}}(D_n)$ is bounded below, so one-tape consistent SOI has bounded loss; (b) $W^{\mathrm{cons}}(D_n) \to 0$ and $W_A(D_n) \to 0$, so coin-free consistent SOI has unbounded loss; (c) plain coin-free SOI on sentences has bounded loss: $W^{\mathrm{all}}(D_n) \geq \mathbf{m}(f) > 0$ for a partial computable f extending D_∞, and every such f is inconsistent.*

Proof. Theorem 4.3 gives $M_F^{\mathrm{cons}} \asymp M_A$, and Theorem 4.2 gives $W^{\mathrm{cons}} \asymp W_A$. (a) $M_F^{\mathrm{cons}}(D_n) \geq 2^{-c} M_A(D_n)$ is bounded below by Theorem 5.1(a). (b) $W^{\mathrm{cons}}(D_n) \leq 2^c W_A(D_n) \to 0$ by Theorem 5.1(c). (c) $M_F(D_n) \geq M_F^{\mathrm{cons}}(D_n)$ is bounded below, so by Proposition C.3(b), applied with inputs $\mathrm{Sent}_{\mathcal{L}}$ and labels $\{0,1\}$, some partial computable f extends D_∞, and $W^{\mathrm{all}}(D_n) \geq \mathbf{m}(f) > 0$. Such an f is inconsistent: a consistent one would give $W^{\mathrm{cons}}(D_n) \geq \mathbf{m}(f) > 0$, against (b). □

C.3 Lotteries in arithmetic

The lottery needs a family of sentences that behave, with respect to consistency with B, like independent atoms. Inside arithmetic with a true background, self-referential ("flexible") formulas provide them. Lemma C.9 is a self-contained proof of the weak form we need, in a version that also prescribes the truth values in $\mathbb{N}$. It is the standard self-referential construction behind Kripke's flexible formulas (Kripke, 1962) and Mostowski's independent sequences (Mostowski, 1961); Woodin's universal algorithm is a much stronger statement of the same kind (Woodin, 2011; Blanck and Enayat, 2017; Hamkins, 2018). We did not find the version with a prescribed truth set stated in this form, but it is a routine variant, and we claim no novelty for it. For a code z let $\varphi_{[z]}$ be the formula with code z.

Lemma C.9 (Flexible formulas with prescribed truth)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true with $B \supseteq \mathsf{Q}$. Write $\mathrm{In}(y,e)$ for a Δ_0 formula saying "y is in the finite set with canonical index e", and for a finite $F \subseteq \mathbb{N}$ with index e_F write $y \in F$ for $\mathrm{In}(y, \underline{e_F})$. For every formula $\Lambda(z,x)$ there is a formula $\Upsilon(x)$ such that:*

(i) $B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ is consistent for every finite $F \subseteq \mathbb{N}$; hence every finite pattern $\{\Upsilon(\underline{n})^{b_n} : n \in F'\}$ is consistent with B;

(ii) $\mathbb{N} \models \Upsilon(\underline{n})$ iff $\mathbb{N} \models \Lambda(\ulcorner\Upsilon\urcorner, \underline{n})$, for every n;

(iii) $\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_\Upsilon(x) \vee (\pi_\Upsilon \wedge \Lambda(\ulcorner\Upsilon\urcorner, x)))$ for a Σ_1 formula $\sigma_\Upsilon(x)$ that is false of every number and a true Π_1 sentence π_Υ.

Proof. Fix a computable enumeration $\tau_0, \tau_1, \ldots$ of the theorems of B. Call a sentence *of form* (z,e) if it is $\neg\forall y\,(\varphi_{[z]}(y) \leftrightarrow \mathrm{In}(y, \underline{e}))$; whether τ_s is of form (z,e) for some e, and for which e, is decidable from (s,z). Let $\sigma_0(z,x)$ be the Σ_1 formula "there are s and e such that τ_s is of form (z,e), no $\tau_{s'}$ with $s' < s$ is of form (z,e') for any e', and $\mathrm{In}(x,e)$" (the computation of $\tau_0, \ldots, \tau_s$ is an existentially quantified witness, checked by a Δ_0 formula), and $\pi_0(z)$ the Π_1 formula "no τ_s is of form (z,e) for any e". By (Diag), applied to $\vartheta(z,x) := \sigma_0(z,x) \vee (\pi_0(z) \wedge \Lambda(z,x))$, take Υ with

$$\mathsf{Q} \vdash \forall x\,(\Upsilon(x) \leftrightarrow \sigma_0(\ulcorner\Upsilon\urcorner, x) \vee (\pi_0(\ulcorner\Upsilon\urcorner) \wedge \Lambda(\ulcorner\Upsilon\urcorner, x))).$$

As Q is true, the same equivalence holds in $\mathbb{N}$. Suppose some τ_s is of form $(\ulcorner\Upsilon\urcorner, e)$; let s_0 be the least such s, with index e_0 and set F_0. In $\mathbb{N}$, $\sigma_0(\ulcorner\Upsilon\urcorner, n)$ holds iff $n \in F_0$, and $\pi_0(\ulcorner\Upsilon\urcorner)$ fails; so $\mathbb{N} \models \forall y\,(\Upsilon(y) \leftrightarrow y \in F_0)$. But τ_{s_0}, the negation of this sentence, is a theorem of the true theory B, a contradiction. Hence no theorem of B is of form $(\ulcorner\Upsilon\urcorner, e)$, which is (i); the pattern follows from $\forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{n \in F' : b_n = 1\}$, since Q decides each $\underline{n} \in F$ correctly by $(\Sigma_1\mathrm{C})$. Moreover $\pi_0(\ulcorner\Upsilon\urcorner)$ is true and $\sigma_0(\ulcorner\Upsilon\urcorner, \underline{n})$ is false for every n, which gives (ii), and (iii) with $\sigma_\Upsilon(x) := \sigma_0(\ulcorner\Upsilon\urcorner, x)$ and $\pi_\Upsilon := \pi_0(\ulcorner\Upsilon\urcorner)$. □

Let $\langle k, j\rangle$ be a Δ_0 pairing with $k, j \le \langle k, j\rangle$.

Proof of Theorem 5.4. The tickets. In (a), Υ_0 is Lemma C.9's formula for $\Lambda(z, x) :\equiv (x \neq x)$. In (b), let $\mathrm{Ref}(z, k, j)$ be the Σ_1 formula "$B \cup \mathsf{W}_k \vdash \neg\varphi_{[z]}(\underline{\langle k, j\rangle})$", and let $\Lambda(z, x)$ be the formula, with all quantifiers outside Ref bounded,

$$\exists k \le x\, \exists t \le x \Big(x = \langle k, t\rangle \wedge t \le N_k \\ \wedge \big[(\mathrm{Ref}(z, k, t) \wedge \forall i < t\, \neg\mathrm{Ref}(z, k, i)) \vee (t = 0 \wedge \forall i \le N_k\, \neg\mathrm{Ref}(z, k, i))\big]\Big);$$

Υ_1 is the lemma's formula for this Λ. In both cases $\mathrm{Tick}_{k,j} := \Upsilon(\underline{\langle k, j\rangle})$ for the formula $\Upsilon \in \{\Upsilon_0, \Upsilon_1\}$, and $\mathrm{Lot}^{\mathrm{ar}}_k := \bigvee_{j \le N_k} \mathrm{Tick}_{k,j}$. In (a), j^*_e is defined as in Theorem 5.1, with $\mathrm{Tick}_{e,j}$ in place of $\mathrm{Win}(\underline{e}, \underline{j})$. In (b), let j°_k be the least $j \le N_k$ with $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$ ($j^\circ_k := 0$ if there is none), and let j^*_k be the least $j \le N_k$ with $j \neq j^\circ_k$ and $B \cup \mathsf{W}_k \nvdash \neg\mathrm{Tick}_{k,j}$ if $B \cup \mathsf{W}_k$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_k$, and the least $j \neq j^\circ_k$ otherwise. The stream is $((\mathrm{Lot}^{\mathrm{ar}}_k, 1), (\mathrm{Tick}_{k,j^*_k}, 0))_{k \ge 0}$.

The random theory. $G^{\mathrm{ar}}_{\mathrm{lot}}$ reads the tape in blocks as G_{lot} does and emits $\mathrm{Lot}^{\mathrm{ar}}_k$, Tick_{k,J_k} and $\neg\mathrm{Tick}_{k,j}$ ($j \le N_k$, $j \neq J_k$) after reading block k. A finite part of its output is implied by $B + \forall y\,(\Upsilon(y) \leftrightarrow y \in F)$ with $F := \{\langle k, J_k\rangle : k \le K\}$ for some K, so by Lemma C.9(i) and compactness its theory is consistent with B on every tape.

The diagonal. If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Lot}^{\mathrm{ar}}_e$, it does not refute every $\mathrm{Tick}_{e,j}$, $j \le N_e$, since $\mathrm{Lot}^{\mathrm{ar}}_e \wedge \bigwedge_j \neg\mathrm{Tick}_{e,j}$ is propositionally contradictory. In (b), j^*_e exists in this case: if W_e refutes some ticket, then j°_e is refuted and every unrefuted ticket differs from j°_e; if it refutes none, then $j^\circ_e = 0$ and $j = 1$ is unrefuted ($N_e \ge 1$). Everything else is the proof of Theorem 5.1 verbatim, with Tick for Win: survival probability $\prod_k (1 - 2^{-(k+1)})$, the diagonal kills every consistent r.e. theory, dominated convergence, and the posterior on deterministic machines.

Truth and complexity. (a) By Lemma C.9(ii) with $\Lambda \equiv (x \neq x)$, every $\mathrm{Tick}_{k,j}$ is false, and by (iii) $\mathsf{Q} \vdash \mathrm{Tick}_{k,j} \leftrightarrow \sigma_{\Upsilon_0}(\underline{\langle k, j\rangle})$, since the second disjunct is Q-refutable: a Σ_1 sentence. So $\mathrm{Lot}^{\mathrm{ar}}_k$ is Q-equivalent to a false Σ_1 sentence and $\neg\mathrm{Tick}_{k,j^*_k}$ to a true Π^*_1 sentence. (b) By Lemma C.9(ii), $\mathrm{Tick}_{k,j}$ is true iff $j = j^\circ_k$: for $z = \ulcorner\Upsilon_1\urcorner$, $\mathrm{Ref}(z, k, j)$ holds iff $B \cup \mathsf{W}_k \vdash \neg\mathrm{Tick}_{k,j}$, and $\Lambda(z, \underline{\langle k, j\rangle})$ says exactly that j is the least refuted ticket, or 0 if none is refuted. Hence $\mathrm{Lot}^{\mathrm{ar}}_k$ is true, and Tick_{k,j^*_k} is false because $j^*_k \neq j^\circ_k$; every datum is true. By (iii), $\mathrm{Tick}_{k,j}$ is Q-equivalent to $\sigma_{\Upsilon_1}(\underline{\langle k, j\rangle}) \vee (\pi_{\Upsilon_1} \wedge \Lambda(\ulcorner\Upsilon_1\urcorner, \underline{\langle k, j\rangle}))$. With a numeral argument, (O) and $(\Sigma_1\mathrm{C})$ unfold the bounded quantifiers of Λ and decide its Δ_0 parts, provably in Q ($t \le N_k$ is Δ_0 because the graph of 2^x is); so $\Lambda(\ulcorner\Upsilon_1\urcorner, \underline{\langle k, j\rangle})$ is Q-equivalent to a Boolean combination of the Σ_1 sentences $\mathrm{Ref}(\ulcorner\Upsilon_1\urcorner, \underline{k}, \underline{i})$, $i \le N_k$. Hence every datum is Q-equivalent to a Boolean combination of Σ_1 sentences. □

Proof of Corollary 5.5. (i) Theorem 5.4(b) with Theorem 5.1(a),(c). (ii) If $\mathbb{U}\{\omega \in E_\infty : B \cup A_\omega \text{ true}\} > 0$, Proposition C.4 gives a true r.e. theory proving the stream, contrary to Theorem 5.1(b). (iii) Fix k with $J := J_k \neq j^\circ_k$. The random lottery theory proves $\mathrm{Tick}_{k,J}$, which is Q-equivalent to $\sigma_{k,J} \vee (\pi_{\Upsilon_1} \wedge \Lambda_{k,J})$ with $\sigma_{k,J} := \sigma_{\Upsilon_1}(\underline{\langle k, J\rangle})$ a false Σ_1 sentence and

$\Lambda_{k,J} := \Lambda(\ulcorner \Upsilon_1 \urcorner, \underline{\langle k, J \rangle})$. Write Ref_i for $\mathrm{Ref}(\ulcorner \Upsilon_1 \urcorner, \underline{k}, \underline{i})$. If some ticket is refuted and $J > j_k^\circ$, then $\mathrm{Ref}_{j_k^\circ}$ is a true Σ_1 sentence, so $\mathsf{Q} \vdash \mathrm{Ref}_{j_k^\circ}$ and $\mathsf{Q} \vdash \neg\Lambda_{k,J}$, and the theory proves the false Σ_1 sentence $\sigma_{k,J}$. If $J < j_k^\circ$, or no ticket is refuted (then $j_k^\circ = 0 < J$), then Q proves $\Lambda_{k,J} \to \mathrm{Ref}_J$ (for $J = 0 < j_k^\circ$ through $\mathsf{Q} \vdash \mathrm{Ref}_{j_k^\circ}$, which refutes the "none refuted" disjunct), and Ref_J is false; so the theory proves $\sigma_{k,J} \vee \mathrm{Ref}_J$, a false Σ_1 sentence. Given survival ($J_k \neq j_k^*$ for all k), the events $J_k = j_k^\circ$ are independent with probabilities $1/N_k$, so all of them hold with probability $\prod_k 1/N_k = 0$. □

A computation (code in the accompanying repository) checks the bookkeeping of Theorem 5.4(b) and the three cases of (iii) exhaustively in a propositional model with up to eight tickets per lottery, for every behaviour of W_k and every winner.

Question C.10. On the stream of Theorem 5.4(b), is $P_\infty(\{\Sigma_1\text{-sound theories}\}) = 0$?

Corollary 5.5(ii) proves that the limit posterior lives on false theories, and (iii) that the random lottery theories among them are Σ_1-unsound; Proposition C.4's majority argument does not transfer to Σ_1-soundness, which is not a constraint of arity one.

C.4 Π_1 data

Throughout this subsection $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and true with $B \supseteq \mathsf{Q}$. Inc_0 is left-c.e. and not computable (Theorem 3.3(a)), hence irrational. Put $I'_k := \lfloor 2^{k+2}\mathrm{Inc}_0 \rfloor 2^{-(k+2)}$, so that $I'_k \leq \mathrm{Inc}_0 < I'_k + 2^{-(k+2)}$; given k, I'_k is given by $k+2$ bits. If D is of Π_1 type, then ψ_D is B-equivalent to $\pi_D \wedge \sigma_D$ with $\pi_D \in \Pi_1^*$ (the class is closed under conjunction) and σ_D a conjunction of true Σ_1 sentences; $B \vdash \sigma_D$ by (Σ_1C); so $B \vdash \psi_D \leftrightarrow \pi_D$, $O_{\psi_D} = O_{\pi_D}$ and $M_A(D) = \mathrm{Bel}_0(\pi_D)$. The definition is invariant under B-equivalence of literals, as it must be, since M_A, W_A and K_A depend on D only through $\mathrm{Cn}(D^\pm)$ (Old).

Theorem C.11 (Threshold theories)**.** *There is c such that for every $k \geq 0$ the set*

$$T_k := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)}\}$$

(i) is r.e., with an index e_k of complexity $K(e_k) \leq k + K(k) + c$; (ii) contains every $\pi \in \Pi_1^*$ with $\mathrm{Bel}_0(\pi) \geq 2^{-k}$; (iii) consists of true sentences of positive prior belief, so $B \cup T_k$ is true, consistent and Σ_1-sound.

Proof. (i) $\mathbb{U}(O_\pi)$ is left-c.e. uniformly in π, and Π_1^* is decidable, so T_k is r.e. uniformly in (k, I'_k); a program consists of a self-delimiting description of k, the $k+2$ bits of I'_k, and a fixed code. (ii) $\mathbb{U}(O_\pi) = \mathrm{Inc}_0 + \mathrm{Bel}_0(\pi) \geq I'_k + 2^{-k} > I'_k + 2^{-(k+2)}$. (iii) If $\mathbb{U}(O_\pi) > I'_k + 2^{-(k+2)} > \mathrm{Inc}_0$, then $\mathrm{Bel}_0(\pi) > 0$: some tape has $B \cup A_\omega$ consistent and proving π, and π is true by (Π_1T). □

Proof of Theorem 5.6. π is emitted on the tape r iff $\mathbb{U}(O_\pi) > 0.r_{1:s} + 2^{-s}$ for some s; as $0.r_{1:s} + 2^{-s}$ decreases strictly to r for non-dyadic r, this holds iff $\mathbb{U}(O_\pi) > r$. If $r < \mathrm{Inc}_0$, every $\pi \in \Pi_1^*$ has $\mathbb{U}(O_\pi) \geq \mathrm{Inc}_0 > r$, including false ones such as $\neg(0 = 0)$, so the theory is inconsistent. If $r > \mathrm{Inc}_0$, every emitted π has $\mathrm{Bel}_0(\pi) > 0$ and is true by (Π_1T); B is true; so $B \cup A_r^{G_{\mathrm{thr}}}$ is true. For D of Π_1 type: if $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$, the theory is consistent and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$; the dyadic tapes form a null set, so $\nu_{G_{\mathrm{thr}}}(D) \geq \mathbb{U}(O_{\pi_D}) - \mathrm{Inc}_0 = \mathrm{Bel}_0(\pi_D) = M_A(D)$. The upper bound is (Dom), with $K(\nu_{G_{\mathrm{thr}}}) \leq K(G_{\mathrm{thr}}) + c$. □

A computation (code in the accompanying repository) checks Theorems 5.6 and C.11 in finite models of the prior.

Proof of Corollary 5.7. The first two inequalities are (Chain). For the last, let $k := \lceil L_A(D) \rceil$, so that $\mathrm{Bel}_0(\pi_D) = M_A(D) \geq 2^{-k}$. By Theorem C.11, $\pi_D \in T_k$, and $B \cup T_k$ is consistent and proves π_D, hence ψ_D. So $K_A(D) \leq K(e_k) + c \leq k + K(k) + c$. □

Remark C.12 (The coin-free version sums over levels). For every D of Π_1 type, $-\log W_A(D) \leq L_A(D) + K^{\mathrm{pad}}(\lceil L_A(D) \rceil + 1) + c$. This is a bound on $-\log W_A$; the proved bound on K_A is the weaker $L_A + K(\lceil L_A \rceil) + c'$ of Corollary 5.7.

Proof. For a pair (j, L) with $L \geq 0$ and $0 \leq j \leq 2^L$ let $Z_{j,L} := \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > j2^{-L}\}$. It is r.e., with an index $e(j, L)$ computable from the *pair* and injective in the pair (write the pair into the index), and $K(e(j, L)) \leq K(j, L) + c \leq L + K(L) + c'$, since j has $L + 1$ bits given L. If $j2^{-L} \in (\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$, then every member of $Z_{j,L}$ has $\mathrm{Bel}_0 > 0$ and is true by (Π_1T), so $B \cup Z_{j,L}$ is true, contains π_D and proves ψ_D. With $k := \lceil L_A(D) \rceil$ the open interval $(\mathrm{Inc}_0, \mathbb{U}(O_{\pi_D}))$ has length $M_A(D) \geq 2^{-k}$, so for each $L \geq k + 1$ it contains at least $2^{L-k} - 1 \geq 2^{L-k-1}$ multiples of 2^{-L}. The indices $e(j, L)$ are distinct for distinct pairs, so

$$W_A(D) \;\geq\; \sum_{L \geq k+1} 2^{L-k-1}\, 2^{-L-K(L)-c'} \;=\; 2^{-k-1-c'} \sum_{L \geq k+1} 2^{-K(L)} \;\geq\; 2^{-k-1-c'-K^{\mathrm{pad}}(k+1)}.$$

Indexing the sets by the rational $j2^{-L}$ instead of by the pair would count $j2^{-L} = 2j2^{-(L+1)}$ once at every finer level, and the displayed sum would not bound W_A as written; a computation (code in the accompanying repository) confirms that the pair-indexed count meets the bound on all test intervals. □

Question C.13. On data of Π_1 type, is $K_A \leq L_A + O(1)$?

The deterministic threshold needs the level written in full; the random threshold of Theorem 5.6 does not.

Corollary C.14 (Bounded loss on a Π_1 stream is one true theory)**.** *If every finite part of D_∞ is of Π_1 type and $\sup_n L_A(D_n) \leq k$, then one true theory, $B \cup T_k$, of complexity at most $k + K(k) + c$, proves every datum; so $\sup_n -\log W_A(D_n) \leq k + K(k) + c$.*

Proof. For each n, $\mathrm{Bel}_0(\pi_{D_n}) = M_A(D_n) \geq 2^{-k}$, so $\pi_{D_n} \in T_k$ by Theorem C.11(ii); T_k does not depend on n. By (iii), $B \cup T_k$ is true, and it proves every ψ_{D_n}; its index has complexity at most $k + K(k) + c$ by (i), which bounds $\sup_n -\log W_A(D_n)$ through Definition 2.7. □

Corollary C.15 (Chaitin's theorem for random theories)**.** *For all m and x, in the semimeasure convention: (a) if $K(x) \leq m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) = \infty$; if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \geq m - K(m) - K(\lceil L_A \rceil) - c \geq m - 2\log m - O(\log\log m)$; (b) if $K(x) > m$, then $L_A(\{(\mathrm{Kgt}_m(x), 1)\}) \leq K_A(\{(\mathrm{Kgt}_m(x), 1)\}) + c \leq m + K(m) + c'$, and one true theory, B plus a single Π_1 sentence of complexity at most $m + K(m) + c$, proves $\mathrm{Kgt}_m(x)$ for every such x at once. The same bounds, with other constants, hold in the three-valued convention, in which the cost of the label 1 on this single datum is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x), 1)\}) + \log(1 - \mathrm{Inc}_0)$.*

Proof. The deterministic bound (Chaitin). If $B \cup \mathsf{W}_e$ is consistent and proves $\mathrm{Kgt}_m(x)$, the program "given e and m, enumerate the theorems of $B \cup \mathsf{W}_e$ and output the first z such that $\mathrm{Kgt}_m(z)$ is among them" halts with some z, and $K(z) \leq K(e) + K(m) + c$; $\mathrm{Kgt}_m(z)$ is true by (Π_1T), so $K(z) > m$, whence $K(e) > m - K(m) - c$. So $K_A(\{(\mathrm{Kgt}_m(x), 1)\}) \geq m - K(m) - c$.

(b) Let s_m be the largest code of a halting computation of V on a program of length at most m (0 if there is none); it exists because there are finitely many such programs and each halting

computation has one code (Uniq). Let $\mathrm{Bd}_m := \forall t\,\forall p\,\forall y\,(p \le \underline{\mathrm{rk}_m} \wedge \mathrm{Comp}(p,t,y) \to t \le \underline{s_m})$, a true Π_1 sentence. For x with $K(x) > m$, the Δ_0 sentence $\forall p \le \underline{\mathrm{rk}_m}\,\forall t \le \underline{s_m}\,\neg\mathrm{Comp}(p,t,\underline{x})$ is true, hence Q-provable, and together with Bd_m it gives $\mathrm{Kgt}_m(x)$ by pure logic. s_m is computable from m and the number $H_m \le 2^m$ of halting programs of length at most m (Kraft's inequality), an $(m+1)$-bit number: run the programs until H_m of them have halted. So $K(\mathrm{Bd}_m) \le K(m) + m + c$. Since $B \cup \{\mathrm{Bd}_m\}$ is true, hence consistent, and proves $\mathrm{Kgt}_m(x)$, Remark A.2(c) gives $K_A \le K_{\mathrm{fin}} + c \le m + K(m) + c'$; and $L_A \le K_A + c$ by (Chain). Without unique codes a halting program could have infinitely many codes, and Bd_m would be false.

(a) For $K(x) \le m$ the datum is a false Π_1 sentence, and $M_A = 0$ by (Π_1T). Otherwise Corollary 5.7 and the deterministic bound give $L_A \ge K_A - K(\lceil L_A \rceil) - c' \ge m - K(m) - K(\lceil L_A \rceil) - c''$. By (b), $\lceil L_A \rceil \le m + K(m) + c$, so $K(\lceil L_A \rceil) \le \log m + O(\log\log m)$, and $K(m) \le \log m + O(\log\log m)$. For the three-valued convention: for a single two-valued datum $M^3_A(\{(\varphi,1)\}) = M_A(\{(\varphi,1)\})$, and $M^3_A(\emptyset) = 1 - \mathrm{Inc}_0$, so the three-valued cost of the label 1 is $-\log \mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x)) = L_A(\{(\mathrm{Kgt}_m(x),1)\}) + \log(1-\mathrm{Inc}_0)$, which differs from L_A by a constant. □

The deterministic step is Chaitin's bound (Chaitin, 1974), used only in its upper-bound form (a consistent extension of Q that proves $\mathrm{Kgt}_m(x)$ has complexity greater than $m - K(m) - c$); it makes no claim about least constants, which depend on the coding of machines (Raatikainen, 1998). One m-bit Π_1 law explains every true complexity lower bound at level m. Yet the datum carries at most one bit of label information: unconstrained SOI predicts it at cost $O(1)$ by always answering "true", and the lower bound of Proposition A.15 is $O(1)$ here. In the normalised convention the m bits disappear:

Proposition C.16 (The normalised convention: deciding, not predicting)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for all m, d and $j \ge 0$, all strings $x \in \{0,1\}^d$ except at most $2^{2m+K(m)+j+c} + 2^{m+1}$ of them satisfy $K(x) > m$ and $\mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{-j}\mathrm{Bel}_\emptyset(\mathrm{Kgt}_m(x))$; so the loss of $(\mathrm{Kgt}_m(x),1)$ in the normalised convention is at most $\log(1+2^{-j}) \le 1.45 \cdot 2^{-j}$.*

Proof. Step 1: few refutable strings per theory. Let $x_1,\dots,x_{2^{m+1}}$ be distinct strings. By (O), $\mathsf{Q} \vdash p \le \underline{\mathrm{rk}_m} \to \bigvee_{q \le \mathrm{rk}_m} p = \underline{q}$, so $\neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to $\bigvee_{q \le \mathrm{rk}_m} \exists t\,\mathrm{Comp}(\underline{q},t,\underline{x_i})$. Hence $\bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$ is Q-equivalent to the disjunction, over all maps $f : \{1,\dots,2^{m+1}\} \to \{0,\dots,\mathrm{rk}_m\}$, of $\bigwedge_i \exists t\,\mathrm{Comp}(\underline{f(i)},t,\underline{x_i})$. There are $\mathrm{rk}_m + 1 = 2^{m+1} - 1$ programs, fewer than the strings, so every f has $f(i) = f(i')$ for some $i \ne i'$, and then (Uniq) gives $x_i = x_{i'}$, which Q refutes. So $\mathsf{PA} \vdash \neg \bigwedge_{i \le 2^{m+1}} \neg\mathrm{Kgt}_m(x_i)$: every consistent extension of B refutes $\mathrm{Kgt}_m(x)$ for at most $2^{m+1} - 1$ strings x. Integrating over the consistent tapes (Tonelli), $\sum_x \mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) \le 2^{m+1}$.

Step 2: a lower bound on belief. For $K(x) > m$, Corollary C.15(b) and (Dom) give $\mathrm{Bel}_0(\mathrm{Kgt}_m(x)) \ge 2^{-m-K(m)-c}$.

Step 3: Markov's inequality. By Step 1, fewer than $2^{m+1} \cdot 2^{j+m+K(m)+c}$ strings have $\mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) > 2^{-j}2^{-m-K(m)-c}$, and fewer than 2^{m+1} strings have $K(x) \le m$. For every other x, Step 2 gives $\mathrm{Bel}_0(\neg\mathrm{Kgt}_m(x)) \le 2^{-j}\mathrm{Bel}_0(\mathrm{Kgt}_m(x))$; and $\mathrm{Dis}_\emptyset$, $\mathrm{Bel}_\emptyset$ are $\mathrm{Bel}_0(\neg\cdot)$, $\mathrm{Bel}_0(\cdot)$ divided by the same $1-\mathrm{Inc}_0$. The normalised probability of the label 1 is $\mathrm{Bel}_\emptyset/(\mathrm{Bel}_\emptyset + \mathrm{Dis}_\emptyset) \ge 1/(1+2^{-j})$, and $\log(1+y) \le y/\ln 2$. □

The m bits are the price of deciding the sentence, not of predicting its truth value: every Π_1 affirmation must be backed by a true theory, while a consistent extension of PA refutes $\mathrm{Kgt}_m(x)$ for fewer than 2^{m+1} strings. So for $d \ge 2m + K(m) + j + c'$ almost every string of length d costs essentially nothing in the normalised convention, while every x with $K(x) > m$ costs $m \pm O(\log m)$ bits in the semimeasure and three-valued conventions (Corollary C.15). In

the three-valued convention most of the remaining mass goes to "independent": by Step 1, $\sum_x \mathrm{Dis}_\emptyset(\mathrm{Kgt}_m(x)) \le 2^{m+1}/(1-\mathrm{Inc}_0)$, so for most x the inducer predicts "independent" with probability $1-2^{-m+O(\log m)}$. One-tape consistent SOI behaves the same way: Theorem 4.3 transfers $M_A(\{(\varphi,b)\})$ to $M_F^{\mathrm{cons}}(\{(\varphi,b)\})$ up to factors $2^{\pm c}$, which leaves the semimeasure statement unchanged and shifts j by a constant in the normalised one. With $B=\mathsf{Q}$ alone, Step 1 needs Q to prove the uniqueness of outputs, which it may not; the normalised behaviour is then open. A computation (code in the accompanying repository) checks the finite bookkeeping of Steps 1 and 3 in finite models in which uniqueness of outputs is imposed.

Proposition C.17 (True Π_1^*-axiomatised theories)**.** *Let T be an r.e. set of true Π_1^* sentences and $k := \lceil -\log \mathbb{U}_A(\mathrm{Ext}T)\rceil$. Then*

$$-\log \mathbb{U}_A(\mathrm{Ext}T) - c \;\le\; -\log \mathbf{m}(\mathrm{Ext}T) \;\le\; -\log \mathbb{U}_A(\mathrm{Ext}T) + K(k) + c.$$

Proof. Right: each $\pi \in T$ has $\mathrm{Bel}_0(\pi) \ge \mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-k}$, since every consistent theory extending $\mathrm{Cn}(T)$ proves π; so $T \subseteq T_k$. $B \cup T_k$ is consistent, so $e_k \in \mathrm{Ext}(T)$ and $\mathbf{m}(\mathrm{Ext}T) \ge \mathbf{m}(e_k) \ge 2^{-k-K(k)-c}$. Left: $\mathbb{U}_A(\mathrm{Ext}T) \ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (Remark A.3; $B \cup T$ is consistent because it is true). □

The same comparison for the exact class $[T]$, and for general T, is part of Question A.19.

Proposition C.18 (One false Σ_1 datum)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, B decidable and true with $B \supseteq \mathsf{Q}$, and Υ_0 the formula of Theorem 5.4(a). For every $d \ge 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(\Upsilon_0(\underline{x_1}),0),(\Upsilon_0(\underline{x_2}),1)\}$ (strings coded by length-lexicographic rank) has $K_A(D) > \log d - 2$ and $L_A(D) \le 2 + c$. The literal $\neg\Upsilon_0(\underline{x_1})$ is Q-equivalent to a true Π_1^* sentence, and $\Upsilon_0(\underline{x_2})$ to a false Σ_1 sentence.*

Proof. Upper bound. The machine that, for every n, reads a fresh bit b_n and emits $\Upsilon_0(\underline{n})^{b_n}$ has a consistent theory on every tape (Lemma C.9(i) and compactness) and proves D with probability $\frac{1}{4}$; apply (Dom). *Lower bound* (the colouring argument of Theorem 3.9). Let $\ell := \lceil \log(d/\log 3)\rceil - 1$, so that $2^\ell < d/\log 3$ and $3^{2^\ell - 1} < 2^d$. Fewer than 2^ℓ programs have length less than ℓ. Colour each $x \in \{0,1\}^d$ by the vector, over these programs p, of "+" if $V(p)$ halts with an index e such that $B \cup \mathsf{W}_e$ is consistent and proves $\Upsilon_0(\underline{x})$, "−" if it is consistent and proves $\neg\Upsilon_0(\underline{x})$, and "∗" otherwise. Two distinct x_1, x_2 share a colour, and then no program shorter than ℓ gives a consistent theory proving $\neg\Upsilon_0(\underline{x_1}) \wedge \Upsilon_0(\underline{x_2})$. So $K_A(D) \ge \ell \ge \log d - \log\log 3 - 1 > \log d - 2$. The complexity claims are Theorem 5.4(a)'s. □

Proposition C.19 (Two true data)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and true with $B \supseteq \mathsf{Q}$. There is a formula Υ_2 such that for every $d \ge 2$ some distinct $x_1, x_2 \in \{0,1\}^d$ give true data $D_d := \{(\Upsilon_2(\underline{x_1}),0),(\Upsilon_2(\underline{x_2}),1)\}$ with $K_A(D_d) > \log d - 2$ and $L_A(D_d) \le 2 + c$. Each datum is Q-equivalent to a Boolean combination of Σ_1 sentences (hence, by prenexing, both to a Σ_2 and to a Π_2 sentence).*

Proof. The colour formulas. For a program p (a numeral) put

- $\mathrm{Halt}_p :\equiv \exists t\, \exists e\, \mathrm{Comp}(\underline{p}, t, e)$, a Σ_1 sentence ($V(p)$ halts);
- $\mathrm{Ok}_p :\equiv \forall t\, \forall e\, \big(\mathrm{Comp}(\underline{p}, t, e) \to \mathrm{Con}(B \cup \mathsf{W}_e)\big)$, a Π_1 sentence, with $\mathrm{Con}(B \cup \mathsf{W}_e)$ the canonical Π_1 formula in e;
- $\mathrm{Prv}_p^b(z,y) :\equiv \exists t\, \exists e\, \big(\mathrm{Comp}(\underline{p}, t, e) \wedge \mathrm{Prov}_{B\cup\mathsf{W}_e}(\ulcorner \varphi_{[z]}(\dot{y})^{b} \urcorner)\big)$, a Σ_1 formula, for $b \in \{0,1\}$.

The colour entry of y at p is "+" iff $\mathrm{Halt}_p \wedge \mathrm{Ok}_p \wedge \mathrm{Prv}_p^1(z,y)$, "−" iff $\mathrm{Halt}_p \wedge \mathrm{Ok}_p \wedge \mathrm{Prv}_p^0(z,y)$, and "$*$" otherwise. In $\mathbb{N}$, where $V(p)$ has at most one output, these say exactly what the colouring in the proof of Proposition C.18 says (with $\varphi_{[z]}$ for Υ_0). The consistency condition is written with a universal quantifier because "$\exists e\,(\mathrm{Comp}(p,t,e) \wedge \mathrm{Con}(\dots))$" would be Σ_2. Let $\Lambda(z,x)$ say: "x is the code of a string of some length $d \geq 2$, and x is the second element of the lexicographically first pair of distinct strings of length d that have the same colour", colours taken over the programs of length less than $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, with all quantifiers outside the colour formulas bounded by x. Let Υ_2 be Lemma C.9's formula for Λ, and (x_1, x_2) that pair for $z = \ulcorner \Upsilon_2 \urcorner$; it exists by the pigeonhole count of Proposition C.18. By Lemma C.9(ii), $\Upsilon_2(\underline{x_2})$ is true and $\Upsilon_2(\underline{x_1})$ is false, so both data are true. The fair-coin machine of Proposition C.18, with Υ_2 in place of Υ_0, gives $L_A(D_d) \leq 2 + c$ by Lemma C.9(i). The pair has the same colour for the actual Υ_2, which gives $K_A(D_d) \geq \ell_d > \log d - 2$ as in Proposition C.18. For a numeral argument, (O) and (Σ_1C) unfold the bounded quantifiers of $\Lambda(\ulcorner \Upsilon_2 \urcorner, \underline{x})$ and decide its Δ_0 parts in Q, leaving a Boolean combination of the Σ_1 sentences Halt_p, Prv_p^b and the Π_1 sentences Ok_p; with Lemma C.9(iii), each datum is Q-equivalent to a Boolean combination of Σ_1 sentences, hence, by prenexing, both to a Σ_2 and to a Π_2 sentence. □

C.5 Gödel phenomena

Proof of Proposition 5.8. (a) The posterior lives on consistent extensions of Q, and all of them prove σ by (Σ_1C). (b) Some conjunct of π is false; its negation is a true Σ_1 sentence, so every consistent extension of Q proves $\neg\pi$. (c) Consider the r.e. theories $D^{\pm} \cup \{\pi\}$, $D^{\pm} \cup \{\neg\pi\}$ and $D^{\pm}$. The first is true in a model of $B \cup D^{\pm}$ whose $\mathcal{L}_{\mathrm{ar}}$-reduct is $\mathbb{N}$, so it is consistent with B, and it proves π. The second is consistent with B because $B \cup D^{\pm} \nvdash \pi$ (by the deduction theorem for the finite set $D^{\pm}$), and it proves $\neg\pi$. The third proves neither: not π by hypothesis, and not $\neg\pi$ because $B \cup D^{\pm}$ is true and $\neg\pi$ is false. All three prove ψ_D and have positive weight by (Dom). (d) Apply (c) to the Π_1^* sentence $\neg\sigma$. □

An extreme case. If the data present $T := \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, a consistent and Σ_1-unsound theory, every consistent extension of T proves $\neg\mathrm{Con}(T)$, since $\mathsf{PA} \vdash \mathrm{Con}(T) \to \mathrm{Con}(\mathsf{PA})$; so $\mathrm{Dis}(\mathrm{Con}(T)) = 1$ although $\mathrm{Con}(T)$ is true. A computation (code in the accompanying repository) checks (a)–(d) and the one-sided error pattern on random instances of a finite model in which (Σ_1C) is the only arithmetical input.

Proof of Theorem 5.9. By the s-m-n theorem there is an injective computable g with $\mathsf{W}_{g(e)} = \mathsf{W}_e \cup \{\neg\mathrm{Con}(B \cup \mathsf{W}_e)\}$, the canonical consistency statement of the presentation e. If $B \cup \mathsf{W}_e$ is consistent, (G2) gives $B \cup \mathsf{W}_e \nvdash \mathrm{Con}(B \cup \mathsf{W}_e)$, so $B \cup \mathsf{W}_{g(e)}$ is consistent; its new axiom is a false Σ_1 sentence; and it proves everything W_e proves. As $\mathbf{m}(g(e)) \geq 2^{-c}\mathbf{m}(e)$ and $K(g(e)) \leq K(e) + c$, summing over the Σ_1-sound compatible e, whose images are distinct, gives the claim, and the posterior bound follows from $W_A(\mathcal{U}; D) \geq 2^{-c} W_A(\overline{\mathcal{U}}; D)$. A true theory is Σ_1-sound, which gives "a fortiori". □

The event that $B \cup A_\omega$ is Σ_1-unsound is $\bigcup_\sigma O_\sigma$ over the false Σ_1 sentences σ, an open set, so $\mathbb{U}_A(\mathcal{U}; D)$ and $P_D(\mathcal{U})$ are defined.

Proposition C.20 (One-tape soundness is controlled by the coding gap)**.** *Under the hypotheses of Theorem 5.9, for every consistent D,*

$$P_D(\mathcal{U}) \;\geq\; 2^{L_A(D) - K_A(D) - c}.$$

Proof. Let e witness $K_A(D)$. Its companion $g(e)$ from the proof of Theorem 5.9 is consistent, Σ_1-unsound, proves ψ_D, and has $K(g(e)) \leq K_A(D) + c$; by (Dom) the tapes that begin with a shortest program for it have measure at least $2^{-K_A(D)-c}$, and their theory is $\mathsf{W}_{g(e)}$. Divide by $M_A(D) = 2^{-L_A(D)}$. □

So one-tape SAI can become confident of soundness only on data where the coding theorem fails badly, which is where randomness dominates (Section 5.3). On data of Π_1 type, Proposition C.20 and Corollary 5.7 give $P_D(\mathcal{U}) \geq 2^{-K(\lceil L_A(D) \rceil)-c}$, a bound that degrades as the loss grows; the threshold machine removes the degradation.

Theorem C.21 (Soundness is never learned from Π_1 data)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. There is c such that for every D of Π_1 type with $M_A(D) > 0$, $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-c} M_A(D)$; so $P_D(\overline{\mathcal{U}}) \leq 1 - 2^{-c}$.*

Proof. For a finite string w let U_w be the set of sentences that G_{thr} (Theorem 5.6) emits on some tape extending w; it is r.e. uniformly in w, and $U_w = \{\pi \in \Pi_1^* : \mathbb{U}(O_\pi) > 0.w\}$, since the emissions under prefixes $v \sqsupseteq w$ have thresholds $0.v + 2^{-|v|}$ with infimum $0.w$, not attained, and those under shorter prefixes $u \sqsubseteq w$ have thresholds at least $0.w + 2^{-|w|}$. Hence U_w is true if $0.w > \mathrm{Inc}_0$, and is all of Π_1^*, an inconsistent set, if $0.w < \mathrm{Inc}_0$ (equality is impossible, since Inc_0 is irrational). Moreover $U_{w'} \subseteq U_w$ for $w \sqsubseteq w'$, provably in PA from the definitions (every prefix of w' is comparable with w). Let $G^{\mathrm{u}}_{\mathrm{thr}}$ emit everything G_{thr} emits and, for every prefix w of the tape, $\chi_w := \neg\mathrm{Con}(B \cup U_w)$.

Consistency. Let $r > \mathrm{Inc}_0$ be a non-dyadic tape and F a finite subset of $A_r^{G^{\mathrm{u}}_{\mathrm{thr}}}$, with companion sentences $\chi_{w_1}, \ldots, \chi_{w_j}$; let w^* be the longest w_i. If $0.w^* < \mathrm{Inc}_0$, every $U_{w_i} \supseteq U_{w^*}$ is inconsistent, every χ_{w_i} is a true Σ_1 sentence, provable from B, and the rest of F lies in the true theory $A_r^{G_{\mathrm{thr}}}$. If $0.w^* > \mathrm{Inc}_0$, then $B \cup U_{w^*}$ is consistent, r.e. and contains PA, so $B \cup U_{w^*} \cup \{\chi_{w^*}\}$ is consistent by (G2); it contains $F \cap A_r^{G_{\mathrm{thr}}}$, because $A_r^{G_{\mathrm{thr}}} = \{\pi : \mathbb{U}(O_\pi) > r\} \subseteq U_{w^*}$ as $r > 0.w^*$; and it proves each χ_{w_i} with $w_i \sqsubseteq w^*$, because $U_{w^*} \subseteq U_{w_i}$ provably, so a proof of $\bot$ from $B \cup U_{w^*}$ is one from $B \cup U_{w_i}$.

Unsoundness. $r > \mathrm{Inc}_0$ has a prefix w with $0.w > \mathrm{Inc}_0$, since the truncations increase to r; for it χ_w is a false Σ_1 sentence.

Measure. For $\mathrm{Inc}_0 < r < \mathbb{U}(O_{\pi_D})$ the theory of $G^{\mathrm{u}}_{\mathrm{thr}}$ is consistent, Σ_1-unsound and contains π_D, and $B \vdash \psi_D \leftrightarrow \pi_D$. So $\nu_{G^{\mathrm{u}}_{\mathrm{thr}}}$ gives the event "consistent, unsound and proving ψ_D" probability at least $M_A(D)$, as in the proof of Theorem 5.6, and (Dom), tape by tape, gives $\mathbb{U}_A(\mathcal{U}; D) \geq 2^{-K(G^{\mathrm{u}}_{\mathrm{thr}})-c} M_A(D)$. □

A computation (code in the accompanying repository) checks the dyadic and prefix facts of this proof in finite models with non-dyadic weights.

What is known about Question 5.10. With B true and $\mathsf{PA} \subseteq \mathrm{Cn}(B)$, the answer is yes for data of Π_1 type (Theorem C.21) and for data with a bounded coding gap $K_A - L_A$ (Proposition C.20). On the Win-lottery stream with $\mathcal{L} = \mathcal{L}_{\mathrm{ar}} \cup \{\mathrm{Win}\}$ it is yes as well: the machine "G_{lot} plus the axiom $\neg\mathrm{Con}(B)$" has the same survival event; its theories are consistent (a model of $B + \neg\mathrm{Con}(B)$ exists by (G2), has distinct numerals as every model of Q does, and is expanded by the winners) and Σ_1-unsound; so $P_{D_n}(\mathcal{U}) \geq 2^{-c} \prod_k (1 - 2^{-(k+1)})$ for every n. On the arithmetic lotteries the random lottery theories are themselves unsound (Corollary 5.5(iii) for (b); in (a) they affirm the false Σ_1 sentences Tick_{k,J_k}). In general we can prove only the following. For a theory machine G let U_w^G be the set of sentences that G emits on some tape extending w, and

$\mathcal{K}^G := \{\omega : B \cup U^G_w$ is consistent for some prefix w of $\omega\}$. Under the hypotheses of Theorem 5.9, for every D,

$$\mathbb{U}_A(\mathcal{U}; D) \;\geq\; 2^{-K(G)-c}\,\mathbb{U}(\{\omega : B \cup A^G_\omega \text{ consistent}, \vdash \psi_D\} \cap \mathcal{K}^G).$$

Indeed, the companion G^{u} emits what G emits and $\neg\mathrm{Con}(B \cup U^G_w)$ for every prefix w of the tape, and $K(G^{\mathrm{u}}) \leq K(G) + c$; on a tape ω on which $B \cup A^G_\omega$ is consistent, the longest-prefix argument above shows that $B \cup A^{G^{\mathrm{u}}}_\omega$ is consistent ($A^G_\omega \subseteq U^G_{w^*}$ because w^* is a prefix of ω), and on $\mathcal{K}^G$ some added sentence is a false Σ_1 sentence; then apply (Dom) tape by tape. This answers the question for no class of data by itself: it fails for coin machines, all of whose sets U^G_w are inconsistent. The difficulty in general is that a companion must add a false Σ_1 sentence that the random theory A_ω does not refute, chosen from finitely much of the tape. $\neg\mathrm{Con}(A_\omega)$ is not a sentence, and A_ω may prove $\mathrm{Con}(Z)$ for every r.e. Z read off a finite prefix (a machine can emit, at each stage, the consistency of everything it has emitted so far). The natural companions need either a consistent U_w, which fails for coin machines, or a fixed false Σ_1 sentence that the random theory leaves open, which fails for such reflective machines.

Proposition C.22 ($\mathrm{Con}(B \cup T)$ given data from T)**.** *Let T be r.e., with $B \cup T$ true and $\mathsf{PA} \subseteq \mathrm{Cn}(B \cup T)$, and let the data present T: $\mathrm{Cn}(D^{\pm}_\infty) = \mathrm{Cn}(B \cup T)$. Write χ_T for $\mathrm{Con}(B \cup T)$ or the Gödel sentence of $B \cup T$. (a) $\lim_n \mathrm{Bel}_{D_n}(\chi_T)$, $\lim_n \mathrm{Dis}_{D_n}(\chi_T)$ and $\lim_n \mathrm{Indep}_{D_n}(\chi_T)$ exist and lie in $(0,1)$. (b) The belief and independent shares certify the truth of χ_T in the sense of Proposition 5.8(b). (c) Which of "true" and "false" gets more mass is a choice of universal machine: for every universal prefix machine V and every j there are universal prefix machines V', V'' under which the ratio of the limits of $\mathrm{Bel}(\chi_T)$ and $\mathrm{Dis}(\chi_T)$ is at least $2^j - 1$, respectively at most $1/(2^j - 1)$.*

Proof. (a) By (Lim) (Proposition A.26(d)), the limit posterior is the prior restricted to the consistent extensions of $B \cup T$, which is incomplete by (G2). Three deterministic theories have positive weight. $T + \mathrm{Con}(B \cup T)$ is true, because $B \cup T$ is true, hence consistent, and $\mathrm{Con}(B \cup T)$ is a true Π_1 sentence. $T + \neg\mathrm{Con}(B \cup T)$ is consistent with B by (G2) for the presentation $B \cup T \supseteq \mathsf{PA}$. T decides neither: not $\mathrm{Con}(B \cup T)$ by (G2), and not its negation because $B \cup T$ is true and $\neg\mathrm{Con}(B \cup T)$ is false. Each of the three limits is therefore positive, and each is less than 1 because the other two are positive. (b) is Proposition 5.8(b) applied to the Π_1 sentence $\mathrm{Con}(B \cup T)$. (c) Let $V'(0^i 1)$ be an index of the deterministic machine for $T + \mathrm{Con}(B \cup T)$ for $i < j$, and $V'(0^j p) := V(p)$; V' is a universal prefix machine. The tapes that begin with some $0^i 1$, $i < j$, have measure $1 - 2^{-j}$ and give a compatible theory proving $\mathrm{Con}(B \cup T)$; all tapes on which $\neg\mathrm{Con}(B \cup T)$ is proved begin with 0^j. So, within E_∞, the masses of the two events are at least $1 - 2^{-j}$ and at most 2^{-j}; dividing both by the common normaliser $\mathbb{U}(E_\infty)$ gives the ratio. V'' is symmetric, with $T + \neg\mathrm{Con}(B \cup T)$. The Gödel sentence of $B \cup T$ is PA-provably equivalent to $\mathrm{Con}(B \cup T)$ (Smoryński, 1977), so the same holds for it. □

The truth hypothesis is needed. With $B := \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, which is decidable and consistent by (G2), and $T := \emptyset$: since $\mathsf{PA} \vdash \mathrm{Con}(B) \to \mathrm{Con}(\mathsf{PA})$, $B \vdash \neg\mathrm{Con}(B)$, so every compatible theory refutes $\mathrm{Con}(B \cup T)$, $\lim \mathrm{Dis} = 1$, and (a) and (c) fail. A computation (code in the accompanying repository) checks the construction of (c) in a finite model.

Proposition C.23 (Old evidence)**.** *If $B \vdash \varphi^b$ for every (φ, b) in D, then $M_A(D) = M_A(\emptyset)$ and the posterior equals the prior; likewise for W_A and K_A. In particular, with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and a Δ_0 formula $P(x)$, the data $\{(P(\underline{n}), 1) : n \leq N\}$ (true, since otherwise they are inconsistent with B) leave $\mathrm{Bel}(\forall x\, P(x))$ at its prior value, for every N.*

Proof. If $B \vdash \varphi^b$ for every $(\varphi, b) \in D$, then $\mathrm{Cn}(D^{\pm}) = \mathrm{Cn}(\emptyset)$, so $M_A(D) = M_A(\emptyset)$, $E_D = E_\emptyset$, and the same for W_A and K_A (Old). With $\mathsf{Q} \subseteq \mathrm{Cn}(B)$ and P a Δ_0 formula, every true instance $P(\underline{n})$ is Q-provable by (Σ_1C), and a false one is Q-refutable, so its datum would be inconsistent with B. □

Instances of a fresh predicate (empirical induction, $B = \emptyset$) do confirm $\forall x\, R(x)$, up to the ω-gap (Proposition 5.11). So the inference from checked instances needs a reasoner for whom computing $P(\underline{n})$ is costly: logical induction (Garrabrant et al., 2016a) is one, and a derivation-charged axiom inducer, as in Section 6 of our earlier report (Hänni and Claude (Anthropic), 2026a), is another. Whether a derivation-charged posterior then concentrates on provers of $\forall x\, P(x)$ depends on competitors that prove the instances cheaply without proving the universal (compare Proposition C.27); we do not settle this.

C.6 The ω-gap

Proof of Proposition 5.11. I is a consistent r.e. theory proving the stream, so $\mathbb{U}(E_\infty) \geq 2^{-K(I)-c} > 0$ (Proposition A.26(c)) and (Lim) applies; by Proposition A.26(d) with $\Gamma = I$, P_∞ is the prior restricted to the consistent extensions of I, whether or not I is complete. In (i) and (ii) every consistent extension of I proves what I proves. In (iii), I, $I + \forall x\, \varphi$ and $I + \neg\forall x\, \varphi$ are consistent r.e. theories with positive weight, which prove $\forall x\, \varphi$, refute it, and decide it neither way, respectively; and a consistent extension of I refutes $\forall x\, \varphi$ iff it is ω-inconsistent at φ, since it proves every instance. □

Two standard cases. With $B = \emptyset$ and the atomic diagram of a computable structure this is Proposition A.29. With $B \supseteq \mathsf{Q}$ and $\varphi \in \Delta_0$ the instances are free (Proposition C.23) and the posterior does not move at all; case (iii) holds iff B decides $\forall x\, \varphi$ neither way. For Σ_1-sound B this is iff $B \nvdash \forall x\, \varphi$: with all instances true, $\neg\forall x\, \varphi$ is logically equivalent to the false Σ_1 sentence $\exists x\, \neg\varphi$, which a Σ_1-sound B does not prove. Without Σ_1-soundness this fails: for $B = \mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ and $\varphi(x)$ the Δ_0 formula "x is not a PA-proof of $\bot$", every instance is true and Q-provable, so $\mathrm{Cn}(I) = \mathrm{Cn}(B)$, and $B \vdash \neg\forall x\, \varphi$: case (ii) holds although $B \nvdash \forall x\, \varphi$.

Proposition C.24 (The Gaifman condition)**.** *Under the hypotheses of Proposition 5.11, enumerate* $\mathrm{CT} = \{t_0, t_1, \dots\}$. *Then* $\lim_n \mathrm{Bel}_0(\bigwedge_{i \leq n} \varphi(t_i))$ *exists, and*

$$\lim_n \mathrm{Bel}_0\Big(\bigwedge_{i \leq n} \varphi(t_i)\Big) - \mathrm{Bel}_0(\forall x\, \varphi) = \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent, proves every } \varphi(t_i),\ \nvdash \forall x\, \varphi\}.$$

This is 0 *iff* $I \vdash \forall x\, \varphi$, *and otherwise at least* $2^{-K(I)-c}$, *where* $K(I)$ *is the complexity of an index of* I. *The same holds for every posterior* Bel_D *with* D *proved by* I, *after dividing by* $M_A(D)$.

Proof. The events $\{\omega : B \cup A_\omega$ consistent and proves $\varphi(t_0), \dots, \varphi(t_n)\}$ decrease in n to the event of proving every instance; continuity from above gives the limit. A theory proving $\forall x\, \varphi$ proves every instance, so the difference is the stated measure. If $I \vdash \forall x\, \varphi$, every consistent theory proving all instances proves $\forall x\, \varphi$, and the difference is 0. Otherwise I itself lies in the event, with weight at least $2^{-K(I)-c}$ by (Dom). For Bel_D, divide by $M_A(D)$. □

"Prove", not "decide", is the right word: in the ω-inconsistent case (ii) of Proposition 5.11 the instances refute the universal, yet the gap equals the whole consistent mass of the extensions of I.

Theorem C.25 (A uniform Gödelian ω-gap)**.** *Let B be decidable and true with $\mathsf{PA} \subseteq \mathrm{Cn}(B)$. For every k there is a Δ_0 formula $\zeta_k(x)$ such that $\forall x\, \zeta_k$ is true, every instance $\zeta_k(\underline{n})$ is a true Δ_0 sentence (so Q-provable, and instance data are free by Proposition C.23), and $2^{-k-K(k)-c} \leq \mathrm{Bel}_0(\forall x\, \zeta_k) < 2^{-k}$. Hence after any instance data the posterior belief is $\mathrm{Bel}_0(\forall x\, \zeta_k)/(1 - \mathrm{Inc}_0) < 2^{-k+c'}$, and $\mathrm{Dis} + \mathrm{Indep} > 1 - 2^{-k+c'}$.*

Proof. Let $\forall x\, \zeta_k$ be the canonical consistency statement $\mathrm{Con}(B \cup T_k)$ of the presentation e_k (Theorem C.11), rewritten with one universal quantifier over a code of a pair (a derivation of $\bot$, an enumeration stage), with membership in B and in T_k certified inside the Δ_0 matrix by computation witnesses and the quantifiers contracted by Δ_0 pairing. For such a natural rewriting $\mathsf{PA} \vdash \forall x\, \zeta_k \leftrightarrow \mathrm{Con}(B \cup T_k)$ (standard). $B \cup T_k$ is true (Theorem C.11(iii)), so $\forall x\, \zeta_k$ is true and its instances are true Δ_0 sentences. If $\mathrm{Bel}_0(\forall x\, \zeta_k) \geq 2^{-k}$, Theorem C.11(ii) puts $\forall x\, \zeta_k$ into T_k, so $B \cup T_k \vdash \forall x\, \zeta_k$, hence, by the PA-provable equivalence, $B \cup T_k \vdash \mathrm{Con}(B \cup T_k)$, contradicting (G2) for the consistent r.e. presentation $B \cup T_k \supseteq \mathsf{PA}$. The lower bound is (Dom) for the true one-sentence theory $\{\forall x\, \zeta_k\}$, whose index is computable from e_k, of complexity at most $K(e_k) + c \leq k + K(\underline{k}) + c'$. The posterior after B-entailed data is the prior normalised by $M_A(\emptyset) = 1 - \mathrm{Inc}_0$, that is $\widehat{\mathrm{Bel}}_0 = \mathrm{Bel}_\emptyset$, so the posterior belief is $\mathrm{Bel}_0(\forall x\, \zeta_k)/(1 - \mathrm{Inc}_0) < 2^{-k+c'}$. □

Remark C.26 (Where the gap closes). (a) Under the ω-rule, with $B \supseteq \mathsf{Q}$, every random theory that is consistent under that rule has $\mathcal{L}_{\mathrm{ar}}$-content $\mathrm{Th}(\mathbb{N})$: the inducer is omniscient about arithmetic a priori. (b) Under full second-order consequence the loss on any presentation of $\mathrm{Th}(\mathbb{N})$ is at most $K(\mathsf{PA}^2) + c$: the unlearnability of $\mathrm{Th}(\mathbb{N})$ (Proposition A.30) rests on effective consequence. (c) The generative variant closes the gap only with the universal sentence among the data: by Theorem 3.9(c1) of the earlier report, with a faithful likelihood and $\forall x\, \varphi$ in the data space, the posterior probability of "proves $\forall x\, \varphi$" tends to $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\, \varphi]$, whether the generator T^* proves it, not to its truth value; for filtered instance data the gap stays open (its part (d1)).

The closures of Remark C.26. (a) *The ω-rule.* Let $\vdash_\omega$ be closure under first-order deduction and the ω-rule (from $\varphi(\underline{n})$ for all n, infer $\forall x\, \varphi$). Then $\mathsf{Q} \vdash_\omega \chi$ for every true $\mathcal{L}_{\mathrm{ar}}$-sentence χ: write χ in prenex form with a Δ_0 matrix and argue by induction on the prefix. True Δ_0 sentences are Q-provable; a true $\exists x\, \psi(x)$ has a true instance $\psi(\underline{n})$, provable by induction, and $\exists$-introduction applies; a true $\forall x\, \psi(x)$ has all instances true and provable, and the ω-rule applies. So with $B \supseteq \mathsf{Q}$ every $\vdash_\omega$-consistent random theory has $\mathcal{L}_{\mathrm{ar}}$-content exactly $\mathrm{Th}(\mathbb{N})$: it proves every truth and, being consistent, no falsehood. The gap closes because the inducer is omniscient about arithmetic a priori. (b) *Full second-order consequence* (given Dedekind's categoricity theorem, standard). Let hypotheses be second-order sentences, and let compatibility mean "has a full model, and every full model satisfies ψ_D". Every full model of second-order Peano arithmetic PA^2 is isomorphic to $\mathbb{N}$, so $\{\mathsf{PA}^2\}$ is compatible with every finite correctly labelled data set. By the dominance argument (the tapes that begin with a program for $\{\mathsf{PA}^2\}$), the loss on any presentation of $\mathrm{Th}(\mathbb{N})$, in any order, is at most $K(\mathsf{PA}^2) + c$, and the limit posterior has $\mathrm{Bel}(\varphi) = \mathbf{1}[\mathbb{N} \models \varphi]$ for every $\mathcal{L}_{\mathrm{ar}}$-sentence, trivially, since every sentence is a datum; the content is the loss bound. This places the unlearnability of $\mathrm{Th}(\mathbb{N})$ (Proposition A.30) on effective consequence, not on the complexity of arithmetical truth. (c) *The generative variant.* Theorem 3.9 of the earlier report concerns the generative inducer over a countable class with i.i.d. data from a generator T^*. Its parts (a), (b): the posterior probability of "proves $\forall x\, \varphi$" tends to the prior share of provers in the class of theories with the generator's law. Its part (c1): in the faithful regime, with $\forall x\, \varphi$ in the data space, the limit is $\mathbf{1}[T^* \vdash_{\mathsf{C}} \forall x\, \varphi]$, whether the generator proves the universal in the calculus C used there, not the truth value of $\forall x\, \varphi$. So this closes the gap only in the sense that the universal sentence is

itself observed. For filtered data the gap stays open: for quantifier-free φ in the minimal class of that paper, the limit is 0 under its plain likelihood and the prior share under its selection-aware likelihood (its part (d1)), and with an unknown filter it is a filter-prior mixture (its part (e)). The generative variant is therefore not a closure of the ω-gap on instance data.

Line budgets. Let hypotheses be single sentences ψ, and call a datum χ *ℓ-compatible* with ψ if $B \cup \{\psi\}$ is consistent and proves χ in a fixed Hilbert calculus by a derivation of at most ℓ lines (lines, not symbols). The data are the closed instances $\varphi(S^n 0)$, $n \in \mathbb{N}$.

Proposition C.27 (Line budgets and a Kreisel-type property)**.** *Assume that some sentence is ℓ-compatible with every datum (for instance $\forall x\, \varphi$, when $B \cup \{\forall x\, \varphi\}$ is consistent and ℓ is at least the constant number of lines of instantiation). Then for any prior with full support on sentences, the limit posterior gives* $\mathrm{Bel}(\forall x\, \varphi) = 1$ *iff every ψ that is consistent with B and proves every $\varphi(S^n 0)$ within ℓ lines also proves $\forall x\, \varphi$ (a* Kreisel property *of (B, φ, ℓ)).*

Proof. The set of sentences compatible with the first n data decreases in n to the set of ψ that are consistent with B and ℓ-compatible with every datum, which is nonempty by hypothesis. If every such ψ proves $\forall x\, \varphi$, the limit belief is 1; a ψ in it that does not prove $\forall x\, \varphi$ has positive weight, and then the limit belief is less than 1. □

With $B = \emptyset$ and $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, the property already fails at a constant budget, but that example is degenerate: every instance is logically valid, so the data are old evidence (Proposition C.23), and the budget plays no role. The following example is not degenerate.

Example C.28 (A non-degenerate failure)**.** Let $B = \emptyset$, $\mathcal{L} = \{0, S, R\}$ with R a fresh unary predicate, and data $R(S^n 0)$; they are not valid, since the one-point structure with R empty refutes $R(0)$. With $\varphi(x) := x = 0 \vee \exists y\, x = Sy$, let $\psi_1 := \forall x\, (\varphi(x) \to R(x))$ and $\psi_2 := \forall x\, R(x)$. Each instance $\varphi(S^n 0)$ has a derivation of at most five lines (an equality axiom $S^n 0 = S^n 0$, the axiom $\chi(t) \to \exists y\, \chi(y)$ with $t = S^{n-1} 0$ and modus ponens, the axiom $\beta \to \alpha \vee \beta$ and modus ponens; three lines for $n = 0$), so ψ_1 derives $R(S^n 0)$ in nine lines (seven for $n = 0$): the hypothesis, its instance at $S^n 0$ by $\forall$-elimination and modus ponens, the derivation of $\varphi(S^n 0)$, and modus ponens. ψ_2 derives each datum in three lines. The structure $\{0, 1, 2, a\}$ with $S(0) = 1$, $S(1) = 2$, $S(2) = 2$, $S(a) = 0$ and $R := \{0, 1, 2\}$ satisfies ψ_1 (the only element outside R is a, and $\varphi(a)$ fails) and every $R(S^n 0)$ (closed terms denote 0, 1 or 2), but not $\forall x\, R(x)$. So with $\ell = 9$ both ψ_1 and ψ_2 stay ℓ-compatible with every datum, ψ_1 is consistent and does not prove $\forall x\, R(x)$, and under any full-support prior the limit belief in $\forall x\, R(x)$ lies strictly between 0 and 1: the Kreisel property fails for $(\emptyset, R, 9)$, with instances that are not valid.

A computation (code in the accompanying repository) checks the derivations line by line for $n \leq 60$ in a small Hilbert-style proof checker, and the countermodel.

Kreisel's conjecture says that if PA proves $\varphi(S^n 0)$ in at most k lines for every n, then $\mathsf{PA} \vdash \forall x\, \varphi$; it is open for the usual Hilbert-style PA, with partial results and counterexamples for nearby theories (Parikh, 1973; Baaz and Pudlák, 1993; Hrubeš, 2007, 2009; Santos and Kahle, 2021). These concern infinite theories with schemata. The property of Proposition C.27 concerns finite extensions $B \cup \{\psi\}$, in which a k-line derivation can use only the finitely many axioms of ψ and logical axioms; we know of no result on it, so the link is a reformulation.

Question C.29. For $B = \mathsf{Q}$ and Δ_0 formulas φ, does every finite ψ that proves all instances of φ in boundedly many lines prove $\forall x\, \varphi$?

Over Q with $\varphi \in \Delta_0$ the instances are old evidence again; the question is which finite theories the line-budget inducer keeps, not what it learns from the instances.

C.7 True arithmetic

Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$, $B \subseteq \mathrm{Th}(\mathbb{N})$ decidable, $(\varphi_i)_{i\geq 1}$ a computable enumeration of all sentences, D_n the first n of them labelled by truth, and $b_{1:n}$ the labels. The loss is unbounded and at most $n + K(n) + c$ (Proposition A.30). It grows at least polynomially, because with binary numerals the sentences "the k-th digit of Ω_V is 1" occur early.

Theorem C.30 (A polynomial lower bound)**.** *Suppose the enumeration lists sentences by length, and within a length in a fixed order, and that (1) at most $C\kappa^L$ sentences have length at most L, for every L, and (2) there are terms $\underline{k}^{\mathrm{b}}$ with value k and length at most $c_1 \log k + c_2$. Put $\xi := 1/(c_1 \log \kappa)$ and $m(n) := \lfloor (n/C')^{\xi} \rfloor$, where C' depends only on the syntax and on a fixed formula for the binary digits of Ω_V. Then for all n, in the semimeasure convention: (a) $K(b_{1:n} \mid n) \geq m(n) - K(n) - c$; (b) $L_A(D_n) \geq C'^{-\xi} n^{\xi} - K(n) - c$; (c) $L_A(D_n) \geq (2C')^{-\xi} n^{\xi} - 2\log\log n - c$ for $n \geq 4$.*

Proof. Let $\Omega_s \uparrow \Omega_V$ be the computable approximations. Since Ω_V is irrational, its k-th binary digit is the eventual value of the k-th digit of Ω_s. Let $\mathrm{Bit}'(y)$ be a formula with $\mathbb{N} \models \mathrm{Bit}'(\underline{k})$ iff that digit is 1: "there is s such that for all $s' \geq s$ the y-th digit of $\Omega_{s'}$ is 1", with the digit predicate expressed by a Δ_0 formula about a computation witness. Put $\mathrm{Bit}(x) := \exists y\, (y = x \wedge \mathrm{Bit}'(y))$, so that x occurs once, and $\mathrm{Bit}_k := \mathrm{Bit}(\underline{k}^{\mathrm{b}})$. Then $|\mathrm{Bit}_k| \leq |\mathrm{Bit}| + c_1 \log k + c_2$, and the position of Bit_k in the enumeration is at most $C\kappa^{|\mathrm{Bit}|+c_2}\kappa^{c_1 \log k} = C' k^{1/\xi}$. The first n sentences therefore contain $\mathrm{Bit}_1, \ldots, \mathrm{Bit}_{m(n)}$, at positions computable from k, and their labels are the digits $\Omega_{1:m(n)}$. So $\Omega_{1:m(n)}$ is computable from n and $b_{1:n}$, and $K(\Omega_{1:m(n)}) \leq K(b_{1:n} \mid n) + K(n) + c$. By Chaitin's theorem $K(\Omega_{1:m}) \geq m - c$. This is (a). (b) is (a) with Proposition A.30(b) and $m(n) \geq (n/C')^{\xi} - 1$. (c) $L_A(D_n)$ is nondecreasing in n, because $E_{D_{n+1}} \subseteq E_{D_n}$. For $n \geq 4$ let $n' := 2^{\lfloor \log n \rfloor} \in (n/2, n]$; then $K(n') \leq K(\lfloor \log n \rfloor) + c \leq 2\log\log n + c'$, and by monotonicity and (b), $L_A(D_n) \geq L_A(D_{n'}) \geq (n/(2C'))^{\xi} - 2\log\log n - c''$. □

The exponent for a stated syntax. Take the Polish-notation syntax with fourteen symbols: terms 0, St, $+tt$, $\cdot tt$ and variables $x' \cdots'$ (x followed by $i \geq 0$ primes); formulas $=tt$, $<tt$, $\neg f$, $\wedge ff$, $\vee ff$, $\to ff$, $\forall vf$, $\exists vf$; and binary numerals $\underline{1}^{\mathrm{b}} = S0$, $\underline{2m}^{\mathrm{b}} = \cdot\, \underline{m}^{\mathrm{b}}\, SS0$, $\underline{2m+1}^{\mathrm{b}} = S \cdot \underline{m}^{\mathrm{b}}\, SS0$, so that $c_1 = 5$ and $c_2 = 2$. Counting all strings ($\kappa = 14$) gives $\xi \geq 1/(5 \log 14) > 0.0525$. Counting formulas gives more. With $V(x) = x/(1-x)$, the generating function of the variables, the generating functions of terms and formulas satisfy

$$T = x + V + xT + 2xT^2, \qquad F = 2xT^2 + xF + 3xF^2 + 2xVF,$$

and every term on the right carries a factor x. If rationals $x_0, T^*, F^* > 0$ satisfy both equations with "$\geq$" in place of "$=$", then the iterates of the right-hand sides from 0 have nonnegative coefficients, agree with the series up to degree k after k steps, and stay below (T^*, F^*) at x_0; so $\sum_L F_L x_0^L \leq F^*$, and the number of formulas, hence of sentences, of length at most L is at most $\frac{F^*}{1-x_0} x_0^{-L}$ for every L. A computation with exact rational arithmetic (code in the accompanying repository) verifies such a super-solution at $x_0 = 1000/5385$, with $T^* \approx 0.800042$ and $F^* \approx 0.609287$; this gives $C = 0.7482$ and $\kappa = 5.385$, hence $\xi \geq 1/(5 \log 5.385) > 0.0823$. The same computation finds no certificate at $\kappa = 5.38$; the growth rate of the formula counts is about 5.383.

Only the numeral length and the growth of the syntax enter ξ; the (large) size of Bit enters only the constant. With unary numerals the position of Bit_k is at most $C''\kappa^k$, so $m(n)$ is only about $\log_\kappa n$; then (b) is vacuous as written, since $K(n)$ can be about $\log n > \log_\kappa n$, but (c)'s monotonicity argument still gives $L_A(D_n) \geq \log_\kappa n - 2\log\log n - c$. The bound is weak in absolute

terms (for $n = 10^6$, $n^{0.0823}$ is about 3.1); it is a statement about the shape of the growth, not a usable estimate.

The digit sentences among the first n carry about $m(n)$ incompressible bits (Chaitin, 1975), and a true theory decides correctly, so:

Corollary C.31 (Quantitative incompleteness)**.** *Let T be r.e. with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$, and let $u_n(T)$ be the number of $i \le n$ such that $B \cup T$ proves neither φ_i nor $\neg\varphi_i$. Then $u_n(T) \ge C'^{-\xi} n^{\xi}/(2 \log n)$ for all large n.*

Proof. $b_{1:n}$ is computable from an index e_T of T, n, the set of undecided positions among the first n and their labels, since a true theory decides correctly. Using $\log \binom{n}{u} \le u \log(en/u)$ and $K(u \mid n) \le \log n + c$, this gives $K(b_{1:n} \mid n) \le K(e_T) + u_n(T)(\log n + 3) + 2 \log n + c$. Compare with Theorem C.30(a), which is the inequality needed here (the statement (b) about L_A would not suffice): $u_n(T)(\log n + 3) \ge (n/C')^{\xi} - K(n) - K(e_T) - 2 \log n - c'$, and the right-hand side grows like n^{ξ} while the subtracted terms are $O(\log n)$. □

Conversely, the loss is small if some true r.e. theory leaves few of the first n sentences undecided:

Proposition C.32 (An upper bound by a reduction)**.** *For every r.e. T with $B \cup T \subseteq \mathrm{Th}(\mathbb{N})$ and every n,*

$$L_A(D_n) \le K(e_T) + \log \tbinom{n}{u_n(T)} + u_n(T) + 2 \log n + O(\log \log n).$$

Hence if some such T has $u_n(T)/n \to 0$, then $L_A(D_n) = o(n)$.

Proof. $B \cup T$ together with the correctly labelled undecided sentences among the first n is a consistent (true) r.e. theory proving D_n, with an index computable from e_T, n, u_n, the positions ($\log \binom{n}{u_n}$ bits, given n and u_n) and the u_n labels; then (Chain) gives $L_A \le K_A + c$. If $u_n/n \to 0$, then $\log \binom{n}{u_n} \le nH(u_n/n) = o(n)$, with H the binary entropy function. □

Question C.33. In such an enumeration of the sentences of $\mathcal{L}_{\mathrm{ar}}$ labelled by truth, is $L_A(D_n) = o(n)$?

By Proposition C.32, a positive answer to Question C.33 would follow if some true r.e. theory decided a set of sentences of asymptotic density one in the enumeration, that is, if the truth of arithmetic were generically decidable in the sense of generic-case complexity (Myasnikov and Rybalov, 2008). Such questions depend on the representation: the halting problem is decidable on a set of asymptotic probability one for one model of Turing machines (Hamkins and Miasnikov, 2006). Calude and Jürgensen (2004) show that the probability that a true sentence of length n is provable tends to 0, in a probability space of statements with n-bit codes under their numbering. If a version of this held for the length enumeration used here, every true r.e. theory would leave a positive fraction of the sentences of each length undecided, and the sufficient condition of Proposition C.32 would fail for every T. It would not settle Question C.33: $K(b_{1:n} \mid n)$ could still be $o(n)$, since truth values along the enumeration are highly redundant (for instance ψ and $\neg\neg\psi$). Corollary C.31 gives only a polynomial lower bound on u_n.

D Proofs for what a theory does and is

This appendix states and proves the results that Section 6 summarises in prose. Appendices D.1 to D.4 concern questions against raw data; Appendices D.5 and D.6 give the precise setting of the flagpole toys, state and prove Proposition D.7, and extend it to a system of laws.

D.1 The answer distribution

Questions, and the answers a theory proves, are defined in Definition 2.10; $[k] := \{1, \dots, k\}$.

Definition D.1 (Answers)**.** A theory T *answers* a question $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ if $B \cup T \vdash \varphi_i$ for some i, and *answers it partially* if $B \cup T \vdash \bigvee_{i \in J} \varphi_i$ for some J with $\emptyset \neq J \subsetneq [k]$.

Proposition D.2 (The answer distribution)**.** *Let $M_A(D) > 0$, let $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ be a question, and let $\mathrm{Un}_D(\mathcal{Q}) := \mathbb{U}(E_D \setminus \bigcup_i O_{\varphi_i})/\mathbb{U}(E_D)$, the posterior probability that the random theory proves no direct answer. (a) $\sum_{i \le k} \mathrm{Bel}_D(\varphi_i) + \mathrm{Un}_D(\mathcal{Q}) = 1$. (b) For $\emptyset \neq J \subsetneq [k]$, $\mathrm{Bel}_D(\bigvee_{i \in J} \varphi_i) \geq \sum_{i \in J} \mathrm{Bel}_D(\varphi_i)$, and the difference is the posterior probability that the random theory answers $\mathcal{Q}$ partially through J but proves no φ_i with $i \in J$; for $J = [k]$, $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$ and the difference is $\mathrm{Un}_D(\mathcal{Q})$. (c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$, (a) is the trichotomy $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi) + \mathrm{Indep}_D(\varphi) = 1$ of Proposition 3.1.*

Proof. On E_D the theory $B \cup A_\omega$ is consistent. If it proved φ_i and φ_j with $i \neq j$, it would prove $\varphi_i \wedge \varphi_j$, while $B \vdash \neg(\varphi_i \wedge \varphi_j)$; so it proves at most one direct answer. Hence the sets $E_D \cap O_{\varphi_i}$, $i \le k$, are pairwise disjoint, and together with $E_D \setminus \bigcup_i O_{\varphi_i}$ they partition E_D. Dividing the measures by $\mathbb{U}(E_D) > 0$ gives (a), since $\mathrm{Bel}_D(\varphi_i) = \mathbb{U}(E_D \cap O_{\varphi_i})/\mathbb{U}(E_D)$.

(b) For $i \in J$, $O_{\varphi_i} \subseteq O_{\bigvee_{j \in J} \varphi_j}$, and the sets $E_D \cap O_{\varphi_i}$ are pairwise disjoint; so

$$\mathbb{U}\Big(E_D \cap O_{\bigvee_{j \in J} \varphi_j}\Big) = \sum_{i \in J} \mathbb{U}(E_D \cap O_{\varphi_i}) + \mathbb{U}\Big(E_D \cap O_{\bigvee_{j \in J} \varphi_j} \setminus \bigcup_{i \in J} O_{\varphi_i}\Big).$$

Divide by $\mathbb{U}(E_D)$: the last term is the posterior probability that the random theory proves $\bigvee_{j \in J} \varphi_j$, that is, answers $\mathcal{Q}$ partially through J, but proves no φ_i with $i \in J$. For $J = [k]$, $B \vdash \bigvee_i \varphi_i$, so $E_D \subseteq O_{\bigvee_i \varphi_i}$ and $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$; by (a) the difference is $1 - \sum_i \mathrm{Bel}_D(\varphi_i) = \mathrm{Un}_D(\mathcal{Q})$.

(c) For $\mathcal{Q} = \{\varphi, \neg\varphi\}$ ($k = 2$), $\mathrm{Bel}_D(\neg\varphi) = \mathrm{Dis}_D(\varphi)$ and $\mathrm{Un}_D(\mathcal{Q})$ is the posterior probability that the random theory decides neither, which is $\mathrm{Indep}_D(\varphi)$. □

D.2 The conservation-law world

For even m, micro-states are $x \in \{0,1\}^m$ with weight $|x|$; the dynamics σ is a uniformly random permutation of each weight class, so the weight is conserved (the law); x_0 is uniform on the class of weight $m/2$, of size $Z := \binom{m}{m/2}$, and $x_{t+1} := \sigma(x_t)$. The raw predictors Raw^+, Raw^- are Bayes for "σ is a uniformly random permutation of each weight class", resp. "of $\{0,1\}^m$", and their step-t loss is that of predicting x_{t+1} from $x_0, \dots, x_t$. The questions are "what is $|x_t|$?", $t = 0, 1, \dots$; the question predictors Qa^+, Qa^-, $\mathsf{Qa}^{\mathrm{bin}}$ are Bayes for "$|x_t| = |x_0|$, with $|x_0|$ uniform on $\{0, \dots, m\}$", "the $|x_t|$ are i.i.d. uniform on $\{0, \dots, m\}$" and "the $|x_t|$ are i.i.d. binomial$(m, \frac{1}{2})$". In each task Mix gives prior $\frac{1}{2}$ to each of the task's two hypotheses ($\mathsf{Raw}^{\pm}$, resp. $\mathsf{Qa}^{\pm}$). Let t^* be the length of the cycle of σ through x_0.

Proposition D.3 (The conservation-law world)**.** *(a) For $t < t^*$ the step-t losses of Raw^+ and Raw^- are $\log(Z - t)$ and $\log(2^m - t)$; for $t \geq t^*$ both are 0. The law's gain $\kappa_t := \log \frac{2^m - t}{Z - t}$ ($t < t^*$) is nondecreasing in t; for $t \le Z/2$, $\log \frac{2^m}{Z} \le \kappa_t \le \log \frac{2^m}{Z} + 1$, where $\log \frac{2^m}{Z} = \frac{1}{2} \log \frac{\pi m}{2} + o(1)$, and the share $\kappa_t / \log(2^m - t)$ of the step loss is $O(\log m / m)$. (b) Qa^+ pays $\log(m+1)$ for $|x_0|$ and 0 afterwards, Qa^- pays $\log(m+1)$ per question; over $t_{\max} + 1$ questions the law's share of the no-law loss is $t_{\max}/(t_{\max} + 1)$. (c) In each task the cumulative loss of Mix exceeds that of the better of its two hypotheses by at most one bit, on every trajectory. (d) For $t < t^*$, $\log(2^m - t) = \kappa_t + \log(Z - t)$ with $\kappa_t = -\log P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \dots, x_t)$: the law's raw gain is exactly the loss of Raw^-'s own answer to "what is $|x_{t+1}|$?". The loss of $\mathsf{Qa}^{\mathrm{bin}}$ on each question is κ_0.*

Lemma D.4 (Next state of a random permutation)**.** *Let σ be a uniformly random permutation of a finite set X, and let $x_0, \dots, x_t \in X$ be distinct with $\sigma(x_s) = x_{s+1}$ for $s < t$. Conditionally on these t values of σ, $\sigma(x_t)$ is uniform on $(X \setminus \{x_0, \dots, x_t\}) \cup \{x_0\}$, a set of $|X| - t$ elements.*

Proof. Conditionally on $\sigma(x_s) = x_{s+1}$ for $s < t$, the restriction of σ to $X \setminus \{x_0, \dots, x_{t-1}\}$ is a uniformly random bijection onto $X \setminus \{x_1, \dots, x_t\}$, since every such bijection extends the given values to exactly one permutation. So $\sigma(x_t)$ is uniform on $X \setminus \{x_1, \dots, x_t\} = (X \setminus \{x_0, \dots, x_t\}) \cup \{x_0\}$, which has $|X| - t$ elements. □

Proof of Proposition D.3. (a) For $t < t^*$ the states $x_0, \dots, x_t$ are distinct, all of weight $m/2$, and x_{t+1} is either unvisited or x_0. Under the hypothesis of Raw^+ the permutations of the weight classes are independent, and the trajectory stays in the class of x_0; Lemma D.4 with X that class ($|X| = Z$) gives the predictive probability $1/(Z - t)$ of the true x_{t+1}, a step loss of $\log(Z - t)$. Under the hypothesis of Raw^- the lemma with $X = \{0,1\}^m$ gives $\log(2^m - t)$. For $t \geq t^*$ the path has returned to x_0, the cycle through x_0 is known to both predictors, and later states are determined: both losses are 0.

Since $2^m > Z \geq t^* > t$, the map $t \mapsto (2^m - t)/(Z - t)$ is increasing on $[0, Z)$ (its derivative is $(2^m - Z)/(Z - t)^2 > 0$), so κ_t is nondecreasing and $\kappa_t \geq \kappa_0 = \log(2^m/Z)$. For $t \leq Z/2$, $(2^m - t)/(Z - t) \leq 2^m/(Z/2)$, so $\kappa_t \leq \log(2^m/Z) + 1$. By Stirling's formula, $Z = \binom{m}{m/2} = 2^m\sqrt{2/(\pi m)}\,(1 + o(1))$, so $\log(2^m/Z) = \frac{1}{2}\log\frac{\pi m}{2} + o(1)$. Finally $\kappa_t \leq \log(2^m/Z) + 1 = O(\log m)$, and $\log(2^m - t) \geq \log(2^m - Z/2) \geq m - 1$ since $Z \leq 2^m$; so the share is $O(\log m/m)$ for $t \leq Z/2$.

(b) Qa^+ predicts $|x_0|$ uniformly on $\{0, \dots, m\}$, a loss of $\log(m + 1)$, and then $|x_t| = |x_0|$ with probability 1, which is true since the weight is conserved: loss 0. Qa^- pays $\log(m + 1)$ on each question. Over the questions $t = 0, \dots, t_{\max}$ the losses are $\log(m + 1)$ and $(t_{\max} + 1)\log(m + 1)$, and the law saves the share $t_{\max}/(t_{\max} + 1)$ of the latter.

(c) In each task Mix is the Bayes mixture with prior $\frac{1}{2}$ on the task's two hypotheses; x_0 has the same law under both raw hypotheses, so the posterior after x_0 is still $(\frac{1}{2}, \frac{1}{2})$. By the chain rule, the cumulative loss of a Bayes mixture is $-\log$ of the mixture probability of the whole sequence. If the two hypotheses give the sequence probabilities 2^{-a} and 2^{-b}, the mixture's loss is $-\log(\frac{1}{2}2^{-a} + \frac{1}{2}2^{-b}) \leq -\log(\frac{1}{2}2^{-\min(a,b)}) = \min(a, b) + 1$, on every trajectory.

(d) Under Raw^-, by Lemma D.4, x_{t+1} is uniform on a set of $2^m - t$ states, of which exactly $Z - t$ have weight $m/2$: the $Z - (t+1)$ unvisited states of the class of x_0, and x_0. So $P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \dots, x_t) = (Z - t)/(2^m - t) = 2^{-\kappa_t}$, and conditionally on that weight x_{t+1} is uniform on those $Z - t$ states, which is exactly the predictive distribution of Raw^+. The chain rule splits the step loss $\log(2^m - t)$ of Raw^- into κ_t, the loss of its prediction of the weight, and $\log(Z - t)$, the loss of Raw^+. Finally, every $|x_t|$ equals $m/2$, and the binomial probability of $m/2$ is $Z2^{-m}$, so $\mathsf{Qa}^{\mathrm{bin}}$ pays $\log(2^m/Z) = \kappa_0$ on each question. □

Computations. A computation (code in the accompanying repository, seeded) gives the following. At the start of a trajectory the law's gain per raw step is 1.871, 2.148, 2.348, 2.837, 3.331, 3.829, 5.326 and 10.292 bits at $m = 8$, 12, 16, 32, 64, 128, 1024 and 10^6, against $\frac{1}{2}\log\frac{\pi m}{2} = 1.826$, 2.118, 2.326, 2.826, 3.326, 3.826, 5.326, 10.292; its share of the raw step loss is 0.23, 0.18, 0.15, 0.089, 0.052, 0.030, 0.0052 and 1.0×10^{-5}. The value $\kappa_{Z/2}$ is 2.659, 2.976, 3.199, 3.732, 4.258, 4.777 and 6.308 bits at $m = 8, \dots, 1024$; the loss of Qa^- per question is 3.170, 3.700, 4.088, 5.044, 6.022, 7.011 and 10.001 bits, about twice κ_0. At $m = 12$ ($Z = 924$), averaging exactly over the cycle length (uniform on $\{1, \dots, Z\}$), the expected raw loss over 300 steps is 2415.6 bits with the law and 3004.9 without (share 0.196), and the mixture's is 2416.6; over the 301 questions the losses are 3.70 and 1113.8 bits (share 0.997). Over a whole cycle the raw share is 0.232 and the

mean gain 2.75 bits per step. In 2000 simulated runs the mean raw losses are within one standard error of these values, and the mixture never exceeds the better hypothesis by more than one bit. The identity of (d) holds at $m = 12$ for every $t < Z$ to within 1.8×10^{-15} bits in floating point.

D.3 Questions from raw data and an extractor

Setting. Let $\mathbb{X}$ be a finite alphabet, μ a computable probability measure on $\mathbb{X}^{\mathbb{N}}$ (the raw world), $(\varphi_q)_{q\in\mathbb{N}}$ a fixed computable family of sentences (the questions), and η a partial computable function (the *extractor*, which reads the answers off the raw data) with $\eta(q, x) \in \{0, 1\}$ computed from a finite prefix of x; put $f_x(\varphi_q) := \eta(q, x)$ and $\mu_\eta(D) := \mu\{x : f_x \supseteq D\}$.

Proposition D.5 (Raw modelling plus an extractor)**.** *If f_x is consistent with B for μ-almost every x, then $M_A(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$ for every finite data set D of question–answer pairs, so $L_A(D) \leq -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$; here c depends also on the family (φ_q). For a computable sequence x with f_x consistent with B, $L_A(D) \leq K(x) + K(\eta) + c$ on every D with $f_x \supseteq D$.*

Proof. Sampling. There is a partial computable map Ψ from tapes to $\mathbb{X}^{\mathbb{N}}$, computable from an index of μ, that is defined for $\mathbb{U}$-almost every tape and satisfies $\Psi_*\mathbb{U} = \mu$. This is the standard construction: read tape bits to refine a dyadic interval, and output the next symbol s as soon as the interval lies inside the μ-interval of s given the symbols already output; it fails only on the null set of tapes that hit interval boundaries. Each output symbol depends on a finite prefix of the tape.

The theory machine. Let G be the theory machine that, on a tape ω, computes $\Psi(\omega)$ as far as needed and, for each q, emits $\varphi_q^{\eta(q,\Psi(\omega))}$ once the tape prefix read so far fixes a prefix of $\Psi(\omega)$ on which the computation of $\eta(q, \cdot)$ halts. Each emission depends on a finite prefix of ω, so G is a c.e. set of pairs (prefix, sentence) as Definition 2.3 requires, and $K(G) \leq K(\mu) + K(\eta) + c$, where c includes an index of the fixed family (φ_q). If $\Psi(\omega)$ is defined, then $A_\omega^G = \Gamma_{f_{\Psi(\omega)}}$, the set of labelled sentences of the labelling $f_{\Psi(\omega)}$.

The bound. Let H be the set of x for which f_x is consistent with B; $\mu(H) = 1$ by hypothesis. On every tape ω with $\Psi(\omega)$ defined, $\Psi(\omega) \in H$ and $f_{\Psi(\omega)} \supseteq D$, the theory $B \cup A_\omega^G$ is consistent and proves ψ_D, so $\omega \in E_D^G$. Hence $\nu_G(D) \geq \mathbb{U}\{\omega : \Psi(\omega) \in H,\ f_{\Psi(\omega)} \supseteq D\} = \mu\{x \in H : f_x \supseteq D\} = \mu_\eta(D)$. By Theorem 3.4, $M_A(D) \geq 2^{-K(\nu_G)}\nu_G(D) \geq 2^{-K(G)}\nu_G(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$, and taking $-\log$ gives the bound on L_A. For a computable sequence x, take $\mu = \delta_x$, with $K(\delta_x) \leq K(x) + c$; the hypothesis is that f_x is consistent with B, and $\mu_\eta(D) = 1$ exactly when $f_x \supseteq D$. □

The constant depends on the question family because the theory machine must write the sentences φ_q. The converse of the proposition fails badly, since answers do not determine raw data; in the conservation-law world the raw loss exceeds the question loss by the whole micro-entropy (Proposition D.3).

D.4 Answering and predicting

The following gives a formal vocabulary for "answering questions is more than predicting data": answering implies predicting, (a), but not conversely, (b), and success at predicting data that present a theory does not settle the questions the data leave open, (c). Part (d) is the coherence that a predictor of separate labels lacks.

Proposition D.6 (Answering and predicting)**.** *(a) If a consistent r.e. theory T proves every datum of a consistent reveal sequence, the cumulative loss of M_A is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T)+c$ in the semimeasure convention, in any order. (b) On the lottery stream of Theorem 5.1, M_A has bounded loss in the semimeasure convention, while no consistent r.e. theory proves every datum and $W_A(D_n) \to 0$. (c) On two-valued data presenting an incomplete consistent r.e. theory T^*, the limit posterior is the prior restricted to the consistent extensions of T^*, and on every sentence independent of $B \cup T^*$ all three values* Bel, Dis, Indep *have positive limits; three-valued data identify the class of T^*. (d) For all D with $M_A(D) > 0$ and all sentences,* $\mathrm{Bel}_D(\bigwedge_{i\le k} \varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$*, and* $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$ *whenever $B \cup \{\varphi\} \vdash \psi$. So the sentences with belief at least $1-\delta$ are closed under entailment and, with error $k\delta$, under k-fold conjunction.*

Proof of Proposition D.6. (a) By Proposition 3.1(d) the cumulative loss in the semimeasure convention is $L_A(D_n) - L_A(\emptyset)$, in any order, and by Theorem 3.7 it is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \le K_{\equiv}(T) + c$.

(b) In the setting of Theorem 5.1, part (a) of that theorem bounds $M_A(D_n)$ below by a positive constant independent of n, so the cumulative loss $L_A(D_n) - L_A(\emptyset)$ is bounded; part (b) says that no consistent r.e. theory proves every datum (with B); and part (c) gives $W_A(D_n) \to 0$.

(c) This is Proposition A.27(b), (c): if $B \cup T^*$ is incomplete and $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, the limit posterior is the prior restricted to the consistent extensions of T^*, and for every ψ independent of $B \cup T^*$ the limits of $\mathrm{Bel}_{D_n}(\psi)$ and $\mathrm{Dis}_{D_n}(\psi)$ lie in $(0,1)$ while $\lim_n \mathrm{Indep}_{D_n}(\psi) \ge \mathbb{U}_A[T^*]/\mathbb{U}_A(\mathrm{Ext}T^*) > 0$; and if every sentence appears with its T^*-trichotomy label, the posterior probability of the class of T^* tends to 1.

(d) A theory that proves φ and ψ proves $\varphi \wedge \psi$, so $O_\varphi \cap O_\psi \subseteq O_{\varphi\wedge\psi}$. Within E_D,

$$\mathbb{U}(O_\varphi \cap O_\psi \cap E_D) \ge \mathbb{U}(O_\varphi \cap E_D) + \mathbb{U}(O_\psi \cap E_D) - \mathbb{U}(E_D);$$

dividing by $\mathbb{U}(E_D)$ gives $\mathrm{Bel}_D(\varphi \wedge \psi) \ge \mathrm{Bel}_D(\varphi) + \mathrm{Bel}_D(\psi) - 1$, and induction on k gives $\mathrm{Bel}_D(\bigwedge_{i\le k}\varphi_i) \ge 1 - \sum_{i\le k}(1 - \mathrm{Bel}_D(\varphi_i))$. If $B \cup \{\varphi\} \vdash \psi$, then $O_\varphi \subseteq O_\psi$, so $\mathrm{Bel}_D(\psi) \ge \mathrm{Bel}_D(\varphi)$. □

D.5 The flagpole

Setting. The language has two sorts, poles and reals; B is the theory of real closed fields on the real sort. There are pole constants $p_1, \dots, p_{N_\mathrm{p}}$, a function symbol ht from poles to reals (the height) and a function symbol sh from poles and reals to reals ($\mathrm{sh}(p,u)$ is the length of the shadow of p when the tangent of the sun's elevation is $u > 0$). Data are sentences $\mathrm{sh}(p_i, u) = v$ and $\mathrm{ht}(p_i) = v$ with rational numerals u, v; interval versions behave in the same way. The two formulations of the shadow law are

$$\Lambda_h := \forall p\, \forall u\, (u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{ht}(p)),$$
$$\Lambda_s := \forall p\, \forall u\, (u > 0 \to \mathrm{sh}(p,u)\cdot u = \mathrm{sh}(p,1)) \ \wedge\ \forall p\, \mathrm{ht}(p) = \mathrm{sh}(p,1),$$

and, with rational numerals a_i for the heights, the four theories are

$$T_H := \Lambda_h + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T_S := \Lambda_h + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}},$$
$$T'_H := \Lambda_s + \{\mathrm{ht}(p_i) = a_i\}_{i\le N_\mathrm{p}}, \qquad T'_S := \Lambda_s + \{\mathrm{sh}(p_i,1) = a_i\}_{i\le N_\mathrm{p}}.$$

In T_S the shadow at 45° "explains" the height. The *intervention vocabulary* consists of $\mathrm{sh}_H(p,u,v)$, the shadow length after the height of p is set to v, and $\mathrm{ht}_S(p,v)$, the height after the reference

shadow $\mathrm{sh}(p,1)$ is set to v by an intervention on the shadow (for instance, by an obstacle). The causal extensions, with the height exogenous and with the shadow exogenous, are

$$C_H := T_H + \{\forall p\,\forall u\,\forall v\,(u>0 \to \mathrm{sh}_H(p,u,v)\cdot u = v),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = \mathrm{ht}(p)\},$$
$$C_S := T_S + \{\forall p\,\forall u\,\forall v\,\mathrm{sh}_H(p,u,v) = \mathrm{sh}(p,u),\ \forall p\,\forall v\,\mathrm{ht}_S(p,v) = v\}.$$

Derivation charges. The cost of a derivation is the number of closed instances it uses of the universally quantified conjuncts of the law axioms (Λ_h has one such conjunct, Λ_s two); steps in B and uses of the initial conditions are free. The *least cost* of a datum for a theory is the least number of such instances that, together with B and the theory's initial conditions, entail it. By completeness this is the least cost of a derivation, so the count is semantic, as in Herbrand's theorem.

Proposition D.7 (The flagpole)**.** *(a) T_H, T_S, T'_H, T'_S are deductively equivalent over B; so every data set is compatible with all four or with none, in the coin-free version the posterior odds of any two programs for them equal their prior odds on every data set compatible with them, and $|K(T_S) - K(T_H)| \le c$. (b) Under derivation charges, every shadow datum $\mathrm{sh}(p_i,u) = v$ with $u \neq 1$ that they entail needs exactly one instance from T_H and T'_S, whose initial conditions sit at the hub of their formulation, and exactly two from T_S and T'_H: which quantity is the hub is a property of the formulation, not of the world. (c) C_H and C_S have the same consequences in the old vocabulary, and one interventional datum separates them.*

The precise form of part (b) is as follows. For a shadow datum $\mathrm{sh}(p_i,u) = a_i/u$ with $u \neq 1$, the least costs for T_H, T_S, T'_H, T'_S are 1, 2, 2, 1; for a height datum $\mathrm{ht}(p_i) = a_i$ they are 0, 1, 0, 1; and for a reference-shadow datum $\mathrm{sh}(p_i,1) = a_i$ they are 1, 0, 1, 0. So the two hub theories T_H and T'_S have equal costs on all shadow data with $u \neq 1$. Their law texts differ by the constant $|\Lambda_s| - |\Lambda_h|$, and in a literal code (Appendix D.6) their initial conditions differ by $N_{\mathrm{p}}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x,1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term; this is 0 if the language has a primitive for the reference shadow, and then the two texts differ by $|\Lambda_s| - |\Lambda_h|$, independently of N_{p} and the data. The precise form of (c) is: C_H and C_S have the same consequences in the language without sh_H and ht_S; $C_H \vdash \mathrm{sh}_H(p_1,1,2a_1) = 2a_1$ ("doubling the pole doubles its 45° shadow"), while $C_S \vdash \mathrm{sh}_H(p_1,1,2a_1) = a_1$, which refutes it when $a_1 \neq 0$; and $C_H \vdash \mathrm{ht}_S(p_1,v) = a_1$ while $C_S \vdash \mathrm{ht}_S(p_1,v) = v$, for every numeral v.

Lemma D.8 (Least costs in a star)**.** *Let T be one of the theories of this appendix without the dark poles: law axioms whose universally quantified conjuncts have closed instances of the form $\pi \to s_1 = \rho\cdot s_2$, with π a condition on real numerals, s_1, s_2 terms of one pole and ρ a real term that B proves nonzero when π holds, and one initial condition $s(p_i) = a_i$ per pole. For a pole p_i, let the* instance graph *have as vertices the closed terms $\mathrm{ht}(p_i)$, $\mathrm{sh}(p_i,u)$ (terms whose real arguments B proves equal are identified) and, in Appendix D.6, $\mathrm{qt}_j(p_i)$, and an edge for each closed instance whose condition B proves, joining the two terms it relates (an instance that relates a term to itself is trivial and is ignored). If the instance graph is a tree, the least cost of a datum $s'(p_i) = v$ that T entails is the distance in the graph from $s'(p_i)$ to the vertex fixed by the initial condition.*

Proof. The instances along the path from the fixed vertex to $s'(p_i)$, together with B and the initial condition, determine the value of $s'(p_i)$; since T entails the datum, that value is v, so the distance is an upper bound. Conversely, let I be a set of fewer instances than the distance. As the graph is a tree, the edges of I do not connect $s'(p_i)$ to the fixed vertex. Fix a model of T over $\mathbb{R}$ (each theory of this appendix has one). Build a model of B, the initial conditions and I

over $\mathbb{R}$ as follows: on the connected component (for the edges of I) of the fixed vertex, and on every term of another pole, keep the values of the fixed model; on the component of $s'(p_i)$, give $s'(p_i)$ a value other than v and propagate it along the edges of I, which is consistent because the component is a tree and each edge is an invertible linear relation; give all other terms arbitrary values. Instances with a false condition hold trivially. This model falsifies the datum, so I does not entail it. □

Proof of Proposition D.7. (a) At $u = 1$, Λ_h gives $\mathrm{sh}(p, 1) = \mathrm{ht}(p)$; given this, $\mathrm{sh}(p, u) \cdot u = \mathrm{ht}(p)$ and $\mathrm{sh}(p, u) \cdot u = \mathrm{sh}(p, 1)$ are equivalent. So $B \vdash \Lambda_h \to \Lambda_s$; conversely Λ_s gives $\mathrm{ht}(p) = \mathrm{sh}(p, 1)$ and hence Λ_h. Given either law, the initial conditions $\mathrm{ht}(p_i) = a_i$ and $\mathrm{sh}(p_i, 1) = a_i$ are interderivable, so the four theories are deductively equivalent over B. Compatibility of a theory with a data set (consistent with B and proving every datum) depends only on its deductive closure, so every data set is compatible with all four or with none. In the coin-free version the posterior weight of a program is $2^{-|p|}$ times the indicator of compatibility, divided by the normaliser, so the odds of two compatible programs are their prior odds. The map on axiom lists that replaces each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i, 1) = a_i$, and its inverse, are fixed computable maps, so $|K(T_S) - K(T_H)| \le c$.

(b) For each pole, the instances of Λ_h join $\mathrm{ht}(p_i)$ to each $\mathrm{sh}(p_i, u)$ (a star centred at the height), and the instances of Λ_s join $\mathrm{sh}(p_i, 1)$ to each $\mathrm{sh}(p_i, u)$ with $u \neq 1$ and to $\mathrm{ht}(p_i)$ (a star centred at the reference shadow); instances with $u \le 0$ have false conditions, and instances at other poles are not connected to p_i. Both graphs are trees, and Lemma D.8 gives the counts: in the height-centred star with the height fixed (T_H), shadows and the reference shadow are at distance 1; with the reference shadow fixed (T_S), a shadow with $u \neq 1$ is at distance 2 and the height at 1; in the shadow-centred star with the height fixed (T'_H), a shadow with $u \neq 1$ is at distance 2 and the reference shadow at 1; with the reference shadow fixed (T'_S), shadows and the height are at distance 1. The law texts of T_H and T'_S are Λ_h and Λ_s; their initial conditions differ only in the terms $\mathrm{ht}(p_i)$ and $\mathrm{sh}(p_i, 1)$, once per pole.

(c) Without the intervention symbols, C_H and C_S are T_H and T_S plus axioms about fresh symbols. Every model of T_H expands to a model of C_H (put $\mathrm{sh}_H(p, u, v) := v/u$ for $u > 0$, arbitrary otherwise, and $\mathrm{ht}_S(p, v) := \mathrm{ht}(p)$), and every model of T_S to a model of C_S (put $\mathrm{sh}_H(p, u, v) := \mathrm{sh}(p, u)$ and $\mathrm{ht}_S(p, v) := v$). So C_H and C_S are conservative over T_H and T_S, whose consequences coincide by (a). The interventional consequences are instances of the added axioms: in C_H, $\mathrm{sh}_H(p_1, 1, 2a_1) \cdot 1 = 2a_1$ and $\mathrm{ht}_S(p_1, v) = \mathrm{ht}(p_1) = a_1$; in C_S, $\mathrm{sh}_H(p_1, 1, 2a_1) = \mathrm{sh}(p_1, 1) = a_1$ and $\mathrm{ht}_S(p_1, v) = v$. If $a_1 \neq 0$ then $2a_1 \neq a_1$, so C_S refutes the datum $\mathrm{sh}_H(p_1, 1, 2a_1) = 2a_1$ that C_H proves. □

Computations. A computation (code in the accompanying repository) confirms the counts by exhaustive search over all subsets of finite instance pools, with exact linear algebra over the rationals, on 500 random data sets with one to four poles (shadows at $u = 2$, and at $u = 3$ in a second implementation); on 2000 random data the four theories prove, refute or leave open exactly the same data. In the literal syntax of that code (numerals counted as one token), $|\Lambda_s| - |\Lambda_h| = 20$ tokens and $\Delta_{\mathrm{sh}} = 1$ token, so T_S is longer than T_H by one token per pole.

D.6 The flagpole with a system of laws

Kitcher's answer to the asymmetry appeals to other laws in which the height figures, which the toy of Proposition D.7 lacks. We add such laws.

Setting. As in Appendix D.5, with further function symbols $\mathrm{qt}_1, \ldots, \mathrm{qt}_k$ from poles to reals (further quantities: weight, the height on the construction plan, and so on), nonzero rational numerals $\theta_1, \ldots, \theta_k$, and data $\mathrm{qt}_j(p_i) = v$. The laws in height form and in shadow form are $\Lambda_j^{\mathrm{ht}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$ and $\Lambda_j^{\mathrm{sh}} := \forall p\, \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p,1)$. The theories are the height hub $T_H^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_{j \le k} + \{\mathrm{ht}(p_i) = a_i\}_i$, the mixed theory $T_S^k := \Lambda_h + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{sh}(p_i,1) = a_i\}_i$, and the shadow hub $T_{SS}^k := \Lambda_s + \{\Lambda_j^{\mathrm{sh}}\}_j + \{\mathrm{sh}(p_i,1) = a_i\}_i$.

Dark variant. Add a predicate Sunlit on poles, restrict the shadow law to sunlit poles, $\Lambda_h^{\mathrm{D}} := \forall p\, (\mathrm{Sunlit}(p) \to \forall u\, (u > 0 \to \mathrm{sh}(p,u) \cdot u = \mathrm{ht}(p)))$, and take sunlit poles $p_1, \ldots, p_{N_\mathrm{p}}$ and dark poles $p'_1, \ldots, p'_{N_\mathrm{d}}$, with the axioms $\mathrm{Sunlit}(p_i)$ and $\neg\mathrm{Sunlit}(p'_l)$ in every theory and rational numerals a'_l for the heights of the dark poles. Laws in shadow form must split cases:

$$\Lambda_j^{\mathrm{sh,D}} := \forall p \Big((\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{sh}(p,1)) \wedge (\neg\mathrm{Sunlit}(p) \to \mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)) \Big).$$

The theories are $T_H^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{ht}}\}_j + \{\mathrm{ht}(p_i) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$ and $T_{SS}^{k,\mathrm{D}} := \Lambda_h^{\mathrm{D}} + \{\Lambda_j^{\mathrm{sh,D}}\}_j + \{\mathrm{sh}(p_i,1) = a_i\}_i + \{\mathrm{ht}(p'_l) = a'_l\}_l$, each with the Sunlit axioms. A *literal code* charges a theory by the number of symbols it writes, each axiom separately, without definitions; $|\cdot|$ denotes this length, a symbol qt_j counts as one token, and $\Delta_{\mathrm{sh}} := |\mathrm{sh}(x,1)| - |\mathrm{ht}(x)|$ is the extra length of the reference-shadow term ($\Delta_{\mathrm{sh}} = 0$ if the language has a primitive for the reference shadow).

Proposition D.9 (The flagpole with a system of laws)**.** *(a)* Class tie. *T_H^k, T_S^k and T_{SS}^k are deductively equivalent over B, and so are $T_H^{k,\mathrm{D}}$ and $T_{SS}^{k,\mathrm{D}}$; so for every k no data set separates them, and their deductive classes coincide. (b)* Universal prior. *$|K(T_{SS}^k) - K(T_H^k)| \le c$ and $|K(T_{SS}^{k,\mathrm{D}}) - K(T_H^{k,\mathrm{D}})| \le c$, with c independent of k, N_p, N_d and the numerals. (c)* Literal codes. *$|T_{SS}^k| - |T_H^k| = (|\Lambda_s| - |\Lambda_h|) + (k + N_\mathrm{p})\Delta_{\mathrm{sh}}$. In the dark variant, $|T_{SS}^{k,\mathrm{D}}| - |T_H^{k,\mathrm{D}}| = k\,\Delta_{\mathrm{split}} + \sum_{j \le k} |\theta_j| + N_\mathrm{p}\Delta_{\mathrm{sh}}$, where $\Delta_{\mathrm{split}} > 0$ is the length of the case-split frame and $|\theta_j|$ the length of the numeral θ_j, which the split law writes twice; so the shadow hub loses by at least $k\,\Delta_{\mathrm{split}}$, an amount growing with k, even when $\Delta_{\mathrm{sh}} = 0$. With one definition the overhead of the laws is a constant: define* $\mathrm{z}(p)$ *once as* $\mathrm{sh}(p,1)$ *on sunlit poles and* $\mathrm{ht}(p)$ *on dark ones, and write every law in* z*; in every model of the theory,* z *is coextensive with* ht*. (d)* Derivation charges *(instances of universally quantified conjuncts, as in Appendix D.5). A datum $\mathrm{qt}_j(p_i) = \theta_j a_i$ has least cost* 1 *for T_H^k and T_{SS}^k, and* 2 *for the mixed theory T_S^k. On shadow data with $u \ne 1$ the two hub theories have least cost* 1 *each; on heights T_H^k has* 0 *and T_{SS}^k has* 1*; on reference shadows the reverse.*

Proof. (a) Each theory contains a form of the shadow law, which gives $B \vdash \mathrm{ht}(p) = \mathrm{sh}(p,1)$ for every pole p (for every sunlit pole in the dark variant), as in the proof of Proposition D.7(a). Substituting, each law in shadow form is equivalent to the law in height form, and each reference-shadow initial condition to the corresponding height condition. In the dark variant, the case split collapses to $\mathrm{qt}_j(p) = \theta_j \cdot \mathrm{ht}(p)$, because on sunlit poles $\mathrm{sh}(p,1) = \mathrm{ht}(p)$ by Λ_h^{D} at $u = 1$.

(b) The syntactic translations and their inverses are fixed computable maps on axiom lists: from T_H^k to T_{SS}^k, replace Λ_h by Λ_s, each Λ_j^{ht} by Λ_j^{sh} and each $\mathrm{ht}(p_i) = a_i$ by $\mathrm{sh}(p_i,1) = a_i$; from $T_H^{k,\mathrm{D}}$ to $T_{SS}^{k,\mathrm{D}}$, replace each Λ_j^{ht} by $\Lambda_j^{\mathrm{sh,D}}$ (reading θ_j and qt_j off the axiom) and the initial condition of each pole declared sunlit by its reference-shadow form. So a program for one list yields a program for the other at a constant extra length, independent of k, N_p, N_d and the numerals.

(c) Counting symbols: each Λ_j^{sh} differs from Λ_j^{ht} by one occurrence of the replaced term, and so does each sunlit initial condition; this gives $(k + N_\mathrm{p})\Delta_{\mathrm{sh}}$ besides the law texts. Each $\Lambda_j^{\mathrm{sh,D}}$ is longer than Λ_j^{ht} by a fixed frame of length $\Delta_{\mathrm{split}} > 0$ (the connectives, the two Sunlit atoms, a

second copy of $\mathrm{qt}_j(p) = \cdot$, and the term $\mathrm{sh}(p, 1)$) and by a second copy of the numeral θ_j; both dark theories contain Λ_h^{D}, the dark initial conditions and the Sunlit axioms. The definition of z is a single axiom of constant length, and laws written in z have the length of the laws written in ht. In every model of the theory, $\mathrm{z}(p) = \mathrm{sh}(p, 1) = \mathrm{ht}(p)$ on sunlit poles by Λ_h^{D} at $u = 1$, and $\mathrm{z}(p) = \mathrm{ht}(p)$ on dark poles by the definition.

(d) For each pole the instance graphs are trees: for T_H^k and T_S^k, a star centred at $\mathrm{ht}(p_i)$ with leaves $\mathrm{sh}(p_i, u)$ and $\mathrm{qt}_j(p_i)$; for T_{SS}^k, a star centred at $\mathrm{sh}(p_i, 1)$ with leaves $\mathrm{sh}(p_i, u)$ $(u \neq 1)$, $\mathrm{ht}(p_i)$ and $\mathrm{qt}_j(p_i)$. Lemma D.8 gives the counts: $\mathrm{qt}_j(p_i)$ is at distance 1 from the fixed centre in T_H^k and T_{SS}^k, and at distance 2 from the fixed leaf $\mathrm{sh}(p_i, 1)$ in T_S^k; the shadow, height and reference-shadow counts are those of Proposition D.7(b) for T_H and T_S'. □

E Proofs for theory shape

Throughout, $\mathcal{S}$ is a law on the base variables $x_1, \dots, x_m$, Φ an explicit or linear theory with m_{aux} auxiliary variables and $\bar{m} := m + m_{\mathrm{aux}}$ variables in all, and $\mathbf{e}_j$ the j-th unit vector. A partial assignment is identified with the set of literals it makes true, and $\hat{x}$ is the conjunction of the base literals true at $x \in \{0,1\}^m$. For a sentence or literal φ, $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

E.1 The model: the listing code, the two encodings, the bridge

Theories that differ only by a renaming of their auxiliary variables are identified, and each class has a fixed canonical representative whose auxiliary variables are $y_1, \dots, y_{m_{\mathrm{aux}}}$ and whose clauses are listed in a fixed canonical order (for instance the lexicographically least listing). Let $\ell_{\mathbb{N}}(n) := |\mathrm{El}(n+1)|$ for $n \geq 0$, with El the Elias delta code of the positive integers; $\ell_{\mathbb{N}}$ is nondecreasing and the lengths of a prefix code of $\{0, 1, 2, \dots\}$. The *listing length* of a canonical explicit theory is

$$\ell_{\mathrm{lst}}(\Phi) := \ell_{\mathbb{N}}(m_{\mathrm{aux}}) + \ell_{\mathbb{N}}(|\Phi|) + \sum_{C \in \Phi} \big(\ell_{\mathbb{N}}(|C|) + |C|(\lceil \log \bar{m} \rceil + 1)\big), \tag{4}$$

each literal being written as a variable index and a sign; a linear theory is listed likewise, each equation by its support and its constant, with $\ell_{\mathbb{N}}(|u|) + |u|\lceil \log \bar{m} \rceil + 1$ bits for an equation of support size $|u|$. Clearly $\ell_{\mathrm{lst}}(\Phi) \geq \|\Phi\|$.

Lemma E.1 (The listing code)**.** $\sum 2^{-\ell_{\mathrm{lst}}(\Phi)} \leq 1$ *over the canonical explicit theories, and* $\ell_{\mathrm{lst}}(\Phi \setminus \{C\}) < \ell_{\mathrm{lst}}(\Phi)$ *for every* $C \in \Phi$*. In particular an* ℓ_{lst}*-shortest* $\mathcal{R}$*-compatible theory is inclusion-minimal among the* $\mathcal{R}$*-compatible theories.*

Proof. Fix m_{aux}. The code of a single clause (its length by $\ell_{\mathbb{N}}$, then that many literals of $\lceil \log \bar{m} \rceil + 1$ bits each) is a prefix code on finite sequences of literals, so $\sum_C 2^{-\ell(C)} \leq 1$ over clauses written in canonical order, where $\ell(C)$ is the summand of (4). The sets of s distinct clauses have total weight at most $(\sum_C 2^{-\ell(C)})^s / s! \leq 1$. The codes $\ell_{\mathbb{N}}(m_{\mathrm{aux}})$ and $\ell_{\mathbb{N}}(s)$ contribute factors whose sums over m_{aux} and over s are at most 1, and canonical theories are a subset of all clause sets; this gives the Kraft inequality. Removing a clause C removes $\ell(C) > 0$ bits and replaces $\ell_{\mathbb{N}}(s)$ by $\ell_{\mathbb{N}}(s-1) \leq \ell_{\mathbb{N}}(s)$; if it removes the last occurrence of an auxiliary variable, then m_{aux} and $\lceil \log \bar{m} \rceil$ do not increase either (after renaming to canonical form, which does not change ℓ_{lst}). □

For two base variables the Kraft sum is 0.2525 (a direct computation over all 256 theories; code in the accompanying repository).

Lemma E.2 (Compatibility with all questions forces the projection)**.** *Let $\mathcal{R}$ be sound and Φ $\mathcal{R}$-compatible with $\mathcal{Q}(\mathcal{S})$. Then* $\operatorname{proj}\Phi \subseteq \mathcal{S}$*; and* $\operatorname{proj}\Phi = \mathcal{S}$ *if every $x \in \mathcal{S}$ has a coordinate that is determined on $\mathcal{S}$ by the others. This holds for every affine law and for the chain law* $\mathrm{Ch}_m := \{x : x_1 \le \cdots \le x_m\}$ *with $m \ge 3$; so for these laws, $\mathcal{R}$-compatibility with $\mathcal{Q}(\mathcal{S})$ and $\mathcal{R}$-usability coincide.*

Proof. Let $x \notin \mathcal{S}$. Choose $I \subsetneq [m]$ maximal such that $\alpha := x|_I$ is consistent with $\mathcal{S}$ ($I = \emptyset$ qualifies since $\mathcal{S} \neq \emptyset$, and $I \neq [m]$ since $x \notin \mathcal{S}$). For $j \notin I$, the extension of α by the value of x_j at x is inconsistent with $\mathcal{S}$, so $\mathcal{S} \wedge \alpha$ entails the literal ℓ on x_j that is false at x, and $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. Compatibility and soundness give $\Phi \cup \alpha \models \ell$; x extends α and falsifies ℓ, so $x \notin \operatorname{proj}\Phi$. Conversely, let $x \in \mathcal{S}$ have its coordinate j determined by $\alpha := x|_{[m]\setminus\{j\}}$. Then $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$ for the literal ℓ on x_j true at x, so $\Phi \cup \alpha$ is satisfiable, by a point of $\operatorname{proj}\Phi \subseteq \mathcal{S}$ extending α, and the only such point of $\mathcal{S}$ is x. For an affine law, a coordinate j is determined by the others if and only if j lies in the support of a nonzero derived equation (Lemma E.7 with $I = [m] \setminus \{j\}$); such a j exists since $r \ge 1$, and the property does not depend on x. For Ch_m, the points are $0^a 1^{m-a}$; if $a \ge 2$, then $x_2 = 0$ forces $x_1 = 0$, and if $m - a \ge 2$, then $x_{m-1} = 1$ forces $x_m = 1$; one of the two holds as $m \ge 3$. □

The hypothesis $m \ge 3$ is needed: $\Phi = \{\neg x_1 \vee x_2,\ x_1 \vee \neg x_2\}$ is UP- and PS_∞-compatible with $\mathcal{Q}(\mathrm{Ch}_2)$, whose only questions are "$x_1 = 1$, so $x_2 = 1$" and "$x_2 = 0$, so $x_1 = 0$", but $\operatorname{proj}\Phi = \{00, 11\} \neq \mathrm{Ch}_2$; the point 01 has no determined coordinate.

Proposition E.3 (The two-valued encoding is degenerate)**.** *Encode a conditional datum (α, ℓ) by the sentence $\bigwedge\alpha \to \ell$ labelled true, with $B = \emptyset$. Then every consistent theory that proves all sentences proved by a compatible theory is compatible. In particular, for every law $\mathcal{S}$ and every $x^0 \in \mathcal{S}$, the theory $T_{x^0} := \{x_i^{x_i^0} : i \le m\}$ proves $\bigwedge\alpha \to \ell$ for every $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$, so $K_A(\{(\bigwedge\alpha \to \ell, 1) : (\alpha, \ell) \in \mathcal{Q}(\mathcal{S})\}) \le K(x^0) + c$; for a homogeneous affine law ($b = 0$, $x^0 = 0$) this is at most $K(m) + c$.*

Proof. Compatibility with a datum labelled true asks only that the theory be consistent and prove the sentence, which passes to consistent theories proving more. If α agrees with x^0, then x^0 is a point of $\mathcal{S}$ extending α, so x^0 satisfies ℓ and $T_{x^0} \vdash \ell$; otherwise $T_{x^0} \vdash \neg\bigwedge\alpha$. Either way $T_{x^0} \vdash \bigwedge\alpha \to \ell$, and T_{x^0} is consistent and computable from x^0. □

Under this encoding the data cannot distinguish a law from a single world that satisfies it. Encoding (E3) of Definition 2.10 makes "given α" hypothetical: a compatible theory may not refute an antecedent, so no single world fits question data with two incompatible antecedents. (E3) is not upward closed, so it cannot be expressed by two-valued proof-requiring labels. For a conditional data set D we write $K_A(D)$ and $W_A(D)$ for the inducers of Definition 2.7 over the language $\mathcal{L}_m$ with $B = \emptyset$, in which an r.e. theory T is compatible with D if it is compatible with every datum of D under (E3).

Proposition E.4 (The bridge to SAI)**.** *(a) With $\mathcal{R} = \mathrm{PS}_\infty$ and as hypotheses the r.e. sets of $\mathcal{L}_m$-sentences, or the r.e. sets of clauses over the base variables, $K_A^{\mathrm{PS}_\infty}(D) = K_A(D) \pm c$ and $W_A^{\mathrm{PS}_\infty}(D) \asymp W_A(D)$ on every conditional data set D. (b) With finite explicit theories, of either class, $K_A(D) - c \le K_A^{\mathrm{PS}_\infty}(D) \le K_A(D) + \ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(2^m) + c$, that is, up to $m + O(\log m)$ bits.*

Proof. (a) For sets of sentences, PS_∞-compatibility of an r.e. set T is (E3), by the completeness theorem: $T \cup \alpha$ is satisfiable if and only if it is consistent, and PS_∞ returns ℓ if and only if $T \cup \alpha \vdash \ell$. For sets of clauses, a fixed program turns an enumeration of T into an enumeration of $\mathrm{Cl}(T)$, the

set of clauses over the base variables derivable from T. For every question, $T \cup \alpha$ is inconsistent iff $T \vdash \neg \bigwedge \alpha$ iff the clause $\neg\alpha$ is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha$ is inconsistent (as $\mathrm{Cl}(T) \subseteq \mathrm{Cn}(T)$); and $T \cup \alpha \vdash \ell$ iff the clause $\neg\alpha \vee \ell$ (non-tautological, as ℓ is on a variable outside $\operatorname{dom}\alpha$) is in $\mathrm{Cl}(T)$ iff $\mathrm{Cl}(T) \cup \alpha \vdash \ell$. So T and $\mathrm{Cl}(T)$ are compatible with the same conditional data. (A theory of $\mathcal{L}_m$ need not consist of Boolean combinations of the atoms, since $\mathcal{L}_m$ has equality and quantifiers; $\mathrm{Cl}(T)$ replaces a normal form.) Conversely, clauses are sentences. The weights compare in the same way, through the fixed maps on programs. (b) *Left.* A fixed prefix replaces a finite explicit theory Φ, whose auxiliary variables are not symbols of $\mathcal{L}_m$, by $\mathrm{PI}(\operatorname{proj}\Phi)$, a finite set of $\mathcal{L}_m$-sentences. It is compatible with the same data: $\Phi \cup \alpha$ is satisfiable iff $\operatorname{proj}\Phi$ meets α, and for a base literal ℓ, $\Phi \cup \alpha \models \ell$ iff every point of $\operatorname{proj}\Phi$ extending α satisfies ℓ. *Right.* Let p enumerate a consistent T that is compatible with D under (E3), and let $\mathcal{S}_T := \{x : T \cup \{\hat{x}\} \text{ consistent}\}$. Then $\mathcal{S}_T \neq \emptyset$, since otherwise T proves $\bigwedge_x \neg\hat{x}$, which is refutable. Since $x \notin \mathcal{S}_T$ iff $T \vdash \neg\hat{x}$, a program given p, m and $|\mathcal{S}_T|$ (self-delimiting, with $\ell_{\mathbb{N}}(m) + \ell_{\mathbb{N}}(|\mathcal{S}_T|) + c$ extra bits) enumerates refutations until $2^m - |\mathcal{S}_T|$ points are refuted, and outputs $\mathrm{PI}(\mathcal{S}_T)$. By completeness, $T \cup \alpha$ is consistent iff $\mathcal{S}_T$ meets α, and $T \cup \alpha \vdash \ell$ iff every point of $\mathcal{S}_T$ extending α satisfies ℓ, as ℓ is a base literal. So $\mathrm{PI}(\mathcal{S}_T)$ is PS_∞-compatible with D exactly when T is, and $\ell_{\mathbb{N}}(|\mathcal{S}_T|) \leq \ell_{\mathbb{N}}(2^m)$. □

So the reasoner-relative inducers extend SAI: with an ideal deducer and (E3) they are SAI on question data, up to the gap between finite and r.e. hypotheses in (b). We do not know whether that gap is attained.

E.2 Program length, listing and time

Lemma E.5 (Prime implicates and one round of propagation)**.** *Let $\mathcal{S}$ be any law. (a) If α is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge \alpha \models \ell$, one round of unit propagation on $\mathrm{PI}(\mathcal{S}) \cup \alpha$ derives ℓ. (b) Every UP_1-usable $\Phi \in \mathcal{E}_m^{\mathrm{base}}$ contains $\mathrm{PI}(\mathcal{S})$.*

Part (a) is folklore; compiling a theory so that unit resolution becomes complete is studied by del Val (1994).

Proof. (a) The clause $\neg\alpha \vee \ell$ is an implicate of $\mathcal{S}$, so it contains a prime implicate C. If $\ell \notin C$, then $C \subseteq \neg\alpha$ is falsified by α, contradicting the consistency of α with $\mathcal{S}$. So $\ell \in C$ and all other literals of C are falsified by α: C fires in round 1. No conflict arises, since $\mathrm{PI}(\mathcal{S}) \cup \alpha$ is satisfiable. (b) Let $C = \neg\alpha' \vee \ell \in \mathrm{PI}(\mathcal{S})$. Since C is prime, $\neg\alpha'$ is not an implicate, so α' is consistent with $\mathcal{S}$ and $(\alpha', \ell) \in \mathcal{Q}(\mathcal{S})$. In one round from α', the clause of Φ that derives ℓ has all its other literals falsified by α', so it is some $C' \subseteq C$ with $\ell \in C'$; C' is an implicate of $\mathcal{S}$ (as Φ is a theory over the base variables with $\operatorname{proj}\Phi = \mathcal{S}$), so $C' = C$ by primality. □

Proof of Theorem 7.2. Let $\mathrm{comp}(\Phi') := \mathrm{PI}(\operatorname{proj}\Phi')$, computable from Φ' (a linear theory is first written as clauses). If Φ' is $\mathcal{R}'$-compatible with D then, by soundness, for each $(\alpha, \ell) \in D$ the set $\Phi' \cup \alpha$ is satisfiable and entails ℓ; so $\operatorname{proj}\Phi'$ meets α, and every point of it extending α satisfies ℓ, as ℓ is a base literal. By Lemma E.5(a) applied to the law $\operatorname{proj}\Phi'$, $\mathrm{comp}(\Phi') \in \mathcal{E}_m^{\mathrm{base}}$ is UP_1-compatible with D. A program for $\mathrm{comp}(\Phi')$ is a fixed prefix followed by a program for Φ', which gives both inequalities. For the equalities: UP_1, UP and PS_∞ are sound, so each is covered by the inequality, within either class; conversely a UP_1-compatible theory is UP- and PS_∞-compatible, since on a satisfiable $\Phi \cup \alpha$ these reasoners derive at least what UP_1 derives and never reach a conflict. □

On the question data of any law, a one-round unit-propagation reasoner over the base variables is therefore as good, under program length, as Gaussian elimination or unbounded proof search with any intermediate concepts, up to a machine constant.

Proof of Theorem 7.3. (a) By Lemma E.2, for an affine law and a sound reasoner, compatibility with $\mathcal{Q}(\mathcal{S})$ is usability. (b) Upper bounds: from $\mathcal{S}$ a fixed program computes $\mathrm{PI}(\mathcal{S}) \in \mathcal{E}_m^{\mathrm{base}}$, which is usable for every UP_k, UP and PS_∞ (Lemma E.5(a)), and a canonical system of equations for $\mathcal{S}$, which is usable for GE. Lower bound: from a program for any compatible Φ a fixed program computes $\operatorname{proj}\Phi = \mathcal{S}$ (Lemma E.2). □

Proposition E.6 (Time-charged bounds)**.** *Let $\mathcal{S}$ be an affine law, $\mathcal{R}$ a sound reasoner, and take either class. Then $Kt_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) \geq \log \min\|\Phi\|$ over the $\mathcal{R}$-usable Φ of the class; and $Kt_A^{\mathrm{UP}_1}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + \log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(r + \log m)$, $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(\log m)$.*

Proof. The lower bound is the printing time: V needs at least $\|\Phi\|$ steps to print $\langle\Phi\rangle$. For the upper bounds, a program runs a program p for (A,b) (time t_p), enumerates the 2^r derived equations, keeps the circuits (time $\mathrm{poly}(m)4^r$), and prints either their clauses, which form $\mathrm{PI}(\mathcal{S})$ (Proposition E.8; time $\mathrm{poly}(m)|\mathrm{PI}(\mathcal{S})|$), or the library of Proposition E.18 with intermediate concepts (at most 2^r circuits, each with at most $4m$ clauses), or, for GE, just the r equations, without enumerating the derived equations. Then log of the total time is at most $\log t_p$ plus $\log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, resp. $O(r + \log m)$, resp. $O(\log m)$, and $|p| + \log t_p = Kt(A,b)$ for an optimal p. (A program of length $K(A,b)$ for (A,b) need not run within any computable time bound, which is why Kt appears.) □

E.3 Linear laws over the base variables

Throughout this subsection $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$ with $A \in \mathrm{GF}(2)^{r\times m}$ of rank $r \geq 1$, $\mathrm{Row}(A)$ is the set of derived equations, $b_u := y \cdot b$ for $u = yA$ (well defined as A has full row rank, and linear in u), and a_* is the number of derived equations of weight d_*. Over GF(2) a vector is determined by its support, so a circuit is the unique nonzero derived equation supported in its support.

Lemma E.7 (Determination)**.** *For $I \subseteq [m]$ and $j \notin I$ the following are equivalent: (i) x_j is a function of x_I on $\mathcal{S}$; (ii) some $u \in \mathrm{Row}(A)$ has $j \in \mathrm{supp}(u) \subseteq I \cup \{j\}$; (iii) some circuit has this property.*

Proof. (ii)⇒(i): $x_j = b_u + \sum_{i\in\mathrm{supp}(u)\setminus\{j\}} x_i$ on $\mathcal{S}$. (i)⇒(ii): if no such u exists, then $\mathbf{e}_j \notin \mathrm{Row}(A) + \mathrm{span}\{\mathbf{e}_i : i \in I\}$, so some z is orthogonal to that space and has $z_j = 1$; then $z \in \ker A$ and $z_i = 0$ on I, and $x, x + z \in \mathcal{S}$ agree on I and differ at j. (ii)⇒(iii): take u of least support among the witnesses of (ii). If a nonzero derived equation u' had $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, then either $j \in \mathrm{supp}(u')$, contradicting minimality, or $u + u'$, with support $\mathrm{supp}(u) \setminus \mathrm{supp}(u') \ni j$, is a smaller witness. So u is a circuit. □

Proposition E.8 (Prime implicates of an affine law)**.** *For $Z \subseteq [m]$, the projection of $\mathcal{S}$ to Z is the affine subspace cut out by the derived equations supported in Z; so $C_{Z,\beta}$ is an implicate of $\mathcal{S}$ if and only if β violates some derived equation supported in Z. The prime implicates of $\mathcal{S}$ are exactly the clauses $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$; there are $\sum_{\text{circuits } u} 2^{|u|-1}$ of them, and $\mathrm{PI}(\mathcal{S})$ is UP_1-usable.*

Proof. An affine subspace is the common solution set of the affine functionals constant on it. The functionals on $\{0,1\}^Z$ that are constant on the projection of $\mathcal{S}$ are the restrictions of the functionals constant on $\mathcal{S}$, that is of $\mathrm{Row}(A)$ (as $\mathcal{S} = x^0 + \ker A$ and $(\ker A)^\perp = \mathrm{Row}(A)$), that vanish off Z. So $C_{Z,\beta}$ is an implicate iff β is outside the projection iff β violates some derived equation u with $\mathrm{supp}(u) \subseteq Z$. Every such u is a sum of circuits supported in $\mathrm{supp}(u)$: if u is not a circuit, some circuit u' has $\mathrm{supp}(u') \subsetneq \mathrm{supp}(u)$, and $u + u'$ has the smaller support $\mathrm{supp}(u) \setminus \mathrm{supp}(u')$; induct. Since $u \mapsto b_u$ is linear, β violates one of these circuits, u_0, and $C_{\mathrm{supp}(u_0),\beta|_{\mathrm{supp}(u_0)}} \subseteq C_{Z,\beta}$ is an implicate. A clause $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ is prime, since a strictly smaller implicate would be violated on a smaller support, by a derived equation supported there, against the minimality of circuits. For each circuit there are $2^{|u|-1}$ such β, and distinct circuits have distinct supports. $\mathrm{PI}(\mathcal{S})$ defines $\mathcal{S}$, and is UP_1-compatible with $\mathcal{Q}(\mathcal{S})$ by Lemma E.5(a). □

Proposition E.9 (Any CNF)**.** *Every implicate of $\mathcal{S}$ has width at least d_*, so every CNF over the base variables equivalent to $\mathcal{S}$ has at least $(1 - 2^{-r})2^{d_*}$ clauses.*

Proof. For $|Z| < d_*$ no nonzero derived equation is supported in Z, so the projection of $\mathcal{S}$ to Z is $\{0,1\}^Z$ (Proposition E.8) and no clause over Z is an implicate. A clause of width $w \geq d_*$ excludes $2^{m-w} \leq 2^{m-d_*}$ points, and the $2^m - 2^{m-r}$ non-solutions must all be excluded. □

Lemma E.10 (Subsumption)**.** *Let $\mathcal{S}$ be any law, Φ a set of implicates of $\mathcal{S}$ over the base variables, and Φ' obtained by replacing each clause C by any prime implicate $C' \subseteq C$. For every α consistent with $\mathcal{S}$ and every k, every literal derived by UP_k on $\Phi \cup \alpha$ is derived by UP_k on $\Phi' \cup \alpha$, and neither run reaches a conflict. So minimum usable theories over the base variables may be sought among subsets of $\mathrm{PI}(\mathcal{S})$, round for round.*

Proof. Both runs derive only literals entailed by $\mathcal{S} \wedge \alpha \neq \emptyset$, so neither conflicts. Induct on rounds. If C fires in round t of the Φ-run, deriving L, all other literals of C are falsified after round $t-1$, hence, by induction, in the Φ'-run. If $L \notin C'$, every literal of $C' \subseteq C$ would be falsified by the state of the Φ-run after round $t-1$; that state holds at every point of $\mathcal{S} \wedge \alpha$, and such a point would falsify the implicate C', which is impossible. So $L \in C'$, and C' fires by round t. □

Only one direction holds: Φ' derives at least what Φ derives, and the converse fails (see the counterexamples after Theorem E.36).

Decomposable circuits. Recall that a circuit u is decomposable if $u = u' + u''$ with $u', u'' \in \mathrm{Row}(A)$ nonzero and $\mathrm{supp}(u') \cap \mathrm{supp}(u'') = \{i\}$ for a single $i \notin \mathrm{supp}(u)$. Equivalently:

$$\begin{gathered}\text{for some, equivalently every, } j \in \mathrm{supp}(u) \text{ there are } u' \in \mathrm{Row}(A) \text{ and } i \notin \mathrm{supp}(u) \\ \text{with } i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}.\end{gathered} \tag{5}$$

Indeed, a decomposition through i gives such a u' for every j, taking u' or u'', whichever does not contain j (they partition $\mathrm{supp}(u)$); conversely such a u' gives the decomposition $u = u' + (u + u')$, whose supports meet in $\{i\}$.

Proposition E.11 (Indecomposable circuits must be written)**.** *For every d_*, every UP-usable explicit theory over the base variables contains all $2^{|u|-1}$ clauses $C_{\mathrm{supp}(u),\beta}$, $u \cdot \beta \neq b_u$, of every indecomposable circuit u.*

Proof. Let Φ be usable (so its clauses are implicates), u indecomposable, $j \in \mathrm{supp}(u)$, $I := \mathrm{supp}(u) \setminus \{j\}$, and α any assignment to I. No nonzero derived equation is supported in I, as u is a circuit, so α is consistent and determines x_j (Proposition E.8 and Lemma E.7). Unit propagation never conflicts on the satisfiable $\Phi \cup \alpha$, and by usability it derives a literal outside I; so round 1 derives one. A literal derived in round 1 on x_i, $i \notin I$, comes from a clause $C \in \Phi$ whose other literals are on I and falsified by α, so $\mathrm{supp}(C) \subseteq I \cup \{i\}$, and C is an implicate. By Proposition E.8 some derived equation u' with $\mathrm{supp}(u') \subseteq \mathrm{supp}(C)$ is violated by the assignment that falsifies C, and $i \in \mathrm{supp}(u')$, since otherwise u' would be supported in I. If $i \neq j$, then $i \notin \mathrm{supp}(u)$ and u' contradicts indecomposability by (5). So $i = j$, $\mathrm{supp}(u') \subseteq \mathrm{supp}(u)$, hence $u' = u$ by the minimality of circuits, and $C = C_{\mathrm{supp}(u),\beta}$ for the assignment β that extends α by the value of x_j opposite to the derived one. The $2^{|u|-1}$ choices of α give all the clauses of u. □

Proposition E.12 (What is forced)**.** *For every affine law, whatever d_*, the clauses contained in every* UP*-usable explicit theory over the base variables are exactly those of $\Phi_{\mathrm{ind}}(\mathcal{S})$.*

Proof. By Proposition E.11 it remains to show that for every clause $C \notin \Phi_{\mathrm{ind}}(\mathcal{S})$ some UP-usable theory over the base variables omits C. Let $\mathrm{Imp}(\mathcal{S})$ be the set of all non-tautological implicates over the base variables; we show that $\mathrm{Imp}(\mathcal{S}) \setminus \{C\}$ is usable. It consists of implicates, so unit propagation on it from a consistent antecedent never conflicts, and once it is compatible with $\mathcal{Q}(\mathcal{S})$ its projection is $\mathcal{S}$ (Lemma E.2). If C is not a prime implicate, the set contains $\mathrm{PI}(\mathcal{S})$ and is usable by Lemma E.5(a) and the monotonicity of unit propagation in the clause set. Otherwise $C = C_{\mathrm{supp}(u),\beta}$ for a decomposable circuit u. Let $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$. In the one-round run on $\mathrm{PI}(\mathcal{S}) \cup \alpha$ some prime implicate $C' \subseteq \neg\alpha \vee \ell$ fires. If $C' \neq C$, it is in our set. If $C' = C$, then ℓ is the literal of C on some $j \in \mathrm{supp}(u)$, and α contains $\beta|_{\mathrm{supp}(u)\setminus\{j\}}$. By (5) there are u' and $i \notin \mathrm{supp}(u)$ with $i \in \mathrm{supp}(u') \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$; put $u'' := u + u'$, so that $j \in \mathrm{supp}(u'') \subseteq (\mathrm{supp}(u) \setminus \mathrm{supp}(u')) \cup \{i\}$. Let v_i be the value of x_i that u' forces given α. The clause on $\mathrm{supp}(u')$ falsified by $\alpha|_{\mathrm{supp}(u')\setminus\{i\}} \cup \{x_i = 1 - v_i\}$ is an implicate (that assignment violates u'), differs from C (its support contains $i \notin \mathrm{supp}(u)$) and fires in round 1, deriving $x_i = v_i$. The clause on $\mathrm{supp}(u'')$ falsified by the restriction of $\alpha \cup \{x_i = v_i\}$ to $\mathrm{supp}(u'') \setminus \{j\}$, extended by the value of x_j opposite to ℓ, is likewise an implicate different from C, and fires in round 2, deriving ℓ ($\alpha \cup \{x_i = v_i\}$ is consistent and entails ℓ). So every question is answered. □

Proof of Theorem 7.4. We show by induction on $|w|$: for every nonzero $w \in \mathrm{Row}(A)$, every $j \in \mathrm{supp}(w)$ and every assignment α to $\mathrm{supp}(w) \setminus \{j\}$ consistent with $\mathcal{S}$, unit propagation on $\Phi_{\mathrm{ind}} \cup \alpha$ derives the value of x_j, and it never conflicts, as $\Phi_{\mathrm{ind}} \cup \alpha$ is satisfiable. By Lemma E.7 and the monotonicity of unit propagation in the antecedent (a larger consistent antecedent derives at least as much), this is usability; the projection is $\mathcal{S}$ by Lemma E.2.

- *w is not a circuit.* By Lemma E.7(iii) with $I = \mathrm{supp}(w) \setminus \{j\}$ there is a circuit $c' \neq w$ with $j \in \mathrm{supp}(c') \subseteq \mathrm{supp}(w)$, hence $|c'| < |w|$; apply the induction hypothesis to c' and the restriction of α.
- *$w = u$ is an indecomposable circuit.* Its clauses are in Φ_{ind}, and one of them fires.
- *$w = u$ is a decomposable circuit.* By (5) write $u = u' + u''$ through $i \notin \mathrm{supp}(u)$ with $j \in \mathrm{supp}(u'')$. If $\mathrm{supp}(u') = (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$, then $u'' = \mathbf{e}_i + \mathbf{e}_j$ has weight $2 < d_*$, which is impossible; so $|u'| \leq |u| - 1$. Also $|\mathrm{supp}(u') \cap \mathrm{supp}(u)| = |u'| - 1 \geq d_* - 1 \geq 2$, so $|u''| = |\mathrm{supp}(u) \setminus \mathrm{supp}(u')| + 1 \leq |u| - 1$. By induction on u' (target i, antecedent the restriction of α to $\mathrm{supp}(u') \setminus \{i\} \subseteq \mathrm{supp}(u) \setminus \{j\}$), unit propagation derives the value v_i of x_i determined by α. The

assignment $\alpha \cup \{x_i = v_i\}$ is consistent, and its restriction to $\mathrm{supp}(u'') \setminus \{j\} \subseteq (\mathrm{supp}(u) \setminus \{j\}) \cup \{i\}$ is an antecedent for u'' and target j; by induction unit propagation derives x_j.

The base case is contained in the induction: a circuit of weight $d_* \geq 3$ is indecomposable, since a decomposition would give, by the two bounds above, $|u'|, |u''| \leq d_* - 1$, against the minimality of d_*. Necessity: every usable theory over the base variables contains Φ_{ind} (Proposition E.11). Sufficiency: an explicit theory over the base variables equivalent to $\mathcal{S}$ consists of implicates, so adding its clauses to Φ_{ind} keeps every derivation and causes no conflict on consistent antecedents. □

Corollary E.13 (Counting, and when the given equations suffice)**.** *Let $d_* \geq 3$. (a) Every circuit of weight d_* is indecomposable, so every* UP*-usable theory over the base variables has at least $a_* 2^{d_*-1}$ clauses. (b) The* core*, the r given equations each written as all of its $2^{|row|-1}$ clauses, is* UP*-usable if and only if every indecomposable circuit is one of the given rows. So a circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one intermediate base quantity, and the core suffices for unit propagation exactly when it already contains every such circuit. (c) One parity law on $m \geq 3$ variables needs exactly its 2^{m-1} clauses over the base variables.*

Proof. (a) As in the base case of the proof of Theorem 7.4. (b) The core is equivalent to $\mathcal{S}$, so by Theorem 7.4 it is usable iff it contains Φ_{ind}. A clause on the support of a circuit u belongs to the core iff some row has that support, that is, as a vector is determined by its support, iff u is a row. (c) $\mathrm{Row}(A) = \{0, 1^m\}$, and 1^m is an indecomposable circuit of weight $m = d_* \geq 3$. □

A derived equation that is not a circuit, for example the sum of two circuits with disjoint supports, never needs to be written.

Corollary E.14 (The simplex family)**.** *Let $r \geq 3$, $m = 2^r - 1$, and let the columns of A be the nonzero vectors of* $\mathrm{GF}(2)^r$*. Every nonzero derived equation has weight $(m+1)/2$ and is an indecomposable circuit; the unique minimum* UP*-usable theory over the base variables is* $\mathrm{PI}(\mathcal{S})$*, with $m\, 2^{(m-1)/2}$ clauses of width $(m+1)/2$, while* GE *uses the $r = \log(m+1)$ equations.*

Proof. Every nonzero $u = yA$ ($y \neq 0$) has $u_c = y \cdot c$ at the column c, which is 1 for exactly half of the 2^r vectors c, none of them 0: so u has weight $2^{r-1} = (m+1)/2$. Equal weights make every nonzero derived equation a circuit, and $d_* = 2^{r-1} \geq 4$, so all m of them are indecomposable by Corollary E.13(a). By Theorem 7.4 the unique minimum is $\Phi_{\mathrm{ind}} = \mathrm{PI}(\mathcal{S})$, with $m 2^{(m-1)/2}$ clauses of width $(m+1)/2$. □

For $r = 3, \ldots, 7$ the minimum is 56, 1,920, 1,015,808, about $1.35 \cdot 10^{11}$ and about $1.2 \cdot 10^{21}$ clauses, against r equations; for $r = 3$ a direct computation (code in the accompanying repository) confirms that $\mathrm{PI}(\mathcal{S})$ answers all 2,352 entailed literals and that deleting any one of its 56 clauses breaks usability.

Remark E.15 (The hypothesis $d_* \geq 3$). The natural extension of Theorem 7.4 and Corollary E.13(a) to $d_* = 2$ fails. The law $x_1 = x_2 = x_3 = x_4$ has $a_* = 6$ derived equations of weight 2, so the bound of Corollary E.13(a) would be 12 clauses, but 4 suffice (a directed cycle; see below); likewise the simplex family with $r = 2$ is the law $x_1 = x_2 = x_3$, with $|\mathrm{PI}| = 6 = a_* 2^{d_*-1}$ and minimum 3. For every d_*, Φ_{ind} is necessary (Proposition E.11) and nothing else is forced (Proposition E.12), but for $d_* = 2$ it need not be usable: a direct computation (code in the accompanying repository) finds, for example, a law with $m = 6$, $r = 3$ and $d_* = 2$ for which Φ_{ind}

has 4 clauses and the least usable theory over the base variables has 8. For $d_* \le 2$ the difficulty is the choice among alternative clauses, not which clauses are forced.

One case is exact. Call $\mathcal{S}$ an *equivalence law* if $\mathrm{Row}(A)$ is spanned by derived equations of weight 2. Then $i \sim j :\iff \mathbf{e}_i + \mathbf{e}_j \in \mathrm{Row}(A)$ partitions the variables that occur in derived equations into classes $J_1, J_2, \ldots$ of sizes at least 2, and *the least* UP*-usable theory over the base variables has exactly* $|J_1| + |J_2| + \cdots$ *clauses: a directed cycle of implications on each class.*

Proof. Every derived equation is a sum of vectors $\mathbf{e}_i + \mathbf{e}_j$ with $i \sim j$, so its restriction to each class has even weight; hence a derived equation of weight at least 3 contains two variables of one class in its support, the circuits are exactly the weight-2 equations, which lie within classes, and there is no derived equation of weight 1. *Lower bound.* Fix a class J, $i \in J$ and $v \in \{0,1\}$. The antecedent $\{x_i = v\}$ is consistent and entails a literal on each $x_{i'}$, $i' \in J \setminus \{i\}$, so unit propagation derives something in round 1, by a clause whose other literals are falsified by $\{x_i = v\}$: a unit clause, which is not an implicate (no variable is fixed), or a 2-clause containing the literal x_i^{1-v}. A 2-clause implicate has as support the support of a derived equation (Proposition E.8), a pair within one class. So each of the $2|J|$ literals on J has its negation in some 2-clause within J; a 2-clause contains two literals, so at least $|J|$ clauses lie within J, and different classes share no clause. *Upper bound.* Fix $x^0 \in \mathcal{S}$; on $\mathcal{S}$, $x_i^{x_i^0} \leftrightarrow x_{i'}^{x_{i'}^0}$ for $i \sim i'$. For $J = \{i_1, \ldots, i_{|J|}\}$ take the $|J|$ implicates $\neg x_{i_t}^{x_{i_t}^0} \vee x_{i_{t+1}}^{x_{i_{t+1}}^0}$ (indices mod $|J|$). A question (α, ℓ) with ℓ on x_j, $j \in J$, has α assigning some x_i with $i \in J$ (Lemma E.7(iii): the circuits are the pairs); if α makes $x_i^{x_i^0}$ true, propagation runs forward around the cycle to x_j, and if false, backward. No conflict arises. □

A direct computation (exhaustive over subsets of PI; code in the accompanying repository) confirms that the minimum is the size of the class for one class of $2, \ldots, 5$ variables, and is 3 for $x_1 = x_2 \oplus 1 = x_3$, and 5 and 6 for two classes of sizes $(2,3)$ and $(3,3)$.

E.4 Intermediate concepts, systems of laws and depth

Proposition E.16 (The chain encoding)**.** *Let $m \ge 4$. Encode $x_1 \oplus \cdots \oplus x_m = b$ by the constraints $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ for $2 \le i \le m-2$, and $y_{m-2} \oplus x_m = b$, each three-variable constraint written as its* 4 *clauses and the last as its* 2 *clauses: $4m-6$ clauses and $m-2$ auxiliary variables. The encoding is* propagation complete on all variables*: for every partial assignment α to the base and auxiliary variables, unit propagation derives every literal entailed by the encoding and α, and reaches a conflict if and only if α is inconsistent with the encoding. Without auxiliary variables 2^{m-1} clauses are necessary (Corollary E.13(c)) and sufficient.*

Proof. Each constraint's clause set is the set of all prime implicates of that constraint, so (Lemma E.5(a)) it derives its last variable when the others are assigned, and gives a conflict when all are assigned inconsistently. Consider a fixpoint of unit propagation without conflict. Every constraint is then either fully assigned, and satisfied, or has at least two unassigned variables; call the latter *active*. Each variable occurs in at most two constraints, and consecutive constraints share exactly one auxiliary variable, so the graph on the active constraints, joined when they share an unassigned variable, is a disjoint union of paths. Fix an unassigned variable z and a value for it. Process each component breadth-first from a root (in the component of z, a constraint containing z, with z set first). When a constraint is processed, at most one of its variables is already set: the one it shares with its parent, or z at the root. It has at least two unassigned variables, so at least one of them is still free: give the other free ones arbitrary values and solve for that one. This yields a solution extending the fixpoint with either value of z, so no unassigned variable is entailed. If α were inconsistent and no conflict arose, the same

construction (without z) would give a solution extending α; so unit propagation reaches a conflict on every inconsistent α, and by soundness only on those. □

The chain encoding is standard. Every CNF encoding of parity with auxiliary variables has at least $3m$ clauses (Emdin et al., 2022), so the chain is within a factor $4/3$ of the least clause count even among encodings that need not be usable.

Proposition E.17 (One round cannot use intermediate concepts)**.** *For any law $\mathcal{S}$, every* UP_1*-usable explicit theory, with any auxiliary variables, contains* $\mathrm{PI}(\mathcal{S})$*: 2^{m-1} clauses for one parity law, $m2^{(m-1)/2}$ for the simplex family.*

Proof. Let $\Phi \in \mathcal{E}_m$ be UP_1-usable and α the antecedent of a question; it assigns base variables only. In round 1 a clause fires only if all its literals but one are falsified; auxiliary literals are unassigned, so a clause containing an auxiliary literal fires only if that literal is its only unfalsified one, and then it derives that auxiliary literal. Hence the base literals derived in round 1 are derived by the clauses of Φ over the base variables, Φ_{base}; no conflict occurs, as $\Phi \cup \alpha$ is satisfiable. Each $C \in \Phi_{\mathrm{base}}$ is an implicate of $\mathcal{S}$, as $\Phi \models C$ and $\operatorname{proj} \Phi = \mathcal{S}$. The proof of Lemma E.5(b), which uses only that the firing clause is an implicate of $\mathcal{S}$, gives $\Phi_{\mathrm{base}} \supseteq \mathrm{PI}(\mathcal{S})$. For one parity law PI consists of the 2^{m-1} clauses of its single circuit; for the simplex family, of $2^{(m-1)/2}$ clauses for each of its m circuits (Corollary E.14). □

A direct computation (code in the accompanying repository) confirms on twelve random laws that deleting any clause of $\mathrm{PI}(\mathcal{S})$, in all its copies, from $\mathrm{PI}(\mathcal{S})$ together with a library with intermediate concepts destroys UP_1-usability, and that the chain-encoded libraries of the simplex family for $r = 3, 4$ are UP-usable but not UP_1-usable.

Proposition E.18 (Libraries with intermediate concepts)**.** *Let* $\mathrm{Circ}_{\mathrm{aux}}(\mathcal{S})$ *write each circuit u of an affine law by the cheaper of its full clause set and its chain encoding with its own auxiliary variables: directly if $|u| \le 4$, and by the chain if $|u| \ge 5$. It is* UP*-usable, of size $\sum_{\text{circuits } u} \mathrm{sz}(|u|)$, where $\mathrm{sz}(w) := 2^{w-1}$ for $w \le 4$ and $\mathrm{sz}(w) := 4w - 6$ for $w \ge 5$. The same holds for the library over all $2^r - 1$ nonzero derived equations, of size at most $(2^r - 1)(4m - 6)$ for $m \ge 4$.*

Proof. A question (α, ℓ), ℓ on x_j, has a circuit u with $j \in \operatorname{supp}(u) \subseteq \operatorname{dom} \alpha \cup \{j\}$ (Lemma E.7). The encoding of u is propagation complete (Proposition E.16, or Lemma E.5(a) for the full clause set of the one-equation law), and α fixes all of $\operatorname{supp}(u)$ but j, so unit propagation on that encoding alone derives ℓ; unit propagation is monotone in the clause set. Each auxiliary variable is a partial sum of base variables, so the union is satisfied exactly by the extensions of the points satisfying every circuit, and its projection is $\mathcal{S}$, since the circuits span $\mathrm{Row}(A)$ (proof of Proposition E.8); a consistent α therefore gives no conflict. For the second library, a derived equation of weight w costs at most $\max(8, 4w - 6) \le 4m - 6$ clauses when $m \ge 4$. □

The second form has size $O(m2^r)$, in line with the fixed-parameter tractability in the number of equations shown by Gwynne and Kullmann (2014).

Remark E.19 (The simplex family with intermediate concepts)*.* For the simplex family with $r \ge 4$, $\mathrm{Circ}_{\mathrm{aux}}$ has $m(2m - 4)$ clauses of width at most 3 and $m(m - 3)/2$ auxiliary variables, and is UP-usable: a theory of polynomial size. Writing each circuit by the balanced-tree encoding of Proposition E.24(b) instead of the chain gives the same size and a $\mathrm{UP}_{2\lceil \log((m+1)/2) \rceil - 1}$-usable theory. So with intermediate concepts and enough rounds the gap to GE is polynomial in written length, and $O(\log m)$ bits under time charging.

Proof. Every circuit has weight $w = 2^{r-1} = (m+1)/2 \geq 8$, so each is chain-encoded with $4w - 6 = 2m - 4$ clauses and $w - 2 = (m-3)/2$ auxiliary variables; usability is Proposition E.18. With balanced trees, each question is answered inside the encoding of one circuit, as in the proof of Proposition E.18, within $2\lceil \log w \rceil - 1$ rounds by Proposition E.24(b), and unit propagation with a fixed number of rounds is monotone in the clause set on satisfiable inputs; the auxiliary variables are again partial sums, so the projection is $\mathcal{S}$. The simplex family is computable from m in polynomial time, so $Kt(A, b) = O(\log m)$, and Proposition E.6 with $r = \log(m+1)$ gives $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) = O(\log m)$. □

A direct computation (code in the accompanying repository) of the listing lengths (4) gives ratios of about 65, 137, 256 and 461 between this library and the r given equations for $r = 4, 5, 6, 7$: polynomial growth, not $2^{(m-1)/2}$; with chains the library at $r = 4$ needs 7 rounds, with balanced trees 5.

Proposition E.20 (Private intermediate concepts add nothing to a system)**.** *Let $\Phi = \bigcup_t \Phi_t$, where Φ_t encodes a constraint c_t on base variables X_t with auxiliary variables Y_t private to it ($Y_t \cap Y_s = \emptyset$ for $t \neq s$, Y_t occurs only in Φ_t, and $\exists Y_t\, \Phi_t \equiv c_t$), and each Φ_t is propagation complete on $X_t \cup Y_t$. For a partial assignment α to the base variables, let its* arc-consistency closure $\mathrm{AC}(\alpha)$ *be the least set of base literals that contains α and is closed under adding, for every t, every base literal on X_t entailed by c_t and the current literals on X_t; call it* contradictory *if it contains two opposite literals or its literals on some X_t are inconsistent with c_t. Then unit propagation on $\Phi \cup \alpha$ reaches a conflict if and only if $\mathrm{AC}(\alpha)$ is contradictory, and otherwise the base literals at its fixpoint are exactly $\mathrm{AC}(\alpha)$. In particular the given equations written directly and the given equations each chain-encoded derive the same base literals and conflict on the same antecedents.*

Proof. Invariant. At every point of the run, the literals on Y_t are entailed by Φ_t and the literals on X_t: they are derived only by clauses of Φ_t, and each derivation step preserves this. Hence a literal derived by a clause of Φ_t is entailed by Φ_t and the current literals on X_t; if it is a base literal, it is entailed by c_t and the current literals on X_t, as $\exists Y_t\, \Phi_t \equiv c_t$. By induction on rounds, every base literal derived before a conflict lies in $\mathrm{AC}(\alpha)$. A conflict arises either from a clause of some Φ_t with all literals falsified, so that the current literals on $X_t \cup Y_t$ are inconsistent with Φ_t and, by the invariant, those on X_t are inconsistent with c_t; or from two opposite derivations, both from one Φ_t if the variable is auxiliary (with the same conclusion), or both base literals of $\mathrm{AC}(\alpha)$. Either way $\mathrm{AC}(\alpha)$ is contradictory. *Conversely,* suppose the run reaches a conflict-free fixpoint γ' with base part β. No clause of Φ_t fires at γ' and Φ_t is propagation complete, so the restriction of γ' to $X_t \cup Y_t$ is consistent with Φ_t and contains every literal it entails with Φ_t. So $\beta|_{X_t}$ is consistent with c_t, and if $c_t \wedge \beta|_{X_t} \models \ell$ then Φ_t and the literals of γ' on $X_t \cup Y_t$ entail ℓ, so $\ell \in \gamma'$. Thus $\beta \supseteq \alpha$ is closed and not contradictory; hence $\beta \supseteq \mathrm{AC}(\alpha)$ and $\mathrm{AC}(\alpha)$ is not contradictory, and with the first part $\beta = \mathrm{AC}(\alpha)$. The direct encoding of one equation (all its clauses) is propagation complete on its own variables by Lemma E.5(a), and the chain encoding by Proposition E.16. □

So intermediate concepts that each belong to one law cannot help a weak reasoner combine laws; the useful intermediate concepts are shared across laws, such as derived equations and their partial sums. On sparse random systems a direct computation (code in the accompanying repository) finds that unit propagation on the given equations fully answers only a part of the sampled questions, between about 0.62 and 0.89 of them depending on the system, while the circuit library answers all of them.

Lemma E.21 (Usability is arc consistency)**.** *If Φ is* UP*-usable for $\mathcal{S}$, then for every partial assignment α to the base variables that is inconsistent with $\mathcal{S}$, unit propagation on $\Phi \cup \alpha$ reaches a conflict. So* UP*-usability is* arc consistency *of a representation in the sense of Gwynne and Kullmann (2014): unit propagation detects every forced base literal and every inconsistency.*

Proof. Take $\alpha' \subseteq \alpha$ minimal inconsistent and a literal x_j^v of α'. Then $\alpha'' := \alpha' \setminus \{x_j^v\}$ is consistent and entails x_j^{1-v}, so unit propagation on $\Phi \cup \alpha''$ derives it. Unit propagation is monotone up to conflict: in the run on the larger $\Phi \cup \alpha$, every clause that fired in the smaller run either fires again or meets a conflict. So the larger run derives x_j^{1-v} against $x_j^v \in \alpha$, or conflicts earlier. □

Lemma E.22 (Unit propagation as a monotone circuit)**.** *Let Φ be an explicit theory with $\bar{m}$ variables, written length $\|\Phi\|$ and clause width at most w, and let $\mathcal{I}$ be a set of literals (the inputs). (a) There is a monotone circuit of $O(\bar{m}(\|\Phi\| + \bar{m}))$ fan-in-2 AND and OR gates, with one input per literal of $\mathcal{I}$, that outputs, for every literal L, whether unit propagation on $\Phi \cup \alpha$ derives L, on every input $\alpha \subseteq \mathcal{I}$ for which $\Phi \cup \alpha$ is satisfiable. (b) For every k there is a monotone circuit of unbounded fan-in AND and OR gates, of depth $2k$, with at most $k(\|\Phi\| + 2\bar{m})$ gates and at most $k(w\|\Phi\| + 2\bar{m})$ wires, with the same inputs, that outputs whether UP_k derives L, on the same inputs.*

The direction "small usable theory gives small monotone circuit" of the equivalence of Bessiere et al. (2009) is (a).

Proof. Define $o_0(L)$ as the input bit if $L \in \mathcal{I}$ and the constant 0 otherwise, and $o_{t+1}(L) := o_t(L) \vee \bigvee_{C \ni L} \bigwedge_{L' \in C,\, L' \neq L} o_t(\neg L')$, where the empty conjunction, for a unit clause $C = (L)$, is the constant 1. On a satisfiable $\Phi \cup \alpha$ unit propagation never conflicts, so a clause whose other literals are all falsified fires (its last literal is unassigned or already true), exactly as in the formula; by induction $o_t(L)$ says whether L is known after t rounds. (a) Prefix and suffix products give the $|C|$ conjunctions of a clause with $3|C|$ fan-in-2 gates, so a round costs $O(\|\Phi\| + \bar{m})$ gates; each round before the fixpoint assigns at least one new variable and no variable is assigned twice, so $\bar{m}$ rounds suffice. (b) Per round, one AND gate for each pair (clause, literal of it), of fan-in $|C| - 1 \leq w - 1$, and one OR gate for each literal, of fan-in one plus the number of clauses containing it: at most $\|\Phi\| + 2\bar{m}$ gates and $w\|\Phi\| + 2\bar{m}$ wires per round. AND layers and OR layers alternate (an OR gate feeding an OR gate of the next round is merged into it), so the depth is $2k$. Constants are removed by simplification without increasing size or depth. □

Proof of Theorem 7.5. The law. $|\Lambda_q| = 2\binom{q}{3} + 2q$, so $m = 2\binom{q}{3} + 2q + 1 = \Theta(q^3)$. The vectors of Λ_q span $\mathrm{GF}(2)^{q+1}$ (already $(\mathbf{e}_v, 0)$ and $(\mathbf{e}_v, 1)$, $v \leq q$, do), so $\dim \mathcal{H}_q = q + 1$ and $\mathcal{H}_q$ is an affine law of rank $r = m - q - 1 \geq 1$. The m equations $x_{(u,a)} = u \cdot y + a\, z_0$, $x_\star = z_0$, with $z = (y, z_0)$ as auxiliary variables, have projection $\mathcal{H}_q$, and GE returns exactly the entailed base literals, so they are GE-usable. *The lower bound.* For $S \subseteq \Lambda_q$ the antecedent $\alpha_S := \{\neg x_v : v \in S\}$ is consistent, as $0 \in \mathcal{H}_q$. On $\mathcal{H}_q$, $x_\star$ is determined by x_S iff $\mathbf{e}_\star \in \mathrm{span}(S)$ (if not, some z has $z \cdot v = 0$ for $v \in S$ and $z \cdot \mathbf{e}_\star = 1$, and z and 0 give two points that agree on S and differ at $x_\star$; if so, $x_\star$ is a sum of the x_v, $v \in S$), iff the XOR system $\{u \cdot y = a : (u, a) \in S\}$ over $y \in \mathrm{GF}(2)^q$ is unsatisfiable (a subset of the equations sums to $0 = 1$, that is, a subset of S sums to $\mathbf{e}_\star$). If $x_\star$ is determined, its value is that at the point 0, so usability gives "unit propagation on $\Phi \cup \alpha_S$ derives $\neg x_\star$"; if not, soundness ($\mathrm{proj}\, \Phi = \mathcal{H}_q$) excludes any literal on $x_\star$. So the output for $\neg x_\star$ of the circuit of Lemma E.22(a), with inputs $\mathcal{I} = \{\neg x_v : v \in \Lambda_q\}$, computes the monotone function $\mathrm{Unsat}_q(S) :=$ [the system S is unsatisfiable], with $O((\|\Phi\| + m)^2)$ gates (auxiliary variables that do not occur in Φ are dropped, so $\bar{m} \leq \|\Phi\| + m$). Göös et al. (2019) show that a monotone variant of XOR-SAT has exponential

monotone circuit complexity. In the form we use, their function $3\mathrm{XorSat}_q$ has one input for each ordered triple $(v_1, v_2, v_3) \in [q]^3$ and each constant $a \in \{0,1\}$ ($2q^3$ inputs), accepts iff the selected constraints $y_{v_1} \oplus y_{v_2} \oplus y_{v_3} = a$ are unsatisfiable, and requires monotone circuits of size 2^{q^ε} for some $\varepsilon > 0$. Each input (v_1, v_2, v_3, a) is the constraint $u \cdot y = a$ with $u = \mathbf{e}_{v_1} + \mathbf{e}_{v_2} + \mathbf{e}_{v_3}$, of weight 1 (if two indices coincide) or 3, so $(u, a) \in \Lambda_q$; hence $3\mathrm{XorSat}_q(Z) = \mathrm{Unsat}_q(\mathrm{pr}(Z))$, where $\mathrm{pr}(Z)_v$ is the OR of the inputs that map to v. A monotone circuit for Unsat_q together with at most $2q^3$ OR gates is a monotone circuit for $3\mathrm{XorSat}_q$. (If the inputs are only the triples of distinct indices, $3\mathrm{XorSat}_q$ is Unsat_q with the weight-1 inputs set to 0, and a circuit for Unsat_q again gives one for it.) Hence $O((\|\Phi\| + m)^2) + 2q^3 \geq 2^{q^\varepsilon}$, so $\|\Phi\| \geq 2^{q^\varepsilon/2 - O(1)} - m$ for $q \geq q_0$, and $q^\varepsilon = \Theta(m^{\varepsilon/3})$. □

The weight-1 functionals matter: with weight-3 functionals only, the function computed would be the restriction of $3\mathrm{XorSat}_q$ that sets the inputs with repeated indices to 0, and a monotone lower bound does not pass to a restriction. A direct computation (code in the accompanying repository) confirms for $q = 3, 4$ that "$x_\star$ determined", "system unsatisfiable" and "$\mathbf{e}_\star \in \mathrm{span}(S)$" agree on all $2^{|\Lambda_q|}$ sets S, and that unit propagation on $\mathrm{PI}(\mathcal{H}_3)$ derives $\neg x_\star$ from α_S exactly when the system is unsatisfiable.

Remark E.23 (XOR systems without auxiliary-variable restrictions). Gwynne and Kullmann (2014) show that systems of XOR constraints have no polynomial-size arc-consistent representation in general, even with auxiliary variables, and that the standard translation is propagation complete for one equation and fails for two, the counterpart with auxiliary variables of Corollary E.13(b). By Lemma E.21 their notion is our UP-usability, so their superpolynomial bound applies to usable theories; Theorem 7.5 gives an explicit family with an exponential bound.

Proposition E.24 (Rounds and size for one parity law)**.** *(a) For $m \geq 4$ the chain encoding is UP_{m-1}-usable and not UP_{m-2}-usable. (b) The* balanced-tree encoding*, along a rooted binary tree of height $\lceil \log m \rceil$ whose leaves are the base variables, with $y_v = y_{v'} \oplus y_{v''}$ (four clauses) at each internal node v other than the root and $y_{v'} \oplus y_{v''} = b$ (two clauses) at the root, has $4m - 6$ clauses ($m \geq 2$) and is $\mathrm{UP}_{2\lceil \log m \rceil - 1}$-usable. (c) There is an absolute constant $c_{\mathrm{H}} > 0$ such that for every $k \geq 1$ and every m large enough given k, every UP_k-usable Φ for Par_m^b, with any auxiliary variables, has $3\|\Phi\| + 2m \geq 2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$; likewise, with $w - 1$ for $m - 1$, for every affine law with a circuit of weight w. (d) Hence, for each fixed k, UP_k-usable theories for one parity law have written length superpolynomial in m; $2\lceil \log m \rceil - 1$ rounds suffice for $4m - 6$ clauses.*

Proof. For one parity law every question has $|\mathrm{dom}\,\alpha| = m - 1$, as the only circuit is 1^m. (a) Write $C_1 = (x_1, x_2, y_1)$, $C_i = (y_{i-1}, x_{i+1}, y_i)$ for $2 \leq i \leq m - 2$, and $C_{m-1} = (y_{m-2}, x_m)$ for the constraints. Each constraint's clauses mention only its own variables and fire only when all but one of them are assigned. *Upper bound:* let x_j be the unknown base variable and $C_{j'}$ its constraint ($j' = \max(1, j - 1)$). The constraints to the left of $C_{j'}$ resolve from the left end (C_1 derives y_1 in round 1, C_2 derives y_2 in round 2, and so on), and those to the right from the right end (C_{m-1} derives y_{m-2} in round 1, and so on); $C_{j'}$ then has all its variables but x_j assigned by round $\max(j' - 1, m - 1 - j')$, and derives x_j one round later, by round $m - 1$. *Lower bound:* for the question with x_1 unknown, after round $t \leq m - 2$ exactly the auxiliary variables $y_{m-2}, \ldots, y_{m-1-t}$ are assigned (induction on t: C_1 has the two unknowns x_1, y_1, and each C_i, $2 \leq i \leq m - 2$, the two unknowns y_{i-1}, y_i, until the boundary reaches it, so in each round exactly one constraint has a single unknown); so x_1 is derived in round $m - 1$ and not before. (b) Let $H := \lceil \log m \rceil$ be the height of the tree. Each constraint's clause set is its set of prime implicates, so it derives its third variable in the round after the other two are known (Lemma E.5(a)). Let x_j be the unknown leaf and $v_0, \ldots, v_s = x_j$ the path from the root, $s \leq H$; for $1 \leq i \leq s$ let w_i be the sibling of v_i. Every

node v not on the path has only known leaves below it, and by induction on its height $\mathrm{ht}(v)$ its value is derived by round $\mathrm{ht}(v)$; w_i is at depth i, so $\mathrm{ht}(w_i) \le H - i$. The root constraint gives y_{v_1} by round $\mathrm{ht}(w_1) + 1 \le H$. If y_{v_i} is known by round $H + i - 1$, the constraint at v_i gives $y_{v_{i+1}}$ by round $\max(H + i - 1, H - i - 1) + 1 = H + i$. So x_j is derived by round $H + s - 1 \le 2H - 1$. No conflict arises, as the antecedent is consistent. Counting: there are $m - 1$ internal nodes, $m - 2$ of them non-root with 4 clauses each, and the root with 2. (c) Let Φ be UP_k-usable, and drop auxiliary variables that do not occur, so that $\bar{m} \le m + \|\Phi\|$. Apply Lemma E.22(b) with inputs the literals on $x_2, \ldots, x_m$ (for a general affine law: on $\mathrm{supp}(u) \setminus \{j\}$ for a circuit u of weight w and some $j \in \mathrm{supp}(u)$). Every assignment to these variables is consistent (no nonzero derived equation is supported among them) and determines x_1 (resp. x_j) as a constant plus their parity (Lemma E.7); so by usability and soundness the output gate for the literal x_1 computes, on the inputs that make exactly one literal of each variable true, the parity of $m - 1$ bits or its negation. This is a circuit of depth $2k$ with unbounded fan-in AND and OR gates over the literals of $m - 1$ variables, with at most $k(\|\Phi\| + 2\bar{m}) \le k(3\|\Phi\| + 2m)$ gates. By Håstad's theorem (Håstad, 1986), in its standard form, every such circuit of depth $d \ge 2$ computing the parity of n variables has at least $2^{c_{\mathrm{H}} n^{1/(d-1)}}$ gates, for an absolute constant $c_{\mathrm{H}} > 0$ and n large enough given d. With $n = m - 1$ (resp. $w - 1$) and $d = 2k$ the claim follows. (For $k = 1$, Proposition E.17 gives the exact value 2^{m-1}.) (d) By (c), for each fixed k the written length is $2^{\Omega(m^{1/(2k-1)})}$, superpolynomial in m. By (b), $2\lceil \log m \rceil - 1$ rounds suffice for $4m - 6$ clauses. □

If Håstad's bound holds uniformly in the depth, with an absolute constant for all $n \ge 2$, then comparing a polynomial size $2^{O(\log m)}$ with $2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$ gives $(m-1)^{1/(2k-1)} = O(\log m)$, that is, $k = \Omega(\log m / \log\log m)$ for polynomial size; we do not use that form.

A direct computation over all $m2^{m-1}$ questions, $m = 3, \ldots, 10$ (code in the accompanying repository), confirms that the chain needs exactly $m - 1$ rounds for $m \ge 4$ and that the balanced tree needs $1, 3, 4, 5, 5, 5, 6, 7$ rounds for $m = 3, \ldots, 10$, within the bound $2\lceil \log m \rceil - 1$. Part (c) is a direct application of Håstad's theorem and very probably folklore.

Proof of Corollary 7.6. First column. One parity law has the single circuit 1^m, indecomposable, so every UP-usable theory over the base variables, and a fortiori every UP_k-usable one, contains its 2^{m-1} clauses (Proposition E.11), and these clauses are UP_1-usable (Corollary E.13(c), Proposition E.8). *Row* UP_1. Proposition E.17, with $\mathrm{PI}(\mathcal{S})$ itself UP_1-usable. *Row* UP_k*, k fixed.* Proposition E.24(c), for parity with $w = m$ and for the simplex family with $w = (m+1)/2$. *Row* UP_k*, $k \ge 2\lceil \log m \rceil - 1$, and* UP. Proposition E.24(b) and Remark E.19 (for $r = 3$, $\mathrm{PI}(\mathcal{S})$ has $56 \le m(2m-4)$ clauses); more rounds never hurt. *Last column.* Theorem 7.5, for every number of rounds. *Row* GE. A linear theory with e equations in $m + m_{\mathrm{aux}}$ variables has a solution set of dimension at least $m + m_{\mathrm{aux}} - e$, whose projection has dimension at least $m - e$; so a theory for a law of rank r has at least r equations, and the given equations (one for parity, $r = \log(m+1)$ for the simplex family) or the m equations of Theorem 7.5 are usable. □

E.5 Directionality

A *directed rule* is $P \Rightarrow \ell$, with P a set of literals (the premise) and ℓ a literal on a variable that does not occur in P; a "rule" whose conclusion's variable occurs in its premise corresponds to a tautological clause and is excluded, as tautological clauses are excluded from explicit theories. Its written length is $|P| + 1$. *Forward chaining* FC_k applies, in each of k parallel rounds, every rule whose premise is satisfied by the current assignment; a conclusion whose opposite is assigned, or two opposite conclusions, give a conflict; $\mathrm{FC} := \mathrm{FC}_\infty$. A rule over the base variables is *sound* for

$\mathcal{S}$ if P is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge P \models \ell$. The projection of a rule set, possibly with auxiliary variables, is that of its clauses $\neg P \vee \ell$; a rule set is FC_k-*usable* for $\mathcal{S}$ if its projection is $\mathcal{S}$ and FC_k derives every entailed base literal from every consistent antecedent. A *rule base* is a set of sound rules over the base variables.

Proposition E.25 (Directed rules against clauses)**.** *(a) For a clause C let* $\mathrm{rules}(C) := \{\neg(C \setminus \{L\}) \Rightarrow L : L \in C\}$*, where $\neg(C \setminus \{L\})$ is the set of negations of the other literals. For every theory Φ, every k and every α,* FC_k *with* $\mathrm{rules}(\Phi)$ *and* UP_k *with Φ derive the same literals and conflict together. Conversely a rule $P \Rightarrow \ell$ gives the clause $\neg P \vee \ell$, and unit propagation with the clause derives at least what forward chaining with the rule derives. Hence, for every law and every round budget k, with or without auxiliary variables, the smallest usable rule set has at least as many rules as the smallest usable clausal theory has clauses, and at most w times as many, w the largest clause width of a smallest usable clausal theory; in written length the factor is likewise at most w. (b)* One parity law on $m \geq 2$ variables, no auxiliary variables. *A sound rule concluding a literal on x_j has all other $m-1$ variables in its premise. So there are exactly $m2^{m-1}$ sound rules, each question needs its own rule, and the only usable rule base is the set of all of them: $m2^{m-1}$ rules of length m, against the unique minimum of 2^{m-1} clauses of length m; a factor m, the number of query directions, in count and in written length. (c)* With auxiliary variables, for every law. *Splitting each clause $(l_1 \vee \cdots \vee l_w)$ with $w \geq 4$ into $(l_1 \vee l_2 \vee z_1), (\neg z_1 \vee l_3 \vee z_2), \ldots, (\neg z_{w-3} \vee l_{w-1} \vee l_w)$ with fresh variables $z_1, \ldots, z_{w-3}$ preserves the projection and* UP*-usability. So every law has an* FC*-usable rule set of written length at most* 9 *times the least written length of a* UP*-usable clausal theory with auxiliary variables. For the chain encoding of parity: $4m-6$ clauses against $3(4m-6)-2$ rules. (d)* One round, any vocabulary. *For one parity law on $m \geq 2$ variables, every FC_1-usable set of sound rules, with any auxiliary variables, contains all $m2^{m-1}$ sound rules over the base variables, while every UP_1-usable theory with any auxiliary variables must contain the 2^{m-1} clauses of $\mathrm{PI}(\mathcal{S})$ (Proposition E.17).*

Proof. (a) Induction on rounds: a clause C fires in round t deriving L exactly when the rule $\neg(C \setminus \{L\}) \Rightarrow L$ has its premise satisfied after round $t-1$ and L is not yet assigned; forward chaining signals a conflict when a conclusion's opposite is assigned, as unit propagation does when a clause is falsified. A rule $P \Rightarrow \ell$ whose premise is satisfied leaves its clause with all literals but ℓ falsified. The bounds follow by translating a smallest usable theory in either direction: $\mathrm{rules}(\Phi)$ has at most $w|\Phi|$ rules and written length $\sum_{C \in \Phi} |C|^2 \leq w\|\Phi\|$, and the clauses of a rule set are at most as many as its rules and are usable for the same round budget. (b) If a sound rule $P \Rightarrow x_j^v$ had a premise missing some x_i, $i \neq j$, then by Lemma E.7 (the only circuit is 1^m) x_j would not be determined by the consistent premise P, and the rule would be unsound. So each sound rule is $\beta|_{[m]\setminus\{j\}} \Rightarrow x_j^v$ with v the forced value, one for each j and each of the 2^{m-1} assignments β to the other variables, and the question with that antecedent is answered only by that rule. The minimum of 2^{m-1} clauses is Corollary E.13(c) for $m \geq 3$, and for $m = 2$ the equivalence-law case of Remark E.15 with one class of two variables. (c) The pieces, with the z's quantified out, are equivalent to the clause. If all literals of the clause but l_t are falsified, propagation from the left derives $z_1, \ldots, z_{t-2}$, from the right $\neg z_{w-3}, \ldots, \neg z_{t-1}$, and the piece containing l_t fires (the end cases are the same with one side empty). Every derivation of the original run is reproduced, in more rounds, and soundness is unchanged. A clause of length $w \geq 4$ becomes $w-2$ clauses of length 3, of total length at most $3w$, and each 3-clause becomes 3 rules of length 3; clauses of width at most 3 become rules of total length at most 3 times their own. The chain encoding has $4m-8$ clauses of width 3 and 2 of width 2, so rules gives $3(4m-8)+4 = 3(4m-6)-2$ rules. (d) From an antecedent α on base variables, a rule fires in round 1 only if its premise is satisfied

by α, so its premise consists of base literals of α. A rule answering the question with antecedent $\beta|_{[m]\setminus\{j\}}$ concludes a literal on x_j, so it is a sound rule over the base variables, and by (b) its premise is that whole antecedent. □

A direct computation (code in the accompanying repository) confirms for $m = 2, \ldots, 7$ that the sound rules of one parity law number exactly $m2^{m-1}$, all with premises of size $m-1$, and that for $m \leq 5$ deleting any one of them breaks usability; on random theories, forward chaining with rules(Φ) and unit propagation with Φ agree in every run.

Theorem E.26 (Directionality is real but cheap)**.** *(1) For one parity law on $m \geq 2$ variables, the only* FC*-usable rule base is the set of all $m2^{m-1}$ sound rules, of written length m each, while the least* UP*-usable theory over the base variables has 2^{m-1} clauses of length m: a factor m, the number of query directions. With any auxiliary variables, every* FC_1*-usable set of sound rules contains these $m2^{m-1}$ rules, while every* UP_1*-usable theory must contain 2^{m-1} clauses. (2) For every law and round budget k, with or without auxiliary variables, the least* FC_k*-usable rule set has at least as many rules as the least* UP_k*-usable theory has clauses and at most w times as many, w the largest clause width of a smallest such theory; in written length the factor is likewise at most w. With auxiliary variables and unbounded rounds, every law has an* FC*-usable rule set of written length at most 9 times the least written length of a* UP*-usable theory (at most 3 times for the chain encoding of parity). A program with internal variables answers all m directions of parity with $O(m)$ operations. (3) $K_A^{\mathrm{FC}_k} = K_A^{\mathrm{UP}_1} \pm c$ on every conditional data set.*

Proof. (1) is Proposition E.25(b), (d). (2) is Proposition E.25(a), (c); the chain encoding has width at most 3, so its rules have written length at most 3 times its own; and a program answers the question with x_j unknown by outputting b plus the sum of the given bits. (3) A fixed program maps an FC_k-compatible rule set R to its clauses and then to PI of their projection, which is UP_1-compatible with the same data by Lemma E.5(a), since forward chaining derives only literals entailed by the clauses; conversely it maps a UP_1-compatible Φ to rules(Φ), which is FC_1-compatible by Proposition E.25(a), hence FC_k-compatible for every $k \geq 1$. □

Remark E.27 (Programs, rule bases and time bounds)*.* A program may consist of a generic solver applied to a declarative law, at $O(1)$ extra length, so "programs lose" cannot be the statement; Theorem 7.2 is the precise form for reasoners: under program length the weakest reasoner together with a compiler is as good as any. The separations above concern what a *fixed* weak reasoner needs written. The translation of Proposition E.25(a) is round for round, so the directionality factor is the same under every round budget. A per-question time bound below that of elimination ($O(r^2 m)$ bit operations for GE on r equations) does not separate directed rules from clauses either: PI($\mathcal{S}$) and its rule translation both answer every question of the simplex family in one round. What a time bound changes is the comparison between reasoners: a reasoner that may not run elimination must be handed a lemma library, which over the base variables is exponentially larger (Appendix E.3); if its time is charged at least linearly in the size of the theory it reads, it cannot even read that library.

Question E.28. In a cell-probe-type model of a reasoner, is there a theory of size polynomial in m that answers every question of the simplex family in time below that of elimination?

E.6 Where the know-how lives

Proposition E.29 (Three homes of know-how)**.** *For an affine law $\mathcal{S} = \{x : Ax = b\}$, the know-how to answer every question of $\mathcal{Q}(\mathcal{S})$ can be placed in any of the following. (a)* In the reasoner:

the r equations with GE*, of listing length $O(rm \log m)$ bits. (b)* In written lemmas: *for* UP *over explicit theories, exactly $\Phi_{\mathrm{ind}}(\mathcal{S})$ over the base variables when $d_* \geq 3$ (Theorem 7.4; exponential for the simplex family), and a library such as $\mathrm{Circ}_{\mathrm{aux}}$ with intermediate concepts (Proposition E.18; polynomial for the simplex family by Remark E.19, exponential for some systems whatever the intermediate concepts by Theorem 7.5, and for one parity law at any fixed depth by Proposition E.24(c)). (c)* In a computation that the cost model does not charge: *either a membership test with free search for derivations: for a suitable polynomial q, the axiom set $\mathrm{PI}(\mathcal{S})$ has a decider of length $K^q(A,b)+c$ that runs in time polynomial in m and in the length of its input, and for every $(\alpha,\ell) \in \mathcal{Q}(\mathcal{S})$ the sentence $\bigwedge \alpha \to \ell$ follows by propositional reasoning from a single member of $\mathrm{PI}(\mathcal{S})$; or a generator of the explicit theory under program length (Theorem 7.2: the compiler $\Phi' \mapsto \mathrm{PI}(\mathrm{proj}\,\Phi')$ costs $O(1)$ bits). Here $K^q(A,b) \leq rm + r + O(\log m)$, and $K^q(A,b) = O(\log m)$ for one parity law and for the simplex family.*

Proof. (a) and (b) collect the results cited. (c) A clause $C_{Z,\beta}$ is in $\mathrm{PI}(\mathcal{S})$ iff the derived equations supported in Z are exactly 0 and the all-ones vector 1_Z on Z, and β violates 1_Z (Proposition E.8; a circuit is the unique nonzero derived equation supported in its support). The derived equations supported in Z are the yA with y in the kernel of the columns of A outside Z, computable by Gaussian elimination in time polynomial in m once (A,b) is available, and a program of length $K^q(A,b)$ produces (A,b) within $q(|(A,b)|)$ steps. A question (α,ℓ) has a prime implicate $C = \neg\alpha' \vee \ell$ with $\alpha' \subseteq \alpha$ (Lemma E.5(a)), and $\bigwedge \alpha \to \ell$ follows from the single instance C by propositional reasoning. Printing the literal matrix and vector, with r and m coded self-delimitingly, takes polynomial time and gives $K^q(A,b) \leq rm + r + O(\log m)$; one parity law and the simplex family are computed from m in polynomial time. □

A direct computation (code in the accompanying repository) confirms that the decider agrees with prime-implicate membership computed by brute force on 9,852 clauses over 30 random laws.

Definition E.30 (Charged deduction with conditional data)**.** We use the readings of charged deduction of Theorem B.31 (derivation size; every instance short, (a1), or every nonlogical instance short, (a2); total material, (b), or nonlogical material, (b2)) at polynomial budgets with the degrees charged, as in Definition E.18 of our earlier report (Hänni and Claude (Anthropic), 2026a). A hypothesis is a triple (p,i,j); $A_{p,i}$ is the set of sentences χ that p accepts within $(|\chi|+2)^i$ steps; the budget for a datum φ is $(|\varphi|+2)^j$; and the weight is $2^{-|\mathrm{El}(p)|-2\lceil\log(i+1)\rceil-2\lceil\log(j+1)\rceil-2}$, with $\mathrm{El}(p)$ a self-delimiting code of p of length $|p| + O(\log|p|)$. The *conditional* version: (p,i,j) is compatible with a conditional data set D if for each $(\alpha,\ell) \in D$ the set $B \cup A_{p,i} \cup \{\alpha\}$ is consistent (this is not charged) and the sentence $\bigwedge \alpha \to \ell$ has a derivation in the calculus K of that report from $B \cup A_{p,i}$ within the budget under the reading. In our setting $\mathcal{L}_m$ is finite, $B = \emptyset$ is decidable in polynomial time, and Assumption B.30 holds vacuously, as the axioms are sentences of $\mathcal{L}_m$, which contain no parameters.

The readings are defined for two-valued data; question data reach them only through the two-valued encoding, which is degenerate (Proposition E.3), or through this definition.

Theorem E.31 (Charged deduction does not charge for the lemma library)**.** *Let $\mathcal{S}$ be an affine law and take any reading of charged deduction in Definition E.30, at polynomial budgets with the degrees charged. (i) If each derivation of Proposition E.29(c) has size at most $(|\bigwedge \alpha \to \ell| + 2)^{j_0}$, the decider of $\mathrm{PI}(\mathcal{S})$ with derivation degree j_0 is a hypothesis compatible with $\mathcal{Q}(\mathcal{S})$ of weight at least $2^{-K^q(A,b)-O(\log m)-2\lceil\log(j_0+1)\rceil}$. (ii) Every hypothesis compatible with $\mathcal{Q}(\mathcal{S})$ has weight at most $2^{-K(\mathcal{S})+c}$.*

Proof. (i) By Proposition E.29(c) the decider runs in time $\mathrm{poly}(m) \cdot \mathrm{poly}(|\chi|) \le (|\chi|+2)^i$ with $i = O(\log m)$, as $|\chi| + 2 \ge 3$, so the degree charge is $O(\log \log m)$; it has length $K^q(A,b) + c$, so $|\mathrm{El}(p)| = K^q(A,b) + O(\log m)$. $\mathrm{PI}(\mathcal{S}) \cup \alpha$ is consistent for every antecedent of $\mathcal{Q}(\mathcal{S})$, and by hypothesis each datum has a derivation from a single member of $\mathrm{PI}(\mathcal{S})$ of size at most $(|\bigwedge \alpha \to \ell| + 2)^{j_0}$, which bounds its instances and its material as well, so it is within the budget under every reading. (ii) Let (p,i,j) be compatible with $\mathcal{Q}(\mathcal{S})$ and $\Gamma := A_{p,i}$. *Claim:* $x \notin \mathcal{S}$ iff there are $I \subsetneq [m]$ and $j' \notin I$ such that $\bigwedge x|_I \to \ell$ has a derivation from Γ within the budget, where ℓ is the literal on $x_{j'}$ false at x. If $x \notin \mathcal{S}$, the proof of Lemma E.2 gives a question $(x|_I, \ell) \in \mathcal{Q}(\mathcal{S})$ with ℓ false at x, and compatibility gives the derivation. If $x \in \mathcal{S}$, then x has a determined coordinate j'' (Lemma E.2), and the datum $(x|_{[m]\setminus\{j''\}}, \ell'')$, with ℓ'' the literal on $x_{j''}$ true at x, makes $\Gamma \cup \{\hat{x}\}$ consistent; a derivation of $\bigwedge x|_I \to \ell$ with ℓ false at x would refute $\Gamma \cup \{\hat{x}\}$. The condition is decidable from (p,i,j): there are finitely many derivations within the budget, and membership in Γ is decided by running p with its clock. So $\mathcal{S}$ is computable from (p,i,j), and $K(\mathcal{S}) \le |\mathrm{El}(p)| + 2\lceil \log(i+1) \rceil + 2\lceil \log(j+1) \rceil + c$, which is the claim. □

Remark E.32 (Derivation sizes, and the two-valued encoding under charged deduction). We do not count the nodes of the derivations below. (1) In the calculus K (connectives $\neg$ and $\to$, with $\vee$ and $\wedge$ defined), the tautology $(\neg\alpha' \vee \ell) \to (\bigwedge \alpha \to \ell)$ has a derivation of size polynomial in $|\alpha| + |\ell| = O(m)$, assembled from the standard derivations of the deduction theorem, $\wedge$-elimination and the double-negation laws, used $O(|\alpha|)$ times; one modus ponens with the axiom $C = \neg\alpha' \vee \ell$ finishes. So the hypothesis of Theorem E.31(i) holds with $j_0 = O(1)$, the decider has weight at least $2^{-K^q(A,b) - O(\log m)}$, and by Theorem E.31(ii) these readings give the full lemma library, whose explicit form has $m2^{(m-1)/2}$ clauses for the simplex family, a cost within $K^q(A,b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible. (2) Under the two-valued encoding these readings are degenerate as SAI is: the point theory T_{x^0} of Proposition E.3 has a polynomial-time decider and is consistent, and the case distinction in the proof of Proposition E.3 gives each $\bigwedge \alpha \to \ell$ from one literal of T_{x^0} by propositional steps of polynomial size.

Charged deduction thus measures the length of a membership test, not the length of what a reader must be handed; the membership test for the full library runs Gaussian elimination, so the know-how lives in the decider. On literal sequences, under Assumption B.30, at polynomial budgets with the degrees charged, charged-deduction SAI corresponds to certificate SOI (Theorem B.31 and Corollary B.32), and the nondeterminism is the free search for derivations.

E.7 Learning know-how pattern by pattern

Let $m \ge 3$ and $\mathrm{Par}_m := \mathrm{Par}_m^0$ (the constant 1 is symmetric). Every question of $\mathcal{Q}(\mathrm{Par}_m)$ has an antecedent on $m-1$ variables; write it as (j,ζ) with $\zeta \in \mathrm{Par}_m$, its *positive point*, antecedent $\zeta|_{[m]\setminus\{j\}}$ and answer $x_j^{\zeta_j}$. Its *pattern* is the violating point $\mathrm{pat}(j,\zeta) := \zeta \oplus \mathbf{e}_j \in \mathcal{P}_m := \{0,1\}^m \setminus \mathrm{Par}_m$. So $|\mathcal{Q}(\mathrm{Par}_m)| = m2^{m-1}$ and $|\mathcal{P}_m| = 2^{m-1}$, and every pattern and every positive point carries exactly m questions. For a conditional data set $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, $\mathrm{pat}(D)$ is the set of *seen* patterns. For $\beta \in \{0,1\}^m$, C_β is the full-width clause falsified exactly by β; in the listing code it costs $\mathrm{len}_m := \ell_{\mathbb{N}}(m) + m(\lceil \log m \rceil + 1)$ bits. A theory *fails* on a question if it errs, abstains, or is inconsistent with the antecedent.

Lemma E.33 (Answers of a theory over the base variables)**.** *For an explicit theory Φ over the base variables and a question (j,ζ) the following are equivalent: (i) Φ is* UP*-compatible with (j,ζ); (ii) Φ answers it correctly (unit propagation derives exactly $x_j^{\zeta_j}$, without conflict); (iii) ζ satisfies Φ, and some clause of Φ with a literal on x_j is falsified by* $\mathrm{pat}(j,\zeta)$.

Proof. Only x_j is unassigned, so unit propagation either finds a clause falsified by the antecedent (a conflict), or derives values of x_j from clauses whose other literals are falsified by it. A clause with a literal on x_j and all other literals falsified by the antecedent derives $x_j^{\zeta_j}$ iff it is falsified by $\mathrm{pat}(j,\zeta)$, and derives the opposite iff it is falsified by ζ; a clause without x_j that is falsified by the antecedent is falsified by ζ. So (iii) gives a derivation of the answer and no conflict, that is (ii), and ζ witnesses satisfiability, so (i) follows. If (i), the satisfying completion of the antecedent is ζ, since the clause that derives $x_j^{\zeta_j}$ excludes $\mathrm{pat}(j,\zeta)$; this gives (iii). (ii)⇒(iii): a correct derivation needs a clause falsified by $\mathrm{pat}(j,\zeta)$, and a clause falsified by ζ would cause a conflict or a wrong derivation. □

Theorem E.34 (A weak reasoner learns its know-how pattern by pattern)**.** *(a)* Full correctness. *An explicit theory over the base variables answers every question of* $\mathcal{Q}(\mathrm{Par}_m)$ *correctly iff it is* $\mathrm{PI}(\mathrm{Par}_m) = \{C_\beta : \beta \in \mathcal{P}_m\}$. *(b)* Minimal theories, any order. *Let* $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$ *and let* Φ *be an inclusion-minimal explicit theory over the base variables that is* UP*-compatible with* D *(for instance any* ℓ_{lst}*-shortest one, Lemma E.1). Then for every unseen pattern* $\beta \notin \mathrm{pat}(D)$*,* Φ *fails on some question of pattern* β*. Hence* Φ *fails on at least* $|\mathcal{P}_m \setminus \mathrm{pat}(D)|$ *questions, and a shortest compatible theory is fully correct only after all* 2^{m-1} *patterns have been seen. (c)* Random order. *If the questions are drawn independently and uniformly from* $\mathcal{Q}(\mathrm{Par}_m)$*, the patterns are independent and uniform on* $\mathcal{P}_m$*; after* n *questions the expected number of unseen patterns is* $2^{m-1}(1-2^{1-m})^n$*, and the expected time until all have been seen is* $2^{m-1}\sum_{i=1}^{2^{m-1}} 1/i = 2^{m-1}\big((m-1)\ln 2 + 0.5772\ldots\big) + \frac{1}{2} + O(2^{-m})$*, where* $0.5772\ldots$ *is Euler's constant. (d)* Bayes, block-complete data (semimeasure convention). *If* D *contains all* m *questions of each seen pattern, every* UP*-compatible explicit theory over the base variables contains* $\{C_\beta : \beta \in \mathrm{pat}(D)\}$*, and* $-\log W_{\mathrm{lst}}^{\mathrm{UP,base}}(D) \geq |\mathrm{pat}(D)|\,\mathrm{len}_m$*. So the cumulative loss is at least* $|\mathrm{pat}(D)|\,\mathrm{len}_m - c$ *bits: about* $m \log m$ *bits per pattern, and* $2^{m-1}\mathrm{len}_m$ *in all. (e)* With the reasoner or with intermediate concepts. *For every* $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$*,* $-\log W_{\mathrm{lst}}^{\mathrm{GE}}(D) \leq \ell_{\mathrm{lst}}(\text{the one equation}) = m\lceil \log m \rceil + O(\log m)$*, and* $-\log W_{\mathrm{lst}}^{\mathrm{UP}}(D) \leq \ell_{\mathrm{lst}}(\text{the chain encoding}) = O(m \log m)$ *(Proposition E.16). So these inducers pay* $O(m \log m)$ *bits in total on every question sequence. (f)* Other measures. *Under program length there is no gap:* $K_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) = K(m) \pm c$*. Under time charging,* $Kt_A^{\mathrm{UP,base}}(\mathcal{Q}(\mathrm{Par}_m)) \geq \log(m2^{m-1}) = m - 1 + \log m$*, while* $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathrm{Par}_m))$ *and* $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathrm{Par}_m))$ *are* $O(\log m)$.

Proof. (a) By Lemma E.33(iii) for all questions: every point of Par_m satisfies Φ, so every clause is an implicate, of width m (Proposition E.9, $d_* = m$), that is, some C_β with $\beta \in \mathcal{P}_m$; and every pattern β needs a clause falsified by it, which among full-width clauses is C_β itself. (b) C_β is falsified only by β, so by Lemma E.33 it serves only questions of pattern β; for unseen β, removing it keeps compatibility with D, so by minimality $C_\beta \notin \Phi$. If for some j no clause of Φ with a literal on x_j is falsified by β, then Φ fails on $(j, \beta \oplus \mathbf{e}_j)$, of pattern β. Otherwise take a clause $C \in \Phi$ falsified by β; $C \neq C_\beta$, so $|C| < m$; take $i \notin \mathrm{supp}(C)$. The question $(i, \beta \oplus \mathbf{e}_i)$ has pattern β, and its positive point $\beta \oplus \mathbf{e}_i$ agrees with β on $\mathrm{supp}(C)$, so it falsifies C, and Φ fails on it (Lemma E.33). Distinct patterns give distinct questions. (c) Each pattern carries m of the $m2^{m-1}$ questions; the rest is the coupon collector's problem, with $\sum_{i \leq N} 1/i = \ln N + 0.5772\ldots + \frac{1}{2N} + O(N^{-2})$. (d) For a seen pattern β, every $\beta \oplus \mathbf{e}_i$ is a seen positive point, so every clause of a compatible Φ is satisfied by all of them (Lemma E.33). A clause falsified by β of width less than m is falsified by $\beta \oplus \mathbf{e}_i$ for i outside its support; so the only admissible clause falsified by β is C_β, and Lemma E.33 requires one. With $R := \{C_\beta : \beta \in \mathrm{pat}(D)\}$ and $\sigma_D := |\mathrm{pat}(D)|$, summing the listing weights of all theories over the base variables that contain R (here $m_{\mathrm{aux}} = 0$) gives, as in the proof of Lemma E.1, at

most $\sum_s 2^{-\ell_{\mathbb{N}}(s)} 2^{-\sigma_D \mathrm{len}_m} \sum_{|\Phi'|=s-\sigma_D} \prod_{C\in\Phi'} 2^{-\ell(C)} \le 2^{-\sigma_D \mathrm{len}_m} \sum_s 2^{-\ell_{\mathbb{N}}(s)}/(s-\sigma_D)! \le 2^{-\sigma_D \mathrm{len}_m}$. The cumulative loss in the semimeasure convention is $-\log W_{\mathrm{lst}}(D) + \log W_{\mathrm{lst}}(\emptyset)$, and $W_{\mathrm{lst}}(\emptyset) \ge 2^{-c}$. (e) The single equation (respectively the chain encoding) is compatible with every $D \subseteq \mathcal{Q}(\mathrm{Par}_m)$, so $W_{\mathrm{lst}} \ge 2^{-\ell_{\mathrm{lst}}}$ of it; the listing lengths are direct from (4). (f) Theorem 7.3(b) and Proposition E.6: the unique usable theory over the base variables has 2^{m-1} clauses of width m, and the chain encoding and the equation are printed from m in polynomial time. □

A simulation (code in the accompanying repository) with uniformly random question sequences for $m = 4, 5, 6$, which computes an exactly ℓ_{lst}-shortest compatible theory at checkpoints, found in each of 330 checkpoints at least as many failures as unseen patterns, as (b) requires, and typically about m times as many; the exact expected times until all patterns are seen are 21.74, 54.09 and 129.87.

Question E.35. Is there a learning theorem of this kind for systems of laws, and for a fixed number of rounds with intermediate concepts? Is there a Bayes lower bound like (d) for arbitrary question sequences that are not block-complete?

E.8 Depth against size for the implication chain

Let $\mathrm{Ch}_m = \{x : x_1 \le \cdots \le x_m\}$, $m \ge 2$. A *k-hop transitive-closure spanner* of the directed path on $[m]$ is a set E of pairs (i, j), $i < j$, such that for all $i < j$ there is a path from i to j in E with at most k edges; $\mathrm{sp}_k(m)$ is the least size of one.

Theorem E.36 (Depth against size: the reduction to spanners)**.** *(i) The prime implicates of Ch_m are exactly the clauses $\neg x_i \vee x_j$, $i < j$. (ii) A 2-CNF of prime implicates, with edge set E, is UP_k-usable iff E is a k-hop transitive-closure spanner of the path. (iii) If an explicit theory Φ over the base variables, with clauses of any width, is UP_k-usable, then for every choice of a prime implicate contained in each clause, the chosen edges form a k-hop transitive-closure spanner of the path. (iv) Hence the least number of clauses of a UP_k-usable theory over the base variables is exactly $\mathrm{sp}_k(m)$, for every k.*

Proof. (i) A partial assignment is consistent with Ch_m iff it sets no $x_i = 1$, $x_j = 0$ with $i < j$; so a clause is an implicate iff it contains some $\neg x_i \vee x_j$ with $i < j$, and these 2-clauses are the prime implicates. (ii) For an edge set E, UP_k from a consistent α derives x_j iff some x_i true in α has a path to j of at most k edges, and $\neg x_i$ iff some x_j false in α is reached from i within k edges (the same clauses read contrapositively); and an entailed literal is entailed by a single literal of α. So usability is the spanner property (take $\alpha = \{x_i\}$ for necessity). (iii) Lemma E.10 holds for every law: the replaced theory derives, round for round, at least what Φ derives, so it is UP_k-usable, and it is a 2-CNF of prime implicates, so (ii) applies (two clauses may give the same edge). (iv) By (iii) a usable Φ has $|\Phi| \ge |E| \ge \mathrm{sp}_k(m)$, and a minimum spanner written as a 2-CNF is usable by (ii). □

The converse of (iii) fails for wide clauses. For $m = 3$, $k = 1$, the theory $\{\neg x_1 \vee x_2,\ \neg x_2 \vee x_3,\ \neg x_1 \vee \neg x_2 \vee x_3\}$ consists of implicates, and choosing $\neg x_1 \vee x_3$ inside the wide clause gives the 1-hop spanner $\{12, 23, 13\}$, but from $x_1 = 1$ one round does not derive x_3. The failure persists when the choice is forced: in $\{\neg x_1 \vee x_2,\ \neg x_1 \vee x_3,\ \neg x_2 \vee x_1 \vee x_3\}$ the wide clause contains only $\neg x_2 \vee x_3$, the edges $\{12, 13, 23\}$ form a 1-hop spanner, but from $x_3 = 0$ one round does not derive $x_2 = 0$. A wide clause propagates in fewer situations than the edge it contains.

The minimum size of a k-hop transitive-closure spanner of the directed path is $\Theta(m\,\mathrm{ia}_k(m))$, with ia_k the k-th row of the inverse Ackermann function: $\mathrm{ia}_2(m) = \log m$, $\mathrm{ia}_3(m) = \log\log m$,

$\mathrm{ia}_4(m) = \log^* m$. This is the bound of Alon and Schieber (1987) for answering product queries in at most k steps, in the language of transitive-closure spanners (Bhattacharyya et al., 2012; Berman et al., 2014); Yao (1982) gave the first inverse-Ackermann trade-off for range queries. A direct computation (integer programming, confirmed by exhaustive search for $m \le 7$; code in the accompanying repository) gives the exact values:

m	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\mathrm{sp}_1(m) = \binom{m}{2}$	1	3	6	10	15	21	28	36	45	55	66	78	91	105	120	136	153
$\mathrm{sp}_2(m)$	1	2	4	6	8	10	13	16	19	22	25	28	31	34	38	42	46
$\mathrm{sp}_3(m)$	1	2	3	5	6	8	10	12	14	16	18	20					
$\mathrm{sp}_4(m)$	1	2	3	4	6	7	9	10	12	14	16						

At these sizes $\mathrm{sp}_3(m) = 2m - 6$ for $6 \le m \le 13$; the $\log\log$ growth is not yet visible.

Proposition E.37 (The midpoint construction)**.** *The midpoint recursion, $S(1) = 0$ and $S(n) = n-1+S(\lceil \frac{n-1}{2} \rceil)+S(\lfloor \frac{n-1}{2} \rfloor)$, gives 2-hop spanners of the path, so $\mathrm{sp}_2(m) \le S(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor = (m+1)L - 2^{L+1} + 2$, where $L := \lfloor \log m \rfloor$.*

Proof. Join a hub vertex h to all other vertices ($i \to h$ for $i < h$, $h \to j$ for $j > h$), with $\lceil \frac{n-1}{2} \rceil$ vertices on one side and $\lfloor \frac{n-1}{2} \rfloor$ on the other, and recurse on both sides: pairs on opposite sides of h, or containing h, are joined within 2 hops, and pairs on one side by the recursion. $\Sigma(m) := \sum_{i \le m} \lfloor \log i \rfloor$ satisfies the same recursion: splitting $i = 2i'$ and $i = 2i' + 1$ ($i \ge 2$), $\lfloor \log i \rfloor = 1 + \lfloor \log i' \rfloor$, so $\Sigma(m) = (m-1) + \Sigma(\lfloor m/2 \rfloor) + \Sigma(\lfloor (m-1)/2 \rfloor)$, and $\lfloor m/2 \rfloor = \lceil (m-1)/2 \rceil$. The closed form is standard. □

The construction attains $\mathrm{sp}_2(m)$ for every $m \le 18$ in the table above.

Conjecture E.38. $\mathrm{sp}_2(m) = \sum_{i=1}^{m} \lfloor \log i \rfloor$ *for every $m \ge 2$.*

We have neither a proof nor a counterexample; a computation confirms it for $m \le 18$. The best lower bound we prove is that of Theorem E.39, about $\frac{1}{2} m \log m$, which holds even with intermediate concepts.

Theorem E.39 (Intermediate concepts and wide clauses gain at most a factor about 2 at depth 2)**.** *Let $m \ge 4$. Every UP_2-usable explicit theory Φ for Ch_m, with any auxiliary variables and clauses of any width, has*

$$|\Phi| \;\ge\; \tfrac{1}{2}\Big((m-2)\log \tfrac{m-2}{2} - m\Big),$$

while $\mathrm{sp}_2(m) \le m \log m$. So at depth 2 the least usable theory has $\Theta(m \log m)$ clauses with or without intermediate concepts, and they save at most a factor of about 2.

Proof. Upper bound. In the midpoint recursion $S(n) = n - 1 + S(a) + S(b)$ with $a + b = n - 1$ and $a, b \le n/2$, so by induction $S(n) \le n - 1 + (n-1)\log(n/2) \le n \log n$. *Simplification.* Let U_0 be the set of literals derived by unit propagation from Φ alone; they are auxiliary, as Ch_m fixes no base variable. Simplifying Φ by U_0 (deleting the clauses satisfied by U_0 and the literals falsified by it) gives Φ' with $|\Phi'| \le |\Phi|$ and the same projection, which does not mention the variables of U_0 and has no unit clauses, and in which every UP_2 derivation of Φ outside U_0 is made no later (induction on rounds: a clause of Φ that fires in a run from a base antecedent contains no literal of U_0, since such a literal would be falsified, that is, its negation derived, against soundness; so its simplification is in Φ' and fires). So we may assume that Φ has no unit clauses. *Single sources.*

For a base index v let $\mathrm{Nb}(v) := \{x_v\} \cup \{L : (\neg x_v \vee L) \in \Phi\}$, the literals known after one round from $\alpha = \{x_v\}$. For a clause C containing x_j let $P(C, j) := \{\neg L' : L' \in C \setminus \{x_j\}\}$, its premises for deriving x_j. From $\{x_u\}$, UP_2 derives x_j iff some $C \ni x_j$ has $P(C, j) \subseteq \mathrm{Nb}(u)$ (round 1 is the case $C = \neg x_u \vee x_j$); say that C *serves* u for j. By soundness, if C serves u for j then $u \leq j$. A clause serving any source for j contains no other positive base literal $x_{w'}$, since its premise $\neg x_{w'}$ would follow from a single x_u, which is unsound. Let $\mathrm{cnt}(j)$ be the number of clauses containing x_j that serve some $u < j$; then $\sum_j \mathrm{cnt}(j) \leq |\Phi|$, and $\sum_v |\mathrm{Nb}(v)| \leq m + |\Phi|$, since a 2-clause $\neg x_v \vee L$ is counted only for v (a 2-clause with two negative base literals is not an implicate). *Random experiment.* Give every literal an independent fair coin, *left* or *right*. Call v *source-alive* if all of $\mathrm{Nb}(v)$ is right, and j *target-alive* if every clause containing x_j that serves some $u < j$ has a left premise. No $u < j$ has u source-alive and j target-alive: by usability some C serves u for j, and all its premises lie in $\mathrm{Nb}(u)$ and are right. *Pairing.* For $a = 1, \ldots, M' := \lfloor m/2 \rfloor$ let $\mathcal{B}_a$ be the event that $2a$ is source-alive and $2a - 1$ target-alive. For $a < a'$ we have $2a < 2a' - 1$, so at most one $\mathcal{B}_a$ occurs and $\sum_a \Pr[\mathcal{B}_a] \leq 1$. Given that $\mathrm{Nb}(2a)$ is all right, a clause serving some $u < 2a - 1$ for $2a - 1$ cannot have all its premises in $\mathrm{Nb}(2a)$ (it would serve $2a$ for $2a - 1$, which is unsound), so it is blocked by a left coin outside $\mathrm{Nb}(2a)$ with probability at least $\frac{1}{2}$, independently of the coins of $\mathrm{Nb}(2a)$; these blocking events are increasing in the set of left coins, so by Harris's inequality they occur together with probability at least $2^{-\mathrm{cnt}(2a-1)}$. Hence $\Pr[\mathcal{B}_a] \geq 2^{-|\mathrm{Nb}(2a)| - \mathrm{cnt}(2a-1)}$, and by the convexity of $x \mapsto 2^{-x}$, $\sum_a (|\mathrm{Nb}(2a)| + \mathrm{cnt}(2a - 1)) \geq M' \log M'$. The shifted pairing ($2a + 1$ source-alive and $2a$ target-alive, for $a \leq M'' := \lfloor (m - 1)/2 \rfloor$; $a < a'$ gives $2a + 1 < 2a'$) gives $\sum_a (|\mathrm{Nb}(2a + 1)| + \mathrm{cnt}(2a)) \geq M'' \log M''$. Adding, and using $M', M'' \geq (m - 2)/2 \geq 1$, $m + 2|\Phi| \geq \sum_v |\mathrm{Nb}(v)| + \sum_j \mathrm{cnt}(j) \geq (m - 2) \log \frac{m-2}{2}$. □

The bound is asymptotic: it is positive only from $m = 8$ on, and gives 11.7 at $m = 16$ against $\mathrm{sp}_2(16) = 38$. The argument adapts Hansel's lemma, that covering the complete graph on n vertices by bicliques needs total size at least $n \log n$ (Hansel, 1964; Katona and Szemerédi, 1967), to ordered pairs and to conjunctive premises: the pairing keeps the diagonal out, and Harris's inequality handles clauses with several premises. For 2-CNFs, after the simplification step the implication graph of a UP_2-usable theory (edges $\neg a \to b$ and $\neg b \to a$ for each clause $a \vee b$) is a Steiner 2-hop transitive-closure spanner of the path, with the positive base literals as its vertices and at most $2|\Phi|$ edges, so lower bounds for Steiner 2-transitive-closure spanners of directed hypergrids (Berman et al., 2014), of which the path is the one-dimensional case, bear on that case; what Theorem E.39 adds is the case of wide clauses and an explicit constant.

Proposition E.40 (Every fixed depth: superlinear with intermediate concepts)**.** *Suppose that for every fixed depth d, every monotone circuit of depth d with unbounded fan-in AND and OR gates that computes all the prefix ORs x_1, $x_1 \vee x_2$, …, $x_1 \vee \cdots \vee x_m$ has $m \cdot \omega(1)$ wires. Then for every fixed k and every bound w on clause width, every UP_k-usable theory Φ for Ch_m with clauses of width at most w, with any auxiliary variables, has $\|\Phi\| \geq m \cdot \omega(1)$.*

Lower bounds of this form for constant-depth prefix circuits are due to Chandra, Fortune and Lipton, in a companion paper to Chandra et al. (1985), which gives the matching constant-depth constructions; we state the property as a hypothesis. Size must be counted in wires: counting gates, prefix OR has depth-1 circuits with m gates.

Proof. On the antecedents $\alpha_{A'} = \{x_i : i \in A'\}$, $A' \subseteq [m]$, all consistent, soundness and usability give: UP_k derives x_j iff $A' \cap [1, j] \neq \emptyset$. So the circuit of Lemma E.22(b), with inputs the literals x_i, is a monotone circuit of depth $2k$ with at most $k(w\|\Phi\| + 2\bar{m})$ wires that computes all prefix ORs, and $\bar{m} \leq m + \|\Phi\|$. Hence $(w + 2)k\|\Phi\| + 2km \geq m \cdot \omega(1)$. □

Question E.41. (a) What is the least UP-usable theory over the base variables when $d_* = 2$, beyond equivalence laws? (b) What is the constant at depth 2 with intermediate concepts, between $\frac{1}{2} m \log m$ (Theorem E.39) and $\sum_{i \le m} \lfloor \log i \rfloor \approx m \log m$ (Conjecture E.38)? What is the effect of intermediate concepts at depth $k \ge 3$ beyond Proposition E.40? (c) How many rounds does one parity law need for usable theories of size polynomial in m? Every fixed number is too few, and $2\lceil \log m \rceil - 1$ suffice (Proposition E.24(d)); if Håstad's bound holds uniformly in the depth, $\Omega(\log m / \log \log m)$ are needed.

E.9 Lemmas in first-order logic

In first-order logic the cost of a lemma-free theory for an analytic reasoner can be non-elementary: there are sentences with proofs of size linear in k in Gentzen's sequent calculus LK with cuts whose cut-free proofs have size not bounded by any Kalmár-elementary function of k (Orevkov, 1982); see also the non-elementary lower bounds of Statman (1979). Cut formulas are lemmas, and cut-free proofs are lemma-free reasoning.

Remark E.42 (Valid lemma sentences restore short cut-free proofs). Let $\mathcal{D}$ be an LK proof of $T_0 \vdash \varphi$, with $T_0 \subseteq T$ finite and cuts numbered $1, \dots, n_{\mathrm{cut}}$, the i-th with left premise $\Gamma_i \vdash \Delta_i, A_i$. Let $L_i := \forall \vec{v} \, (\bigwedge \Gamma_i \to \bigvee \Delta_i \vee A_i)$, the universal closure over the free variables. Then (a) each L_i is valid, since the sequent $\Gamma_i \vdash \Delta_i, A_i$ is derivable and LK is sound, so $T \cup \{L_1, \dots, L_{n_{\mathrm{cut}}}\}$ is deductively equivalent to T; and (b) $L_1, \dots, L_{n_{\mathrm{cut}}}, T_0 \vdash \varphi$ has a cut-free proof of size polynomial in the size of $\mathcal{D}$.

Proof sketch. We do not write out the bookkeeping of contexts in the chosen sequent calculus. Replace the i-th cut, with premises $\Gamma \vdash \Delta, A$ and $A, \Gamma' \vdash \Delta'$, as follows. From the right premise and identity sequents $\delta \vdash \delta$ ($\delta \in \Delta$), $\vee$-left gives $\bigvee \Delta \vee A, \Gamma' \vdash \Delta, \Delta'$; from identity sequents $\gamma \vdash \gamma$, $\wedge$-right gives $\Gamma \vdash \bigwedge \Gamma$; $\to$-left gives $\bigwedge \Gamma \to \bigvee \Delta \vee A, \Gamma, \Gamma' \vdash \Delta, \Delta'$; and $\forall$-left steps, instantiating the bound variables by the free variables of the cut's premises, give $L_i, \Gamma, \Gamma' \vdash \Delta, \Delta'$. The subproof of the left premise is dropped. Carry L_i down to the end sequent, by weakening where branches meet and contraction for duplicates; L_i is closed, so no eigenvariable condition below the former cut is affected. Identity sequents for composite formulas have cut-free proofs of size polynomial in the formula. □

The added sentences are valid: the theory's deductive closure, and so everything the idealised inducer sees, is unchanged, and only its written form changes. If every cut's left premise has the form $T_i \vdash A_i$ with $T_i \subseteq T$, the closures $\forall \vec{v} \, A_i$ of the cut formulas are theorems of T and serve directly; in general they are not, since a cut formula can be arbitrary.

Proposition E.43 (When a lemma pays). *Let* $\mathrm{Obj}(T) := \mathrm{lh}(T) + \sum_{\varphi \in D} \rho_T(\varphi)$, *with* lh *additive over axioms and* $\rho_T(\varphi)$ *the cost of the cheapest derivation of the datum* φ *from* T, *non-increasing when axioms are added. Adding a theorem* $L \notin T$ *of* T *as an axiom lowers* Obj *if and only if* $\sum_{\varphi \in D} (\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi)) > \mathrm{lh}(L)$*: a lemma enters iff the derivation cost it saves on the data exceeds its own length.*

Proof. By additivity, $\mathrm{Obj}(T \cup \{L\}) - \mathrm{Obj}(T) = \mathrm{lh}(L) - \sum_{\varphi \in D} (\rho_T(\varphi) - \rho_{T \cup \{L\}}(\varphi))$. □

This is the general form of the observation in Remark E.9(iv) of the earlier report that a time-bounded likelihood favours adopting theorems with long proofs as axioms.

F Proofs for theoretical terms

Throughout, B is decidable and consistent, "consistent" means consistent with B, and Cn has B built in (Section 2). For a sentence ψ, $\psi^{\wedge 1} := \psi$ and $\psi^{\wedge(m+1)} := \psi^{\wedge m} \wedge \psi$, where $\alpha \wedge \beta$ abbreviates $\neg(\alpha \to \neg\beta)$ in the syntax of K; so $|\psi^{\wedge m}| \geq m|\psi|$. On literal data (Definition 8.2) the literal setting of Definition 2.2 is assumed: R does not occur in B, and some model of B interprets the terms t_w as pairwise distinct elements. Sentence length is the symbol count. The computations quoted below check constructions and counts on finite fragments; the theorems are proved in the text, and the code is in the accompanying repository.

Definition F.1 (The named objects)**.** $\mathrm{Th}_{\mathfrak{T}}$ is the first-order theory of the named structure $(\mathfrak{T}; \mathtt{e}, \mathtt{s}_0, \mathtt{s}_1)$. A theory in a language containing $\mathcal{L}_O$ is *about the named objects* if it has a model whose $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct is $\mathfrak{T}$, and *compatible with the named structure* if it is consistent with $\mathrm{Th}_{\mathfrak{T}}$.

The second notion allows objects other than the named ones, but they obey every first-order law of the named objects (for instance, $\mathtt{s}_0, \mathtt{s}_1$ are injective with disjoint ranges that miss $\mathtt{e}$, and $\mathtt{s}_u x \neq x$ for every nonempty word u). "Named object" is our formal stand-in for "observable thing".

F.1 In principle

Corollary 3.6 showed that new vocabulary changes old-vocabulary weights by a constant factor. Part (i) of Theorem 8.3 needs more: classes of contents, a theoretical background, the penalised versions, and the identity tape by tape.

Definition F.2 (Class-restricted observational laws)**.** For a Borel set Θ of observational contents (subsets of $\mathrm{Sent}_{\mathcal{L}_O}$, product topology), $\mathcal{C}_\Theta := \{T : T \cap \mathrm{Sent}_{\mathcal{L}_O} \in \Theta\}$, and $\mathbb{U}_A(\mathcal{C}_\Theta; D)$, $M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$, $W_A(\mathcal{C}_\Theta; D)$, $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$ are the class-restricted laws and weights (Definition 2.4), with $M_A^{\mathcal{L}_O}$ and $W_A^{\mathcal{L}_O}$ as in Definition 8.1. Observational restrictions of the penalised versions of Theorem B.9 are their sub-sums over hypotheses whose axioms are $\mathcal{L}_O$-sentences on every tape.

Theorem F.3 (Theoretical vocabulary costs $O(1)$ bits, class by class)**.** *There is c, depending on V, the coding, B and $\mathcal{L}_O \subseteq \mathcal{L}$, such that for every finite observational D and every Borel set Θ of observational contents: (a) $2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D) \leq M_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D) \leq 2^c \mathbb{U}_A(\mathcal{C}_\Theta; D)$; (b) the same holds for $W_A(\mathcal{C}_\Theta; D)$ and $W_A^{\mathcal{L}_O}(\mathcal{C}_\Theta; D)$, and $|K_A^{\mathcal{L}_O}(D) - K_A(D)| \leq c$; (c) for each time-constructible $t(m) \geq m^{a^*}$ there is c_t such that, class by class, the observational restrictions of $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$ (all forms) agree with W_A within $2^{\pm c_t}$ under Assumption B.5, and those of $M_A^{\mathrm{chk},t}$ and $M_A^{\mathrm{gen},t}$ agree with M_A within $2^{\pm c_t}$ under Assumption B.6; (d) for every theory machine G in $\mathcal{L}$, $G^O := \{(w, \varphi) : \varphi \in \mathrm{Sent}_{\mathcal{L}_O},\ B \cup A_w^G \vdash \varphi\}$ is a theory machine emitting only $\mathcal{L}_O$-sentences, and on every tape ω: $A_\omega^{G^O} = T_O(A_\omega^G)$; $B \cup A_\omega^{G^O}$ is consistent iff $B \cup A_\omega^G$ is; and $B \cup A_\omega^{G^O} \vdash \varphi^b$ iff $B \cup A_\omega^G \vdash \varphi^b$, for all observational φ and $b \in \{0, 1\}$. Hence, for every set Θ, the events defining $\nu_{G^O}(\mathcal{C}_\Theta; D)$ and $\nu_G(\mathcal{C}_\Theta; D)$ coincide tape by tape, and the two laws are equal.*
Consequently, in each version, posterior odds between two classes of observational contents and predictive probabilities of observational sentences agree within $2^{\pm 2c}$, and cumulative semimeasure losses on observational reveal sequences within $2c$.

In outline: (d) is compactness; (a) applies (d) to G_U and the mixture identity of Theorem 3.4 both ways; (b) maps e to indices of $T_O(\mathsf{W}_e)$ and of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; (c) pads the observational content as in Theorem B.9, the only place where Craig's padding enters. G^O is the pull-back machine of Theorem 3.5 for the inclusion $\mathrm{Sent}_{\mathcal{L}_O} \subseteq \mathrm{Sent}_{\mathcal{L}}$.

Proof. (d) G^O is c.e. because G is and derivability from the decidable B is c.e. By compactness, and because A^G_w grows with w along the prefixes of ω (Lemma A.1(a)), $A^{G^O}_\omega = \bigcup_{w \sqsubseteq \omega} \{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B \cup A^G_w \vdash \varphi\} = T_O(A^G_\omega)$. Since $T_O(A) \subseteq \mathrm{Cn}(A)$, consistency of $B \cup A$ passes to $B \cup T_O(A)$; if $B \cup A$ is inconsistent, $T_O(A)$ contains the observational sentence $\neg\forall x\, x = x$. Again from $T_O(A) \subseteq \mathrm{Cn}(A)$, $\mathrm{Cn}(T_O(A)) \cap \mathrm{Sent}_{\mathcal{L}_O} = T_O(A)$, so an observational φ^b follows from $B \cup T_O(A)$ iff it lies in $T_O(A)$ iff it follows from $B \cup A$. The observational content of G^O on ω is $T_O(A^G_\omega)$ itself, so membership in $\mathcal{C}_\Theta$ is preserved, and the two events coincide tape by tape.

(a) By the mixture identity of Theorem 3.4, on the cylinder of a program p with $V(p)$ an index of a machine G, the universal machine emits $A^G_{\omega'}$ on the rest ω' of the tape, hence G^O_U emits $A^G_{\omega'} \cap \mathrm{Sent}_{\mathcal{L}_O}$, and $\mathbb{U}(\,\cdot \cap [p]) = 2^{-|p|}\mathbb{U}(\cdot)$ on the shifted event; here the measurability of $\mathcal{C}_\Theta$ is used (the map $\omega \mapsto T_O(A_\omega)$ is Borel, since each coordinate "$\varphi \in T_O(A_\omega)$" is the effectively open event O_φ). *Lower bound.* Apply (d) to $G := G_U$. $(G_U)^O$ is a fixed machine emitting only $\mathcal{L}_O$-sentences, so on the tapes $p\omega'$ with $V(p)$ an index of $(G_U)^O$ the filtered machine G^O_U emits exactly $A^{(G_U)^O}_{\omega'}$; hence $M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D) \geq 2^{-K((G_U)^O)}\nu_{(G_U)^O}(\mathcal{C}_\Theta; D) = 2^{-c}\mathbb{U}_A(\mathcal{C}_\Theta; D)$. *Upper bound.* G^O_U is a theory machine of $\mathcal{L}$, so the same identity in $\mathcal{L}$ gives $\mathbb{U}_A(\mathcal{C}_\Theta; D) \geq 2^{-K(G^O_U)}\nu_{G^O_U}(\mathcal{C}_\Theta; D) = 2^{-c}M^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$. The program-less tapes contribute $(1 - \Omega_V)\mathbf{1}[B \vdash \psi_D,\ T_O(\emptyset) \in \Theta]$ to both sides.

(b) Let $o(e)$ be an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}_O} : B \cup \mathsf{W}_e \vdash \varphi\}$ and $f(e)$ one of $\mathsf{W}_e \cap \mathrm{Sent}_{\mathcal{L}_O}$; both maps are computable. By the argument of (d), $o(e)$, read in $W^{\mathcal{L}_O}_A$, is compatible with D and has content in Θ iff e, read in W_A, is; and $f(e)$ read in W_A behaves as e read in $W^{\mathcal{L}_O}_A$. The pushforward of $\mathbf{m}$ under a computable map is a lower semicomputable semimeasure, hence at most $2^c\mathbf{m}$. So, with P the compatibility-and-class predicate, $W_A(\mathcal{C}_\Theta; D) = \sum_e \mathbf{m}(e)\mathbf{1}[P(o(e))] \leq 2^c \sum_j \mathbf{m}(j)\mathbf{1}[P(j)] = 2^c W^{\mathcal{L}_O}_A(\mathcal{C}_\Theta; D)$, and symmetrically with f. Since $K(o(e)) \leq K(e) + c$ and $K(f(e)) \leq K(e) + c$, the statement on K_A follows.

(c) *Coin-free.* For an index e, let f^O_e be proof search from $B \cup \mathsf{W}_e$ restricted to observational inputs: accept φ if $B \cup \mathsf{W}_e \vdash \varphi$, reject if $B \cup \mathsf{W}_e \vdash \neg\varphi$, abstain on non-observational inputs. Then $\Gamma_{f^O_e} = T_O(\mathsf{W}_e)$, which is consistent iff W_e is. Theorem 6.14 of our earlier report (Hänni and Claude (Anthropic), 2026a), which is Theorem B.9(b), maps every consistent labeller f, injectively and at length cost c_t, to the decider (respectively enumerator) of the padded set of Lemma E.12 there, whose members are conjunction-powers of members of Γ_f, with the same consequences as Γ_f and membership (respectively generation) within t, under Assumption B.5. For $f = f^O_e$ the padded set is observational and has the observational content of W_e. Summing over e with the pushforward bound gives the lower bound. The upper bound holds because the observational restriction is a sub-sum of $W^{\mathrm{chk},t}_A$ (respectively $W^{\mathrm{gen},t}_A$), and a penalised hypothesis is an ordinary one with the same theory (Proposition B.28 for the class form). *One tape.* The padded random decider in the proof of Theorem B.9(a), applied to the random decider program that says yes on χ iff χ is observational and $B \cup A^G_\omega \vdash \chi$ (dovetailed proof search over the tape), has on every tape the padded copy of $T_O(A^G_\omega)$ as its yes-set; this set is observational, has the observational content of A^G_ω, and meets the time bound uniformly in ω under Assumption B.6. The program-less tapes are matched by the decider of $\emptyset$, as in Proposition B.10. Sum as in (a). For the upper bound, a random decider or enumerator is read as a theory machine through a constant-size wrapper, and Proposition B.10 bounds the penalised mixture by M_A class by class.

The consequences. Divide the class bounds for Θ_1 and Θ_2, and the bounds for $D + (\varphi, b)$ and D; the semimeasure loss is $-\log M(D_n) + \log M(\emptyset)$ for each version M. □

A direct computation on a propositional analogue (three observational atoms, two theoretical atoms, the theoretical background $(t_0 \to o_0) \wedge (t_1 \to \neg o_2)$, 40 random theory machines on 16

tapes) finds the law of G^O equal to that of G on all 24480 pairs of machine and observational data set and on 21830 class checks, while dropping the theoretical axioms instead of projecting changes the law of 37 of the 40 machines; an independent implementation with three backgrounds agrees.

F.2 Craig's construction

Lemma F.4 (Re-checking a literal)**.** *On literal data, let A be a set of sentences each logically equivalent to a literal $\pm\varphi_{w'}$, with $B \cup A$ consistent. If $B \cup F \vdash \varphi_w^b$ for a finite $F \subseteq A$, then F contains a member equivalent to φ_w^b.*

Proof. Suppose not. The members of F are equivalent to literals; no two of them are complementary, since $B \cup F$ is consistent; and those about w, if any, are equivalent to φ_w^{1-b}. Take a model of B in which the $t_{w'}$ are pairwise distinct and reinterpret R, which does not occur in B: put $t_{w'}$ into R iff $R(t_{w'})$ is among the literals of F, and in addition put t_w into R iff $b = 0$. This model satisfies B, every member of F and $\neg\varphi_w^b$. □

Lemma F.5 (Craig's padding writes a run time in unary)**.** *Let $B \cup T$ be consistent.* Canonical proof search f_T^{can} *tests, on input φ, the binary strings in length-lexicographic order as codes of* K*-derivations from $B \cup T$, in a code in which a derivation of symbol size s has code length at least s, spending at least one step per string, and stops at the first derivation of φ or of $\neg\varphi$. Let* $\mathrm{ds}(\varphi^b)$ *be the least symbol size of a derivation of whichever of $\varphi, \neg\varphi$ is provable from $B \cup T$. (a) The running time of f_T^{can} on φ is at least $2^{\mathrm{ds}(\varphi^b)} - 1$. (b) On literal data, let $A^{\mathrm{Cr}} := \{(\varphi_w^b)^{\wedge(k+1)} :$ f_T^{can} decides φ_w with b in exactly k steps$\}$ be Craig's set of f_T^{can} restricted to literal inputs, or its version padded by a work counter (Lemma E.12 of the earlier report). Every* K*-derivation of φ_w^b from $B \cup A^{\mathrm{Cr}}$ contains a line of size at least $2^{\mathrm{ds}(\varphi_w^b)}|\varphi_w^b|$; so its size, its material and its largest axiom instance are all at least that. (c) With Craig's original padding by the Gödel number* gn *of a derivation (the number whose binary expansion is* 1 *followed by the code), $A^{\mathrm{gn}} := \{(\varphi_w^b)^{\wedge\mathrm{gn}}\}$, the same bound holds.*

Proof. (a) Every derivation of φ^b has size at least ds, hence code length at least ds, and no derivation of φ^{1-b} exists. So all $2^0 + 2^1 + \cdots + 2^{\mathrm{ds}-1} = 2^{\mathrm{ds}} - 1$ strings of length less than ds are tested and fail. (b) By Lemma F.4 the derivation cites a member of A^{Cr} equivalent to φ_w^b. The only such member is $(\varphi_w^b)^{\wedge(k+1)}$, with $k \geq 2^{\mathrm{ds}} - 1$ by (a); for the padded version the counter is at least the simulated time. That member is a line of the derivation, of size at least $(k+1)|\varphi_w^b|$. (c) Every member of A^{gn} equivalent to φ_w^b is a power with $\mathrm{gn} \geq 2^\ell$ factors, where $\ell \geq \mathrm{ds}$ is the code length; apply Lemma F.4. □

Theorem F.6 (Run-padded axiomatisations fit exactly P)**.** *On literal data, with run-padded sets as in Section 8.3: (a) if A is run-padded, with membership decidable in polynomial time, and is compatible with every D_n^X under reading (a1), (a2), (b) or (b2) with a polynomial budget h, or with derivations of size at most h, then $X \in \mathsf{P}$; (b) every $X \in \mathsf{P}$ is fitted under all these readings, at polynomial budgets with polynomial-time membership, by a run-padded set.*

Proof. (a) Under each reading, the member of A that a derivation of φ_w^b cites and that is equivalent to φ_w^b (Lemma F.4) has size at most $h(|\varphi_w|)$: it is an axiom line, a nonlogical instance, a part of the material, and a line of a derivation of size at most h. Its form is $(\varphi_w^b)^{\wedge m}$ with $m \leq h(|\varphi_w|)$. Decide w as follows: for $m = 1, \ldots, h(|w| + 2)$ test whether $(\varphi_w)^{\wedge m} \in A$, and accept iff one is. These are polynomially many tests of polynomial time. If a power of φ_w is in A, then $B \cup A \vdash \varphi_w$,

so $w \in X$ (otherwise $B \cup A$ would also prove the datum $\neg\varphi_w$ and be inconsistent). If $w \in X$, the derivation within budget cites such a power with $m \leq h$. (b) Clock a polynomial-time decider f of X at $\tau(m) := (m+2)^k$. Its Craig set $\{(\varphi_w^b)^{\wedge(j+1)} : f \text{ decides } \varphi_w \text{ with } b \text{ in exactly } j \text{ steps}\}$ is parameter-free, has polynomial-time membership (read the word off a candidate with m factors and run f for $m-1$ steps), and derives each datum by citing $(\varphi_w^b)^{\wedge(j+1)}$, one substitution instance of a fixed propositional derivation of $(\beta \wedge \gamma) \to \gamma$, and modus ponens; size, material and largest instance are $O((\tau(m)+1)(m+4))$. This is Lemma E.22 of the earlier report. □

A run-padded axiom carries $O(\log)$ bits, its length, while the axiom $\theta_\gamma \to \varphi^b$ of Theorem B.31 carries the bits of a certificate γ. So the natural strengthening of Theorem F.6(a) to all sets whose members are each equivalent to a literal fails: certificate sets are such sets and fit $\mathsf{NP} \cap \mathsf{coNP}$ under (b). Theorem F.6 does not contradict the standard fact that every r.e. theory has an axiomatisation decidable in polynomial time, by a variant of Craig's construction (Krajíček, 2025): it concerns what such sets fit when derivations are charged. Against all hypotheses (Corollary B.32), run-padded sets lose unconditionally under (a1), (a2), (b2), and under (b) iff $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$ (Corollary F.8).

Example F.7 (The size of a Craig axiom)**.** Take the seven-sentence Horn theory with theoretical predicates Ev, Od (an even, respectively odd, number of 1s): $\mathrm{Ev}(\mathsf{e})$; $\mathrm{Ev}(x) \to \mathrm{Ev}(\mathsf{s}_0 x)$; $\mathrm{Ev}(x) \to \mathrm{Od}(\mathsf{s}_1 x)$; $\mathrm{Od}(x) \to \mathrm{Od}(\mathsf{s}_0 x)$; $\mathrm{Od}(x) \to \mathrm{Ev}(\mathsf{s}_1 x)$; $\mathrm{Ev}(x) \to R(x)$; $\mathrm{Od}(x) \to \neg R(x)$. In a small proof system, a derivation is a sequence of ground literals over $\mathrm{Ev}, \mathrm{Od}, R$ with terms t_u, $|u| \leq |w|$, each justified by a rule whose premise occurs earlier; a literal has $|u|+2$ symbols, one more if negated; canonical search tests sequences by number of lines, then lexicographically. For a word w of length n there are $N_{\mathrm{cand}} = 4(2^{n+1}-1)$ candidate lines; a shortest derivation has $L^* = n+2$ lines ($\mathrm{Ev}(\mathsf{e})$, the parity atoms of the n nonempty suffixes of w, and the literal), and none is shorter, since each rule application extends the term by one letter and needs its premise earlier; its size is $\mathrm{ds}_T = \sum_{j=0}^{n}(j+2) + (n+2) + \mathbf{1}[\#_1 w \text{ odd}]$; and the number k of sequences tested is at least $\sum_{L=1}^{L^*-1} N_{\mathrm{cand}}^L$, since every sequence with fewer lines is tested and fails. The Craig axiom has at least $(k+1)|\varphi^b|$ symbols; the certificate axiom $\theta_\gamma \to \varphi^b$, with γ coding the shortest derivation in $L^*(2+\lceil \log N_{\mathrm{cand}} \rceil)$ bits, has $|\gamma|+15+|\varphi^b|$; the Craig axiom of a fast parity labeller has $(n+3)|\varphi^b|$.

At $n = 24$ these closed forms give $N_{\mathrm{cand}} = 134217724$, $L^* = 26$, $\mathrm{ds}_T = 376$, a Craig axiom of more than $10^{204.6}$ symbols, and 795 and 702 symbols for the certificate axiom and for the fast labeller's Craig axiom (a direct computation, which also reproduces, for $|w| \leq 4$, an exact count by exhaustive search; code in the accompanying repository).

Corollary F.8 (The practical horn)**.** *Under Assumption B.30, on literal sequences, at polynomial budgets with polynomial membership and degrees charged, the fit classes are:*

hypotheses	*(b)*	*(a1)*	*(a2), (b2)*
run-padded observational sets (Craig type)	P	P	P
observational certificate sets	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$	$\supseteq \mathsf{NP} \cap \mathsf{coNP}$
all observational hypotheses†	$\mathsf{NP} \cap \mathsf{coNP}$	$\mathsf{EXPTIME}$	*all decidable*
finite theories in $\mathcal{L}_O$*, unary predicates, constants**†	*regular*	*regular*	*regular*
all hypotheses	$\mathsf{NP} \cap \mathsf{coNP}$	$\mathsf{EXPTIME}$	*all decidable*

(* *compatible with the named structure;* † *with* $B = \emptyset$*.) Hence under (a1) the run-padded sets fit strictly less than the axiom sets with theoretical symbols and, with* $B = \emptyset$*, than the observational hypotheses that are not run-padded, unconditionally; under (a2) and (b2) also unconditionally; and under (b) iff* $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$*.*

Proof. Row 1 is Theorem F.6. Row 2: the certificate sentence $\theta_\gamma := Z \to (C_\gamma \to Z)$ of Theorem B.31, with Z a closed tautology and C_γ a valid sentence coding γ, uses only logical symbols, so the certificate sets $\{\theta_\gamma \to \varphi_w^b\}$ are observational; they fit $\mathsf{NP} \cap \mathsf{coNP}$ under every reading by the third inequality of Theorem B.31(b) (material $2|\gamma|+29+|\varphi^b|$, largest instance $|\gamma|+15+|\varphi^b|$, every nonlogical instance a member). Row 3 is Proposition F.9 for (b), Proposition F.11 for (a1) and Theorem F.14 for (a2), (b2), each bounded above by row 5. Row 4 is Theorems F.16 and F.17. Row 5 is Corollary B.32. The separations: $\mathsf{P} \subsetneq \mathsf{EXPTIME}$ by the time hierarchy theorem; some decidable set is not in P; and under (b) the classes are P and $\mathsf{NP} \cap \mathsf{coNP}$. □

F.3 Eliminating vocabulary

Proposition F.9 (Elimination by proof padding)**.** *Let $\mathcal{L}$ be finite and B decidable in polynomial time and closed under permutations of the parameters; assume Assumption B.30. With the constants k_0, β of Theorem B.31, put $\bar{g}(m) := (g(m)+m+1)(g(m)+m+2)/2$, $\bar{g}'(m) := \beta\bar{g}(m)\log(\bar{g}(m)+2)$ and $t^{+\#}(m) := m^{k_0} + m^{k_0}(t(m)+1)$. For every hypothesis p of $W_A^b[t,g]$ there is a hypothesis p' whose axiom set is observational, with $|p'| \le |p| + c$ and $p \mapsto p'$ injective, such that p' is compatible with every observational D with which p is, under (b) with budget $2\bar{g}'(m)+m+31$ and under (a1) with budget $\bar{g}'(m)+m+16$, with membership time $t^{+\#}$. Hence, on observational data,*

$$W_A^b[t,g](D) \le 2^c\, W_A^{\mathcal{L}_O,b}[t^{+\#}, 2\bar{g}'+m+31](D), \qquad W_A^b[t,g](D) \le 2^c\, W_A^{\mathcal{L}_O,a1}[t^{+\#}, \bar{g}'+m+16](D),$$

where $W_A^{\mathcal{L}_O,b}$ and $W_A^{\mathcal{L}_O,a1}$ are the sub-sums over hypotheses with observational axiom sets.

Proof. By the second inequality of Theorem B.31(b) (Proposition E.21 of the earlier report), p maps to the verifier $\mathcal{V}_p(\varphi,\gamma)$ that outputs acc if γ codes a derivation of φ from $B \cup A_p$ of material at most $g(|\varphi|)$, rej if it codes one of $\neg\varphi$, and $\perp$ otherwise; it runs in time $O(t^+)$, and every datum that p derives has a certificate of length at most $\bar{g}'(|\varphi|)$ (this step uses Assumption B.30). Restrict $\mathcal{V}_p$ to observational φ, with output $\perp$ elsewhere (a linear-time syntax test). The third inequality maps a verifier $\mathcal{V}$ to $A_\mathcal{V} := \{\theta_\gamma \to \varphi : \mathcal{V}(\varphi,\gamma) = \text{acc}\} \cup \{\theta_\gamma \to \neg\varphi : \mathcal{V}(\varphi,\gamma) = \text{rej}\}$ (certificates of length at most $\bar{g}'$), a parameter-free set with membership time $t^{+\#}$. Each member is observational and logically equivalent to an observational theorem of $B \cup A_p$, so $B \cup A_\mathcal{V} \subseteq \mathrm{Cn}(A_p)$ is consistent. Each observational datum derived by p within material g has the derivation θ_γ, $\theta_\gamma \to \varphi^b$, φ^b, of material $2|\gamma|+29+|\varphi^b|$ and largest instance $|\gamma|+15+|\varphi^b|$. Lengths add constants, and the maps are injective. □

Under (a1) we compare only fit classes, through Proposition F.11 below; a hypothesis-by-hypothesis translation is not claimed. Under (a2) and (b2), at a fixed budget, an observational substitute may need a table of the data below the size of its own members, so its description length need not be $|p| + O(1)$; again only fit classes are compared (Theorem F.14).

The pairing structure. $\mathfrak{U}$ is the least set that contains the words (the elements t_w of $\mathfrak{T}$) and is closed under a formal pairing $(a,b) \mapsto \mathrm{pr}(a,b)$, where pr is injective and its values are new (not words). $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$ act on words as in $\mathfrak{T}$, and on pairs by $\mathtt{s}_0\,\mathrm{pr}(a,b) := a$ and $\mathtt{s}_1\,\mathrm{pr}(a,b) := \mathrm{pr}(b,a)$; R is left free. In $\mathcal{L}_O$ let $\mathrm{Pair}(z) := \mathtt{s}_1\mathtt{s}_1 z = z$, $\mathrm{Std}(y) := \neg\mathrm{Pair}(y)$ and

$$\theta_2(a,b) := \forall y_\theta\,(\mathrm{Pair}(y_\theta) \to (\mathtt{s}_0 y_\theta = a \to (\mathtt{s}_0\mathtt{s}_1 y_\theta = b \to R(y_\theta)))),$$

where the bound variable y_θ occurs in no other formula below. It must be fresh: $\theta_2(a,b)$ is substituted for atoms inside formulas with their own quantified variables, and a bound variable named like one of them would capture it.

Lemma F.10 (Pairing)**.** *In $\mathfrak{U}$, for every interpretation of R: (i)* Pair *holds exactly on the pairs, so* Std *holds exactly on the words, and $\mathfrak{U} \models \mathrm{Std}(\mathsf{e})$ and $\mathfrak{U} \models \forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; (ii) for all $a, b \in \mathfrak{U}$, $\mathrm{Pair}(z) \wedge \mathsf{s}_0 z = a \wedge \mathsf{s}_0\mathsf{s}_1 z = b$ holds iff $z = \mathrm{pr}(a, b)$; (iii) for all $a, b \in \mathfrak{U}$, $\theta_2(a, b)$ holds iff $R(\mathrm{pr}(a, b))$.*

Proof. (i) On words, $\mathsf{s}_1\mathsf{s}_1 t_v = t_{11v} \neq t_v$; on pairs, $\mathsf{s}_1\mathsf{s}_1\,\mathrm{pr}(a, b) = \mathsf{s}_1\,\mathrm{pr}(b, a) = \mathrm{pr}(a, b)$; and words are closed under $\mathsf{s}_0, \mathsf{s}_1$. (ii) A pair $z = \mathrm{pr}(c, d)$ has $\mathsf{s}_0 z = c$ and $\mathsf{s}_0\mathsf{s}_1 z = \mathsf{s}_0\,\mathrm{pr}(d, c) = d$; by injectivity of pr it equals $\mathrm{pr}(a, b)$ iff $(c, d) = (a, b)$. (iii) By (ii) the premise singles out $y_\theta = \mathrm{pr}(a, b)$. □

A computation on a fragment of $\mathfrak{U}$ with 800 elements (583200 triples), and independently on a fragment of 552 elements including pairs of pairs, finds no exception.

Proposition F.11 (Ground pairing translation, reading (a1))**.** *Let $B = \emptyset$, and let $\mathcal{M}$ be an alternating machine with branching at most 2 that decides X in space $s(m) \geq m$ and halts on every path, and let $A_{\mathcal{M}}$ be the ground axioms of Proposition E.23(b) of the earlier report in $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R, \mathsf{C}, \mathsf{C}'\}$, where $\mathsf{C}(t_w, t_\varsigma)$ says that the configuration ς of $\mathcal{M}$ on w accepts and $\mathsf{C}'(t_w, t_\varsigma)$ that it rejects. Let $A^O_{\mathcal{M}}$ replace every atom $\mathsf{C}(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_0 t_\varsigma)$ and every $\mathsf{C}'(t_w, t_\varsigma)$ by $\theta_2(t_w, \mathsf{s}_1 t_\varsigma)$. Then $A^O_{\mathcal{M}}$ is observational and parameter-free with polynomial-time membership (by a decider of length $|\mathcal{M}| + c$); $A^O_{\mathcal{M}} \cup \Gamma_{f_X}$ is consistent; and every datum has a derivation from $A^O_{\mathcal{M}}$ by modus ponens alone with $2^{O(s(|w|)+1)}$ lines, each of size $O(s(|w|) + 1)$. Hence the fit class of the observational hypotheses of W^{a1}_A at polynomial budgets is* EXPTIME*, as for all hypotheses.*

Proof. Membership: parse the fixed shape of the θ_2-atoms, read off t_w, the tag letter and t_ς, and run the check of Proposition E.23(b) there. *Derivations:* the derivations there use modus ponens only, between ground atoms and ground members; modus ponens is syntactic in the major premise, so translating atom by atom gives derivations from $A^O_{\mathcal{M}}$, and each atom grows by $O(1)$ symbols. *Consistency:* in $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(t_w, t_{0\varsigma})) := \mathbf{1}[\varsigma \text{ accepts}]$, $R(\mathrm{pr}(t_w, t_{1\varsigma})) := \mathbf{1}[\varsigma \text{ rejects}]$, and R false on all other pairs. By Lemma F.10(iii) each translated atom has the truth value of its original in the model of Proposition E.23(b) (the term algebra, with C, C' the least fixed points), so $A^O_{\mathcal{M}}$ and Γ_{f_X} are true in $\mathfrak{U}$. *Fit class:* $\supseteq$ EXPTIME by Corollary 6.17(b) there, with $A^O_{\mathcal{M}}$ in place of $A_{\mathcal{M}}$ and EXPTIME = APSPACE (Chandra et al., 1981); $\subseteq$ by Proposition E.23(a) there, which holds for every hypothesis satisfying (F1). □

A computation on an alternating machine for an interval game (positions are intervals of the word, the players alternately delete the first or last letter, and the existential player wins iff the last letter is 1) finds valid derivations, against an independent parsing decider, for all 2046 data with $1 \leq |w| \leq 10$, and all 14322 members with $|w| \leq 7$ recognised and true in $\mathfrak{U}$.

F.4 One binary relation: two-stack programs stated backwards

For a word $u = u_1 \cdots u_k$ and a term τ write $u\tau$ for $\mathsf{s}_{u_1}(\cdots \mathsf{s}_{u_k}(\tau) \cdots)$; so $t_w = w\mathsf{e}$, and a pair of words is a pair of closed terms.

Definition F.12 (Two-stack programs and their backward theories)**.** A *pattern* is $\pi = (r_\pi, \epsilon_r;\ l_\pi, \epsilon_l)$ with words r_π, l_π and $\epsilon_r, \epsilon_l \in \{\mathrm{var}, \mathrm{exact}\}$. A pair of words (r, l) *matches* π with values (Z_r, Z_l) if $r = r_\pi Z_r$ when $\epsilon_r = \mathrm{var}$, and $r = r_\pi$ (with Z_r absent) when $\epsilon_r = \mathrm{exact}$, and likewise $l = l_\pi Z_l$ or $l = l_\pi$. A *rule* $\pi \Rightarrow \pi'$ is a pair of patterns with the same ϵ's; it is *linear* (each variable occurs once on each side), and applied to $(r_\pi Z_r, l_\pi Z_l)$ it gives $(r_{\pi'} Z_r, l_{\pi'} Z_l)$. A *two-stack program* $\mathcal{P}$ is a finite set of rules; it is *deterministic* if no pair of words matches the

current patterns of two rules. Programs carry a *tag* letter $\mathrm{tg} \in \{0,1\}$ at the front of l (0 for an acceptance run, 1 for a rejection run), which every rule preserves, followed by a fixed-length *state code* $\mathrm{st}(q)$; the *halting prefixes* are $H_{\mathrm{acc}} := 0\,\mathrm{st}(\mathrm{ACC})$ and $H_{\mathrm{rej}} := 1\,\mathrm{st}(\mathrm{REJ})$, and the *start prefixes* are $\mathrm{tg}\,\mathrm{st}(\mathrm{start})$.

The *backward theory* of $\mathcal{P}$ in $\mathcal{L}_O \cup \{\mathsf{C}\}$, C a binary relation symbol, is

$$T_{\mathcal{P}}^{\mathsf{C}} := \{\forall Z_r \forall Z_l\,(\mathsf{C}(\pi') \to \mathsf{C}(\pi)) : (\pi \Rightarrow \pi') \in \mathcal{P}\} \cup \{\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l),\ \forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)\}$$
$$\cup\, \{\forall x \forall z\,(\mathsf{C}(x, 0\,\mathrm{st}(\mathrm{start})\,z) \to R(x)),\ \forall x \forall z\,(\mathsf{C}(x, 1\,\mathrm{st}(\mathrm{start})\,z) \to \neg R(x))\},$$

where $\mathsf{C}(\pi)$ is $\mathsf{C}(r_\pi Z_r, l_\pi Z_l)$ (without a variable where the pattern is exact) and the quantifiers range over the variables present. The two *output rules* carry a free certificate variable z; they are used only for programs with a certificate phase that consumes z (Lemma F.13), never for the alternating programs in the proof of Theorem F.14(iv), which use exact output rules. The *observational translation* $T_{\mathcal{P}}^{O}$ replaces each atom $\mathsf{C}(a,b)$ by $\theta_2(a,b)$, prefixes the matrix of every member by $\mathrm{Std}(v) \to \cdots$ for each of its universally quantified variables v, and adds $\mathrm{Std}(\mathsf{e})$ and $\forall y\,(\mathrm{Std}(y) \to \mathrm{Std}(\mathsf{s}_a y))$ for $a = 0, 1$; it is a finite set of $\mathcal{L}_O$-sentences.

A forward theory (a start axiom, rules $\mathsf{C}(\pi) \to \mathsf{C}(\pi')$, an output rule reading the input from a component) must keep the input word in the configuration until the end, which needs a relation of higher arity. Stated backwards, C means "this configuration leads to a halting configuration of the right kind", and the input is read only by the output rules, at the start configuration.

Lemma F.13 (Every verifier is a deterministic two-stack program)**.** *Let $\mathcal{V}$ be a verifier: a deterministic multitape machine that halts on every pair (x, z) of words with output acc, rej or $\perp$, such that no x has both an accepting and a rejecting certificate ($f_{\mathcal{V}}$ is consistent); let $X_{\mathrm{acc}} := \{x : \exists z\ \mathcal{V}(x,z) = \mathrm{acc}\}$ and X_{rej} likewise. There is a deterministic two-stack program $\mathcal{P}_{\mathcal{V}}$ such that for all words x, z and both tags: (i) every rule is linear, the current patterns are pairwise disjoint, and no rule applies to a pair whose l begins with a halting prefix or with the code of the state $\perp$; (ii) the run from the* start pair *$(x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$ reaches a pair with prefix H_{acc} (for $\mathrm{tg} = 0$) iff $\mathcal{V}(x,z) = \mathrm{acc}$, and one with prefix H_{rej} (for $\mathrm{tg} = 1$) iff $\mathcal{V}(x,z) = \mathrm{rej}$; (iii) the run has $N_{\mathrm{run}} = O(T_1 + |x| + |z|)$ steps, every pair on it has length $O(S_1 + |x| + |z|)$, and each step changes $O(1)$ letters, where T_1 and S_1 are the time and space of a one-tape simulation of $\mathcal{V}$ (so T_1 is polynomial in the time of $\mathcal{V}$ and $S_1 = O(S_{\mathcal{V}} + |x| + |z|)$).*

Proof. (1) The one-tape machine. By the one-tape simulation of multitape machines (standard; cf. Hartmanis and Stearns, 1965) there is a deterministic one-tape machine $\mathcal{M}_1$ that, started on the tape $\mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1} \# x_1 \cdots x_{|x|}$ (the certificate reversed, in letters $\mathtt{Z}_0, \mathtt{Z}_1$, then a separator, then the input) with the head on cell 0, halts in state ACC, REJ or $\perp$ according to $\mathcal{V}(x,z)$; rearranging the input format costs $O((|x| + |z|)^2)$ steps, and the simulation is polynomial in the time of $\mathcal{V}$ and linear in its space. Bring $\mathcal{M}_1$ to a normal form: its tape alphabet $\Sigma_{\mathcal{M}}$ (without the blank $\square$) contains $\#, 0, 1, \mathtt{Z}_0, \mathtt{Z}_1$ and a symbol $\mathtt{B}$ on which $\mathcal{M}_1$ has exactly the transitions it has on $\square$, and wherever $\mathcal{M}_1$ would write $\square$ it writes $\mathtt{B}$; so every transition writes a symbol of $\Sigma_{\mathcal{M}}$. Moving left on cell 0 leaves the head there. Then on every run from a start tape the non-blank cells form an initial segment $[0, \mathrm{nb})$ and the head position hp satisfies $\mathrm{hp} \le \mathrm{nb}$: a step at $\mathrm{hp} < \mathrm{nb}$ keeps both properties, and a step at $\mathrm{hp} = \mathrm{nb}$ writes a symbol of $\Sigma_{\mathcal{M}}$ there, so nb grows by one.

(2) Coding. Fix injective codes $\mathrm{st} : Q \to \{0,1\}^{k_Q}$ (the states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{cert}, \mathrm{sep}, \mathrm{inp}, \mathrm{rew}$) and $\mathrm{blk} : \Sigma_{\mathcal{M}} \to \{0,1\}^{k_B}$. A configuration of $\mathcal{M}_1$ with state q, tape $a_0 \cdots a_{\mathrm{nb}-1} \square\square \cdots$ and head $\mathrm{hp} \le \mathrm{nb}$ is coded, with tag tg, as

$$[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}} := \big(\mathrm{blk}(a_{\mathrm{hp}}) \cdots \mathrm{blk}(a_{\mathrm{nb}-1}),\quad \mathrm{tg}\,\mathrm{st}(q)\,\mathrm{blk}(a_{\mathrm{hp}-1}) \cdots \mathrm{blk}(a_0)\big) :$$

r holds the cells from the head on, l the tag, the state and the cells to the left of the head, nearest first; $r = \varepsilon$ means that the head reads $\square$.

(3) Rules, for each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$; a pattern is written $(\rho; \lambda)$, and "$\varepsilon \,|\, \mathrm{exact}$" or "$\mathrm{cd}(q) \,|\, \mathrm{exact}$" marks an exact component:

- *start:* $(Z_r;\, \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$, which pushes the separator onto r, above the raw input;
- *phase cert:* $(Z_r;\, \mathrm{cd}(\mathrm{cert})aZ_l) \Rightarrow (\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{cert})Z_l)$ for $a \in \{0,1\}$ (a raw certificate letter below the state code becomes a block on r), and $(Z_r;\, \mathrm{cd}(\mathrm{cert}) \,|\, \mathrm{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep}) \,|\, \mathrm{exact})$;
- *phase sep:* $(\mathrm{blk}(\mathtt{Z}_a)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{sep})\mathrm{blk}(\mathtt{Z}_a)Z_l)$, and $(\mathrm{blk}(\#)Z_r;\, \mathrm{cd}(\mathrm{sep})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)Z_l)$;
- *phase inp:* $(aZ_r;\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0,1\}$, and $(\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \,|\, \mathrm{exact};\, \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\, \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\mathtt{Z}_0, \mathtt{Z}_1, \#, 0, 1\}$, and $(Z_r;\, \mathrm{cd}(\mathrm{rew}) \,|\, \mathrm{exact}) \Rightarrow (Z_r;\, \mathrm{cd}(q_0) \,|\, \mathrm{exact})$;
- *the machine:* for each transition $\delta(q,a) = (q', b, \mathrm{dir})$, with read pattern $\rho := \mathrm{blk}(a)Z_r$ if $a \in \Sigma_{\mathcal{M}}$ and $\rho := \varepsilon \,|\, \mathrm{exact}$ if $a = \square$, and ρ' the remainder (Z_r, respectively $\varepsilon \,|\, \mathrm{exact}$): for $\mathrm{dir} = R$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\rho';\, \mathrm{cd}(q')\mathrm{blk}(b)Z_l)$; for $\mathrm{dir} = S$, $(\rho;\, \mathrm{cd}(q)Z_l) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$; for $\mathrm{dir} = L$, for each $e \in \Sigma_{\mathcal{M}}$, $(\rho;\, \mathrm{cd}(q)\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)\mathrm{blk}(b)\rho';\, \mathrm{cd}(q')Z_l)$, and at the left end $(\rho;\, \mathrm{cd}(q) \,|\, \mathrm{exact}) \Rightarrow (\mathrm{blk}(b)\rho';\, \mathrm{cd}(q') \,|\, \mathrm{exact})$.

(i) Linearity is visible in the list. Rules with different tags or state codes differ in a fixed-length prefix of l. Within one state, the read patterns $\mathrm{blk}(a)Z_r$ (fixed length, injective code) and $\varepsilon \,|\, \mathrm{exact}$ are pairwise disjoint, as are, within one left-moving transition, the neighbour patterns $\mathrm{blk}(e)Z_l$ and the exact left-end pattern; in phase cert the raw-letter patterns $\mathrm{cd}(\mathrm{cert})aZ_l$ and the exact $\mathrm{cd}(\mathrm{cert})$ are disjoint, and in phase inp the raw-letter patterns aZ_r and the exact ε. There are no rules for ACC, REJ and $\bot$, and fixed-length state codes prevent a spurious halting prefix. (ii) *Phases.* From $(x, \mathrm{cd}(\mathrm{start})z)$ the start rule gives $(\mathrm{blk}(\#)x, \mathrm{cd}(\mathrm{cert})z)$; phase cert moves the letters of z, first letter first, onto r, ending in $(\mathrm{blk}(\mathtt{Z}_{z_m}) \cdots \mathrm{blk}(\mathtt{Z}_{z_1})\mathrm{blk}(\#)x,\ \mathrm{cd}(\mathrm{sep}))$; phase sep moves these blocks to l, the separator last, ending in $(x,\ \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#)\mathrm{blk}(\mathtt{Z}_{z_1}) \cdots \mathrm{blk}(\mathtt{Z}_{z_m}))$; phase inp converts the raw letters of x; and phase rew moves all blocks back to r and ends exactly at the left end in $[\![q_0,\ \mathtt{Z}_{z_m} \cdots \mathtt{Z}_{z_1}\#x,\ 0]\!]_{\mathrm{tg}}$, the code of the start configuration of $\mathcal{M}_1$. *Steps.* For a configuration with $\mathrm{hp} \le \mathrm{nb}$ and a non-blank initial segment, exactly one rule applies to its code, namely the rule of the transition taken, with the neighbour case given by the cell $\mathrm{hp} - 1$ or the left end, and its result is the code of the successor: for $\mathrm{dir} = R$ the written b moves to the top of l; for S it replaces the top of r; for L the left neighbour e moves from l to r above b; and at $a = \square$ ($r = \varepsilon$) the written b becomes the new last cell. Reading $\mathtt{B}$ is reading the block $\mathrm{blk}(\mathtt{B})$, with the transitions of $\square$. By (1) the invariant is preserved. So the run of $\mathcal{P}_{\mathcal{V}}$ from the start pair passes through the codes of the run of $\mathcal{M}_1$ and stops exactly when $\mathcal{M}_1$ halts. The tag is preserved, so a 0-tagged run can reach H_{acc} only through ACC, and a 1-tagged run H_{rej} only through REJ. (iii) The phases take $2|z| + |x| + (|z| + |x| + 1) + O(1)$ steps, the machine T_1; every rule changes $O(k_Q + k_B)$ letters. $\square$

Theorem F.14 (One binary theoretical relation suffices under charged deduction)**.** *Let $B = \emptyset$, let $\mathcal{V}$ be a verifier as in Lemma F.13, $\mathcal{P} := \mathcal{P}_{\mathcal{V}}$, and let $\mathsf{C}^{\mathfrak{T}} \subseteq \mathfrak{T}^2$ be the least set of pairs of words that contains all pairs with l-prefix H_{acc} or H_{rej} and is closed under the rules read backwards (if the result of a rule instance is in the set, so is its current pair). Write $T^{\mathsf{C}}_{\mathcal{V}} := T^{\mathsf{C}}_{\mathcal{P}}$ and $T^{O}_{\mathcal{V}} := T^{O}_{\mathcal{P}}$.*

(i) Relation version. $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X_{\mathrm{acc}}) \models T^{\mathsf{C}}_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$ *and* $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := \{w : w \notin X_{\mathrm{rej}}\}) \models T^{\mathsf{C}}_{\mathcal{V}}$. *So* $T^{\mathsf{C}}_{\mathcal{V}}$ *is a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *and* $T^{\mathsf{C}}_{\mathcal{V}} \vdash R(t_w)$ *iff* $w \in X_{\mathrm{acc}}$, $T^{\mathsf{C}}_{\mathcal{V}} \vdash \neg R(t_w)$ *iff* $w \in X_{\mathrm{rej}}$.

(ii) Observational version. *In* $\mathfrak{U}$ *put* $R(t_w) := \mathbf{1}[w \in X_{\mathrm{acc}}]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ *for words* a, b, *and* R *false on all other pairs. Then* $\mathfrak{U} \models T^{O}_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$, *and with* $R(t_w) := \mathbf{1}[w \notin X_{\mathrm{rej}}]$ *instead,* $\mathfrak{U} \models T^{O}_{\mathcal{V}}$. *So* $T^{O}_{\mathcal{V}}$ *is a finite set of* $\mathcal{L}_O$*-sentences, consistent with* $\Gamma_{f_{\mathcal{V}}}$, *proving exactly the literals* φ_w *with* $w \in X_{\mathrm{acc}}$ *and* $\neg\varphi_w$ *with* $w \in X_{\mathrm{rej}}$.

(iii) Derivations. *If* $\mathcal{V}(w,z) \in \{\mathrm{acc}, \mathrm{rej}\}$, *the corresponding literal has a* K*-derivation from* $T^{\mathsf{C}}_{\mathcal{V}}$ *(respectively* $T^{O}_{\mathcal{V}}$*) by instances of the quantifier axiom A4 and modus ponens only, with* $O(N_{\mathrm{run}})$ *(respectively* $O(N_{\mathrm{run}} + \ell)$*) lines,* N_{run} *the number of steps of Lemma F.13(iii) and* ℓ *the largest length of a pair on the run, each line of size* $O(\ell + 1)$*; its nonlogical instances are members of the theory, each of size at most* $\mathrm{mx}_{\mathcal{V}}$ *and together at most* $\mathrm{tot}_{\mathcal{V}}$.

(iv) Fit classes. *Hence, on literal sequences, finite theories of either kind are compatible with all of* D^X *for every decidable* X *under (a2) and (b2), at every nondecreasing unbounded budget with* $h(m) \geq m+1$ *(with a finite table of the data below the budget threshold added as literals; with a budget of at least* $\mathrm{mx}_{\mathcal{V}}$*, respectively* $\mathrm{tot}_{\mathcal{V}}$*, everywhere no table is needed); and, at polynomial budgets, for every* $X \in \mathsf{NP} \cap \mathsf{coNP}$ *under (b) (material* $O(N_{\mathrm{run}}\ell)$*, polynomial for polynomial-time verifiers), the fit class of all hypotheses (Corollary B.32). Under (a1): for every* $X \in \mathsf{EXPTIME}$ *there are a finite theory in* $\mathcal{L}_O \cup \{\mathsf{C}\}$ *about the named objects and a finite set of* $\mathcal{L}_O$*-sentences, each consistent with* Γ_{f_X}, *from which every datum* $\varphi_w^{\mathbf{1}[w \in X]}$ *has a* K*-derivation whose every line, logical axiom instances included, has size* $O(s(|w|) + |w|)$ *for a polynomial* s*; so at polynomial budgets the (a1) fit class of both kinds of finite theory is* $\mathsf{EXPTIME}$*, the class of all hypotheses. Membership is linear-time (finite sets of sentences) and Assumption B.30 holds (sentences).*

Proof. (i) The rules hold in $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}})$ by the closure of $\mathsf{C}^{\mathfrak{T}}$, the halting axioms by its base. For the output rules we show: *a start pair* $\sigma = (x, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,z)$, z *any word, lies in* $\mathsf{C}^{\mathfrak{T}}$ *iff its run reaches a pair with prefix* H_{acc} *(*$\mathrm{tg} = 0$*) or* H_{rej} *(*$\mathrm{tg} = 1$*).* Let S be the set of pairs whose deterministic run reaches a halting prefix. S contains the base and is closed under the backward rules: if the result $\pi'(Z_r, Z_l)$ of a rule instance is in S, then $\pi(Z_r, Z_l)$ matches that rule and, by disjointness (Lemma F.13(i), on all pairs of words), no other, so its run moves to $\pi'(Z_r, Z_l)$ and continues into the base; so $\mathsf{C}^{\mathfrak{T}} \subseteq S$. Conversely $S \subseteq \mathsf{C}^{\mathfrak{T}}$ by induction on the number of steps to the base. With Lemma F.13(ii): $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{acc}$, which implies $x \in X_{\mathrm{acc}}$, so the first output rule holds with $R = X_{\mathrm{acc}}$; and $(x, 1\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{V}(x,z) = \mathrm{rej}$, which implies $x \in X_{\mathrm{rej}}$, hence $x \notin X_{\mathrm{acc}}$ by the consistency of $f_{\mathcal{V}}$, so the second holds. $\Gamma_{f_{\mathcal{V}}}$ holds since $X_{\mathrm{acc}} \cap X_{\mathrm{rej}} = \emptyset$. The second model is checked in the same way. The two models give that $T^{\mathsf{C}}_{\mathcal{V}} \vdash R(t_w)$ only if $w \in X_{\mathrm{acc}}$ and $T^{\mathsf{C}}_{\mathcal{V}} \vdash \neg R(t_w)$ only if $w \in X_{\mathrm{rej}}$; the converses are (iii).

(ii) For values of its variables in $\mathrm{Std}^{\mathfrak{U}} = \mathfrak{T}$ (Lemma F.10(i)), each member of $T^{O}_{\mathcal{V}}$ says what the corresponding member of $T^{\mathsf{C}}_{\mathcal{V}}$ says with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ (Lemma F.10(iii)); terms built from words by $\mathtt{s}_0, \mathtt{s}_1$ stay words and are evaluated as in $\mathfrak{T}$. For values outside $\mathfrak{T}$ a Std premise fails. The Std axioms hold by Lemma F.10(i).

(iii) Let $\sigma_0, \ldots, \sigma_{N_{\mathrm{run}}}$ be the run from $\sigma_0 = (t_w, \mathrm{tg}\,\mathrm{st}(\mathrm{start})\,t_z)$. Instantiate the halting axiom at $\sigma_{N_{\mathrm{run}}}$ (two A4 instances and two modus ponens steps) to get $\mathsf{C}(\sigma_{N_{\mathrm{run}}})$; for $i = N_{\mathrm{run}} - 1, \ldots, 0$, instantiate the rule that applies to σ_i at its values (one A4 instance and one modus ponens step per variable) and apply modus ponens to $\mathsf{C}(\sigma_{i+1})$, getting $\mathsf{C}(\sigma_i)$; finally instantiate the output rule at $x := t_w$, $z := t_z$ and apply modus ponens. In $T^{O}_{\mathcal{V}}$ the instantiated members carry Std premises

for the values, all words: derive $\mathrm{Std}(t_v)$ for every suffix v of every component from $\mathrm{Std}(\mathsf{e})$ and the propagation axioms (one A4 instance and two modus ponens steps per letter, keeping the set of words with a derived Std fact closed under suffixes, each step adding $O(1)$ letters), and discharge them by modus ponens. Every line is a member, an instance of a member at the values of one pair on the run, or an atom about such a pair, so it has size $O(\ell + 1)$.

(iv) For a decidable X take $\mathcal{V}(x, z) :=$ the decision of X on x (ignoring z). Under (a2) and (b2) every nonlogical instance is a member of the finite theory, so the budget needed is a constant; data below it are listed as table literals, of size $|\varphi_w|$ or $|\varphi_w| + 1$, which keeps the theory finite and true in the models above. Under (b) the material is at most the number of distinct instances, $O(N_{\mathrm{run}})$, times their size, $O(\ell)$; for $X \in \mathsf{NP} \cap \mathsf{coNP}$ combine polynomial-time verifiers for X and its complement into one consistent verifier, and the upper bound is that of all hypotheses (Corollary B.32).

Reading (a1). Step 1 (the machine). An alternating one-tape machine has states partitioned into existential and universal states and the halting states ACC and REJ, and a transition relation giving, for each non-halting state q and symbol $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, a list $\delta(q, a)$ of one or two moves $(q_i, b_i, \mathrm{dir}_i)$. A configuration *accepts* if its state is ACC, or it is existential and some successor accepts, or it is universal and all successors accept (least fixed point); when every computation path halts, the computation tree from a configuration is finite (König's lemma), acceptance is decided by induction on it, and a configuration that does not accept *rejects*. Normal form: (1) as in Lemma F.13(1) ($\mathtt{B}$ has the transitions of $\Box$, only symbols of $\Sigma_{\mathcal{M}}$ are written, a left move at cell 0 stays), so $\mathrm{hp} \le \mathrm{nb}$ on every path from a start tape; (2) $\delta(q, a)$ is nonempty for every non-halting q and every a (add a move to REJ at an existential state, and to ACC at a universal state, where none exists; this does not change acceptance); (3) every computation path from a start configuration (tape $\#x$, head on cell 0, state q_0) halts. Such a machine exists for every $X \in \mathsf{EXPTIME}$. By $\mathsf{EXPTIME} = \mathsf{APSPACE}$ (Chandra et al., 1981), X is decided by a multitape alternating machine $\mathcal{M}$ in polynomial space $S(m)$. *Clock.* $\mathcal{M}$ has at most $N_{\mathrm{cf}}(m) = 2^{c_{\mathcal{M}}(S(m)+\log m)}$ configurations on inputs of length m. Let A_j be the set of configurations with an accepting computation tree of height at most j; $A_j \subseteq A_{j+1}$, the accepting configurations are $\bigcup_j A_j$, and the sequence stabilises after at most $N_{\mathrm{cf}}(m)$ strict increases, so every accepting configuration has an accepting tree of height at most $N_{\mathrm{cf}}(m)$. Add a binary step counter of $c_{\mathcal{M}}(S(m) + \log m) + 1$ bits on an extra tape, and go to REJ when it overflows. If $\mathcal{M}$ accepts x, the clocked machine follows an accepting tree of height at most $N_{\mathrm{cf}}(m)$ and accepts; if the clocked machine accepts, its accepting tree contains no overflow (overflow leads to REJ, which an accepting tree does not contain) and is an accepting tree of $\mathcal{M}$. Every path of the clocked machine halts. *One tape.* Simulate the tapes on tracks of one tape, each step of the clocked machine by a deterministic sequence of steps (existential states with one successor) that ends with the branching step; the computation trees correspond, so acceptance is preserved and all paths halt, in space $O(S + \log m)$. The normal form (1)–(2) is a local change. Call the result $\mathcal{M}_1$, and split moves so that $|\delta(q, a)| \le 2$.

Step 2 (the compile). Fix codes st (states of $\mathcal{M}_1$ and the phase states $\mathrm{start}, \mathrm{inp}, \mathrm{rew}$) and blk as in Lemma F.13(2). For each tag, with $\mathrm{cd}(q) := \mathrm{tg}\,\mathrm{st}(q)$:

- *start:* $(Z_r;\ \mathrm{cd}(\mathrm{start})Z_l) \Rightarrow (Z_r;\ \mathrm{cd}(\mathrm{inp})\,\mathrm{blk}(\#)Z_l)$: the separator goes onto l, and there is no certificate phase;
- *phase inp:* $(aZ_r;\ \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (Z_r;\ \mathrm{cd}(\mathrm{inp})\mathrm{blk}(a)Z_l)$ for $a \in \{0, 1\}$, and $(\varepsilon \,|\, \mathrm{exact};\ \mathrm{cd}(\mathrm{inp})Z_l) \Rightarrow (\varepsilon \,|\, \mathrm{exact};\ \mathrm{cd}(\mathrm{rew})Z_l)$;
- *phase rew:* $(Z_r;\ \mathrm{cd}(\mathrm{rew})\mathrm{blk}(e)Z_l) \Rightarrow (\mathrm{blk}(e)Z_r;\ \mathrm{cd}(\mathrm{rew})Z_l)$ for $e \in \{\#, 0, 1\}$, and

$(Z_r;\,\mathrm{cd}(\mathrm{rew})\,|\,\mathrm{exact}) \Rightarrow (Z_r;\,\mathrm{cd}(q_0)\,|\,\mathrm{exact})$;

- *the machine:* for each non-halting q and $a \in \Sigma_{\mathcal{M}} \cup \{\Box\}$, with read pattern ρ and remainder ρ' as in Lemma F.13(3), the *neighbour cases* are $\nu \in \{\mathrm{blk}(e)Z_l : e \in \Sigma_{\mathcal{M}}\} \cup \{|\,\mathrm{exact}\}$ if some move in $\delta(q,a)$ goes left, and the single case $\nu = Z_l$ otherwise. For each case the current pattern is $\pi := (\rho;\,\mathrm{cd}(q)\nu)$, and each move $(q_i, b_i, \mathrm{dir}_i)$ gives a next pattern π_i': $(\rho';\,\mathrm{cd}(q_i)\mathrm{blk}(b_i)\nu)$ for R; $(\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)\nu)$ for S; $(\mathrm{blk}(e)\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)Z_l)$ for L when $\nu = \mathrm{blk}(e)Z_l$; and $(\mathrm{blk}(b_i)\rho';\,\mathrm{cd}(q_i)\,|\,\mathrm{exact})$ for L at the left end. The rule is *conjunctive*, $\pi \Rightarrow \{\pi_1', \pi_2'\}$ with backward form $\forall Z_r \forall Z_l\,(\mathsf{C}(\pi_1') \to (\mathsf{C}(\pi_2') \to \mathsf{C}(\pi)))$, when all successors are needed: at universal states for tag 0 and at existential states for tag 1. Otherwise there is one rule $\pi \Rightarrow \pi_i'$ per move, with backward form $\forall Z_r \forall Z_l\,(\mathsf{C}(\pi_i') \to \mathsf{C}(\pi))$; a single move gives a single rule;
- *halting axioms:* $\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{acc}} Z_l)$ and $\forall Z_r \forall Z_l\,\mathsf{C}(Z_r, H_{\mathrm{rej}} Z_l)$;
- *exact output rules:* $\forall x\,(\mathsf{C}(x,\, 0\,\mathrm{st}(\mathrm{start})) \to R(x))$ and $\forall x\,(\mathsf{C}(x,\, 1\,\mathrm{st}(\mathrm{start})) \to \neg R(x))$, where $\mathrm{tg\,st}(\mathrm{start})$ stands for the closed term $t_{\mathrm{tg\,st}(\mathrm{start})}$. In the observational version they are $\forall x\,(\mathrm{Std}(x) \to (\theta_2(x, t_{0\,\mathrm{st}(\mathrm{start})}) \to R(x)))$ and its rejecting twin.

Call the result $\mathcal{A}$, $T_{\mathcal{A}}^{\mathsf{C}}$ its backward theory (rules, halting axioms, exact output rules) and $T_{\mathcal{A}}^{O}$ its observational translation (Definition F.12). The current patterns of distinct cases (state, read symbol, neighbour case, tag) are pairwise disjoint, by the argument of Lemma F.13(i); the two rules of a non-conjunctive branching share one current pattern.

Step 3 (the meaning of C *at reachable pairs).* Let $\mathsf{C}^{\mathfrak{T}}$ be the least set of pairs of words that contains the pairs with a halting prefix and is closed under the backward rules (a conjunctive rule adds π when all its next pairs are in, the others when their one next pair is in). Let Reach be the set of pairs reachable by the rules from the *exact* start pairs $(x,\, \mathrm{tg\,st}(\mathrm{start}))$, x a word. Every pair in Reach is a phase pair of the run from a start pair, or the code $[\![q, a, \mathrm{hp}]\!]_{\mathrm{tg}}$ of a configuration of $\mathcal{M}_1$ reachable from $\#x$: the start rule gives $(x,\, \mathrm{cd}(\mathrm{inp})\mathrm{blk}(\#))$, phase inp gives $(\varepsilon,\, \mathrm{cd}(\mathrm{rew})\mathrm{blk}(x_{|x|}) \cdots \mathrm{blk}(x_1)\mathrm{blk}(\#))$, phase rew gives $(\mathrm{blk}(\#)\mathrm{blk}(x_1) \cdots \mathrm{blk}(x_{|x|}),\, \mathrm{cd}(q_0)) = [\![q_0, \#x, 0]\!]_{\mathrm{tg}}$, and from a configuration code the patterns that match are exactly those of its case, whose next pairs are the codes of its successors (the step argument of Lemma F.13). For $\sigma \in$ Reach let $\mathrm{acc}(\sigma)$ mean that the configuration of σ accepts (for a phase pair: the start configuration it leads to accepts). *Claim: for* $\sigma \in$ Reach *with tag* 0, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff* $\mathrm{acc}(\sigma)$*; with tag* 1, $\sigma \in \mathsf{C}^{\mathfrak{T}}$ *iff not* $\mathrm{acc}(\sigma)$. ($\Leftarrow$, tag 0.) By induction on the height of the finite computation tree of σ, phase steps counting as deterministic steps: a halting accepting configuration has prefix H_{acc}, so it is in the base; at an existential configuration some successor accepts and the rule for that move puts σ in; at a universal one all successors accept and the conjunctive rule puts σ in; a deterministic or phase step uses its single rule. ($\Rightarrow$, tag 0.) The set $S_0 := \{\sigma \in \mathrm{Reach} : \text{tag } 0,\ \mathrm{acc}(\sigma)\} \cup \{0\text{-tagged pairs outside Reach}\} \cup \{\text{all 1-tagged pairs}\}$ contains the base (a reachable pair with prefix H_{acc} is a configuration in state ACC) and is closed under the backward rules. Take a rule instance with current pair σ whose next pairs (all of them for a conjunctive rule, its one next pair otherwise) lie in S_0. If σ is 1-tagged or outside Reach, $\sigma \in S_0$. Otherwise $\sigma \in$ Reach is 0-tagged, the rule belongs to the case of σ, and its next pairs are codes of successors of σ, hence in Reach and 0-tagged, hence accepting. A conjunctive rule at tag 0 sits at a universal or deterministic configuration, all of whose successors are then accepting; a non-conjunctive one sits at an existential configuration with an accepting successor. Either way $\mathrm{acc}(\sigma)$, so $\mathsf{C}^{\mathfrak{T}} \subseteq S_0$. (Tag 1.) The same, with acceptance and rejection, existential and universal exchanged: at tag 1 the conjunctive rules sit at existential states. So for the exact start pairs, $(x, 0\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $\mathcal{M}_1$ accepts $\#x$ iff $x \in X$, and $(x, 1\,\mathrm{st}(\mathrm{start})) \in \mathsf{C}^{\mathfrak{T}}$ iff $x \notin X$.

Step 4 (theories, truth, derivations). By Step 3, $(\mathfrak{T}, \mathsf{C}^{\mathfrak{T}}, R := X) \models T^{\mathsf{C}}_{\mathcal{A}} \cup \Gamma_{f_X}$: the rules by closure, the halting axioms by the base, the exact output rules by the Claim. So $T^{\mathsf{C}}_{\mathcal{A}}$ is about the named objects, and it proves φ^b_w only if $b = \mathbf{1}[w \in X]$. In $\mathfrak{U}$ put $R(t_w) := \mathbf{1}[w \in X]$, $R(\mathrm{pr}(a,b)) := \mathbf{1}[(a,b) \in \mathsf{C}^{\mathfrak{T}}]$ for words a, b, and R false on other pairs; as in (ii), $\mathfrak{U} \models T^{Q}_{\mathcal{A}} \cup \Gamma_{f_X}$ (the ground term $t_{\mathrm{tg\,st(start)}}$ of the exact output rule is a word, so $\theta_2(x, t_{\mathrm{tg\,st(start)}})$ holds iff $(x, \mathrm{tg\,st(start)}) \in \mathsf{C}^{\mathfrak{T}}$). For $w \in X$ fix an accepting computation tree of $\mathcal{M}_1$ on $\#w$ (all successors at universal nodes, one accepting successor at existential nodes) and the phase pairs before it; derive $\mathsf{C}(\sigma)$ for its nodes bottom-up, a leaf by the halting axiom (two A4 instances, two modus ponens steps), an inner node by the instance of its rule at its values and one modus ponens step per premise; then apply the exact output rule at $x := t_w$ and modus ponens. Rejection is the same with a rejecting tree and tag 1. In $T^{Q}_{\mathcal{A}}$ the Std premises are discharged as in (iii). The number of lines may be exponential, but every line is a member, an instance of a member at the values of one reachable pair, or an atom about one, so every line, logical instances included, has size $O(S_1 + |w|)$, a polynomial. That is reading (a1) at a polynomial budget; the upper bound $\subseteq \mathsf{EXPTIME}$ is Corollary B.32. □

Why the output rules must be exact for alternating programs. Without a certificate phase nothing consumes the variable z of the output rules of Definition F.12. Phase rew rewinds every block-aligned z onto the tape, so $\mathcal{M}_1$ runs on $z'\#x$, a tape about which the normal form says nothing. If $\mathcal{M}_1$ has, at q_0, transitions on the letters 0 and 1 (legal, since cell 0 of a start tape always holds #), the theory with free-z output rules proves both $R(t_x)$ and $\neg R(t_x)$ for every x. A computation confirms this on three machines with such transitions (checked derivations of $R(t_{0110})$ and $\neg R(t_{0110})$), and confirms the exact rules on the same machines. In Lemma F.13 phase cert consumes z, so the free-z output rules are correct there.

Relativisation to Std **is needed.** In $\mathfrak{U}$ every element is in the range of $\mathtt{s}_0$ and $\mathtt{s}_0$ is not injective, so unrelativised patterns uZ_l match pairs, and $u\,\mathrm{pr}(a,b)$ can be a word with the wrong prefix. A computation on the backward theory of a decider for the words with as many 0s as 1s, with $\mathsf{C}(a,b)$ read as $R(\mathrm{pr}(a,b))$ in $\mathfrak{U}$, finds all 3017 instances at word values true and 318 of 12519 instances at pair values false. So without Std the structure $\mathfrak{U}$ is not a model; whether the unrelativised theory is consistent we have not examined.

Computations. For that decider (140 rules) and a verifier whose accepting certificate is a position (274 rules), the compilation has no non-linear rule, no overlapping current patterns (checked symbolically, on all pairs of words) and no rule in a halting state, and the runs from all start pairs checked ($|x| \le 10$, $|z| \le 3$, respectively $|x| \le 6$, $|z| \le 5$) end as specified. All 511 data with $|w| \le 8$ have derivations from the relation version (144 sentences) and the observational version (147 sentences), checked by an independent derivation checker; corrupted derivations and a corrupted machine are rejected. For an alternating machine for the words in which every 1 has a 0 to its right, and three further machines with unreachable transitions out of q_0 (one universal and one existential with left moves, one deterministic that writes and reads B), all data with $|x| \le 6$ have checked derivations in both versions, and the largest line grows linearly in $|x|$, as reading (a1) requires. An independent implementation agrees.

F.5 Vocabulary or ontology

Residuals and the relational form. For $X \subseteq \{0,1\}^*$ and a word v, $X_v := \{u : uv \in X\}$. In the term model $R(t_{uv}) = R(u\,t_v)$, so X_v is what the subtree below t_v sees. X is regular iff it

has finitely many residuals: the X_v are the right quotients of X, that is, the reversals of the left quotients of the reversal of X, and the Myhill–Nerode theorem applies to the reversal (standard). Let $\mathcal{L}^{\mathrm{rel}}$ have a unary Root, binary $\mathrm{Gr}_0, \mathrm{Gr}_1$, unary $R, P_1, \ldots, P_k$ and unary $\mathrm{Cst}_1, \ldots, \mathrm{Cst}_j$. A structure $\mathfrak{A}$ for $\mathcal{L}_O \cup \{P_i\} \cup \{\mathtt{c}_i\}$ gives $\mathfrak{A}^{\mathrm{rel}}$ with Root $= \{\mathtt{e}\}$, Gr_a the graph of $\mathtt{s}_a$ ($\mathrm{Gr}_a(x, y)$ iff $\mathtt{s}_a x = y$) and $\mathrm{Cst}_i = \{\mathtt{c}_i\}$; every sentence ψ has a relational translation ψ^{rel}, obtained by unnesting terms, with $\mathfrak{A} \models \psi$ iff $\mathfrak{A}^{\mathrm{rel}} \models \psi^{\mathrm{rel}}$ (standard). $\equiv_q$ is agreement on all sentences of quantifier rank at most q. Over a finite relational signature there are finitely many $\equiv_q$-classes, and $\equiv_q$ holds iff the duplicator wins the q-round Ehrenfeucht–Fraïssé game (Ehrenfeucht, 1961; Fraïssé, 1954); see Ebbinghaus and Flum (1999) for the textbook form. In Theorem F.16, ncl_q is the number of $\equiv_q$-classes of $\mathcal{L}^{\mathrm{rel}} \cup \{\mathrm{Rt}, \mathrm{Att}\}$-structures, Rt and Att unary.

Lemma F.15 (Composition)**.** *(a) For relational structures in one finite signature, $\mathfrak{A}_1 \equiv_q \mathfrak{B}_1$ and $\mathfrak{A}_2 \equiv_q \mathfrak{B}_2$ imply $\mathfrak{A}_1 \sqcup \mathfrak{A}_2 \equiv_q \mathfrak{B}_1 \sqcup \mathfrak{B}_2$, where $\sqcup$ is the disjoint union with each relation interpreted as the union. (b) If $\mathfrak{A}, \mathfrak{B}$ arise from $\mathfrak{A}', \mathfrak{B}'$ (on the same domains) by defining each relation by a quantifier-free formula, then $\mathfrak{A}' \equiv_q \mathfrak{B}'$ implies $\mathfrak{A} \equiv_q \mathfrak{B}$.*

Proof. (a) The duplicator answers a move in a component with her winning strategy for that component's game. Within a component the chosen tuples form a partial isomorphism; no relation holds across components on either side, and equality across components fails on both sides. (b) Substituting the defining formulas turns a rank-q sentence about $\mathfrak{A}$ into a rank-q sentence about $\mathfrak{A}'$. □

Theorem F.16 (Unary theoretical vocabulary sees only regular regularities)**.** *Let $B = \emptyset$, and let T be a finite set of sentences in $\mathcal{L}_O$ plus unary predicate symbols $P_1, \ldots, P_k$ and constants $\mathtt{c}_1, \ldots, \mathtt{c}_j$ ($k, j \geq 0$), compatible with the named structure. If $T \vdash \varphi_w^{\mathbf{1}[w \in X]}$ for every word w, then X is regular. Quantitatively, X has at most $\mathrm{ncl}_q + |\mathrm{Suf_c}|$ residuals, where q is the quantifier rank of the relational translation of $\bigwedge T$, ncl_q the number of classes of rank-q elementary equivalence in the relational signature defined above, and $\mathrm{Suf_c}$ the set of suffixes of the words w_i with $\mathtt{c}_i = t_{w_i}$ in a fixed model of $T \cup \mathrm{Th}_{\mathfrak{T}}$, together with the empty word.*

In outline: by $\mathrm{Th}_{\mathfrak{T}}$ a single edge joins the subtree below t_v to the rest of a model, and unary symbols are labels; so by the composition lemma (Lemma F.15) the labels of one subtree can be replaced by those of another of the same rank-q type without changing the truth of T. As T proves the literals, the two residuals coincide.

Proof. Take $\mathfrak{A} \models T \cup \mathrm{Th}_{\mathfrak{T}}$. The closed terms denote pairwise distinct elements, and $R^{\mathfrak{A}}(t_w) = \mathbf{1}[w \in X]$ because T proves the literals. For a word $v \neq \varepsilon$, $v = a\hat{v}$ with a a letter, let $\mathrm{Sub}_v := \{t_{uv} : u \in \{0,1\}^*\}$ (standard elements), let $\mathfrak{S}_v$ be the restriction of $\mathfrak{A}^{\mathrm{rel}}$ to Sub_v with a new unary $\mathrm{Rt} := \{t_v\}$ (and $\mathrm{Att} := \emptyset$), and let $\mathfrak{C}_v$ be the restriction to the rest of the domain with a new unary $\mathrm{Att} := \{t_{\hat{v}}\}$ (and $\mathrm{Rt} := \emptyset$).

One edge between the parts. Sub_v is closed under $\mathtt{s}_0, \mathtt{s}_1$, and $\mathtt{e} \notin \mathrm{Sub}_v$, as $\mathtt{e}$ is not in the range of $\mathtt{s}_0, \mathtt{s}_1$. If $\mathtt{s}_b y = t_{uv}$ with $y \notin \mathrm{Sub}_v$, then $u = \varepsilon$: otherwise $u = b'u'$ and $t_{uv} = \mathtt{s}_{b'} t_{u'v}$, so $b = b'$ by disjoint ranges and $y = t_{u'v} \in \mathrm{Sub}_v$ by injectivity, both in $\mathrm{Th}_{\mathfrak{T}}$. So $\mathtt{s}_b y = t_v = \mathtt{s}_a t_{\hat{v}}$, hence $b = a$ and $y = t_{\hat{v}}$. Hence $\mathfrak{A}^{\mathrm{rel}}$ is obtained from $\mathfrak{C}_v \sqcup \mathfrak{S}_v$ by redefining $\mathrm{Gr}_a(x, y)$ as $\mathrm{Gr}_a(x, y) \vee (\mathrm{Att}(x) \wedge \mathrm{Rt}(y))$ and forgetting Att and Rt, a quantifier-free redefinition.

The subtrees are alike up to labels. For $v, v' \notin \mathrm{Suf_c}$ the map $t_{uv} \mapsto t_{uv'}$ is an isomorphism of the $\{\mathrm{Root}, \mathrm{Gr}_0, \mathrm{Gr}_1, \mathrm{Rt}, \mathrm{Cst}_1, \ldots, \mathrm{Cst}_j\}$-reducts of $\mathfrak{S}_v$ and $\mathfrak{S}_{v'}$: Root and the Cst_i are empty on both, because a constant lies in Sub_v only if it is some t_{w_i} with v a suffix of w_i, and nonstandard denotations lie in no Sub_v.

Transplanting. Suppose X has more than $\mathrm{ncl}_q + |\mathrm{Suf_c}|$ residuals. Choose one word per residual; more than ncl_q of them lie outside $\mathrm{Suf_c}$, so two of them, $v \neq v'$, have $X_v \neq X_{v'}$ and $\mathfrak{S}_v \equiv_q \mathfrak{S}_{v'}$. Let $\mathfrak{A}^*$ be $\mathfrak{A}$ with the labels on Sub_v replaced by those copied from $\mathrm{Sub}_{v'}$: $R^*(t_{uv}) := R^{\mathfrak{A}}(t_{uv'})$ and $P_i^*(t_{uv}) := P_i^{\mathfrak{A}}(t_{uv'})$, everything else unchanged (in particular the $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1\}$-reduct and the constants). Then $\mathfrak{S}_v^* \cong \mathfrak{S}_{v'} \equiv_q \mathfrak{S}_v$ and $\mathfrak{C}_v^* = \mathfrak{C}_v$, so by Lemma F.15, $(\mathfrak{A}^*)^{\mathrm{rel}} \equiv_q \mathfrak{A}^{\mathrm{rel}}$ and $\mathfrak{A}^* \models T$. Its reduct still satisfies $\mathrm{Th}_{\mathfrak{T}}$, so, as T proves the literals, $R^*(t_{uv}) = \mathbf{1}[uv \in X]$ for all u. But $R^*(t_{uv}) = \mathbf{1}[uv' \in X]$. Hence $X_v = X_{v'}$, a contradiction. Overlapping subtrees, $\mathrm{Sub}_v \supset \mathrm{Sub}_{v'}$, are harmless, since the labels are copied from the original structure. □

Unary predicates and constants are labels: they do not connect a subtree to its context, so the composition lemma applies. A binary theoretical relation can connect any two subtrees (Theorem F.14(i)); objects that violate $\mathrm{Th}_{\mathfrak{T}}$ (pairs, with $\mathtt{s}_1\mathtt{s}_1 z = z$) can carry information between subtrees through $\mathtt{s}_0$, which is not injective on them (Theorem F.14(ii)); and a finite set is essential, since infinite sets of padded literals are true of the named objects and fit P (Theorem F.6). The proof uses no tree automata and allows nonstandard elements and constants. Theorem F.16 is in the spirit of the reported example of Craig and Vaught that unary additional predicates do not suffice in general, in a different setting; we have not established that it is new.

Theorem F.17 (Every regular regularity has a short observational theory)**.** *For every regular X there is a finite $T \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, such that every datum has a K-derivation from T with $O(|w|)$ lines, each of size $O(|w|)$, the constants depending on X. So the budgets needed are linear under (a1); constant under (a2) and (b2) (at least the largest member, respectively the total size of T; with smaller budgets, a finite table of short data as in Theorem F.14(iv)); and quadratic under (b).*

Proof. Let k_X be the number of residuals of X, and write $A/a := \{u : ua \in A\}$, so $X_v/a = X_{av}$. Let $\approx_j$ be the equivalence "agree on all words of length less than j" on the finite set $\{X_v\}$. Then $A \approx_{j+1} A'$ iff $(\varepsilon \in A \Leftrightarrow \varepsilon \in A')$ and $A/a \approx_j A'/a$ for $a = 0, 1$, so $\approx_{j+1}$ is a fixed function of $\approx_j$; once $\approx_j = \approx_{j+1}$ the sequence is constant; $\approx_0$ has one class and each strict refinement adds a class; so $\approx_{k_X - 1}$ is equality. Hence the window $\{0,1\}^{<\mathrm{wd}_X}$ with $\mathrm{wd}_X := \max(k_X - 1, 1)$, or any smaller window that contains ε and separates the residuals, determines X_v. Put $\mathrm{sig}(v) := X_v \cap \{0,1\}^{<\mathrm{wd}_X}$; then $\mathrm{sig}(av)$ is a function $\sigma \mapsto \sigma_a$ of $\mathrm{sig}(v)$. With $\mathrm{Sg}_\sigma(x) := \bigwedge_{|u|<\mathrm{wd}_X} R(ux)^{\sigma(u)}$ let

$$T := \{\mathrm{Sg}_{\mathrm{sig}(\varepsilon)}(\mathtt{e})\} \cup \{\forall x\, (\mathrm{Sg}_\sigma(x) \to \mathrm{Sg}_{\sigma_a}(\mathtt{s}_a x)) : \sigma \text{ realised},\ a \in \{0,1\}\}.$$

$\mathrm{Sg}_\sigma(t_v)$ holds in $(\mathfrak{T}, X)$ iff $\mathrm{sig}(v) = \sigma$, so T is true there. Derive $\mathrm{Sg}_{\mathrm{sig}(v)}(t_v)$ for the suffixes v of w from the inside out (one A4 instance and modus ponens steps per letter) and read off the conjunct $u = \varepsilon$, which is $R(t_w)^{\mathbf{1}[w \in X]}$. In a curried form with one literal per conclusion only A4 and modus ponens occur. □

The bound $\max(k_X - 1, 1)$ on the window is attained: on random automata the least separating window equals $k_X - 1$ in many cases (90 of 1500 random automata with $k_X \geq 3$, and never above $\max(k_X - 1, 1)$; a computation, which also checks the theories of two regular languages on all data with $|w| \leq 9$, and three more, in an independent implementation, on all data with $|w| \leq 8$).

Corollary F.18 (Vocabulary or ontology)**.** *On literal sequences with $B = \emptyset$ (membership is linear-time for finite sets), the fit classes of finite theories are:*

finite theories	*(a2), (b2)*†	*(b): polynomial*	*(a1): polynomial*
$\mathcal{L}_O$ *+ unary predicates + constants, compatible with the named structure*	*regular*	*regular*	*regular*
$\mathcal{L}_O$ *+ one binary relation, about the named objects*	*all decidable*	NP ∩ coNP	EXPTIME
$\mathcal{L}_O$ *only, quantifying over unnamed objects*	*all decidable*	NP ∩ coNP	EXPTIME
all finite theories, any vocabulary	*all decidable*	NP ∩ coNP	EXPTIME

(†*at every nondecreasing unbounded budget with* $h(m) \geq m+1$*.) The theories of row 3 are inconsistent with* $\mathrm{Th}_{\mathfrak{T}}$ *(Theorem F.16).*

Proof. Row 1: Theorem F.16 gives the upper bound (at any budget, and also with free deduction) and Theorem F.17 the lower bound. Rows 2 and 3: Theorem F.14. Row 4: the upper bounds are those of all hypotheses (Corollary B.32), the lower bounds rows 2 and 3. □

Corollary F.19 (With free deduction too)**.** *Let* $B = \emptyset$*. For a decidable non-regular* X*, let* $K_{\mathrm{fin}}^{\mathrm{named}}(D)$ *be the least* $K(\psi)$ *over single sentences* ψ *in* $\mathcal{L}_O$ *plus any finitely many unary predicates and constants that are compatible with the named structure and prove* ψ_D*. Then* $K_{\mathrm{fin}}^{\mathrm{named}}(D_n^X) \to \infty$*, while* $K_{\mathrm{fin}}^{\mathcal{L}_O \cup \{\mathsf{C}\}}(D_n^X) \leq K(\mathcal{V}) + c$ *and* $K_{\mathrm{fin}}^{\mathcal{L}_O}(D_n^X) \leq K(\mathcal{V}) + c$ *for a decider* $\mathcal{V}$ *of* X*.*

Proof. By Theorem F.16 no admissible sentence proves all of D^X; each fails some datum, and only finitely many sentences have $K \leq j$. The upper bounds are given by the conjunctions of $T_{\mathcal{V}}^{\mathsf{C}}$ and $T_{\mathcal{V}}^{O}$ (Theorem F.14), computable from $\mathcal{V}$. □

The upper bound with C is an effective instance of the theorem of Kleene and of Craig and Vaught (Kleene, 1952; Craig and Vaught, 1958), as stated in Pakhomov and Visser (2019): the axioms of $\mathrm{Th}_{\mathfrak{T}}$ listed after Definition F.1 together with Γ_{f_X} form an r.e. theory without finite models, which therefore has a finitely axiomatised extension with one additional binary relation symbol into which every model of it, $(\mathfrak{T}, X)$ included, expands.

Proposition F.20 (A forced ontological commitment)**.** *Let* $B = \emptyset$*. (a) A finite theory in* $\mathcal{L}_O$ *plus unary predicates and constants that proves every literal datum of a non-regular* X *proves some* $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$*-sentence false in the named structure. (b) For a verifier* $\mathcal{V}$*,* $T_{\mathcal{V}}^{O} \vdash \exists z\, \mathsf{s}_1\mathsf{s}_1 z = z$*, and* $T_{\mathcal{V}}^{O} \vdash \neg\, \mathsf{s}_1\mathsf{s}_1 t_w = t_w$ *for every word* w*. So* $T_{\mathcal{V}}^{O} \cup \mathrm{Th}_{\mathfrak{T}}$ *is inconsistent:* $T_{\mathcal{V}}^{O}$ *is consistent with all the data but proves an observational sentence that is false of every named object and refutable from the observational theory of the named structure. In particular* $T_{\mathcal{V}}^{O}$ *is not conservative over* $\Gamma_{f_{\mathcal{V}}}$*, although* $\Gamma_{f_{\mathcal{V}}} \cup \mathrm{Th}_{\mathfrak{T}}$ *is consistent.*

Proof. (a) By Theorem F.16 the theory is inconsistent with $\mathrm{Th}_{\mathfrak{T}}$, so by compactness it proves the negation of some $\{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1\}$-sentence true in $\mathfrak{T}$. (b) In a model of $T_{\mathcal{V}}^{O} \cup \{\forall z\, \neg\, \mathsf{s}_1\mathsf{s}_1 z = z\}$, Std holds everywhere and every $\theta_2(a, b)$ holds vacuously, so the two output rules give $\forall x\, R(x)$ and $\forall x\, \neg R(x)$; no such model exists. $\mathrm{Std}(t_w)$ follows from $\mathrm{Std}(\mathsf{e})$ and the propagation axioms, and $\forall z\, \neg\, \mathsf{s}_1\mathsf{s}_1 z = z \in \mathrm{Th}_{\mathfrak{T}}$. Finally $(\mathfrak{T}, X_{\mathrm{acc}}) \models \mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$. □

A computation confirms that without pairs the two output rules of the observational theory cannot both hold (for 200 of 200 random interpretations of R).

Proposition F.21 (Conservativity of the relation version)**.** *Let* $B = \emptyset$*. For a verifier* $\mathcal{V}$*,* $T_{\mathcal{V}}^{\mathsf{C}}$ *is conservative over* $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$*: for every* $\mathcal{L}_O$*-sentence* χ*,* $T_{\mathcal{V}}^{\mathsf{C}} \cup \mathrm{Th}_{\mathfrak{T}} \vdash \chi$ *iff* $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}} \vdash \chi$*.*

Proof. ($\Leftarrow$) $T^{\mathsf{C}}_{\mathcal{V}} \vdash \Gamma_{f_{\mathcal{V}}}$ by Theorem F.14. ($\Rightarrow$) Every model $\mathfrak{M}$ of $\mathrm{Th}_{\mathfrak{T}} \cup \Gamma_{f_{\mathcal{V}}}$ expands to a model of $T^{\mathsf{C}}_{\mathcal{V}}$. Let $\mathsf{C}^{\mathfrak{M}}$ be the least subset of M^2 that contains the pairs with a halting prefix and is closed under the backward rules, interpreted in $\mathfrak{M}$. The rules and halting axioms hold by construction. In $\mathfrak{M}$ the current patterns are still pairwise disjoint and the values (Z_r, Z_l) of a match are unique, since disjointness and injectivity are first-order consequences of $\mathrm{Th}_{\mathfrak{T}}$ (for instance $p\,Z \neq p'\,Z'$ for incomparable words p, p'); so, as in Theorem F.14(i), $\mathsf{C}^{\mathfrak{M}}$ is the set of pairs whose deterministic run in $\mathfrak{M}$ reaches a halting prefix. Consider a start pair $(x, \mathrm{tg\,st}(\mathrm{start})\,z)$. If z is nonstandard, phase cert never ends: a nonstandard $z \neq \mathtt{e}$ is $\mathtt{s}_a z'$ for some letter a (a sentence of $\mathrm{Th}_{\mathfrak{T}}$) with z' nonstandard, so each step leaves a nonstandard remainder. If z is standard and x nonstandard, phase inp never ends, for the same reason. If both are standard, the run stays in the standard part, a copy of $\mathfrak{T}$, and is the run of Lemma F.13. So $(x, 0\,\mathrm{st}(\mathrm{start})z) \in \mathsf{C}^{\mathfrak{M}}$ implies $x = t_w$ with $\mathcal{V}(w, z') = \mathrm{acc}$ for some z', hence $\mathfrak{M} \models R(t_w)$ by $\Gamma_{f_{\mathcal{V}}}$; likewise for rejection. Completeness gives the claim. □

The design of Lemma F.13, which reads both raw words to their ends before the machine starts, is what makes this work. Proposition F.21 is the conservative-extension form of the theorem of Kleene and of Craig and Vaught for this particular theory, with a derivation-bounded extension.

Proposition F.22 (Richer observational languages)**.** *(a) Let $B = \emptyset$, and let the observational language have constants $\mathtt{e}_1, \ldots, \mathtt{e}_{r_0}$, unary function symbols $\mathtt{f}_1, \ldots, \mathtt{f}_{r_1}$ and unary observable predicates $R_1, \ldots, R_{r_2}$, the named structure being the free term algebra $\mathfrak{T}'$ on these constants and function symbols and the data all literals $R_i(\tau)^{\pm}$, τ closed. If a finite theory in this language plus unary theoretical predicates and constants is compatible with $\mathrm{Th}(\mathfrak{T}')$ and proves every datum, then each R_i has finitely many residuals $\{u \in \{1, \ldots, r_1\}^* : R_i(\mathtt{f}_u\tau)\}$ (τ closed); in particular $\{u : R_i(\mathtt{f}_u\mathtt{e}_l)\}$ is regular for each root $\mathtt{e}_l$. (b) Let the observational language be $\{\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, R, J\}$ with J a binary observable relation, the data being all literals of R and J on closed terms, and let the world be $J := \{(t_u, t_v) : \#_0(u) = \#_1(v)\}$, $R := \mathrm{EQ} := \{w : \#_0(w) = \#_1(w)\}$, which is not regular. The 13 observational sentences $J(\mathtt{e}, \mathtt{e})$; $\forall y\, \neg J(\mathtt{e}, \mathtt{s}_1 y)$; $\forall x\, \neg J(\mathtt{s}_0 x, \mathtt{e})$; the four sentences $\forall y\, (J(\mathtt{e}, y) \to J(\mathtt{e}, \mathtt{s}_0 y))$, $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_1 y))$, $\forall x \forall y\, (J(\mathtt{s}_0 x, y) \to J(\mathtt{s}_0 x, \mathtt{s}_0 y))$, $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$, each also with both J-atoms negated; and $\forall x\, (J(x, x) \to R(x))$, $\forall x\, (\neg J(x, x) \to \neg R(x))$ are true in $(\mathfrak{T}, R, J)$, so they form a theory about the named objects with no theoretical vocabulary, and they derive every datum with $O(|u| + |v|)$ lines. So with a binary observable, finite observational theories about the named objects fit a non-regular target.*

Proof. (a) Use, from $\mathrm{Th}(\mathfrak{T}')$: each $\mathtt{f}_i$ is injective, the ranges of distinct $\mathtt{f}_i$ are disjoint, no root is in any range, and distinct closed terms are distinct. For a non-root closed term $\tau = \mathtt{f}_a\tau'$ put $\mathrm{Sub}_\tau := \{\mathtt{f}_u\tau : u \in \{1, \ldots, r_1\}^*\}$. By these facts an element $y \notin \mathrm{Sub}_\tau$ with $\mathtt{f}_b y \in \mathrm{Sub}_\tau$ must be τ' with $b = a$ (the one-edge argument in the proof of Theorem F.16, with $\{0, 1\}$ replaced by $\{1, \ldots, r_1\}$), so $\mathfrak{A}^{\mathrm{rel}}$ is $\mathfrak{C}_\tau \sqcup \mathfrak{S}_\tau$ with one quantifier-free redefined edge. The exceptional set is the set of closed subterms of the denotations of the constants, together with the roots (roots are never the top of a Sub_τ); for τ, τ' outside it, $\mathfrak{S}_\tau$ and $\mathfrak{S}_{\tau'}$ are isomorphic up to labels, possibly under different roots. The transplant, which copies all labels $R_1, \ldots, R_{r_2}, P_1, \ldots$ at once, then bounds the number of distinct residuals over all closed terms, for each R_i, by ncl_q plus the size of the exceptional set. (b) Truth: $\#_0$ and $\#_1$ change as the letters say. Every pair (u, v) is $(\varepsilon, \varepsilon)$, $(\varepsilon, 1v')$, $(\varepsilon, 0v')$, $(0u', \varepsilon)$, $(1u', v)$ (v arbitrary, including ε), $(0u', 1v')$ or $(0u', 0v')$, and each case is an instance of a sentence or reduces by one sentence, in its positive or negative form, to a pair of smaller total length (for $(1u', v)$ the sentence $\forall x \forall y\, (J(x, y) \to J(\mathtt{s}_1 x, y))$ at $y := t_v$). The two R-sentences finish. □

A computation finds the 13 sentences true for $|x|, |y| \leq 6$, and checked derivations of all 3969 J-literals with $|u|, |v| \leq 5$ and all 1023 R-literals with $|w| \leq 9$; the windowed residual count of J grows ($5, 7, 9$ at depths $2, 3, 4$).

Naming composites without observing them. If the observational language has a binary function symbol $\mathtt{p}$ but the data concern only the words t_w, then $T_{\mathcal{V}}^{\mathsf{C}}$ with every atom $\mathsf{C}(a, b)$ replaced by $R(\mathtt{p}(a, b))$, with no relativisation, is about the named objects (the term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1, \mathtt{p}$, with $R(\mathtt{p}(a, b)) := \mathbf{1}[(a, b) \in \mathsf{C}^{\mathfrak{T}}]$ for words a, b, false for other composites, and $R := X_{\mathrm{acc}}$ on words), and fits every decidable X under (a2) and (b2). In the free term algebra matching is syntactic, and a start pair with a non-word component gets stuck when phase cert or phase inp reaches the $\mathtt{p}$-term, since no rule matches a term whose head is $\mathtt{p}$; so the output rules hold as in Theorem F.14. Theorem F.16 thus depends on the composites being unnamed, or, if named, observed.

Question F.23 (Binary observables)**.** What is the fit class of finite observational theories about the named objects when the observational language contains a binary observable relation J, as in Proposition F.22(b)? Is it every decidable target? The answer must depend on the observable: with $J = \emptyset$ it is the class of regular sets, by the proof of Theorem F.16.

Proposition F.24 (Reading (a1) for infinite hypotheses about the named objects)**.** *Let $X \in$ EXPTIME be* EXPTIME*-hard under polynomial-time many-one reductions. There is a parameter-free set $A \subseteq \mathrm{Sent}_{\mathcal{L}_O}$, true in $(\mathfrak{T}, X)$, with polynomial-time membership, such that every datum $\varphi_w^{\mathbf{1}[w \in X]}$ has a derivation from A by modus ponens and instances of the axiom scheme A1 of* K *in which every line has size polynomial in $|w|$. Hence, at polynomial budgets under (a1), the fit class of observational hypotheses about the named objects contains every* EXPTIME*-hard set in* EXPTIME*, besides* NP $\cap$ coNP.

Proof. Let $\mathcal{M}$ be an alternating polynomial-space machine for X with all paths halting (Step 1 of the proof of Theorem F.14(iv)), and let Acc $\in$ EXPTIME be the set of pairs (w, ς) with ς an accepting configuration of $\mathcal{M}$ on w. Let red be a polynomial-time reduction of Acc (pairs coded as words) to X; since every configuration accepts or rejects, ς rejects iff $\mathrm{red}(w, \varsigma) \notin X$. For a word u let θ_u be the valid certificate sentence of Theorem B.31 encoding u, and $\langle \cdot \rangle$ a pairing of words. A consists of, for all w and configurations ς of $\mathcal{M}$ on w: $\theta_{\langle w,\varsigma \rangle} \to R(t_{\mathrm{red}(w,\varsigma)})$ for halting accepting ς and $\theta_{\langle w,\varsigma \rangle} \to \neg R(t_{\mathrm{red}(w,\varsigma)})$ for halting rejecting ς; at an existential ς with successors ς_1, ς_2, $\theta_{\langle w,\varsigma,\varsigma_i \rangle} \to (R(t_{\mathrm{red}(w,\varsigma_i)}) \to R(t_{\mathrm{red}(w,\varsigma)}))$ for $i = 1, 2$ and $\theta_{\langle w,\varsigma,\varsigma_1,\varsigma_2 \rangle} \to (\neg R(t_{\mathrm{red}(w,\varsigma_1)}) \to (\neg R(t_{\mathrm{red}(w,\varsigma_2)}) \to \neg R(t_{\mathrm{red}(w,\varsigma)})))$, and at a universal ς the duals; and $\theta_{\langle w \rangle} \to (R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to R(t_w))$ and $\theta_{\langle w \rangle} \to (\neg R(t_{\mathrm{red}(w,\mathrm{init}(w))}) \to \neg R(t_w))$. Every member is true in $(\mathfrak{T}, X)$, since θ is valid and $R(t_{\mathrm{red}(w,\varsigma)})$ says that ς accepts. Membership: read the coded tuple off the θ-part, recompute the red-values and check the transition relation, in polynomial time. A datum is derived along an accepting (rejecting) computation tree: each step cites a member, the valid sentence θ_u (an A1 instance), and modus ponens; every line has polynomial size, as configurations and red-values do. □

For infinite observational axiom sets true of the named objects, with polynomial membership, the fit class at polynomial budgets is exactly NP $\cap$ coNP under (b): certificate sets are true of the named objects, since θ_γ is valid, and the class of all hypotheses bounds it (Corollary B.32). Under (a1) it lies between the classes of Proposition F.24 and EXPTIME.

Question F.25 (Reading (a1) for infinite hypotheses)**.** Is the (a1) fit class of infinite observational hypotheses about the named objects, at polynomial budgets, all of EXPTIME? For an X that is neither easy nor hard, the facts about the named objects that a derivation can use are the values of X itself and sentences about $(\mathfrak{T}, X)$, and it is not clear that they can store an exponential computation. If they cannot, theoretical vocabulary or entities would be needed under (a1) even for infinite hypotheses.

Question F.26 (A unary theoretical function; soft elimination)**.** (a) Does one unary theoretical function symbol over the named objects suffice for every decidable X? A function is not a label, so the proof of Theorem F.16 does not apply; a two-stack configuration cannot be read off a single word by prefix patterns, so the construction of Theorem F.14 does not transfer. (b) Is there a soft version of Proposition F.9, with rate factor 1? Soft axiom induction and soft certificate induction admit no two-sided constant-regret correspondence at any pair of rates (Proposition B.33), which suggests that the answer is no.

F.6 Finite theories with free deduction

In this subsection $B = \emptyset$, and $K^{\mathcal{L}}_{\mathrm{fin}}$, $W^{\mathcal{L}}_{\mathrm{fin}}$ are the finite-axiomatisation versions of Definition 2.7 in the language $\mathcal{L}$. Recall $K^{\mathrm{pad}}(n) = \min_{m \geq n} K(m)$, which is nondecreasing and unbounded. $\mathcal{L}_{\mathrm{mon}}$, $\mathcal{L}_f$, Inf_n, D_n and Dinf_f are as in Section 8.5: $\mathcal{L}_{\mathrm{mon}} := \{=\} \cup \{P_i : i \in \mathbb{N}\}$, $\mathcal{L}_f := \mathcal{L}_{\mathrm{mon}} \cup \{f\}$ with f unary, $\mathrm{Inf}_n := \exists x_1 \cdots \exists x_n \bigwedge_{i<j} x_i \neq x_j$, $D_n := \{(\mathrm{Inf}_m, 1) : m \leq n\}$, so that $\psi_{D_n} \equiv \mathrm{Inf}_n$, and $\mathrm{Dinf}_f := \forall x \forall y\, (fx = fy \to x = y) \wedge \exists y \forall x\, fx \neq y$.

Lemma F.27 (A quantitative finite model property of monadic logic)**.** *If $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$ uses k predicate symbols, has quantifier rank q and has a model, then it has a model with at most $2^k \max(q, 1)$ elements. For $k = 0$, all sets of size at least $\max(q, 1)$ satisfy the same rank-q sentences.*

Proof. Let $\mathfrak{M} \models \psi$ and $q' := \max(q, 1)$. Partition $\mathfrak{M}$ into the 2^k cells Cell_ϵ, $\epsilon \in \{0,1\}^k$, by the values of the k predicates, and let $\mathfrak{M}' \subseteq \mathfrak{M}$ keep $\min(|\mathrm{Cell}_\epsilon|, q')$ elements of each cell; $\mathfrak{M}' \neq \emptyset$. The duplicator wins the q-round game on $(\mathfrak{M}, \mathfrak{M}')$, keeping the invariant "chosen pairs lie in the same cell, and $a_i = a_j \iff b_i = b_j$": a repeated element is answered by its partner, and a fresh element in cell ϵ by a fresh element of that cell on the other side, which exists. If $|\mathrm{Cell}_\epsilon| \geq q'$, both sides have at least $q' \geq q$ elements of the cell and fewer than q are chosen; if $|\mathrm{Cell}_\epsilon| < q'$, both sides have exactly $|\mathrm{Cell}_\epsilon|$ elements of the cell, and the chosen ones are matched bijectively. The invariant gives a partial isomorphism for $=$ and the P_i, so $\mathfrak{M}' \equiv_q \mathfrak{M}$ and $\mathfrak{M}' \models \psi$. □

This is a quantitative form of the finite model property of monadic first-order logic, whose decidability goes back to Löwenheim (1915) and Behmann (1922). A computation by exact Ehrenfeucht–Fraïssé games confirms that truncation at $\max(q, 1)$ preserves $\equiv_q$ in all 185 structures tried, that truncation at $q - 1$ breaks it in all 40 tried, and that the bound $2^k q$ is attained.

Theorem F.28 (Infinity needs a theoretical function)**.** *Let $\mathrm{mb}(\psi) := 2^{k(\psi)} \max(q(\psi), 1)$ for $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$, with $k(\psi)$ the number of predicate symbols in ψ and $q(\psi)$ its quantifier rank. (a) If $\psi \in \mathrm{Sent}_{\mathcal{L}_{\mathrm{mon}}}$ is consistent and $\psi \vdash \mathrm{Inf}_n$, then $n \leq \mathrm{mb}(\psi)$. No consistent finite monadic theory proves all of D_∞. (b) $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) = K^{\mathrm{pad}}(n) \pm c$. So $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) \to \infty$, and $K^{\mathcal{L}_{\mathrm{mon}}}_{\mathrm{fin}}(D_n) = o(u(n))$ for every computable unbounded nondecreasing $u : \mathbb{N} \to \mathbb{N}$. (c) Dinf_f is consistent and proves every Inf_n, so $K^{\mathcal{L}_f}_{\mathrm{fin}}(D_n) \leq K(\mathrm{Dinf}_f) + c$. (d) Under the prior $\mathbf{m}$ on $\mathrm{Sent}_{\mathcal{L}_f}$, the posterior probability, given D_n, that the finite hypothesis does not use f is at most $2^c \sum_{m \geq n} \mathbf{m}(m) / \mathbf{m}(\mathrm{Dinf}_f)$, which tends to 0. (e) $K^{\{=\}}_A(D_n) \leq c$, by the r.e. theory $\{\mathrm{Inf}_m : m \geq 1\}$.*

Proof. (a) By Lemma F.27, ψ has a model with at most $\mathrm{mb}(\psi)$ elements. (b) $\mathrm{mb}(\psi) \geq n$ is computable from ψ, so $K(\psi) \geq K(\mathrm{mb}(\psi)) - c \geq K^{\mathrm{pad}}(n) - c$. Conversely, Inf_m for $m \geq n$ is consistent, proves Inf_n and has $K(\mathrm{Inf}_m) \leq K(m) + c$. Divergence: K^{pad} is unbounded because only finitely many m have $K(m) \leq j$. Slow growth: for a computable unbounded nondecreasing u_1 let $h_{u_1}(j)$ be the least n' with $u_1(n') \geq j$; then $h_{u_1}(u_1(n)+1) > n$, so $K^{\mathrm{pad}}(n) \leq K(h_{u_1}(u_1(n)+1)) + c \leq 2\log(u_1(n)+1) + c'$. With $u_1 := \lfloor 2^{u/r} \rfloor$ this is at most $2u(n)/r + c''$ for every r, hence $o(u)$. (c) $(\mathbb{N}, \text{successor}) \models \mathrm{Dinf}_f$. In any model, with y_0 outside the range of f, the elements $y_0, fy_0, \ldots, f^{n-1}y_0$ are distinct: $f^i y_0 = f^j y_0$ with $i < j$ gives $y_0 = f^{j-i}y_0$ by injectivity, which lies in the range. (d) The pushforward of $\mathbf{m}$ under the computable partial map $\psi \mapsto \mathrm{mb}(\psi)$ (on $\mathcal{L}_{\mathrm{mon}}$) is at most $2^c\mathbf{m}$; compatible monadic sentences have $\mathrm{mb}(\psi) \geq n$; and the compatible mass with f is at least $\mathbf{m}(\mathrm{Dinf}_f)$. (e) is immediate. □

A computation confirms that Dinf_f has no model with at most 7 elements.

Remark F.29 (The same with named objects). In the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ (R unary, c_j constants), with $B = \emptyset$, the data sets $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$ have $K_A(D_n) \leq c$ and $K_{\mathrm{fin}}(D_n) = K^{\mathrm{pad}}(n) \pm c$. So $K_{\mathrm{fin}} - K_A$ is unbounded, and finite axiomatisability is a different simplicity measure.

Proof. $T := \{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$ is r.e. with an index of complexity $O(1)$ and is consistent (domain $\mathbb{N} \cup \{*\}$, $c_j \mapsto j$, $R := \mathbb{N}$), so $K_A(D_n) \leq c$. *Upper bound on* K_{fin}: for $N \geq n$ with $K(N) = K^{\mathrm{pad}}(n)$, the sentence $\bigwedge_{j<N} R(c_j) \wedge \exists x\, \neg R(x)$ is consistent, proves ψ_{D_n}, and has complexity at most $K(N) + c$. *Lower bound*: let ψ be consistent with $\psi \vdash \psi_{D_n}$. If some c_j, $j < n$, does not occur in ψ, then $\psi \vdash R(c_j)$ and the theorem on constants give $\psi \vdash \forall x\, R(x)$, which contradicts $\psi \vdash \exists x\, \neg R(x)$ and the consistency of ψ. So ψ mentions $c_0, \ldots, c_{n-1}$, and $1 + \max\{j : c_j \text{ occurs in } \psi\} \geq n$ is computable from ψ; hence $K(\psi) \geq K^{\mathrm{pad}}(n) - c$. □

Proposition F.30 (The cost of "at least n things" in symbols and in bits)**.** *Let $|\psi|$ be the number of symbols of ψ (each P_i one symbol), and let ψ range over* consistent *sentences with $\psi \vdash \mathrm{Inf}_n$. (a) In pure equality, $q(\psi) \geq n$ for $n \geq 2$, so $|\psi| \geq n$; Inf_n itself has $\Theta(n^2)$ symbols. (b) For monadic ψ, $|\psi| \geq \log n - \log\log n - O(1)$, and $|\psi| = O(\log n)$ is attained by the* carry counter

$$\mathrm{Cnt}'_k := \forall x\, \mathrm{Cr}_0 x \;\wedge\; \bigwedge_{i<k} \forall x\, (\mathrm{Cr}_{i+1}x \leftrightarrow (\mathrm{Cr}_i x \wedge P_i x)) \;\wedge\; \exists x \bigwedge_{i<k} \neg P_i x \;\wedge\; \forall x \Big(\neg \mathrm{Cr}_k x \to \exists y \bigwedge_{i<k} (P_i y \leftrightarrow (P_i x \oplus \mathrm{Cr}_i x)) \Big),$$

with $\mathrm{Cr}_0, \ldots, \mathrm{Cr}_k$ further unary predicates, which holds exactly in the monadic structures in which the Cr_i are the carries of the P-vector and all 2^k cells of $P_0, \ldots, P_{k-1}$ are nonempty; so $\mathrm{Cnt}'_{\lceil \log n \rceil} \vdash \mathrm{Inf}_n$. (c) With f, $O(1)$ symbols suffice (Dinf_f). (d) Code sentences in Polish notation, symbol by symbol, with a prefix-free code on the symbols in which each of the finitely many logical symbols and variables used costs between 1 and c_0 bits and P_i costs $\ell(i)$ bits, with ℓ the length function of a prefix-free code on $\mathbb{N}$ with $\ell(i) \leq 2\log(i+1) + O(1)$. Then the codes of sentences form a prefix-free set, so $\sum_\psi 2^{-\mathrm{bits}(\psi)} \leq 1$ and $2^{-\alpha\,\mathrm{bits}(\psi)}$, $\alpha \geq 1$, is a sub-probability on sentences. In this code pure equality costs at least n bits, f costs $O(1)$ bits, and monadic sentences cost $\Theta(\log n \cdot \log\log n)$ bits.

Proof. (a) Let $\mathfrak{M} \models \psi$; then $|\mathfrak{M}| \geq n$. If $|\mathfrak{M}| \geq \max(q, 1)$, Lemma F.27 with $k = 0$ makes every set of size $\max(q, 1)$ a model of ψ, so $\max(q, 1) \geq n$; otherwise $\max(q, 1) > |\mathfrak{M}| \geq n$. Either way $q \geq n$ when $n \geq 2$, and each quantifier is a symbol. (b) *Lower bound*: Theorem F.28(a) gives $n \leq 2^k \max(q, 1)$ with $k, q \leq |\psi|$. *Correctness of* Cnt'_k: the definitional conjuncts force

$\mathrm{Cr}_i x \leftrightarrow \bigwedge_{j<i} P_j x$, the carry into bit i of the binary increment. If all P-vectors are realised, the remaining conjuncts hold, since every non-maximal vector has its increment. Conversely, the vector 0 is realised, and if the vector of value $v < 2^k - 1$ is realised, so is that of $v+1$ ($\neg\mathrm{Cr}_k x$ says that the vector of x is not all ones, and bit i of $v+1$ is $P_i \oplus \mathrm{Cr}_i$); by induction all 2^k vectors are realised, so there are at least 2^k elements. Each conjunct indexed by i has constant size. (c) Theorem F.28(c). (d) *Prefix-freeness*: a sentence in Polish notation is determined, symbol by symbol, by its own beginning, so no sentence code is a proper prefix of another. *Lower bound, monadic*: a consistent $\psi \vdash \mathrm{Inf}_n$ with k distinct predicate symbols and rank q contains at least q quantifier symbols, of at least one bit each, and k distinct predicate codewords. Sort their lengths $\ell_{(1)} \le \cdots \le \ell_{(k)}$; by Kraft's inequality $j\,2^{-\ell_{(j)}} \le \sum_{i\le j} 2^{-\ell_{(i)}} \le 1$, so $\ell_{(j)} \ge \log j$ and the predicate codewords cost at least $\log k! \ge k\log k - k\log e$ bits. With $n \le 2^k \max(q,1)$: if $k \ge \frac{1}{2}\log n$, the cost is at least $\frac{1}{2}\log n\,(\log\log n - 1) - \frac{\log e}{2}\log n$; if $k < \frac{1}{2}\log n$, then $q \ge n/2^k > \sqrt{n}$. Either way $\mathrm{bits}(\psi) = \Omega(\log n \log\log n)$. *Upper bound*: Cnt'_k with $k = \lceil \log n \rceil$, its $2k+1$ predicates numbered $1, \ldots, 2k+1$, has $O(k)$ symbols of at most $\max(c_0, 2\log(2k+2) + O(1))$ bits each, so $O(\log n \log\log n)$ bits. *Pure equality*: at least $q \ge n$ symbols of at least one bit. *With f*: Dinf_f is a fixed sentence. □

So the single best theory with f outweighs the best monadic theory by a factor $n^{\Theta(\alpha)}$ and the best pure-equality theory by $2^{\Omega(\alpha n)}$ under the weight $2^{-\alpha|\psi|}$, and by $2^{\Theta(\alpha \log n \log\log n)}$ and $2^{\Omega(\alpha n)}$ under the normalisable weight of (d); with each P_i one symbol, the weight $2^{-\alpha|\psi|}$ is not normalisable, so the first comparison is between single theories, not under a prior. A computation gives $64k + 13$ symbols for Cnt'_k in one convention (linear in k in three conventions), checks that Cnt'_k holds iff all cells are nonempty on all 6560 carry-labelled structures with $k = 3$ and cell counts at most 2, and finds the ratios of the lower and upper bit bounds to $\log n \log\log n$ bounded for $n = 2^8, \ldots, 2^{1024}$. The counter is elementary, and we make no novelty claim for it.

Theorem F.31 (The theorems of PA as data)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{ar}}$ and let D_∞ list the theorems of* PA*, labelled true, in the order of an effective enumeration of* PA*-derivations, so that every axiom of* PA *occurs. (a) $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to \infty$ and $W_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \to 0$; and $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}}(D_n) \le K(n) + c$. (b) $K_A^{\mathcal{L}_{\mathrm{ar}}}(D_n) \le K(\mathsf{PA}) + c = O(1)$. (c) In the three-sorted extension $\mathcal{L}_{\mathrm{ar}}^+$ of Theorem F.33, $K_{\mathrm{fin}}^{\mathcal{L}_{\mathrm{ar}}^+}(D_n) \le c$, and the posterior over finite $\mathcal{L}_{\mathrm{ar}}^+$-theories puts mass tending to 1 on theories that use the new sorts. The winning theory adds no $\mathcal{L}_{\mathrm{ar}}$-content: its $\mathcal{L}_{\mathrm{ar}}$-consequences are exactly those of* PA*.*

Proof. We use the theorem of Ryll-Nardzewski (1952): no consistent extension of PA in $\mathcal{L}_{\mathrm{ar}}$ is finitely axiomatisable; equivalently, no consistent $\mathcal{L}_{\mathrm{ar}}$-sentence proves every axiom of PA. (a) So every consistent ψ fails to prove some axiom α_ψ of PA, which occurs in D_∞, and ψ is incompatible with D_n for large n. Only finitely many ψ have $K(\psi) \le j$, which gives $K_{\mathrm{fin}} \to \infty$, and dominated convergence ($\sum \mathbf{m} \le 1$) gives $W_{\mathrm{fin}} \to 0$. The conjunction of the first n data is consistent, given $\mathrm{Con}(\mathsf{PA})$, and computable from n. (b) PA is r.e. (c) Theorem F.33 with $\mathsf{W}_e = \mathsf{PA}$ gives a single $\mathcal{L}_{\mathrm{ar}}^+$-sentence Fin_e whose $\mathcal{L}_{\mathrm{ar}}$-consequences are those of PA; with (a), the posterior mass of the $\mathcal{L}_{\mathrm{ar}}$-sentences tends to 0 while that of Fin_e is a fixed positive number. □

A classical instance of (c) is ACA_0, in the two-sorted language of second-order arithmetic, which is finitely axiomatisable and has the same first-order consequences as PA (see Simpson, 2009).

Remark F.32 (A truth predicate for PA). A theoretical *predicate* on the same domain also appears to give $O(1)$: $\mathrm{I}\Sigma_1$ plus compositional truth clauses for a unary predicate Tr plus $\forall\varphi\,(\text{PA-Ax}(\varphi) \to \mathrm{Tr}(\varphi))$ is a finite theory, true in $\mathbb{N}$, that proves every axiom of PA.

Proof sketch. We do not write out the derivation of the Tarski biconditionals from the clauses over $\mathrm{I}\Sigma_1$. For each standard sentence α the theory proves $\mathrm{Tr}(\ulcorner\alpha\urcorner) \leftrightarrow \alpha$ by external induction on α from the compositional clauses, using that $\mathrm{I}\Sigma_1$ proves the needed facts about the coding (that numerals evaluate correctly needs Σ_1-induction); with $\mathrm{Tr}(\ulcorner\alpha\urcorner)$ for every axiom α of PA this proves every axiom. $\mathrm{I}\Sigma_1$ is finitely axiomatisable (standard; see Hájek and Pudlák, 1993). So in arithmetic a unary theoretical predicate appears to suffice, because the domain already carries $+$ and $\cdot$, which code syntax; in pure equality unary predicates never help (Theorem F.28(a)), nor over observation sequences (Theorem F.16). We do not claim that unary predicates help only where the domain codes syntax.

Theorem F.33 (r.e. axiom induction is finite-theory induction with theoretical sorts)**.** *Let $\mathcal{L}$ be a finite language (relation, function and constant symbols, with equality) and $B \subseteq \mathrm{Sent}_{\mathcal{L}}$ decidable. Let $\mathcal{L}^+$ be the three-sorted extension (many-sorted first-order logic, every sort with a nonempty domain) with a sort o carrying $\mathcal{L}$; a sort n with $0, S, +, \cdot, <$; and a sort a of assignments, with* $\mathrm{val} : \mathsf{a} \times \mathsf{n} \to \mathsf{o}$, $\mathrm{upd} : \mathsf{a} \times \mathsf{n} \times \mathsf{o} \to \mathsf{a}$, $\mathrm{den} : \mathsf{n} \times \mathsf{a} \to \mathsf{o}$ *and a relation* $\mathrm{Sat} \subseteq \mathsf{n} \times \mathsf{a}$*. There is a computable injective map $e \mapsto \mathrm{Fin}_e$ from indices to single $\mathcal{L}^+$-sentences such that for every e*

$$\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\mathrm{Fin}_e\} \vdash \chi\} = \mathrm{Cn}(\mathsf{W}_e),$$

both sides being inconsistent together. Consequently, for every finite data set D in $\mathcal{L}$, $K^{\mathcal{L}^+}_{\mathrm{fin}}(D) = K^{\mathcal{L}}_A(D) \pm c$ and $2^{-c}W^{\mathcal{L}}_A(D) \le W^{\mathcal{L}^+}_{\mathrm{fin}}(D) \le 2^c W^{\mathcal{L}}_A(D)$.

Proof. Construction. Fix a Gödel numbering of the $\mathcal{L}$-terms and $\mathcal{L}$-formulas (variables $x_0, x_1, \dots$) and Σ_1 formulas of $\mathcal{L}_{\mathrm{ar}}$ that define in $\mathbb{N}$ the syntactic relations: $\mathrm{Var}(c, v)$, "c codes x_v"; for each function symbol g of arity k (constants: $k = 0$), $\mathrm{App}_{\mathsf{g}}(c, d_1, \dots, d_k)$, "the d_i code terms τ_i and c codes $\mathsf{g}(\tau_1, \dots, \tau_k)$"; for each relation symbol r, $\mathrm{At}_{\mathsf{r}}(c, \vec{d})$, and $\mathrm{Eq}(c, d, d')$, for atomic formulas over coded terms; $\mathrm{Neg}(c, d)$, "d codes a formula χ and c codes $\neg\chi$"; $\mathrm{Imp}(c, d, d')$; $\mathrm{All}(c, v, d)$, "d codes χ and c codes $\forall x_v \chi$"; and $\mathrm{Enum}_e(c)$, "$c \in \mathsf{W}_e$ and c codes an $\mathcal{L}$-sentence". The bounded quantifiers in these formulas use $<$. Fin_e is the conjunction of:

- (Ar) Robinson's Q on the sort n, including the defining axiom $\forall x \forall y\,(x < y \leftrightarrow \exists z\,(Sz + x = y))$, so that Q proves every true Σ_1 sentence whose bounded quantifiers use $<$ (standard; see Boolos et al., 2007);
- (Up) $\forall\alpha \forall v \forall x\, \mathrm{val}(\mathrm{upd}(\alpha, v, x), v) = x$ and $\forall\alpha \forall v \forall v' \forall x\,(v \neq v' \to \mathrm{val}(\mathrm{upd}(\alpha, v, x), v') = \mathrm{val}(\alpha, v'))$;
- (Den) $\forall c \forall v\,(\mathrm{Var}(c, v) \to \forall\alpha\, \mathrm{den}(c, \alpha) = \mathrm{val}(\alpha, v))$ and, for each g, $\forall c \forall \vec{d}\,(\mathrm{App}_{\mathsf{g}}(c, \vec{d}) \to \forall\alpha\, \mathrm{den}(c, \alpha) = \mathsf{g}(\mathrm{den}(d_1, \alpha), \dots, \mathrm{den}(d_k, \alpha)))$;
- (Sat) for each r, $\forall c \forall \vec{d}\,(\mathrm{At}_{\mathsf{r}}(c, \vec{d}) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \mathsf{r}(\mathrm{den}(d_1, \alpha), \dots)))$; the same for Eq with $=$; $\forall c \forall d\,(\mathrm{Neg}(c, d) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \neg\mathrm{Sat}(d, \alpha)))$; the same for Imp with $\to$; and $\forall c \forall v \forall d\,(\mathrm{All}(c, v, d) \to \forall\alpha\,(\mathrm{Sat}(c, \alpha) \leftrightarrow \forall x\, \mathrm{Sat}(d, \mathrm{upd}(\alpha, v, x))))$, with x of sort o;
- (Ax) $\forall c\,(\mathrm{Enum}_e(c) \to \forall\alpha\, \mathrm{Sat}(c, \alpha))$, with e written as a numeral, so that $e \mapsto \mathrm{Fin}_e$ is injective.

($\supseteq$) Let $\varphi \in \mathsf{W}_e$. Q proves the true Σ_1 sentence $\mathrm{Enum}_e(\underline{\ulcorner\varphi\urcorner})$, so $\mathrm{Fin}_e \vdash \forall\alpha\, \mathrm{Sat}(\underline{\ulcorner\varphi\urcorner}, \alpha)$. By external induction on terms τ and formulas χ whose variables are among $x_{v_1}, \dots, x_{v_j}$: $\mathrm{Fin}_e \vdash \forall\alpha\, \mathrm{den}(\underline{\ulcorner\tau\urcorner}, \alpha) = \tau^\alpha$ and $\mathrm{Fin}_e \vdash \forall\alpha\,(\mathrm{Sat}(\underline{\ulcorner\chi\urcorner}, \alpha) \leftrightarrow \chi^\alpha)$, where $^\alpha$ replaces each x_v by $\mathrm{val}(\alpha, \underline{v})$. Each step instantiates one clause at the numerals of the immediate subterms or subformulas, whose guard is a true Σ_1 sentence and hence provable in Q. The quantifier step uses (Up) and $\mathsf{Q} \vdash \underline{v} \neq \underline{v'}$ for $v \neq v'$, which give $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v}) = x$ and $\mathrm{val}(\mathrm{upd}(\alpha, \underline{v}, x), \underline{v'}) = \mathrm{val}(\alpha, \underline{v'})$,

so the induction hypothesis at $\mathrm{upd}(\alpha, \underline{v}, x)$ turns $\forall x\, \mathrm{Sat}(\ulcorner \underline{\chi} \urcorner, \mathrm{upd}(\alpha, \underline{v}, x))$ into $(\forall x_v \chi)^\alpha$. Extra, nonstandard witnesses of the guards in other models only add constraints and cannot block these derivations. For the sentence φ, $\varphi^\alpha = \varphi$, and the sort a is nonempty, so $\mathrm{Fin}_e \vdash \varphi$. ($\subseteq$) Let $\mathfrak{M}$ be an $\mathcal{L}$-structure with $\mathfrak{M} \models B \cup \mathsf{W}_e$. Expand it: the sort n is $\mathbb{N}$, with the true $<$; the sort a is $M^{\mathbb{N}}$, with $\mathrm{val}(\alpha, v) := \alpha(v)$ and $\mathrm{upd}(\alpha, v, x) := \alpha[v \mapsto x]$; $\mathrm{den}(c, \alpha)$ is the value under α of the term coded by c (a fixed element if c codes no term); and $\mathrm{Sat}(c, \alpha)$ holds iff c codes a formula χ with $\mathfrak{M} \models \chi[\alpha]$. The guards are evaluated in $\mathbb{N}$, where the Σ_1 formulas mean what they say, so each clause holds (the guards of Neg, Imp and All require the components to code formulas, so no clause compares a non-formula with a formula), and (Ax) holds because $\mathfrak{M} \models \mathsf{W}_e$. Thus every model of $B \cup \mathsf{W}_e$ expands to a model of $B \cup \{\mathrm{Fin}_e\}$, and by completeness every $\mathcal{L}$-consequence of $B \cup \{\mathrm{Fin}_e\}$ is in $\mathrm{Cn}(\mathsf{W}_e)$. If $B \cup \mathsf{W}_e$ is inconsistent, so is $B \cup \{\mathrm{Fin}_e\}$, by ($\supseteq$); if it is consistent, a model expands, so $B \cup \{\mathrm{Fin}_e\}$ is consistent. *Complexities.* $K(\mathrm{Fin}_e) \leq K(e) + c$, and Fin_e is compatible with D iff e is. Conversely, the map $\chi \mapsto \mathrm{ct}(\chi)$, an index of $\{\varphi \in \mathrm{Sent}_{\mathcal{L}} : B \cup \{\chi\} \vdash \varphi\}$, is computable; $\mathrm{ct}(\chi)$ is compatible with D iff χ is (the set contains $\neg \forall x\, x = x$ iff $B \cup \{\chi\}$ is inconsistent); and $K(\mathrm{ct}(\chi)) \leq K(\chi) + c$. For the weights, $\mathbf{m}(\mathrm{Fin}_e) \geq 2^{-c}\mathbf{m}(e)$ because $e \mapsto \mathrm{Fin}_e$ is computable and injective, and $\sum_{\chi : \mathrm{ct}(\chi) = j} \mathbf{m}(\chi) \leq 2^c \mathbf{m}(j)$ because pushforwards of $\mathbf{m}$ under computable maps are dominated by $\mathbf{m}$. □

Theorem F.33 is an effective, many-sorted form of the theorem of Kleene (1952) and Craig and Vaught (1958) (as stated in Pakhomov and Visser, 2019). Using extra sorts to remove the condition that the theory have no finite models is routine; what we add is the Bayesian consequence, which needs an effective construction with an injective map and two-sided complexity bounds. We do not treat the one-sorted version.

Remark F.34 (Not one tape). Let $\mathcal{L}$ be finite and satisfy the hypotheses of Theorem 5.1, and for the universal theory machine of $\mathcal{L}^+$ let $M^{\mathrm{fin}}(D) := \mathbb{U}\{\omega : A_\omega \text{ is finite, } B \cup A_\omega \text{ is consistent and } \vdash \psi_D\}$. On the lottery stream D_∞ of Theorem 5.1, $M^{\mathrm{fin}}(D_n) \to 0$, while $M_A(D_n)$ is bounded below by a positive constant. So the equivalence of Theorem F.33 is not one-tape: random r.e. theories are strictly stronger than random finite theories with any amount of theoretical apparatus.

Proof. $\mathcal{L}^+$ is finite, so it has computable syntax, and for finite A_ω the $\mathcal{L}$-content $\{\chi \in \mathrm{Sent}_{\mathcal{L}} : B \cup A_\omega \vdash \chi\}$ is r.e. and, if $B \cup A_\omega$ is consistent, consistent. By Theorem 5.1(b) it does not prove every datum of D_∞. So the events $\{\omega : A_\omega \text{ finite, consistent with } B, \vdash \psi_{D_n}\}$ decrease in n to the empty set, and by continuity from above their measure tends to 0. The lower bound on $M_A(D_n)$ is Theorem 5.1(a). □

Proposition F.35 (Infinity: Ramsey, Craig and a theoretical function)**.** *In pure equality: (a) the Ramsey sentence $\exists f\, \mathrm{Dinf}_f$ of the theory $\{\mathrm{Dinf}_f\}$, a single existential second-order sentence with no theoretical constant, has in standard semantics exactly the infinite sets as models; (b) $\{\mathrm{Inf}_m : m \geq 1\}$ has exactly the infinite sets as models, and its consequences form a complete theory; (c) so $\exists f\, \mathrm{Dinf}_f$ and $\{\mathrm{Inf}_m : m \geq 1\}$ (a first-order observational axiomatisation, necessarily infinite by Theorem F.28(a)) have the same first-order $\{=\}$-consequences, and no finite first-order theory in $\{=\}$, nor in any monadic extension, has them; (d) the first-order theory $\{\mathrm{Dinf}_f\}$ in the extended language $\{=, f\}$ is finite and has them as its $\{=\}$-consequences.*

Proof. (a) If f is injective and not surjective on A, then A is infinite (Theorem F.28(c)); conversely an infinite A has a countable subset $\{a_0, a_1, \dots\}$ (countable choice), and the map $a_i \mapsto a_{i+1}$, the identity elsewhere, is injective and misses a_0. (b) The first claim is clear. Completeness: by Lemma F.27 with $k = 0$, all sets of size at least q satisfy the same rank-q sentences, so all infinite

sets are elementarily equivalent. (c), (d) The same models give the same first-order consequences; the rest is Theorem F.28(a), (c). □

A computation on 1200 random pure-equality sentences of rank 1 to 4 finds their truth constant on sets of sizes q to $q+2$.

Proposition F.36 (Factoring an inference graph)**.** *Let $\mathcal{L}$ contain unary predicates $\mathrm{A}_1, \dots, \mathrm{A}_{n_A}$ and $\mathrm{B}_1, \dots, \mathrm{B}_{n_B}$; let $T_{\mathrm{dir}} := \{\forall x\,(\mathrm{A}_i x \to \mathrm{B}_j x)\}_{i,j}$ and, in $\mathcal{L} \cup \{\mathrm{Mid}\}$, $T_{\mathrm{Mid}} := \{\forall x\,(\mathrm{A}_i x \to \mathrm{Mid}\,x)\}_i \cup \{\forall x\,(\mathrm{Mid}\,x \to \mathrm{B}_j x)\}_j$. An* atom-to-atom implication *is a sentence $\forall x\,(\mathrm{P}x \to \mathrm{P}'x)$ with P, P' distinct unary predicate symbols. (a) $\mathrm{Cn}(T_{\mathrm{Mid}}) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$. (b) Every theory of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j\}$ equivalent to T_{dir} has at least $n_A n_B$ members. (c) In full first-order logic, T_{dir} is equivalent to the single sentence $\forall x\,((\mathrm{A}_1 x \vee \cdots \vee \mathrm{A}_{n_A} x) \to (\mathrm{B}_1 x \wedge \cdots \wedge \mathrm{B}_{n_B} x))$, of length $O(n_A + n_B)$, so* Mid *saves only a constant factor there. (d) Every theory S of atom-to-atom implications over $\{\mathrm{A}_i, \mathrm{B}_j, \mathrm{Mid}\}$ with $\mathrm{Cn}(S) \cap \mathrm{Sent}_{\mathcal{L}} = \mathrm{Cn}(T_{\mathrm{dir}})$ has at least $\min(n_A n_B, n_A + n_B)$ members, and T_{dir} or T_{Mid} attains the bound. So the new predicate shortens the theory iff $(n_A - 1)(n_B - 1) > 1$, by $\max(0, n_A n_B - n_A - n_B)$ members.*

Proof. (a) $T_{\mathrm{Mid}} \vdash \mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$, and every model of T_{dir} expands to one of T_{Mid} with $\mathrm{Mid} := \bigcup_i \mathrm{A}_i \subseteq \bigcap_j \mathrm{B}_j$. (b) For a set S of atom-to-atom implications over any set of unary predicates, $S \vdash \forall x\,(\mathrm{P}x \to \mathrm{P}'x)$ iff P' is reachable from P in the graph S: if not, the one-element structure in which exactly the atoms reachable from P hold satisfies S, P and $\neg\mathrm{P}'$. In models of T_{dir} no $\mathrm{A}_i \to \mathrm{A}_{i'}$, $\mathrm{B}_j \to \mathrm{B}_{j'}$ ($i \neq i'$, $j \neq j'$) or $\mathrm{B}_j \to \mathrm{A}_i$ holds (take the one-element structures with only A_i true together with all B's, respectively only B_j true). So an equivalent S over $\{\mathrm{A}_i, \mathrm{B}_j\}$ contains only edges $\mathrm{A}_i \to \mathrm{B}_j$, and reachability from A_i to B_j then needs the edge itself: all $n_A n_B$ of them. (c) Propositional logic. (d) S proves every $\mathrm{A}_i \to \mathrm{B}_j$ and, by the equality of $\mathcal{L}$-consequences, none of the implications excluded in (b). Take a shortest path from A_i to B_j in the graph S; it has no repeated vertex. An interior vertex $\mathrm{A}_{i'}$ would give $S \vdash \mathrm{A}_i \to \mathrm{A}_{i'}$ ($i' \neq i$), and an interior $\mathrm{B}_{j'}$ would give $S \vdash \mathrm{B}_{j'} \to \mathrm{B}_j$ ($j' \neq j$). So the path is the edge $\mathrm{A}_i \to \mathrm{B}_j$ or the path $\mathrm{A}_i \to \mathrm{Mid} \to \mathrm{B}_j$. With $I_A := \{i : (\mathrm{A}_i \to \mathrm{Mid}) \in S\}$ and $I_B := \{j : (\mathrm{Mid} \to \mathrm{B}_j) \in S\}$, every pair outside $I_A \times I_B$ needs its own edge, so $|S| \geq |I_A| + |I_B| + n_A n_B - |I_A||I_B|$. This function is affine in each of $|I_A|$ and $|I_B|$, so its minimum over $[0, n_A] \times [0, n_B]$ is at a corner: $n_A n_B$, $n_A n_B + n_A$, $n_A n_B + n_B$ or $n_A + n_B$. Finally $n_A n_B - (n_A + n_B) = (n_A - 1)(n_B - 1) - 1$. □

A computation that decides the consequence relation exactly (a set of atom-to-atom implications says that the type of every element lies in a set of types, and the $\mathcal{L}$-reducts of its models are the structures whose types lie in the projection of that set) confirms the least sizes $n_A n_B$ without Mid for the nine pairs with $n_A n_B \leq 6$, and $\min(n_A n_B, n_A + n_B)$ with it for all ten pairs up to $(3, 3)$; Mid saves nothing except for $(2, 3)$, $(3, 2)$ (one member) and $(3, 3)$ (three members).

G Proofs for approximate laws and the best system

This appendix states in full the results of Section 9 that the main text states in words, and proves every result of that section. The notation is that of Definitions 9.1 and G.1. We measure a finite theory by $\|T\|$, without the stop bit of a prefix-free code of theories; using the prefix-free length $\|T\| + 1$ instead changes every statement below by an additive constant at most, and shifts J_λ by the same amount λ on every theory, so that it leaves the λ-MAPs unchanged. Computed numbers come from seeded scripts in the accompanying repository; they are never part of a statement.

G.1 The setting in full

Definition G.1 (The setting in full, continuing Definition 9.1)**.**

(a) *Answers and objectives.* $\operatorname{dom} D$ lists the questions of a data set D in a canonical order, and y_D their answers. $A_T(\mathcal{Q}) := \{a' \in \mathcal{Q} : B \cup T \nvdash \neg a'\}$ is the set of answers that T allows. J^Q_λ, J^F_λ and J^R_λ are J_λ with Exc^Q, Exc^F and Exc^R.

(b) *Interval laws.* B = RCF is decidable and complete (standard; Tarski, 1951), and R is rational. The total computable maps $\underline{f} \le \overline{f} : I \to \mathbb{Q} \cap [0, \mathsf{R}]$ of a pointwise interval law are named by a codeword of a fixed decidable prefix-free presentation language, and $\|T\|$ is the codeword's length. The honest law T^ε has $\underline{f} = f - \varepsilon$ and $\overline{f} = f + \varepsilon$, clipped to $[0, \mathsf{R}]$. The questions of interval data partition $[0, \mathsf{R})$ into finitely many cells with rational endpoints. With C_t the intersection of the cells that D assigns to the time t,

$$P^R_T(D) := \prod_t \frac{|C_t \cap [\underline{f}(t), \overline{f}(t)]|}{\overline{f}(t) - \underline{f}(t)}$$

($|\cdot|$ the length; the factor is $\mathbf{1}[\underline{f}(t) \in C_t]$ if $\underline{f}(t) = \overline{f}(t)$), the probability that values drawn uniformly and independently from the law's intervals lie in the cells of the data.

(c) *Regime G.* Datum j is "$q(t_j) \in [v_j - \delta, v_j + \delta]$", with $v_j = q(t_j) + \zeta_j$ rational and $|\zeta_j| \le \delta_0 \le \delta$.

(d) *Regime R.* q lives on a circle of circumference R. Datum j is about a *fresh* time t_j (the t_j form an injective computable sequence of numerals); the errors $e_j := e(t_j)$ are i.i.d.; and datum j is the cell of width Δ (R/Δ an integer) that contains $q(t_j)$ in a grid shifted by a dither u_j, uniform and independent of everything else.

Technical points. (i) *Rational cells.* In regime R the dither u_j is uniform on a grid $\{k/N_\mathrm{d} : 0 \le k < N_\mathrm{d}\}$, so that cell endpoints are rational. Every event used below has the form "u_j lies in an interval of length ℓ'", whose grid probability is within $1/N_\mathrm{d}$ of ℓ'; the formulas are the limit $N_\mathrm{d} \to \infty$ (equivalently, endpoints may be real algebraic numbers). A cell that crosses the zero point of the circle is a disjunction of two interval sentences. (ii) *Fresh times.* Data are sentences, so they mention numerals, a countable set; no atomless law lives on it. Times drawn i.i.d. from a law on numerals repeat: the number of distinct times among n draws is $o(n)$ almost surely (take a finite set of mass at least $1 - \epsilon$ and apply the strong law of large numbers). A repeat changes the cost of every form except an exact law: a patched law withdraws the law at a whole time, so every later datum at a refuted time is an exception, and the cells of an honest law at one time intersect, which narrows its band. A computation with times drawn uniformly from the 4383 days of a twelve-year window (sinusoidal error, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$, $c_\mathrm{time} = \log 4383 + 2$) gives a patched law's cost per datum of 0.40, 1.96 and 3.23 bits at $n = 10^3$, 10^5 and 2×10^5, against the rate 0.240 at fresh times, and a code-R cost of the honest law falling from 0.037 to 0.022, against 0.030. This is why regime R takes fresh times: an injective computable sequence of numerals, with i.i.d. errors and dithers. Fresh times have names that grow, and Proposition G.17 says what that costs. (iii) *Presentations.* No decidable set of programs covers all total computable maps, so the class of pointwise interval laws depends on the presentation language (for example, primitive recursive definitions, or programs paired with an explicit clock); every statement holds for each fixed choice. (iv) *A model of the data.* In regime R the error at a fresh time is a draw from a fixed law; for the Kepler residuals this law is the law of $e(t)$ for t uniform in the window. Answered data with a question $\mathcal{Q}$ are read through Definition 2.10: for consistent $B \cup T$, T proves $(\mathcal{Q}, a)$ iff

$A_T(\mathcal{Q}) = \{a\}$, since $T \vdash a$ refutes every other answer by incompatibility, and refuting every other answer proves a because B proves $\bigvee \mathcal{Q}$.

Lemma G.2 (Decidability)**.** *Let T be a pointwise interval law and C a cell, a finite union of intervals with rational endpoints. Then T proves the datum "$q_t \in C$" iff $[\underline{f}(t), \overline{f}(t)] \subseteq C$, and refutes it iff $[\underline{f}(t), \overline{f}(t)] \cap C = \emptyset$; and $B \cup T \cup D^{\pm}$ is consistent iff, for every time t of D, $[\underline{f}(t), \overline{f}(t)]$ meets the intersection C_t of the cells that D assigns to t. All three are decidable from the presentation of T.*

Proof. Every choice of reals $q_t \in [\underline{f}(t), \overline{f}(t)]$ gives a model of $B \cup T$ in $\mathbb{R}$, and if the q_t also lie in the assigned cells, a model of $B \cup T \cup D^{\pm}$. This gives the "only if" of the first two claims (choose q_t outside the cell, respectively inside it) and the "if" of the third. Inclusion and disjointness of finite unions of rational intervals are Boolean combinations of inequalities between rational numerals, which RCF proves or refutes; this gives the "if" of the first two claims. If for some t the interval misses C_t, the finitely many inequalities involved are jointly refutable in RCF, which gives the "only if" of the third. The values $\underline{f}(t), \overline{f}(t)$ are computable from the presentation, so all three conditions are decidable. □

Proposition G.3 (Laws plus listed exceptions are deterministic hypotheses)**.**

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $L_A(D) - c \le K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$.

(b) For an r.e. T and $E \subseteq D$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_D$, $L_A(D) \le K_{\equiv}(T) + K(E) + c$.

Proof. (a) The finite set $T' := T \cup \{a : (\mathcal{Q}, a)$ a datum of D that T does not prove$\}$ is contained in the consistent set $T \cup D^{\pm}$, so $B \cup T'$ is consistent, and T' proves every datum. Concatenating the codewords of the sentences of T, each after a continuation bit, the codewords $c_S(a)$ of the exceptions, each after a continuation bit, and a stop bit gives a prefix-free code of T' of length $\|T\| + \mathrm{Exc}_D^F(T) + 1$, and a fixed program turns it into an index of T'. So $K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$. The instance form of dominance (Theorem 3.4) gives $M_A(D) \ge 2^{-K(e)-c}$ for every index e of a theory consistent with B that proves ψ_D, hence $L_A(D) \le K_A(D) + c$. (b) is Proposition A.10(b). □

So code F pays for *which* questions are exceptions, since the answer sentence names its question; a deterministic theory cannot avoid this, because it does not see the questions.

G.2 Randomised theories and the best two-part code

Proposition G.4 (Randomised theories pay question-relative costs)**.** *Let T be a pointwise interval law and D a set of interval data consistent with T.*

(a) $L_A(D) \le K(T) + \mathrm{Exc}_D^R(T) + c \le \|T\| + \mathrm{Exc}_D^R(T) + c$.

(b) If D comes from a computable question family with one question $\mathcal{Q}_t$ per time, and $k_t(T) := |A_T(\mathcal{Q}_t)| \le |\mathcal{Q}_t|$, then $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \sum \log k_t(T) + c$, the sum over the data that T does not prove; in particular $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \mathrm{Exc}_D^Q(T) + c$.

(c) Code R charges an exception of a patched law, at a time where $\underline{f} = 0$ and $\overline{f} = \mathsf{R}$, exactly $\log(\mathsf{R}/|C_t|)$, which is the code-Q cost for equal cells; it charges the honest law $-\log(|C_t \cap [f(t) - \varepsilon, f(t) + \varepsilon]|/2\varepsilon)$, which is 0 when the band lies inside the cell.

Proof. (a) *The machine.* Enumerate I as $t_1, t_2, \ldots$ and fix a computable injection $(i,k) \mapsto \langle i,k\rangle$ into tape positions, increasing in k. On a tape ω put $U_i := \sum_{k\geq 1} \omega_{\langle i,k\rangle} 2^{-k}$; under $\mathbb{U}$ the U_i are independent and uniform on $[0,1]$. Let G_T be the theory machine that emits the sentences of T at the empty prefix and, once the prefix read contains the positions $\langle i,1\rangle, \ldots, \langle i,k\rangle$, whose bits form the binary numeral of an integer m, also emits

$$\underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, m 2^{-k} \;\leq\; q_{t_i} \;\leq\; \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, (m+1) 2^{-k}.$$

This set of (prefix, sentence) pairs is c.e., and an index of it is computable from the presentation of T, so $K(G_T) \leq K(T) + c$.

The random theory is always consistent. Put $q_{t_i} := v_i := \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i) U_i$. Each emitted closed interval contains v_i, so these values satisfy every emitted sentence, and $\mathbb{R}$ with them is a model.

It proves a datum "$q_{t_i} \in C$" whenever v_i lies in the interior of C. The emitted intervals contain v_i and have lengths $(\overline{f} - \underline{f})(t_i) 2^{-k} \to 0$, so one of them lies in the interior of C, and RCF proves the inclusion. (If $\underline{f}(t_i) = \overline{f}(t_i)$, T alone decides the datum.) The times of D are distinct indices i, the U_i are independent, and endpoints have probability 0, so

$$\nu_{G_T}(D) \;\geq\; \mathbb{U}\{\omega : v_t \in \operatorname{int} C_t \text{ for every time } t \text{ of } D\} \;=\; P_T^R(D).$$

Dominance (Theorem 3.4), $M_A \geq 2^{-K(\nu)}\nu$ for every theory-machine law ν, gives $M_A(D) \geq 2^{-K(G_T)-c}\nu_{G_T}(D)$, which is (a); and $K(T) \leq \|T\| + c$.

(b) Let the machine use U_i to pick the $\lfloor k_{t_i}(T) U_i \rfloor$-th element a of $A_T(\mathcal{Q}_{t_i})$, which is computable by Lemma G.2, and emit a as soon as enough bits have been read (almost surely after finitely many). The cell a meets $[\underline{f}(t_i), \overline{f}(t_i)]$, so the random theory has a model in $\mathbb{R}$; it proves the true answer with probability $1/k_{t_i}(T)$, independently across times, and $k_t(T) = 1$ when T proves the datum. Since D is consistent with T, T refutes no datum. The machine's index is computable from T and the question family, and dominance gives the bound; the last claim follows from $\log k_t(T) \leq \log|\mathcal{Q}_t| = \gamma(\mathcal{Q}_t)$. (c) is the definition of P_T^R. □

Proposition G.5 (SAI is the best two-part code over theory machines)**.**

(a) For every data set D, $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$, the minimum over theory machines G; it is attained, up to c, by G_U.

(b) Restricting G gives upper bounds: $L_A(D) \leq \min_T J_1^R(T;D) + c$ over pointwise interval laws, $L_A(D) \leq \min_T J_1^F(T;D) + c$ over finite theories, and $L_A(D) \leq K_A(D) + c$.

(c) Neither K_A nor $\min_T J_1^Q(T;\cdot)$ over finite theories is within an additive constant of L_A in general; nor, therefore, is $\min_T J_1^F(T;\cdot)$ when $\gamma \leq \ell$.

(d) Over all theory machines steeper selection collapses. Take the point family of programs $p \in \operatorname{dom} V$ for theory machines, with length $|p|$ and fit $-\log \nu_{G_{V(p)}}(D)$, and let ℓ_U be the length of a program for G_U. For $\lambda > 1$, every λ-MAP p has $|p| \leq \ell_U + \ell_U/(\lambda - 1)$.

Proof. (a) Dominance gives $L_A(D) \leq K(G) - \log \nu_G(D) + c$ for every theory machine G, and $M_A = \nu_{G_U}$ gives $K(G_U) - \log \nu_{G_U}(D) = L_A(D) + O(1)$. (b) is Propositions G.3 and G.4 with $K_{\equiv}$ and K of a finite theory at most its length plus a constant.

(c) That $K_A - L_A$ is unbounded is the absence of a coding theorem (Theorem 3.9). For $\min_T J_1^Q(T;\cdot)$, take the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ with $B = \emptyset$ and the yes/no data $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$, so that every datum has literal cost $\gamma = 1$. By Remark F.29, applied to the r.e. theory $\{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$, $K_A(D_n) \leq c$, hence $L_A(D_n) \leq K_A(D_n) +$

$c \le c'$. Let T be finite with $B \cup T \cup D_n^{\pm}$ consistent. If c_j does not occur in T and $T \vdash R(c_j)$, then $T \vdash \forall x\, R(x)$ by the theorem on constants, which contradicts consistency with $\exists x\, \neg R(x)$. So T proves at most the data $R(c_j)$ with c_j occurring in T, together with $\exists x\, \neg R(x)$. In a syntax code in which each symbol occurrence costs at least one bit, each constant occurring in T costs at least one bit of $\|T\|$, and each unproved datum one bit of Exc^Q; so $J_1^Q(T; D_n) \ge n$. Finally $\mathrm{Exc}^F \ge \mathrm{Exc}^Q$ when $\gamma \le \ell$, since then each exception costs at least as much in code F.

(d) Proposition 5.7(ii) of the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d) states: if a point family contains a member π_U, of length ℓ_U, with fit $L(\pi_U) \le L(\pi) + \ell(\pi) + \kappa_0$ for every member π, then for every $\lambda > 1$ every λ-MAP has length at most $\ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$. The mixture identity of Theorem 3.4 gives $M_A(D) \ge \sum_{p \in \mathrm{dom}\, V} 2^{-|p|} \nu_{G_{V(p)}}(D) \ge 2^{-|p|} \nu_{G_{V(p)}}(D)$ for each p, so a program p_U for G_U has fit $-\log \nu_{G_U}(D) = L_A(D) \le -\log \nu_{G_{V(p)}}(D) + |p|$ for every p. This is the hypothesis with $\kappa_0 = 0$. □

So over all theory machines a steeper penalty collapses onto the universal machine, and the best-system analyses of Section 9.5 use laws with private literal facts instead.

Remark G.6 (Pointwise interval laws: the separation is open). For every set D of interval data, both $L_A(D)$ and $\min_T J_1^R(T; D)$ over pointwise interval laws are at least $K(y_D \mid \mathrm{dom}\, D) - c$. For L_A: the answer conjunctions of the questions in $\mathrm{dom}\, D$ form a finite set S of pairwise B-incompatible sentences, computable from $\mathrm{dom}\, D$, and $M_A(D) = M_A(\{(\psi_D, 1)\})$ (Proposition 3.1); so the coding lower bound of Proposition 3.8 gives $L_A(D) \ge K(\psi_D \mid S) - c \ge K(y_D \mid \mathrm{dom}\, D) - c'$. For J_1^R: Proposition G.9(b). So neither bound separates the two. The natural candidate for a separation, data in which pairs of times share a random cell, has $L_A \approx n\gamma/2$; but it does not separate them either, because a presentation may store values: the program that stores the $n/2$ shared cells and returns each for both times of its pair has length $n\gamma/2 + O(\log n)$ and $\mathrm{Exc}^R = 0$, so $\min_T J_1^R \le n\gamma/2 + O(\log n)$ as well.

Question G.7. Fix a presentation language for pointwise interval laws. Is $\min_T J_1^R(T; D) - L_A(D)$ bounded over all sets D of interval data? (For K_A and for $\min_T J_1^Q$ over finite theories the answer is no, by Proposition G.5(c).)

Proposition G.8 (Which hypotheses of the companion paper on steeper priors hold here)**.**

(a) Pointwise interval laws in code R *(a fixed presentation language; one question family with finitely many cells). The codewords form a decidable prefix-free set; P_T^R is computable uniformly in T, is the law ν_{G_T} of a theory machine (a shared hypothesis in the sense of Definition 3.4 of the companion paper on steeper priors) and, on data at distinct times, a product hypothesis (Definition 3.3 there); Kraft's inequality holds, lengths are integers, and finitely many laws have each length. So Lemmas A.1, 3.8, 3.9 and 5.3, Corollaries 5.4 and 5.5 and Theorem 5.6 of the companion paper on steeper priors apply, with $J_\lambda = J_\lambda^R$.*

(b) Finite theories over a decidable consistent B, in code Q, *with $P_T(D) := 2^{-\mathrm{Exc}_D^Q(T)} \mathbf{1}[B \cup T \cup D^{\pm}$ consistent$]$. P_T satisfies the semimeasure inequalities of Definition 3.4 of the companion paper on steeper priors, and it is a product over questions when consistency can be checked question by question (pointwise laws at distinct times). It is not lower semicomputable in general (for $B = \emptyset$ and a language with a binary relation symbol), so this class is not a hypothesis class of the companion paper. Its hull lemma (Lemma 3.8), kink lemma (Lemma 5.3), Corollary 5.4, Theorem 5.6 and Lemma A.1(i) use only the point family $(\|T\|, \mathrm{Exc}_D^Q(T))_T$ and apply. Its coding bound (Lemma 3.9) is replaced by Proposition G.9,*

with which Corollary 5.5 holds with offset $b+\kappa+n$ in general, and $b+\kappa$ when provability of the answers from the theories is decidable (κ the constant of Proposition G.9).

Proof. Which hypotheses are used. Each listed property is checked in the statement: for (a), $P_T^R = \nu_{G_T}$ on interval data by the proof of Proposition G.4(a) (the random theory proves "$q_t \in C$" iff $v_t \in \operatorname{int} C$, up to null events), and the product form holds because the U_i are independent. That the hull, kink and converse results need only the point family can be read off their statements, which concern countable point families, and their proofs, which use only lengths, fits, and the finiteness of the set of programs of each length.

The semimeasure inequalities in (b). $P_T(\emptyset) = \mathbf{1}[B \cup T \text{ consistent}] \le 1$. Let $\mathcal{Q} \notin \operatorname{dom} D$ and $P_T(D) > 0$. If T proves an answer a_0, every other answer is refuted by T and contributes 0, while a_0 adds no cost and keeps consistency; so the sum is $P_T(D)$. If T proves no answer, each of the at most $|\mathcal{Q}|$ answers consistent with $B \cup T \cup D^{\pm}$ contributes at most $2^{-\gamma(\mathcal{Q})} P_T(D) = P_T(D)/|\mathcal{Q}|$. If $P_T(D) = 0$, every term is 0.

Not lower semicomputable. Take $B = \emptyset$ and a language with a binary relation symbol. If $T \mapsto P_T(\emptyset)$ were lower semicomputable uniformly in T, the set of finite theories with $P_T(\emptyset) > \frac{1}{2}$, that is the consistent ones, would be r.e. The inconsistent ones are r.e. as well, so consistency of finite theories would be decidable, contrary to Church's theorem (standard; Church, 1936). □

Proposition G.9 (Coding bound for laws with literal exceptions)**.** *Let $n := |D|$.*

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $\|T\| + n + \operatorname{Exc}_D^Q(T) \ge K(y_D \mid \operatorname{dom} D) - c$.

(b) Suppose that provability of answers from T is decidable uniformly in T, for T ranging over a class of presented theories with $K(T) \le \|T\| + c$ (finite theories over a decidable complete B, or pointwise interval laws). Then $\|T\| + \operatorname{Exc}_D^Q(T) \ge K(y_D \mid \operatorname{dom} D) - c$ and $\|T\| + \operatorname{Exc}_D^R(T) \ge K(y_D \mid \operatorname{dom} D) - c$ for every member T consistent with D.

Proof. (a) For a question $\mathcal{Q}$ put

$$w_T(a \mid \mathcal{Q}) := \tfrac{1}{2}\mathbf{1}[a \text{ is the first answer of } \mathcal{Q} \text{ found provable from } B \cup T] + \tfrac{1}{2}|\mathcal{Q}|^{-1},$$

where "first" refers to a fixed dovetailed proof search. Then $\sum_a w_T(a \mid \mathcal{Q}) \le 1$, and w_T is lower semicomputable uniformly in $(T, \mathcal{Q})$, because the first answer found never changes. So $P'_T(y) := \prod_j w_T(y_j \mid \mathcal{Q}_j)$ is a semimeasure on the answer lists for $\operatorname{dom} D$, lower semicomputable uniformly in $(T, \operatorname{dom} D)$. If $B \cup T$ is consistent and proves y_j, it proves no other answer of $\mathcal{Q}_j$, so y_j is the first found and $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}$; otherwise $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}|\mathcal{Q}_j|^{-1}$. Hence $-\log P'_T(y_D) \le n + \operatorname{Exc}_D^Q(T)$. The conditional coding theorem ($K(y \mid z) \le -\log P(y \mid z) + c_P$ for a semimeasure $P(\cdot \mid z)$ lower semicomputable uniformly in z; standard, see Li and Vitányi, 2008) gives $K(y_D \mid T, \operatorname{dom} D) \le n + \operatorname{Exc}_D^Q(T) + c$, and $K(y_D \mid \operatorname{dom} D) \le K(T) + K(y_D \mid T, \operatorname{dom} D) + c \le \|T\| + n + \operatorname{Exc}_D^Q(T) + c'$.

(b) With decidable provability put $w_T(a \mid \mathcal{Q}) := 1$ if a is the first answer, in a fixed order of $\mathcal{Q}$, that T proves; $:= 0$ for the other answers that T proves; and $:= |\mathcal{Q}|^{-1}$ for every answer if T proves none. This sums to at most 1 for every T, consistent or not, and it is computable. For T consistent with D the cost bound of (a) holds without the flag bit, and the same coding argument gives the first inequality. For code R, $P_T^R(\cdot \mid \operatorname{dom} D)$ is itself a semimeasure on answer lists, computable uniformly in $(T, \operatorname{dom} D)$ (Proposition G.8(a)), and the same argument gives the second. □

So the right line of slope 1 around a sufficient law (Corollary 5.5(a) of the companion paper on steeper priors) holds in both classes; in code Q without decidability it carries the flag cost n,

which makes it useless, and for that reason the analyses of Section 9 use laws whose provability on the data is decidable.

G.3 Deterministic theories and precision

Proposition G.10 (Compatibility is monotone in logical strength)**.**

(a) Let $B \cup T$ be consistent and $\mathrm{Cn}(T) \supseteq \mathrm{Cn}(T')$. On every data set with which T' is compatible ($B \cup T'$ consistent and proving every datum), T is compatible. So among the deterministic hypotheses of W_A and M_A the posterior odds of T against T' equal their prior odds on every data set with which T' is compatible, and T' is incompatible whenever T is: data never move the odds towards the weaker theory.

(b) $T_0 \vdash T^\varepsilon$ for every $\varepsilon \geq 0$, so every data set compatible with the honest law T^ε is compatible with the exact law T_0, at every sample size and every precision. With prior weights $2^{-\|T\|}$ in a code in which the tolerance clause costs at least one bit, the posterior odds of T_0 against T^ε are at least $2^{\|T^\varepsilon\| - \|T_0\|} \geq 2$ on every such data set.

(c) If $\sup_t |e(t)| > 0$, then T_0 is false and, on a question at precision $\delta' < \sup_t |e(t)|$, proves a false answer at some times; T^ε with $\varepsilon \geq \sup_t |e(t)|$ is true and is silent wherever its band straddles a cell boundary.

Proof. A consistent theory that proves everything T' proves proves every datum that T' proves; this is (a), and (b) is (a) with $T = T_0$, $T' = T^\varepsilon$. (c) is immediate from the definitions. □

Proposition G.11 (Precision monotonicity)**.** *If $B \cup T$ proves "$q(t) \in [v - \delta', v + \delta']$", then it proves "$q(t) \in [v - \delta, v + \delta]$" for every $\delta \geq \delta'$. So compatibility with data at a fixed set of centres is monotone in the precision.*

Proof. The first interval is contained in the second. □

Proposition G.12 (Guaranteed tolerances)**.** *In regime G:*

(a) T_0 proves datum j iff $|e(t_j) - \zeta_j| \leq \delta$, and refutes it otherwise.

(b) If $\delta \geq \varepsilon^* + \delta_0$, then T_0 is compatible, at every n, with every such data set whose errors lie in the essential range (so almost surely).

(c) Suppose the pairs $(e(t_j), \zeta_j)$ are i.i.d., $e(t_j)$ is independent of ζ_j, both $\pm\delta_0$ lie in the support of the law of ζ_j, and $\delta < \varepsilon^* + \delta_0$. Then the per-datum refutation probability $r(\delta) := \mathbb{P}[|e - \zeta| > \delta]$ is positive, T_0 survives n data with probability $(1 - r(\delta))^n$, and it is almost surely refuted eventually.

So the switch point $\delta = \varepsilon^* + \delta_0$ is sharp.

Proof. (a) T_0 proves $q(t_j) = f(t_j) = v_j + e(t_j) - \zeta_j$. This point lies in $[v_j - \delta, v_j + \delta]$ iff $|e(t_j) - \zeta_j| \leq \delta$, and if it does not, T_0 proves the negation of the datum. (b) $|e - \zeta| \leq |e| + |\zeta| \leq \varepsilon^* + \delta_0$ almost surely. (c) Pick $\varepsilon' < \varepsilon^*$ and $s > 0$ with $\varepsilon' + \delta_0 - s > \delta$. Since ε^* is the essential supremum of $|e|$, $\mathbb{P}[e > \varepsilon'] > 0$ or $\mathbb{P}[e < -\varepsilon'] > 0$. In the first case $\mathbb{P}[\zeta < -\delta_0 + s] > 0$ (support), and independence gives $\mathbb{P}[e - \zeta > \delta] > 0$; the second case is symmetric, with $\zeta > \delta_0 - s$. Refutations of distinct data are independent, so survival has probability $(1 - r(\delta))^n \to 0$, and a refuted theory stays refuted. □

Proposition G.13 (Dithered rounding)**.** *In regime R, given $e(t_j)$, T_0 proves datum j with probability $(1-|e(t_j)|/\Delta)^+$ and refutes it otherwise. So T_0 survives n data with probability $(1-p_\Delta)^n$, its first refutation has a geometric index with mean $1/p_\Delta$, and it is almost surely refuted eventually iff $\mathbb{P}[e \neq 0] > 0$, for every Δ, however coarse. For $\Delta \geq \varepsilon^*$, $p_\Delta = \mathbb{E}|e|/\Delta$: the exact law survives about $\Delta/\mathbb{E}|e|$ data.*

Proof. T_0 proves "$q(t_j)$ lies in the cell of $f(t_j)$" and so refutes every other cell. Two points at distance $|e| < \Delta$ fall into different cells of a uniformly shifted grid of width Δ with probability $|e|/\Delta$, and at distance at least Δ they always do; this gives the conditional proof probability $(1 - |e(t_j)|/\Delta)^+$. The refutation indicators of distinct data are independent (they depend only on each datum's own error and dither), which gives the product $(1-p_\Delta)^n$ and the geometric law of the first refutation; and $p_\Delta > 0$ iff $\mathbb{P}[e \neq 0] > 0$. For $\Delta \geq \varepsilon^*$, $\min(|e|/\Delta, 1) = |e|/\Delta$ almost surely. □

This corrects the natural expectation that coarse data protect a false exact law: precision belongs to the data set, not to the datum. An exact law with a sinusoidal error of amplitude $1'$ is refuted by positions rounded to a dithered grid of 1° after about 94 data (the computations below), and Proposition G.14 compares several exact laws. We have not found these two regimes stated elsewhere; precision-dependent selection as such is familiar from accounts based on AIC, MDL and MML (cf. Forster and Sober, 1994; Rissanen, 1978; Wallace, 2005).

Proposition G.14 (Which exact theory a proof-requiring inducer settles on)**.** *Let $T_1, \ldots, T_k$ be exact pointwise laws for the same quantity, with lengths $\|T_1\| < \cdots < \|T_k\|$ and errors $e_1, \ldots, e_k$, and let the inducer choose the shortest compatible theory in this pool (the MAP of any prior that decreases with length, or K_A restricted to the pool). Let the data be i.i.d.*

(G) Under the hypotheses of Proposition G.12(c) at precision δ, almost surely the choice is eventually $T_{i^*(\delta)}$, $i^*(\delta) := \min\{i : \varepsilon_i^* + \delta_0 \leq \delta\}$; if no i qualifies, every theory is eventually refuted. As δ decreases past the thresholds $\varepsilon_i^* + \delta_0$, the eventual choice moves to longer and more exact theories.

(R) In regime R, for every $\Delta > 0$, almost surely the choice is eventually the shortest T_i with $\mathbb{P}[e_i \neq 0] = 0$, whatever Δ is; if there is none, every theory is eventually refuted. At sample size n, T_i survives with probability $(1 - p_{\Delta,i})^n$, and for $\Delta \geq \varepsilon_i^*$ it is abandoned after about $\Delta/\mathbb{E}|e_i|$ data.

Proof. Apply Proposition G.12 and Proposition G.13 to each T_i. The choice at n is the first survivor in order of length, and a finite pool has a last refutation time almost surely. □

Computations. In regime G, with $\varepsilon^* = 1'$, $\delta_0 = 0.5'$ and ζ uniform on $[-\delta_0, \delta_0]$, the refutation probability is exactly $r(\delta) = 0.218$ at $\delta = 1'$, 6.0×10^{-4} at $\delta = 1.49'$ and 0 for $\delta \geq 1.5'$ for a sinusoidal error, and 0.025, 5.0×10^{-4} and 0 for an error equal to $\pm\varepsilon^*$ on 5% of the times and 0 otherwise: positive below $\varepsilon^* + \delta_0$ and zero above it for both. In regime R, simulated survival after 1, 10 and 100 data matches $(1 - p_\Delta)^n$ at every Δ from $0.5'$ to $600'$; for the sinusoid with $\varepsilon^* = 1'$ the mean first-refutation index is 94.1 against the exact $1/p_\Delta = \pi\Delta/2 = 94.25$ at $\Delta = 60'$, and 942.7 against 942.48 at $\Delta = 600'$. For the pool T_1 (length 100, $\varepsilon^* = 10'$), T_2 (length 150, $\varepsilon^* = 1'$), T_3 (length 400, exact), with sinusoidal errors and 2000 runs, every run chooses the predicted theory at $n = 10^4$ in regime G with $\delta_0 = 0.5'$ at each of $\delta = 20'$, $10.5'$, $5'$, $1.5'$, $1.2'$ (predictions T_1, T_1, T_2, T_2, T_3); in regime R every run ends at T_3, T_1 being abandoned after about 94 data at $\Delta = 10^\circ$ and 9.4 at $\Delta = 1^\circ$, and T_2 after about 942 and 94.

G.4 Honest laws, laws with exceptions, laws with error models

Forms and pricings. Work in regime R. The forms are: the empty law $\emptyset$; the *patched tolerance law* $T^{\varepsilon,-Z}$, the tolerance law T^{ε} withdrawn at the set Z of times of the data it refutes (fewer patches are inconsistent with the data, more only add cost); for $\varepsilon = 0$ the exact law, written T_0 while it has no patch and T_0^{-Z} after; for $\varepsilon \geq \varepsilon^*$ the honest law T^{ε}, which is true and never patched; and a true exact theory T_{tr}. Their lengths, relative to a common part, are $\|\emptyset\| = 0$, $\|T_0\| = \ell_0$, $\|T^{\varepsilon}\| = \ell_0 + h$ (h the length of the tolerance clause and its numeral) and $\|T_{\mathrm{tr}}\| = \ell_0 + h_{\mathrm{tr}}$. A pricing fixes the price γ_{si} of a datum on which a form is silent and the price γ_{ex} of an exception:

- *Q/det*: code Q, each patch naming its time inside the law at c_{time} bits; $\gamma_{\mathrm{si}} = \gamma$ and $\gamma_{\mathrm{ex}} = \lambda c_{\mathrm{time}} + \gamma$. This is a finite-horizon cost model (Proposition G.17).
- *Q/rand*: code Q with randomised exceptions: a theory machine withdraws the law at each time independently with probability ρ, and the fit is charged $-\log(1-\rho)$ per kept datum and $-\log\rho + \gamma$ per exception. The share ρ can be stated in $O(\log n)$ bits or integrated out: drawing ρ uniformly from the tape and then withdrawing independently is a single theory machine, whose cost for k exceptions among n data is $\log\left((n+1)\binom{n}{k}\right) + k\gamma$. With $\rho = p_\Delta$, or integrated out, the rate is $H_2(p_\Delta) + p_\Delta\gamma$, where H_2 is the binary entropy, so $\gamma_{\mathrm{ex}} = \gamma + H_2(p_\Delta)/p_\Delta$. These are exact theory-machine costs for every n.
- *R/det* and *R/rand*: as above, with a silent datum priced by random completion inside the band (code R); an exception of a patched law is priced γ in code R too (Proposition G.4(c)).

SAI's pricing is R/rand: the honest law with random completion and the exact law with random exceptions are theory machines, and R/rand compares their weights in M_A's mixture.

Lemma G.15 (What a tolerance law does with one datum)**.** *In regime R, fix a datum with error e and let $\varepsilon \geq 0$. Over the dither, T^{ε} refutes the datum with probability* $\min\left((|e|-\varepsilon)^+/\Delta, 1\right)$*, proves it with probability* $\left(1-(\varepsilon+\max(|e|,\varepsilon))/\Delta\right)^+$*, and is silent otherwise. In the* coarse regime $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ *it is silent with probability exactly $2\varepsilon/\Delta$, whatever e is, and refutes with probability $(|e|-\varepsilon)^+/\Delta$. For $\varepsilon = 0$ it refutes with probability* $\min(|e|/\Delta, 1)$ *and is never silent.*

Proof. Place q at 0. The cell containing it is $[-u\Delta, (1-u)\Delta)$ with u uniform on $[0,1)$, and the band is $[e-\varepsilon, e+\varepsilon]$, since $f = q + e$; by symmetry let $e \geq 0$. *Refutation* means that the band misses the cell. If $e \leq \varepsilon$, the band contains 0, so this is impossible. If $e > \varepsilon$, the band lies in $(0,\infty)$, and it misses the cell iff $(1-u)\Delta \leq e - \varepsilon$, of probability $\min((e-\varepsilon)/\Delta, 1)$. *Proof of the datum* means that the band lies in the cell: $-u\Delta \leq e - \varepsilon$ and $e + \varepsilon < (1-u)\Delta$. If $e \leq \varepsilon$, this is $u \in [(\varepsilon - e)/\Delta, (\Delta - \varepsilon - e)/\Delta)$, of probability $(1 - 2\varepsilon/\Delta)^+$; if $e > \varepsilon$, the first condition holds automatically and the second is $u < (\Delta - e - \varepsilon)/\Delta$. Both cases equal $(1 - (\varepsilon + \max(e,\varepsilon))/\Delta)^+$. When $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ the positive parts are inactive for every e in the essential range, and silence has probability $1 - (1 - (\varepsilon + \max(e,\varepsilon))/\Delta) - (\max(e,\varepsilon) - \varepsilon)/\Delta = 2\varepsilon/\Delta$. □

The coarse-regime condition cannot be weakened to $\Delta \geq \varepsilon + \varepsilon^*$ when $\varepsilon > \varepsilon^*$: with $|e| = \varepsilon^* = 1$, $\varepsilon = 2$ and $\Delta = 3$ the formula $2\varepsilon/\Delta$ would give a silence probability of $4/3$.

Proposition G.16 (Rates)**.** *In regime R, write $z_j := (e_j, u_j)$. Each form X has $J_\lambda(X; D_n) = a_X + \sum_{j \leq n} c_X(z_j)$ with a constant a_X and i.i.d. per-datum costs $c_X(z_j)$ (under Q/rand and R/rand with ρ stated; integrating ρ out adds $O(\log n)$). In the coarse regime $\Delta \geq 2\varepsilon^*$ the rates $m_X := \mathbb{E}c_X$ are: $m_\emptyset = \gamma$; $m_{T_{\mathrm{tr}}} = 0$; $m_{T_0} = \infty$ if $\mathbb{P}[e \neq 0] > 0$ (an unpatched exact law is eventually refuted); $m_{T_0^{-z}} = p_\Delta\gamma_{\mathrm{ex}}$, that is $p_\Delta(\lambda c_{\mathrm{time}} + \gamma)$ under the det pricings and $H_2(p_\Delta) + p_\Delta\gamma$ under the rand pricings, with $p_\Delta = \mathbb{E}|e|/\Delta$; and $m_{T^{\varepsilon^*}} = s_\Delta\gamma_{\mathrm{si}}$ with $s_\Delta = 2\varepsilon^*/\Delta$, that is $s_\Delta\gamma$ in code Q and $s_\Delta\beta_e$ in code R, β_e as in Lemma G.20.*

Proof. For the patched exact law at its cheapest patch set, a datum costs γ_{ex} if T_0 refutes it and 0 otherwise: under det the patch names the time ($\lambda c_{\mathrm{time}}$) and the datum is then literal (γ); under rand it is the flag code. Lemma G.15 with $\varepsilon = 0$ gives the probability p_Δ. The honest law costs γ_{si} on a datum iff it is silent, which in the coarse regime has probability $2\varepsilon^*/\Delta$ independently of e (Lemma G.15); in code R the expected cost given silence, averaged over the errors, is β_e (Lemma G.20(i)). □

Proposition G.17 (The price of naming exceptions deterministically)**.** *In regime R, let $Z_n \subseteq \{1, \dots, n\}$ be the indices of the data that the exact law refutes, so that the events $j \in Z_n$ are independent with probability $p_\Delta \in (0, 1)$.*

(a) For every prefix-free code of the finite subsets of $\{1, \dots, n\}$, even one that depends on n, the expected code length of Z_n is at least $nH_2(p_\Delta)$; so the expected cost per exception is at least $H_2(p_\Delta)/p_\Delta$ bits, the cost of randomised exceptions.

(b) If the patched law names each patched time t_j by a numeral in a prefix-free code ℓ_{num} of numerals, then $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$, so the cost per exception is at least $\log n$ on average. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$ for all j (any standard numeral code), the cost per datum $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j)$ tends to ∞ almost surely.

(c) Hence Q/det and R/det with a constant c_{time} are cost models valid only while the times used cost about c_{time} bits to name, that is for $n \lesssim 2^{c_{\mathrm{time}}}$. Asymptotically, at fresh times, deterministic patching costs at least as much per exception as randomised exceptions, and literal patching costs unboundedly more; so the patched exact law under det is eventually beaten by the exact law with random exceptions.

Proof. (a) Let P_Z be the law of Z_n and ℓ' a prefix-free code. By Kraft's inequality $\sum_Z 2^{-\ell'(Z)} \leq 1$, and by Jensen's inequality, with H the entropy,

$$\mathbb{E}\ell'(Z_n) - H(Z_n) = -\sum_Z P_Z(Z) \log \frac{2^{-\ell'(Z)}}{P_Z(Z)} \geq -\log \sum_Z 2^{-\ell'(Z)} \geq 0,$$

where $H(Z_n) = nH_2(p_\Delta)$. The expected number of exceptions is $p_\Delta n$. (b) The numerals $t_1, \dots, t_n$ are distinct, so Kraft gives $\sum_{j\leq n} 2^{-\ell_{\mathrm{num}}(t_j)} \leq 1$, and Jensen gives $2^{-\frac{1}{n}\sum_j \ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}\sum_j 2^{-\ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}$, that is $\sum_{j\leq n} \ell_{\mathrm{num}}(t_j) \geq n \log n$. Since $\mathbb{P}[j \in Z_n] = p_\Delta$ for every j, $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$, then $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq \frac{1}{n}\#\{j \in Z_n : j > n/2\}(\log n - 1 - c)$, and the count over n tends to $p_\Delta/2$ almost surely by the strong law of large numbers. (c) follows from (a), (b) and Proposition G.16. □

A computation (code in the accompanying repository) gives, at $p_\Delta = 0.01$, 15.0, 23.0 and 30.3 bits per exception for literal listing by Elias delta numerals at $n = 10^3$, 10^5 and 10^7, and 9.02, 8.13 and 8.08 bits for an enumerative code of the set, against $H_2(p_\Delta)/p_\Delta = 8.08$.

Proposition G.18 (The peakedness criterion)**.** *Let $\Delta \geq 2\varepsilon^*$, under any of the four pricings (the deterministic ones as a cost model).*

(a) The honest law T^{ε^*} has a lower rate than the exact law with exceptions T_0^{-Z} iff $\mathrm{pk}(e) < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

(b) If the exact law with exceptions comes from a law f_1 with error e_1, and the honest law from a law f_2 of the same length up to $O(1)$, with tolerance $\varepsilon_2 \geq \operatorname{ess\,sup}|e_2|$ and $\Delta \geq 2\varepsilon_2$, then the honest law has the lower rate iff $\varepsilon_2/\mathbb{E}|e_1| < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

Proof. By Proposition G.16, $s_\Delta \gamma_{\mathrm{si}} < p_\Delta \gamma_{\mathrm{ex}}$ iff $2\varepsilon^* \gamma_{\mathrm{si}} < \mathbb{E}|e|\,\gamma_{\mathrm{ex}}$. In (b), $s_\Delta = 2\varepsilon_2/\Delta$ (Lemma G.15) and $p_\Delta = \mathbb{E}|e_1|/\Delta$. □

Under Q/det the criterion reads $\lambda c_{\mathrm{time}} > \gamma(2\mathrm{pk}(e) - 1)$. Simulated rates for a sinusoid ($\mathrm{pk} = \pi/2$) with $c_{\mathrm{time}} = 22$ pick the exact law with exceptions at $\lambda = 1$ and the honest law at $\lambda = 2$ for $\Delta = 2'$, $3'$ and $10'$, and the honest law at both for $\Delta = 60'$, as the threshold $(\lambda c_{\mathrm{time}} + \gamma)/(2\gamma)$ predicts; for an error with $\mathrm{pk} = 20$ they pick the exact law with exceptions everywhere. Under Q/rand it reads $\mathrm{pk}(e) < \frac{1}{2} + H_2(p_\Delta)/(2p_\Delta\gamma)$; since $H_2(p)/p \approx \log(\mathrm{e}/p)$ for small p, the threshold grows only logarithmically in $1/p_\Delta$: at $\gamma = \log 360$ it is 1.17 at $p_\Delta = 10^{-3}$, 1.76 at 10^{-6} and 2.93 at 10^{-12}. For the sinusoid it is 0.637, 0.742 and 0.971 at $\Delta = 3'$, $10'$ and $60'$, so the exact law with random exceptions wins although the error is spread. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand the threshold is $(\gamma + H_2(p_\Delta)/p_\Delta)/(2\beta_e)$, roughly 5 to 30 for typical cell widths (9.2–9.3 for the sinusoid, 20.4–20.5 for the error with $\mathrm{pk} = 20$). This comparison is relative to its pool: the exact law with random exceptions completes its exceptions over the whole circle, while the equally short two-point completion $\mathrm{HY}(\rho)$ completes them inside the band, and against it the error with $\mathrm{pk} = 20$ is held as "exact except in about 10% of cases" at a third of the uniform band's rate (Proposition 9.2(d) and the computation after its proof).

Proposition G.19 (Optimal tolerance)**.** *Under Q/det in the regime $\Delta \geq 2\varepsilon^*$, the patched tolerance law $T^{\varepsilon,-Z}$ with $0 \leq \varepsilon \leq \varepsilon^*$ has rate*

$$\bar{m}(\varepsilon) = \frac{2\varepsilon\gamma + (\lambda c_{\mathrm{time}} + \gamma)\,\mathbb{E}(|e| - \varepsilon)^+}{\Delta}.$$

$\bar{m}$ is convex, with right derivative $\big(2\gamma - (\lambda c_{\mathrm{time}} + \gamma)\mathbb{P}[|e| > \varepsilon]\big)/\Delta$, and its minimisers on $[0, \varepsilon^*]$ are the ε with $\mathbb{P}[|e| > \varepsilon] \leq \theta_{\mathrm{ex}}$ and, if $\varepsilon > 0$, $\theta_{\mathrm{ex}} \leq \mathbb{P}[|e| \geq \varepsilon]$, where $\theta_{\mathrm{ex}} := 2\gamma/(\lambda c_{\mathrm{time}} + \gamma)$. So the optimal law states the $(1 - \theta_{\mathrm{ex}})$-quantile of $|e|$ as its tolerance and lists the remaining share θ_{ex} of cases as exceptions. If $\theta_{\mathrm{ex}} \geq 1$ ($\lambda c_{\mathrm{time}} \leq \gamma$), the exact law with patches is optimal. The pure honest law ($\varepsilon = \varepsilon^*$, no exceptions) is optimal iff $\mathbb{P}[|e| \geq \varepsilon^*] \geq \theta_{\mathrm{ex}}$, which never holds for an error law without an atom at its maximum.

Proof. A datum costs γ if it is not proved, and in addition $\lambda c_{\mathrm{time}}$ if it is refuted (the patch). By Lemma G.15, with $\Delta \geq 2\varepsilon^* \geq \varepsilon + \varepsilon^*$, $\mathbb{P}[\text{not proved}] = (2\varepsilon + \mathbb{E}(|e| - \varepsilon)^+)/\Delta$ and $\mathbb{P}[\text{refuted}] = \mathbb{E}(|e| - \varepsilon)^+/\Delta$, which gives $\bar{m}$. The function $\varepsilon \mapsto \mathbb{E}(|e| - \varepsilon)^+$ is convex, with right derivative $-\mathbb{P}[|e| > \varepsilon]$ and left derivative $-\mathbb{P}[|e| \geq \varepsilon]$. This gives convexity and the derivatives of $\bar{m}$, and a convex function is minimised exactly where its right derivative is non-negative and its left derivative non-positive (at $\varepsilon = 0$ only the right derivative matters), which is the stated condition. The last two claims are the cases $\varepsilon = 0$ and $\varepsilon = \varepsilon^*$ of it. □

Minimising $\theta\varepsilon + \mathbb{E}(|e| - \varepsilon)^+$ over ε is the quantile-loss fact, and in ν-support-vector regression the parameter ν likewise bounds the share of points outside an ε-tube whose width is a quantile of the residuals (cf. Schölkopf et al., 2000). What is specific here is the reading for axiom inducers: the tube is a tolerance clause, the points outside it are patched exceptions, and θ_{ex} is a ratio of literal costs. The steeper the prior, or the dearer the naming of an exception, the smaller θ_{ex}: the law states a wider tolerance and lists fewer exceptions. A computation at $\Delta = 10'$, $c_{\mathrm{time}} = 22$ finds the grid minimiser at 0.52, 0.80 and 0.94 for a sinusoid of amplitude 1 at $\lambda = 1, 2, 4$, against the predicted quantiles 0.496, 0.807 and 0.939.

Lemma G.20 (The code-R cost of the honest law, exactly)**.** *Let $\varepsilon \geq \varepsilon^*$ and let $w := (\varepsilon - e)/(2\varepsilon) \in [0,1]$ be the position of the true value in the band. Put $\vartheta(z) := 1 - z + z\ln z$ for $0 \leq z < 1$ ($\vartheta(0) = 1$) and $\vartheta(z) := 0$ for $z \geq 1$. The expected code-R cost of a datum for T^ε, over the dither, is*

$$\bar{c}_\Delta(w) = \begin{cases} \dfrac{2\varepsilon}{\Delta}\big(\log \mathrm{e} - H_2(w)\big) & \text{if } \Delta \geq 2\varepsilon, \\ \log\dfrac{2\varepsilon}{\Delta} + \log \mathrm{e}\left[\vartheta\Big(\dfrac{2\varepsilon w}{\Delta}\Big) + \vartheta\Big(\dfrac{2\varepsilon(1-w)}{\Delta}\Big)\right] & \text{if } \Delta \leq 2\varepsilon. \end{cases}$$

(i) In the coarse regime the cost given silence is $\log \mathrm{e} - H_2(w)$. Averaged over the errors, $\gamma_{\mathrm{si}} = \beta_e := \log \mathrm{e} - \mathbb{E}H_2(w) \in [\log \mathrm{e} - 1, \log \mathrm{e}] = [0.443, 1.443]$ bits.

(ii) In the fine regime T^ε is silent on every datum, and $\log(2\varepsilon/\Delta) \leq \bar{c}_\Delta(w) \leq \log(2\varepsilon/\Delta) + \log \mathrm{e}$; the upper bound is approached as $w \to 0$ at $\Delta = 2\varepsilon$.

(iii) For every w, $\bar{c}_\Delta(w)$ is continuous and strictly decreasing in $\Delta \in (0, \infty)$.

Proof. Coarse regime. The band straddles a cell boundary with probability $2\varepsilon/\Delta$ (Lemma G.15, since $\varepsilon \geq \varepsilon^*$); otherwise it lies in a cell and the datum costs 0. Given a straddle, exactly one boundary lies in the band, at a relative position v uniform on $[0,1]$ (a uniformly shifted lattice of spacing at least the band's length). The true cell contains the part of the band on the true value's side, of relative length v if $w < v$ and $1 - v$ otherwise, and random completion inside the band puts that much probability on it. So the expected cost given a straddle is, in nats,

$$\int_w^1 -\ln v \, dv + \int_0^w -\ln(1-v)\, dv = 1 + w\ln w + (1-w)\ln(1-w),$$

which is $\log \mathrm{e} - H_2(w)$ in bits.

Fine regime. Let y be the length of the part of the true cell inside the band; the cost is $\log(2\varepsilon/y) = \log(2\varepsilon/\Delta) - \log(y/\Delta)$. With $a := 2\varepsilon w/\Delta$ and $b := 2\varepsilon(1-w)/\Delta$, so that $a + b = 2\varepsilon/\Delta \geq 1$, the band is $[-a\Delta, b\Delta]$ around the true value, and with dither u the true cell is $[-u\Delta, (1-u)\Delta)$, so $y/\Delta = \min(u, a) + \min(1-u, b)$. If $a, b \geq 1$, then $y = \Delta$ and the excess $-\ln(y/\Delta)$ is 0. If $a < 1 \leq b$, then $y/\Delta = 1$ for $u < a$ and $1 + a - u$ for $u \geq a$, so the expected excess is $\int_a^1 -\ln(1 + a - u)\, du = \int_a^1 -\ln s \, ds = \vartheta(a)$; the case $b < 1 \leq a$ is symmetric. If $a, b < 1$, then $1 - b \leq a$, and $y/\Delta = u + b$ for $u < 1 - b$, 1 on $[1-b, a]$, and $a + 1 - u$ for $u > a$; the expected excess is $\int_b^1 -\ln s \, ds + \int_a^1 -\ln s \, ds = \vartheta(a) + \vartheta(b)$.

(i) is the coarse case, and $\mathbb{E}H_2(w) \in [0,1]$. (ii) In the fine regime $2\varepsilon \geq \Delta$, so T^ε is silent on every datum by Lemma G.15. $\vartheta \geq 0$ and ϑ is decreasing on $[0,1]$ ($\vartheta'(z) = \ln z$). In the last case $\vartheta(a) + \vartheta(b)$ is largest on $a + b = 1$, where it equals $1 - H_2(a)\ln 2 \leq 1$; in the middle case $\vartheta(a) \leq \vartheta(0) = 1$. The value 1 is approached as $a \to 0$, $b \to 1$. (iii) Write $r := 2\varepsilon/\Delta$ and the cost in nats. In the coarse regime ($r \leq 1$) it is $r(1 - H_2(w)\ln 2)$, increasing in r because $1 - H_2(w)\ln 2 \geq 1 - \ln 2 > 0$. In the fine regime ($r \geq 1$) it is $\ln r + \vartheta(wr) + \vartheta((1-w)r)$, and on each piece its derivative in r is $\frac{1}{r}(1 + \sum z\ln z)$, the sum over $z \in \{wr, (1-w)r\}$ with $z < 1$; since $z\ln z \geq -1/\mathrm{e}$, this is at least $\frac{1}{r}(1 - 2/\mathrm{e}) > 0$. The cost is continuous in r (ϑ is continuous at 1), and at $r = 1$ the two forms agree, since $\vartheta(w) + \vartheta(1-w) = 1 - H_2(w)\ln 2$. So it is strictly increasing in r, that is strictly decreasing in Δ. □

A computation (code in the accompanying repository) confirms the closed form against a 5000-point quadrature over the dither to 10^{-4} bits on 36090 cells, with largest fine-regime excess $1.442695 = \log \mathrm{e}$; the coarse averages are $\beta_e = 0.885$ for a sinusoid and 0.493 for an error equal to $\pm\varepsilon^*$ on 5% of the times.

Band-confined completions. The machine $\mathrm{BC}_{\mathcal{G}}$ of Section 9.3 draws $\hat{w}_i \sim \mathcal{G}$ by the inverse distribution function from the uniform variables U_i of the proof of Proposition G.4, and emits the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ by narrowing intervals for the continuous part of $\mathcal{G}$ and exactly at an atom. A computable $\mathcal{G}$ is, for instance, piecewise linear with rational knots plus finitely many atoms at rational points. The argument of Proposition G.4(a) gives $K(\mathrm{BC}_{\mathcal{G}}) \le \|T^{\varepsilon}\| + K(\mathcal{G}) + c$, the random theory is always consistent, and on data at fresh times $\nu_{\mathrm{BC}_{\mathcal{G}}}(D) = \prod_j \mathcal{G}(C'_j)$ (cell endpoints are null events for continuous dithers).

Proof of Proposition 9.2. (a) As in Lemma G.20: a cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a relative position v uniform on $[0,1]$ and independent of w. With no boundary in the band the cell contains the band and the datum costs 0. With a boundary at v, the true cell contains $[0,v)$ if $w < v$, of $\mathcal{G}$-mass $\mathcal{G}(v)$, and $[v,1]$ if $w > v$, of mass $1 - \mathcal{G}(v)$; this gives $c_{\mathcal{G}}$, which does not depend on Δ.

(b) Given v, the event $w < v$ has probability $\mathcal{F}(v)$, so the conditional expected cost is $\mathcal{F}(v)(-\log \mathcal{G}(v)) + (1-\mathcal{F}(v))(-\log(1-\mathcal{G}(v))) \ge H_2(\mathcal{F}(v))$, with equality iff $\mathcal{G}(v) = \mathcal{F}(v)$ (Gibbs' inequality). Integrate over v. The uniform band has $\mathcal{G}(v) = v$, which equals $\mathcal{F}$ on $[0,1]$ iff w is uniform, that is iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) Let $P^{\star}$ be the true conditional law of the answer list given the question list. The errors are independent of the questions, and $q = f - e$, so the conditional probability of the observed cell of $q(t_j)$ is $\mathcal{F}(C'_j)$; hence $P^{\star}(D_n) = \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n) = \prod_{j \le n} \mathcal{F}(C'_j)$, and $-\log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$ is a sum of i.i.d. terms with finite mean $r_{\mathcal{F}}(\Delta)$; the strong law of large numbers gives the limit. Let G' be any theory machine. Answers to one question are pairwise B-incompatible, so for a fixed question list the events "the random theory is consistent and proves the answer list y" are disjoint across y; hence $\sum_y \nu_{G'}(y) \le 1$ and $\mathbb{E}_{P^{\star}}[\nu_{G'}(D_n)/P^{\star}(D_n) \mid \text{questions}] \le 1$. By Markov's inequality $\mathbb{P}[-\log \nu_{G'}(D_n) < -\log P^{\star}(D_n) - 2\log n] \le n^{-2}$, and by the Borel–Cantelli lemma, almost surely $-\log \nu_{G'}(D_n) \ge -\log P^{\star}(D_n) - 2\log n$ for all large n. Dividing by n gives the $\liminf$ bound, and $G' = G_U$ gives it for L_A. If $\mathcal{F}$ is computable, dominance gives $L_A(D_n) \le K(\mathrm{BC}_{\mathcal{F}}) + c - \log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$, hence the limit. In the coarse regime $r_{\mathcal{F}}(\Delta) = (2\varepsilon/\Delta) c_{\mathcal{F}}$ by (a) and the equality case of (b).

(d) Write $\sigma := 1 - \rho$ for the weight of the atom at $\frac{1}{2}$. The family is symmetric, $\mathcal{G}(1-v) = 1 - \mathcal{G}(v)$, so the cost given w equals the cost given $1-w$, and we may take $w \le \frac{1}{2}$. *Convexity.* For fixed (w,v) the probability assigned to the true side is affine in σ and positive for $\sigma \in [0,1)$, and $-\ln$ of a positive affine function is convex; so $c_{\mathrm{HY}(\rho)}$, an expectation of convex functions, is convex in σ, hence in $\rho \in (0,1]$. *Right derivative at $\sigma = 0$, in nats.* For $v < w$ the true side has probability $1 - (1-\sigma)v$, and the derivative of $-\ln$ of it at $\sigma = 0$ is $-v/(1-v) \in [-1,0]$; for $w < v < \frac{1}{2}$ the probability is $(1-\sigma)v$, with derivative $+1$; for $v > \frac{1}{2}$ it is $\sigma + (1-\sigma)v$, with derivative $-(1-v)/v \in [-1,0]$. Difference quotients of convex functions decrease monotonically as $\sigma \downarrow 0$, so by monotone convergence the derivative of the expectation is the expectation of the derivatives:

$$c'(0^+) = \mathbb{E}\Big[-\int_0^w \tfrac{v}{1-v}\, dv + (\tfrac{1}{2} - w) - \int_{1/2}^1 \tfrac{1-v}{v}\, dv \Big] = \mathbb{E}\big[\ln(1-w) + 1 - \ln 2\big],$$

using $\int_0^w \frac{v}{1-v} dv = -\ln(1-w) - w$ and $\int_{1/2}^1 \frac{1-v}{v} dv = \ln 2 - \frac{1}{2}$. For $w \le \frac{1}{2}$, $1 - w = (1 + |e|/\varepsilon)/2$, so $c'(0^+) = \mathbb{E}\ln(1 + |e|/\varepsilon) + 1 - 2\ln 2$. *The criterion.* If $c'(0^+) < 0$, some $\sigma > 0$ beats $\sigma = 0$, which is the uniform band; if $c'(0^+) \ge 0$, convexity gives $c(\sigma) \ge c(0)$ for every σ. *Necessity.* $\ln(1+z) \ge z \ln 2$ on $[0,1]$ (a concave function lies above its chord), so $\mathbb{E}\ln(1 + |e|/\varepsilon) \ge \ln 2/\mathrm{pk}_{\varepsilon}(e)$, and $c'(0^+) < 0$ forces $\mathrm{pk}_{\varepsilon}(e) > \ln 2/(2\ln 2 - 1)$. *Sufficiency.* Jensen's inequality gives

$\mathbb{E}\ln(1+|e|/\varepsilon) \le \ln(1+1/\mathrm{pk}_\varepsilon(e))$, which is below $2\ln 2-1$ iff $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4-\mathrm{e})$. *Boundary.* $\int_0^1 \ln(1+z)\,dz = 2\ln 2 - 1$. □

The following computation (synthetic errors, 2×10^5 samples, band $\varepsilon = \varepsilon^*$; code in the accompanying repository) gives per-straddle costs in bits in the coarse regime. SH is a 40-bin smoothed histogram fitted on a random half of the samples and evaluated on the other half, $c_{\mathcal{F}}$ is the bound of Proposition 9.2(b) on that half, and $c'(0^+)$ (in nats) is the derivative that decides whether some HY(ρ) with $\rho < 1$ beats the uniform band.

error	pk	uniform	best HY	ρ	SH	$c_{\mathcal{F}}$	$c'(0^+)$
sinusoid	1.57	0.8854	0.8854	1	0.8514	0.8510	$+0.087$
$\pm\varepsilon^*$ on 5%	20.1	0.4924	0.1790	0.099	0.2006	0.1704	-0.352
$\lvert e\rvert$ uniform	2.00	0.7213	0.7213	1	0.7227	0.7227	≈ 0
0.25 or 1 (90/10%)	3.08	0.5832	0.5543	0.68	0.4797	0.4640	-0.116

So the two-point form beats the uniform band exactly when $c'(0^+) < 0$, and the fitted histogram comes within about 0.02 bits of the Gibbs bound of the half it did not see.

Proposition G.21 (Fine data)**.** *Let $\Delta \le 2\varepsilon^*$. The honest law T^{ε^*} is silent on every datum.*

(a) In code Q its rate is $\gamma = m_\emptyset$, so it never beats the empty law, and T_0^{-Z} beats the empty law iff $p_\Delta \gamma_{\mathrm{ex}} < \gamma$.

(b) In code R its rate is at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$, which is below $m_\emptyset = \gamma = \log(\mathsf{R}/\Delta)$ by at least $\log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e}$ bits per datum; this is positive iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, about 66.2° for a longitude.

(c) If moreover $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$ and $\mathbb{P}[|e| < \Delta]\log(1/\Delta) \to 0$ as $\Delta \to 0$ (for instance, if $|e|$ has a bounded density near 0), then in code R the honest law has the least rate among $\emptyset$, T_0^{-Z} and T^{ε^*} for all small Δ.

Proof. Silence and (a) are Lemma G.15 and Proposition G.16's argument with silence probability 1. (b) is Lemma G.20(ii). (c) $m_{T_0^- z} \ge p_\Delta \gamma$, and $1 - p_\Delta = \mathbb{E}(1-|e|/\Delta)^+ \le \mathbb{P}[|e| < \Delta]$, so $m_{T_0^- z} - m_{T^{\varepsilon^*}} \ge \log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e} - \mathbb{P}[|e| < \Delta]\log(\mathsf{R}/\Delta)$, which is positive for small Δ; and (b) handles the empty law. □

So under SAI's pricing an approximate law keeps its value at precisions finer than its error: it narrows each answer to its band, which is worth $\log(\mathsf{R}/2\varepsilon^*)$ bits per datum (about 13 bits for a $1'$ error on a full circle of longitude), and by Proposition 9.2 a completion shaped like the error narrows it further.

Theorem G.22 (Almost-sure selection)**.** *Let $\mathcal{X}$ be a finite set of forms with $J_\lambda(X; D_n) = a_X + \sum_{j\le n} c_X(z_j) + r_X(n)$, where the z_j are i.i.d., $c_X(z) \in [0,\infty]$, and $r_X(n) = o(n)$ almost surely ($r_X = 0$ for deterministic forms and for stated ρ; $O(\log n)$ when ρ is integrated out). Let $m_X := \mathbb{E}c_X \in [0,\infty]$, $m_{\min} := \min_X m_X < \infty$ and $\mathcal{X}^* := \{X : m_X = m_{\min}\}$.*

(i) Almost surely, for all large n every λ-MAP within $\mathcal{X}$ lies in $\mathcal{X}^*$. If $\mathcal{X}^* = \{X^*\}$, the λ-MAP is eventually X^* alone.

(ii) Let $X, Y \in \mathcal{X}^*$ with $r_X = r_Y = 0$. If $c_X - c_Y$ is almost surely finite, with finite variance, and not almost surely constant, then almost surely $J_\lambda(X; D_n) < J_\lambda(Y; D_n)$ for infinitely many n and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ for infinitely many n: there is no eventual winner. If $c_X - c_Y = 0$ almost surely, a_X against a_Y decides at every n.

(iii) Let $r_X = r_Y = 0$, $m_Y < m_X < \infty$, $\mathbb{E}|c_X - c_Y| < \infty$ and $x := a_Y - a_X > 0$. Let τ_x be the first n with $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$, ς_x the last n with $J_\lambda(Y; D_n) \geq J_\lambda(X; D_n)$, and $\mu := m_X - m_Y$. Then $\mathbb{E}\tau_x > x/\mu$; if moreover $c_X - c_Y \leq b$ almost surely, then $\mathbb{E}\tau_x \leq (x+b)/\mu$; and on almost every sample path, $\tau_x/x \to 1/\mu$ and $\varsigma_x/x \to 1/\mu$ as $x \to \infty$.

Proof. The probabilistic facts used are standard (see Durrett, 2019). (i) Let $X^* \in \mathcal{X}^*$; then $\frac{1}{n} J_\lambda(X^*; D_n) \to m_{\min}$ almost surely. Let $X \notin \mathcal{X}^*$. If $\mathbb{P}[c_X = \infty] > 0$, almost surely some datum has infinite cost, after which $J_\lambda(X; D_n) = \infty$. If $c_X < \infty$ almost surely and $m_X = \infty$, then for each K, $\frac{1}{n}\sum_{j \leq n} \min(c_X(z_j), K) \to \mathbb{E}\min(c_X, K)$, and $\mathbb{E}\min(c_X, K) \to \infty$ as $K \to \infty$, so $\frac{1}{n} J_\lambda(X; D_n) \to \infty$. If $m_X < \infty$, then $\frac{1}{n}(J_\lambda(X; D_n) - J_\lambda(X^*; D_n)) \to m_X - m_{\min} > 0$. Since $\mathcal{X}$ is finite, the claim follows. (ii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ is a random walk with mean-zero, non-degenerate increments of finite variance. For each K, $\mathbb{P}[S_n > K] \to \frac{1}{2}$ by the central limit theorem, so $\mathbb{P}[S_n > K \text{ infinitely often}] \geq \limsup_n \mathbb{P}[S_n > K] = \frac{1}{2}$. This event is invariant under finite permutations of the increments, so by the Hewitt–Savage 0–1 law its probability is 1. Intersecting over $K \in \mathbb{N}$ gives $\limsup_n S_n = +\infty$ almost surely, and likewise $\liminf_n S_n = -\infty$; so $J_\lambda(X; D_n) - J_\lambda(Y; D_n) = a_X - a_Y + S_n$ changes sign infinitely often. The degenerate case is immediate. (iii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ has i.i.d. integrable increments with mean $\mu > 0$, and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ iff $S_n > x$. So $\tau_x = \min\{n : S_n > x\}$ is a stopping time with $\mathbb{E}\tau_x < \infty$ (a random walk with positive drift), and Wald's identity gives $\mathbb{E}S_{\tau_x} = \mu\mathbb{E}\tau_x$ with $S_{\tau_x} > x$; if the increments are at most b, also $S_{\tau_x} \leq x + b$. For the path statement fix $\epsilon \in (0, \mu)$; by the strong law there is $N(\omega)$ with $|S_n - n\mu| \leq \epsilon n$ for $n \geq N$. For $x > \max_{n \leq N} S_n$ we have $\tau_x - 1 \geq N$, and $S_{\tau_x} > x \geq S_{\tau_x - 1}$ gives $x/(\mu + \epsilon) < \tau_x \leq x/(\mu - \epsilon) + 1$. Similarly $\varsigma_x \geq \tau_x - 1 \geq N$ and $S_{\varsigma_x} \leq x$ give $\varsigma_x \leq x/(\mu - \epsilon)$. Divide by x, let $x \to \infty$ and then $\epsilon \to 0$. □

The code-R cost of a silent datum is $-\log$ of a share that is uniform near 0 (the proof of Lemma G.20), so it has all moments, and (i)–(iii) cover every application in this section. The bounded-increment bound of (iii) applies to the race between T_0^{-Z} and T^{ε^*}, whose cost differences are bounded above by the largest cost of T_0^{-Z}; for the race between T^{ε^*} and T_{tr} in code R the increments are unbounded, and only the lower bound and the path limits apply.

Corollary G.23 (The phases of selection, coarse data)**.** *Take a pricing with i.i.d. per-datum costs as in Proposition G.16: exactly Q/rand and R/rand (with ρ stated), or Q/det and R/det as a constant-c_{time} cost model (Proposition G.17). Assume $\Delta \geq 2\varepsilon^*$, $\mathbb{P}[e \neq 0] > 0$, $\varepsilon^* > 0$ and $h_{\mathrm{tr}} > h$. The pool is $\{\emptyset, T_0$ (and $T_0^{-Z}), T^{\varepsilon^*}, T_{\mathrm{tr}}\}$, with $a_\emptyset = 0$, $a_{T_0} = \lambda\ell_0$, $a_{T^{\varepsilon^*}} = \lambda(\ell_0 + h)$ and $a_{T_{\mathrm{tr}}} = \lambda(\ell_0 + h_{\mathrm{tr}})$. Almost surely:*

- *(a) the exact law is refuted for the first time at a geometric index of mean $1/p_\Delta$, and until then T_0 and T_0^{-Z} coincide;*
- *(b) T_{tr} is eventually the unique λ-MAP; the first time it beats T^{ε^*} has mean above $\lambda(h_{\mathrm{tr}} - h)/m_{T^{\varepsilon^*}}$, and at most $(\lambda(h_{\mathrm{tr}} - h) + \gamma)/m_{T^{\varepsilon^*}}$ under code Q, where the costs of T^{ε^*} are bounded by γ;*
- *(c) without T_{tr}, the eventual λ-MAP is the form of least rate among $m_\emptyset$, $m_{T_0^{-z}}$ and $m_{T^{\varepsilon^*}}$ (T^{ε^*} against T_0^{-Z} by Proposition G.18, ties by Theorem G.22(ii));*
- *(d) the expected switch points are: $\emptyset \to T_0$ at $\lambda\ell_0/\gamma$ (deterministic, if T_0 is not yet refuted); $T_0 \to T_0^{-Z}$ at the first refutation, of mean $1/p_\Delta$; $T_0^{-Z} \to T^{\varepsilon^*}$, if $m_{T_0^{-z}} > m_{T^{\varepsilon^*}}$, at a time of mean above $\lambda h/(m_{T_0^{-z}} - m_{T^{\varepsilon^*}})$, and at most $(\lambda h + \gamma_{\mathrm{ex}})/(m_{T_0^{-z}} - m_{T^{\varepsilon^*}})$ when γ_{ex} is bounded; and $T^{\varepsilon^*} \to T_{\mathrm{tr}}$ as in (b).*

When these expected switch points increase in this order, the predicted order of the λ-MAPs is $\emptyset \to T_0 \to T_0^{-Z} \to T^{\varepsilon^*} \to T_{\mathrm{tr}}$.

Proof. Theorem G.22 with the costs of Proposition G.16 and Lemma G.15: refutation by T_0 has probability p_Δ per datum, independently, and while unrefuted T_0 proves every datum (it is never silent), so $J_\lambda(T_0; D_n) = \lambda\ell_0 < n\gamma = J_\lambda(\emptyset; D_n)$ once $n > \lambda\ell_0/\gamma$. T_{tr} has rate 0 and every other form a positive rate ($p_\Delta > 0$, $s_\Delta > 0$, $\gamma_{\mathrm{si}} \geq \log \mathrm{e} - 1 > 0$), which gives (b) with Theorem G.22(iii) at level $\lambda(h_{\mathrm{tr}} - h)$. The bound in (d) for $T_0^{-Z} \to T^{\varepsilon^*}$ is Theorem G.22(iii) applied to the race between these two forms from $n = 0$, at level λh. □

A simulation of the Q/det cost model ($\ell_0 = 200$, $h = 20$, $h_{\mathrm{tr}} = 2000$, $c_{\mathrm{time}} = 22$; 400 runs per setting; code in the accompanying repository) shows the five phases in this order when the predicted switch points are ordered: for a sinusoidal error at $\Delta = 60'$ and $\lambda = 1$ (rates 0.323 and 0.283) the mean switch time to the honest law is 529.8, inside the interval $[495.6, 1251.3]$ given by (d), the mean switch time to the true theory is 7010.1, inside $[6994.9, 7024.9]$, and the mean first refutation is 91.8 against 94.25; when the two rates are close, the patched and the honest law alternate several times before the honest law settles. With rates tuned to be equal, the sign of $J_\lambda(T_0^{-Z}) - J_\lambda(T^{\varepsilon^*})$ changes 148 times on average up to $n = 2 \times 10^5$, and 43% of runs still change sign after $n = 10^5$. Under SAI's pricing with exact theory-machine costs (R/rand, sinusoid, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$; rates 0.1749 and 0.02953), the mean first time the honest law is the λ-MAP is 133.6 against a predicted switch near 138, and the mean first time the true theory is is 67060 against 67055; all 400 runs reach it.

Proposition G.24 (Humility)**.** *Let* $\Pi_\lambda(T \mid D) \propto 2^{-J_\lambda(T;D)}$ *be the power-prior posterior (Definition 3.6 of the companion paper on steeper priors) restricted to a finite pool of deterministic forms (code Q) containing* $\emptyset$, T_0^{-Z}, T^{ε^*} *and possibly* T_{tr}*, and ask a new question at a fresh time* t'*, with cell width* $\Delta' \leq 2\varepsilon^*$.

(a) The pool's three-valued prediction (the analogue, for the pool, of the trichotomy of Proposition 3.1) puts mass $\Pi_\lambda(T^{\varepsilon^*} \mid D) + \Pi_\lambda(\emptyset \mid D)$ *on "independent", mass* $\Pi_\lambda(T_0^{-Z} \mid D)$ *on the cell of* $f(t')$*, which is wrong with probability* $p_{\Delta'}$ *over the dither, and mass* $\Pi_\lambda(T_{\mathrm{tr}} \mid D)$ *on the true cell.*

(b) Without T_{tr}*, if the least rate at the training cell width* Δ *is attained by one form only, the posterior concentrates on it almost surely: the inducer ends humble (independent mass tending to* 1 *on finer questions) if* T^{ε^*} *or* $\emptyset$ *is selected, and overconfident (a wrong cell with probability* $p_{\Delta'}$*) if* T_0^{-Z} *is; Proposition G.18 at* Δ *and Proposition G.21 decide which. If two forms tie in rate with a non-degenerate cost difference of finite variance, the posterior oscillates between them forever.*

(c) In a randomised pool (code R) a band-confined completion $\mathrm{BC}_{\mathcal{G}}$ *is not silent: it puts on each cell its* $\mathcal{G}$*-mass, a spread over the cells its band meets and never probability one on a wrong cell; the exact law with random exceptions puts probability* $1 - \rho$ *on the cell of* $f(t')$.

Proof. (a) T^{ε^*} is silent on every question with $\Delta' \leq 2\varepsilon^*$ (Lemma G.15), and $\emptyset$ is silent. T_0^{-Z} proves the cell of $f(t')$, since t' is fresh and so not a patched time, and T_{tr} proves the true cell. (b) Theorem G.22(i), (ii), with $\Pi_\lambda(X \mid D)/\Pi_\lambda(Y \mid D) = 2^{-(J_\lambda(X;D) - J_\lambda(Y;D))}$, whose exponent diverges linearly when the rates differ. (c) The definition of $\mathrm{BC}_{\mathcal{G}}$, and the withdrawal probability ρ. □

body	ε^*, minimax fit (12 / 100 / 2000 yr)	pk, minimax fit	pk, mean fit
Mercury	$0.38'$ / $2.02'$ / $37.4'$	2.80–3.19	3.29–4.71
Venus	$0.36'$ / $0.47'$ / $3.6'$	2.27–3.65	2.81–4.07
Earth–Moon barycentre	$0.27'$ / $0.62'$ / $7.3'$	2.34–3.95	2.91–4.03
Mars	$1.44'$ / $3.04'$ / $49.9'$	2.71–3.38	3.33–4.47
Jupiter	$0.70'$ / $6.57'$ / $34.0'$	1.60–2.65	3.00–7.08
Saturn	$0.17'$ / $7.46'$ / $81.9'$	1.66–2.39	3.21–6.91
Moon (1 month / 1 yr / 18.6 yr)	$30.9'$ / $156'$ / $480'$	1.89–2.21	1.89–3.17

Table 7: Longitude errors of one fixed Kepler ellipse per body and window, against the plan94 and moon98 ephemerides (not the sky): the largest error of the minimax fit, and the peakedness of each fit over the windows of 12 to 2000 years (for the Moon, one month to 18.6 years). Largest longitude errors of the minimax fit; the least-squares fits of Table 8 measure angular separation and differ. Computed; code in the accompanying repository.

The Kepler residuals. Table 7 summarises the error laws. For each body and window, one fixed Kepler ellipse (seven elements) was fitted four ways (three-dimensional least squares, least squares on longitude and latitude, a robust fit for small mean error, and an L_p continuation for small maximum error), and two series of longitude errors at 8000 uniform times were kept: that of the fit with the least sample maximum (the *minimax fit*) and that of the fit with the least sample mean (the *mean fit*). The ephemerides are plan94 for the planets and moon98 for the Moon. An honest law whose tolerance is the sample maximum is slightly false if the window's true supremum is larger. A local parabola through the three time-sorted samples around the maximum estimates the excess at a median 2×10^{-4} (relative) and at most 1.9% in 33 of the 39 series apart from one, Mercury over 2000 years, where the estimate is unreliable because the sampling interval of about 91 days is comparable to Mercury's 88-day period. The band machines therefore use $\varepsilon = (1 + \kappa_\mathrm{b})\varepsilon^*$ with $\kappa_\mathrm{b} \in \{0, 0.01, 0.05\}$, and every comparison below comes out the same for the three values. The rates are computed at eleven cell widths from $0.1'$ to $30°$, so that there are 429 (body, window, cell width) cells. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand (and under R/det at $\lambda = 1$ and 4), the uniform band has the least rate in all 429 cells, with exact closed-form costs; in that pool's tolerance family with whole-circle exceptions the optimum is a tolerance of about 0.7–$1.0\,\varepsilon^*$ with at most 1% of the data as exceptions. With band-confined completions, the histogram machine SH has the least rate in all 429 cells at each κ_b, between 0.4% and 25.7% below the uniform band (median 6.6%), and lies within 0.0011 bits of the Gibbs bound of its test half in every coarse-regime pair; the best two-point form beats the uniform band in 379 of the 429 cells at $\kappa_\mathrm{b} = 0.01$, and in the coarse regime in 34 of the 39 body–window pairs, with best ρ between 0.52 and 0.997, exactly where the criterion of Proposition 9.2(d) says. The five pairs in which the uniform band beats every two-point form are Jupiter over 12 and 30 years, Saturn over 100 years and the Moon over one month and 18.6 years, whose minimax-fit peakedness is 1.60–1.96. The gain persists at fine precision: for the Earth–Moon barycentre over 100 years at $\Delta = 1'$ the rates of the uniform band, the best two-point form and SH are 0.651, 0.560 and 0.503 bits per datum, and for Mars over 12 years 1.660, 1.558 and 1.496. Under Q/det (naming a time costs $c_\mathrm{time} = \log(\text{days in the window}) + 2$ bits, 14.1–21.5 for the planets and 6.8, 10.5 and 14.7 for the Moon's windows; a finite-horizon cost model) the threshold $(\lambda c_\mathrm{time} + \gamma)/(2\gamma)$ grows as the cells coarsen. At $\lambda = 1$ the exact law with exceptions wins from cell widths of about ε^* upwards, the honest law only at the coarsest cells ($30°$, and from $10°$ for Jupiter over 30 years and Saturn over 2000 years) and not even there for 17 of the 39 pairs, and at finer cells no law pays; the best

member of the tolerance family is then a small tolerance (0.05–0.5 ε^*) with exception shares up to tens of per cent. At $\lambda = 2$ the honest law takes over at cell widths from $8.1'$ to $962'$ (median $134'$), and at $\lambda = 4$ from $0.72'$ to $962'$ (median $13.4'$). Under Q/rand the exact law with random exceptions wins almost everywhere. The residuals are deterministic functions of time, so a better theory (Newton's, or the ephemeris itself) eventually beats every Kepler-based machine.

G.5 Derivation horizons and sound reasoning with approximate models

Work over $B = \mathrm{RCF}$ with constants $x_0, x_1, \ldots$ for the states of a system in $\mathbb{R}^m$ (one constant per coordinate), a rational initial state a, a map Ψ definable in B, and a norm $|\cdot|$ for which B proves that Ψ is L-Lipschitz (for a linear $\Psi(x) = Ax$ with rational A and the sup-norm, $L = \|A\|_\infty$, provably in RCF). The *honest dynamical law* is $T^\varepsilon_\Psi := \{|x_0 - a| \le \varepsilon^{(0)}\} \cup \{|x_{s+1} - \Psi(x_s)| \le \varepsilon : s \ge 0\}$, and the *exact law* is $T^0_\Psi := \{x_0 = a\} \cup \{x_{s+1} = \Psi(x_s) : s \ge 0\}$.

Proposition G.25 (Derivation horizon)**.**

(a) For every $L \ge 0$, $T^\varepsilon_\Psi \vdash |x_r - \Psi^r(a)| \le \varepsilon^{(r)}$, with $\varepsilon^{(r)} := L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ for $L \ne 1$ and $\varepsilon^{(r)} := \varepsilon^{(0)} + r\varepsilon$ for $L = 1$.

(b) Let $\Psi(x) = Ax$ be linear, with sup-norm tolerances. The values of the i-th coordinate $x_{r,i}$ over the models of $B \cup T^\varepsilon_\Psi$ form exactly the interval $(A^r a)_i \pm h_{r,i}$, where $h_{r,i} := \varepsilon^{(0)}\|(A^r)_{i\cdot}\|_1 + \varepsilon \sum_{s<r}\|(A^s)_{i\cdot}\|_1$ (the projection of a zonotope). So T^ε_Ψ proves "$x_{r,i} \in [v - \delta, v + \delta]$" iff that interval contains $(A^r a)_i \pm h_{r,i}$. In dimension one, $h_r = \varepsilon^{(r)}$ with $L = |A|$, so (a) is tight.

(c) Put $r^*(\delta) := \max\{r : \varepsilon^{(r)} \le \delta\}$. For $\varepsilon^{(0)} = 0$: $r^*(\delta) = \lfloor \delta/\varepsilon \rfloor$ for $L = 1$; $r^*(\delta) = \lfloor \log_L(1 + \delta(L - 1)/\varepsilon) \rfloor$ for $L > 1$; and for $L < 1$, $r^*(\delta) = \infty$ iff $\delta \ge \varepsilon/(1 - L)$, and $r^*(\delta) = \lfloor \log_L(1 - \delta(1 - L)/\varepsilon) \rfloor$ otherwise. For every L-Lipschitz Ψ, T^ε_Ψ proves a datum at precision δ about x_r, centred at $\Psi^r(a)$, if $r \le r^*(\delta)$; for linear Ψ in dimension one with $L = |A|$, also only if.

Proof. (a) By induction, $|x_{s+1} - \Psi^{s+1}(a)| \le |x_{s+1} - \Psi(x_s)| + |\Psi(x_s) - \Psi(\Psi^s(a))| \le \varepsilon + L\varepsilon^{(s)}$; each step is an instance of the axioms and of the provable Lipschitz inequality, and the recursion $\varepsilon^{(s+1)} = L\varepsilon^{(s)} + \varepsilon$ has the stated solution. (b) $x_r - A^r a = A^r(x_0 - a) + \sum_{s<r} A^{r-1-s}u_s$ with $u_s := x_{s+1} - Ax_s$, and in the real model the deviations $x_0 - a$ and u_s range freely over sup-norm boxes of half-widths $\varepsilon^{(0)}$ and ε. A linear image of a product of boxes is a zonotope, which is convex; its projection on coordinate i is the interval of half-width the sum of the ℓ_1-norms of the corresponding rows, with the extreme values attained by sign choices. So exactly the stated interval of values occurs in $\mathbb{R}$. Since RCF is complete and the claim about $x_{r,i}$ involves only the finitely many axioms with $s < r$, T^ε_Ψ proves an interval statement about $x_{r,i}$ iff it holds in every real model, that is iff the interval contains $(A^r a)_i \pm h_{r,i}$. (c) Solve $\varepsilon^{(r)} \le \delta$; the "if" is (a), and the "only if" for linear Ψ in dimension one is (b). □

For non-linear Ψ the "only if" fails. Let $\Psi(x) = 0$ on $[-1, 1]$ and $\Psi(x) = 2(x - \operatorname{sign} x)$ outside, so that $L = 2$, with $a = 0$, $\varepsilon^{(0)} = 0$ and $\varepsilon = 1/10$. Then $|x_s| \le 1/10 \le 1$ gives $\Psi(x_s) = 0$ and so $|x_{s+1}| \le 1/10$; by induction T^ε_Ψ proves "$x_r \in [-1/10, 1/10]$" at every depth, while $\varepsilon^{(r)} = (2^r - 1)/10$ gives $r^*(1/10) = 1$. A computation (code in the accompanying repository) confirms the closed form against exact interval propagation for $r \le 200$ and several L between 0.5 and 2; at $\delta/\varepsilon = 10^4$ the horizons are 10^4, 463, 72 and 13 for $L = 1$, 1.01, 1.1 and 2, and infinite for $L \le 0.99$. In dimensions 3 and 10, for random matrices of spectral radius 0.95–1.05, propagating boxes by interval arithmetic overestimates the exact tolerance $h_{r,i}$ by a median factor of 5–300 at $r = 10$

and up to about 10^{10} at $r = 30$, and the ∞-norm Lipschitz bound by up to about 7×10^{13}: a reasoner that propagates boxes has a much shorter effective horizon than the theory.

Remark G.26 (Typical against worst case). With $L = 1$ and $r = 100$ steps the propagated tolerance is 100ε, while with independent random-sign errors $\pm\varepsilon$ the mean deviation is $\varepsilon r\binom{r}{r/2}2^{-r} \approx 7.96\varepsilon$: a probabilistic error model answers much further out, with a stated confidence instead of a guarantee.

Proposition G.27 (The horizon of an exact approximation)**.** *Suppose the true trajectory satisfies T^{ε}_{Ψ}, but the inducer holds the exact law T^{0}_{Ψ} with the same a. Then T^{0}_{Ψ} is consistent and proves $x_r = \Psi^r(a)$, which is within $\varepsilon^{(r)}$ of the true x_r. If $r \le r^*(\delta)$, the depth-r conclusions of T^{0}_{Ψ} are δ-correct for every trajectory compatible with the honest law. For linear Ψ in dimension one with $L = |A|$, some trajectory compatible with the honest law attains the deviation $\varepsilon^{(r)}$, so in that case they are δ-correct for all such trajectories only if $r \le r^*(\delta)$. And T^{0}_{Ψ} together with one true datum about x_r whose interval excludes $\Psi^r(a)$ is inconsistent, and so proves every sentence.*

Proof. The exact trajectory is a model of T^{0}_{Ψ}, and the true trajectory is a model of T^{ε}_{Ψ}. The middle claims are Proposition G.25(a), (b). For the last, the datum and $x_r = \Psi^r(a)$ are jointly refutable in B. □

Proposition G.28 (Sound reasoning with approximate models)**.** *Let $T_i := \{\forall t \in \mathrm{Dom}_i : \ |q(t) - f_i(t)| \le \varepsilon_i\}$, $i \le k$, be honest pointwise models of one quantity q (scalar, or vector-valued with a norm), with q otherwise unconstrained, and let T^{0}_{i} be their exact versions ($\varepsilon_i = 0$).*

(a) Let B be true in the intended structure. If every T_i is true, then $\bigcup_i T_i$ is consistent and every sentence derivable from $B \cup \bigcup_i T_i$ is true; every derived statement about $q(t)$ is a set containing the true value.

For (b)–(d) assume moreover that B decides every sentence built from the f_i, the Dom_i and the ε_i at the times concerned (for example $B = \mathrm{RCF}$, with semialgebraic f_i and Dom_i and rational ε_i).

(b) $\bigcup_i T^{0}_{i}$ is inconsistent, and so proves every sentence, iff $f_i(t) \ne f_j(t)$ for some $t \in \mathrm{Dom}_i \cap \mathrm{Dom}_j$.

(c) $\bigcup_i T_i$ is consistent iff for each t the balls $\{v : |v - f_i(t)| \le \varepsilon_i\}$ with $t \in \mathrm{Dom}_i$ have a common point. For scalar q, and for vector q with sup-norm (box) tolerances, this holds iff it holds pairwise, $|f_i(t) - f_j(t)| \le \varepsilon_i + \varepsilon_j$ on every overlap. For Euclidean tolerances in $\mathbb{R}^m$, $(m+1)$-wise intersection is necessary and sufficient, and pairwise intersection is not sufficient.

(d) If B proves that a map Ψ is L_Ψ-Lipschitz, then $T_i \vdash |\Psi(q(t)) - \Psi(f_i(t))| \le L_\Psi\varepsilon_i$.

Proof. (a) The intended structure is a model, and derivations preserve truth. (b) If two exact models disagree at a common time t, together they prove $f_i(t) = f_j(t)$, which B refutes by the hypothesis. If they agree on all overlaps, then in every model of B (in which B's decisions fix the values $f_i(t)$) setting $q(t) := f_i(t)$ for $t \in \mathrm{Dom}_i$, and arbitrarily elsewhere, gives a model. (c) The constants $q(t)$ for distinct t are distinct and otherwise unconstrained, so by compactness $B \cup \bigcup_i T_i$ is consistent iff for each t, B is consistent with "some v lies in all the balls at t". By the hypothesis B decides this sentence, and B is true in the intended structure, so it is consistent with B iff it holds there, that is iff the balls have a common point. Intervals that meet pairwise have $\max_i$(left end) $\le \min_j$(right end), which is a common point (Helly's theorem in dimension one), and boxes are products of intervals. Helly's theorem in $\mathbb{R}^m$ (standard; Helly, 1923) gives the

$(m+1)$-wise criterion. Three discs of radius 1.1 centred at the vertices of an equilateral triangle of side 2 meet pairwise but have no common point, since the circumradius is $2/\sqrt{3} \approx 1.155$. (d) is the Lipschitz inequality in B. □

Without the hypothesis on B, (b) and (c) fail: take $B = \emptyset$, $f_i(t)$ and $f_j(t)$ distinct constant symbols and $\mathrm{Dom}_i = \mathrm{Dom}_j$ a single time; then $T_i^0 \cup T_j^0$ is consistent (interpret the two constants as equal), although the values differ in the intended structure. A computation finds pairwise and joint overlap in agreement on 10^5 random families of 2 to 6 intervals and on 2×10^4 random families of boxes in $\mathbb{R}^2$.

G.6 Best systems

Definition G.29 (Explanatory power)**.** For a theory T and a module $\mathcal{A} \subseteq T$, $\mathrm{EP}_D(\mathcal{A}; T) := \big(\mathrm{Exc}_D(T \setminus \mathcal{A}) - \mathrm{Exc}_D(T)\big)/\|\mathcal{A}\|$: the bits of data the module explains per bit of its statement, in the context T.

Proposition G.30 (Marginal slopes)**.** *Let T^* be a λ-MAP within a family of theories that contains $T^* \setminus \mathcal{A}$ and $T^* \cup \mathcal{B}$ for the modules considered. For every module $\mathcal{A} \subseteq T^*$, $\mathrm{Exc}_D(T^* \setminus \mathcal{A}) - \mathrm{Exc}_D(T^*) \geq \lambda\|\mathcal{A}\|$; and for every $\mathcal{B}$ disjoint from T^*, $\mathrm{Exc}_D(T^*) - \mathrm{Exc}_D(T^* \cup \mathcal{B}) \leq \lambda\|\mathcal{B}\|$.*

Proof. $J_\lambda(T^*; D) \leq J_\lambda(T^* \setminus \mathcal{A}; D)$ and $J_\lambda(T^*; D) \leq J_\lambda(T^* \cup \mathcal{B}; D)$, with $\|T^*\| - \|T^* \setminus \mathcal{A}\| = \|\mathcal{A}\|$ and $\|T^* \cup \mathcal{B}\| - \|T^*\| = \|\mathcal{B}\|$. □

Proof of Proposition 9.3. By the hypotheses (Proposition G.8) the hull lemma (Lemma 3.8(v) of the companion paper on steeper priors) applies: for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are exactly the theories at v_i, and at $\lambda = \sigma_j$ $(j \geq 1)$ they are the theories on the closed edge $[v_{j-1}, v_j]$. (b)⇒(a): $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$. (a)⇒(b): if some edge slope σ_j lay in (τ, σ), the λ-MAP set at $\lambda = \sigma_j$, the theories on $[v_{j-1}, v_j]$, would differ from the set just below σ_j, the theories at v_j only, since $v_{j-1} \neq v_j$. So no edge slope lies in (τ, σ), and (τ, σ) lies in one interval (σ_{i+1}, σ_i), which is (b). (b)⇔(c) is Corollary 5.4(c) of the companion paper on steeper priors: a zero-window (σ, τ)-kink with $\tau < \sigma$ is a vertex v_i with $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$, and conversely every vertex v_i is a zero-window (σ_i, σ_{i+1})-kink, hence a (σ, τ)-kink whenever $\sigma \leq \sigma_i$ and $\tau \geq \sigma_{i+1}$. □

A computation on 400 random point families with 10 random intervals each, testing "the λ-MAP set is constant on (τ, σ)" exactly at every crossing point and between consecutive ones, agrees with (b) in all 4000 cases.

Proposition G.31 (The trivialisation is blocked)**.** *Let $\mathcal{L}' \supseteq \mathcal{L}$ be any extension of the language, for example by a predicate* Act *true of exactly the actual things, with the axiom $\forall x\, \mathrm{Act}(x)$; fix a syntax code for $\mathcal{L}'$, and let the data be base-language answers whose provability from the theories considered is decidable. Then every finite theory T in $\mathcal{L}'$ consistent with D has $\|T\| + \mathrm{Exc}_D^Q(T) \geq K(y_D \mid \mathrm{dom}\, D) - K(\mathcal{L}') - c$, where $K(\mathcal{L}')$ is the complexity of a decoder for $\mathcal{L}'$ and its syntax code. Without decidability, add n on the left. If $\mathcal{L}'$ and its code are fixed independently of D, $K(\mathcal{L}')$ is a constant.*

Proof. The weights in the proof of Proposition G.9 search for, or decide, proofs from T of base-language answers. Given a decoder for the syntax of $\mathcal{L}'$ they are computable (respectively lower semicomputable) from T and the questions, and they do not otherwise depend on the language. So the argument of Proposition G.9(b) (respectively (a)) gives $K(y_D \mid \mathrm{dom}\, D) \leq K(\mathcal{L}') + K(T \mid \mathcal{L}') + \mathrm{Exc}_D^Q(T) + c$ (respectively with n added), and $K(T \mid \mathcal{L}') \leq \|T\| + c$. □

Proposition G.32 (Facts used once never enter for $\lambda > 1$)**.**

(a) Let T^* be a λ-MAP with $\lambda > 1$ and χ a sentence. Suppose that, in the context $T^* \setminus \{\chi\}$, adding χ turns exactly the exceptions in a set $\mathcal{D}_\chi \subseteq D$ into proved data and changes nothing else, and that $\sum_{(\mathcal{Q},a)\in\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi)$. Then $\chi \notin T^*$. In particular this applies to a datum's answer a stated as an axiom whenever $\gamma(\mathcal{Q}) \le \ell(a)$ and stating it proves no other datum in that context.

*(b) Let provability on the data be decidable in the class, with κ the constant of Proposition G.9, and let T_s be b-*sufficient*: $\|T_s\| + \mathrm{Exc}_D(T_s) \le K(y_D \mid \mathrm{dom}\, D) + b$. Every theory T has $\mathrm{Exc}_D(T) \ge \mathrm{Exc}_D(T_s) - (\|T\| - \|T_s\|) - (b+\kappa)$; a module $\mathcal{A}$ added to T_s has $\mathrm{EP}_D(\mathcal{A}; T_s \cup \mathcal{A}) \le 1 + (b+\kappa)/\|\mathcal{A}\|$; and every λ-MAP T with $\lambda > 1$ has $\|T\| \le \|T_s\| + (b+\kappa)/(\lambda-1)$.*

Proof. (a) If $\chi \in T^*$, deleting it raises Exc_D by exactly $\sum_{\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi) = \lambda\|\{\chi\}\|$, contradicting Proposition G.30. For an answer stated as an axiom, $\mathcal{D}_\chi$ is the one datum, and $\gamma(\mathcal{Q}) \le \ell(a) < \lambda\ell(a)$. (b) Proposition G.9(b) gives $\|T\| + \mathrm{Exc}_D(T) \ge K(y_D \mid \mathrm{dom}\, D) - \kappa \ge \|T_s\| + \mathrm{Exc}_D(T_s) - b - \kappa$, which is the first claim, a right line of slope 1 through the point of T_s. For a module $\mathcal{A}$ disjoint from T_s, $\|T_s \cup \mathcal{A}\| - \|T_s\| = \|\mathcal{A}\|$, which gives the second. For a λ-MAP T, $\lambda\|T\| + \mathrm{Exc}_D(T) \le \lambda\|T_s\| + \mathrm{Exc}_D(T_s)$; combined with the first claim this gives $(\lambda-1)(\|T\| - \|T_s\|) \le b + \kappa$, which is Corollary 5.5(a) of the companion paper on steeper priors in this class. □

So at $\lambda = 1$ a fact that explains only itself is excluded only by the code's per-axiom overhead, while every $\lambda > 1$ excludes every module whose explanatory power is below λ.

The solar-system toy. Bodies $b \le k_\mathrm{b}$ have n_b observations each, each of literal cost γ (code Q), and $n_\mathrm{tot} := \sum_b n_b$. The modules are the law Λ (L bits), body b's elements El_b (k bits), and Kepler's third law K3 (L_3 bits), which shortens every elements module by s bits, $0 < s < k$, because it derives the period from the semi-major axis. $\Lambda \cup \mathrm{El}_b$ proves all of b's observations, and no other combination proves any observation: laws alone answer no position question, and neither do elements alone. The theories are Λ or not, K3 or not, and a set S of bodies whose elements are axioms.

Proposition G.33 (Laws and initial conditions in the solar-system toy)**.** *Put $k_\Lambda := k$, $k_{\Lambda+\mathrm{K3}} := k - s$ and $S_x(\lambda) := \{b : \gamma n_b > \lambda k_x\}$ for $x \in \{\Lambda, \Lambda + \mathrm{K3}\}$. The optimal values of the three law choices are*

$$J^*_\emptyset = \gamma n_\mathrm{tot}, \qquad J^*_\Lambda = \lambda L + \sum_b \min(\gamma n_b, \lambda k), \qquad J^*_{\Lambda+\mathrm{K3}} = \lambda(L + L_3) + \sum_b \min(\gamma n_b, \lambda(k-s)),$$

*attained by $S = S_x(\lambda)$ (bodies with $\gamma n_b = \lambda k_x$ may go either way); the λ-MAP is the best of the three (*K3 *without Λ is useless).*

(a) Λ beats the empty theory iff $\sum_{b\in S_\Lambda(\lambda)}(\gamma n_b - \lambda k) > \lambda L$.

(b) $J^*_{\Lambda+\mathrm{K3}} - J^*_\Lambda = \lambda L_3 - \sum_b \big[\min(\gamma n_b, \lambda k) - \min(\gamma n_b, \lambda(k-s))\big]$. If no body has $\lambda(k-s) < \gamma n_b \le \lambda k$, so that S is the same with and without K3*, then $J^*_{\Lambda+\mathrm{K3}} < J^*_\Lambda$ iff $s|S| > L_3$, whatever λ is.* K3 *is in the λ-MAP iff in addition $J^*_{\Lambda+\mathrm{K3}} < J^*_\emptyset$.*

(c) In the λ-MAP $\Lambda \cup \{\mathrm{El}_b : b \in S\}$ (with or without K3*), $\mathrm{EP}(\Lambda) = \gamma \sum_{b\in S} n_b/L$ and $\mathrm{EP}(\mathrm{El}_b) = \gamma n_b/k_x$.*

Proof. Without Λ no observation is proved, so the best such theory is empty, with $J = \gamma n_{\text{tot}}$. For a law choice x with Λ, J_λ is a constant plus a sum over bodies, where including El_b costs λk_x and excluding it costs γn_b; minimise each term. (a) is the comparison of J^*_Λ with $J^*_\emptyset$. (b) The bracket is λs when $\gamma n_b > \lambda k$, $\gamma n_b - \lambda(k-s)$ when $\lambda(k-s) < \gamma n_b \le \lambda k$, and 0 otherwise; without bodies in the middle range the difference is $\lambda(L_3 - s|S|)$. The λ-MAP is the best of the three law choices, the empty one included. (c) Deleting Λ (respectively El_b) makes exactly the observations of the bodies in S (respectively of b) literal. □

Part (b) compares law choices; it is not membership of the λ-MAP. With two bodies of one observation each, $\gamma = 10$, $L = 1000$, $k = 5$, $s = 4$, $L_3 = 1$ and $\lambda = 1$, S is both bodies with and without K3 and $s|S| = 8 > L_3$, but the λ-MAP is the empty theory: $J^*_\emptyset = 20$ against $J^*_{\Lambda+\text{K3}} = 1003$ and $J^*_\Lambda = 1010$. The middle-range condition of (b) matters too: in a population of eight planets with 655 to 3838 observations, sixty asteroids with a median of about five and thirty comets with one or two ($L = 300$, $L_3 = 150$, $k = 60$, $s = 15$, $\gamma = 12.4$), K3 is in the λ-MAP at $\lambda = 6.75$ although $s|S_\Lambda(\lambda)| = 135 < L_3$, because with K3 more bodies pay for their elements. A computation (code in the accompanying repository) confirms the closed form against brute force over all theories in 2100 random cases, and the criterion of (b) in all 13250 random cases with the same S under both law choices.

In the same population the elements of a well-observed body explain hundreds of bits per bit (Proposition G.33(c)) and sit on the steep side of the hull with the law, and a computation (code in the accompanying repository) finds that the λ-MAP is "the law plus the elements of all eight planets" for every $\lambda \in (7.03, 135.37)$, while below $\lambda \approx 7$ it changes with every asteroid. Inside such an interval the worry that the best system depends on an arbitrary choice of balance (cf. Woodward, 2014) is moot, and robustness to the balance (Proposition 9.3) is a checkable property of a language, a family of candidates and a data set.

Proposition G.34 (Laws and initial conditions in the question format)**.** *Let the data be conditional questions "given the elements α_b of body b, is ℓ the case?", with compatibility as in* (E3)*: a datum is compatible with T when $B \cup T \cup \{\alpha_b\}$ is consistent and proves ℓ. In the toy, every theory of the family (the law,* K3*, and elements modules stating the true elements) is consistent with every true antecedent. Then an elements module El_b with the same content as α_b adds nothing, so $\text{EP}(\text{El}_b) = 0$ and no λ-MAP contains elements; and Λ alone answers every question, with $\text{EP}(\Lambda) = \gamma n_{\text{tot}}/L$ however the n_{tot} observations are spread over bodies.*

Proof. Immediate from the definition of compatibility: $T \cup \{\alpha_b\} \cup \text{El}_b$ and $T \cup \{\alpha_b\}$ prove the same sentences, and $\Lambda \cup \{\alpha_b\}$ proves all of b's observations. Adding El_b raises J_λ by $\lambda k > 0$ and lowers no exception cost. □

Without the consistency condition the encoding is degenerate: a theory that refutes the antecedents would prove every datum (Proposition E.3). In the question format the distinction between laws and initial conditions is the distinction between theory and question antecedent.

G.7 Explanatory power, unification and correction

Proposition G.35 (Explanatory power as a likelihood ratio; unification)**.** *Write $P_T(D) := 2^{-\text{Exc}_D(T)}$ (code Q or R). For a hypothesis h, a background theory $T_\flat$, data sets D_1, D_2 on disjoint questions and a theory x, put $I(h; D \mid T_\flat) := \log(P_{T_\flat \cup h}(D)/P_{T_\flat}(D))$, $I_x(D_1; D_2) := \log \frac{P_x(D_1 \cup D_2)}{P_x(D_1)P_x(D_2)}$ and $U(D_1, D_2; h \mid T_\flat) := I(h; D_1 \cup D_2 \mid T_\flat) - I(h; D_1 \mid T_\flat) - I(h; D_2 \mid T_\flat)$.*

(a) $\|\mathcal{A}\| \, \text{EP}_D(\mathcal{A}; T) = I(\mathcal{A}; D \mid T \setminus \mathcal{A})$.

(b) $U(D_1, D_2; h \mid T_\flat) = I_{T_\flat \cup h}(D_1; D_2) - I_{T_\flat}(D_1; D_2)$. *If P is a product over questions for both theories, then $U = 0$ for every split.*

(c) *Let h be a law family with a k-bit parameter θ under the uniform prior, $P_h(D) := 2^{-k} \sum_\theta P_{h \cup \{\theta\}}(D)$, and $P_\emptyset$ a product. If each of D_1, D_2 is proved by $h \cup \{\theta^*\}$ and has likelihood 0 under every other θ, then $U(D_1, D_2; h) = k$. If $\theta = (\theta_1, \theta_2)$ with D_i depending only on θ_i, then $U = 0$. If $\theta_i = (\theta_{\mathrm{sh}}, \theta_i')$ share a k_{sh}-bit constant θ_{sh} and each D_i identifies $(\theta_{\mathrm{sh}}, \theta_i')$, then $U = k_{\mathrm{sh}}$.*

Proof. (a) is the definition of EP. (b) Expand the six logarithms: the first term of the difference is $\log P_{T_\flat \cup h}(D_1 \cup D_2) - \log P_{T_\flat \cup h}(D_1) - \log P_{T_\flat \cup h}(D_2)$, and the second is the same expression for $T_\flat$; a product law makes $I_x(D_1; D_2) = 0$. (c) Use (b) with $I_\emptyset = 0$. In the first case $P_h(D_1) = P_h(D_2) = P_h(D_1 \cup D_2) = 2^{-k}$, so $I_h(D_1; D_2) = k$. In the second, $P_h(D_1 \cup D_2) = P_h(D_1) P_h(D_2)$. In the third, with θ_i' of k' bits, $P_h(D_i) = 2^{-k_{\mathrm{sh}} - k'}$ and $P_h(D_1 \cup D_2) = 2^{-k_{\mathrm{sh}} - 2k'}$, so $I_h(D_1; D_2) = k_{\mathrm{sh}}$. □

A computation with $k = 6$ and observations given by a random table gives $U = 0.00$ for a stated parameter, 6.00 for a marginalised parameter and one body, 0.00 for two bodies with independent parameters, and 3.00 for two bodies sharing a 3-bit constant; the identity (b) holds to 3×10^{-15} on 1000 random joint laws. Myrvold describes unification as the degree to which a hypothesis renders phenomena informationally relevant to one another, a contribution to evidential support (Myrvold, 2003); by (b), U is the change in the mutual information between D_1 and D_2 that adding h brings about, which has that form. We do not claim that his formula coincides with U in every detail. At $\lambda = 1$ a module pays iff its explanatory power exceeds 1, which is the compression criterion of inductive logic programming (cf. Muggleton, 1995; Srinivasan et al., 1994).

Copernicus against Ptolemy (for the history, see Kuhn, 1957) is unification without better prediction. The lengths in the following comparison are not measured.

Proposition G.36 (Unification without better prediction)**.** *Consider three theories with a common part: Pt_f, Ptolemy's system with free epicycle phases, which pays for the coincidence of each superior planet p's retrograde motion with its opposition in its fit, a_p bits per planet; Pt_s, the same with the coincidences stated as axioms, $\sum_p a_p$ law bits more than Pt_f and with the fit of* Cop*; and* Cop*, Copernicus's system, H law bits more than Pt_f and with the fit of Pt_s.*

(a) Cop *beats Pt_s, which makes the same predictions, at every λ iff $H < \sum_p a_p$.*

(b) Cop *is Pt_f plus a module with $\mathrm{EP} = \sum_p a_p / H$, so it beats Pt_f iff $\lambda < \sum_p a_p / H$.*

(c) Pt_s *beats Pt_f iff $\lambda < 1$.*

Hence if $H < \sum_p a_p$, Cop *is the best of the three exactly for $\lambda < \sum_p a_p / H$, and Pt_f above; if $H \geq \sum_p a_p$,* Cop *is never strictly best: Pt_s is best for $\lambda < 1$ (tying* Cop *if $H = \sum_p a_p$) and Pt_f for $\lambda > 1$.*

Proof. Measure lengths and fits relative to Pt_f and write $a := \sum_p a_p$. Then $J_\lambda(\mathrm{Pt}_f) = 0$, $J_\lambda(\mathrm{Pt}_s) = \lambda a - a$ and $J_\lambda(\mathrm{Cop}) = \lambda H - a$. (a) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_s) = \lambda(H - a)$, negative for every $\lambda > 0$ iff $H < a$. (b) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_f) = \lambda H - a$, and the module of H bits lowers the fit by a. (c) $J_\lambda(\mathrm{Pt}_s) - J_\lambda(\mathrm{Pt}_f) = (\lambda - 1)a$. If $H < a$, then $a/H > 1$: for $\lambda < a/H$, Cop beats both others; for $\lambda > a/H > 1$, Pt_f beats both. If $H \geq a$: for $\lambda < 1$, Pt_s beats Pt_f and is at least as good as Cop, with equality iff $H = a$; for $\lambda > 1 \geq a/H$, Pt_f beats both; at $\lambda = 1$, Pt_s and Pt_f tie and Cop is no better. □

Newton's theory does not entail Kepler's laws; with the planets' masses and initial conditions it contradicts their exact form (cf. Duhem, 1954) and derives corrected statements, and Kepler's

laws within tolerances on windows, as in the model of reduction to a corrected reduced theory (cf. Schaffner, 1967); deriving "Kepler within ε on a window" from Newton's theory itself needs validated perturbation bounds (Section 10.5). The following toy makes the structure provable. Over $B = \mathrm{RCF}$ with constants $x_{t,1}, x_{t,2}$ for $t \le \bar{t}$ (a planar state x_t per step), let Rot be the rotation with cosine $3/5$ and sine $4/5$, $\mathbf{P}$ a rational 2×2 matrix with operator norm $\|\mathbf{P}\|$, $\eta_c := 1/100$, $A := \mathrm{Rot} + \eta_c \mathbf{P}$ and $a := (1, 0)$; $|\cdot|$ is the Euclidean norm, and $|v| \le \varepsilon$ is stated as $|v|^2 \le \varepsilon^2$ with ε^2 rational. The *old exact law* is $T_0 := \{x_t = \mathrm{Rot}^t a : t \le \bar{t}\}$ (uniform circular motion), the *successor* is $T_{\mathrm{new}} := \{x_0 = a\} \cup \{x_{t+1} = A x_t : t < \bar{t}\}$, the *honest old law* is $T^{\varepsilon} := \{|x_t - \mathrm{Rot}^t a| \le \varepsilon : t \le \bar{t}\}$, and a *time-dependent form* is $T^{(\varepsilon_\cdot)} := \{|x_t - \mathrm{Rot}^t a| \le \varepsilon_t : t \le \bar{t}\}$.

Proposition G.37 (Explanation by approximate correction)**.**

(a) $T_{\mathrm{new}} \vdash T^{\varepsilon}$ *for every rational* $\varepsilon \ge \varepsilon(\bar{t}) := ((1 + \eta_c \|\mathbf{P}\|)^{\bar{t}} - 1)|a|$.

(b) $T_{\mathrm{new}} \vdash T^{\varepsilon}$ *iff* $\varepsilon^2 \ge \varepsilon_{\min}(\bar{t})^2 := \max_{t \le \bar{t}} |A^t a - \mathrm{Rot}^t a|^2$*, a rational number. So* $\varepsilon_{\min}(\bar{t}) \le \varepsilon(\bar{t})$*, and* $T^{\varepsilon_{\min}(\bar{t})}$ *is the strongest uniform-tolerance form of the old law on the window that the successor proves.*

(c) If $\eta_c \mathbf{P} a \ne 0$ *and* $\bar{t} \ge 1$*, then* $T_{\mathrm{new}} \vdash \neg \bigwedge T_0$.

(d) Among uniform-tolerance forms, T^{ε} *is a common consequence of* T_0 *and* T_{new} *iff* $\varepsilon \ge \varepsilon_{\min}(\bar{t})$*; the strongest common one is* $T^{\varepsilon_{\min}(\bar{t})}$.

(e) The time-dependent form with $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$ *is a common consequence of both theories, strictly stronger than* $T^{\varepsilon_{\min}(\bar{t})}$ *whenever* $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ *for some* t.

(f) The strongest common consequence of the two finite theories is their disjunction $\bigwedge T_0 \vee \bigwedge T_{\mathrm{new}}$*, which is not a tolerance law; if* $\varepsilon_{\min}(\bar{t}) > 0$*,* $T^{\varepsilon_{\min}(\bar{t})}$ *does not entail it.*

Proof. Write $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$; ε_t^2 is rational because $A^t a$ and $\mathrm{Rot}^t a$ are rational vectors. In every model of $B \cup T_{\mathrm{new}}$, that is in any real closed field with interpretations of the constants, the recurrence determines $x_t = A^t a$, a rational vector, since $\mathbb{Q}$ embeds uniquely into every ordered field; instantiating the recurrence t times gives $T_{\mathrm{new}} \vdash x_t = A^t a$.

(a) $\|\mathrm{Rot}\| = 1$ and $\|A\| \le 1 + \eta_c \|\mathbf{P}\|$, so $\varepsilon_{t+1} = |A(A^t a - \mathrm{Rot}^t a) + \eta_c \mathbf{P}\, \mathrm{Rot}^t a| \le (1 + \eta_c \|\mathbf{P}\|)\varepsilon_t + \eta_c \|\mathbf{P}\| |a|$ with $\varepsilon_0 = 0$, and by induction $\varepsilon_t \le ((1 + \eta_c \|\mathbf{P}\|)^t - 1)|a| \le \varepsilon(\bar{t})$ for $t \le \bar{t}$. Since T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, it proves each conjunct of T^{ε} for rational $\varepsilon \ge \varepsilon(\bar{t})$.

(b) Over $B \cup T_{\mathrm{new}}$ each conjunct of T^{ε} is equivalent to the inequality $\varepsilon_t^2 \le \varepsilon^2$ between rationals, which RCF proves or refutes. So $T_{\mathrm{new}} \vdash T^{\varepsilon}$ iff all $\bar{t} + 1$ inequalities hold, iff $\varepsilon^2 \ge \max_{t \le \bar{t}} \varepsilon_t^2 = \varepsilon_{\min}(\bar{t})^2$; and $\varepsilon_{\min}(\bar{t}) \le \varepsilon(\bar{t})$ by the bound in (a).

(c) $T_{\mathrm{new}} \vdash x_1 = Aa$ and $T_0 \ni (x_1 = \mathrm{Rot}\, a)$, with $Aa - \mathrm{Rot}\, a = \eta_c \mathbf{P} a \ne 0$.

(d) T_0 proves $x_t = \mathrm{Rot}^t a$, hence T^{ε} for every $\varepsilon \ge 0$; combine with (b).

(e) T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, and T_0 proves $|x_t - \mathrm{Rot}^t a|^2 = 0 \le \varepsilon_t^2$. If $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ at some t, choose a real point x_t with $\varepsilon_t < |x_t - \mathrm{Rot}^t a| \le \varepsilon_{\min}(\bar{t})$ and put $x_s := \mathrm{Rot}^s a$ for $s \ne t$: this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ that violates the time-dependent form.

(f) For finite theories, a sentence follows from both iff it follows from their disjunction (proof by cases), so the disjunction is the strongest common consequence. Put $x_t := \mathrm{Rot}^t a + (\varepsilon_{\min}(\bar{t}), 0)$ for every $t \le \bar{t}$: if $\varepsilon_{\min}(\bar{t}) > 0$, this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ with $x_0 \ne a$, so it is a model of neither theory, and $T^{\varepsilon_{\min}(\bar{t})}$ does not entail the disjunction; in particular the disjunction is not equivalent to a uniform-tolerance law. □

Remark G.38 (Derivations of polynomial size)*. Proof sketch.* What is not written out is the size of derivations of arithmetic on binary numerals from the field axioms, which is standard. The

natural derivation of T^{ε} from T_{new} instantiates the recurrence t times, writes each x_t as an exact rational vector and checks $\bar{t}+1$ numerical inequalities; the numerals it writes have total bit size $\sum_{t\le\bar{t}} \text{bits}(A^t a) = O(\bar{t}^2 \log \text{ht})$, where ht is the height of the entries of A. If the numeral arithmetic has derivations of size polynomial in the numerals' size, the whole derivation has size polynomial in $\bar{t}$. For $\mathbf{P} = \mathbf{J}\,\text{Rot}$, with $\mathbf{J}$ the rotation by 90°, the numerals have 1894 bits at $\bar{t} = 10$ and about 1.8×10^5 at $\bar{t} = 100$.

Exact rational arithmetic (code in the accompanying repository) gives, for three perturbations with $\|\mathbf{P}\| = 1$ ($\mathbf{J}$ is the rotation by 90°):

$\mathbf{P}$	$\varepsilon_{\min}(\bar{t})$ at $\bar{t} = 1, 10, 100, 200$	bound $\varepsilon(\bar{t})$
$\mathbf{J}\,\text{Rot}$ (outward spiral with precession)	0.0100, 0.0999, 0.961, 1.69	0.0100, 0.105, 1.70, 6.32
$\text{diag}(1,-1)\,\text{Rot}$ (distortion)	0.0100, 0.0127, 0.0186, 0.0243	the same
identity (spiral)	0.0100, 0.103, 1.33, 3.49	the same

In each case T_{new} refutes T_0 at $t = 1$. The bound $\varepsilon(\bar{t})$ is nearly tight for small windows when the perturbation accumulates, and very loose for a bounded distortion (ratio 0.004 at $\bar{t} = 200$). For $\mathbf{P} = \mathbf{J}\,\text{Rot}$ the time-dependent tolerances have mean 0.0500 at $\bar{t} = 10$ and 0.490 at $\bar{t} = 100$, about half of $\varepsilon_{\min}(\bar{t})$. A circle refitted on the window (radius and angular speed; floating point, for comparison only) absorbs the precession almost entirely in the first case (2.5×10^{-4} and 2.5×10^{-3} at $\bar{t} = 10, 100$) and the growth not at all in the third (3.7×10^{-2} and 0.80): the toy analogue of the fitted ellipses of Table 7.

G.8 Domains of applicability and retention

Let a law Λ with per-system elements prove the data of every system but one, $\star$, whose $n_\star$ data it fits, with $\star$'s elements ($k_\star$ bits), at the per-datum rate $m^\star_{\text{pat}}$ of the exact law with exceptions, or at the rate $m^\star_{\text{hon}}$ of an honest tolerance for $\star$, at a one-off cost of $h_\star$ bits. The *domain-restricted law* $\Lambda_{\text{Dom}} := \forall b\,(\text{Dom}(b) \to \text{law}(b))$, with Dom a definable predicate that B proves of the other systems and refutes of $\star$, costs $\|\text{Dom}\|$ bits more than Λ (a few bits if "planet" is a base predicate, about the logarithm of the number of systems to name $\star$) and leaves $\star$'s data literal, at rate γ. So $\star$'s data have three treatments: restriction (rate γ, constant $\lambda\|\text{Dom}\|$), patched (rate $m^\star_{\text{pat}}$, constant $\lambda k_\star$) and honest (rate $m^\star_{\text{hon}}$, constant $\lambda(k_\star + h_\star)$).

Proposition G.39 (Domain restriction)**.**

(a) $J_\lambda(\Lambda_{\text{Dom}}) - J_\lambda(\Lambda \cup \text{El}_\star$ *with* $\star$ *'s exceptions*$) = \lambda\|\text{Dom}\| - \lambda k_\star + \sum_{\star\text{'s data}}(\gamma - c_{\text{pat}})$*, and likewise against the honest treatment of* $\star$*, with* $\lambda(k_\star + h_\star)$ *and* c_{hon}*, where* $c_{\text{pat}}, c_{\text{hon}}$ *are the per-datum costs.*

(b) With i.i.d. data and $n_\star \to \infty$*, the restriction is eventually the* λ*-MAP's treatment of* $\star$ *iff* $\gamma < \min(m^\star_{\text{pat}}, m^\star_{\text{hon}})$*, or* $\gamma = m^\star_{\text{hon}} < m^\star_{\text{pat}}$ *with the honest cost almost surely* γ *and* $\|\text{Dom}\| < k_\star + h_\star$*. If the least rate is shared by two treatments whose costs differ non-degenerately, there is no eventual winner.*

(c) Under SAI's pricing (code R), $m^\star_{\text{hon}} < \gamma$ *whenever* $\varepsilon^*_\star < \mathsf{R}/(2\mathrm{e})$ *(for* $\Delta \le 2\varepsilon^*_\star$*) or* $\Delta < \mathsf{R}/\mathrm{e}$ *(for* $\Delta \ge 2\varepsilon^*_\star$*). So a system that the law confines within about* 66° *of longitude (a sufficient bound) is never cut out asymptotically at any cell width below* R/e*: it is kept with its own, wider tolerance. Under code Q with* $\Delta \le 2\varepsilon^*_\star$ *the honest treatment costs exactly* γ *per datum, and the restriction is eventually selected iff* $m^\star_{\text{pat}} > \gamma$ *and* $\|\text{Dom}\| < k_\star + h_\star$*.*

Proof. (a) The two theories differ in the domain clause, in $\star$'s elements, and in the treatment of $\star$'s data; the other systems' data are proved by both. (b) Theorem G.22(i), (ii) applied to the three treatments (code-Q costs are bounded and code-R costs have all moments): a unique least rate wins eventually; a tie with an almost surely zero cost difference is decided by the constants, which is the case of the restriction and the honest treatment under code Q in the fine regime, where the honest law is silent on every datum and costs exactly γ; and a tie with a non-degenerate difference has no eventual winner. (c) In the fine regime Lemma G.20(ii) gives $m^{\star}_{\mathrm{hon}} \le \log(2\varepsilon^{*}_{\star}/\Delta) + \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $2\mathrm{e}\varepsilon^{*}_{\star} < \mathsf{R}$. In the coarse regime $m^{\star}_{\mathrm{hon}} = s_{\Delta}\beta_{e} \le \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $\Delta < \mathsf{R}/\mathrm{e}$. Under code Q with $\Delta \le 2\varepsilon^{*}_{\star}$ the honest law is silent on every datum (Lemma G.15), and (b) applies. □

For the Moon over one year, with one fixed ellipse ($\varepsilon^{*} = 156.3' = 2.6^{\circ}$, $\mathbb{E}|e| = 61.7'$ for the mean fit, $k_{\star} = 60$, $h_{\star} = 20$, $\|\mathrm{Dom}\| = 4$, $c_{\mathrm{time}} = 10.5$), an exact computation (code in the accompanying repository) gives the eventually selected treatment of the Moon's data, with rates in bits per datum ("excluded" is the domain restriction, chosen at an exact tie between the honest and the literal treatment):

Δ	γ	code Q, $\lambda = 1$	code Q, $\lambda = 2$	code R, $\lambda = 1, 2$
$10'$	11.08	excluded (tie; patched 20.52)	excluded (patched 30.51)	honest (4.973)
$60'$	8.49	excluded (tie; patched 13.66)	excluded (patched 21.22)	honest (2.445)
$300'$	6.17	patched (3.430)	patched (5.592)	honest (0.686)
$600'$	5.17	patched (1.612; honest 2.694)	patched (2.6933; honest 2.6939)	honest (0.343)
$1800'$	3.58	patched (0.483; honest 0.623)	honest (0.623; patched 0.843)	honest (0.114)

At the exact ties the constants decide ($\|\mathrm{Dom}\| = 4$ against $k_{\star} + h_{\star} = 80$). In code R the honest treatment overtakes the restriction after about $\lambda \cdot 76/(\gamma - m^{\star}_{\mathrm{hon}})$ of the Moon's data in expectation, about 12 at $\lambda = 1$ and 25 at $\lambda = 2$ at $\Delta = 10'$ and $60'$. More generally, for every planet and window of Table 7, and for the Moon over a month, a year and 18.6 years, a law with its own tolerance has a lower rate under code R than leaving the body's data literal at every cell width computed, by Proposition G.39(c); a sharp boundary is selected only when even an honest tolerance is worse than literal data, or under literal pricing at fine precision.

For retention, let a datum proved from T cost in addition $r_{\mathrm{d}}\,\mathrm{dl}_{T}$ bits, where dl_{T} is the size of the derivation used, and let $J^{\mathrm{der}}_{\lambda}(T; D) := \lambda\|T\| + \mathrm{Exc}_{D}(T) + \sum_{\text{data proved}} r_{\mathrm{d}}\,\mathrm{dl}_{T}$. Compare an honest old law T_{old} (tolerance $\varepsilon_{\mathrm{old}}$ on a window) with an exact successor T_{new} that proves every datum, with per-datum derivation charges $c_{\mathrm{old}} < c_{\mathrm{new}}$.

Proposition G.40 (Retention of a superseded theory)**.**

(a) T_{old} *beats* T_{new} *on* D *iff*

$$\lambda(\|T_{\mathrm{new}}\| - \|T_{\mathrm{old}}\|) + (c_{\mathrm{new}} - c_{\mathrm{old}})\,N_{\mathrm{p}} + c_{\mathrm{new}}\,(n - N_{\mathrm{p}}) > \mathrm{Exc}_{D}(T_{\mathrm{old}}),$$

where N_{p} *is the number of data that* T_{old} *proves and* $n = |D|$*.*

(b) With i.i.d. data the rates are $m_{\mathrm{old}} = s_{\mathrm{old}}\gamma_{\mathrm{si}} + (1 - s_{\mathrm{old}})c_{\mathrm{old}}$*, with* s_{old} *the silence probability of* T_{old}*, and* $m_{\mathrm{new}} = c_{\mathrm{new}}$*. If* $m_{\mathrm{old}} < m_{\mathrm{new}}$ *(respectively* $>$*), then almost surely* T_{old} *(respectively* T_{new}*) is eventually the* λ*-MAP of the two, whatever* λ *and the lengths are. At* $m_{\mathrm{old}} = m_{\mathrm{new}}$ *the lengths decide if the cost difference is almost surely constant, and otherwise there is no eventual winner.*

(c) *Let* $s_{\mathrm{old}} = \min(2\varepsilon_{\mathrm{old}}/\Delta, 1)$. *Under code Q,* m_{old} *is non-increasing in* Δ *on* $\{\Delta : \gamma(\Delta) \geq c_{\mathrm{old}}\}$*; under code R it is non-increasing in* Δ *on* $(0, \infty)$ *if* $\beta_e \geq c_{\mathrm{old}}$. *In either case the cell widths in that range at which* T_{old} *is retained form an interval with no upper bound inside the range; call its lower end* Δ^*. *For* $\Delta \geq 2\varepsilon_{\mathrm{old}}$ *under code Q,* Δ^* *solves* $2\varepsilon_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})/\Delta = c_{\mathrm{new}} - c_{\mathrm{old}}$.

Proof. (a) $J_\lambda^{\mathrm{der}}(T_{\mathrm{new}}) = \lambda\|T_{\mathrm{new}}\| + c_{\mathrm{new}} n$ and $J_\lambda^{\mathrm{der}}(T_{\mathrm{old}}) = \lambda\|T_{\mathrm{old}}\| + \mathrm{Exc}_D(T_{\mathrm{old}}) + c_{\mathrm{old}} N_{\mathrm{p}}$; compare term by term. (b) T_{old} is honest, so it never refutes a datum; a datum costs c_{old} when it proves it and γ_{si} when it is silent. Apply Theorem G.22. (c) *Code Q.* In the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})$, and s_{old} and $\gamma(\Delta) - c_{\mathrm{old}} \geq 0$ are both non-negative and non-increasing in Δ, so their product is too; in the fine regime $m_{\mathrm{old}} = \gamma(\Delta)$, which is decreasing. *Code R.* In the fine regime T_{old} is silent on every datum and $m_{\mathrm{old}} = \mathbb{E}\bar{c}_\Delta(w)$, strictly decreasing by Lemma G.20(iii); in the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\beta_e - c_{\mathrm{old}})$, non-increasing when $\beta_e \geq c_{\mathrm{old}}$. The two forms agree at $\Delta = 2\varepsilon_{\mathrm{old}}$. A non-increasing rate stays below c_{new} once it is below it, which gives the interval, and the equation for Δ^* is $m_{\mathrm{old}} = c_{\mathrm{new}}$ in the coarse regime under code Q. □

With derivations from the successor assumed a thousand times larger ($\mathrm{dl}_{T_{\mathrm{new}}}/\mathrm{dl}_{T_{\mathrm{old}}} = 1000$, as for a validated integration against solving Kepler's equation), exact code-R costs and bisection (code in the accompanying repository) give the least retained cell width Δ^* for $c_{\mathrm{new}} = 0.01$, 0.1 and 1 bit per datum:

body, window	ε^*	code Q	code R
Mercury, 100 yr	2.02′	1542′, 258.5′, 37.2′	229.8′, 22.97′, 2.33′
Mars, 12 yr	1.44′	1202′, 195.6′, 27.7′	174.2′, 17.42′, 1.744′
Mars, 2000 yr	49.9′	10450′, 2895′, 533′	5717′, 572′, 57.8′
Jupiter, 100 yr	6.57′	3468′, 661′, 101.6′	867′, 86.7′, 8.61′
Saturn, 2000 yr	81.9′	never, 3994′, 784′	10320′, 1032′, 102.9′
Moon, 1 month	30.9′	8414′, 2086′, 364′	4473′, 447′, 44.4′

Under random completion the retention region reaches precisions 1.8 to 6.9 times finer than under literal pricing at $c_{\mathrm{new}} = 0.01$, 3.9 to 11 times at 0.1, and 7.6 to 16 times at 1, because the honest law's silence is cheap.

Question G.41. What is the posterior over an explicit pool of domain predicates (base predicates such as "planet", windows, and their conjunctions, with stated syntax costs) on the Kepler residuals? By Proposition G.39 we expect the answer to depend on the pricing: tolerances with error models under SAI's pricing, and sharp domains under literal pricing at fine precision.

G.9 What log loss does and does not measure

This subsection states and proves the results behind Section 9.3. Losses are in bits, and for every score lower is better. For a real random variable X, $\mathrm{h}(X) := -\int f \log f$ is the differential entropy of its density f (possibly $-\infty$), and $\mathrm{h}(X) := -\infty$ if X has no density; H is the Shannon entropy and H_2 the binary entropy. For a forecast P of a real quantity and an outcome x, $\mathrm{CRPS}(P, x) := \int (P(X \leq t) - \mathbf{1}[x \leq t])^2 \, dt$ is the continuous ranked probability score, a proper score that grows with the distance of P's mass from x (cf. Gneiting and Raftery, 2007). In regime R we call the rate $r_{\mathcal{F}}(\Delta)$ of Proposition 9.2(c), the expected cost per datum of $\mathrm{BC}_{\mathcal{F}}$ when the errors have the law that $\mathcal{F}$ determines, the *price* of the calibrated completion of a law with that error law.

Proposition G.42 (What questions about one quantity encode)**.** *Let $B \supseteq$ RCF have a constant q; for a finite union C of rational intervals let σ_C be the sentence "$q \in C$". For a theory machine G let $\beta_G(C) := \nu_G(\{(\sigma_C, 1)\})$, the probability that G's random theory is consistent and proves $q \in C$, and define the cumulative loss of G along a reveal sequence as in the semimeasure convention of Definition 2.5, with ν_G in place of M_A.*

(a) Let D be a two-valued data set of sentences σ_C with $\nu_G(D) > 0$, and C_D the Boolean combination of their sets that the labels determine. Then $B \vdash \psi_D \leftrightarrow \sigma_{C_D}$, $\nu_G(D) = \beta_G(C_D)$, and on every consistent reveal sequence of D, in any order and with repeats, the cumulative loss of G is $-\log \beta_G(C_D) + \log \nu_G(\emptyset)$. If D answers every question of a finite family of interval questions about q truly, C_D is the cell of the true value in the partition that their intervals generate, and β_G is a sub-probability on the cells of that partition: two machines that give the true cell the same mass pay the same, wherever the rest of their mass lies.

(b) Let the true value be 0, let $C = [a_0, a_0 + \Delta)$ be a fixed cell containing it, and let P be a probability law of q. For $P = N(d, \sigma^2)$, $-\log P(C) = d^2/(2\sigma^2 \ln 2) + O(|d|)$ as $|d| \to \infty$; for the Laplace law with location d and scale b, $-\log P(C) = |d|/(b \ln 2) + O(1)$. The same holds, averaged over the dither, for the cell of 0 in a uniformly dithered grid of width Δ.

(c) Let q live on a circle of circumference R (an integer multiple of Δ, at least 2Δ), let the question be the cell of q in a grid of width Δ with a uniform offset, and let $P_{d,s} := (1 - s)\delta_d + s\,U$ with U uniform on the circle and $0 < s < 1$. With x the true value and $\operatorname{sep}(x, d) := \min(|x - d|/\Delta, 1)$, $|x - d|$ the distance on the circle, the expected loss of $P_{d,s}$ over the offset is $\operatorname{sep}(x, d) \log(\mathsf{R}/(s\Delta)) + (1 - \operatorname{sep}(x, d))(-\log(1 - s + s\Delta/\mathsf{R}))$.

Proof. (a) RCF proves the Boolean identity, and $\nu_G(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$ (Proposition 3.1(c)), so $\nu_G(D) = \beta_G(C_D)$. ν_G is monotone ($D' \subseteq D$ gives $E^G_{D'} \supseteq E^G_D$), so every prefix D_{t-1} has $\nu_G(D_{t-1}) \geq \nu_G(D) > 0$; the conditional of a new datum is $\nu_G(D_t)/\nu_G(D_{t-1})$ and that of a repeat is 1 (Definition 2.1), and the product telescopes to $\nu_G(D)/\nu_G(\emptyset)$. True answers to all the questions determine the cell of the true value. Distinct cells are B-incompatible and a consistent theory proves at most one of them, so the events defining β_G on the cells are disjoint. (b) Let $d \to +\infty$ and put $z_1 := (d - a_0 - \Delta)/\sigma < z_2 := (d - a_0)/\sigma$, so that $P(C) = \Phi(-z_1) - \Phi(-z_2)$ with Φ the standard normal distribution function. Since $z_2 - z_1 = \Delta/\sigma$ is fixed, $\Phi(-z_2)/\Phi(-z_1) \to 0$, and the Mills-ratio asymptotics $\ln \Phi(-z) = -z^2/2 - \ln z + O(1)$ give $-\ln P(C) = (d - a_0 - \Delta)^2/(2\sigma^2) + O(\ln d) = d^2/(2\sigma^2) + O(d)$, uniformly for a_0 in a bounded set. For the Laplace law and $d \geq a_0 + \Delta$, $P(C) = \frac{1}{2} e^{-(d - a_0 - \Delta)/b}(1 - e^{-\Delta/b})$. The case $d \to -\infty$ is symmetric, and the bounds are uniform in the offset, so they survive averaging. (c) x and d lie in different cells iff a grid point falls on the shorter arc between them, which has probability $\operatorname{sep}(x, d)$. If they do, $P_{d,s}$ gives the cell of x the mass $s\Delta/\mathsf{R}$; otherwise $1 - s + s\Delta/\mathsf{R}$. □

So under log loss misses are graded, without bound, but by each hypothesis's own error model; a family of questions about one quantity fixes only a resolution, and asking each of many values one randomly offset question grades near-point misses by sep, a bounded metric that is constant beyond one cell width. Regime R is such a design. A direct computation (code in the accompanying repository) gives, for Gaussian forecasts with $\sigma = 1$ and $\Delta = 0.1$, an extra cost of 25.95 bits at $d = 6$ over $d = 0$, against $d^2/(2 \ln 2) = 25.97$, on a fixed and on a dithered grid alike. For a single yes/no datum every scoring rule is local; for three or more outcomes see Section 6.3.

Proposition G.43 (Fitted error models rank laws by error norms)**.** *Let $e_1, \ldots, e_n$ be the errors of a law on n data, $M_p := (\frac{1}{n} \sum_j |e_j|^p)^{1/p}$ and $M_\infty := \max_j |e_j|$. With the scale fitted by maximum*

likelihood, the least total log loss of the error densities is $\frac{n}{2}\log(2\pi\mathrm{e}M_2^2)$ *for a Gaussian error model,* $n\log(2\mathrm{e}M_1)$ *for a Laplace one,* $n\log(2M_\infty)$ *for the uniform law on* $[-\delta,\delta]$ *(at* $\delta = M_\infty$*; for smaller* δ *the loss is infinite), and* $n\log\left(2\Gamma(1+\frac{1}{p})(p\mathrm{e})^{1/p}M_p\right)$ *for the density proportional to* $\exp(-|z/s|^p)$*. So these error models rank laws by RMS, mean absolute, maximum and* L^p *error.*

Proof. In nats the total loss of the last family is $nM_p^p/s^p + n\ln\left(2s\Gamma(1+\frac{1}{p})\right)$, minimised at $s^p = pM_p^p$; $p=2$ and $p=1$ are the Gaussian and Laplace cases. The uniform case is monotone in $\delta \geq M_\infty$. □

Lemma G.44 (Dithered cells)**.** *Let* e *be a real random variable,* $\tilde{u}$ *uniform on* $(0,1]$ *and* u *uniform on* $[0,1)$*, both independent of* e*, and* $C_u(e)$ *the cell of the grid* $\{(k-u)\Delta : k\in\mathbb{Z}\}$ *that contains* e*. Then* $H(C_u(e)\mid u) = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$*. Here* $e+\Delta\tilde{u}$ *has the density* $f(y) = \mathbb{P}[y-\Delta\leq e<y]/\Delta \leq 1/\Delta$*, so* $\mathrm{h}(e+\Delta\tilde{u}) \geq \log\Delta$*.*

Proof. Given u, the cell $[(k-u)\Delta,(k+1-u)\Delta)$ has probability $\Delta f((k+1-u)\Delta)$. Average $\sum_k -\Delta f\log(\Delta f)$ at these points over u; the substitution $y=(k+1-u)\Delta$ turns the average into $\int -f(y)\log(\Delta f(y))\,dy = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$ (Tonelli's theorem, all terms being non-negative). □

This is the known rate of entropy-coded subtractively dithered quantisation; the short proof is included for completeness.

Theorem G.45 (The best rate in regime R is a smoothed entropy)**.** *In regime R, with* $\mathsf{R} \geq 2\varepsilon^* + \Delta$*, let* $\varepsilon \geq \varepsilon^*$ *and* $\mathcal{F}$ *be as in Proposition 9.2, and* $\tilde{u}$ *as in Lemma G.44. Then* $r_{\mathcal{F}}(\Delta) = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$*.*

(a) If the support of e *lies in an interval of length at most* Δ*, then* $r_{\mathcal{F}}(\Delta) = \mathrm{disp}(e)/\Delta$*, where* $\mathrm{disp}(e) := \int H_2(F_e(t))\,dt = 2\varepsilon c_{\mathcal{F}}$ *and* F_e *is the distribution function of* e*;* disp *is translation invariant and* $\mathrm{disp}(\lambda e) = |\lambda|\,\mathrm{disp}(e)$*.*

(b) As $\Delta\to 0$*,* $r_{\mathcal{F}}(\Delta) - \log(1/\Delta) \to \mathrm{h}(e) \in [-\infty,\infty)$*.* $\mathrm{h}(e)$ *is invariant under measure-preserving rearrangements of the density of* e*, and* $\mathrm{h}(\lambda e) = \mathrm{h}(e) + \log|\lambda|$*.*

(c) If e *takes finitely many values,* $r_{\mathcal{F}}(\Delta) = H(e)$ *for every* Δ *up to the least gap between them. If the law of* e *has a non-atomic part,* $r_{\mathcal{F}}(\Delta)\to\infty$ *as* $\Delta\to 0$*.*

Proof. By the proof of Proposition 9.2(c), $r_{\mathcal{F}}(\Delta)$ is the conditional entropy of the answer given the question. Given the time t_j and the dither, the answer is the cell of $q(t_j) = f(t_j) - e_j$; translation by $f(t_j)$ keeps the dither uniform, and $\mathsf{R} \geq 2\varepsilon^* + \Delta$ keeps the cells that meet the band distinct on the circle, so $r_{\mathcal{F}}(\Delta) = H(C_u(-e)\mid u) = \mathrm{h}(-e+\Delta\tilde{u}) - \log\Delta$ by Lemma G.44. Since $-e+\Delta\tilde{u} = -(e+\Delta(1-\tilde{u})) + \Delta$ and $1-\tilde{u}$ has the law of $\tilde{u}$ up to a null set, this is $\mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$, and also $H(C_u(e)\mid u)$. (a) With the support in $[a,b]$, $b-a\leq\Delta$, the density of $e+\Delta\tilde{u}$ is $F_e(y^-)/\Delta$ on $[a,b]$, $1/\Delta$ on $[b,a+\Delta]$ and $(1-F_e((y-\Delta)^-))/\Delta$ on $[a+\Delta,b+\Delta]$; integrating $-f\log(\Delta f)$ gives $\frac{1}{\Delta}\int_a^b H_2(F_e(t))\,dt$. The substitution $t=\varepsilon-2\varepsilon v$ and $H_2(1-x)=H_2(x)$ give $\mathrm{disp}(e) = 2\varepsilon c_{\mathcal{F}}$. (b) *Lower bound.* If e has a density, $\mathrm{h}(e+\Delta\tilde{u}) \geq \mathrm{h}(e+\Delta\tilde{u}\mid\tilde{u}) = \mathrm{h}(e)$; otherwise $\mathrm{h}(e) = -\infty$. *Upper bound.* Fix a compact interval I of length ℓ that contains the support of $e+\Delta\tilde{u}$ for every $\Delta\leq 1$, and let Q be the uniform law on I. For a law P on I, $\mathrm{h}(P) = \log\ell - \mathrm{KL}(P\|Q)$, with $\mathrm{KL}=\infty$ when P has no density. As $\Delta\to 0$, $e+\Delta\tilde{u}\to e$ weakly, and relative entropy is lower semicontinuous under weak convergence (by the Donsker–Varadhan formula it is a supremum, over bounded continuous g, of the continuous functions $P\mapsto \mathbb{E}_P g - \log\mathbb{E}_Q 2^g$). So $\limsup_{\Delta\to 0}\mathrm{h}(e+\Delta\tilde{u}) \leq \log\ell - \mathrm{KL}(e\|Q) = \mathrm{h}(e)$. The invariances hold because h depends on the density only through the measures of its level sets. (c) With

atoms p_i at least Δ apart, the density of $e + \Delta\tilde{u}$ is p_i/Δ on disjoint intervals of length Δ, so $\mathrm{h}(e+\Delta\tilde{u}) = H(e) + \log\Delta$. If the law of e has a non-atomic part of mass $w_\mathrm{c} > 0$, let Z indicate it. Then $H(C_u(e) \mid u) \geq H(C_u(e) \mid u, Z) \geq w_\mathrm{c}\,\mathbb{E}_u[-\log\max_k p_k^\mathrm{c}(u)] \geq w_\mathrm{c}\log(1/Q_\mathrm{c}(\Delta))$, where $p_k^\mathrm{c}(u)$ are the cell probabilities of the normalised non-atomic part and $Q_\mathrm{c}(\Delta) := \sup_a P_\mathrm{c}([a, a+\Delta))$, which tends to 0 as $\Delta \to 0$ for a non-atomic law. □

So in the coarse regime the pricing is a dispersion of the error: sensitive to distance within one cell width, and blind to a constant bias, which the calibrated error model absorbs. In the fine limit it is the entropy of the error: sensitive to the volume the errors occupy and blind to where that volume lies; and at fine enough resolution a law whose error takes finitely many values beats every law whose error law has a non-atomic part, however small its errors. The convergence in (b) can be slow: for the sinusoidal error of Proposition G.13 with amplitude 1 (the arcsine law on $[-1, 1]$, whose density is unbounded), a direct computation (code in the accompanying repository) gives $r_\mathcal{F}(\Delta) - \log(1/\Delta) - \mathrm{h}(e) = 0.8542$, 0.1935, 0.0274 and 8.66×10^{-4} bits at $\Delta = 1$, 0.05, 10^{-3} and 10^{-6}, decaying about as $0.866\sqrt{\Delta}$. Part (b) of Proposition 9.2 does not depend on the logarithmic score: the errors are independent of the questions, so the cell law of $\mathrm{BC}_\mathcal{F}$ is the true conditional law of the answer given the question, which every proper score rewards at least as much as any other band-confined completion. What is specific to log loss is the ranking across laws.

Proposition G.46 (Rival laws in regime R)**.** *Let f_A and f_B be laws for one quantity q on a circle of circumference R, whose errors $e_A(t) = f_A(t) - q(t)$ and $e_B(t) = f_B(t) - q(t)$ lie in arcs shorter than $\mathsf{R}/2$. If the errors of f_A at the data times are i.i.d., with law $\mathcal{F}_A$ and independent of the questions, and the errors of f_B at the data times are identically distributed, then $f_A(t_j) - f_B(t_j)$ is the same for every j (modulo R), and the two laws' calibrated completions have the same price at every cell width.*

Proof. $e_B(t_j) = e_A(t_j) - d_j$ with $d_j := f_A(t_j) - f_B(t_j)$, so $\mathcal{F}_A$ rotated by d_j is the law of $e_B(t_j)$ for every j, and $\mathcal{F}_A$ is invariant under rotation by $d_j - d_1$. A law on an arc shorter than half the circle is invariant under no non-zero rotation: invariance under a rotation by an irrational fraction of R would make its closed support the whole circle, and invariance under a rotation by a rational fraction p/k in lowest terms, $k \geq 2$, gives invariance under rotation by R/k, so that the support contains k points spaced R/k apart, which lie in no arc shorter than $\mathsf{R} - \mathsf{R}/k \geq \mathsf{R}/2$. So $d_j = d_1$, and the prices agree because h and disp are translation invariant. □

So when two rival laws are compared by the prices of their calibrated completions, at least one of them has errors that are not random in the sense of regime R, and SAI's own rate on such data can be lower than either price; if one law's errors are random, its calibrated completion is never out-priced (Proposition 9.2(c)).

Example G.47 (Rival laws, priced)**.** (a) Law A has errors uniform on $[-1, 1]$. Law B has errors $-M$, 0, M with probabilities 0.1, 0.8, 0.1, $M = 1000$: exactly right on 80% of the times and off by 1000 otherwise. By Theorem G.45 the prices of their calibrated completions are $r_A(\Delta) = 1 + \Delta/(4\ln 2) - \log\Delta$ for $\Delta \leq 2$ and $(\log\mathrm{e})/\Delta$ for $\Delta \geq 2$, and $r_B(\Delta) = H(0.1, 0.8, 0.1)$ for $\Delta \leq M$ and $2MH_2(0.1)/\Delta$ for $\Delta \geq 2M$. So B is priced below A exactly for the cell widths below the root Δ_c of $r_A(\Delta) = H(0.1, 0.8, 0.1)$. A direct computation (code in the accompanying repository) gives $H(0.1, 0.8, 0.1) = 0.92193$ bits and $\Delta_c = 1.558542$. Yet the maximum, RMS and mean absolute errors (1, 0.577 and 0.5 for A; 1000, 447.2 and 200 for B), the fitted error models of

Proposition G.43 and the CRPS of the calibrated forecasts (0.333 against 180) prefer A at every resolution.

(b) Law B' has error $s(t) \in \{-1, 1\}$, a fair sign, and law A' has error $s(t)U(t)$, with $U(t)$ uniform on $[0, 1]$ and independent of $s(t)$. At every time the prediction of A' lies between the truth and that of B'. The price of B''s calibrated completion is 1 bit for $\Delta \le 2$ and that of A' is $r_A(\Delta)$, so B' is priced below A' exactly for Δ below the smaller root Δ'_c of $\Delta = 4 \ln \Delta$, and a direct computation gives $\Delta'_c = 1.429612$. The calibrated CRPS prefers A' (1/3 against 1/2).

(c) On a circle of circumference $\mathsf{R} \ge 4M$ let Z be uniform and independent of (e_A, e_B), with e_A and e_B independent and distributed as in (a); put $f_A := Z$, $q := f_A - e_A$ and $f_B := q + e_B$. Then $f_B = Z - e_A + e_B$ is uniform and independent of (e_A, e_B), so e_A is independent of f_A and e_B of f_B, and each calibrated forecast is the conditional law of q given that law's own prediction: an ideal forecast for its own information. The two information sets are not nested, and the pair determines q, since e_B is the element of $\{-M, 0, M\}$ nearest to $f_B - f_A$ (as $|e_A| \le 1 < M/2$). The logarithmic score on dithered cells of width $\Delta < \Delta_c$ and the CRPS rank the two ideal forecasts oppositely, by (a).

By Proposition G.46, neither pair of prices in (a) and (b) is SAI's rate for both laws on the same data. When both laws are computable and, as in (c), the pair of predictions determines q by a computable rule, a theory machine that states both laws and the rule proves every answer, at rate 0. Non-nested information sets are one of the known cases in which rankings of forecasts depend on the loss (for point forecasts, Patton, 2020); (c) is an exact instance in which no forecaster is miscalibrated relative to its own information.

Example G.48 (Local and distance-sensitive scores; misspecification)**.** (a) With the outcome 0, the uniform forecasts on $\{-2, \dots, 2\}$ and on $\{-20, -10, 0, 10, 20\}$ have the same logarithmic score, $\log 5$ bits, and CRPS 0.4 and 4.0; a forecast with mass 0.21 on 0 and 0.1975 on each of ± 10, ± 20 has a lower logarithmic score (2.2515 bits) than the first and a higher CRPS (3.9006). (b) Let the truth be $\frac{1}{2}N(-1, 0.1^2) + \frac{1}{2}N(1, 0.1^2)$ and the forecasts $N(0, s^2)$. The expected logarithmic score is least at $s = (\mathbb{E}X^2)^{1/2} = 1.004988$, the Kullback–Leibler projection, and the expected CRPS at $s = 1.202984$. The first forecast has expected logarithmic score 2.05427 bits (of the density) against 2.09580 for the second, and expected CRPS 0.60447 against 0.59725: each proper score prefers its own projection, although every strictly proper score ranks the truth first. All numbers are from a direct computation (code in the accompanying repository).

The log-loss posterior over a misspecified family concentrates near the Kullback–Leibler projection (Kleijn and van der Vaart, 2006; cf. Berk, 1966), so asymptotically its predictions are those of the first forecast of (b), whatever the purpose.

Proposition G.49 (Threshold purposes and dominance)**.**

(a) Let $B \supseteq$ RCF have constants q_t for the times t of a finite domain W, let f be a computable law with rational values on W, $\delta \ge 0$ rational and $k \le |W|$. Then "for every $t \in W$, $|q_t - f(t)| \le \delta$" (the honest law T^δ on W) and "for at least k of the $t \in W$, $|q_t - f(t)| \le \delta$" are sentences, so SAI holds posterior credences in them (Bel_D*).*

(b) Fix a family of questions, each about one real quantity, and say that law A dominates *law B if for every question A's answer lies in the closed interval between the true value and B's answer. Then A is at least as good as B for every* accuracy-monotone *purpose: per-question losses $\mathsf{L}_Q(a, x)$ that do not increase as the answer a moves towards the true value x from either side, aggregated by any rule that is non-decreasing in each per-question loss (a sum, a maximum, a quantile, a count of failures).*

(c) *Dominance is invariant under every strictly monotone reparametrisation of each quantity separately, but not under a change of the quantities: with truth* $(0,0)$*,* $A=(1,0)$ *dominates* $B=(2,-2)$ *in the quantities* (x,y)*, while in the quantities* $(x,x+y)$ *the predictions are* $(1,1)$ *and* $(2,0)$*, and* B *is exact on the second.*

Proof. (a) Each condition is a finite Boolean combination (for (a)'s second sentence, a disjunction over the k-subsets of W) of order relations between the constants q_t and rationals. (b) Per question $\mathsf{L}_Q(a_A,x)\le\mathsf{L}_Q(a_B,x)$ by monotonicity; then aggregate. (c) A strictly monotone map preserves betweenness; the example is direct. □

A graded purpose, a real-valued loss of the answers or its expectation, is a functional, not a sentence, and SAI has no place for it. Given the quantities, dominance is purpose-neutral, and a dominant law is best for every accuracy-monotone purpose; purposes enter in completing dominance to a ranking and in the choice of the quantities, as in the longitudes and distances of Mars (Section 10.4). That closeness rankings of predictions can reverse under a change of quantities is a classical observation (cf. Miller, 1975). The calibrated pricing can reverse dominance: in Example G.47(b) A' dominates B', yet below Δ'_c the pricing prefers B', so there it is the ranking of no accuracy-monotone purpose.

Rival fits of Kepler's laws. For each of the 39 series of Table 7 we kept the longitude errors of all four fits, 234 pairs of fits in all, and compared the fits by the maximum, RMS and mean absolute error and by the calibrated pricing. The pricing was estimated three ways: by the 40-bin histogram machine SH fitted and scored on disjoint halves of the times, by the same machine with 160 bins on further splits, and by the exact rate $r_{\mathcal{F}}(\Delta)$ of each empirical error law through Lemma G.44, at cell widths from $0.1'$ to $1800'$; a difference is counted only when it is significant by the estimator's rule (for SH, more than two paired standard errors with the same sign on five random half-splits). A computation (code in the accompanying repository) finds the following. The classical criteria disagree with each other: the maximum against the mean absolute error on 204 of the 234 pairs, the maximum against the RMS on 121, the RMS against the mean absolute error on 101. The calibrated pricing agrees with none of them; in the coarse regime it is close to, but not the same as, a ranking by the standard deviation of the errors (with the exact rate at $1800'$ it disagrees with the maximum error on 67 of the 165 decided pairs, with the mean absolute error on 85, with the RMS on 14 and with the standard deviation on 13). Its own ranking of the fits reverses between cell widths for 11 to 16 of the 234 pairs, depending on the estimator and the split; for 13 of the 14 reversals found with the exact rate the coarse-end difference is 0.7% to 6.8% of the rate. Where its significantly best fit is none of the classical winners, the case is a near-tie between the two least-squares fits (0.01% to 0.1% of the rate). Wherever SH and the exact rate both name a significantly best fit, they name the same one.

Definition G.50 (Purpose-SAI)**.** Let $\mathsf{L}:\mathsf{A}\times[-\mathsf{Y},\mathsf{Y}]\to[0,\infty]$ be a loss on a set A of acts, *η-mixable* for an $\eta>0$: for every probability π on A some $\alpha\in\mathsf{A}$ has $\mathsf{L}(\alpha,o)\le-\frac{1}{\eta}\ln\int e^{-\eta\mathsf{L}(\alpha',o)}\pi(d\alpha')$ for every o. The data are answered questions $(\mathcal{Q}_t,a_t)$ (Definition 9.1), each answer a carrying a value $m(a)\in[-\mathsf{Y},\mathsf{Y}]$ (for instance the midpoint of an interval cell), and the outcome at step t is $o_t:=m(a_t)$. After data D the *answer distribution* of a theory machine G on $\mathcal{Q}$ is $P_G(a\mid D,\mathcal{Q}):=\nu_G(D+(a,1))/\sum_{a'\in\mathcal{Q}}\nu_G(D+(a',1))$ when the denominator is positive, and the *act* of G is a Bayes act of L under the induced law of $m(a)$ (for square loss, its mean), or a fixed default act otherwise. *Purpose-SAI* is the aggregating algorithm (Vovk, 1990, 2001) over the theory machines G_i with these acts, learning rate η and prior weights $\mathbf{m}(i)$: at step t it

plays an act α_t that satisfies the mixability inequality for the law π_t that puts mass proportional to $\mathbf{m}(i)e^{-\eta \mathsf{L}_{t-1}(G_i)}$ on the act of G_i, where $\mathsf{L}_t(G)$ is the cumulative loss of G's acts on the first t data.

Proposition G.51 (Regret of purpose-SAI)**.** *If L is η-mixable, then on every sequence of answered questions, for every theory machine G and every n, purpose-SAI has cumulative loss at most $\mathsf{L}_n(G) + K(G)\ln 2/\eta$. Square loss on $[-\mathsf{Y}, \mathsf{Y}]$ is η-mixable for $\eta = 1/(2\mathsf{Y}^2)$, and the regret is then at most $2\mathsf{Y}^2 K(G)\ln 2$.*

Proof. At step t mixability gives $e^{-\eta \mathsf{L}(\alpha_t, o_t)} \geq \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_t(G_i)} / \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_{t-1}(G_i)}$. The product over $t \leq n$ telescopes, and since $\sum_i \mathbf{m}(i) \leq 1$ the cumulative loss is at most $-\frac{1}{\eta}\ln \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_n(G_i)} \leq \mathsf{L}_n(G_i) + \ln(1/\mathbf{m}(i))/\eta$ for every i (the known bound of the aggregating algorithm with prior weights; Vovk, 2001). Take i with $G_i = G$ and $K(i) = K(G)$, so that $\mathbf{m}(i) \geq 2^{-K(G)}$. Square loss on $[0, 1]$ is 2-mixable (cf. Haussler et al., 1998); rescaling $[0, 1]$ to $[-\mathsf{Y}, \mathsf{Y}]$ multiplies the loss by $4\mathsf{Y}^2$ and divides η by the same factor. □

The bound needs an a priori range and grows with its square: for longitudes ($\mathsf{Y} = 10800'$) one bit of $K(G)$ costs about 1.6×10^8 square arcminutes of cumulative square loss, so it is informative only for losses on a range comparable to the errors. Absolute loss is mixable for no η: for outcomes in $\{0, 1\}$ and $\pi = \frac{1}{2}(\delta_0 + \delta_1)$ an act α would need $|\alpha| + |1 - \alpha| \leq -\frac{2}{\eta}\ln \frac{1+e^{-\eta}}{2} < 1$. Like SAI, purpose-SAI is not computable, since its acts need ν_G. After n data its weights are proportional to $\mathbf{m}(i)e^{-\eta \mathsf{L}_n(G_i)}$, the form of the generalised Bayesian posteriors of Bissiri et al. (2016): credences, if anything, about which machine minimises the expected purpose loss, not about which machine is true. For the logarithmic loss with $\eta = \ln 2$ the aggregating algorithm is Bayes. With universal weights over predictors of sequences, the aggregating algorithm underlies predictive complexity (cf. Vovk and Watkins, 1998); only the version over theory machines is new here.

Remark G.52 (A decision layer). Accepting or using an approximate law is an act. Given a posterior predictive of the true answers and a purpose loss on a distribution of questions, laws can be ranked by their posterior-expected purpose loss. This ranks approximate laws by purpose without changing credence, and, if the posterior concentrates on the truth, by their purpose loss against the truth: by maximum error for a tolerance purpose, by RMS error for squared error. For a bounded inducer with a restricted, misspecified pool the posterior predictive is that of the Kullback–Leibler projection (Example G.48(b)), and its acts need not serve the purpose; then the purpose loss must enter learning, as in purpose-SAI or in generalised Bayesian updating. SAI as defined has no act layer, and adding one is not immediate: its predictions are belief functions, so a Bayes act needs a choice, such as the lower expectation or a completion P_D between Bel_D and Pl_D (Proposition 3.2(c)). For Solomonoff's mixture in sequence prediction, acting Bayes-optimally under a bounded loss has cumulative expected loss close to that of the informed predictor when the true distribution is in the class (Hutter, 2003); for M_A this is open (Question G.53). A user may also prefer a simple approximate law for the cost of computing, communicating or teaching with it; that trade-off is outside both routes.

Question G.53. Can a purpose enter SAI? (i) How should acts be scored under SAI's belief-function predictions (for example by lower expectation), and do the resulting acts serve every bounded purpose loss when the data come from a computable law, as for Solomonoff's mixture? (ii) Which properties of SAI survive in purpose-SAI, for example its equivalence with consistent SOI (Theorem 4.3), and is there an upper semicomputable universal element: a superloss process over theory machines whose loss is within $O(K(G))$ of every theory machine's loss, as predictive complexity is for sequence prediction?

H The Kepler computations

This appendix gives the proofs of the Kepler propositions, the definitions behind each computation, the full tables, and the elements that witness each reported fit. All computations take the plan94 analytical ephemeris of the planets and the moon98 ephemeris of the Moon, as implemented in the ERFA library, as the data; every positional error is against the plan94 ephemeris (moon98 for the Moon), not against the sky.

H.1 Proofs

The background B of Definition 10.1 cannot be the whole theory of the reals with sin, which is undecidable ($\sin(\pi x) = 0$ defines $\mathbb{Z}$); the numerical bounds it must prove are rational-function bounds of the kind MetiTarski proves (Akbarpour and Paulson, 2010).

The Ramsey sentence $R(T)$ of a theory T is obtained by replacing its theoretical function symbols by second-order variables and quantifying them existentially; its consequences are taken in second-order semantic consequence.

Proposition H.1 (The Ramsey sentence of Kepler's laws)**.**

(a) $R(T)$ has the same $\mathcal{L}_{\mathrm{hel}}$-consequences as T.

(b) For every B true in $\mathbb{R}$, $R(\mathrm{K12} \cup \mathrm{K3}) \cup B$ entails no datum $\mathrm{Near}(b, t, \mathbf{v}, \delta)$ with $\delta < \pi$.

Proof. (a) A model of $R(T)$ is an $\mathcal{L}_{\mathrm{hel}}$-structure that expands to a model of T, so $R(T)$ and T have the same $\mathcal{L}_{\mathrm{hel}}$-reducts of models. (b) Fix b, t, $\mathbf{v}$ and $\delta < \pi$, and a model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ on the standard reals. If b is a planet, keep $a(b)$ and $P(b)$, so that K3 is untouched, and choose $i(b), \Omega(b), \varpi(b)$ and $M_0(b)$ so that the Keplerian direction of b at t is $-\mathbf{v}$: the orbital plane can be any plane through the Sun (the three angles give every rotation), and the phase can place the body at any point of its orbit, which meets every direction of its plane. If b is not a planet, $\mathrm{K12} \cup \mathrm{K3}$ does not constrain its direction at t (for the Sun it requires only $r(\mathrm{Sun}, t) = 0$), and we set that direction to $-\mathbf{v}$. Defining θ, β, r from the positions gives a model of $B \cup \mathrm{K12} \cup \mathrm{K3}$, hence of $R(\mathrm{K12} \cup \mathrm{K3}) \cup B$, in which the direction of b at t makes the angle $\pi > \delta$ with $\mathbf{v}$. □

So the laws alone answer only structural questions; the laws together with the elements answer position questions.

Proof of Proposition 10.2. The position $\mathbf{r}(b, t)$ that K12 gives is $a(b)$ times an expression free of a, so a Kepler direction depends on $(e, i, \Omega, \varpi, M_0, P)$ and not on a. Given a model of $B \cup \mathrm{K12}$, put $a'(b) := (k^2 P(b)^2 / 4\pi^2)^{1/3}$ for a fixed $k > 0$ (cube roots exist in a real closed field) and leave every other element unchanged. Positions and distances scale by $a'(b)/a(b) > 0$, directions are unchanged, and K3 holds because $a'(b)^3 / P(b)^2 = k^2 / 4\pi^2$ is the same for every body. So an $\mathcal{L}_{\mathrm{dir}}$-sentence true in every model of $B \cup \mathrm{K12} \cup \mathrm{K3}$ is true in every model of $B \cup \mathrm{K12}$; the converse is trivial. □

Proposition H.2 (Content in geocentric directions)**.** *Let χ_{opp} be the sentence $\exists t\ \mathrm{dir}_g(\mathrm{Ma}, t) \neq -\mathrm{dir}_g(\mathrm{Sun}, t)$ of $\mathcal{L}_{\mathrm{geo}}$ ("Mars is not permanently in opposition").*

(a) χ_{opp} holds in every model of $\mathrm{K12} \cup \mathrm{K3}$ whose real sort is $\mathbb{R}$. Hence $B \cup \mathrm{K12} \cup \mathrm{K3} \vdash \chi_{\mathrm{opp}}$ for every B that contains the two true sentences ζ_{K3} and ζ_{pos} of the real field with sin *and* cos *given in the proof below.*

(b) For every B true in $\mathbb{R}$, $B \cup \mathrm{K12} \nvdash \chi_{\mathrm{opp}}$.

The sentence χ_{opp} is odd (its negation describes a fantastic world); it only shows that the geocentric language can register K3. The natural form of K3's content is Proposition 10.3.

Proof idea. (b) Circular coplanar orbits with equal periods and phases, Mars at distance $\rho > 1$, satisfy K12 (which relates no bodies), violate K3 and keep Mars in opposition. (a) Over $\mathbb{R}$, a Mars permanently in opposition stays on the ray from the Sun through the Earth, and Kepler's second law forces $\mathbf{r}(\text{Ma}, t) = \kappa\, \mathbf{r}(\text{Ea}, t)$ with a constant $\kappa > 1$: equal periods, different semi-major axes, against K3. ζ_{K3} states this about the elements, and ζ_{pos} (the Earth is never at the Sun) lets the argument cover non-standard models.

Proof. (b) Let $\mathcal{M}_0$ have the Earth with $a = 1$, $e = 0$, $i = 0$, period P_0 and some phase, and Mars with $a = \rho > 1$, $e = 0$, $i = 0$, the same period P_0 and the same phase. Then $\mathbf{r}(\text{Ma}, t) = \rho\, \mathbf{r}(\text{Ea}, t)$ for all t, so $\mathbf{r}(\text{Ma}, t) - \mathbf{r}(\text{Ea}, t) = (\rho - 1)\mathbf{r}(\text{Ea}, t) \neq 0$ and $\text{dir}_g(\text{Ma}, t) = \mathbf{r}(\text{Ea}, t)/|\mathbf{r}(\text{Ea}, t)| = -\text{dir}_g(\text{Sun}, t)$ for every t. So $\mathcal{M}_0 \models \neg\chi_{\text{opp}}$. $\mathcal{M}_0$ satisfies K12, since each body moves on a Kepler orbit and K12 imposes no relation between bodies, and it violates K3 (equal periods, unequal semi-major axes). It is a model of every B true in $\mathbb{R}$, so $B \cup \text{K12} \nvdash \chi_{\text{opp}}$.

(a), standard models. Let $\mathcal{M}$ be a model of $\text{K12} \cup \text{K3}$ whose real sort is $\mathbb{R}$, and suppose $\mathcal{M} \models \neg\chi_{\text{opp}}$. The Earth moves in the ecliptic ($i(\text{Ea}) = 0$), so $\mathbf{u}(t) := \mathbf{r}(\text{Ea}, t)/|\mathbf{r}(\text{Ea}, t)|$ lies in the ecliptic and differs from $\mathbf{z}$. By $\neg\chi_{\text{opp}}$, $\text{dir}_g(\text{Ma}, t) = \mathbf{u}(t) \neq \mathbf{z}$ for all t, so the bridge gives $\mathbf{r}(\text{Ma}, t) \neq \mathbf{r}(\text{Ea}, t)$ and $\mathbf{r}(\text{Ma}, t) = \mathbf{r}(\text{Ea}, t) + c(t)\mathbf{u}(t) = (r_{\text{E}}(t) + c(t))\, \mathbf{u}(t)$ with $c(t) > 0$, where $r_b(t) := |\mathbf{r}(b, t)|$. So Mars is always on the ray from the Sun through the Earth, beyond the Earth: it moves in the ecliptic, its heliocentric direction equals the Earth's at every t, and $r_{\text{M}}(t) > r_{\text{E}}(t)$. Take continuous polar angles $\theta_{\text{M}} = \theta_{\text{E}}$ in the ecliptic. Kepler's second law, a consequence of K12, gives $r_b(t)^2\dot{\theta}_b(t) = h_b$ with a nonzero constant h_b for each body (from $r = a(1 - e\cos\psi)$, $\tan\frac{\nu}{2} = \sqrt{(1+e)/(1-e)}\,\tan\frac{\psi}{2}$ for the true anomaly ν, and $\dot{M} = 2\pi/P$, one gets $r^2\dot{\nu} = 2\pi a^2\sqrt{1-e^2}/P$; standard). Hence $r_{\text{M}}^2/r_{\text{E}}^2 = h_{\text{M}}/h_{\text{E}}$ is constant: $r_{\text{M}}(t) = \kappa\, r_{\text{E}}(t)$ with a constant $\kappa > 1$, so $\mathbf{r}(\text{Ma}, t) = \kappa\, \mathbf{r}(\text{Ea}, t)$ for all t. The period of a Kepler orbit is the least period of $t \mapsto \mathbf{r}(b, t)$, and its semi-major axis is half the sum of its largest and smallest distances from the Sun ($r = a(1 - e\cos\psi)$). So $P(\text{Ma}) = P(\text{Ea})$ and $a(\text{Ma}) = \kappa\, a(\text{Ea}) \neq a(\text{Ea})$, contradicting K3.

(a), arbitrary models. Let ζ_{K3} be the following sentence of the language of the real field with sin and cos:

$$\forall\vec{\varepsilon}_{\text{Ea}}\,\forall\vec{\varepsilon}_{\text{Ma}}\Big(\text{Adm}(\vec{\varepsilon}_{\text{Ea}}, \vec{\varepsilon}_{\text{Ma}}) \to \exists t\,\forall\psi_{\text{Ea}}\,\forall\psi_{\text{Ma}}\big(\text{Kep}_{\text{Ea}}(t, \psi_{\text{Ea}}) \wedge \text{Kep}_{\text{Ma}}(t, \psi_{\text{Ma}}) \to \Delta(t, \psi_{\text{Ea}}, \psi_{\text{Ma}})\big)\Big).$$

Here $\vec{\varepsilon}_b$ are the seven elements of body b; Adm says that $0 \le e < 1$ and $a, P > 0$ for both bodies, that $i_{\text{Ea}} = 0$, and that K3 holds for the pair; $\text{Kep}_b(t, \psi)$ is Kepler's equation $\psi - e_b\sin\psi = M_{0,b} + 2\pi t/P_b$; and Δ says that the positions $\boldsymbol{\xi}_b := \text{Rot}_b(a_b(\cos\psi_b - e_b),\, a_b\sqrt{1-e_b^2}\sin\psi_b,\, 0)$ either coincide or satisfy $(\boldsymbol{\xi}_{\text{Ma}} - \boldsymbol{\xi}_{\text{Ea}})/|\boldsymbol{\xi}_{\text{Ma}} - \boldsymbol{\xi}_{\text{Ea}}| \neq \boldsymbol{\xi}_{\text{Ea}}/|\boldsymbol{\xi}_{\text{Ea}}|$. Let ζ_{pos} be the sentence $\forall a\,\forall e\,\forall\psi\,(a > 0 \wedge 0 \le e < 1 \to a(1 - e\cos\psi) > 0)$. Both are true in $\mathbb{R}$: ζ_{pos} obviously, and ζ_{K3} because, for standard element values satisfying Adm, Kepler's equation has exactly one solution for each t (its left side is strictly increasing in ψ when $0 \le e < 1$), so the antecedent of the inner implication fixes the true positions, and the standard case above gives a t at which they coincide or the directions differ.

Now let $\mathcal{M}$ be any model of $B \cup \text{K12} \cup \text{K3}$ with $\zeta_{\text{K3}}, \zeta_{\text{pos}} \in B$; its real sort may be non-standard. Instantiate ζ_{K3} at the elements of the Earth and Mars in $\mathcal{M}$ (Adm holds by K12, K3 and the frame convention) and obtain t. K12 supplies, for this t, values $\psi_{\text{Ea}}, \psi_{\text{Ma}}$ that solve Kepler's equations and give $\mathbf{r}(b, t) = \boldsymbol{\xi}_b$; the universal quantifiers over $\psi_{\text{Ea}}, \psi_{\text{Ma}}$ apply to them, so Δ holds of the actual

positions. By ζ_{pos}, $|\mathbf{r}(\mathrm{Ea},t)| = a_{\mathrm{Ea}}(1 - e_{\mathrm{Ea}}\cos\psi_{\mathrm{Ea}}) > 0$, so $-\mathrm{dir}_g(\mathrm{Sun},t) = \mathbf{r}(\mathrm{Ea},t)/|\mathbf{r}(\mathrm{Ea},t)|$. If the positions coincide, the bridge gives $\mathrm{dir}_g(\mathrm{Ma},t) = \mathbf{z}$, which differs from $\mathbf{r}(\mathrm{Ea},t)/|\mathbf{r}(\mathrm{Ea},t)|$, whose third coordinate is 0 since $i_{\mathrm{Ea}} = 0$. Otherwise the bridge gives $\mathrm{dir}_g(\mathrm{Ma},t) \neq \mathbf{r}(\mathrm{Ea},t)/|\mathbf{r}(\mathrm{Ea},t)|$. Either way $\mathcal{M} \models \chi_{\mathrm{opp}}$, and by the completeness theorem $B \cup \mathrm{K12} \cup \mathrm{K3} \vdash \chi_{\mathrm{opp}}$. □

The universal quantifier over ψ in ζ_{K3} is what makes the instantiation valid in a model in which Kepler's equation might have several solutions; an existential quantifier would leave that case open. The sentence ζ_{pos} supplies $\mathbf{r}(\mathrm{Ea},t) \neq 0$ in non-standard models, the only fact about the reals used outside ζ_{K3}.

Remarks on Proposition H.2. (i) The bridge needs a convention at coincidence: without it, a K3-model in which Mars coincides with the Earth would leave $\mathrm{dir}_g(\mathrm{Ma},t)$ unconstrained there and could imitate $\mathcal{M}_0$. (ii) For an inner co-moving planet ($\rho < 1$) the analogue fails: a K3-model with the planet on the Earth's orbit, diametrically opposite (behind the Sun), is always in the direction of the Sun; so χ_{opp} is stated for an outer planet. (iii) A computation (code in the accompanying repository) illustrates the difference in the circular coplanar case with $\rho = 1.5237$: $\mathcal{M}_0$ keeps Mars at opposition to rounding error (1.2×10^{-6} degrees in floating point), while the K3-model with the same Earth, the same ρ and the same initial phase leaves opposition by 1.3° within a day, 13° within 10 days and 87° within 100 days. A second, independently written implementation finds, over 300 random $\mathrm{K12} \cup \mathrm{K3}$ models (an outer planet, the Earth in the ecliptic, the planet's plane random), a largest angle from opposition over 20 years of at least 175° in every model, and over 50 random models of the kind of $\mathcal{M}_0$ never more than 6×10^{-14} degrees. Whether finitely many interval data true in $\mathcal{M}_0$ refute every K3-model was not examined.

Proof of Proposition 10.3. (a) K3 with $a(\mathrm{Ea}) = 1$ gives $a(\mathrm{Ma})^3 = P(\mathrm{Ma})^2/P(\mathrm{Ea})^2$, so $\rho = (P(\mathrm{Ma})/P(\mathrm{Ea}))^{2/3}$; then K12 and Num^- determine $\mathbf{r}(\mathrm{Ma},t)$ and $\mathbf{r}(\mathrm{Ea},t)$, and the bridge determines the geocentric direction. A datum $\mathrm{Near}_g(\mathrm{Ma},t,\mathbf{v},\delta)$ satisfied by the computed direction then follows once B proves the numerical bound that places the computed direction within δ of $\mathbf{v}$.

(b) Fix t with $\mathbf{r}(\mathrm{Ea},t)$ and $\mathbf{F}(t)$ not parallel. Every $\rho > 0$ gives a model of $\mathrm{K12} \cup \mathrm{Num}^-$ over $\mathbb{R}$, since K12 constrains a only by $a > 0$. The geocentric vector is $\mathbf{g}(\rho) = \rho\mathbf{F}(t) - \mathbf{r}(\mathrm{Ea},t)$; as ρ runs over $(0,\infty)$ it runs along the open ray from $-\mathbf{r}(\mathrm{Ea},t)$ in the direction $\mathbf{F}(t)$, inside the plane spanned by the two vectors. The ray does not pass through the origin because the two vectors are not parallel. Seen from the origin, its direction therefore moves continuously and monotonically along the great circle through $-\mathbf{u}(t)$ and $\mathbf{F}(t)/|\mathbf{F}(t)|$, tends to $-\mathbf{u}(t)$ as $\rho \to 0$ and to $\mathbf{F}(t)/|\mathbf{F}(t)|$ as $\rho \to \infty$, and covers the open arc between them, whose length $\alpha(t)$ lies in $(0,\pi)$. If $B \cup \mathrm{K12} \cup \mathrm{Num}^-$ proved $\mathrm{Near}_g(\mathrm{Ma},t,\mathbf{v},\delta)$ for a B true in $\mathbb{R}$, the whole arc would lie in the closed cap of angular radius δ about $\mathbf{v}$; its two end points, at angular distance $\alpha(t)$ from each other, would then lie in that cap too, so $\alpha(t) \leq 2\delta$. □

A computation over 1580–1600 (daily, plan94) gives the distribution of $\alpha(t)$ quoted in Example H.4: percentiles $0, 5, 25, 50, 75, 95, 100$ of 0.11°, 9.52°, 46.68°, 90.22°, 133.91°, 170.34°, 179.96°; no sampled day is within $2'$ of conjunction or opposition. Along the arc the direction moves monotonically as ρ grows on every one of 21 sampled days (a second implementation finds the same on 60 seeded days, with the great-circle identity to 9×10^{-15} radians), and the sensitivity of the geocentric direction to ρ at the true ρ has median $969'$ and maximum $3555'$ per unit of ρ.

Table 8: How wrong Kepler's laws are, against plan94: largest angular separation (arcminutes) between one fixed Kepler ellipse, fitted by least squares, and the plan94 positions, over windows centred on J2000; for the Moon (moon98, geocentric), in degrees. Last column: the fitted third-law constant $q := (2\pi/P)^2 a^3/k^2$ (k Gauss's constant) over the windows of 30 years and more, with the Newtonian two-body value $1 + m/M$ in brackets. Least-squares fits, angular separation; the minimax longitude errors of the same windows, used in Section 9, are in Table 7.

window (years)	12	30	100	300	1000	2000	third-law constant q
Mercury	0.46	0.89	2.17	6.04	19.3	38.6	
Mars	1.58	1.86	3.61	9.17	26.2	51.8	
Jupiter	1.05	4.19	7.64	7.74	31.8	38.5	1.00075–1.00099 (1.00095)
Saturn	1.70	3.32	9.73	11.7	91.3	92.4	1.0013–1.0019 (1.00029)
Moon	1.12° (one month), 3.0° (one year), 9.6° (18.6 years)						

H.2 Data, fits and the error budget

Residuals of one fixed ellipse. For each body and window, centred on J2000, one fixed Kepler ellipse (seven elements) is fitted by least squares to the plan94 heliocentric positions (for the Moon, the moon98 geocentric positions) in the J2000 ecliptic, and the table reports the largest angular separation between the fit and the data. The fitted third-law constant is $q := (2\pi/P)^2 a^3/k^2$, with k Gauss's constant. Over the 12-year window the fits give $q = 1.00062$ (Jupiter) and 1.00487 (Saturn); over the windows of 30 years and more they give the ranges of the table. The Newtonian two-body value is $1 + m/M$; for Saturn, a secular, circular and coplanar estimate that treats the orbit-averaged pull of each other planet as that of a ring gives $q = 1.00154$, of which Jupiter supplies 1.25×10^{-3}. This is a heuristic estimate, not a perturbation theory. A second computation, with other sample times and longitude errors only, agrees with the table to within about 8% in every planetary cell. The same fits, with a precessing ellipse, give Mercury's line of apsides a rotation of $575.4''$ per century in the fixed J2000 frame (fits over 100–2000 years), and the 2000-year longitude residuals of Saturn and Jupiter carry oscillations of period about 900 and 910 years and amplitude $43.8'$ and $18.3'$.

The third law's own error. With $q - 1 = 9.5 \times 10^{-4}$ (Jupiter, the two-body value) and 1.5×10^{-3} (Saturn, the middle of its fitted range), an exact K3 calibrated on the Earth changes the semi-major axes by the relative amounts 3.170×10^{-4} and 4.985×10^{-4} and misplaces Jupiter and Saturn geocentrically by at most $0.233'$ and $0.193'$ over 1580–1600 (medians $0.150'$ and $0.129'$).

The error budget. The figures of this paragraph are those quoted in the documentation of the ERFA routine `eraPlan94` (which pyerfa wraps): from comparisons with JPL's DE102 ephemeris over 1800–2050, maximum errors in heliocentric longitude of $17''$ (Mars), $71''$ (Jupiter) and $81''$ (Saturn); an accuracy over 1000–3000 no worse than 1.5 times that, and, against DE200 over 1800–2100, $26''$, $78''$ and $87''$. Hence: (i) over 1580–1600 plan94's longitude error for Mars is at most about $25'' \approx 0.43'$, the size of the gap $1.41' - 1.07' = 0.34'$ between the two best forms of the Mars pool, so the ordering of those two forms is established against plan94, and against the sky it rests on the errors against an exact ellipse in Table 9; (ii) the 12-year entries for Jupiter ($1.05'$) and Saturn ($1.70'$) lie inside 1800–2050, where plan94's own errors reach $1.18'$ and $1.35'$, so they measure the ellipse against plan94; (iii) plan94's Mars distance error (at most 7700 km over 1800–2050, about 3×10^{-5} of Mars's distance) is far below the distance tolerances of Appendix H.6.

The data of the Mars pool. The 7306 daily heliocentric positions of Mars over 1580.0–1600.0 are reduced to the best plane through the Sun (inclination 1.8828° and node 50.757° to the J2000 ecliptic; the largest angle of a position out of this plane is $0.070'$), and v_i is the longitude in that plane. The geocentric data of Appendix H.5 are the ecliptic longitudes of plan94 Mars minus plan94 Earth–Moon barycentre (J2000 ecliptic): geometric positions, without aberration or light time, and taken from the barycentre rather than from the Earth.

H.3 The Mars pool: forms, law texts and Occam factors

The forms. Write $\mathbf{u}(\varphi) := (\cos\varphi, \sin\varphi)$ and $\Lambda(t) := L + n(t - t_{\mathrm{mid}})$ for the mean longitude, with t_{mid} the middle of the window, L the mean longitude at t_{mid} and n the mean motion; w is the longitude of perihelion and $M(t) := \Lambda(t) - w$ the mean anomaly. Each form gives the heliocentric longitude $\theta(t)$ in the plane of the data; in longitude the forms are scale-free, so the circle has radius 1. The forms are heliocentric reconstructions: the historical models before Kepler were geocentric and also represented the Earth's motion.

- *Circle about the Sun* (elements L, n): $\theta(t) = \Lambda(t)$.
- *Circle with centre* $C = -e_1\mathbf{u}(w)$ *and equant point* $Q = -(e_1 + e_2)\mathbf{u}(w)$, both on the line of apsides on the aphelion side of the Sun: the planet moves on the unit circle about C so that its direction seen from Q is $\Lambda(t)$, and $\theta(t)$ is its direction seen from the Sun. The *eccentric circle* is the case $e_2 = 0$ (uniform motion about the centre; elements e, w, L, n with $e = e_1$), the *bisected equant* the case $e_1 = e_2 = e$ (centre halfway between the Sun and the equant point; elements e, w, L, n), and the *vicarious hypothesis* the case of free e_1, e_2 (elements e_1, e_2, w, L, n).
- *Ellipse* (elements e, w, L, n): $\theta(t) = w + \nu$, where ν is the true anomaly that Kepler's equation $\psi - e\sin\psi = M(t)$ gives for eccentricity e.

The element ranges are $e, e_1, e_2 \in [0, 1)$, $w, L \in [0, 2\pi)$ and $n \in [0, 0.1)$ radians per day. Each element is written as a fixed-length 64-bit binary numeral for the midpoint of a cell of its range. The weight of the compatible numerals of a form is then $\mathrm{Vol}_{\mathsf{F}}(\delta)/\prod(\text{ranges})$ up to a grid error of order 2^{-64}, which gives the posterior weight $2^{-(|\mathrm{LAW}_{\mathsf{F}}|_S + \mathrm{Occ}_{\mathsf{F}}(\delta))}$ of Section 10.4.

Minimax errors. Each form is fitted by least squares from several starts, refined by L_p fits with increasing p, and its largest residual is then minimised by sequential linear programming. Table 9 lists the minimax errors and the elements that witness them; each value against plan94 is an upper bound on the true minimax error. A second implementation, an exchange algorithm written independently and started elsewhere, finds the same values to four decimals. The circle's value is exact: its minimax problem is a one-dimensional convex minimisation. For the others, global optimality is not proved. The column "exact" gives the minimax errors against an exact, unperturbed Kepler ellipse (the ellipse's own least-squares elements on these data, $e = 0.0930$, whose largest deviation from plan94 is $1.25'$); so more than half of the vicarious hypothesis's $1.41'$ is perturbation error shared with the ellipse. The eccentric's error against the exact ellipse agrees with the second-order expansions of the two equations of centre, $2e^2\sin 2M$ for the eccentric and $\frac{5}{4}e^2\sin 2M$ for the ellipse, whose difference $\frac{3}{4}e^2$ is about $22.5'$ for Mars.

The law texts. A law text is a sentence $\forall b\,\forall t\,(\theta(b,t) = \ldots)$, or with an existential quantifier as below, in which the elements are unary function symbols $L(b), n(b), w(b), e(b), e_1(b), e_2(b)$ and the mean longitude is written out as $L(b) + n(b)\cdot t$, with t measured from the middle of the window, wherever it occurs (below we abbreviate it by Λ, and $\Lambda - w(b)$ by M; the abbreviations

Table 9: Minimax errors of the Mars pool, 1580–1600 (arcminutes): against plan94, upper bounds witnessed by the elements shown (w and L in degrees, L at the middle of the window; n in radians per day), and against an exact ellipse; k is the number of elements.

form	k	plan94	exact	witnessing elements
circle	2	640.7908	–	$L = 309.45$, $n = 0.00914611$
eccentric	4	23.0508	22.34	$e = 0.184516$, $w = 283.471$, $L = 309.448$, $n = 0.0091461$
equant	4	8.1653	7.43	$e = 0.0933767$, $w = 283.465$, $L = 309.446$, $n = 0.00914613$
vicarious	5	1.4127	0.62	$e_1 = 0.115724$, $e_2 = 0.0707907$, $w = 283.462$, $L = 309.446$, $n = 0.00914609$
ellipse	4	1.0733	0	$e = 0.0930389$, $w = 283.445$, $L = 309.442$, $n = 0.00914609$

are expanded in the counted texts). The syntaxes are: *S1*, with $+, -, \cdot, /, \sin, \cos, \mathrm{atan2}, \sqrt{}$, the numerals $0, 1, 2$, $\forall$, $\exists$, $\wedge$, $=$, $>$; *S2*, S1 with a unit-vector primitive $\mathrm{uc}(\varphi) = \mathbf{u}(\varphi)$, a direction primitive $\mathrm{dir}(\mathbf{x})$ (the polar angle of $\mathbf{x}$) and $\mathrm{rc}(Q, C, \varphi)$, the point where the ray from Q at angle φ meets the unit circle about C; and *S3*, S1 with $\mathrm{kep}(e, M)$, the true anomaly for eccentricity e and mean anomaly M. The texts are (arguments b of the element symbols omitted):

- circle (all syntaxes): $\theta(b,t) = L + n \cdot t$;
- eccentric: in S1 and S3, $\theta(b,t) = \mathrm{atan2}(\sin\Lambda - e \cdot \sin w,\ \cos\Lambda - e \cdot \cos w)$; in S2, $\theta(b,t) = \mathrm{dir}(\mathrm{uc}(\Lambda) + (-e) \cdot \mathrm{uc}(w))$;
- vicarious hypothesis: in S1 and S3, $\exists s\,(s > 0 \wedge s \cdot s - 2 \cdot (e_2 \cdot (s \cdot \cos M)) + e_2 \cdot e_2 = 1 \wedge \theta(b,t) = \mathrm{atan2}(-((e_1+e_2)\cdot\sin w) + s\cdot\sin\Lambda,\ -((e_1+e_2)\cdot\cos w) + s\cdot\cos\Lambda))$; in S2, $\theta(b,t) = \mathrm{dir}(\mathrm{rc}(-(e_1 + e_2) \cdot \mathrm{uc}(w),\ (-e_1) \cdot \mathrm{uc}(w),\ \Lambda))$;
- bisected equant: the vicarious texts with e for e_1 and e_2 in S1 and S3, and with $2 \cdot e$ for $e_1 + e_2$ and e for e_1 in S2;
- ellipse: in S1 and S2, $\exists E\,(E - e \cdot \sin E = M \wedge \theta(b,t) = w + \mathrm{atan2}(\sqrt{1 - e \cdot e} \cdot \sin E,\ \cos E - e))$; in S3, $\theta(b,t) = w + \mathrm{kep}(e, M)$;
- K3 (all syntaxes; with $P = 2\pi/n$): $\forall b\,\forall c\,((n(b) \cdot n(b)) \cdot (a(b) \cdot (a(b) \cdot a(b))) = (n(c) \cdot n(c)) \cdot (a(c) \cdot (a(c) \cdot a(c))))$.

A text is serialised with one opening and one closing bracket per compound, and its length is its number of tokens times $\lceil \log_2(\text{alphabet size}) \rceil$. The alphabets have 31 (S1), 34 (S2) and 32 (S3) symbols, so a token costs 5, 6 and 5 bits. This gives the text lengths of Table 5: 31, 89, 198, 198, 109 and 75 tokens in S1 for the circle, eccentric, equant, vicarious hypothesis, ellipse and K3; 31, 57, 75, 78, 109, 75 in S2; 31, 89, 198, 198, 52, 75 in S3. Every text was evaluated numerically and compared with the model it is attached to, at 40 random elements and times each: all agree to within 1.4×10^{-14} radians. A second serialisation, Polish notation with fixed arities and a separate symbol for unary minus, gives about half these lengths: in S1 75, 195, 440, 440, 245 and 165 bits, in S2 90, 150, 198, 204, 294, 198 and in S3 75, 195, 440, 440, 120, 165; a compact Polish code for K3 with a power symbol and the numeral 3 has 23 tokens, 115 bits.

Occam factors. $\mathrm{Occ}_{\mathsf{F}}(\delta)$ is estimated by hit-or-miss Monte Carlo with the exact model on all 7306 data: points are drawn uniformly in a parallelepiped built from the whitened linearisation of the residuals at the minimax point, enlarged where hits come close to a face, and capped at the element ranges; about 2000 hits per entry give a standard error of about 0.03 bits. The table below gives every value. Two cross-checks support them where the region is far from linear.

Hit-or-miss sampling in an axis-aligned box with the full range of the eccentricities gives, for the vicarious hypothesis, 22.73 ± 0.03, 33.36 ± 0.03, 37.87 ± 0.07 and 43.02 ± 0.24 bits at $720'$, $120'$, $60'$ and $30'$, within about one combined standard error of the table. An independent implementation, with boxes found by its own random-walk explorer, gives 21.16 (ellipse, $720'$), 20.21 (eccentric, $720'$), 21.17 (equant, $720'$), 31.72 (equant, $120'$), 31.69 (ellipse, $120'$), 35.67 (ellipse, $60'$), each ± 0.03 or better, and 20.929 (circle, $720'$, an exact one-dimensional integral): all within 0.06 bits of the table. At $720'$ the compatible regions have a thin part with the apsidal angle opposite the true perihelion and small eccentricity (since the circle fits to $640.8' < 720'$, every apsidal angle is compatible for small e); the parallelepiped at $720'$ spans 6.5 radians in w, and a full-range rerun shows the effect of this part to be within the noise.

δ	$720'$	$120'$	$60'$	$30'$	$20'$	$12'$	$9'$	$6'$	$4'$	$3'$	$2'$	$1.6'$	$1.3'$
circle	20.93	–	–	–	–	–	–	–	–	–	–	–	–
eccentric	20.24	31.17	36.34	45.38	–	–	–	–	–	–	–	–	–
equant	21.20	31.70	35.88	40.35	43.63	49.49	57.34	–	–	–	–	–	–
vicarious	22.77	33.34	37.93	42.83	45.77	49.56	51.78	54.98	58.56	61.53	67.56	74.54	–
ellipse	21.18	31.63	35.68	39.65	42.01	45.01	46.81	49.21	51.86	53.88	57.41	59.93	63.81

Occam factors $\mathrm{Occ}_{\mathsf{F}}(\delta)$ in bits ("–": δ below the form's minimax error).

The MAP form. At these 13 precisions the MAP form is: in S1, the circle at $720'$, the eccentric at $120'$–$30'$ and the ellipse from $20'$ on; in S2, the circle at $720'$, the eccentric at $120'$–$30'$, the bisected equant at $20'$–$9'$, the vicarious hypothesis at $6'$–$1.6'$ and the ellipse at $1.3'$; in S3, the circle at $720'$ and the ellipse from $120'$ on. Below $1.07'$ no form is compatible. In every row the runner-up has at most $2^{-12.4}$ of the MAP's weight. Each K3 variant of a form has exactly $2^{-|\mathrm{K3}|}$ times the form's weight: with one planet K3 is vacuous, and in heliocentric direction data it has no content (Proposition 10.2).

The data without the text prior. With a uniform prior over the five forms, the element prior above and no text prior, the MAP is the eccentric at $720'$ and $120'$ (margins 0.69 and 0.46 bits) and the ellipse at every $\delta \le 60'$, with margins 0.20, 0.70, 1.62, 4.48, 4.97, 5.77, 6.70, 7.65, 10.15 and 14.61 bits at $60'$, $30'$, $20'$, $12'$, $9'$, $6'$, $4'$, $3'$, $2'$ and $1.6'$. These margins are largest just above a competitor's minimax error, where its compatible volume shrinks to zero. They depend on the element ranges: halving the ranges of the eccentricities (the compatible regions lie inside $[0, 0.5)$ at every $\delta \le 120'$) adds one bit of prior density per eccentricity, which moves the vicarious hypothesis by one bit against the four-element forms and the circle by one bit the other way.

Text against data. Form F beats form G iff $|\mathrm{LAW}_{\mathsf{F}}|_S - |\mathrm{LAW}_{\mathsf{G}}|_S < \mathrm{Occ}_{\mathsf{G}}(\delta) - \mathrm{Occ}_{\mathsf{F}}(\delta)$. For the epicyclic syntax's equant and vicarious texts (18 bits apart), the Occam-factor difference moves from 2.14 bits in the equant's favour at $20'$ to 5.56 bits in the vicarious hypothesis's favour at $9'$, so the equant keeps a margin of 12.4 bits at $9'$; text and data contributions are of the same order there. For every other pair of forms the text difference is between 100 and 835 bits, an order of magnitude larger than any Occam-factor difference, except that in S1 and S3 the equant and vicarious texts have equal length; both lose to the ellipse there by more than 400 bits.

Robustness. A variable-length numeral code (an Elias-gamma length prefix per element) adds about 5–11 bits per element; this moves the vicarious hypothesis against the four-element forms by 6–11 bits and the circle against them by 12–20 bits, and changes none of the MAP rows. In

Polish notation the text differences shrink by about half and no MAP row changes; the closest case is the epicyclic syntax's equant and vicarious texts, which then differ by 6 bits (198 against 204), against an Occam-factor difference of at most 5.56 bits. A natural form outside the pool, the equation-of-centre series truncated at second or third order (four elements, like the ellipse), has S1 texts of 680 and 1345 bits, against the ellipse's 545, and minimax errors $4.11'$ and $1.31'$, so it changes no S1 row.

H.4 Graded fit

Example H.3 (Graded fit; a computation)**.** Under dithered rounding every form in the Mars pool of Section 10.4, the ellipse included, is eventually refuted (Proposition G.13); but inside M_A a theory may be a sampler, the law with a Gaussian error model of stated σ (Proposition G.4). At $\sigma = 2'$, roughly Tycho's precision, the expected penalty per independent observation relative to the ellipse is 45.05 bits (eccentric), 4.96 (equant) and 0.035 (vicarious hypothesis), so the text advantages over the ellipse are overcome after 2 and 7 observations (eccentric, S1 and S2), 41 (equant, S2) and about 5,300 (vicarious hypothesis, S2); a simulation agrees. Honest uniform-tolerance theories give 23 to 71 and about 470 data, a conservative estimate, since SAI's mixture prefers a law with a calibrated error model (Proposition 9.2; details below). So with graded fit the data, not the syntax, decide every step within tens of observations, except the epicyclic syntax's choice between the vicarious hypothesis and the ellipse on longitudes, which needs thousands of longitudes at Tycho-level noise.

Gaussian noise. Least-squares fits to the 7306 longitudes have root-mean-square residuals $454.4'$ (circle), $15.81'$ (eccentric), $5.26'$ (equant), $0.628'$ (vicarious hypothesis) and $0.446'$ (ellipse). For a sampler with Gaussian errors of standard deviation σ, the expected log-likelihood penalty per independent observation relative to the ellipse is $(\mathrm{RMS}_{\mathsf{F}}^2 - \mathrm{RMS}_{\mathrm{ell}}^2)/(2\sigma^2 \ln 2)$ bits. A form's text advantage over the ellipse is overcome after (text advantage)/(penalty) observations, which omits the parameter-fitting terms:

form	penalty (bits per observation) $\sigma = 1'$	$\sigma = 2'$	$\sigma = 4'$	observations needed to overcome the text advantage at $\sigma = 1', 2', 4'$
eccentric	180.2	45.05	11.26	S1 (100 bits): 0.6, 2.2, 8.9; S2 (312 bits): 1.7, 6.9, 27.7
equant	19.84	4.96	1.24	S2 (204 bits): 10.3, 41.1, 164.5
vicarious	0.142	0.035	0.0089	S2 (186 bits): 1313, 5253, 21013

In S1 and S3 the ellipse already has the shorter text against the equant and the vicarious hypothesis, by 445 and 730 bits, and in S3 also against the eccentric.

A simulation. For n days drawn at random from 1580–1600 (seeded; 20 runs per cell), the plan94 longitude plus Gaussian noise is fitted by each form by least squares, and the evidence is computed by the Laplace approximation under the element prior above; the noise model's text is common to all forms and cancels. At $\sigma = 2'$, in S1 and S3 the MAP is the ellipse in 20 of 20 runs for every $n \geq 10$ (at $n = 5$, S1's MAP is the eccentric in all runs); in S2 it is the vicarious hypothesis in 20 of 20 runs for $n = 20$ to 2560, in 10 of 20 at $n = 5120$, and the ellipse in 20 of 20 at $n = 7306$. At $\sigma = 1'$ the S2 switch to the ellipse comes between $n = 640$ (vicarious hypothesis in 20 of 20) and $n = 2560$ (ellipse in 20 of 20). The median evidence ratio of the ellipse over the vicarious hypothesis, without text prior, grows from $2^{2.2}$ at $n = 5$ to 2^{276} at $n = 7306$ ($\sigma = 2'$). The simulation agrees with the expected-penalty estimates.

Honest uniform-tolerance theories. Give each form the honest law with tolerance equal to its minimax error (Proposition G.4). For cells much narrower than the tolerances, the cost per datum against the ellipse is about $\log_2(\varepsilon_{\mathsf{F}}/\varepsilon_{\mathrm{ell}})$: 4.43 bits (eccentric), 2.93 (equant) and 0.40 (vicarious hypothesis). The text advantages are then overtaken after 23 (S1) and 71 (S2) data for the eccentric, 70 for the equant (S2) and about 470 for the vicarious hypothesis (S2). These are estimates from the pricing of Section 9, not a separate posterior computation, and they are conservative: SAI's mixture prefers a law with a calibrated error model to the uniform band (Proposition 9.2).

H.5 What K3 is worth

Example H.4 (K3's worth against its cost; a computation)**.** By Proposition 10.3, given the other elements, K3's geocentric content is exactly ρ, and it is large: over 1580–1600 the median of $\alpha(t)$ is 90° (5th and 95th percentiles 9.5° and 170°). Take as data the daily geocentric longitudes of Mars over 1580.0–1600.0 at a common tolerance δ, and Kepler ellipses for the Earth and Mars with $\rho \in [0, 4)$ free or given by K3. With fixed-length numerals for the elements, the posterior odds for K3 are $2^{s_{\mathrm{K3}}(\delta)-|\mathrm{K3}|}$, where $|\mathrm{K3}|$ is the length of its text and $s_{\mathrm{K3}}(\delta) := \log(4\,\mathrm{Vol}_8(\delta)/\mathrm{Vol}_9(\delta))$ ($\mathrm{Vol}_8, \mathrm{Vol}_9$ the volumes of the compatible element regions with and without K3) is what K3 saves (the method is described below). A computation gives $s_{\mathrm{K3}} = 11.3$, 12.3, 13.4, 14.7 bits at $\delta = 8'$, $4'$, $2'$, $1'$, about one more bit per halving of δ, as for any real-valued element; K3 costs $0.005'$ of fit. Its text costs 375 bits in the syntax S1 of Section 10.4 (brackets counted), 165 in Polish notation and 115 in a compact Polish code; it is paid once and saves $s_{\mathrm{K3}}(\delta)$ bits per body whose size it fixes. By Proposition G.33 (with the law in place, at exchange rate 1; a comparison of law choices, not membership of a steeper MAP), K3 pays once it fixes the sizes of about 25–33, 11–15 or 8–10 bodies in the three codes, while the five planets other than the Earth give about 55–75 bits if each saves about as much as Mars (an extrapolation; we computed Mars only). So in none of these codes does compression alone select K3 at Kepler's precision. This is conditional on the code: under the universal prior the relevant cost is the complexity of K3 given the rest of the theory, which these computations do not determine.

Setting. The data are those of Example H.4 (Appendix H.2). The hypotheses are Kepler ellipses for the Earth (in the ecliptic, $a = 1$) and for Mars (its plane fixed at the best heliocentric plane of Appendix H.2), each with elements e, w, L, n, and either $\rho = a(\mathrm{Ma})/a(\mathrm{Ea})$ as a ninth free element in $[0, 4)$ (without K3) or $\rho := (n_{\mathrm{Ea}}/n_{\mathrm{Ma}})^{2/3}$ (with K3, eight elements). As in Appendix H.3, the numerals are of fixed length, which gives the posterior odds $2^{s_{\mathrm{K3}}(\delta)-|\mathrm{K3}|}$ of Example H.4, with Vol_8 and Vol_9 the volumes of the exact compatible regions with and without K3.

Fit. The minimax geocentric error is $0.627'$ with ρ free (fitted $\rho = 1.523676$, against the K3 value 1.523687 from the fitted periods) and $0.632'$ with K3; an independent fit gives $0.6266'$ and $0.6318'$ with the same values of ρ. So K3 costs $0.005'$ of fit and is compatible with the data at every $\delta \geq 0.632'$. These minimax values are upper bounds witnessed by elements. The data are values of a model, so plan94's own error does not affect the comparison.

Method. With $y := \rho - (n_{\mathrm{Ea}}/n_{\mathrm{Ma}})^{2/3}$, a shear of unit Jacobian turns the volume ratio into a density: $\mathrm{Vol}_8/\mathrm{Vol}_9 = p_y(0)$, the density at 0 of y under the uniform distribution on the nine-dimensional compatible region. That distribution is sampled by hit-and-run (Smith, 1984), with stepping-out brackets in the manner of slice sampling (Neal, 2003) and the exact membership

test on all data; $p_y(0)$ is estimated as the fraction of samples with $|y| < h$ divided by $2h$, for $h = 0.05$, 0.1 and 0.2 standard deviations of y. Box sampling cannot be used here: the linearised compatible region fills a fraction of 10^{-5} or less of its bounding box.

δ	$8'$	$4'$	$2'$	$1'$
$s_{\mathrm{K3}}(\delta)$ (bits)	11.28 ± 0.04	12.31 ± 0.03	13.39 ± 0.03	14.73 ± 0.04
second implementation	11.27–11.32	12.28–12.32	13.32–13.35	14.69–14.71
standard deviation of y	$6.47 \cdot 10^{-4}$	$3.10 \cdot 10^{-4}$	$1.47 \cdot 10^{-4}$	$5.54 \cdot 10^{-5}$
$\log_2\left(4/(\sqrt{2\pi}\,\mathrm{sd}(y))\right)$	11.27	12.33	13.41	14.81

The fourth row is the value for a Gaussian y, as a check. The errors in the first row are one standard error over the 492–583 independent chains (of 600) that start at exactly compatible points. The three kernel widths agree to within 0.03 bits, and the first and second halves of the kept steps agree to within 0.06 bits except at $8'$ (11.33 against 11.23), where we quote 11.3 ± 0.1. The second implementation, written independently and run with three kernel widths, is a hit-and-run on the linearised compatible polytope with exact chords (60,000 samples per δ); its shares of linearised samples that are exactly compatible, 0.819, 0.901, 0.958, 0.968, match the first implementation's 0.820, 0.895, 0.948, 0.972. The sampler failed to bracket or to accept in 38, 5, 1 and 0 of about 128,000–152,000 chain steps per row; the direct effect is at most about 0.01 bits at $8'$. A cruder estimate from the single most sensitive day (9.8, 10.8, 11.8 bits at $8'$, $4'$, $2'$) is lower, as it should be. s_{K3} includes $\log_2$ of the range of ρ ($[0, 4)$, 2 bits); the range $[0, 2)$ would lower every entry by one bit.

Against the text. |K3| is 375 bits in S1 with brackets counted (450 in S2), 165 bits in Polish notation and 115 bits in the compact Polish code (Appendix H.3). For one pair of planets the posterior odds against K3 at $8'$ are therefore about 2^{-364} (S1) to 2^{-104} (compact code). By the criterion of Proposition G.33(b), when the same bodies' elements are in the theory with and without K3, K3 pays, with the law in place, iff the number $|S|$ of bodies whose size it fixes satisfies $s_{\mathrm{K3}}|S| > |\mathrm{K3}|$; at $s_{\mathrm{K3}} = 11.3$ and 14.7 bits per body this needs $|S| > 33.2$ and 25.5 (375 bits), 14.6 and 11.2 (165 bits), 10.2 and 7.8 (115 bits). The extrapolation to the five planets other than the Earth (about 55–75 bits) assumes that each saves about as much as Mars; the other planets were not computed.

H.6 Distances as data

The data are those of the Mars pool together with, for every day, the sentence "$r(t_i)$ lies within a factor $1 \pm \tau_{\mathrm{d}}$ of r_i", with r_i the plan94 heliocentric distance in the best plane. Each form gets one more element, its scale. An entry of Table 10 is the least longitude tolerance found at which a form, with suitable elements and scale, proves every longitude datum and every distance datum. The search combines sequential linear programming from many starts: earlier optima, perturbations of them, the optimum at the neighbouring τ_{d} in a decreasing and in an increasing sweep, and six seeded random starts; a second, independently written search was run as well, and each entry is the best value found by either. Every entry is witnessed by printed elements and was re-checked with the same residual function; none is proved minimal. The circle about the Sun is not in this computation. The equant and ellipse rows come from a coarser search; their entries at 3%, 1.5% and 0.75% follow by monotonicity (elements feasible at a smaller τ_{d} are feasible at a larger one, and no form beats its longitude minimax), and the independent search confirms them:

at its longitude minimax the ellipse needs only a distance tolerance of 0.0153% and the equant 0.2487%.

Table 10: Distances as data: the least longitude tolerance found (arcminutes, against plan94) at which a form proves every longitude datum and every distance datum at relative distance tolerance τ_{d}; "none": no elements meeting τ_{d} were found; r^*: the least distance tolerance found with no longitude constraint. Every entry is an upper bound on the least tolerance, witnessed by elements; minimality is not proved.

form	r^*	5%	3%	2%	1.5%	1%	0.75%	0.5%	0.2%	0.1%	0.05%
eccentric	0.226%	150.73	218.31	252.45	269.56	286.70	295.32	305.33	none	none	none
equant	0.222%	8.17	8.17	8.17	8.17	8.17	8.17	8.17	none	none	none
vicarious	0.0164%	1.41	1.41	2.01	3.54	5.12	5.97	6.95	23.74	94.95	132.04
ellipse	0.0117%	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07	1.07

What the distances decide. At $\tau_{\mathrm{d}} = 1\%$ the least longitude tolerance at which the vicarious hypothesis fits all the data is at most $5.12'$ (witnessed by elements; two independent searches found nothing lower), while the ellipse fits both kinds of data at $1.07'$. So for δ below the former only the ellipse is compatible, in every syntax; with longitudes alone, only below $1.41'$. Compatibility alone thus eliminates the circle-based vicarious hypothesis once the data language contains distances: this reproduces the abandonment of circular orbits, not Kepler's last step from an oval to the ellipse, which is outside the pool.

The witness at $\tau_{\mathrm{d}} = 1\%$. The elements $e_1 = 0.102835781$, $e_2 = 0.0837969992$, $w = -1.33587144$, $L = -0.882335245$ (radians), $n = 0.00914611825$ radians per day and scale 1.52509743 fit every distance of the vicarious hypothesis to within 1.00000% and every longitude to within $5.119'$. The fit is sensitive to rounding: with the elements rounded to six decimals (the rounding of n to 0.009146 is the largest change) the longitude error rises to $6.2'$. The vicarious entry at 0.05% ($132.04'$) is the independent search's; the other search found $132.35'$ there.

Reading. Distances order the forms differently from longitudes. The bisected equant fits the distances to within 0.5% without giving up any longitude accuracy, and to 0.22% at best. The eccentric circle fits distances alone to 0.23%, but not with the elements its longitudes need: the distance fit wants a smaller eccentricity ($e = 0.142$ at 5%, against 0.185 for the longitude fit). The vicarious hypothesis fits distances alone to 0.016% (with $e_1 \approx e_2$ it is close to the equant), but not together with its longitude fit. The ellipse fits both at once, at its longitude minimax, for every τ_{d} down to 0.05%. Not computed: the Occam factors with distance data, and hence the MAP form above the thresholds; a lower bound on any least tolerance (it would need, for example, interval arithmetic over a covering of the element space); and the precision of Kepler's own distance determinations, so the table does not say which τ_{d} was historically available.

H.7 Reproducing the computations

The scripts in the accompanying repository are deterministic or seeded (seed 20261015 for the residual fits and the Mars pool, 20261016 for the graded-fit and distance computations) and were run with Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 and pyerfa 2.0.1.5. Their saved outputs contain every computed number of Section 10 and of this appendix, except the two sampler

figures (the bounding-box fraction and the number of chain steps), which follow from the script's parameters; the other figures are from the literature cited. Re-running reproduces them apart from timing lines; the Mars-pool posterior with the value of K3 takes about 80 minutes on one core, the distance search about 36 minutes, the residual fits about 2 minutes and the graded-fit computations under a minute. The independently written implementations quoted above are in the same repository.

I Notation

The table lists the symbols used in more than one place, with the place where each is defined. Symbols local to one proof, example or subsection are defined where they occur. Logarithms are to base 2 and losses are in bits; $c, c', \dots$ are constants that depend only on the fixed universal machines, the coding of syntax and the background B, unless a statement says otherwise.

symbol	meaning	where
Algorithmic information and probability		
V; K, $K(\cdot \mid \cdot)$	the fixed universal prefix machine; prefix complexity	§ 2.1
$\mathbf{m}$	universal discrete semimeasure, $\mathbf{m}(i) = \sum_{V(p)=i} 2^{-\lvert p\rvert}$	§ 2.1
Ω_V; $\Omega_V^{\geq \ell}$	halting probability of V; halting mass of the V-programs of length at least ℓ	§ 2.1
$K^{\mathrm{pad}}(n)$	$\min_{n' \geq n} K(n')$, nondecreasing and unbounded	§ 2.1
Kt; K^q	Levin's complexity $Kt(x) = \min\{\lvert p\rvert + \log t : V(p) = x \text{ within } t \text{ steps}\}$; the least $\lvert p\rvert$ with $V(p) = x$ within $q(\lvert x\rvert)$ steps, q a polynomial	§ 7.2; § 7.6
$\mathbb{U}$; $[w]$; $\omega_{1:n}$	fair-coin measure on tapes $\omega \in \{0,1\}^\infty$; cylinder of a finite string w; first n bits of ω	§ 2.1
$\mathbb{P}$, $\mathbb{E}$; $\mathbf{1}[\cdot]$; KL	law of a data process and its expectation; indicator; Kullback–Leibler divergence in bits	Def. 9.1; Thm. A.12
$\asymp$	equality within a factor $2^{\pm c}$ on every finite data set	§ 4
Logic		
$\mathcal{L}$, Sent; B; $\mathrm{Cn}(\Gamma)$	language, its sentences; background theory (decidable, consistent); $\{\varphi : B \cup \Gamma \vdash \varphi\}$	Def. 2.1
φ^1, φ^0; "decides"	φ, $\neg\varphi$; Γ decides φ if it proves φ or $\neg\varphi$	Def. 2.1
W_e	the e-th r.e. set of sentences	§ 2.1
$\mathcal{L}_{\mathrm{ar}}$, $\underline{n}$; Q, PA	the language $\{0, S, +, \cdot, <\}$, numerals; Robinson and Peano arithmetic	§ 5
Π_1^*; "B true"	conjunctions of Π_1 sentences and negations of Σ_1 sentences; B has a model with $\mathcal{L}_{\mathrm{ar}}$-reduct $\mathbb{N}$	§ 5
Ind_B	sentences that B decides neither way	§ 4.4
$[T]$; $K_{\equiv}(T)$; $\mathbf{m}_{\equiv}(T)$	indices e with $\mathrm{Cn}(\mathsf{W}_e) = \mathrm{Cn}(T)$; $\min_{e \in [T]} K(e)$; $\sum_{e \in [T]} \mathbf{m}(e)$	Def. 2.8
$\mathrm{Ext}(T)$; $\mathbf{m}(\mathrm{Ext}T)$	indices of the consistent theories that extend $\mathrm{Cn}(T)$; their total $\mathbf{m}$-weight	Def. 2.8
K_{enum}, $\mathbf{m}_{\mathrm{enum}}$, $\mathbb{U}_{\mathrm{enum}}$	enumeration complexity of a set, its universal discrete semimeasure, its probability under the universal theory machine	App. A.6

symbol	meaning	where
Data, conventions and questions		
D, $\operatorname{dom} D$; Y_3, ι	data set (finite partial function to $\{0,1\}$); three-valued labels $\{1,0,\iota\}$, ι "independent"	Def. 2.1
$D^{\pm}$; ψ_D; $D+(\varphi,y)$	the literals of D; their conjunction; one more datum	Def. 2.1
D_t, D_∞; $\nu(y \mid D,\varphi)$	prefixes of a reveal sequence, a stream; conditional, with the repeated-input convention	Def. 2.1
fresh atoms; $\mathtt{e}$, $\mathtt{s}_0$, $\mathtt{s}_1$, R, t_w, φ_w	the literal setting and language; $\varphi_w := R(t_w)$	Def. 2.2
D^X, f_X; $\mathfrak{T}$; fit class	literal sequence and labeller of a set of words X; the named structure; the sets X on whose D^X an inducer has bounded loss	Def. 2.2
conventions	semimeasure (primary), normalised and three-valued loss conventions	Def. 2.5
$\mathcal{Q}$; (α,ℓ); (E3)	question (pairwise B-incompatible direct answers); conditional question; its encoding	Def. 2.10
$\mathrm{Un}_D(\mathcal{Q})$	posterior probability that the random theory proves no direct answer	Prop. D.2
Theory machines and the one-tape law		
G; A^G_w, A^G_ω	theory machine (c.e. set of pairs (prefix, sentence)); axioms emitted under a prefix, on a tape	Def. 2.3
O^G_χ; Cons^G; E^G_D	tapes on which the theory proves χ; on which it is consistent; consistent and proving ψ_D	Def. 2.3
ν_G, ν^3_G; tri^G_ω; $K(G)$, $K(\nu)$	law and three-valued law of G; trichotomy function; complexities of a machine and a law	Def. 2.3
G_U, A_ω, O_χ, Cons, E_D, E^3_D	the universal theory machine and its events	Def. 2.4
M_A; M^3_A; L_A	SAI, the one-tape proof-requiring law ν_{G_U}; its three-valued law; $-\log M_A$	Def. 2.4
Inc_0	probability that the random theory is inconsistent	Def. 2.4
Bel_D, Dis_D, Indep_D	posterior probabilities that the random theory proves, refutes, decides neither	Def. 2.4
Bel_0; $\widehat{\mathrm{Bel}}_0 = \mathrm{Bel}_\emptyset$; Pl_0, Pl_D	prior probability of provability (unnormalised); its normalisation; plausibilities	Def. 2.4
$\mathbb{U}_A(\mathcal{C};D)$; $\mathbb{U}_A[T]$; $\mathbb{U}_A(\mathrm{Ext}T)$	class-restricted laws: a class of theories; the class of T; the consistent extensions of T	Def. 2.4
E_∞; P_∞	$\bigcap_n E_{D_n}$; the limit posterior $\mathbb{U}(\cdot \mid E_\infty)$	§ 3.7
$\varepsilon_\emptyset$	mass of the empty theory	Def. 2.9
M^{nc}_A, $\mathrm{Bel}^{\mathrm{nc}}_D$, $\mathrm{Dis}^{\mathrm{nc}}_D$	consistency-requiring weight and predictor	Def. 2.9
$M^{1/2}_A$; Ξ	graded consistency-requiring weight; generative law of unlabelled data	Def. A.20; Def. A.36
Versions of SAI		
W_A, K_A; W^{dec}_A, K^{dec}_A	coin-free and Kolmogorov versions over enumerators; over deciders	Def. 2.7
K_{fin}, W_{fin}	the same over single sentences (finite axiomatisations)	Def. 2.7
$K^{\mathrm{gen},t}_A$	least length of a delay-form enumerator compatible with the data	Def. 2.7
M^{dec}_A	one-tape mixture over random deciders	Def. 4.1

symbol	meaning	where
$M_A^{\mathrm{chk},t}$, $M_A^{\mathrm{gen},t}$; $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$; $W_A^{\mathrm{Kt},t}$	laws with membership-time or generation-time penalties on axioms, one-tape and coin-free; Levin's penalty	§ 4.2, Def. B.7, and Prop. B.23
W_A^3	coin-free three-valued SAI	§ 4.4
(S), ($\mathrm{S}_{\mathrm{tape}}$), (F1)	self-simulation assumptions; closure under renaming of parameters	Ass. B.5, B.6 and B.30
$W_A[t,s]$; W_A^{a1}, W_A^{a2}, W_A^b, W_A^{b2}	charged deduction: derivation size s; readings (a1), (a2), (b), (b2)	§ 4.6 and App. B.6
$M_A^{\mathcal{L}_O}$, $W_A^{\mathcal{L}_O}$, $K_A^{\mathcal{L}_O}$	hypotheses restricted to observational axioms	Def. 8.1
The operator-induction side		
F, f_ω^F, ν_F; F_U, f_ω; M_F	function machine, its random labeller and law; universal ones; one-tape SOI	Def. 4.1
Γ_f; consistent, closed	$\{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$; labellers consistent with B, closed under consequence	Def. 4.1
M_F^{cons}, M_F^{cl}; M_F^{mw}	one-tape consistent and closed SOI; machines consistent on every tape	Def. 4.1; Prop. B.21
W^{cons}, W^{cl}, W^{all}; K_F^{cons}, K_F^{cl}	coin-free consistent, closed and plain SOI on sentences; Kolmogorov versions	Def. 4.1
W, K_F	the companion paper's coin-free and Kolmogorov versions on strings	Def. 4.1
$M_F^{3,\mathrm{cons}}$, $W^{3,\mathrm{cons}}$	three-valued consistent SOI, one-tape and coin-free	§ 4.4
$W^{\mathrm{cons},\tau}$, $W^{\mathrm{all},\tau}$; $M_F^{\mathrm{cons},\tau}$, $M_F^{\mathrm{all},\tau}$	clocked consistent and plain SOI, coin-free and one-tape	§ 4.5
W^{cert}; M_{prod}	certificate SOI; the companion paper's product mixture	App. B.6; Cor. B.12
Random theories and arithmetic		
Win, Lot_k; G_{lot}	lottery relation and sentences; the random lottery theory machine	§ 5.1
$\mathrm{Tick}_{k,j}$, $\mathrm{Lot}_k^{\mathrm{ar}}$	arithmetic tickets and lotteries	Thm. 5.4
G_{thr}	the threshold machine	Thm. 5.6
$\mathrm{Kgt}_m(x)$	the Π_1 sentence "$K(x) > m$"	§ 5.3
$\mathcal{U}$; $P_D(\mathcal{C})$	the Σ_1-unsound theories; posterior probability of a class	§ 5
Bit_k; ξ	sentences for the binary digits of Ω_V; the exponent of the lower bound on true arithmetic	App. C.7; Thm. C.30
Theory shape		
m; x_i, y_j	number of base variables; base variables, auxiliary variables (intermediate concepts)	Def. 7.1
Φ; $\mathrm{proj}\,\Phi$; $\mathcal{E}_m^{\mathrm{base}}$, $\mathcal{E}_m$	explicit or linear theory; its projection; classes without and with auxiliary variables	Def. 7.1
$\mathcal{S}$; $\mathcal{Q}(\mathcal{S})$	law; its conditional questions	Def. 7.1
$\mathcal{R}$; UP_k, UP, GE, PS_∞; FC_k	reasoner; unit propagation, elimination, proof search; forward chaining	Def. 7.1; Prop. E.25
$K_A^{\mathcal{R}}$, $W_A^{\mathcal{R}}$; $K_{\mathrm{lst}}^{\mathcal{R}}$, $W_{\mathrm{lst}}^{\mathcal{R}}$; $Kt_A^{\mathcal{R}}$	program length; listing (the theory charged as written); time-charged	Def. 7.1
$\mathrm{PI}(\mathcal{S})$; d_*	prime implicates; least weight of a nonzero derived equation	§ 7.2; § 7.3

symbol	meaning	where
Par_m^b; $\mathcal{H}_q$; ia_k	parity law; hidden-state law; inverse-Ackermann rows	§§ 7.4 to 7.6
Theoretical terms		
$\mathcal{L}_O$, $\mathcal{L}_T$; $T_O(A)$; $\mathcal{C}_\Theta$	observational and theoretical vocabulary; observational content; class of theories with content in Θ	Def. 8.1; Def. F.2
$\mathrm{Th}_{\mathfrak{T}}$; about, compatible with the named structure	the theory of $(\mathfrak{T}; \mathsf{e}, \mathsf{s}_0, \mathsf{s}_1)$; two relations of a theory to it	Def. F.1
C; $\mathcal{L}^+$; Inf_n	binary theoretical relation; three-sorted extension; "there are at least n things"	§ 8.3; Thm. F.33; Thm. F.28
Approximate laws, explanation, the Kepler example		
$(\mathcal{Q}, a)$; $\gamma(\mathcal{Q})$	answered datum; literal cost $\log\lvert\mathcal{Q}\rvert$	Def. 9.1
$\ell(\chi)$, $\lVert\Gamma\rVert$; Exc^Q, Exc^F, Exc^R	code lengths; exception codes	Def. 9.1
λ; J_λ; λ-MAP	steepness; $\lambda\lVert T\rVert + \mathrm{Exc}_D(T)$; its minimisers	Def. 9.1
T_0, T^ε; $e(t)$, ε^*	exact and honest laws; error and its essential supremum	Def. 9.1
δ, δ_0; Δ; $\mathrm{pk}(e)$	data precision, measurement-error bound; dithered cell width; peakedness	Def. 9.1
regimes G, R; p_Δ, s_Δ	guaranteed tolerances, dithered rounding; $\mathbb{E}\min(\lvert e\rvert/\Delta, 1)$, $\min(2\varepsilon^*/\Delta, 1)$	Defs. 9.1 and G.1
P_T^R; A_T; J_λ^Q, J_λ^F, J_λ^R	probability of the data under uniform completion of the law's intervals; answers T allows; J_λ with the three exception codes	Def. G.1
$\mathrm{BC}_\mathcal{G}$, $\mathrm{HY}(\rho)$	band-confined completions	§ 9.3
$r_\mathcal{F}(\Delta)$; h; CRPS	best rate in regime R, the price of a calibrated completion; differential entropy in bits; continuous ranked probability score	Prop. 9.2; App. G.9
$\mathrm{EP}_D(\mathcal{A}; T)$	explanatory power of a module	Def. G.29
μ, η, μ_η	raw world, extractor, induced law on question data	App. D.3
$\mathcal{L}_{\mathrm{hel}}$, $\mathcal{L}_{\mathrm{dir}}$, $\mathcal{L}_{\mathrm{geo}}$; K12, K3	Kepler data languages; Kepler's first two laws, third law	Def. 10.1
$\mathrm{Occ}_\mathsf{F}(\delta)$; S1, S2, S3	Occam factor of a form of the Mars pool; syntaxes of the law texts	§ 10.4